Advanced Corporate Finance for MBA Leaders

Advanced Valuation, Credit, Governance, Private Markets and Transactions

Volume 2 of a Three-Volume Textbook for MBA Students

About This Volume

Volume 2 assumes the material of Volume 1 and goes deeper: advanced valuation and real options, credit and structured finance, governance and incentives, private markets, complex transactions, tax, climate finance and quantitative methods. All companies, figures and cases are fictional and illustrative unless stated; they are for teaching only and are not investment, legal or tax advice. Figures were computed by the authors' working and should be verified before use in professional decisions; legal and regulatory descriptions are general and should be checked against current rules.

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Part I — Advanced Valuation and Real Options

Chapter 1: Advanced DCF: APV, Capital Cash Flow and Valuation in Complex Settings

Learning Objectives

After completing this chapter you will be able to:

- Derive the adjusted present value (APV), weighted average cost of capital (WACC) and capital cash flow (CCF) methods from a single set of assumptions, and explain exactly when they give identical answers and when they must diverge.
- Select the correct discount rate for interest tax shields under the Modigliani-Miller, Harris-Pringle and Miles-Ezzell financing policies, and relever betas consistently with the policy chosen.
- Value a business whose capital structure changes over time, including a scheduled debt paydown, using APV with year-by-year tax shields rather than a single constant WACC.
- Incorporate interest-deductibility limits, disallowed-interest carryforwards and similar tax frictions into a tax shield forecast and judge how much of the nominal shield is really worth.
- Diagnose and resolve circularity in valuation models, distinguishing circularity that is only a spreadsheet nuisance from circularity that signals a conceptual inconsistency.
- Extend APV to financial distress costs, issuance costs and subsidized financing, and use it to frame practical capital structure decisions.

Opening Vignette

Illustrative scenario. Brightwater Logistics is a fictional company.

Brightwater Logistics, a regional freight and warehousing group, has agreed in principle to buy a cold-chain operator for what its board calls “a fair price.” The deal team has done what Volume 1 taught: forecast free cash flow, estimate a WACC from peer data using a 35 percent target debt-to-value ratio, and discount. The answer is a little over 2.4 billion. Then the sponsor’s bankers arrive with a financing plan that looks nothing like the target ratio. Brightwater will borrow heavily on day one, repay roughly a fifth of the debt every year from operating cash flow, and return to a leaner, steady-state structure by year five. The bankers’ own model values the same business at a different number, and the two models disagree by more than the entire projected tax shield.

The corporate development director, Imani, asks the question that exposes the problem. “If the debt balance falls every year, what is our WACC in year three? In year four? Why would a single number apply to all of them?” Nobody has a good answer. The treasury analyst adds a second complication: the credit agreement limits deductible interest to a fraction of EBITDA, so in the early, highly levered years some interest will not be deductible at all. A third complication arrives from the modeling team. Their spreadsheet shows a circular reference warning every time the debt schedule links to the valuation, and the usual workaround, hard-coding the debt-to-value ratio, quietly assumes the answer.

Each complaint is legitimate, and each shows the same underlying fact: WACC is a shortcut that bundles an operating valuation and a financing valuation into one discount rate. When financing is simple and stable, the shortcut is accurate and convenient. When financing is complicated, the bundle comes apart. This chapter takes the bundle apart deliberately, values

the operations and the financing side effects separately, shows how to put them back together, and equips you to know which tool to trust in each setting.

1.1 The Limits of the Standard WACC Model

Volume 1, Chapter 11 presented WACC as the discount rate that converts expected free cash flow to the firm (FCFF) into enterprise value:

```
V = Σ FCFF_t / (1 + WACC)^t
WACC = (E/V) · ke + (D/V) · kd · (1 − Tc)
```

The formula is neat, but it hides three assumptions that are easy to forget because they are never written on the page.

Assumption 1: the weights are stable. WACC uses market-value weights E/V and D/V. If those weights change over the forecast, so does the WACC, and a single constant rate is wrong. In an acquisition financed by a leveraged loan that is paid down, the weights drift every period.

Assumption 2: the debt policy is one the formula understands. Every WACC derivation assumes some rule for how debt moves with value. The textbook formula assumes debt is rebalanced to keep D/V constant, so that the debt balance rises and falls with firm value and the tax shield is as risky as the firm's operating assets. If instead the company commits to a fixed debt schedule, the tax shield is nearly as safe as the debt itself and should be discounted at a much lower rate. Using the textbook WACC on a fixed-schedule firm gives the wrong value even if every input is perfectly estimated.

Assumption 3: the tax shield is simple. The term `kd · (1 − Tc)` assumes every dollar of interest is deductible at the full marginal rate in the year it is paid. Real tax codes impose caps on deductible interest, require taxable income to absorb the deduction, and sometimes allow carryforwards. A shield that cannot be used is worth nothing, and a shield that is deferred is worth less than a shield taken now.

None of this means WACC is wrong. It means WACC is a derived formula, and a derived formula is only as general as its derivation. The deeper method is to ask what the tax shield is worth, directly, rather than to infer it through an adjusted discount rate. That is the logic of APV.

1.1.1 Valuation as additive pieces

Modigliani and Miller's first proposition says that, absent taxes and frictions, firm value is independent of capital structure. Their second paper added corporate taxes and showed that the tax deductibility of interest creates a gain from leverage. The natural valuation decomposition follows:

```
Value of levered firm = Value of unlevered firm
                        + PV(interest tax shields)
                        − PV(financial distress and agency costs)
                        + PV(other financing side effects)
```

The first term is an operating valuation: what the assets are worth if financed entirely by equity, discounted at the unlevered cost of capital ku. The remaining terms are financing effects, each valued with a discount rate appropriate to its own risk. This is the APV method, associated with Stewart Myers, who proposed it in the mid-1970s as a way to value projects with complex financing. Its virtue is modularity: each piece answers one question, can be audited separately, and can be changed without rebuilding the whole model.

1.1.2 The three engines and the one idea behind them

The WACC method discounts FCFF at a rate adjusted for the tax shield. The APV method discounts FCFF at ku and adds the separately valued shield and other side effects. The capital cash flow (CCF) method adds the shield to the cash flow, CCF = FCFF + interest tax shield, and discounts at a rate matched to the combined stream. All three rest on one idea: the interest tax shield is a cash flow to the firm's capital providers like any other, and the methods differ only in where it is placed: the denominator (WACC), a separate present value (APV) or the numerator (CCF). With consistent assumptions they must agree, and verifying that agreement is the best audit of a complex valuation.

1.2 Unlevered Value and the Foundation of APV

The first step of APV is always the unlevered value, which requires an unlevered cost of equity ku. This number is not directly observable, so it is built from the observed betas of comparable firms.

1.2.1 Estimating ku

The unlevered cost of capital is the expected return on the firm's operating assets, which are financed by both debt and equity holders. It equals the weighted average of the expected returns on the firm's debt and equity before any tax effect:

```
ku = (E/V) · ke + (D/V) · kd        (pre-tax weights; V = E + D)
```

The same relation holds for betas: βu = (E/V) · βe + (D/V) · βd. In practice analysts often assume βd = 0 or use a small positive number reflecting credit spread risk. At high leverage the assumption matters, because a debt beta of zero forces all of the firm's systematic risk onto the equity.

Two refinements deserve mention. First, the comparable firms' leverage is itself changing, so unlevering with current market D/E is only an approximation to the average over the period. Second, the unlevering formula depends on the debt policy assumed, a point we return to in Section 1.4 when we compare financing policies.

1.2.2 Why ku is the right rate for operating cash flows

Leverage re-slices the same operating risk between debt and equity holders; it does not change the risk of the assets, so it does not change the rate that discounts operating cash flows. Comparing a division's unlevered return on capital with ku isolates operating performance from the parent's financing policy, whereas comparing it with a WACC blurs the two.

Example 1.1 – Constant debt in perpetuity: APV, WACC and CCF in agreement. Fiddlewood Components, a fictional firm, generates an expected free cash flow to the firm of 100 per year forever (no growth). Its unlevered cost of equity is ku = 10%. It issues permanent debt of D = 400 at kd = 5% and pays a 25% corporate tax rate. Because the debt amount is fixed, the tax shield is as safe as the debt, so we discount it at kd (Modigliani-Miller 1963 assumption).

Step 1: Unlevered value.

```
VU = FCFF / ku = 100 / 0.10 = 1,000
```

Step 2: Tax shield. Annual interest is 0.05 × 400 = 20; the annual shield is 0.25 × 20 = 5. Discounted at kd as a perpetuity:

```
PV(TS) = 5 / 0.05 = 100        (equivalently Tc × D = 0.25 × 400)
```

Step 3: APV.

```
VL = 1,000 + 100 = 1,100
E  = 1,100 − 400 = 700
```

Step 4: Cost of equity. With fixed debt, the Modigliani-Miller formula with taxes gives

```
ke = ku + (1 − Tc)(ku − kd)(D/E)
   = 0.10 + 0.75 × 0.05 × (400/700)
   = 0.10 + 0.02143 = 12.143%
```

Step 5: WACC.

```
WACC = (700/1,100) × 12.143% + (400/1,100) × 5% × 0.75
     = 7.727% + 1.364% = 9.091%
```

Step 6: Check by WACC valuation. VL = 100 / 0.09091 = 1,100.0. The two methods agree exactly.

Step 7: CCF. The capital cash flow is 100 + 5 = 105 per year. Its risk is a blend: 100 at ku and 5 at kd. The value is 100/0.10 + 5/0.05 = 1,000 + 100 = 1,100. Discounting the whole 105 at a single rate would require that rate to be 105/1,100 = 9.545%, which is neither ku nor kd but a weighted blend. This is the first hint that the CCF method needs a discount rate matched to the mixed stream whenever the shield is safer than the operations.

A common mistake. Suppose an analyst uses WACC = 9.091% correctly but, forgetting that ke contains the (1 − Tc) term, computes ke = ku + (ku − kd)(D/E) = 12.857%. The resulting WACC is 0.636 × 12.857% + 0.364 × 3.75% = 8.182% + 1.364% = 9.545%, giving V = 100 / 0.09545 = 1,047.6, which understates the tax benefit by more than half. Internal consistency between ke and WACC formulas is not optional.

Takeaway. In the simplest case all three engines agree. The value of APV is not that it gives a different answer in this case; it is that its logic remains valid when the WACC shortcut stops being valid.

1.3 Debt Policy and the Discount Rate for Tax Shields

The central technical question in APV is the discount rate for the interest tax shield. The correct answer depends on one thing only: how the firm sets its debt level. The tax shield in year t equals Tc × kd × D(t−1), and its riskiness is the riskiness of the debt balance D(t−1) as a forecast variable, plus the risk that the firm will not be able to use the deduction.

1.3.1 Policy one: fixed debt schedule (Modigliani-Miller 1963)

If the firm commits to a predetermined debt schedule, independent of how the business performs, then the future debt balances are known today and the tax shields are known cash flows as safe as the interest payments themselves. They should be discounted at the cost of debt kd. This is the right policy for a leveraged buyout with an amortization schedule written into the credit agreement, for a project financed by a bond with a sinking fund, or for a firm that issues a long-dated bond and does not touch it. The perpetuity result Tc × D is the special case of a permanent fixed balance.

The risk adjustment here is subtle. The shield is not literally riskless: if the firm defaults, the shield disappears, and in practice the firm must have taxable income to use the deduction. The cost of debt kd already contains a default premium, so discounting at kd, rather than at the risk-free rate, partially captures this. For most investment-grade firms the approximation is reasonable.

1.3.2 Policy two: continuous rebalancing to a target ratio (Harris-Pringle)

If the firm adjusts debt continuously to keep D/V at a target L, then the debt balance moves with firm value. When firm value falls, the firm repays debt; when it rises, the firm borrows more. The tax shield therefore rises and falls with the operating performance of the firm, and it inherits the systematic risk of the operating assets. Harris and Pringle (1985) argued that the appropriate discount rate for the shield is then ku. Under this assumption:

```
WACC = ku − kd · Tc · L
ke   = ku + (ku − kd) · D/E
```

Note the difference from the Modigliani-Miller formula: the tax term does not appear in the cost of equity because the tax shield has the same risk as the assets, so leverage does not compound its effect through a tax-adjusted spread. The WACC reduction is simply the tax shield per dollar of value, kd × Tc × L.

1.3.3 Policy three: annual rebalancing (Miles-Ezzell)

Miles and Ezzell (1980) considered a more realistic variant. Debt is set at the start of each year as a fixed fraction L of the then-current value, and held fixed during the year. The shield during any given year is therefore known at the start of that year (hence safe over one period, discounted at kd for that one period) but depends on the value path before that year (hence risky for all earlier periods, discounted at ku). The resulting WACC is:

```
WACC_ME = ku − L · kd · Tc · (1 + ku) / (1 + kd)
```

Because (1 + ku)/(1 + kd) exceeds one when ku > kd, the WACC reduction under Miles-Ezzell is slightly larger than under Harris-Pringle, so ME gives a slightly higher value. The reason is intuitive: discounting the shield at kd for its final period, instead of at ku, makes each year's shield worth a little more.

1.3.4 Which policy is right?

No firm follows any of these policies exactly, but the choice should reflect the facts of the case. Table 1.1 summarizes.

Table 1.1 – Financing policies and tax shield discount rates

Policy	Debt rule	Discount rate for shield	WACC formula	When to use
Modigliani-Miller	Fixed schedule of amounts	kd	Needs adjusted ke formula; WACC varies with time if schedule is not perpetual	LBOs, bonds with sinking funds, project finance with scheduled amortization
Harris-Pringle	Continuous rebalancing to constant D/V	ku	ku − kd·Tc·L	Mature firms with a stated target leverage ratio; terminal value periods
Miles-Ezzell	Annual rebalancing to constant D/V	ku (then kd for the final year)	ku − L·kd·Tc·(1+ku)/(1+kd)	Same as HP, with discrete annual debt resets; a slightly higher value
Hybrid	Fixed schedule for N years, then target ratio	kd during schedule, ku afterward	Time-varying	Acquisitions where deleveraging precedes steady state

Three practical guidelines follow. First, if the debt balance in the model is a fixed schedule, use kd. If the debt balance in the model is computed as a percentage of a value, use ku. Second, if the answer between the two policies is large relative to the deal premium, the financing policy is itself a negotiating variable and must be stated explicitly in the valuation memo. Third, for the terminal value period, where firms are assumed to settle into a stable target, the Harris-Pringle approach is standard.

The choice also has an economic meaning: tax shields that depend on firm value are riskier because they vanish in bad states, and a firm that cuts debt as value falls is gaining safety for its creditors but giving up shield exactly when the firm most needs it. In the opposite policy, a rigid schedule makes the shield safer but exposes the firm to distress in downturns. The valuation consequence of this trade-off is just an application of discounting by risk.

1.3.5 Relevering betas consistently

The policy chosen has direct consequences for unlevering and relevering betas, which is where most practical errors originate. For the three policies the relationship between levered equity beta βe and unlevered beta βu is:

```
Harris-Pringle:        βe = βu + (βu − βd) · D/E
Modigliani-Miller:     βe = βu + (βu − βd) · (1 − Tc) · D/E
Hamada (βd = 0, MM):   βe = βu · [1 + (1 − Tc) · D/E]
```

The Hamada formula, familiar from Volume 1, is the MM formula with a zero debt beta. It is the most commonly used and the most commonly misapplied: it assumes a fixed-debt-amount policy even when the firm targets a constant ratio.

Example 1.2 – Brightwater: WACC, APV and CCF for a target-ratio firm. Return to the Brightwater acquisition target. Expected FCFF for the next five years is 120, 132, 145, 157 and 168, growing at 3% perpetually thereafter. Unlevered cost of equity ku = 9.5%; cost of debt kd = 5%; tax rate 25%. The firm maintains a target D/V = 35%, rebalanced to value (Harris-Pringle).

Step 1: WACC.

```
WACC = 0.095 − 0.05 × 0.25 × 0.35 = 0.095 − 0.004375 = 9.0625%
```

Step 2: Terminal value at year 5 (WACC method). FCFF in year 6 = 168 × 1.03 = 173.04.

```
V5 = 173.04 / (0.090625 − 0.03) = 173.04 / 0.060625 = 2,854.27
```

Step 3: Discount at WACC. Present values of the explicit cash flows are 110.03, 110.97, 111.77, 110.97 and 108.88, totaling 552.62. The terminal value contributes 2,854.27 / 1.090625^5 = 1,849.77. Enterprise value:

```
V0 = 552.62 + 1,849.77 = 2,402.39
```

Step 4: APV, step one (unlevered value at ku = 9.5%). The explicit FCFF present values are 109.59, 110.09, 110.44, 109.21 and 106.72, a total of 546.04. The unlevered terminal value is 173.04 / (0.095 − 0.03) = 2,662.15, whose present value is 2,662.15 / 1.095^5 = 1,691.07. Hence VU = 2,237.12.

Step 5: APV, step two (tax shields). Debt at the start of each year equals 35% of that year's opening value. Using the value path implied by Step 3, the year-end values V0 through V5 are 2,402.39, 2,500.11, 2,594.68, 2,684.82, 2,771.13 and 2,854.27, so opening debt D0 through D5 is 840.84, 875.04, 908.14, 939.69, 969.90 and 998.99. The tax shield in year t is Tc × kd × D(t−1) = 0.0125 × D(t−1). Discounted at ku (Harris-Pringle):

Table 1.2 – Brightwater tax shield schedule (Harris-Pringle)

Year	Opening value	Opening debt (35%)	Tax shield	Discount factor at 9.5%	PV of shield
1	2,402.39	840.84	10.51	0.9132	9.60
2	2,500.11	875.04	10.94	0.8340	9.12
3	2,594.68	908.14	11.35	0.7617	8.65
4	2,684.82	939.69	11.75	0.6956	8.17
5	2,771.13	969.90	12.12	0.6352	7.70
Total, years 1-5					43.24

The terminal shield stream begins in year 6 at 0.0125 × 998.99 = 12.49 and grows at 3%; at year 5 it is worth 12.49 / (0.095 − 0.03) = 192.11 (this uses the unrounded debt; the figures match to the second decimal), and its present value is 192.11 / 1.5742 = 122.04. The total PV of tax shields is 43.24 + 122.04 = 165.27.

Step 6: APV total.

```
V0 = 2,237.12 + 165.27 = 2,402.39
```

APV equals the WACC value to the penny, as theory demands.

Step 7: CCF. Add the shield to the FCFF: 130.51, 142.94, 156.35, 168.75 and 180.12 in years 1-5, and a year-6 capital cash flow of 173.04 + 12.49 = 185.53 in the terminal period. Discounting at ku = 9.5%: terminal value = 185.53/0.065 = 2,854.27 (the same as the levered terminal value, as it should be) and the PV equals 2,402.39 again.

Step 8: Equity value and cost of equity. Opening debt is 840.84, so equity value is 2,402.39 − 840.84 = 1,561.55. Cost of equity under Harris-Pringle:

```
ke = 0.095 + (0.095 − 0.05) × 0.35/0.65 = 0.095 + 0.02423 = 11.923%
```

Two incorrect variants. (a) Discounting the same shield stream at kd = 5% instead of ku assigns the whole shield a value of 538.11 and a firm value of 2,775.23. This overstates value by 372.84, more than 15%, because the shield is attached to debt that moves with firm value and is not safe. (b) Miles-Ezzell with its own value path gives WACC = 0.095 − 0.35 × 0.05 × 0.25 × (1.095/1.05) = 9.04375% and V0 = 2,410.01, about 7.6 higher than Harris-Pringle. The sensitivity to the financing policy is modest between HP and ME and enormous between HP and a kd-discounted shield.

Takeaway. The tax shield discount rate is not a technicality. In this problem, the rate choice moves the answer from 2,402 to 2,775 depending only on the assumption of how debt behaves.

Example 1.3 – Relevering a comparable's beta under different policies. A comparable firm has an observed equity beta of 1.20, a market D/E of 0.50, a debt beta of 0.20 and a 25% tax rate. The risk-free rate is 4.0% and the market risk premium is 5.5%. We want the equity beta for our target at D/E = 1.00.

Step 1: Unlever under each policy.

```
Hamada (βd = 0, MM):        βu = 1.20 / (1 + 0.75 × 0.5) = 1.20 / 1.375 = 0.8727
Harris-Pringle:             βu = (1.20 × 1 + 0.20 × 0.5) / 1.5 = 1.30 / 1.5 = 0.8667
MM with debt beta:          βu = (1.20 + 0.20 × 0.75 × 0.5) / 1.375 = 1.275 / 1.375 = 0.9273
```

Step 2: Relever to D/E = 1.

```
Hamada:            βe = 0.8727 × (1 + 0.75 × 1) = 1.5273
Harris-Pringle:    βe = 0.8667 + (0.8667 − 0.20) × 1 = 1.5333
MM with debt beta: βe = 0.9273 + (0.9273 − 0.20) × 0.75 × 1 = 1.4727
```

Step 3: Costs of capital under Harris-Pringle. ku = 4.0% + 0.8667 × 5.5% = 8.767%. kd = 4.0% + 0.20 × 5.5% = 5.1%. At D/V = 50%: ke = 4.0% + 1.5333 × 5.5% = 12.433%. WACC = 0.5 × 12.433% + 0.5 × 5.1% × 0.75 = 6.217% + 1.913% = 8.129%. The shortcut ku − kd × Tc × L = 8.767% − 5.1% × 0.25 × 0.5 = 8.129% gives the same number, confirming consistency.

The three methods give relevered betas between 1.47 and 1.53, a range of 0.06, which at a 5.5% premium is only about 0.3 percentage points of cost of equity. The choice matters less here than it did for the tax shield discount rate, but it should still be consistent: unlever and relever with the same formula, and never unlever with one and relever with another.

1.4 Changing Capital Structure and Debt Paydown Schedules

Leveraged acquisitions, project financings, newly privatized utilities and turnarounds all share one feature: the debt balance is not stable. It is high at the start and falls as cash flow repays it, or it rises as the firm borrows to fund growth. The constant-WACC model cannot represent such a firm. A single rate that is correct in year 1 is wrong in year 4 because the weights have changed.

1.4.1 Why a time-varying WACC is awkward and APV is natural

You can compute a different WACC for each year. The year-t WACC applies to the transition from V(t−1) to V(t) and uses the capital structure at the start of the year. Doing so requires valuing the firm at every date first, because the weights depend on V(t−1), which is the very quantity you are solving for. Thus the time-varying WACC method is circular by construction, and unless you take care it is easy to get the algebra wrong. The APV method has no such problem. The unlevered value path does not depend on leverage at all, and the shield depends only on the debt schedule, which is an input. The method is easy to explain: operations plus shield.

When the debt schedule is a fixed commitment, discount the shield at kd. The unlevered valuation uses ku. The resulting levered values, V(t), at every date can then be used to back out the implied WACC if a stakeholder insists on seeing it. The implied rate is an output of the valuation, not an input.

1.4.2 Terminal value when leverage is still changing

The most common error in changing-structure valuations is the terminal value. Two clean choices exist:

1. **Terminal at target structure.** Assume that after the explicit horizon the firm settles at a stable target ratio, then value the terminal period as the unlevered terminal value plus a Harris-Pringle tax shield at that ratio. This is the standard approach for an LBO exit or a mature steady state.
2. **Terminal unlevered.** If the debt will be fully repaid, the terminal value is the unlevered value at ku. This is conservative and keeps the example transparent, but ignores the possibility that a rational firm will relever.

What is never defensible is to compute a WACC at today’s high leverage and apply it to a terminal value that assumes a very different future leverage.

Example 1.4 – Valuing a business with a scheduled debt paydown. Larkspur Packaging, a fictional firm, is being acquired with debt that amortizes on a fixed schedule. Expected FCFF is 90, 105, 120, 135 and 150 in years 1 to 5. Beyond year 5 the firm is debt-free (assume the debt is fully repaid) and FCFF grows at 2.5%. The unlevered cost of equity is ku =

10%, the cost of debt is kd = 7% and the tax rate is 25%. The opening debt in years 1 to 5 is 600, 480, 360, 240 and 120, and it is zero thereafter. The debt amortization is contractual, so the shield is discounted at kd.

Step 1: Unlevered terminal value at year 5.

```
VU5 = 150 × 1.025 / (0.10 − 0.025) = 153.75 / 0.075 = 2,050.00
```

Step 2: Unlevered value. The present value of the explicit FCFF at 10% is 444.10, and the terminal value contributes 2,050.00 / 1.1^5 = 1,272.89. So VU = 1,716.99.

Step 3: Tax shields. Interest in each year is kd times opening debt, so the shield equals 0.25 × 0.07 × D(t−1) = 0.0175 × D(t−1):

Table 1.3 – Larkspur tax shield and implied WACC path

Year	Opening debt	Interest	Tax shield	PV at kd = 7%	Opening value V(t−1)	D/V	Implied WACC
1	600	42.0	10.50	9.81	1,743.98	34.4%	9.35%
2	480	33.6	8.40	7.34	1,817.07	26.4%	9.51%
3	360	25.2	6.30	5.14	1,884.82	19.1%	9.65%
4	240	16.8	4.20	3.20	1,946.67	12.3%	9.78%
5	120	8.4	2.10	1.50	2,001.96	6.0%	9.89%

Step 4: APV.

```
PV(TS) = 9.81 + 7.34 + 5.14 + 3.20 + 1.50 = 26.99
V0     = 1,716.99 + 26.99 = 1,743.98
```

The opening equity value is 1,743.98 − 600 = 1,143.98.

Step 5: The implied WACC path. Compute the value at each date by running the same APV from that date forward (for example V1 = PV at 10% of the remaining FCFF and terminal value, plus PV at 7% of the remaining shields = 1,817.07). Then the year-t WACC is the discount rate that satisfies V(t−1) = [V(t) + FCFF(t)] / (1 + WACC(t)). For year 1: (1,817.07 + 90) / 1,743.98 − 1 = 9.35%. The path is 9.35%, 9.51%, 9.65%, 9.78%, 9.89%, converging on ku = 10% as the debt disappears. The WACC rises monotonically because the tax advantage shrinks with the debt.

Step 6: Compare with a constant WACC. Suppose the analyst, working from today's capital structure, uses 9.35% for every year including the terminal period. Then:

```
V = Σ FCFF_t / 1.0935^t + [153.75 / (0.0935 − 0.025)] / 1.0935^5 = 1,887.41
```

This overstates the value by 143.43, or 8.2 percent, almost entirely because it applies a tax-advantaged rate to a terminal period in which the firm carries no debt. Even applying the correct 5-year rates to the explicit period and 10% to the terminal period is just APV done the hard way.

Step 7: Sensitivity to the discount rate on the shield. Discounting the same shields at ku instead of kd gives PV(TS) = 25.39 and V0 = 1,742.38, a difference of only 1.60. When the shield is small relative to firm value, the discount rate debate matters little; when the shield is large, as in the next sections, it matters a great deal.

Takeaway. The shield is only 1.5 percent of value in this deal, so the more important error was the constant-WACC terminal value, not the shield rate.

1.4.3 Managerial interpretation

A paydown schedule front-loads the shield, and the implied discount rate rises as leverage falls, so the same operating forecast supports a different price under a different debt schedule. For an auditor or impairment reviewer, a single WACC applied to a business with a changing structure is a red flag.

1.5 Interest Deductibility Limits and Other Tax Frictions

The APV formula Tc × kd × D implicitly assumes that interest is fully deductible at a constant marginal rate, in the year paid, with sufficient taxable income to use it. Since the early 2010s many jurisdictions have introduced rules that restrict this. Without naming specific statutes, the common designs are:

- **Earnings-based caps.** Deductible net interest is limited to a percentage of a profit measure, often EBITDA or EBIT (30% is a common benchmark in several regimes). Interest above the cap is disallowed in the current year.
- **Carryforwards.** Disallowed interest may be carried forward indefinitely or for a limited period, to be deducted in years when capacity exists.
- **Thin capitalization rules.** Debt above a ratio to equity may generate non-deductible interest, especially on related-party loans.
- **Anti-hybrid and withholding rules** that deny deductions or add taxes in cross-border structures.
- **Limits on use of tax losses.** Net operating losses (NOLs) can usually offset only a fraction of taxable income in a given year, and the deduction may be delayed.

Rules vary by country and change frequently, so the principle matters more than any detail: model the deduction actually available, not the interest actually paid.

1.5.1 Modeling a capped shield

The correct method builds the tax shield from a deductibility schedule, not from the debt schedule alone. For each year:

```
Interest capacity_t  = cap% × EBITDA_t
Interest available_t = Interest_t + Carryforward_(t−1)
Deducted_t           = min(Interest available_t, Interest capacity_t)
Carryforward_t       = Interest available_t − Deducted_t
Tax shield_t         = Tc × Deducted_t
```

Then discount the shield. The discount rate question reappears with a twist: the shield now depends on EBITDA, an operating variable, as well as on debt. A shield whose size is capped by EBITDA is positively correlated with the firm's operating performance, so it deserves a rate between kd and ku. A common compromise is a rate midway between the two, or discounting the capped portion at ku and any portion not subject to the cap at kd. Whatever you choose, document it.

1.5.2 What the cap does to optimal leverage

An earnings-based cap changes the shape of the benefit of debt. Below the cap, each additional dollar of debt generates a full shield. Above it, the marginal interest dollar is disallowed in the current year and valued, at best, at the discounted value of a deduction far in the future. The marginal shield falls to near zero. Optimal leverage in the presence of a binding cap therefore lies near the point where interest equals the cap, in contrast with the unconstrained trade-off, where the optimum depends on the distress cost curve. For a firm whose interest burden is well within the cap, the cap is irrelevant, but for LBOs with 6x leverage and 7 percent interest, it often binds in the early years.

Example 1.5 – Valuing a capped shield with carryforward. Tallgrass Foods, a fictional firm, carries permanent debt of 1,200 at kd = 7%, so annual interest is 84. The tax rate is 25%. Deductible net interest is limited to 30% of EBITDA. EBITDA is 180, 195, 210, 225 and 240 in years 1 to 5, growing at 2% thereafter. Disallowed interest carries forward indefinitely.

Step 1: Naive shield. Without the cap, the shield is 0.25 × 84 = 21.00 each year, and its present value at 7% is 21.00 / 0.07 = 300.00 (equal to Tc × D).

Step 2: Cap schedule, years 1 to 5.

Table 1.4 – Tallgrass interest limitation schedule

Year	EBITDA	Interest	Cap at 30%	Deducted	Carryforward (end)	Shield (actual)	Shield (naive)
1	180.0	84.0	54.0	54.00	30.00	13.50	21.00
2	195.0	84.0	58.5	58.50	55.50	14.63	21.00
3	210.0	84.0	63.0	63.00	76.50	15.75	21.00
4	225.0	84.0	67.5	67.50	93.00	16.88	21.00
5	240.0	84.0	72.0	72.00	105.00	18.00	21.00

The cap binds in every year, and in all of them interest exceeds capacity. The carryforward balance grows to 105 in year 5 and, because EBITDA grows at only 2%, continues to grow until EBITDA reaches the level at which the cap (30% × EBITDA) equals 84, which is EBITDA = 280. It then declines slowly. The carryforward is fully exhausted only in year 26.

Step 3: Present value. Discounting the actual shield at 7% over all years gives 280.53, compared with the naive 300.00. The cap costs 19.47, or 6.5% of the shield. If the carryforward did not exist, so that disallowed interest vanished, the value would be 271.68, a loss of 28.32 or 9.4%.

Step 4: Sensitivity to the discount rate. Because the shield depends on EBITDA, an analyst may reasonably discount at a higher rate. At 8.5%, the actual shield is worth 226.75 against a naive 247.06, a loss of 20.31. At 10%, 189.32 against 210.00, a loss of 20.68. The cap's cost is stable at about 20 across rates, but the level of the shield falls substantially: the rate matters more than the cap.

Step 5: Debt capacity. In year 1 the cap of 54 supports debt of only 54 / 0.07 = 771 at 7% interest. The firm's 1,200 is 55 percent above that. A deal team that sized debt to the year-1 cap would borrow about 36 percent less, cut the nominal shield to 0.25 × 771 = 192.86, and avoid paying, for decades, interest that earns no deduction; whether that trade is attractive depends on the cost of the extra equity, which is why the cap turns a tax question into a capital structure question.

Takeaway. The cap converts a safe, immediate cash flow into a delayed, uncertain one. The loss is modest here because growth in EBITDA eventually releases the carryforward, but for a firm with flat or falling EBITDA the carryforward would never be used, and the value of the shield would approach the 271.68 case or lower.

1.5.3 Taxable income and loss limits

A deduction is valuable only if taxable income exists to absorb it. If Tallgrass earns EBIT of only 60 against interest of 84, taxable income is −24, the year's shield is zero, and a later year's taxes fall by 25% of 24 = 6, discounted. For a stable profitable firm the timing cost is trivial; for a start-up or a highly levered firm in a down year it can eliminate most of the

shield. The robust method is to forecast taxable income and compute the shield as the difference between taxes with and without interest, which captures caps, losses and carryforwards automatically; accumulated NOLs and disallowed-interest carryforwards are then valued as separate assets at a rate between kd and ku.

1.6 Circularity in Valuation Models

Circular references are the most common technical frustration in applied valuation. They appear in three distinct forms, and the right response differs.

1.6.1 Form one: the WACC circularity

If the capital structure is specified as a dollar amount of debt, D/V depends on V, WACC depends on D/V and V depends on WACC. The WACC method with a fixed debt amount is therefore circular. The circularity is real in the sense that the answer depends on both, but it is well behaved: the mapping from a guess of V to an updated V is a contraction, and simple iteration converges rapidly. Analysts often cut the circle by hard-coding a target D/V, which converts the problem into the standard target-ratio case but silently changes the debt policy. A better resolution is to solve the circle algebraically (the Harris-Pringle model with D given today has a closed form) or to use APV, which has no circle at all.

Example 1.6a – Iterating to the fixed point. A firm has FCFF of 120 in year 1, growing at 3%. ku = 10%, kd = 5.5%, tax rate 25%. Today's debt is 600 and thereafter it is rebalanced to a constant ratio of value (Harris-Pringle).

Iteration. Start with a guess V = 1,500. The implied D/V = 0.40, WACC = 0.10 − 0.055 × 0.25 × 0.40 = 9.45%, so the updated V = 120 / (0.0945 − 0.03) = 1,860.47. Repeat:

Table 1.5 – Convergence of the WACC circularity

Iteration	V used	D/V	WACC	Updated V
0	1,500.00	40.00%	9.4500%	1,860.47
1	1,860.47	32.25%	9.5570%	1,830.23
2	1,830.23	32.78%	9.5490%	1,832.28
3	1,832.28	32.75%	9.5500%	1,832.13
4	1,832.13	32.75%	9.5500%	1,832.14

The iteration converges to V = 1,832.14 in four steps. The closed form from APV confirms it: VU = 120 / 0.07 = 1,714.29; the shield is Tc × kd × D / (ku − g) = 0.25 × 0.055 × 600 / 0.07 = 117.86; V = 1,714.29 + 117.86 = 1,832.14. APV found in one line what the WACC method needed a loop to find. The converged D/V of 32.75% differs from the starting 40% used in the first pass, which shows how misleading it is to compute WACC from a debt amount divided by an unadjusted or book value: the first pass overstated the leverage benefit by about 28 in value.

1.6.2 Form two: the interest-cash-sweep circularity

In financing models, interest is calculated on average debt balance, the balance depends on cash sweep, and the sweep depends on cash flow after interest. This circularity is operational, not conceptual. It exists because interest is paid during the year while the balance changes during the year.

Example 1.6b – Interest on average balance with a cash sweep. Opening debt is 800, interest rate 8%, tax 25%. EBITDA is 200, depreciation 50, capital expenditure 40, increase in working capital 10. All free cash after interest and tax is used to repay debt. Interest is charged on the average of opening and closing balances.

Step 1: Pre-interest free cash flow. Unlevered after-tax cash flow = (200 − 50) × 0.75 + 50 − 40 − 10 = 112.50.

Step 2: Express the circularity. Cash available for repayment = 112.50 − interest × (1 − 0.25). Closing debt = 800 − repayment. Interest = 0.08 × (800 + closing) / 2.

Step 3: Iterate. Guess closing = 800 → interest 64.00 → repayment 112.50 − 48.00 = 64.50 → closing 735.50. Next: interest = 0.04 × (800 + 735.50) = 61.42 → repayment 66.435 → closing 733.565. Next: interest 61.3426 → repayment 66.493 → closing 733.507. Then 733.505, and the process has converged.

Step 4: Closed form. Let k = 0.08 × 0.75 / 2 = 0.03. Then closing = [800 × (1 + k) − 112.50] / (1 − k) = (824 − 112.50) / 0.97 = 733.505. Interest = 61.34, repayment 66.49.

Spreadsheet implementations cut this circle three ways: enabling iterative calculation, computing interest on opening balance (which understates the repayment benefit slightly), or using the algebraic solution. The algebraic solution is the most robust because it never fails to converge when a model is broken by an error elsewhere.

1.6.3 Form three: the conceptual circularity

The third form is not a spreadsheet problem. It arises when inputs already embody the answer: a beta relevered at a target D/V derived from the valuation itself, book-value weights, a debt amount set as a multiple of EBITDA that reflects a prior price assumption, or a terminal growth rate inconsistent with terminal reinvestment. Iteration converges to a self-consistent but unjustified answer. The remedy is to state the financing policy in observable terms (debt schedule, target rating, coverage ratio) and let value be the output.

Practical rules: if debt is a dollar schedule, use APV (no circle); if debt is a target ratio, use WACC with the ratio as an input (no circle); if you must combine a dollar amount with WACC, iterate or solve algebraically and report the converged ratio; and always reconcile APV against WACC, with a visible convergence check on any iterative switch.

1.7 Financial Distress, Issuance Costs and Subsidized Financing in APV

The strength of APV is that any financing side effect can be added as its own line. The tax shield is only the first. Three others matter in practice.

1.7.1 Expected costs of financial distress

Debt creates a tax shield, but it also raises the probability that the firm will fall into distress, where costs are incurred: legal and administrative fees, lost customers and suppliers, asset fire sales, underinvestment and managerial distraction. The trade-off theory of capital structure (Volume 1, Chapter 15) balances these two effects. APV makes the trade-off computational:

```
PV(distress costs) ≈ Σ P(distress in year t) × Cost of distress × discount factor
```

A tractable approximation uses a constant annual probability of distress p, that depends on leverage, and a cost equal to a fraction c of unlevered value. Treat distress as a perpetual hazard. The expected annual loss is p × c × VU, and the present value, using a discount rate of ku + p to reflect the fact that the process stops once distress occurs, is:

```
PV(distress) = c × VU × p / (ku + p)
```

This formula is a model of the form of the cost, not an estimate of its level. Both p and c must come from evidence: a rating-based default probability for p, and studies or case experience for c. Estimates in the academic literature vary widely, and the honest practice is to present a range.

Example 1.7 – APV optimal leverage with distress costs. A fictional firm has unlevered value VU = 1,000 (ku = 10%) and a tax rate of 25%. Debt is permanent, so the shield is Tc × D. The annual probability of distress rises with leverage, and the cost of distress is 30% of unlevered value.

Step 1: Compute the shield and distress cost at each leverage level. For example at D = 600 with p = 3%: shield = 0.25 × 600 = 150; distress = 0.30 × 1,000 × 0.03 / (0.10 + 0.03) = 9 / 0.13 = 69.23; net benefit = 150 − 69.23 = 80.77.

Table 1.6 – Net value of leverage with distress costs

Debt D	Annual p(distress)	PV tax shield	PV distress cost	Levered value	Net gain vs. unlevered
0	0.0%	0.0	0.0	1,000.0	0.0
200	0.3%	50.0	8.7	1,041.3	41.3
400	1.0%	100.0	27.3	1,072.7	72.7
600	3.0%	150.0	69.2	1,080.8	80.8
800	7.0%	200.0	123.5	1,076.5	76.5
1,000	14.0%	250.0	175.0	1,075.0	75.0

Step 2: Read the optimum. Value peaks at D = 600 (60% of unlevered value, about 55.5% of levered value) with a net gain of 80.8, which is about 8% of unlevered value. The curve is flat: between 400 and 1,000 of debt, the levered value stays between 1,072.7 and 1,080.8. The flatness is the real lesson. The precise optimum is unknowable, but the loss from being roughly right is small, whereas the loss from being very wrong (for example, borrowing to the point where distress costs include fire sales and lost franchise) is not captured by an approximation with a constant cost fraction.

Step 3: Interpret. Managers should regard the APV leverage curve as a way to see the gradient of tax benefit and distress cost, not as a precise target. The firm with volatile cash flows, intangible assets or high growth options has a steeper distress cost curve and a lower optimum, which matches the intuition that biotech firms carry little debt and regulated utilities carry a lot.

1.7.2 Issuance costs and subsidized financing

Raising external capital costs money. Underwriting fees, legal fees and, for equity, discounts to market price are real cash outflows. APV subtracts them directly: if issuing 500 of debt costs 1.5% in fees, subtract 7.50 (or the fee amortized if tax-deductible, which adds back a small tax shield on the amortization). In WACC models, issuance costs are usually either ignored or hidden by adjusting the cost of capital upward, which is imprecise because fees are a one-time outflow, not a continuing rate.

Subsidized financing is the second example. A government development bank that lends at 3% to a project whose market borrowing cost would be 6% provides a benefit equal to the present value of the interest saving, discounted at the market rate:

```
PV(subsidy) = Loan − PV of loan payments at market rate kd(market)
```

For a bullet loan of 1,000 for 10 years at 3% against a market rate of 6%, the annual interest is 30, the principal is 1,000, and the PV at 6% is 30 × 7.3601 + 1,000 × 0.5584 = 220.80 + 558.39 = 779.19. The subsidy is 1,000 − 779.19 = 220.81, which is added to the APV (and taxed if the interest saving is taxable). A WACC model would typically embed the 3% cost of debt in the rate and mix the subsidy with the operating value, making it impossible to evaluate whether the project is attractive on its own merits or only because of the subsidy.

1.7.3 Other side effects

APV can also carry tax-loss carryforwards, investment tax credits, government guarantees and hedging costs, each as a cash flow discounted at its own risk-appropriate rate.

1.8 Implementation: Choosing Methods and Avoiding Mistakes

1.8.1 Which method when?

Use WACC for mature firms with a stable target D/V and simple taxes. Use APV, with the shield at kd, when debt follows a schedule (LBOs, project finance); APV or CCF with an explicit tax computation when caps, losses or carryforwards matter; and APV with extra lines for divisional valuation, subsidized or cross-border financing. FCFE or CCF serve as a cross-check from the equity holders' perspective.

1.8.2 Common mistakes

1. **Using the book or target weights without checking they are consistent with the debt schedule.** Weights must be market values at each date.
2. **Applying a constant WACC to a firm whose leverage is changing.**
3. **Discounting a value-linked tax shield at kd.** Example 1.2 showed an error of more than 15 percent.
4. **Discounting a fixed-schedule tax shield at ku.** The error is the opposite and usually smaller: it understates the shield slightly.
5. **Using the same tax rate for the shield as for operations without checking the marginal rate.** The shield should be valued at the marginal rate on the deducted interest, which may differ from the average rate on operating income.
6. **Double-counting the tax shield.** If FCFF is calculated from NOPAT (taxed on operating income, which excludes the shield) and then discounted at WACC, the shield is counted once through WACC. Adding a separate shield to a WACC valuation counts it twice.
7. **Ignoring debt beta when leverage is high.** At D/V above 50 percent, a zero debt beta forces implausibly high unlevered betas.
8. **Mismatching nominal and real, or currency, between cash flows and discount rates.**
9. **Letting terminal value dominate silently.** In Example 1.2, 77 percent of enterprise value is the terminal value. Terminal assumptions about growth, reinvestment and leverage deserve separate attention.

1.8.3 Reconciliation and managerial implications

The most useful habit in advanced DCF is reconciliation: value operations at ku, value the shield under the stated policy, compute the implied WACC path, check that CCF equals APV, and test the implied cost of equity against comparables (a cost of equity below the cost of debt signals an error). A failed check means an internal inconsistency that no sensitivity analysis will cure.

APV also makes the financing contribution of a decision visible: of a bid of 2,400, about 2,237 is operating value any buyer could capture and 165 is a financing effect specific to the bidder's structure. That component is the most assumption-dependent, so a negotiator should be most cautious about sharing it with the seller. A bidder who pays for a shield it cannot use, because of an interest cap or a loss-making business, transfers value to the seller for nothing.

Case Study: Halvorsen Industrial Coatings

Illustrative case. All names and figures are fictional.

Halvorsen Industrial Coatings makes protective coatings for pipelines and storage tanks. A strategic buyer, Kestrel Materials, is considering acquiring it. Halvorsen's owners have set a price expectation of "around 1,300." Kestrel's investment banker has delivered a WACC-based valuation, and Kestrel's treasurer has proposed a financing plan that borrows 450 at closing, amortizes 50 per year and has the business settle at a debt ratio of 25 percent of value after year 5. Kestrel's CFO, Dara Okonkwo, wants the finance team to produce a valuation she can defend to her board and to the lender, one that explains how much of any premium comes from operations and how much comes from financing.

Forecasts. Expected FCFF is 70, 78, 86, 93 and 100 in years 1 to 5, growing 2.5% a year thereafter. EBITDA is 130, 138, 146, 153 and 160 in years 1 to 5. Kestrel estimates the unlevered cost of capital at 9.5% using comparable firms. The acquisition debt costs 6.5%. The tax rate is 25%. The credit agreement and the tax code both limit deductible interest to 30 percent of EBITDA, with indefinite carryforward of disallowed interest. The debt balances at the start of years 1 to 5 are 450, 400, 350, 300 and 250. After year 5, debt is rebalanced to 25% of value.

The banker's WACC valuation. The banker computed WACC using the closing debt of 450 over an estimated enterprise value of about 1,334, a D/V of 33.7 percent. WACC = 9.5% − 6.5% × 0.25 × 0.337 = 8.95%. Applied to the full forecast including the terminal value, this gives an enterprise value of 1,362.40.

The APV analysis.

1. *Unlevered value.* The FCFF discounted at 9.5% gives 1,252.85 (explicit period plus a terminal value of 1,464.29 at year 5).
2. *Shield during amortization.* The interest is 29.25, 26.00, 22.75, 19.50 and 16.25, all below the cap in the base case (cap 39.0, 41.4, 43.8, 45.9 and 48.0). The shield is 7.31, 6.50, 5.69, 4.88 and 4.06, with present value 24.06 at 6.5%.
3. *Terminal shield.* After year 5, the Harris-Pringle shield at 25% leverage lifts the year-5 value from 1,464.29 to 1,554.50, a shield of 90.22 at year 5 and 57.31 today.
4. *APV total:* 1,252.85 + 24.06 + 57.31 = 1,334.22. Equity value at closing is 1,334.22 − 450 = 884.22.

Cross-checks and stresses. A constant WACC at the 25% target ratio (9.094%) gives 1,332.30, close to APV, but only because the time-varying shield in the amortization period is small and the terminal ratio matches the WACC; the banker's use of today's 33.7% leverage for all years, including the terminal period, overstates the value by 28.18. In a downside where EBITDA is only 70% of forecast, the cap (27.3 in year 1) falls below interest of 29.25 for one year, and APV falls by only 0.03. If EBITDA is 50% of forecast, the cap binds in all five years (interest exceeds capacity in the first three years, and the carryforward then absorbs the later headroom), a carryforward of 4.70 remains at year 5, and APV falls by 0.84 to 1,333.38. The caps do not matter much because the debt is modest relative to EBITDA, with 3.5 times opening leverage on year-1 EBITDA.

Dara's team notes that the financing contribution to value is small: the shield is 81 out of 1,334, or 6 percent, and 70 percent of it is the terminal shield, which depends on a leverage policy Kestrel will choose and not on anything Halvorsen's owners have created. Her concern is the price. The owners expect 1,300, and APV says the business is worth 1,334 to Kestrel under the planned financing but 1,253 to a buyer who uses no debt. Any bidder with a similar structure can capture similar shields, so competitive pressure will tend to push the price toward the shield-inclusive value. Kestrel's incremental benefit comes from operating synergies, which are not in these forecasts.

Discussion Questions

1. Explain why the banker's 1,362.40 and the APV value of 1,334.22 differ, and identify which assumption in the banker's model is the least defensible. Would the answer change if the banker used a 25% leverage ratio throughout?
2. Kestrel's treasurer proposes paying down debt faster. How would you use the APV framework to estimate the value effect, and what would you need to know about the cost of the equity or cash used to repay?
3. The base case cap is not binding but the 50% EBITDA case causes a carryforward. How should the valuation team decide whether to discount the shield at kd, ku or something in between, and what is the effect on the 1,334.22?
4. Halvorsen's owners argue that the tax shield belongs to them because it exists "inside" their company. Develop the argument for and against sharing the shield with the seller, including which portion of the shield is attributable to Kestrel's own financing decisions.

Key Takeaways

- WACC, APV and CCF are three arrangements of the same valuation logic. With consistent assumptions they must agree, and a disagreement is a diagnostic signal rather than a matter of preference.
- The unlevered value discounts operating cash flows at ku, which reflects only operating risk, and is independent of financing policy by construction.
- The discount rate for the tax shield follows from the debt policy: kd for a fixed debt schedule, ku for debt rebalanced continuously to a target ratio, and ku with a one-period kd adjustment for annual rebalancing.
- Using a shield rate inconsistent with the debt policy can change value by more than 15 percent in a levered target, which is far larger than most disputes about the cost of equity.
- When the capital structure changes, as in an LBO, build the valuation from APV with year-by-year shields and treat the implied WACC as an output that rises as debt is repaid.
- A constant WACC applied to a changing structure, and especially to a terminal value with a different leverage, is the most common and most costly error in acquisition valuation.
- Interest limitation rules convert the shield from a function of debt alone to a function of debt and earnings, with a delay and a lower value; model the deduction actually available, and value carryforwards as separate assets.
- Circularity comes in three forms: a real but convergent WACC loop, an operational interest-on-average-balance loop, and a conceptual loop in which inputs embody the answer; only the first two can be solved numerically.
- APV extends naturally to distress costs, issuance costs and subsidies, and its leverage curve is typically flat near the optimum, so practical policy should aim for a sensible range, not a point.
- Always reconcile two methods and test implied costs of equity for reasonableness before relying on a valuation.

Review Questions

1. Why does the value of the unlevered firm not depend on the firm's capital structure, and what discount rate does this imply for operating cash flows?
2. Explain how the debt policy determines the riskiness of the interest tax shield. Why is a shield tied to a fixed debt schedule less risky than one tied to a constant D/V ratio?
3. Write the WACC formulas under Modigliani-Miller (fixed debt) and Harris-Pringle (constant D/V) policies. What is the economic reason that the cost of equity formula differs between them?
4. What is the Miles-Ezzell assumption, and why does it produce a slightly higher value than Harris-Pringle?
5. Why is a single WACC inappropriate for valuing a leveraged buyout with a scheduled debt paydown? How does APV avoid the problem?
6. Describe how you would treat the terminal value in a valuation where leverage declines during the forecast period and the firm then relevers to a target ratio.
7. What is the capital cash flow method and how does it differ in the placement of the tax shield from APV and WACC?
8. How does an earnings-based interest cap affect the value of the tax shield and the optimal level of debt? What additional information is needed to value a disallowed-interest carryforward?
9. Distinguish the three forms of circularity in valuation models and give a resolution for each.
10. Why is the APV leverage curve flat near its maximum, and what does this imply about how precisely managers should try to target a capital structure?

Problems

Problem 1. A firm has FCFF of 60 per year forever, ku = 12%, permanent debt of 250 at kd = 6%, and a 30% tax rate. Compute (a) the unlevered value, (b) the PV of the tax shield, (c) levered value and equity value, (d) the cost of equity using the Modigliani-Miller formula, and (e) the WACC, and verify that discounting FCFF at WACC reproduces the levered value.

Problem 2. A firm expects FCFF of 50 next year, growing at 4% forever. ku = 10%, kd = 5%, tax rate 25%, and the firm rebalances debt to 40% of value. Compute the WACC, the enterprise value, the unlevered value, the value of the tax shields and the opening debt balance.

Problem 3. A target's FCFF is 40, 45 and 50 in years 1 to 3, with 3% growth afterward. It is financed with debt of 300, 200 and 100 at the start of years 1 to 3 and no debt afterward. ku = 9%, kd = 6%, tax rate 25%. Compute the unlevered value, the PV of shields discounted at kd, the APV, and the APV if shields are discounted at ku. Assume that the firm is unlevered after year 3.

Problem 4. A firm has EBITDA of 100 each year forever, permanent debt of 500 at 8%, a tax rate of 25% and a cap of 30% of EBITDA on deductible interest, with disallowed interest lost permanently (no carryforward). Compute the annual shield actually obtained, its PV at 8%, the naive PV, and the value lost to the cap.

Problem 5. A firm has FCFF1 = 90 growing at 2%, ku = 9%, kd = 5%, tax rate 30%, and a current debt of 500, rebalanced thereafter to a constant fraction of value (Harris-Pringle). Starting from a guess V = 1,200, iterate the WACC circularity three times, and then find the exact value using APV.

Problem 6. A comparable has βe = 1.10, D/E = 0.40, βd = 0.15 and a 30% tax rate. The risk-free rate is 3.5% and the market premium 5%. (a) Unlever under Harris-Pringle. (b) Relever to D/E = 1.0. (c) Compute ku, kd, ke and the WACC at D/V = 50%. (d) Compare with relevering using the Hamada formula (βd = 0).

Problem 7. Using the data in Problem 2, compute the year-1 capital cash flow (FCFF plus the year-1 tax shield) and show that discounting the CCF stream at ku with growth of 4% reproduces the enterprise value from Problem 2.

Problem 8. A firm has unlevered value 800 (ku = 10%) and a 25% tax rate. With permanent debt of D, the PV of distress costs is 0.35 × 800 × p / (0.10 + p). Debt levels and annual distress probabilities are: 0 (0%), 200 (0.4%), 400 (2%), 600 (6%), 800 (20%). Compute levered value at each level and identify the best level.

Answers to Selected Problems

Problem 1. (a) VU = 60 / 0.12 = 500. (b) Shield = 0.30 × 250 = 75. (c) VL = 575, E = 325. (d) ke = 12% + 0.70 × (12% − 6%) × 250/325 = 12% + 3.23% = 15.23%. (e) WACC = (325/575) × 15.23% + (250/575) × 6% × 0.70 = 8.61% + 1.83% = 10.43%; 60 / 0.10435 = 575.

Problem 2. WACC = 10% − 5% × 0.25 × 0.40 = 9.5%. V = 50 / (0.095 − 0.04) = 909.09. VU = 50 / (0.10 − 0.04) = 833.33. Shield value = 75.76. Opening debt = 0.40 × 909.09 = 363.64.

Problem 3. Unlevered V3 = 50 × 1.03 / 0.06 = 858.33; VU = 775.97. Shields: 4.50, 3.00, 1.50; PV at 6% = 8.17; APV = 784.15. At ku: PV = 7.81; APV = 783.78.

Problem 4. Interest = 40; cap = 30; deducted 30; shield = 7.50 per year; PV = 93.75; naive PV = 0.25 × 40 / 0.08 = 125.00; lost value = 31.25.

Problem 5. Iteration: V = 1,200 gives D/V = 0.4167, WACC = 8.375%, V = 1,411.76. Next: D/V = 0.3542, WACC = 8.469%, V = 1,391.30. Next: D/V = 0.3594, WACC = 8.461%, V = 1,392.99. APV: VU = 90 / 0.07 = 1,285.71; shield = 0.30 × 0.05 × 500 / 0.07 = 107.14; V = 1,392.86.

Problem 6. (a) βu = (1.10 + 0.15 × 0.4) / 1.4 = 0.8286. (b) βe = 0.8286 + (0.8286 − 0.15) × 1 = 1.5071. (c) ku = 3.5% + 0.8286 × 5% = 7.643%; kd = 3.5% + 0.15 × 5% = 4.25%; ke = 3.5% + 1.5071 × 5% = 11.036%; WACC = 0.5 × 11.036% + 0.5 × 4.25% × 0.70 = 5.518% + 1.4875% = 7.005%, which equals ku − kd × Tc × L = 7.643% − 4.25% × 0.30 × 0.5. (d) Hamada: βu = 1.10 / (1 + 0.7 × 0.4) = 0.8594; relevered = 0.8594 × (1 + 0.7 × 1) = 1.4609, lower by 0.046.

Problem 7. Opening debt = 363.64; shield = 0.25 × 0.05 × 363.64 = 4.545; CCF1 = 54.545. Value = 54.545 / (0.10 − 0.04) = 909.09, matching Problem 2.

Problem 8. Distress costs: 0, 10.77, 46.67, 105.00, 186.67; shields: 0, 50, 100, 150, 200. Levered values: 800.00, 839.23, 853.33, 845.00, 813.33. The best level is D = 400.

Chapter 2: Real Options Valuation in Depth

Learning Objectives

After completing this chapter you will be able to:

- Distinguish market risk from private (technical) risk in real option problems, and explain how risk-neutral valuation, the replicating portfolio and the "twin security" assumption justify discounting each differently.
- Build and interpret a recombining binomial lattice for a real option, including calibration of up and down moves, cash-flow leakage, early exercise and the treatment of discretization error.
- Value R&D and pharmaceutical pipelines as compound options with technical-success probabilities, and compare the result with a risk-adjusted NPV (rNPV) decision tree.
- Value natural resource options (develop, defer, abandon) using convenience yields and operating leverage, and explain why the optimal trigger price lies far above the NPV breakeven price.
- Value platform and staged investments as sequential (compound) options, and analyze competitive timing with simple option games.
- Apply Monte Carlo and least-squares Monte Carlo simulation to options that lattices handle poorly, estimate project volatility from a cash flow model, and recognize the pitfalls and governance risks of real options analysis.

Opening Vignette

Illustrative scenario. Marrowfield Therapeutics is a fictional company.

Marrowfield Therapeutics is a mid-sized drug developer with one promising compound entering Phase II. Its finance team has done what every textbook instructs. They have forecast the cash flows from launch six years away, discounted them at a company cost of capital of 10 percent, multiplied by the probability that the compound survives each clinical phase, and subtracted the development costs. The rNPV comes out at roughly zero: slightly negative in the base case. The CFO, Teodora Valdis, is prepared to shelve the program.

Then the head of research makes an objection that the spreadsheet cannot answer. "The model assumes we will spend the money and launch the drug no matter what we learn along the way. We will not. If Phase II fails we stop, which your tree already captures. But if Phase III succeeds and the market for this indication has shrunk, or a competitor has launched a better drug, we will not pay a quarter of a billion for a launch. And if the market has grown, we will launch bigger. Your model has probabilities for the science and nothing for the market. Doesn't that flexibility have a value?"

Teodora knows he is right, and she also knows the objection can be abused. Every project champion in the company claims that the project is "strategic," that it creates "options," and that the NPV therefore understates its worth. A valuation method that justifies anything justifies nothing. She needs to know how much the flexibility is worth, under what assumptions, which part of the value is real and which part is an artifact of generous inputs, and what evidence the board should demand.

This chapter gives her the tools. Starting from the logic of risk-neutral valuation, we build binomial lattices, then apply them to pharmaceutical pipelines, natural resource development, staged platform investments and competitive timing games. We use simulation for settings that lattices cannot handle, and we devote a full section to the ways in which real options analysis goes wrong, because a technique this powerful is also easy to misuse.

2.1 Foundations: Why Real Options Need More Than Black-Scholes

Volume 1, Chapter 18 introduced options and the Black-Scholes model, and Volume 1's capital budgeting chapters showed how static NPV can understate the value of projects with flexibility. This chapter assumes that background and asks the next question: how do we put numbers on flexibility when the project looks nothing like a listed stock option?

2.1.1 The real option analogy and its limits

A real option is the right, but not the obligation, to take a business action at a future date at a predetermined cost. The analogy to a financial call option is direct:

Table 2.1 – Mapping real options to financial option parameters

Financial option parameter	Real option counterpart	Typical difficulty
Underlying asset price S	PV of the cash flows from the opportunity if undertaken now (gross of the investment cost)	Not traded; must be estimated from a DCF model
Exercise price K	Investment outlay required to exercise (launch cost, development capex)	May itself be uncertain or staged
Time to expiration T	Window of opportunity (patent life, lease, license, competitive window)	Often ambiguous; may be shortened by competitors
Volatility σ	Volatility of the percentage changes in the value of the underlying project	Must be inferred; the single most debated input
Risk-free rate r	Risk-free rate	Straightforward
Dividend yield δ	Cash-flow leakage: value lost per year by waiting (forgone profits, convenience yield, competitive erosion)	Easy to ignore; very important

Three differences from listed options deserve emphasis. First, the underlying is not traded, so there is no market price to observe and no volatility to read from an options screen. Second, exercise often has consequences for other players (competitors respond), which Black-Scholes ignores. Third, many real options are interrelated: exercising one changes the value of another, and the joint value is not the sum of the parts. The remainder of the chapter is largely about handling these three differences.

2.1.2 Risk-neutral valuation and replication

The logic of option pricing does not require that investors be risk-neutral. It requires only that the option's payoff can be replicated by a portfolio of the underlying asset and risk-free borrowing. If it can, the option's value must equal the cost of the replicating portfolio, otherwise arbitrage would be possible. Cox, Ross and Rubinstein showed that this replication value can be computed by an equivalent shortcut: take expected payoffs using *risk-neutral probabilities* and discount at the risk-free rate.

In a one-period binomial setting with up factor u, down factor d, and risk-free gross return R = 1 + r, the risk-neutral probability of the up move is

```
q = (R − d) / (u − d)
```

and the value of a claim with payoffs Cu and Cd is

```
C = [q · Cu + (1 − q) · Cd] / R
```

If the underlying pays a continuous leakage yield δ, the risk-neutral drift is r − δ, and q becomes (e^((r−δ)Δt) − d)/(u − d). The probability q is not a forecast. It is a pricing device that absorbs the market's risk aversion into the probabilities, which is why discounting at r is then correct. Applied to a project, it means we do not need the project's risk-adjusted discount rate to value the option on it: once the value of the underlying project is taken as given, the option is valued by replication.

2.1.3 The twin security assumption and the marketed asset disclaimer

Replication needs a traded asset that moves with the project's value. Copeland and Antikarov proposed the marketed asset disclaimer (MAD): take the project's own present value without flexibility as the best unbiased estimate of its market value and as its own twin, estimating volatility from the simulated variability of that value. The assumption is strong and only as good as the DCF behind it. When a genuine twin exists, such as a commodity with liquid futures, prefer it: natural resources are the best case and pure R&D the worst.

2.1.4 Two kinds of uncertainty

The most important conceptual step in this chapter is separating the two kinds of uncertainty that real projects face.

Market (systematic) uncertainty. Commodity prices, demand levels, interest rates and the prices of competing products move with the economy and are correlated with the market portfolio. Investors demand compensation for bearing them, so they must be priced by risk-neutral valuation. A lattice for the underlying value captures this kind of risk.

Private (technical, idiosyncratic) uncertainty. Whether a compound passes Phase II, whether an exploration well strikes oil, whether an engineering prototype works: these outcomes are largely uncorrelated with the market, can be diversified by investors, and command no risk premium. They enter the valuation as ordinary probabilities, with cash flows weighted by the probability of the outcome and discounted at the risk-free rate (or, equivalently, at the rate appropriate to the market-risk component).

The practical rule: use risk-neutral probabilities for market risk, and real-world probabilities for private risk, in the same model. Many errors in pharma valuation arise from blending the two, for instance by applying a high discount rate to cash flows that have already been probability-weighted for technical risk, thereby penalizing the same uncertainty twice.

2.2 Binomial Lattices in Depth

The binomial lattice is the workhorse of real options analysis because it handles early exercise, changing costs, staged decisions and non-standard payoffs, all of which Black-Scholes cannot, and because it is transparent: each node answers the question "what would we do here, and what is it worth?"

2.2.1 Building the lattice

For an underlying value S following a geometric process with volatility σ over a time step Δt, the Cox-Ross-Rubinstein calibration uses

```
u = e^(σ√Δt)      d = 1/u = e^(−σ√Δt)
q = (e^((r − δ)Δt) − d) / (u − d)
discount factor per step = e^(−rΔt)
```

The recombining property (an up move followed by a down move equals a down move followed by an up move) keeps the number of nodes at step i equal to i + 1, so that a 1,000-step lattice has only about half a million nodes and runs instantly on a computer. The algorithm is backward induction:

1. At the final date, compute the payoff at each node.
2. Step back one date. At each node compute the continuation value, [q · V(up) + (1 − q) · V(down)] · e^(−rΔt).
3. Compare the continuation value with the value of exercising now (for an American-style option). Take the larger.
4. Repeat until the first node. That value is the option value today.

A strength of the lattice is that step 3 provides a decision rule as a by-product: the set of nodes at which exercise is optimal is the exercise boundary, the policy the firm should follow.

2.2.2 Discretization error and convergence

A lattice with few steps is a crude approximation, and values converge with a saw-tooth pattern as steps increase. Run several step counts, report the converged value, and for discrete decision dates (stage gates at year-ends) set the steps so that decision dates coincide with lattice dates.

2.2.3 Cash-flow leakage

Waiting is not free. If the project would generate cash flows now that are forgone by waiting, or competitors may preempt, or the patent clock is running, the underlying "pays a dividend." In the lattice, the leakage rate δ reduces the risk-neutral drift and makes early exercise attractive. For a real option, δ is often the most influential input after σ: with δ = 0 a call option on a non-dividend-paying asset is never exercised early, whereas with a positive δ it may be optimal to invest as soon as the value is high enough.

Example 2.1 – Option to defer an investment: three-step lattice. Aldermoor Industrial, a fictional firm, can build a plant that costs I = 110. The present value of the plant's operating cash flows is S0 = 100, so the NPV of investing now is −10 and a static analysis says reject. The firm holds a license that lets it invest at any time in the next three years. The volatility of the plant's value is σ = 30%, the risk-free rate is r = 5%, and the value leaks at δ = 4% a year (forgone early profits and expected competitive entry). Value the option to wait.

Step 1: Calibrate the lattice (annual steps).

```
u = e^0.30 = 1.34986      d = 1/u = 0.74082
q = (e^(0.05 − 0.04) − 0.74082) / (1.34986 − 0.74082)
  = (1.01005 − 0.74082) / 0.60904 = 0.44206
discount factor = e^(−0.05) = 0.95123
```

Step 2: Project values.

Year	Node values of S
0	100.00
1	74.08, 134.99
2	54.88, 100.00, 182.21
3	40.66, 74.08, 134.99, 245.96

Step 3: Terminal payoffs, max(S − 110, 0). At year 3: 0, 0, 24.99, 135.96.

Step 4: Roll back, comparing continuation with immediate exercise. At year 2, node S = 182.21: continuation = 0.95123 × [0.44206 × 135.96 + 0.55794 × 24.99] = 0.95123 × [60.10 + 13.94] = 70.43; immediate exercise = 182.21 − 110 = 72.21. Exercise is better, so the node value is 72.21: this is early exercise driven by the leakage. At year 2, node S = 100: continuation = 0.95123 × [0.44206 × 24.99 + 0.55794 × 0] = 10.51; exercise value 0, so the value is 10.51. The lowest node

is worthless. At year 1, S = 134.99: continuation = 0.95123 × [0.44206 × 72.21 + 0.55794 × 10.51] = 0.95123 × [31.92 + 5.86] = 35.94; exercise = 24.99; the option is worth more alive: 35.94. At year 1, S = 74.08: continuation = 0.95123 × [0.44206 × 10.51 + 0] = 4.42. At year 0: 0.95123 × [0.44206 × 35.94 + 0.55794 × 4.42] = 0.95123 × [15.89 + 2.47] = 17.46; immediate exercise is −10, so the option is worth 17.46.

Step 5: Convergence. The three-step value of 17.46 is a coarse estimate. Re-running with more steps gives 16.44 (6 steps), 16.49 (12), 16.33 (50), 16.27 (200) and 16.25 (1,000). The converged value is about 16.25, so the three-step answer overstates by about 7 percent. The qualitative conclusion is stable: a plant with NPV of −10 is part of a license worth about 16.3.

Step 6: Interpret. The decision rule from the lattice is: do not invest now, and invest early only if the plant value rises well above the cost (at year 2, node S = 182.21, where the NPV is 72.21). The option value of 16.25 is the maximum the firm should pay for the license, and also a fair measure of what is lost if a competitor preempts the opportunity.

Takeaway. The option value is the product of volatility, time and the cost of waiting. Remove any of the three and the value shrinks; with σ = 0 the value is zero because the NPV is negative and nothing will change.

2.2.4 Calibrating volatility

Choosing σ is the central judgment call. The main methods are a market proxy (unlevered equity volatility of pure-play comparables, which measures firm rather than project risk), commodity volatility scaled by operating leverage (Section 2.4), simulation of project value (Copeland-Antikarov; Section 2.7) and scenario analysis. All are noisy, so the honest practice is to show value at low, base and high σ and ask whether the decision survives. If the case rests on a volatility no one can defend, it is a bet on the estimate rather than on the project.

2.3 R&D and Pharmaceutical Pipelines

Pharmaceutical development is the textbook case for real options because it combines staged investment, high technical risk, and large commercial upside whose size is uncertain until launch. The structure is a sequence of gates: Phase I, Phase II, Phase III, regulatory filing, launch. At each gate management can abandon, continue or (occasionally) accelerate. A decision tree already captures one part of this flexibility: abandonment after a failed trial. What it usually omits is the second part: the flexibility to respond to *market* information that arrives while the science is progressing.

2.3.1 Risk-adjusted NPV (rNPV): the industry baseline

The standard approach in pharma and biotech is rNPV. For each stage it computes the probability that the program reaches that stage, multiplies the costs and revenues by those probabilities, and discounts. Costs at a later stage are incurred only if the program survives to it; revenues occur only after launch. The probabilities are phase transition rates that companies estimate from their own history and from industry studies; this chapter uses illustrative values, not statistics for any particular therapeutic area.

Example 2.2 – rNPV of a four-stage program. Marrowfield's compound has the following profile (all figures in millions, illustrative):

Table 2.2 – Marrowfield development plan

Stage	Timing of cost	Cost	Probability of success	Cumulative probability of reaching next stage
Phase I	t = 0	20	60%	0.600
Phase II	t = 1	50	35%	0.210
Phase III	t = 3	150	60%	0.126
Filing and approval	t = 5	10	90%	0.1134

If the drug is approved, it launches at t = 6, and the expected net value of commercial cash flows at launch (after launch costs) is 1,100. The discount rate is 10%.

Step 1: Probability-weighted costs and discounting. A cost is paid only if the program has survived to that date:

```
Phase I:    20 × 1.000 / 1.10^0 = 20.00
Phase II:   50 × 0.600 / 1.10^1 = 27.27
Phase III: 150 × 0.210 / 1.10^3 = 23.67
Filing:     10 × 0.126 / 1.10^5 =  0.78
Total PV of expected costs      = 71.72
```

Step 2: Probability-weighted commercial value.

```
1,100 × 0.1134 / 1.10^6 = 124.74 / 1.771561 = 70.41
```

Step 3: rNPV.

```
rNPV = 70.41 − 71.72 = −1.31
```

For contrast, the unrisked NPV that ignores technical failure is $-20 - 50/1.1 - 150/1.1^3 - 10/1.1^5 + 1{,}100/1.1^6 = 436.56$. Technical risk, not discounting, is what destroys the value: the program has only an 11.3 percent chance of reaching the market.

Step 4: Interpret. The rNPV is marginally negative. A launch value of 1,120.5 would bring it to zero. Taken at face value, Marrowfield should shelve the compound. But note what the calculation assumes: the net launch value of 1,100 is a single number, and the firm launches in every approved scenario, whatever the market looks like. This is the gap the real options approach fills.

2.3.2 Where rNPV and real options differ

rNPV has strengths: it is simple, it uses firm-specific technical probabilities, and it respects the staged abandonment option implicitly. Its weaknesses are the ones the vignette identified:

1. **A single discount rate for everything.** A 10 percent rate is applied to costs, which are nearly certain, and to revenues, which are very uncertain. Costs should be discounted at close to the risk-free rate; applying a high rate to costs understates their burden. Revenues deserve a rate tied to market risk, not to the technical risk already handled by the probabilities.
2. **Static commercial value.** The launch value is one number, so no distinction is made between a market that has grown and one that has shrunk. Launch is assumed to occur whatever the value.
3. **No value for market information.** The program takes six years. During this time, competitor results, reimbursement policy and epidemiology change, and management can respond.

2.3.3 The compound option structure

The program is a *compound option*: an option on an option on an option. Paying the Phase I cost buys the right to pay the Phase II cost, which buys the right to pay the Phase III cost, and so on, until the last right is the right to launch by paying the launch cost I in return for the commercial value V. Geske derived a closed form for a compound option with two stages. For more stages, with mixed market and technical risk, the lattice is the practical tool.

The valuation uses both types of uncertainty from Section 2.1.4:

- **Market uncertainty** is carried by the lattice for the value V of the commercial opportunity, with risk-neutral probabilities and discounting at the risk-free rate.
- **Technical uncertainty** enters at the gates through ordinary probabilities: at the end of each phase, the node value is multiplied by the probability of success, and the next phase cost is paid only if the continuation is worth more than its cost.

Example 2.3 – Marrowfield as a compound option on a lattice. Use the same program, but model the commercial opportunity explicitly. The market value today of the gross commercial opportunity, the present value of launch-phase cash flows before the launch investment, is V0 = 900. (This corresponds to an expected gross value at launch of 1,400 if growth is at an annual rate of 7.6%, a rate reflecting only the market risk of commercial cash flows; the 10% used in Example 2.2 also tried to cover technical risk, which the probabilities already cover.) The launch investment is I = 300 at t = 6, so the net expected launch value is 1,100 as in Example 2.2. Market volatility is σ = 40%, and the risk-free rate is 4%. Annual lattice steps with six periods. The gates are:

- t = 0: pay 20 to start Phase I (decision).
- t = 1: Phase I succeeds with probability 0.60; if so, decide whether to pay 50 for Phase II.
- t = 3: Phase II succeeds with probability 0.35; if so, decide whether to pay 150 for Phase III.
- t = 5: Phase III succeeds with probability 0.60; if so, decide whether to pay 10 for filing.
- t = 6: approval with probability 0.90; then decide whether to launch: payoff max(V6 − 300, 0).

Step 1: Calibrate.

```
u = e^0.40 = 1.49182      d = 0.67032
q = (e^0.04 − 0.67032) / (1.49182 − 0.67032) = (1.04081 − 0.67032) / 0.82150 = 0.45099
```

Step 2: Terminal values at t = 6. The lattice at t = 6 ranges from V6 = 900 × d^6 ≈ 82 to 900 × u^6 ≈ 9,918. The payoff at each node is 0.90 × max(V6 − 300, 0).

Step 3: Roll back with gate decisions. Between gates, node values are discounted risk-neutral expectations. At each gate the node value is multiplied by the success probability and the next phase cost is paid only if continuation exceeds that cost. For instance, at t = 5 the five of the six nodes whose continuation is above 10 pay the filing cost; at t = 3 the lowest node, V3 = 900 × d^3 = 271, is abandoned; at t = 1, the down node (V1 = 603.3) is abandoned because its continuation value is less than the Phase II cost of 50, while the up node (V1 = 1,342.6) continues with a probability-weighted value of 64.80.

Step 4: Value today. The value at the first node is

```
Value = −20 + e^(−0.04) × [0.45099 × 64.80 + 0.54901 × 0] = −20 + 0.96079 × 29.22 = 8.08
```

Marrowfield's compound is worth +8.08 under the lattice, against −2.49 if the firm were committed to all stages and the launch (the corresponding "static" value in this framework), and against a marginally negative rNPV of −1.31.

Step 5: Where does the 10.6 of option value come from? Decompose by switching off flexibility:

Table 2.3 – Decomposing the option value (σ = 40%)

Case	Abandon at technical gates?	Launch only if profitable?	Value
Fully committed	No	No	−2.49
Launch option only	No	Yes	−0.54
Gate abandonment only	Yes	No	7.95
Full flexibility	Yes	Yes	8.08

Almost all of the option value comes from the *gates*, not from the final launch option. This makes sense: the gates give the firm the right to stop spending after bad news (bad commercial news as well as bad science), while the launch decision matters only in a narrow range, because the launch payoff has only an 11.3% chance of being reached. The distinction is useful for management: the important flexibility to preserve is the ability to stop after each phase, which implies preserving contract terms (no minimum commitments to manufacturers or partners) that keep gates free.

Step 6: Sensitivity to volatility. The commercial volatility is not observable. Value as a function of σ:

Table 2.4 – Compound option value versus market volatility

σ	Fully committed	Launch option only	Full flexibility
30%	−2.49	−2.01	3.59
40%	−2.49	−0.54	8.08
50%	−2.49	1.09	12.58
60%	−2.49	3.56	17.36

The committed value is independent of σ because, without flexibility, payoffs are linear in V and the expected value under risk-neutral probabilities does not depend on dispersion. Flexibility is what converts volatility into value. At σ = 25% the full flexibility value falls to 1.45, and at σ = 20% or below it is zero (the program is not started at all), so the decision to proceed depends on a volatility estimate that no one can verify. The right conclusion for Teodora's board is not "the program is worth 8.08" but "the program is worth between roughly 4 and 17 under volatilities between 30 and 60 percent, and nothing if the commercial market is as stable as 20 percent volatility implies; and the value arises mainly from the right to stop after each gate."

2.3.4 Managerial implications for pipelines

Value the gates, not the program: at each gate the question is whether continuation value exceeds the next cost. Anything that raises the cost of stopping (manufacturing commitments, minimum purchase contracts, hiring ahead of evidence) destroys option value. Low-cost experiments that resolve uncertainty before a large commitment are worth more than their direct NPV. Licensing a compound early sells the option, so the price should be compared with the option value, not with the rNPV; a partner with a lower cost of capital or better commercial capabilities may value the same option higher, which creates room for a deal. Finally, technical risk diversifies across independent programs but shared commercial uncertainty does not.

2.4 Natural Resource Options

Mining, oil and gas, forestry and agriculture were the first home of real options theory (Brennan and Schwartz, and Paddock, Siegel and Smith, are the classic references) for two reasons. The underlying has a traded price that supplies a volatility estimate and, through futures markets, a risk-neutral drift. And the managerial options are plain: an owner of an undeveloped reserve can wait, develop, expand, curtail, temporarily close or abandon, depending on the price.

2.4.1 Convenience yield and the risk-neutral drift

A commodity owner receives a benefit from holding the physical good that a futures holder does not: the ability to keep production running, to avoid stock-outs and to profit from temporary shortages. This is the *convenience yield*, δ. It is analogous to a dividend, and it determines the slope of the futures curve. In the risk-neutral world, the spot price of a commodity grows at r − δ, and the futures price for delivery in τ years is F(τ) = S × e^((r − δ)τ). If the futures curve is downward-sloping (backwardation), δ exceeds r; if upward-sloping (contango), δ is below r. Extracting δ from the futures curve is therefore a market-based calibration: it gives the risk-neutral drift without requiring any view about the expected price.

For an undeveloped reserve, δ plays a second role. The opportunity to produce the commodity is the underlying of the development option, and its value decays at a rate linked to δ because waiting delays cash flows that depend on the price. A high convenience yield makes waiting costly and favors earlier development.

2.4.2 Valuing the developed mine

If the mine, once developed, produces Q units a year for N years at a unit extraction cost c, and costs are fixed in nominal terms, then under risk-neutral valuation the value of the developed mine at spot price S is

```
V(S) = Σ(τ=1..N) e^(−rτ) × Q × [ S·e^((r−δ)τ) − c ]
     = Q × [ S × A_δ − c × A_r ]
where A_δ = Σ e^(−δτ),  A_r = Σ e^(−rτ)
```

The expression separates into a price-sensitive term and a cost term. Two consequences matter. First, the *operating leverage* of the mine is the ratio of the price-sensitive term to the net value, which amplifies price volatility into value volatility. Second, the fixed-cost term means that V can be small or negative even when S is substantially above zero: the mine is a levered long position in the commodity.

Example 2.4 – When to develop an undeveloped copper deposit. Redstone Mining, a fictional company, holds a deposit that can produce Q = 60 million pounds of copper a year for N = 10 years once developed, at a unit cost of c = 1.80 per pound (constant). Developing costs I = 650 million, paid at the time of development. The spot price is S0 = 3.00 per pound, with volatility 25%, a convenience yield δ = 4% (derived from the futures curve) and a risk-free rate r = 5%. The development license expires in T = 3 years. All values in millions.

Step 1: Annuity factors.

```
A_δ = Σ e^(−0.04τ), τ = 1..10 = 8.0783
A_r = Σ e^(−0.05τ), τ = 1..10 = 7.6743
```

Step 2: Developed value and NPV now.

```
V(3.00) = 60 × (3.00 × 8.0783 − 1.80 × 7.6743) = 60 × (24.235 − 13.814) = 625.26
NPV now = 625.26 − 650 = −24.74
```

Static NPV says reject. The NPV breakeven price is S* = (650/60 + 1.80 × 7.6743)/8.0783 = (10.833 + 13.814)/8.0783 = 3.05 per pound.

Step 3: Operating leverage. The elasticity of V with respect to S is 60 × 3.00 × 8.0783 / 625.26 = 2.33. So a 25% price volatility becomes about 2.33 × 25% = 58% volatility in the value of the developed mine, which is the effective volatility the option is exposed to.

Step 4: American option to develop within three years (lattice). With quarterly steps (n = 12): Δt = 0.25, u = e^(0.25 × √0.25) = e^0.125 = 1.1331, d = 0.8825, and the risk-neutral probability is q = (e^((0.05 − 0.04) × 0.25) − 0.8825) / (1.1331 − 0.8825) = (1.0025 − 0.8825)/0.2506 = 0.479. At each node the mine value V(S) − I is compared with continuation. The lattice gives:

```
n = 3:    250.06
n = 6:    226.43
n = 12:   230.13
n = 36:   232.22
n = 300:  232.69
```

The converged option value is 232.7 against an NPV of −24.7. Even though the project is under water at today's prices, the right to develop is worth about 37% of the developed mine's value.

Step 5: The exercise trigger. Find the lowest spot price at which exercising today beats waiting. Solving the lattice numerically gives S ≈ 5.96 per pound, nearly twice the NPV breakeven of 3.05. In other words, Redstone should not develop at 3.50, where the NPV is +217.6, because the option value at 3.50 is 381.2, well above the NPV. Only above roughly 5.96 does the cost of waiting exceed the value of waiting.

Step 6: Sensitivity.

Table 2.5 – Redstone development option: sensitivities (base: S = 3.00, σ = 25%, δ = 4%, I = 650, T = 3)

Variable	Value of variable	Option value	Comment
Price volatility σ	15%	142.6	Value falls but remains positive
	25% (base)	232.7	
	40%	365.0	
Convenience yield δ	0%	604.7	Waiting is costless; value skyrockets
	4% (base)	232.7	
	8%	73.3	Waiting is very costly
Development cost I	550	273.5	Cheaper exercise raises value
	850	167.0	
Option life T	1 year	135.2	
	3 years (base)	232.7	
	5 years	289.9	Longer window, higher value

Step 7: Interpret. Three things are visible. The option value is highly sensitive to the convenience yield: δ is observed (from futures) and should be taken seriously, because a mine owner who ignores it will overvalue waiting. The value rises with the volatility of the commodity, an uncomfortable conclusion for the risk-averse manager but a correct one for the owner of

the *right* to develop, because the downside is limited to forgoing development. And the exercise trigger shows why mining companies sit on large undeveloped reserves through periods in which a naive NPV would say develop; the opposite, rushing to develop when prices are only moderately above breakeven, destroys option value.

Takeaway. The correct investment rule is "develop when the price rises above the trigger," not "develop when the NPV is positive."

2.4.3 Other natural resource options and cautions

The same logic covers temporary shutdown and reopening (optimal policy has a hysteresis band: a mine shuts below variable cost and restarts only well above it, because switching costs money), abandonment (a put on project value), modular expansion (a series of calls) and exploration (a staged, pharma-style structure feeding the development option). Two cautions apply. Convenience yield and volatility change over time, and spot-based volatility reflects short-term noise that does not apply to long-dated cash flows. And costs often correlate with prices (high prices raise input costs), which dampens effective operating leverage. A defensible valuation tests mean-reverting price processes and cost correlation before concluding that waiting is valuable.

2.5 Platform and Staged Investments

Many strategic investments are not single decisions. A company enters a new market with a pilot; if the pilot succeeds, it builds regional capacity; if the region succeeds, it scales nationally. Each step is small relative to the whole, buys information and keeps the next option alive. These are *platform investments* or *staged expansions*, and they are the situation in which "strategic value" claims are most frequent and most easily abused.

2.5.1 Sequential options

A sequence of stages is a compound option. Stage 1's payoff is the right to undertake stage 2, whose payoff is the right to undertake stage 3, and so on. Two properties distinguish compound options from simple ones.

1. **Interdependence.** The value of stage 2 depends on stage 3. Valuing each stage on its own and adding them up is wrong in both directions: it ignores the information flowing from earlier stages, and it ignores that later stages' exercise costs must be paid to keep the chain alive.
2. **The cost of the chain.** Each intermediate payment is a *fee for the next option*. If the fee exceeds the value of the next option in some states, management declines, which is why abandonment appears naturally in the lattice.

2.5.2 What makes a platform valuable

A platform is worth more than its direct NPV when uncertainty is high enough for information to matter, the initial outlay is small relative to later commitments, the stages are separable so that stopping leaves little stranded cost, and a proprietary advantage makes the follow-on rights hard to replicate. If any condition fails, the "platform" is an ordinary investment with a strategic label.

Example 2.5 – Pilot, regional rollout and national scale-up. Greenfield Retail, a fictional chain, can launch a convenience-store format. The decision sequence is:

- t = 0: open a pilot costing 10, which preserves the right to continue.
- t = 2: if the pilot looks promising, pay 40 for a regional rollout that preserves the right to scale.

- t = 4: pay 120 to scale nationally, receiving the national business, whose present value S follows a lognormal process with S0 = 100 today, volatility 35%, risk-free rate 5%, and no leakage.

Step 1: Static NPV. If the firm committed to all three payments today, the NPV is

```
100 − 10 − 40/1.05² − 120/1.05⁴ = 100 − 10 − 36.28 − 98.72 = −45.01
```

The project fails by a wide margin. A manager who accepts it as "strategic" is effectively ignoring a 45 loss.

Step 2: The terminal option. At t = 4 the firm will scale only if S4 > 120. Viewed from today, a European call on S with strike 120 and T = 4 (Black-Scholes with σ = 0.35, r = 0.05) is worth 28.01.

Step 3: The stage-2 decision at t = 2. At t = 2 the firm holds an option expiring in two years with strike 120. Its value is a Black-Scholes call on S2. The firm pays 40 to keep it only if the call is worth more than 40. Computing the call at a range of S2 values:

Table 2.6 – The stage-2 decision at t = 2

S at t = 2	Value of call (K = 120, 2 years)	Net of 40 fee	Decision
60	2.20	0	Stop
100	16.37	0	Stop
120	28.49	0	Stop
150	51.02	11.02	Continue
200	95.16	55.16	Continue

The threshold is S2 = 136.02: below it the continuation is worth less than the fee. The risk-neutral probability that S2 exceeds 136.02 is 25.2% (with a real-world drift of 10% it would be 32.1%, but the probability used in pricing is the risk-neutral one).

Step 4: Value today (lattice, 400 steps). Discounting the stage-2 payoff back to t = 0 gives a compound option value of 10.69. Subtracting the pilot cost:

```
Platform value = 10.69 − 10 = +0.69
```

Step 5: Compare. Static NPV: −45.01. Platform with staged options: +0.69. The flexibility is worth about 45.7.

Step 6: Why not 28.01? The single call (28.01) would apply if the firm could buy the scale-up right for nothing at t = 0 and wait. The firm must instead pay 10 at t = 0 and 40 at t = 2 (when it chooses to continue) to keep the right alive. The compound option's value of 10.69 is the 28.01 call less the expected cost of the fees on the paths where the firm continues and the lost value on paths where it stops. This is the cost of the chain.

Step 7: Sensitivity to volatility.

Table 2.7 – Platform value (net of pilot cost) by volatility

σ	Compound option value	Net of pilot (10)	Single call, K = 120, T = 4
20%	2.62	−7.38	16.60
30%	7.68	−2.32	24.26
35%	10.69	+0.69	28.01
45%	17.15	+7.15	35.31
55%	23.88	+13.88	42.28

The pilot pays for itself only if volatility is at least about 34 percent (interpolating between the 30% and 35% rows). The conclusion is a disciplined version of the "strategic" argument: the pilot is justified only if the market is uncertain enough, the staging cheap enough, and the follow-on rights credible. The analysis gives managers a specific, testable claim: under these assumptions the pilot breaks even at roughly 34% volatility. If experience in comparable markets suggests that volatility is lower, the pilot should not go ahead on option grounds.

2.5.3 Interacting options and the limits of additivity

When options coexist on the same underlying (for example, an option to expand and an option to abandon), their values interact. Having the abandonment right makes expansion less risky, and having the expansion right makes abandonment less likely. The combined value is less than the sum of the separate values, because exercising one option reduces the value of the other. A lattice that carries both choices at each node handles this correctly. Adding up separately computed option values overstates, and this is one of the most common sources of inflated "strategic value."

2.6 Option Games: Real Options with Competition

The models so far treat the investment decision as a single-agent problem against nature. In many industries, however, the value of waiting depends on what rivals do. If a rival invests first and captures the market, the value of waiting evaporates. Option games combine real options with game theory (the work of Smit and Trigeorgis is the standard reference) to analyze such situations. The core tension is a trade-off: uncertainty favors waiting to learn, but competition punishes waiting.

2.6.1 Structure of option games

An option game needs the following ingredients: a stochastic state (market value V), a timing choice for each player (invest or wait), payoffs that depend on whether one or both firms are in the market, and a solution concept. The most common concept is subgame-perfect Nash equilibrium: solve the last decision date first, then work backward, treating each date's decisions as a simultaneous-move game with continuation values filled in from later dates.

The payoff structure determines the type of game:

- **First-mover advantage** (the leader earns more than the follower): leads to *preemption*. Each firm wants to invest first, and competition may drive investment earlier than a monopolist would choose. The game resembles chicken or the war of attrition.
- **Second-mover advantage** (waiting to see the market is more valuable than being first): leads to *waiting equilibria*. Both firms wait, and investment is delayed.
- **Mutually destructive simultaneous entry:** neither firm wants to enter at the same moment as the other, which generates chicken-type games with several equilibria.

Example 2.6 – A two-firm timing game. Two firms, A and B, can each build a plant costing I = 60. The market value if the plant is in operation and the firm is alone is V; V today is 100, and next year it will be 150 (up) or 60 (down), with a risk-free rate of 5%, so the risk-neutral probability of the up move is (1.05 × 100 − 60)/(150 − 60) = 0.50. Competition reduces payoffs: if both firms are in the market at the same time, each earns 0.55 V. If one firm enters first and the other follows later, the early entrant keeps 0.70 V after the follower arrives and the follower earns 0.45 V. The decision dates are t = 0 and t = 1.

Step 1: The last date, t = 1. Suppose neither firm has invested.

- State up (V = 150): if both invest, each earns 0.55 × 150 − 60 = 22.5. Investing alone yields 150 − 60 = 90, so each firm has a dominant incentive to invest, and the outcome is that both invest and earn 22.5.
- State down (V = 60): even a monopolist earns 60 − 60 = 0, and duopoly gives 0.55 × 60 − 60 = −27; neither invests; payoff zero.

Step 2: Value of the "wait–wait" outcome at t = 0. If both wait at t = 0, each earns (0.50 × 22.5 + 0.50 × 0)/1.05 = 10.71.

Step 3: Leader and follower values when one invests at t = 0. Suppose firm A invests now and B waits. At t = 1, B follows only if 0.45 V − 60 > 0. In the up state, 0.45 × 150 − 60 = 7.5 > 0, so B follows and A keeps 0.70 V = 105. In the down state, B does not follow: 0.45 × 60 − 60 < 0. A is then a monopolist worth V = 60.

```
Leader (A) value = [0.5 × 105 + 0.5 × 60] / 1.05 − 60 = 82.5 / 1.05 − 60 = 78.57 − 60 = 18.57
Follower (B) value = [0.5 × 7.5 + 0.5 × 0] / 1.05 = 3.57
```

Step 4: Both invest at t = 0. Each earns 0.55 × 100 − 60 = −5.

Step 5: The payoff matrix at t = 0.

Table 2.8 – Payoffs at t = 0 (firm A, firm B)

	B invests	B waits
A invests	(−5.00, −5.00)	(18.57, 3.57)
A waits	(3.57, 18.57)	(10.71, 10.71)

Step 6: Equilibria. If B waits, A prefers to invest (18.57 > 10.71). If B invests, A prefers to wait (3.57 > −5). The game has two pure-strategy equilibria, (invest, wait) and (wait, invest), and a mixed-strategy equilibrium. In the mixed equilibrium each firm invests with probability q such that the other is indifferent:

```
q × (−5) + (1 − q) × 18.57 = q × 3.57 + (1 − q) × 10.71
18.57 − 23.57q = 10.71 − 7.14q  →  7.86 = 16.43q  →  q = 0.478
```

Each firm's expected value in the mixed equilibrium is 0.478 × 3.57 + 0.522 × 10.71 = 7.30, lower than the 10.71 that cooperative waiting would deliver. With probability 0.478^2 = 22.9% both invest simultaneously and each loses 5.

Step 7: Compare with the single-firm benchmark. A monopolist can invest now for an NPV of 100 − 60 = 40, or wait and earn (0.5 × 90 + 0.5 × 0)/1.05 = 42.86. The monopolist therefore waits. Competition has cut the value from about 42.9 to between 7.3 and 18.6 per firm and has created a coordination problem.

Step 8: Sensitivity to the advantage of the first mover. Vary the leader's retained share and the follower's share.

Table 2.9 – Equilibrium at t = 0 for different degrees of first-mover advantage

Leader share / follower share	Leader value	Follower value	Wait-wait value	Equilibrium
0.55 / 0.55 (no advantage)	7.86	10.71	10.71	Waiting is dominant: both wait
0.60 / 0.50	11.43	7.14	10.71	Chicken; mixed probability of investing now 5.6%
0.70 / 0.45	18.57	3.57	10.71	Chicken; mixed probability of investing now 47.8%
0.80 / 0.40	40.00	0.00	10.71	Chicken; mixed probability of investing now 85.4%

(In the last row the follower does not find entry profitable even in the up state, since $0.40 \times 150 - 60 = 0$, so the leader is effectively a monopolist and keeps the whole value.)

Takeaway. Without competition the optimal policy is to wait. As the first-mover advantage grows, the probability that a rival invests early rises from zero to almost one, and the option to wait is destroyed. The managerial task is to assess the size of the first-mover advantage honestly. Overestimating it leads to premature, value-destroying races; underestimating it leads to being preempted.

2.6.2 Practical use of option games

Option games rarely yield a unique answer to impose on a manager. Their value is to identify which payoff structure (preemption or waiting) describes the industry, to price the cost of waiting under competition, and to show how commitment devices (patents, exclusive contracts, capacity announcements) shift the equilibrium. Payoff parameters are hard to estimate, equilibria are often multiple, and real competition involves signaling that stylized games omit.

2.7 Simulation-Based Option Valuation

Lattices are ideal when one or two state variables drive the value and decisions occur on a modest number of dates. They become unwieldy when the payoff depends on the whole *path* (cumulative production, minimum price along the way), when there are several sources of uncertainty (price, cost, demand, exchange rate), or when the stochastic process is not simple (mean reversion with jumps). Monte Carlo simulation handles all of these by generating many random scenarios and averaging discounted payoffs. Its weakness is that, in its basic form, it can only value European-style claims, because the value of waiting at a future date requires knowing the future continuation value, which a forward simulation does not provide.

2.7.1 Monte Carlo basics

To value a European-style option by simulation under risk-neutral valuation:

1. Simulate the state variable forward with risk-neutral drift $r - \delta$, for example $S(t + \Delta t) = S(t) \times \exp[(r - \delta - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t} \times Z]$, with Z standard normal.
2. At expiry, compute the payoff of each path.
3. Discount the payoff at the risk-free rate and average over paths.
4. Report the standard error: SE = (sample standard deviation of discounted payoffs)/$\sqrt{n}$.

Accuracy improves only with √n: to cut the standard error by a factor of 10 requires 100 times as many paths. *Variance reduction* is therefore valuable. Antithetic variates pair each path with its mirror image (using −Z), halving the effective variance for monotone payoffs. Control variates subtract a correlated quantity with a known expectation, such as the Black-Scholes value of a similar European option. Low-discrepancy sequences (Sobol) fill the space more evenly than pseudo-random numbers.

2.7.2 The early-exercise problem and least-squares Monte Carlo

Real options are often American-style: the firm can invest or abandon at any time. Longstaff and Schwartz solved this with *least-squares Monte Carlo* (LSM). The algorithm works backward through the simulated paths:

1. Simulate all paths forward and record the state at each decision date.
2. At the last decision date, the payoff is the exercise value.
3. At the previous date, consider only the paths for which exercise is in the money. Regress the *discounted future cash flow along each path* on a polynomial of the current state (for example 1, S, S^2). The fitted value is the estimated continuation value.
4. For each path, exercise if the immediate exercise value exceeds the fitted continuation value, and update that path's cash flow and exercise date.
5. Repeat back to date 1, then discount each path's cash flow to time zero and average.

The regression is the clever part: it estimates the conditional expectation that would otherwise require a nested simulation. LSM produces a slightly low-biased estimate (because the exercise policy is estimated and therefore suboptimal) and its accuracy depends on the number of paths and the richness of the basis functions.

Example 2.7 – Abandonment option by LSM, checked against a lattice. A project has present value S0 = 100 with volatility 30%, leakage δ = 3%, and risk-free rate 5%. The firm may abandon the project at any quarter-end during the next three years and receive a salvage value K = 100. This is an American put on the project value with 12 exercise dates (Bermudan). We value it three ways.

Step 1: European benchmark. The Black-Scholes-type value of a European put with S = 100, K = 100, T = 3, σ = 30%, r = 5%, δ = 3% is 15.65. The option can be exercised only at the end; this is a lower bound.

Step 2: Lattice benchmark. A binomial lattice with 1,000 steps (American, exercisable at every node) gives 16.84. With 12 steps (matching the exercise dates) it gives 16.62. The early exercise premium over the European value is therefore about 1.0 to 1.2.

Step 3: LSM. Simulate 100,000 paths (50,000 draws plus their antithetic mirror images) with quarterly dates, using a degree-2 polynomial of S/S0 as the regression basis, regressing only on in-the-money paths. A run with random seed 42 gives:

```
Option value = 16.75     standard error = 0.053
```

Repeating with seeds 1, 2 and 3 gives 16.67, 16.75 and 16.73, so the estimate is stable at about 16.7 ± 0.1. Using 10,000 paths (5,000 plus antithetic) gives 16.73 with a standard error of 0.170, and using 500,000 paths gives 16.72 with a standard error of 0.024, which shows the √n law in action.

Step 4: Interpret. The LSM, lattice and discrete-date lattice results agree to within about 0.2 (the 12-step lattice carries its own discretization error, so LSM's expected small downward bias is not visible against it), and all exceed the European value by one to one and a half. The abandonment right is worth 16.7 on a project whose present value is 100. For a manager, the useful output is not only the value but the *policy*: the regression identifies for each date the project value below which abandonment is optimal, which can be turned into a trigger for monitoring.

Takeaway. Use the lattice when it fits the problem, and simulate when it does not, but always benchmark the simulation against a lattice or closed form in a simplified version of the problem. If the two disagree, the simulation code is the first suspect.

2.7.3 Estimating project volatility by simulation

The most common use of simulation in real options practice is not to value the option directly, but to *estimate σ*. In the Copeland-Antikarov approach, the analyst builds a cash flow model with the main uncertain variables (price, volume, cost), simulates one period ahead, and computes the value of the project at time 1 *conditional on the first-period outcome*, including the first period's cash flow. The volatility is the standard deviation of the log return z = ln[(PV1 + CF1) / PV0].

Example 2.8 – Volatility from a cash flow model with operating leverage. A project has eight years of cash flows. Revenue in year t has expected value R_t = 200 × 1.04^t. The contribution margin is 40% of revenue, and there is a fixed cost F per year. The discount rate for the project's own cash flows is 9%. The revenue shock in year 1 is lognormal with a standard deviation of 20% and mean one, and it persists: expected revenues in years 2 to 8 are scaled by the same factor. (Taxes are ignored to keep the arithmetic visible.)

Step 1: PV0. With F = 30:

```
PV0 = Σ(t=1..8) [0.40 × 200 × 1.04^t − 30] / 1.09^t = 355.06
```

Step 2: Year-1 outcome. If the realized year-1 revenue is R1 = 208 × e^ε (where ε has mean −0.02 and standard deviation 0.20 so that the mean of e^ε is one), then

```
CF1 = 0.40 × R1 − 30
PV1 = 0.40 × R1 × 5.8269 − 30 × 5.0330
```

where 5.8269 = Σ(t=2..8) $1.04^{(t-1)/1.09}$(t−1) and 5.0330 = Σ(t=2..8) 1/1.09^(t−1). For the mean revenue R1 = 208, CF1 = 53.20, PV1 = 333.81, and PV1 + CF1 = 387.01, which equals PV0 × 1.09, as required by consistency.

Step 3: Simulate. Draw 200,000 values of ε and compute z = ln[(PV1 + CF1)/PV0] for each. The standard deviation of z is 0.3032. A first-order approximation gives (elasticity of project value to revenue) × 0.20 = 0.294, with the difference due to the nonlinearity of the logarithm.

Step 4: Operating leverage. Repeat for different fixed costs:

Table 2.10 – Operating leverage and project volatility (revenue volatility 20%)

Fixed cost F	PV0	Volatility of log return σ_z	Amplification vs. revenue volatility
0	521.10	19.98%	1.00
20	410.40	25.76%	1.29
30	355.06	30.32%	1.52
40	299.71	37.34%	1.87

Step 5: Interpret. The project's value is more volatile than its revenue because fixed costs create operating leverage: each unit of revenue shock falls through to profit, but costs do not move. With F = 30 the project volatility is about 30%, which would be the σ for the lattice. An analyst who instead used the 20% revenue volatility would understate the option value substantially (compare the sensitivity tables in Sections 2.3 and 2.4). Equally, an analyst who used equity volatility from a highly levered comparable firm would overstate σ, because financial leverage is not present in the project's unlevered value.

2.7.4 Choosing between methods

Use closed forms (Black-Scholes, Geske) for European claims with one risk source; lattices for American exercise, stage gates and one or two risk factors, where each node is auditable; and Monte Carlo or LSM for path dependence and many uncertain variables, with a fixed seed and a benchmark against a simplified lattice.

2.8 Pitfalls and Governance of Real Options Analysis

Real options analysis can raise the value of almost any project, which makes it dangerous as well as powerful. The pitfalls fall into four groups.

Conceptual pitfalls. An option exists only if the firm holds an exclusive or protected right and is genuinely free to decline. A growth opportunity open to every competitor is not a proprietary option, and a right the firm cannot decline for contractual or political reasons is an obligation. Volatility creates value only if management can respond to it; otherwise it is merely risk. Risk can be double-counted by discounting probability-weighted cash flows at a rate that already contains a technical-risk premium, or by using risk-neutral probabilities and then discounting at a high rate. And a deferral option that ignores leakage will recommend waiting forever.

Input pitfalls. Option value rises with σ, so a promoter who wants a higher value picks a higher σ. Defensible practice documents the source, shows a range, and reports the break-even volatility at which the project becomes worthwhile (as in Example 2.5), turning the board's question into "is a volatility of at least 34 percent plausible for this market?" Option lives taken from patent terms or leases ignore competitors who may act sooner. Exercise costs that inflate or hide stranded costs must be modeled. And errors in the underlying DCF (inflated growth, understated reinvestment) are magnified by the option.

Model pitfalls. Interacting options are not additive (Section 2.5.3). Volatility usually falls as uncertainty resolves, for instance as a drug moves through trials. Geometric Brownian motion suits project values and equities but not mean-reverting commodities or jump-prone technologies. Too few lattice steps give coarse answers, and values quoted to four significant figures from inputs known to ±50 percent convey false precision.

Governance. Firms that use real options well document the option (who holds the right, when it expires, what it costs, what information will arrive, who decides); report static NPV and option value separately so the board sees how much of the case rests on flexibility; stress-test σ, δ, T and the underlying; record the exercise triggers the valuation implies and review actual decisions against them; build cheap flexibility into contracts (break clauses, modular designs); and keep the promoter from setting the volatility. The value of real options thinking lies at least as much in the questions it forces (what could we learn, what could we do about it, what does it cost to wait?) as in the numbers it produces.

Case Study: Calder Biosciences and the Phase II Decision

Illustrative case. All names and figures are fictional.

Calder Biosciences is a small company with one asset, CBX-201, a candidate for a chronic metabolic condition that has just completed preclinical work and is ready to enter Phase II. The board is deciding whether to fund the program internally or to sell the asset to a larger company, Meridian Pharma, that has offered a one-time payment of 4 million today.

The program. Phase II costs 60 million now and takes two years, with a 40 percent chance of success. If it succeeds, Calder can pay 180 million at t = 2 for a three-year Phase III trial with a 55 percent chance of success at t = 5. If that succeeds, Calder pays 15 million to file, and approval at t = 6 has an 85 percent probability. After approval, Calder decides whether to launch by paying a commercialization investment of 250 million. The cumulative probability of reaching approval is 0.40 × 0.55 × 0.85 = 18.7 percent. The gross value of the commercial opportunity, defined as the present value today of the launch-phase cash flows before the 250 million investment, is estimated at 850 million, with a risk-free rate of 4 percent.

The committee's view. Calder's finance committee used rNPV, treating the opportunity as a fixed value and committing to every stage, and obtained a program value of −7.0 million (that is, −6.99). On this basis, it recommends accepting Meridian's offer of 4 million.

The real options analysis. The business development team argued that the commercial value is uncertain, that the committee's calculation commits Calder to launch in every approved scenario, and that the program can be stopped at the end of Phase II, at the end of Phase III or at launch. They built a lattice with annual steps for six years, risk-neutral probabilities for the commercial value and the technical probabilities at the gates. The volatility of the commercial value is not observable; they considered comparable biotechnology firms' unlevered volatilities (about 45 percent) but also noted that the volatility of a single drug market might be lower, around 30 percent, or higher, around 60 percent.

The results (in millions):

Commercial volatility	Fully committed	Launch option only	Gates only	Full flexibility
30%	−6.99	−6.72	0.00	0.00
45%	−6.99	−3.60	8.39	8.69
60%	−6.99	−0.08	18.23	18.23

At 45 percent volatility the program is worth 8.7 million, more than twice Meridian's offer, and nearly all of the value comes from the gates (the right to stop after Phase II or Phase III) and not from the launch decision. At 30 percent the program is worth nothing, so Meridian's 4 million would look generous. At 60 percent it is worth 18.2 million.

Further considerations. The CFO notes three complications. First, the technical probabilities are Calder's own estimates, not industry statistics, and a small deviation (for instance a Phase II success probability of 0.30 instead of 0.40) changes the answer materially. Second, Calder must finance 60 million now from a cash balance of 90 million, and funding Phase III later will require new capital whose availability depends on Phase II data; the model assumes frictionless financing. Third, Meridian has a lower cost of capital and a commercial infrastructure that Calder lacks, so it may value the same asset more highly than Calder does and 4 million may simply be an opening bid.

The board must decide whether to proceed internally, accept the offer, or negotiate. The chair asks the team to return with a recommendation that explains what must be true about volatility, financing and Meridian's alternatives for each course to be right.

Discussion Questions

1. Explain why the committee's rNPV of −6.99 and the lattice value of 8.69 differ. Which of the committee's assumptions is replaced in the lattice, and is the replacement justified when the volatility cannot be observed?
2. The table shows that the program value is zero at 30 percent volatility and 18.2 million at 60 percent. How would you advise the board to use this range, and what evidence about volatility would you seek before the decision?
3. Nearly all of the value comes from the gates. What contractual or operational choices could Calder make to protect these gates, and what choices might destroy them?
4. Evaluate Meridian's offer of 4 million. Consider the value to Calder, the value to Meridian, the financing risk Calder bears, and how Calder might use the option analysis in negotiation, including the idea of selling a share of the program or retaining a milestone-based royalty.

Key Takeaways

- A real option is a right, not an obligation, to take an action at a cost, and it has value only if the firm has an exclusive right, real flexibility, and uncertainty that will be at least partly resolved before the decision.
- Use risk-neutral probabilities and the risk-free rate for market risk, and ordinary probabilities for private technical risk. Never penalize the same risk twice with both a high discount rate and a probability weight.
- The binomial lattice is the practical workhorse: it handles early exercise, staged costs and gates, and it returns a decision rule (an exercise boundary) as well as a value. Always test convergence by increasing the number of steps.
- Cash-flow leakage (the cost of waiting) is as important as volatility. A deferral option analysis that sets leakage to zero will recommend waiting indefinitely.
- In pharmaceutical pipelines, rNPV captures technical failure but omits market flexibility; treating the pipeline as a compound option shows that most of the option value comes from the right to stop at each gate.
- In natural resources, operating leverage multiplies price volatility into value volatility, and the optimal development trigger can be nearly twice the NPV breakeven price.
- Platform and staged investments are compound options whose stages interact; adding up separately valued options overstates the total, and the chain of fees is a real cost.
- Option games show that competition can destroy the value of waiting: strong first-mover advantage produces preemption, weak or negative advantage produces waiting equilibria, and multiple equilibria are common.
- Monte Carlo handles path dependence and many risk factors, and least-squares Monte Carlo extends it to early exercise; always benchmark a simulation against a simple lattice or closed form, and report standard errors.
- Real options analysis requires governance: document the option, show the static NPV separately, stress-test volatility, report break-even assumptions and tie the analysis to explicit exercise triggers.

Review Questions

1. Explain why a real option can be valued with risk-neutral probabilities even when the project's own cost of capital is unknown. What assumption makes this legitimate for a project that is not traded?
2. Distinguish market risk from private risk in a drug development program. How should each be treated in a valuation, and what error results from treating both with a single high discount rate?
3. List the real option analogues of the five Black-Scholes inputs (S, K, T, σ, δ) for an undeveloped oil field. Which are observable and which must be estimated?

4. Why can early exercise be optimal for a real call option on a project with cash-flow leakage but not for a call on a non-dividend-paying stock? Interpret the exercise boundary in managerial terms.
5. Describe the main limitations of rNPV for pharmaceutical valuation, and explain how a compound option lattice addresses them.
6. Why does the option value of the Marrowfield program come mostly from the technical gates and not from the launch option? What are the implications for contract design?
7. Explain why the optimal development trigger for a mine exceeds the NPV breakeven price. How do operating leverage and the convenience yield affect the gap?
8. What is the difference between a platform investment and an ordinary investment that is called “strategic”? List the conditions under which a platform has genuine option value.
9. In an option game with a first-mover advantage, why might competition lead firms to invest earlier than a monopolist would? What does the mixed-strategy equilibrium imply about the likelihood of simultaneous entry?
10. When would you choose least-squares Monte Carlo over a lattice? Describe what the regression step estimates and why LSM is biased low.

Problems

Problem 1. A project has present value 100 and requires an investment of 105 (NPV −5). Next year the project value will be 140 or 70, with equal risk-neutral probability to be determined from a risk-free rate of 5%. The firm can wait one year and invest only if worthwhile. Find the risk-neutral probability and the value of the option to wait.

Problem 2. A drug candidate requires 10 now (Phase I, success probability 0.50). If successful, Phase II costs 40 at t = 1 and succeeds with probability 0.40 at t = 3, after which the product launches with a net value at launch of 600. Discount at 12% (rNPV approach) with costs paid only if the program has reached the stage. Compute the rNPV.

Problem 3. A mine, once developed, produces 20 million units a year for 5 years at a cost of 2.00 per unit. The spot price is 3.00, the risk-free rate is 5% and the convenience yield 3%. Development costs 110. (a) Compute the developed value V(S) = Q × [S × A_δ − c × A_r] and the NPV now. (b) Find the breakeven spot price. (c) With a one-year option to develop and annual price factors u = 1.30, d = 1/1.30, find the risk-neutral probability and the value of the option to develop in one year (exercise only at year 1, discounting at 5%).

Problem 4. A project is worth 100 today. In each of two periods the value rises by 25% or falls by 20%. The risk-free rate is 5% per period and there is no leakage. The firm may abandon at any date for a salvage value of 100. Compute the value of the abandonment option using a two-step lattice, and compare with the value if abandonment were possible only at the end of period 2.

Problem 5. A pilot project gives the right to build a full-scale plant. The plant’s present value is S = 80, volatility 40%, risk-free rate 4%, no leakage. The plant costs K = 90 and the right lasts 3 years. (a) Compute the Black-Scholes value of the option and the NPV of investing today. (b) What is the hedge ratio N(d1), and what does it imply for the replicating portfolio?

Problem 6. Two firms face a timing game with V0 = 100, up and down values 150 and 60 at t = 1, risk-free rate 5% and investment cost 70 each. If both invest at the same date each earns 0.55 V. If one invests first and the other follows, the leader keeps 0.75 V and the follower earns 0.50 V. Compute the payoffs of leader, follower, wait-wait and invest-invest at t = 0 and find the mixed-strategy probability of investing now.

Problem 7. A project's value comprises a variable-margin component with present value 400 and fixed costs with present value 100, so that V = 300. Revenue volatility is 25%. Estimate the volatility of project value and explain the difference.

Problem 8. A Monte Carlo valuation of an option uses 40,000 paths and finds a mean discounted payoff of 18.4 with a sample standard deviation of 31.2. (a) Compute the standard error and a 95% confidence interval. (b) How many paths are required for a 95% confidence interval half-width of 0.10?

Answers to Selected Problems

Problem 1. With u = 1.4 and d = 0.7, q = (1.05 − 0.7)/(1.4 − 0.7) = 0.50. Up payoff = 140 − 105 = 35; down payoff = 0. Value = 0.5 × 35 / 1.05 = 16.67, versus NPV of −5.

Problem 2. Costs: −10 − 0.5 × 40/1.12 = −10 − 17.86 = −27.86. Launch value: 0.5 × 0.4 × 600/1.12³ = 120/1.4049 = 85.41. rNPV = 85.41 − 27.86 = 57.56.

Problem 3. (a) A_δ = Σ e^(−0.03τ), τ = 1..5 = 4.5738; A_r = Σ e^(−0.05τ) = 4.3143. V(3.00) = 20 × (3.00 × 4.5738 − 2.00 × 4.3143) = 20 × (13.721 − 8.629) = 101.85; NPV = 101.85 − 110 = −8.15. (b) S* = (110/20 + 2.00 × 4.3143)/4.5738 = (5.5 + 8.629)/4.5738 = 3.09. (c) q = (e^0.02 − 0.7692)/(1.30 − 0.7692) = (1.0202 − 0.7692)/0.5308 = 0.473. Up price 3.90 gives V = 184.18 and payoff 74.18; the down price 2.31 gives V = 38.52 and payoff zero. Value = e^(−0.05) × 0.473 × 74.18 = 33.37.

Problem 4. q = (1.05 − 0.80)/(1.25 − 0.80) = 0.5556. Terminal values: 156.25, 100, 64; abandonment payoffs: 0, 0, 36 (salvage 100). Period-1 continuation, up node (125): 0; down node (80): 0.4444 × 36/1.05 = 15.24, versus immediate abandonment 100 − 80 = 20, so abandon: node value 20. Value today = (0.4444 × 20)/1.05 = 8.47. If abandonment is possible only at the end: continuation values 0 and 15.24 give value (0.4444 × 15.24)/1.05 = 6.45. The early-exercise premium is 2.02.

Problem 5. (a) d1 = [ln(80/90) + (0.04 + 0.08) × 3]/(0.4 × 1.7321) = (−0.1178 + 0.36)/0.6928 = 0.3496; d2 = 0.3496 − 0.6928 = −0.3432; N(d1) = 0.6367, N(d2) = 0.3657. C = 80 × 0.6367 − 90 × e^(−0.12) × 0.3657 = 50.93 − 29.19 = 21.74. NPV now = 80 − 90e^(−0.12) = 80 − 79.82 = 0.18. (b) The hedge ratio is 0.637: replicate the option by holding 0.637 units of the project (worth 50.93) and borrowing 29.19.

Problem 6. Follower enters at t = 1 in the up state only (0.50 × 150 − 70 = 5 > 0). Leader: [0.5 × 0.75 × 150 + 0.5 × 60]/1.05 − 70 = 82.14 − 70 = 12.14. Follower: 0.5 × 5/1.05 = 2.38. Wait-wait: both invest in the up state at t = 1, earning 0.55 × 150 − 70 = 12.5 each; value 0.5 × 12.5/1.05 = 5.95. Invest-invest: 0.55 × 100 − 70 = −15. Mixed probability: q = (L − W)/[(L − W) + (F − S)] = (12.14 − 5.95)/[(12.14 − 5.95) + (2.38 + 15)] = 6.19/23.57 = 0.263. Expected value in the mixed equilibrium is 5.01.

Problem 7. The elasticity of V to revenue is 400/300 = 1.33, so volatility is about 25% × 1.33 = 33.3%. Fixed costs create operating leverage, which amplifies the revenue volatility.

Problem 8. (a) SE = 31.2/√40,000 = 0.156; 95% CI = 18.4 ± 1.96 × 0.156 = [18.09, 18.71]. (b) n = (1.96 × 31.2/0.10)² ≈ 373,957, about 374,000 paths (nearly ten times as many).

Chapter 3: Valuing Intangible-Rich and High-Growth Firms

Learning Objectives

After studying this chapter, you should be able to:

- Explain why standard discounted cash flow practice, earnings multiples and book-value ratios mislead when a firm has negative earnings, heavy intangible investment and a very wide range of possible futures.
- Capitalize research and development (and other intangible spending) to restate operating income, invested capital and return on invested capital, and interpret the adjusted figures.
- Build a unit-economics model, including customer lifetime value, acquisition cost, payback and a cohort retention model, and connect it to enterprise value.
- Construct a scenario-weighted DCF that incorporates the probability of failure, dilution and funding needs, and distinguish it from a single "base case" valuation and from a simulation.
- Apply terminal-value discipline: link terminal growth, reinvestment and return on capital, test implied multiples, and decide when a terminal value is doing too much of the work.
- Value startups and platform businesses using the venture capital method, option-like reasoning and network-effect arguments, and explain what the resulting numbers do and do not mean.

Opening Vignette

Illustrative scenario; the company is fictional.

Brightline Systems sells workflow software to mid-sized logistics companies. In its most recent fiscal year it reported revenue of 100 million, up 60 percent from the year before, and an operating loss of 30 million. It spends 38 percent of revenue on sales and marketing, 30 percent on research and development, and runs a gross margin of 74 percent. It has 55 million in cash, has been burning roughly 3 million a month, and has just received a term sheet for a new equity round.

Three people at Brightline's board meeting hold three different views of what the company is worth. The chief financial officer argues from the income statement: with negative earnings there is no price-to-earnings multiple, and a discounted cash flow model over a ten-year horizon is "mostly terminal value anyway," so she prefers a revenue multiple taken from a handful of listed software firms. An investor on the board argues from unit economics: each customer costs 12,000 to acquire and produces about 9,000 of gross profit per year with an annual churn of 12 percent, so the company is "obviously worth far more than it spends." A third director, a former operator, says both are wrong. Brightline's accounting profit is depressed by spending that builds assets (software, customer relationships, brand) that the balance sheet does not show, so the loss overstates the economic problem. But, he adds, at least one in four companies in this category runs out of cash before reaching scale, and a single "expected" forecast hides that fact.

All three are partly right. The revenue multiple ignores margin structure and risk; the unit-economics argument ignores overhead, the discount rate, and the cohort decay of retention; the accounting adjustment is necessary but says nothing about survival. This chapter builds the tools to combine them. We will restate the accounts to treat intangible spending as

investment, model customers as cohorts, weigh scenarios including failure, discipline the terminal value, and finally translate everything into a price per share that a board could defend.

3.1 Why Standard Valuation Breaks Down

In Volume 1 (Chapters 5 and 21) we valued firms with a simple recipe: forecast free cash flow to the firm, discount at the weighted average cost of capital, add a terminal value, subtract debt and divide by shares. That recipe works well for mature companies whose current earnings are a reasonable base from which to extrapolate. It strains in three ways when applied to intangible-rich and high-growth firms.

Negative or meaningless current earnings

A firm that is losing money has no stable earnings base. Price-to-earnings is undefined; growth rates applied to a negative number produce nonsense (a loss that "grows 20 percent" gets bigger). Even when earnings are positive but tiny, a multiple of a depressed number overstates value and a multiple of a peak number understates the risk. The fix is not to give up on fundamentals, but to move the anchor from current earnings to a **target state**: estimate what the firm's revenue, margins and reinvestment will look like when it has matured, and work backward to today.

Intangible investment that accounting expenses

Under most accounting standards, outlays for research, brand building, customer acquisition and employee training are expensed immediately, while spending on a machine or a building is capitalized and depreciated. This asymmetry made sense when most economic assets were physical and measurable. It is distorting when the main assets are knowledge, data, software and relationships. Three consequences follow. Operating income is understated in years of rising intangible spending, because the investment is charged against current revenue while its benefits arrive later. Book equity and invested capital are understated, so return on invested capital looks spuriously high for a firm that has expensed its way to its capital base. And ratios such as price-to-book lose comparability across companies with different mixes of tangible and intangible assets.

An enormous dispersion of outcomes

The third problem is the most fundamental. For a mature utility, the distribution of possible future cash flows is narrow, and a single forecast is a decent summary. For a young company, the distribution is wide, skewed and contains a mass at zero. A single base-case forecast cannot represent such a distribution, for two reasons. First, if the probability of failure is meaningful, an expected cash flow that ignores it is biased upward. Second, because the payoff is convex (a successful company is worth many multiples of an average one, while the downside is limited to the investment), the mean of the distribution can sit very far from the "most likely" outcome. In Section 3.5 we compute an example in which the most likely scenario is worth 306 and the probability-weighted value is 288, and the value is dominated by a 15 percent chance of an outcome worth 1,202.

The framework of this chapter

We keep the DCF skeleton because value is still the present value of the cash flows that investors will ultimately receive. What changes is how each element is estimated.

Element	Mature firm	Intangible-rich / high-growth firm
Earnings base	Current normalized earnings	Target-state margin and revenue in a future year
Reinvestment	Capex plus working capital less depreciation	Includes capitalized R&D, acquisition spending, and sales-to-capital ratios
Growth	A single stable rate	A fade from very high growth to a stable rate over many years
Discount rate	Roughly constant WACC	Declines as the firm matures and risk falls
Uncertainty	Sensitivity analysis	Explicit scenarios, probability of failure, simulation
Terminal value	60-80 percent of value, tied to a steady state	May exceed 100 percent of value; requires strict discipline

The numbers should always be traceable to a story (how large the market is, what share the firm can take, how durable its pricing is, how much it must reinvest), and each story should be stress-tested against competitors and history.

3.2 Intangible Assets and the Capitalization Adjustment

The logic of capitalizing R&D

If R&D produces benefits over several years, then economically it is capital expenditure, not an operating cost. The adjustment treats the past R&D spend as an asset with a finite life, amortizes it over that life, and adds back the current year's spend to operating income. The procedure has four steps.

1. Choose an amortizable life, based on how long it takes research to turn into products and how long those products remain valuable. Pharmaceuticals might use 10 years or more; software and consumer electronics perhaps 3 to 5.
2. Gather R&D expenses for as many past years as the life requires.
3. Compute the **research asset** (the unamortized portion of each year's expense) and the current-year **amortization**.
4. Adjust operating income and invested capital:

```
Adjusted EBIT = Reported EBIT + R&D(current) − Amortization of research asset
Adjusted invested capital = Reported invested capital + Research asset (unamortized)
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With straight-line amortization over a life of L years, and with the convention that spending in year t begins amortizing in year t+1, the current amortization equals the average of the prior L years' R&D, and a dollar spent k years ago has a remaining fraction of (1 − k/L) unamortized.

Example 3.1 – Capitalizing R&D. Orion Devices reports EBIT of 300 million and invested capital (book equity plus debt less cash) of 1,500 million. Its tax rate is 25 percent. R&D spending for the last six years (millions), most recent first, was: year 0: 140; year −1: 125; year −2: 110; year −3: 100; year −4: 90; year −5: 80. We assume a five-year life with straight-line amortization.

Step 1: unamortized research asset. Each year's spend carries the fraction (1 − k/5) forward, where k is the number of years elapsed.

Year	R&D	Fraction unamortized	Research asset
0 (current)	140	1.00	140.0
−1	125	0.80	100.0
−2	110	0.60	66.0
−3	100	0.40	40.0
−4	90	0.20	18.0
−5	80	0.00	0.0
Total			**364.0**

Step 2: current amortization. The five prior years' spend is amortized over five years each, so amortization = (125 + 110 + 100 + 90 + 80) / 5 = 505 / 5 = 101.0 million.

Step 3: adjusted EBIT. 300 + 140 − 101 = 339 million. The adjustment raised operating income by 39 million, because current R&D (140) exceeds amortization (101) in a growing R&D budget.

Step 4: adjusted invested capital. 1,500 + 364 = 1,864 million.

Step 5: returns. Reported after-tax ROIC = 300 × 0.75 / 1,500 = 225 / 1,500 = 15.0 percent. Adjusted ROIC = 339 × 0.75 / 1,864 = 254.25 / 1,864 = 13.6 percent.

Interpretation. The firm's operating profit is higher than reported (339 versus 300), but the capital it has actually invested is also much higher (1,864 versus 1,500), and the return on that fuller base is lower than the headline. If a valuation uses the reported 15 percent ROIC to justify high future reinvestment value, it overstates the value of growth. The adjustment also changes the reinvestment rate used in the terminal value: net reinvestment now includes R&D minus amortization (39 million) in addition to ordinary capex net of depreciation. This is a simplification, because tax effects are ignored: in practice R&D is tax deductible when spent, so a rigorous adjustment tracks a deferred tax effect. For comparison across firms, the simplified version is usually adequate.

Beyond R&D: SG&A and customer acquisition

The same logic applies to customer acquisition spending in subscription businesses, where a share of SG&A creates a customer relationship that yields revenue for years. The risk is overreach: if every unprofitable expense is relabeled investment, losses vanish by assertion. Two tests keep the adjustment honest: evidence of durability (cohort retention data show a multi-year benefit) and marginal return (a rising ratio of acquisition cost to first-year gross profit signals impairment).

Valuation consequences of the adjustment

Capitalization changes the earnings base (adjusted income may be positive where reported income is negative), the reinvestment rate (growth equals reinvestment rate times ROIC, both measured on the adjusted base), and the comparability of multiples across R&D-intensive and asset-heavy firms. Capitalized R&D is an operating asset, not a financial one, so it does not offset debt in the bridge from enterprise value to equity value. A common mistake is to adjust earnings but not capital, or vice versa: adding back R&D to EBIT without adding the research asset to invested capital inflates ROIC and makes growth look free.

Stock-based compensation

Stock compensation is a real cost because new shares dilute existing holders. Either deduct it as an operating expense in cash flows and use the current share count, or exclude it and model future dilution through the share count; ignoring it in both places double-counts the benefit. Options already granted should be valued (Volume 1, Chapter 18) and subtracted from equity value.

3.3 Valuing Firms with Negative Earnings

From losses to a path to profitability

Trace the path: how revenue grows, how the margin improves from negative to target, and how much capital the firm must raise.

Revenue trajectory. Begin from the market: total addressable market, plausible share and time to reach it. A path starting at 60 percent and fading toward long-run economic growth over ten years is typical; check the cumulative revenue against the market-size assumption.

Margin convergence. The target operating margin should come from mature peers adjusted for the business model, and is usually the most influential assumption, so give it a range. A linear glide is the simplest rule; converge faster where costs are mostly fixed (software, platforms).

Reinvestment through sales-to-capital. Each incremental dollar of revenue requires 1/(S/C) dollars of net reinvestment; a ratio of 2.0 means adding 100 of revenue needs 50 of reinvestment. Asset-light software firms often exceed 2, capital-intensive businesses fall below 1. The ratio ties growth to the cash the firm must raise.

Tax and loss carryforwards. Early losses create net operating losses that shelter later profits, subject to statutory limits. Model them explicitly (accumulate, apply against taxable income until exhausted) rather than adding a lump sum.

Funding, dilution and the survival constraint

A firm with negative free cash flow needs financing. Two consequences follow for valuation.

First, **enterprise value is not the same as the value of today's shares**. Enterprise value discounts all future cash flows including the negative ones, which implicitly assumes that the cash needed to fund the burn will be provided at a fair price. If the firm has to issue equity at a depressed valuation, or cannot raise money at all, the value to the current owners is different. In practice, valuation analysts handle this in two ways: by discounting the negative cash flows at the firm's cost of capital (assuming fair-priced financing), or by modelling explicit financing rounds and dilution. The first is simpler and defensible in the context of enterprise value; the second is essential for pricing a specific financing round.

Second, a **runway constraint** introduces a survival probability. If the firm has 24 months of cash and requires 48 months to reach cash break-even, it must raise capital at least once, and whether it can depends on progress milestones and market conditions. The probability of failure is partly endogenous: capital markets finance firms that demonstrate traction, and withhold it from those that do not. This is why unit economics (Section 3.4) are the key diagnostic: they predict whether additional growth spending will create or destroy value, and therefore whether new capital will be forthcoming.

Cash-burn arithmetic

Three early-warning measures are runway (cash divided by monthly net burn), the burn multiple (net burn divided by net new recurring revenue; below about 1.5 is usually efficient for software, above 3 suggests expensive growth), and the gap between months to cash break-even and runway. They calibrate the sales-to-capital ratio and the survival probability.

Relative valuation with negative earnings

Where there are no earnings, analysts use revenue or gross-profit multiples. These are fast but fragile, because a multiple embeds unstated assumptions about margin, growth and risk. Improve the comparison by using gross profit where gross margins differ, by regressing the multiple on growth and target margin across peers, and by using a forward multiple for a year in which the firm is expected to be profitable, discounted back and adjusted for the probability of getting there. The last is a compact DCF in which the exit multiple stands in for the terminal value, with the warnings of Section 3.6.

3.4 Unit Economics, Cohorts and Lifetime Value

Intangible-rich consumer and subscription businesses are best understood from the bottom up: a customer is an asset that costs money to acquire and yields a stream of contribution margin until the customer leaves. The aggregate firm is a collection of such assets.

The basic metrics

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ARPU      = average revenue per user per period
Gross margin (GM) = (Revenue − Cost of service) / Revenue
Churn (c)         = fraction of customers lost per period
Expected lifetime = 1 / c periods
LTV (undiscounted)= ARPU × GM / c
CAC       = (Sales and marketing spend) / (New customers acquired)
Payback   = CAC / (ARPU × GM)   (periods)
```

The ratio LTV/CAC is the standard summary. A rule of thumb in subscription software sets 3 as healthy and below 1 as value-destroying, but the rule hides three weaknesses that this section corrects: it ignores discounting, it assumes a constant churn rate, and it ignores the overhead that gross profit must also cover.

Example 3.2 – Basic LTV and the effect of discounting. A subscription service charges 60 per month. Cost of service is 25 percent of revenue, so gross margin is 75 percent. Monthly churn is 2 percent and the cost of acquiring a customer is 500. The annual discount rate is 10 percent, or 0.10/12 = 0.833 percent per month.

Undiscounted lifetime. 1 / 0.02 = 50 months.

Monthly gross profit per customer. 60 × 0.75 = 45.

Undiscounted LTV. 45 / 0.02 = 2,250.

LTV/CAC. 2,250 / 500 = 4.5.

Payback. 500 / 45 = 11.1 months.

Discounted LTV. A perpetuity of 45 per month, surviving with probability (1 − c) each month, is discounted at the combined rate. In the form LTV = m / (c + r), with m = 45 and r = 0.00833: LTV = 45 / (0.02 + 0.00833) = 45 / 0.02833 = 1,588.2. (This treats the first margin as arriving one period after acquisition and the survival-and-discount rate as c + r, a standard approximation for small rates.)

Discounted LTV/CAC. 1,588.2 / 500 = 3.18.

Interpretation. The headline ratio falls from 4.5 to 3.2 when discounting is included, a 29 percent reduction. In a business where customers last 50 months on average, the time value of money matters; for a high-churn business it matters less. Notice also that payback of 11 months means the firm’s cash is tied up for almost a year per customer; faster growth

increases the cash drain even though each customer is profitable. This is the arithmetic by which a firm with an excellent LTV/CAC can still run out of money: growth spending is paid up front, the gross profit comes later.

Contribution, not gross margin

Gross margin is the wrong margin for a customer-level asset if serving the customer involves further variable costs (support, payment processing, onboarding); use **contribution margin**. Fixed overhead is excluded from LTV but must be covered by aggregate contribution, which is why a firm with healthy LTV/CAC can lose money until it reaches scale. The test has two parts: is each cohort worth more than it costs, and are there enough customers for aggregate contribution to exceed the fixed cost base?

Cohort analysis

Churn is rarely constant: early churn is high and later churn low as survivors prove loyal, so constant-churn LTV usually understates value for loyal customers and overstates it for fragile ones. A **cohort model** tracks customers acquired in the same period and measures the fraction active each period. A reliable model has a measured retention curve (a plateau near 35 percent indicates a durable core; a curve sinking toward zero indicates a leaky product), revenue per surviving customer (net revenue retention above 100 percent means survivors' spending offsets churn), and allowance for cohort quality drift, since later cohorts are usually costlier and weaker than early ones.

Example 3.3 – Cohort model of customer value. A software firm acquires a cohort of customers at a cost of 700 per customer. In year 0 (the first year), the average customer pays 600 in revenue, growing 5 percent per year per surviving customer. The contribution margin is 70 percent. Retention (fraction of the original cohort still active in each year) is 100, 70, 55, 47, 42, 39, 37, 36, 35.5 and 35 percent for years 0 to 9. The discount rate is 12 percent, and we assume flows arrive on average at mid-year, so year t flow is discounted by (1.12)^(t + 0.5). We truncate the horizon at ten years (a conservative assumption; the retention plateau suggests value beyond that).

Calculations for each year, per original customer. Contribution = retention × revenue per survivor × 0.70.

Year	Retention	Revenue per survivor	Contribution	Discount factor	PV
0	1.000	600.0	420.0	0.9449	396.9
1	0.700	630.0	308.7	0.8436	260.4
2	0.550	661.5	254.7	0.7532	191.8
3	0.470	694.6	228.5	0.6725	153.7
4	0.420	729.3	214.4	0.6004	128.8
5	0.390	765.8	209.1	0.5361	112.1
6	0.370	804.1	208.3	0.4787	99.7
7	0.360	844.3	212.8	0.4274	90.9
8	0.355	886.5	220.3	0.3816	84.1
9	0.350	930.8	228.0	0.3407	77.7
Total					**1,596.1**

Customer value. PV of contribution = 1,596.1. Net of acquisition cost: 1,596.1 − 700 = 896.1 per customer. LTV/CAC = 1,596.1 / 700 = 2.28.

Payback. Cumulative undiscounted contribution is 420.0 after year 0 and 728.7 after year 1, so the 700 cost is recovered during year 1: payback is about 0.9 years into the second year (280 / 308.7 = 0.91), roughly 1.9 years from acquisition on a year-end convention.

Contrast with a constant-churn shortcut. The average annual churn over the first four years is high (30 percent in year 1), but only about 1 percent per year after year 7. A constant churn estimated from the first year (30 percent) would give a lifetime of 3.3 years and would understate value; one estimated from the plateau (about 1.4 percent) would give a lifetime of 71 years and badly overstate it. The cohort shows that neither is right, and it is a reminder that the shape of the curve is the model.

Interpretation. Each customer is worth about 1,596 in present-value contribution against an acquisition cost of 700, a surplus of 896 per customer. The firm should keep acquiring customers while marginal CAC stays below marginal LTV; if CAC climbs above about 1,596 growth destroys value. Because the retention plateau extends beyond year 9, the truncated sum omits some value from the 35 percent of customers who remain; leaving it out is a conservative choice.

From unit economics to enterprise value

With a cohort model, value can be built bottom-up: forecast new customers (constrained by market size and rising CAC), apply retention and revenue-per-user curves to each cohort, add variable and fixed costs (fixed costs scale much more slowly than revenue), compute taxes, reinvestment and free cash flow, and discount with the terminal-value discipline of Section 3.6. The advantage over a top-down growth rate is that scenarios are defined by operating variables managers understand, and growth that creates value (high LTV/CAC) is separated from growth that merely consumes capital.

Common errors in unit-economics work

Typical errors are survivorship bias in churn estimates, averaging across heterogeneous segments (enterprise customers may be hugely profitable and small customers unprofitable), using blended rather than marginal CAC as channels saturate, counting revenue rather than contribution, and omitting the cost of capital, so that an undiscounted LTV/CAC of 3 may hide an economic return near zero for a slow-payback business.

3.5 Scenario-Weighted DCF and the Probability of Failure

Why a single forecast is biased

Suppose a firm will become worth 1,200 with 15 percent probability, 306 with 30 percent, 45 with 30 percent, and 10 (salvage) with 25 percent. A “base case” valuation might choose the 306 scenario because it is a central outcome; but the expected value is 288 and it contains a lottery component: the top scenario contributes 180 of the 288 total. A single forecast can be constructed to match the mean by judiciously mixing optimistic and pessimistic assumptions, but analysts rarely do so consistently, and they rarely disclose the implied probability of failure. Scenario weighting makes the distribution explicit.

Constructing scenarios

Good scenario analysis follows five rules. First, scenarios are coherent stories, not independent sensitivities: in a breakout scenario growth, margin and reinvestment move together. Second, include a failure scenario with a salvage value; for young companies the probability of failure within five years is often 20 to 50 percent, anchored to base rates for the category and stage. Third, assign probabilities that sum to one and document them; a table without probabilities is a sensitivity analysis. Fourth, keep failure risk in the cash flows through the weights, not in the discount rate, which should

reflect systematic risk only; inflating the rate double-counts and penalizes distant cash flows more than the early-concentrated failure risk. Fifth, check scenario values against market evidence: a breakout worth more than the largest comparable firm needs strong justification.

Cash flows in a young firm: a worked structure

For each scenario, the forecast has the following structure, which we apply in Example 3.4:

```
Revenue(t)  = Revenue(t−1) × (1 + g(t))
EBIT(t)     = Revenue(t) × margin(t), margin gliding from the current to the target
Tax(t)      = max(0, EBIT − NOL used) × t
Reinvest(t) = (Revenue(t) − Revenue(t−1)) / (Sales-to-capital)
FCFF(t)     = EBIT − Tax − Reinvest
TV(10)      = NOPAT(11) × (1 − g / ROIC) / (WACC − g)
```

Example 3.4 – Scenario-weighted DCF for a software start-up. Brightline-style firm; amounts in millions. Current revenue is 100 and the operating margin is −30 percent. Sales-to-capital is 2.0 (each 1 of reinvestment supports 2 of incremental revenue), the tax rate is 25 percent with loss carryforwards applied, terminal growth is 3 percent, and the terminal WACC is 9 percent. The operating margin glides linearly from −30 percent to the target over ten years. The discount rate falls over time from 14 percent to 9 percent as the firm matures (the table of rates is cumulative: each year's factor compounds the year's rate). Four scenarios are modelled; the figures below were computed with a Python script, and the base case is shown in full.

Base scenario inputs. Growth path: 60, 45, 35, 28, 22, 17, 13, 9, 6, 4 percent. Target margin 20 percent. Terminal ROIC 15 percent. Discount rates by year: 14, 14, 14, 13, 13, 12, 11, 10, 9.5, 9 percent.

Year	Revenue	Margin	EBIT	Tax	Reinvestment	FCFF	Cumulative discount factor
1	160.0	−25%	−40.0	0.0	30.0	−70.0	1.140
2	232.0	−20%	−46.4	0.0	36.0	−82.4	1.300
3	313.2	−15%	−47.0	0.0	40.6	−87.6	1.482
4	400.9	−10%	−40.1	0.0	43.9	−83.9	1.674
5	489.1	−5%	−24.5	0.0	44.1	−68.5	1.892
6	572.2	0%	0.0	0.0	41.6	−41.6	2.119
7	646.6	5%	32.3	0.0	37.2	−4.9	2.352
8	704.8	10%	70.5	0.0	29.1	41.4	2.587
9	747.1	15%	112.1	4.2	21.1	86.7	2.833
10	777.0	20%	155.4	38.9	14.9	101.6	3.088

Tax is zero through year 8 because the accumulated loss pool (about 198 after year 5) shelters profits in years 7 and 8 and most of year 9; the pool is exhausted in year 9, and tax of 4.2 is paid on the residual 17.0 of taxable income.

PV of explicit FCFF. Summing each FCFF divided by its cumulative factor gives −212.5.

Terminal value. NOPAT in year 11 = 777.0 × 1.03 × 0.20 × 0.75 = 120.0. Reinvestment rate = g / ROIC = 0.03 / 0.15 = 20 percent. TV = 120.0 × (1 − 0.20) / (0.09 − 0.03) = 96.0 / 0.06, which is 1,600.6 using unrounded inputs. Discounted by the year-10 cumulative factor 3.088: PV(TV) = 518.4.

Base enterprise value. −212.5 + 518.4 = 305.9.

The other scenarios were computed by the same method, with different growth paths, target margins, terminal returns and discount rates:

Scenario	Probability	Target margin	Year-10 revenue	Terminal ROIC	PV explicit FCFF	PV terminal	Enterprise value
Failure	25%	n/a	n/a	n/a	n/a	n/a	10.0 (salvage)
Niche	30%	16%	479.8	12%	−190.9	235.9	45.0
Base	30%	20%	777.0	15%	−212.5	518.4	305.9
Breakout	15%	28%	1,391.0	22%	−200.7	1,402.5	1,201.8

Probability-weighted value. 0.25 × 10 + 0.30 × 45.0 + 0.30 × 305.9 + 0.15 × 1,201.8 = 2.5 + 13.5 + 91.8 + 180.3 = **288.0**.

Comparison. The base case alone gives 305.9, 6 percent above the weighted value. Ignoring failure and renormalizing the surviving scenarios (dividing the three non-failure contributions, 285.5, by 0.75) gives 380.7, which overstates value by 32 percent relative to the properly weighted number. And 63 percent of the expected value (180.3 of 288.0) comes from the 15 percent breakout scenario.

Interpretation. The weighted value is lower than the base case, but not dramatically so, because the convex upside in the breakout scenario compensates for the high failure probability. Three points should be explicit in a valuation memo. First, the discount rate is the same in all scenarios (declining along the same path) because the failure risk is carried by the probabilities. Second, in every non-failure scenario the explicit-period cash flows are negative, and the terminal value is more than 100 percent of the enterprise value: in the base case, PV(TV) = 518.4 against an EV of 305.9, i.e. 169 percent. Third, the weights are judgment, and the analyst should show how the value changes with them. If the failure probability rises from 25 to 40 percent and the other three scenario probabilities are scaled down proportionally (each multiplied by 0.60/0.75 = 0.8), the weighted value falls to 0.40 × 10 + 0.8 × 285.5 = 232.4, a drop of 19 percent.

Simulation: when scenarios are not enough

A discrete scenario set shows the structure but is a coarse representation of a continuous distribution. A **Monte Carlo simulation** draws each uncertain input (growth, margin, reinvestment, survival, terminal return) from a probability distribution, computes the value for each draw, and reports the distribution. It adds three things: a more complete picture of the distribution (percentiles, probability of loss); a way to capture correlations; and an honest accounting of how wide the uncertainty is. It adds one danger: spurious precision. The distributions are assumptions, and the simulation merely propagates them.

Example 3.5 – Monte Carlo valuation of a young firm. We model a firm with current revenue 100. Inputs are drawn independently as follows: survival over the horizon is Bernoulli with probability 0.75; the ten-year revenue CAGR is Normal with mean 17 percent and standard deviation 6 percent; the terminal operating margin is Triangular(8 percent, 18 percent, 30 percent) with mode 18 percent; terminal ROIC is Uniform(9 percent, 20 percent); the present value of interim cash burn is Uniform(−260, −150); the tax rate is 25 percent; the terminal growth rate is 3 percent; the terminal WACC is 9 percent, and the terminal value is discounted at 12 percent for ten years to reflect the blended rate over the horizon. Value is bounded below at a salvage of 10 (limited liability; failed firms are worth 10). The script, with a fixed seed so the result can be reproduced, is:

```
import numpy as np
rng = np.random.default_rng(42)
N = 100000
surv = rng.random(N) < 0.75
cagr = rng.normal(.17, .06, N)
rev10 = 100 * (1 + cagr) ** 10
margin = rng.triangular(.08, .18, .30, N)
roic = rng.uniform(.09, .20, N)
g, w = .03, .09
nopat11 = rev10 * (1 + g) * margin * .75
tv = nopat11 * (1 - g / roic) / (w - g)
pvtv = tv / 1.12 ** 10
burn = -rng.uniform(150, 260, N)
val = np.where(surv, pvtv + burn, 10)
val = np.maximum(val, 10)
```

Output (100,000 draws). Mean value 110.7; median 15.3; 5th percentile 10.0; 25th percentile 10.0; 75th percentile 159.0; 95th percentile 446.0. The probability that value sits at the floor of 10 is 48.9 percent (25 percent from failure and about 24 percent of all draws from survivors whose terminal value does not cover the burn). The probability of a value above 500 is 3.7 percent and above 1,000 is 0.28 percent. The mean revenue after ten years is 540.8 and the median is 481.3.

Interpretation. The mean (110.7) is more than seven times the median (15.3), a signature of a highly skewed distribution. Neither the median nor the mean is “the value”: the mean is the right expected value for a diversified investor, but an owner who holds a single position is exposed to a 49 percent chance of near-total loss. Venture investors build this skew into their return targets (Section 3.7). The simulation also shows how expensive the interim burn is: a survivor needs both strong growth and a decent terminal margin before its terminal value overtakes the cumulative burn. And the survival variable and the value are only moderately correlated (a correlation coefficient of 0.35 in the output), because survival alone does not guarantee that the firm is worth more than its salvage.

The two examples differ in design. Example 3.4 has a coherent story for each scenario and discloses all inputs; Example 3.5 has continuous distributions but treats inputs as independent, which understates the joint risk (in a real downturn, growth, margin and ROIC all fall together). A careful analysis uses both: scenarios for communication and stress-testing, simulation for the distribution, and correlations between key drivers.

Real options

A young firm owns options to expand, pivot or abandon. The failure scenario (floored at salvage) and the breakout scenario already embed the abandonment and expansion options, so adding a separate option value on top double-counts unless it prices something the scenarios omit, such as a specific staged-investment decision.

3.6 Terminal Value Discipline

In Example 3.4 the terminal value was 169 percent of enterprise value. This is normal for young firms, and it should make the analyst nervous: the value is dominated by the assumptions about a state that may be a decade away. Terminal-value discipline means that each assumption is internally consistent, economically plausible and transparent.

The value-driver formula

For a firm that grows at rate g forever and earns return on new capital ROIC, the reinvestment rate needed is g / ROIC. The terminal value is then:

```
TV(N) = NOPAT(N+1) × (1 − g / ROIC) / (WACC − g)
```

where NOPAT(N+1) = NOPAT(N) × (1 + g). Compare this with the simpler “growing perpetuity of NOPAT” that ignores reinvestment: TV = NOPAT(N+1) / (WACC − g). The simple version implicitly sets ROIC to infinity, so growth is free. The value-driver formula forces the analyst to say what the firm earns on the capital it adds.

Three limiting cases clarify the formula.

- If ROIC = WACC in the terminal period, then (1 − g / WACC) / (WACC − g) = 1 / WACC, and the terminal value is NOPAT(N+1) / WACC *regardless of g*. Growth adds nothing because each dollar invested earns only its cost of capital. This is the competitive-equilibrium assumption: in the long run, excess returns are competed away.
- If ROIC > WACC, growth adds value, and the value rises with g.
- If ROIC < WACC, growth destroys value, and the firm would be worth more by not growing.

Example 3.6 – How terminal ROIC drives value. A firm has NOPAT of 200 in year N. WACC is 9 percent and terminal growth is 4 percent. NOPAT in year N+1 is 200 × 1.04 = 208.

No-reinvestment perpetuity (implicitly infinite ROIC). TV = 208 / (0.09 − 0.04) = 208 / 0.05 = 4,160. TV/NOPAT(N) = 20.8.

ROIC = 30 percent. Reinvestment rate = 0.04 / 0.30 = 13.33 percent; payout = 86.67 percent. TV = 208 × 0.8667 / 0.05 = 180.27 / 0.05 = 3,605.3. Multiple of current NOPAT = 18.0.

ROIC = 15 percent. Reinvestment rate = 26.67 percent. TV = 208 × 0.7333 / 0.05 = 152.53 / 0.05 = 3,050.7. Multiple = 15.3.

ROIC = WACC = 9 percent. Reinvestment rate = 44.44 percent. TV = 208 × 0.5556 / 0.05 = 115.56 / 0.05 = 2,311.1. Check: NOPAT(N+1) / WACC = 208 / 0.09 = 2,311.1. Multiple = 11.6.

Terminal assumption	Reinvestment rate	TV	TV / NOPAT(N)
No reinvestment	0%	4,160.0	20.8
ROIC 30%	13.3%	3,605.3	18.0
ROIC 15%	26.7%	3,050.7	15.3
ROIC = WACC (9%)	44.4%	2,311.1	11.6

Interpretation. Moving from a naive perpetuity to the competitive-equilibrium assumption lowers the terminal value by 44 percent (4,160 to 2,311). Many published valuations of fast-growing firms effectively use the first row. The disciplined analyst picks the row that matches the firm’s competitive position: ROIC = WACC for commodity-like businesses, a positive spread that fades for firms with durable advantages (brands, network effects, regulation). The difference between rows is the value of the competitive advantage in perpetuity, and the analyst should be able to defend it.

Growth cap and the economy

Terminal growth cannot exceed the growth of the economy in which the firm operates; a practical cap is the risk-free rate or somewhat below it. Growth of 3 percent with 4 percent nominal rates is defensible, 7 percent is not. A ten-year fade should converge smoothly: “40 percent in year 10, 3 percent in year 11” places the terminal value where the firm is nowhere near steady state.

Tests for terminal value

Apply these checks to every terminal value. (1) **Implied multiples:** convert TV to EV/NOPAT or EV/EBITDA and compare with mature firms; a terminal EV/NOPAT of 21 for a mature firm requires a very high ROIC or very low WACC. (2) **Terminal share of value:** above 100 percent, as in Example 3.4, the valuation is a bet on the end state, so lean on cross-checks. (3) **Implied market share:** terminal revenue of 777 is a 15 percent share of a 5,000 market but more than 50 percent of a 1,500 market. (4) **Terminal margin** should not exceed the best mature peers without a structural reason. (5) **Reinvestment consistency:** reinvestment must match growth and ROIC. (6) **Discount rate transition:** WACC should converge to a mature value, since the beta and cost of debt of a mature firm differ from those of a young one.

Exit multiples versus perpetuity growth

Practitioners often estimate terminal value with an exit multiple of EBITDA or revenue. The method is easy to explain, and it embeds market pricing, but it imports the market's current sentiment into a value that is supposed to be fundamental, and it hides the growth, ROIC and WACC assumptions. A good practice is to compute both and reconcile: derive the growth rate implied by the exit multiple, and the multiple implied by the perpetuity formula. If the two differ widely, one of them is wrong. For example, an exit multiple of 25 times NOPAT with a WACC of 9 percent and a ROIC of 15 percent implies a growth rate that solves $25 = (1 + g)(1 - g / 0.15) / (0.09 - g)$, which is about 6.6 percent: not credible as a perpetual rate.

3.7 Startups and Platforms

The venture capital method

Venture investors value a young company by working backward from an expected exit. The method has four steps: (1) estimate the exit value at the horizon (typically 4 to 7 years) using a multiple of the projected metric; (2) choose a target return (internal rate of return), which is high because it must cover the failures in the portfolio; (3) compute the required ownership at exit; (4) adjust for dilution from future funding rounds to determine the ownership needed today.

```
Required value at exit = Investment × (1 + target IRR)^years
Required ownership at exit = Required value / Exit value
Required ownership today   = Required ownership at exit / Retention
Post-money valuation = Investment / Required ownership today
```

where Retention is the fraction of today's ownership that remains after future dilutive rounds (the product of one minus each round's dilution).

Example 3.7 – Venture capital method. A venture fund considers investing 20 million. It projects revenue of 300 million in five years and expects an exit multiple of 4.0 times revenue, so the exit value is 1,200 million. The fund's target IRR is 50 percent. It expects two further financing rounds that will dilute existing holders by 20 percent and 15 percent respectively.

Required value at exit. 20 × (1.5)^5 = 20 × 7.59375 = 151.9 million.

Required ownership at exit. 151.9 / 1,200 = 12.66 percent.

Retention. (1 − 0.20) × (1 − 0.15) = 0.80 × 0.85 = 0.68.

Required ownership today. 12.66 / 0.68 = 18.6 percent.

Post-money valuation. 20 / 0.1861 = 107.5 million; the pre-money valuation is therefore 87.5 million.

Sensitivity to the target IRR. At a 30 percent IRR, the required value at exit is 20 × 1.3^5 = 74.3 million and required exit ownership 6.2 percent; at 70 percent it is 20 × 1.7^5 = 284.0 million and 23.7 percent. Without dilution adjustments, the post-money valuation at the 50 percent target would be 20 / 0.1266 = 158 million; ignoring dilution overstates value by roughly 47 percent.

Interpretation. The post-money price of 107.5 million is the value *at which the investor earns the target return if the projections materialize*. It is not a measure of intrinsic value: the target IRR of 50 percent is far above any plausible cost of capital because it must compensate for the many ventures that fail. Equivalently, if the success probability is 25 percent and failures return nothing, a 50 percent IRR on the success case (a 7.6 times money multiple) corresponds to an expected multiple of 0.25 × 7.59 = 1.90 and an expected annual return of about 13.7 percent, a figure much closer to a plausible cost of equity for risky assets. The venture method is therefore a pricing convention for negotiating equity splits, not a replacement for DCF. A sound use is to compute both: the DCF value (with explicit probabilities, as in Example 3.4) and the venture-method price, and compare. A large gap indicates either that the exit assumptions are inconsistent with the DCF story, or that the target IRR embeds a failure probability that is not in the forecast.

Preferred stock and the capital structure

Startups issue preferred equity with liquidation preferences and anti-dilution protection, so common stock is not worth the post-money valuation divided by the share count. The standard approach is the **option-pricing method**: treat each class as a call option on enterprise value with strikes at the waterfall breakpoints, and price them with Black-Scholes (Volume 1, Chapter 18). For the same enterprise value, preferred is worth more per share than common, so using the preferred price for employee option grants overstates their value.

Platforms and network effects

Platforms (marketplaces, payment networks, social networks) gain value as participants join, and conventional models mis-specify several features.

Two-sided structure and take rate. A marketplace needs both buyers and sellers and often subsidizes one side to attract the other, so unit economics must be built per side. Revenue equals take rate times gross merchandise value (GMV); the take rate is capped by the alternatives available to participants, and high rates invite competitors and disintermediation.

Shape of growth. Network effects imply a slow cold-start phase, rapid growth after a tipping point and then saturation. Metcalfe’s observation that network value scales with the square of users is a stylized overstatement, since not all connections are equally valuable; a concave value-per-user function checked against observed retention is safer.

Winner-take-most. Strong network effects concentrate markets: the leader earns sustained excess returns, the follower little. This supports the scenario structure of Section 3.5, with the breakout scenario carrying much of the value, and it affects terminal ROIC, which may stay above WACC for long only if the analyst can say what stops competitors (data advantages, switching costs, high multi-homing costs) and models the fade. Where both sides multi-home cheaply, network effects are weak and take rates are pushed down. Include a regulatory scenario that constrains take rates or practices.

Relative valuation for platforms

For listed or recently funded platforms the practical benchmarks are EV/GMV, EV/gross profit and EV/revenue, adjusted for growth. Compare on gross profit or contribution, because a marketplace reports net revenue while a reseller reports gross sales. A regression of the multiple on growth and margin across the peer set gives a fitted value, but it fits current market sentiment and is therefore a cross-check on the DCF rather than a substitute.

Valuation when there is no revenue

For firms before revenue (a biotechnology company with an early-stage drug, a battery start-up with a prototype), the cash flows are nowhere to be found in the accounts, and the main unit of analysis is the **project pipeline**. The standard approach is the risk-adjusted net present value (rNPV): for each program, forecast the cash flows conditional on success, multiply by the cumulative probability of reaching each stage (industry phase-transition probabilities give a starting point), and discount at a cost of capital that reflects systematic risk only. As in the scenario method, technical risk goes into the probabilities and not into the discount rate. The sum across programs, less unallocated corporate costs, gives the value of the pipeline. This is a scenario-weighted DCF with a stage-gate structure.

3.8 Putting It Together

A valuation workflow

1. **Diagnose accounting distortions**: capitalize R&D and other intangible spending, treat stock compensation as a cost, remove one-offs.
2. **Write the story** (market size, share, pricing power, competitive position) before the numbers.
3. **Model unit economics by cohort and segment**, and identify the operating drivers and their ranges.
4. **Forecast the path**: growth fading to a stable rate, margin converging to a target, reinvestment via sales-to-capital, NOL usage and funding needs.
5. **Define scenarios, including failure**, with probabilities; use one discount path that declines to the mature level.
6. **Discipline the terminal value** with the value-driver formula, implied multiples and market-share checks.
7. **Cross-check** against relative valuation and venture-method pricing, convert to value per share after options, preferred stock and dilution, and report the distribution and the three assumptions that matter most.

Managerial implications

Founders and managers should track unit economics by cohort, spend on acquisition only while discounted marginal LTV/CAC is comfortably above 1, raise capital when the firm is strong, and communicate adjusted metrics (capitalized R&D, contribution margin, net revenue retention) so that the market prices economics rather than accounting. Investors and acquirers should avoid paying for the breakout scenario with certainty: milestone payments, earn-outs and staged investment make the price rise only if favorable states materialize. Boards of established firms should recognize that expensed R&D and marketing are capital allocation decisions that must clear the same hurdle rate as capital expenditure (Volume 1, Chapter 6).

Case Study

Illustrative case; Halcyon Health Technologies is fictional.

Halcyon Health Technologies sells a subscription platform that helps independent medical clinics manage scheduling, billing and patient messaging. After six years it has 2,400 clinic customers and annual recurring revenue of 96 million, growing 48 percent. Gross margin is 71 percent. Reported operating loss is 22 million on 96 million of revenue: sales and marketing is 41 percent of revenue, R&D 27 percent and administration 14 percent. Cash is 64 million and net burn is 2.6 million a month. The company has a convertible note of 20 million outstanding and a pool of employee options equal to 11 percent of fully diluted shares. Stock compensation is 9 percent of revenue and is not included in the reported loss.

Halcyon's chief executive wants to raise 60 million at a pre-money valuation of 450 million, which is 4.7 times current revenue. The lead investor, Marlowe Partners, has offered 380 million. Both parties have pointed to a group of listed health-software firms that trade at 5 to 8 times revenue, but those firms are larger, are profitable, and grow at 15 to 25 percent.

The board has asked for a valuation that will support the financing negotiation, a view on whether the terms are fair, and a list of operating metrics that should be reported quarterly.

Management's plan assumes growth fading from 48 to 5 percent over ten years, a long-run margin of 24 percent, sales-to-capital of 1.8, terminal growth of 3.5 percent and a terminal ROIC of 28 percent "because that is what our best clinics earn," in a market with two similar private rivals and a price-cutting division of a large records vendor. The plan needs about 1,000 net new clinics a year from an addressable base of roughly 30,000, while the win rate has fallen for three quarters. Cohort data: blended CAC is 21,000 (29,000 for paid channels in the last four quarters); a new clinic pays 3,000 per month; annual logo retention is 82, 90, 93 and 94 percent in years 1 to 4 and later; net revenue retention among survivors is 108 percent; contribution margin is 71 percent. R&D has grown from 11 to 26 million over five years, with a product life of about five years.

Discussion questions

1. Restate Halcyon's income and invested capital using a five-year capitalization of R&D, and explain which direction each adjustment moves and why stock compensation must be handled consistently with the share count. What data would you need to complete the calculation?
2. Using the cohort data, compute the discounted contribution per clinic over a ten-year life at a 12 percent discount rate, and compare it with blended CAC (21,000) and paid-channel CAC (29,000). What does the gap imply for the pace of growth spending?
3. Design a four-scenario valuation (failure, niche, base, breakout) with probabilities. Which inputs (growth, margin, terminal ROIC, probability of failure) would differ most across scenarios, and how would you justify the CFO's terminal ROIC of 28 percent against the competitive facts in the case?
4. Compute the venture-method price for Marlowe Partners at a 35 percent target IRR over five years with one further round diluting 15 percent. Is 380 million a fair pre-money value, and which terms (milestones, liquidation preference, tranching) would bridge the gap with the 450 million ask?

Key Takeaways

- Negative earnings do not remove the need for fundamentals; they move the anchor from current earnings to a target state (revenue, margin, reinvestment) from which value is discounted back.
- Expensing R&D and customer acquisition understates operating income and invested capital; capitalizing them changes both, and adjusted ROIC is often lower than the headline figure.
- Unit economics should use contribution margin, discounting and cohort retention curves; the shape of the retention curve is the model, and fast payback matters as much as LTV/CAC for cash needs.
- A single base-case forecast is biased when failure is likely and payoffs are convex; scenario weighting with an explicit failure state makes the distribution visible.
- Failure risk belongs in the probabilities, not the discount rate; raising the discount rate to "reflect risk" while also weighting scenarios double-counts.

- Terminal value in a young firm often exceeds 100 percent of enterprise value, so the value-driver formula TV = NOPAT × (1 − g/ROIC) / (WACC − g) and tests of implied multiples and market share are essential.
- When terminal ROIC equals WACC, growth adds no value; assuming a high ROIC in perpetuity is a claim about durable competitive advantage that must be defended.
- Stock compensation is a real cost: deduct it from cash flow or model dilution, but never ignore both.
- The venture capital method prices an equity split to deliver a target return if projections hold; its high target IRR embeds failure risk and it is not a measure of intrinsic value.
- Platform value depends on take rates, network-effect durability and the chance of winning; report a distribution, not a point estimate, and name the assumptions that drive it.

Review Questions

1. Why does an expensed-R&D income statement understate operating income for a firm with growing R&D, and when would the adjustment go the other way?
2. Why is it a mistake to add back R&D to EBIT without also adding the research asset to invested capital?
3. Explain the difference between LTV/CAC and discounted LTV/CAC, and give conditions under which the gap is large.
4. How does a cohort retention curve with a plateau change the interpretation of “average churn”?
5. Why is contribution margin, rather than gross margin, the right basis for customer lifetime value?
6. Explain why failure probability should be placed in scenario weights rather than in the discount rate.
7. In Example 3.4, why is the probability-weighted value close to the base case even though 25 percent of outcomes are failures?
8. What does the value-driver formula imply when ROIC equals WACC, and what economic assumption does that represent?
9. List three checks on a terminal value, and explain what each would reveal about an aggressive forecast.
10. Why can the venture capital method deliver a post-money valuation that differs sharply from a DCF value, and how should the difference be interpreted?

Problems

1. **R&D capitalization.** A firm reports EBIT of 120 and invested capital of 700, with a 25 percent tax rate. R&D spending in years 0 to −4 is 90, 80, 70, 60 and 50 (year 0 is current). Amortize straight-line over four years, with spending beginning to amortize in the following year. Compute the research asset, current amortization, adjusted EBIT, adjusted invested capital and adjusted after-tax ROIC.
2. **Basic unit economics.** ARPU is 40 per month, gross margin 80 percent, monthly churn 3 percent and CAC 600. Compute undiscounted LTV, LTV/CAC and the payback period in months.
3. **Discounted LTV.** For the firm in Problem 2, use a monthly discount rate of 1 percent and the formula LTV = m / (c + r). Compute discounted LTV and LTV/CAC, and say whether growth creates value.
4. **Scenario weighting.** A start-up has four outcomes with values and probabilities of: failure 0 (30 percent), niche 60 (35 percent), base 260 (25 percent) and breakout 900 (10 percent). Compute the expected value, and the share of expected value contributed by the breakout case.

5. **Terminal value and ROIC.** NOPAT is 150 in year N, WACC is 10 percent and terminal growth is 3 percent. Compute the terminal value (a) ignoring reinvestment, (b) with ROIC of 15 percent and (c) with ROIC equal to WACC, and the multiple of year-N NOPAT in each case.
6. **Venture method.** A fund invests 15 at a target IRR of 40 percent over five years. The exit value is 900. Future rounds leave existing holders with 75 percent of their stake. Compute required ownership at exit, required ownership today and the post-money valuation.
7. **Cohort value.** A cohort retains 100, 60, 45 and 40 percent in years 0 to 3. Each surviving customer yields contribution of 300 per year (500 revenue at a 60 percent margin), received at the start of each year, and the discount rate is 10 percent. Compute the present value of contribution per original customer.
8. **Runway and dilution.** A firm has cash of 24 million and net burn of 3 million per month. It raises 36 million at a pre-money valuation of 120 million. Compute the runway before the raise and the ownership percentage sold in the round.

Answers to Selected Problems

1. Research asset = 90 + 80 × 0.75 + 70 × 0.50 + 60 × 0.25 + 50 × 0 = 200. Amortization = (80 + 70 + 60 + 50) / 4 = 65. Adjusted EBIT = 120 + 90 − 65 = 145. Adjusted capital = 700 + 200 = 900. Adjusted ROIC = 145 × 0.75 / 900 = 12.1 percent (reported: 120 × 0.75 / 700 = 12.9 percent).
2. Monthly gross profit 32; LTV = 32 / 0.03 = 1,066.7; LTV/CAC = 1.78; payback = 600 / 32 = 18.75 months.
3. LTV = 32 / (0.03 + 0.01) = 800; LTV/CAC = 1.33. Growth creates value, but with a thin margin of safety and 19 months of payback; adverse changes in churn or CAC would reverse it.
4. EV = 0.35 × 60 + 0.25 × 260 + 0.10 × 900 = 21 + 65 + 90 = 176. The breakout contributes 90 / 176 = 51 percent.
5. a. 150 × 1.03 / 0.07 = 2,207.1, multiple 14.7. (b) 154.5 × (1 − 0.03/0.15) / 0.07 = 154.5 × 0.8 / 0.07 = 1,765.7, multiple 11.8. (c) 154.5 × (1 − 0.3) / 0.07 = 1,545.0, equal to 154.5 / 0.10; multiple 10.3.
6. Required value at exit = 15 × 1.4^5 = 80.7; ownership at exit = 80.7 / 900 = 8.96 percent; today = 8.96 / 0.75 = 11.95 percent; post-money = 15 / 0.1195 = 125.5.
7. PV = 300 × (1 + 0.6/1.1 + 0.45/1.21 + 0.40/1.331) = 300 × 2.2179 = 665.4.
8. Runway = 24 / 3 = 8 months. Ownership sold = 36 / (120 + 36) = 23.1 percent.

Chapter 4: Valuing Financial, Cyclical and Commodity Firms

Learning Objectives

After studying this chapter, you should be able to:

- Explain why enterprise-value DCF is inappropriate for banks and insurers, and value them with equity-based methods: excess returns (residual income), the P/B-ROE relationship and dividend (free cash flow to equity) models constrained by regulatory capital.
- Analyze an insurer's economics through float, the combined ratio and underwriting versus investment returns, and link them to a justified price-to-book ratio.
- Estimate normalized (mid-cycle) earnings for a cyclical firm and avoid the peak-earnings and trough-earnings traps in multiples.
- Build a commodity price deck, value a resource project or mine with it, and quantify sensitivity, mean reversion and the option to curtail production.
- Construct a net asset value (NAV) for a resource company and interpret price-to-NAV, including risk-weighting of development assets.
- Perform a sum-of-the-parts valuation, treat corporate costs, pensions and net debt consistently, and judge whether a conglomerate discount is justified.

Opening Vignette

Illustrative scenario; all companies are fictional.

Linden Capital Partners runs a concentrated equity fund, and its analyst, Priya, has been asked to defend three holdings at the annual investor meeting. The first is Harbor Trust Bancorp, a regional bank that trades at 1.2 times book value. A colleague has built a DCF of the bank's "free cash flow" by adding back interest expense and subtracting capital expenditure, arriving at a value three times the market price. Priya knows something is wrong but cannot say what: the bank's debt is its raw material, not a financing choice, and its "capex" is lending.

The second holding is Ironvale Steel. In last year's trough the company earned almost nothing, the trailing price-to-earnings ratio was above 60, and a screen had labeled the stock "expensive." Two years earlier, at peak earnings, the ratio was 4, and the same screen called it "cheap." Priya bought near the trough and the stock has doubled, but she cannot explain to investors what the stock is actually worth.

The third is Cordillera Copper, which has two producing mines, a development project and an exploration portfolio. The sell-side's price target depends on a copper price assumption that differs by 40 percent from the next analyst's. A fourth question hangs over all three: a conglomerate in the fund's watch list, Karst Industries, trades at a discount to what its parts would fetch, and an activist has appeared on its register.

Each question has an answer, but not the one standard enterprise DCF supplies. Financial firms are valued through equity, because liabilities are operating inputs. Cyclical firms are valued through normalized earnings, because the current year is a poor sample of the cycle. Commodity producers are valued through explicit price decks and asset-by-asset NAV, because

value is driven by a price the firm does not control. Diversified groups are valued by parts, because each part has different economics. This chapter develops each approach, with numbers, and shows how to reconcile them with the market.

4.1 Why These Firms Need Different Tools

Volume 1 (Chapters 11 and 21) valued industrial firms by discounting free cash flow to the firm at WACC, then subtracting net debt to reach equity. That architecture assumes that operating assets can be separated from financing, that debt is a choice of capital structure, and that current earnings are a reasonable starting point for forecasting. Each assumption fails for the firms in this chapter, in a different way.

Firm type	What breaks	Valuation response
Banks and insurers	Debt and deposits are inputs to production; “operating” and “financing” cannot be separated; capital is regulated	Value equity directly: excess returns, P/B-ROE, dividend discount with a capital constraint
Cyclical firms	Current earnings are far from average; multiples whipsaw	Normalize earnings over the cycle; use mid-cycle margins and multiples; value through the trough
Commodity producers	Revenue depends on an exogenous price; reserves deplete; operating leverage is extreme	Explicit price decks, project NPV, scenario and simulation, NAV by asset
Conglomerates	Different segments have different growth, risk and multiples	Sum of the parts, with consistent treatment of corporate costs and net debt

There is a common thread. In every case the headline accounting number (net income, EBITDA, book value) is a poor proxy for value unless adjusted for the structure of the business: the regulatory capital that a bank must retain, the point in the cycle at which earnings were measured, the price at which a commodity was sold, the cost of running a corporate center. The rest of the chapter takes the adjustment for each in turn.

4.2 Valuing Banks

Why not enterprise DCF

For a bank, interest on deposits and wholesale funding is the principal cost of goods sold. Treating it as a financing cost, as in a free-cash-flow-to-the-firm model, adds back the main operating expense and makes “operating” income meaningless. Net debt is not well defined (a bank’s liabilities are mostly deposits and trading balances), and the notion of WACC does not apply because the capital structure is largely determined by regulation and not by a value-maximizing trade-off. The correct target is the value of equity, obtained either by discounting cash flows to equity (dividends or free cash flow to equity) at the cost of equity, or by the residual-income (excess-return) method, which discounts earnings above a capital charge.

The economics of a bank in one equation

A bank’s profitability reduces to its **return on equity**, which decomposes as:

```
ROE = ROA × Equity multiplier
ROA = (Net interest income + Fee income − Operating costs − Loan-loss provisions) / Assets
```

Net interest income is the spread between loan yields and funding costs, times earning assets. Fee income depends on payments, advisory and asset-management businesses. The cost-to-income ratio captures operating efficiency, and loan-loss provisions reflect credit quality over the cycle. The leverage is not free: regulation sets a minimum ratio of capital to risk-weighted assets (RWA), so growth in RWA must be matched by growth in equity.

In current regulatory frameworks the key measure is the Common Equity Tier 1 (CET1) ratio, CET1 capital divided by RWA. A bank must hold a minimum plus buffers, and in practice management holds a margin above the requirement. This leads to the central modeling principle: **a bank can pay out as dividends only the earnings it does not need to retain to keep its target capital ratio as it grows.**

Excess returns and the P/B-ROE relationship

In the residual-income method, value of equity equals book value plus the present value of expected excess returns:

$$\text{Value of equity} = BV_0 + \Sigma\ [(ROE_t - r_e) \times BV_{t-1}] / (1 + r_e)^t$$

where r_e is the cost of equity. The term $(ROE - r_e) \times BV$ is the **residual income**: earnings above the charge for the equity capital used. If ROE equals the cost of equity forever, the bank is worth exactly its book value, regardless of growth. If ROE exceeds r_e, growth adds value; if ROE is below r_e, growth destroys it.

In steady state, with constant ROE and growth g, the formula collapses to the relationship used throughout bank analysis:

$$P/B = (ROE - g) / (r_e - g)$$

This is the banking counterpart of the value-driver formula of Chapter 3: replace NOPAT and WACC with ROE and r_e. A bank with a 14 percent ROE, 10 percent cost of equity and 4 percent growth should trade at 10/6 = 1.67 times book. Two comparable banks with the same P/B but different ROE are not equally priced: the P/B-ROE regression across a peer group (price-to-book on the vertical axis, ROE on the horizontal) identifies the cheap and expensive ones relative to the fundamentals the market itself pays for.

Example 4.1 – Residual-income valuation of a bank with fading excess returns. Harbor Trust has book equity of 10,000 million and a cost of equity of 10 percent. Analysts expect ROE to fade over five years as competition erodes the spread, and book value to grow with lending.

Steady-state benchmark. With constant ROE 14 percent and g = 4 percent: P/B = (0.14 − 0.04) / (0.10 − 0.04) = 1.667, so equity value = 16,667.

Fading ROE forecast. ROE is 15.0, 14.5, 14.0, 13.0 and 12.0 percent in years 1 to 5, and book value grows 6, 6, 5, 5 and 4 percent. Residual income in year t = $(ROE_t - 0.10) \times$ beginning book value. Amounts in millions.

Year	Beginning BV	ROE	Net income	Capital charge (10%)	Residual income	PV at 10%	Ending BV
1	10,000	15.0%	1,500.0	1,000.0	500.0	454.5	10,600
2	10,600	14.5%	1,537.0	1,060.0	477.0	394.2	11,236
3	11,236	14.0%	1,573.0	1,123.6	449.4	337.7	11,798
4	11,798	13.0%	1,533.7	1,179.8	353.9	241.7	12,388
5	12,388	12.0%	1,486.5	1,238.8	247.8	153.8	12,883

PV of explicit residual income. 454.5 + 394.2 + 337.7 + 241.7 + 153.8 = 1,581.9 (1,582.0 with unrounded inputs).

Terminal value of excess returns. Assume residual income after year 5 decays with persistence ω = 0.7 per year ($RI_6 = 0.7 \times RI_5$, $RI_7 = 0.7 \times RI_6$, and so on). The value at year 5 of that stream is $RI_5 \times \omega / (1 + r_e - \omega) = 247.8 \times 0.7 / (1.10 - 0.70) = 173.5 / 0.40 = 433.6$. Discounted: $433.6 / 1.10^5 = 433.6 / 1.6105 = 269.3$.

Equity value. 10,000 + 1,582.0 + 269.3 = 11,851 million, a P/B of 1.185.

Alternatives. If excess returns vanish immediately after year 5 (ω = 0), value is 10,000 + 1,582 = 11,582 (P/B 1.158). If the 14 percent ROE persisted at 4 percent growth (steady state), value would be 16,667.

Interpretation. The bank deserves a P/B of 1.19, not 1.67, because the market must be paying for a fade. If it trades at 1.2 times book, as Harbor Trust does in the vignette, the market price is consistent with ROE falling from 15 to 12 percent over five years and excess returns dying out gradually. The only way to justify a higher price is to argue that the fade is shallower. Residual income makes that debate explicit: roughly 86 percent of the value above book comes from the first five years (1,582 of 1,851), so the argument is about the next five years' ROE, not about a distant terminal value. This is why residual-income models are less sensitive to terminal assumptions than dividend models: book value, which is already on the balance sheet, delivers most of the value up front.

Dividend and free-cash-flow-to-equity models with a capital constraint

The alternative is to forecast distributable cash flow to equity. For a bank:

```
FCFE = Net income − (Required growth in equity)
Required growth in equity = Target CET1 ratio × Growth in RWA
```

If the bank holds its CET1 target constant, equity must grow at the same rate as RWA. Net income not retained for that purpose is distributable via dividends and buybacks. In steady state the payout ratio equals 1 − g / ROE; for example, ROE 14 percent and growth 5 percent imply a payout of 64 percent.

Example 4.2 – FCFE valuation with a CET1 constraint. A bank has RWA of 80,000 million and targets CET1 of 12 percent, so equity is 9,600 million. ROE is 14 percent over the explicit period, the cost of equity is 10 percent, and RWA growth is 8, 7, 6, 5 and 4 percent over years 1 to 5. After year 5 growth is 3 percent and ROE is 12 percent (so the payout is 1 − 0.03/0.12 = 75 percent).

Mechanics. Equity at the end of year t equals equity at the start times (1 + growth), since CET1 and RWA both scale. Net income = 14 percent × beginning equity. FCFE = net income − (ending equity − beginning equity).

Year	Beginning equity	Net income	Equity retained	FCFE	PV at 10%
1	9,600.0	1,344.0	768.0	576.0	523.6
2	10,368.0	1,451.5	725.8	725.8	599.8
3	11,093.8	1,553.1	665.6	887.5	666.8
4	11,759.4	1,646.3	588.0	1,058.3	722.9
5	12,347.4	1,728.6	493.9	1,234.7	766.7

PV of explicit FCFE. 523.6 + 599.8 + 666.8 + 722.9 + 766.7 = 3,279.8.

Terminal value. Equity at end of year 5 = 12,347.4 × 1.04 = 12,841.2. Year-6 net income = 12 percent × 12,841.2 = 1,541.0; payout 75 percent gives dividends of 1,155.7. TV = 1,155.7 / (0.10 − 0.03) = 16,510.2. PV = 16,510.2 / 1.6105 = 10,251.5.

Equity value. 3,279.8 + 10,251.5 = 13,531 million; P/B = 13,531 / 9,600 = 1.41.

Interpretation. Here 76 percent of the value comes from the terminal value, the typical weakness of dividend models. Compare with Example 4.1, where most value sat in the book and the first years' excess returns. The two methods should be reconciled; if they give very different answers, the difference is in the ROE and persistence assumptions embedded in each, and the analyst should find it. Note also the role of growth: in year 1 the bank retains 768 of its 1,344 earnings (57 percent) to support 8 percent growth, leaving a payout of only 43 percent. Growth is not free for a bank.

Capital requirements and the cost of regulation

Higher capital requirements reduce ROE for a given return on risk-weighted assets, and hence value, unless the cost of equity falls proportionally (a Modigliani-Miller argument that holds only partly given deposit insurance and implicit guarantees). Writing ROE = (NI/RWA) / CET1, the steady-state value per unit of RWA is:

```
Value per unit of RWA = (NI/RWA − g × CET1) / (r_e − g)
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Example 4.2b – Sensitivity to the CET1 requirement. Use the bank of Example 4.2 with NI/RWA = 1.68 percent (14 percent ROE × 12 percent CET1), g = 4 percent and r_e = 10 percent, holding the return on RWA constant.

- CET1 10 percent: value per RWA = (1.68% − 0.4%) / 6% = 21.33 percent, i.e. 17,067 on RWA of 80,000; ROE = 16.8 percent; P/B = 2.13.
- CET1 12 percent: (1.68% − 0.48%) / 6% = 20.00 percent, i.e. 16,000; ROE 14.0 percent; P/B = 1.67.
- CET1 14 percent: (1.68% − 0.56%) / 6% = 18.67 percent, i.e. 14,933; ROE 12.0 percent; P/B = 1.33.

Each percentage point of additional CET1 reduces equity value by about 1,067 million (6.7 percent) if the return on RWA is unchanged, because equity that must be held earns the same RWA return but is charged at the cost of equity. In practice banks respond by shrinking low-return RWA, repricing loans or cutting costs, so the true cost is lower than the arithmetic suggests. The calculation does show why capital regulation is a first-order valuation input and why bank valuations move on stress-test announcements.

Credit cycle, provisions and normalization

Bank earnings are procyclical: provisions are low at the top and high after a downturn, so current ROE overstates value at the peak and understates it at the trough. Normalize provisions to a through-the-cycle loss rate scaled to the current loan mix, and examine coverage and concentration to judge whether reported provisions are adequate. Where book value is doubtful, adjust it for unrecognized losses such as unrealized losses on held-to-maturity securities, which are not in reported equity but would reduce capital in a stress.

Other financial firms

Asset managers and exchanges have small capital needs and can be valued on fee-related earnings and cash flow. Broker-dealers and lenders with large balance sheets resemble banks and are valued by P/B-ROE. In every case, ask which liabilities are operating inputs and which are financing.

4.3 Valuing Insurers

Float and the two sources of profit

An insurer collects premiums first and pays claims later. The accumulated liabilities (loss reserves and unearned premiums) are called **float**: money held for policyholders and invested for the insurer's benefit. An insurer therefore has two sources of profit: **underwriting** (premiums minus claims and expenses) and **investment income** on float and capital. For a property-casualty (P&C) insurer, the underwriting result is summarized by the **combined ratio**:

```
Combined ratio = (Claims incurred + Underwriting expenses) / Premiums earned
```

A combined ratio below 100 percent is an underwriting profit; above 100 percent, a loss. If the loss is smaller than the investment return on the float, float is cheap funding; if the insurer consistently runs a combined ratio above 100 percent, float has a **cost** equal to the underwriting loss divided by float, and it adds value only if that cost is below the investment yield (or the cost of alternative funding).

The ROE of a P&C insurer

ROE = (Underwriting profit + Investment income) × (1 − tax) / Equity. The elements are premium-to-equity leverage (written premiums divided by equity), float-to-equity leverage, the combined ratio and the investment yield. The relationship shows why insurance is a leverage business: a firm that writes premiums equal to 1.25 times equity and holds float equal to 1.9 times equity earns investment income on a large asset base relative to equity, at the cost of exposing equity to reserve errors and catastrophe losses.

Example 4.3 – Underwriting, float and ROE. A P&C insurer writes and earns premiums of 1,000 million, has equity of 800 million and float (reserves) of 1,500 million, so investable assets are 2,300 million. The portfolio yields 4 percent; the tax rate is 21 percent. Investment income is 2,300 × 4% = 92 million. We examine four combined ratios.

Combined ratio	Underwriting result	Investment income	Pre-tax profit	Net income	ROE	Cost of float
94%	+60	92	152	120.1	15.0%	−4.0%
98%	+20	92	112	88.5	11.1%	−1.3%
100%	0	92	92	72.7	9.1%	0.0%
103%	−30	92	62	49.0	6.1%	+2.0%

Calculations (98 percent row). Underwriting result = 1,000 × (1 − 0.98) = 20; pre-tax profit = 20 + 92 = 112; net income = 112 × 0.79 = 88.5; ROE = 88.5 / 800 = 11.06 percent; cost of float = (0.98 − 1) × 1,000 / 1,500 = −1.33 percent (negative means the insurer is paid to hold float).

Break-even combined ratio. For a 10 percent ROE, net income must be 80, pre-tax profit 80 / 0.79 = 101.3, underwriting result = 101.3 − 92 = 9.3, so the combined ratio must be 1 − 9.3/1,000 = 99.1 percent.

Justified P/B. With cost of equity 10 percent and growth 3 percent, P/B = (ROE − g) / (r_e − g): 94% → (15.0 − 3)/7 = 1.72; 98% → 1.15; 100% → 0.87; 103% → 0.45.

Interpretation. A swing of nine points in the combined ratio (94 to 103) takes the ROE from 15 to 6 percent and the justified P/B from 1.72 to 0.45, a 74 percent decline. The insurer's value hinges on underwriting discipline, which the market can observe only with a lag, because reserves for long-tailed lines take years to prove adequate. This is why analysts

test reserve development (are prior-year reserves repeatedly released or strengthened?), why a single good year proves little, and why a combined ratio should be normalized for catastrophe losses (a benign year flatters; a heavy year distorts the other way) using a multi-year average catastrophe load.

Life insurers and embedded value

Life insurers hold long-duration liabilities with embedded guarantees and options, and earnings recognition is spread over decades, so accounting book value understates the economics. Practitioners use **embedded value**: the value of net assets (adjusted net worth) plus the present value of future profits from in-force policies (value of in-force business), less the cost of holding required capital. **New business value** adds the value created by the year's new sales, discounted at a risk discount rate. Because the discount rate, lapse rates, mortality and the valuation of guarantees (interest rate options embedded in policies, valued with the option methods of Volume 1, Chapter 18) drive the result, embedded value is only as good as its assumptions. Price to embedded value is the life-insurance analogue of P/B, and a discount to embedded value reflects market doubt about the assumptions or the sustainability of new business.

Reserve risk, catastrophe risk and capital

Two features distinguish insurance from banking. Insurance liabilities are estimates, and the estimate can be wrong in either direction: valuation must adjust for reserve adequacy. And losses are fat-tailed (hurricanes, earthquakes, pandemics), so capital must be sized to a tail event, typically a one-in-200-year loss in regulatory frameworks, which constrains growth the way CET1 constrains a bank. Reinsurance transfers some of the tail; its cost belongs in the underwriting result. As for banks, the principal valuation tools are P/B-ROE, residual income and dividend models with a solvency constraint.

4.4 Valuing Cyclical Firms: Normalized Earnings

The cyclical trap

A cyclical firm's profits swing with the economic cycle, and often far more than revenue because of **operating leverage**: fixed costs (plant, labor, maintenance) do not fall when volume does. A steelmaker whose revenue falls 25 percent from peak to trough may see its operating profit fall by more than 90 percent. Multiples applied to such earnings are misleading in a systematic way. At the **peak**, earnings are high, the price-to-earnings ratio looks low, and the stock seems cheap, just when the cycle is about to turn. At the **trough**, earnings are depressed, the P/E ratio looks high (or is undefined), and the stock seems expensive, just when recovery is likely. Markets often anticipate this, so cyclical stocks typically trade at the highest P/E near the trough and the lowest near the peak, an inversion that confuses the unwary.

Normalization

The remedy is to value the firm on **normalized (mid-cycle) earnings**: the earnings the firm would generate at an average point in the cycle. There are four approaches, usually combined.

1. **Average over a full cycle.** Average margins (not absolute profits) over at least one full cycle, apply them to normalized revenue, and deduct the average depreciation. Average margins, not average earnings, because the scale of the business changes across the cycle.
2. **Mid-cycle capacity and price.** For industries driven by utilization and a commodity-like price, estimate mid-cycle utilization and price (for example, the long-run marginal cost of production plus a return) and compute earnings from the cost curve.

3. **Return-on-capital approach.** Estimate a through-the-cycle ROIC and apply it to invested capital: normalized NOPAT = ROIC × capital. This links earnings to the asset base and is useful when the capital base is stable.
4. **Replacement-cost cross-check.** Cyclical industries with excess capacity trade below the cost of building new plants, and the market value of the firm should tend toward replacement cost in a boom and fall below it in a bust. Comparing enterprise value to replacement cost (a Tobin's q logic) tells you where in the cycle the market is pricing the firm.

Normalization has a risk: the cycle may have structurally changed. A decade of low margins may reflect permanent overcapacity, Chinese competition, a technology shift or regulation, and not a cycle at all. The analyst must distinguish cyclical from structural, which requires understanding the industry's supply side (capacity additions, exit barriers, cost curve) and not just the history of earnings.

Example 4.4 – Normalizing a steelmaker. Ironvale Steel has ten years of data (millions of currency units). D&A is 700 per year, net debt 3,000 and there are 100 million shares.

Year	Revenue	EBITDA margin	EBITDA
1	8,000	14%	1,120
2	9,500	19%	1,805
3 (peak)	11,000	24%	2,640
4 (trough)	7,500	3%	225
5	6,800	5%	340
6	8,200	12%	984
7	9,600	18%	1,728
8	10,800	22%	2,376
9	9,000	16%	1,440
10	7,000	8%	560
Total / average	**87,400 / 8,740**	**15.1% (weighted)**	**13,218 / 1,321.8**

Normalized EBITDA. Average revenue 8,740 × weighted-average margin 15.12 percent = 1,321.8. (Identical to the average of annual EBITDA because the margin is revenue-weighted.)

Normalized EBIT. 1,321.8 − 700 = 621.8.

Valuation using a mid-cycle EV/EBITDA multiple of 6.0. EV = 6.0 × 1,321.8 = 7,930.8. Equity = 7,930.8 − 3,000 = 4,930.8. Per share = 49.31.

The traps. Apply what a naive investor would do at each point in the cycle.

Point in cycle	EBITDA	Multiple the market would pay	EV	Equity per share
Peak (year 3)	2,640	4.5x	11,880	88.80
Trough (year 4)	225	14.0x	3,150	1.50
Normalized	1,321.8	6.0x	7,930.8	49.31

Interpretation. Trough EBITDA of 225 is below depreciation of 700 (EBIT of −475), so no P/E exists, and valuing the firm at 6 times trough EBITDA gives $6 \times 225 - 3{,}000 < 0$ for a company that earns 621.8 of EBIT in an average year. Peak EBITDA at even a low multiple gives 88.80, well above the mid-cycle value, because the multiple is applied to earnings that will not last. The relevant comparison is between the normalized value (49.31) and the market price: if the stock trades at 20 at the trough, the market is pricing a permanent decline or fearing the leverage (net debt of 3,000 is 13 times trough EBITDA). The result is sensitive to inputs: a mid-cycle margin of 12 percent instead of 15.1 percent lowers normalized EBITDA to 1,048.8 and equity per share to 32.9.

Leverage and the survival constraint

Net debt of 3,000 is 2.3 times normalized EBITDA but 13 times trough EBITDA. If a covenant breach or refinancing wall arrives at the trough, equity may be wiped out or diluted before the cycle turns, so the equity of a levered cyclical is an option on the cycle (Volume 1, Chapter 15 on distress costs and Chapter 18 on options). Practitioners test the balance sheet against a downside cycle (two years at trough margins), compute liquidity and covenant headroom, and judge the probability of an equity raise at depressed prices. Mid-cycle multiples should come from peers on the same basis (EV divided by each peer's normalized EBITDA) and be cross-checked against EV/invested capital plotted against normalized ROIC minus WACC.

4.5 Commodity Producers: Price Decks, Project Value and Uncertainty

The role of the price deck

For a commodity producer, the key driver of value is a price the firm cannot influence. The standard practice is to construct a **price deck**: a year-by-year forecast of the commodity price (and often of exchange rates and input costs) used in the valuation. The deck must reflect three considerations.

- **Spot versus long-run price.** Commodity prices mean-revert in the long run toward the marginal cost of supply (the incentive price that brings on enough new production to meet demand). A deck usually starts near the futures curve for the first years (where forward markets are liquid), and blends toward a long-run price assumption derived from the cost curve.
- **Consistency.** Within an investment bank or a company, the deck should be common across assets and analyses. Different analysts using different decks produce uncomparable values.
- **Real versus nominal.** The deck must be consistent with the discount rate: a real price deck with a real WACC, or nominal with nominal.

Futures prices are risk-adjusted market expectations for the near term and are a useful anchor, though liquidity declines beyond a few years, and the term structure reflects storage costs and convenience yield as well as expectations. Cost-curve analysis anchors the long run: the price must be at least the cash cost of the marginal producer in the long run to sustain supply, and it will eventually reflect the full cost (including a return on capital) of the incentive project needed to meet demand.

Operating leverage and the sensitivity of value

A mine has low variable costs relative to price, and fixed costs and capital are large, so a small price change produces a large change in cash flow. In the example below a 6.7 percent change in the copper price (0.25 on 3.75) changes value by nearly 50 percent. The key sensitivity tools are:

- **Price sensitivity table**, which shows NPV at several price decks, and the **break-even price** at which NPV is zero.
- **Cost-curve position.** A producer in the lowest quartile of the cost curve survives price troughs and earns excess returns in booms; a high-cost producer is the marginal supplier and has little margin of safety.
- **Hedging.** Hedging locks in the price for a portion of output. It changes the cash-flow profile and reduces financial distress risk, but it does not change the value of the asset in an efficient market. A valuation should use the unhedged deck for the asset, and add or subtract the mark-to-market value of the hedge book separately.

Example 4.5 – Valuing a copper mine under different price decks. A mine will produce 440 million pounds of payable copper per year for eight years. Cash cost is 2.40 per pound, sustaining capital expenditure is 120 million per year, and the initial capital of 1,500 million (spent today) is depreciated straight-line over eight years (187.5 per year). The tax rate is 30 percent, applied to positive taxable income, and the discount rate is 8 percent (cash flows at year-end). Amounts in millions except prices.

Mean-reverting deck. The price is 4.50 in year 1, 4.25 in year 2, 4.00 in year 3, and the long-run price of 3.75 from year 4 on.

Year	Price (per lb)	Revenue	EBITDA	Tax	FCF	PV at 8%
1	4.50	1,980.0	924.0	221.0	583.0	539.9
2	4.25	1,870.0	814.0	188.0	506.0	433.9
3	4.00	1,760.0	704.0	155.0	429.0	340.6
4	3.75	1,650.0	594.0	122.0	352.0	258.8
5	3.75	1,650.0	594.0	122.0	352.0	239.6
6	3.75	1,650.0	594.0	122.0	352.0	221.9
7	3.75	1,650.0	594.0	122.0	352.0	205.4
8	3.75	1,650.0	594.0	122.0	352.0	190.2

(Calculations are shown to one decimal and use unrounded intermediate values, so a row may differ by 0.1.) Example, year 1: EBITDA = 440 × (4.50 − 2.40) = 924.0; taxable income = 924.0 − 187.5 = 736.5; tax = 0.30 × 736.5 = 221.0 (220.95 unrounded); FCF = 924.0 − 220.95 − 120 = 583.05; PV = 583.05 / 1.08 = 539.9.

Present value of cash flows. 2,430.1. *NPV* = 2,430.1 − 1,500 = **930.1**.

Other decks. Flat spot price of 4.50 for all eight years: NPV = 1,850.6. Flat long-run price of 3.75: NPV = 523.1.

Sensitivity of the mean-reverting deck to a parallel shift in price (per lb):

Shift	−0.50	−0.25	0	+0.25	+0.50
NPV	45.2	487.7	930.1	1,372.6	1,815.1

Break-even price. The flat price at which NPV is zero is 3.45 per pound (computed by solving NPV(P) = 0).

Interpretation. Using the spot price throughout would value the mine at 1,850.6, double the 930.1 from a deck that reverts to the long-run price, which is why equity analysts' price targets can differ by 40 percent on the same asset. Each 0.25 per pound is worth about 442.5 million of NPV, 48 percent of the base value (each 0.25 adds 110 of annual pre-tax revenue, 77 after tax, times the 8-year annuity factor of 5.747). The break-even price of 3.45 is only 8 percent below the long-run assumption of 3.75, so the margin of safety is thin: a project valued at 930 that would be worthless at a 3.45 price. The

1,500 of up-front capital makes it a levered bet on the price. A prudent investment committee would require that the project is positive-NPV at a conservative deck (for example, the lower quartile of the cost curve) before approving the project, or that the upside be shared with a partner.

Mean reversion and stochastic prices

A deterministic deck is the expected path but not the uncertainty around it. Commodity prices mean-revert: a high price spurs supply and a low price curtails it. A standard model (Schwartz-style) makes the log of the price follow an **Ornstein-Uhlenbeck** process: the price is pulled toward a long-run level at speed κ and subject to volatility σ:

```
d ln P = κ (μ − ln P) dt + σ dW
```

The half-life of a deviation is ln 2 / κ; with κ = 0.35 per year the half-life is about two years. Simulation of this process lets us value the **operating flexibility** that a deterministic deck omits. A producer is not obliged to produce at a loss: if the price falls below the cash cost plus the cost of keeping the mine on care and maintenance, it can **curtail** production, an option that truncates the downside. (Restarting costs money and takes time, a complication we ignore.)

Example 4.6 – Monte Carlo valuation with a curtailment option. The mine of Example 4.5 is valued with 20,000 simulated eight-year price paths. The log price follows the exact discretization of the mean-reverting process, starting at 4.50, with long-run median 3.75 (μ = ln 3.75), κ = 0.35 and σ = 0.22 per year. Each year, free cash flow is computed as in Example 4.5. With the option, if free cash flow would be below −60 million (the cost of care and maintenance), the mine is curtailed and the cash flow is −60. The script (seed 7) is:

```
import numpy as np
rng = np.random.default_rng(7)
N, T = 20000, 8
kappa, mu, sig, p0 = 0.35, np.log(3.75), 0.22, 4.50
prod, cost, capex, tax, dep, w = 440, 2.40, 120, .30, 1500/8, .08
x = np.full(N, np.log(p0))
pv = np.zeros(N); pv_no = np.zeros(N)
for t in range(1, T + 1):
    x = mu + (x - mu)*np.exp(-kappa) + sig*np.sqrt((1-np.exp(-2*kappa))/(2*kappa))*rng.standard_normal(N)
    p = np.exp(x)
    ebitda = prod*(p - cost)
    fcf = ebitda - np.maximum(ebitda - dep, 0)*tax - capex
    pv_no += fcf/(1+w)**t
    pv += np.where(fcf < -60, -60, fcf)/(1+w)**t
npv_opt, npv_no = pv - 1500, pv_no - 1500
```

Output. Mean NPV without the option: 1,108.5. Mean NPV with the option: 1,132.0. Value of the curtailment option: 23.4. Distribution of NPV with the option: 5th percentile −632.1; 25th 289.2; median 1,036.6; 75th 1,865.9; 95th 3,236.6. The probability that NPV is negative is 17.0 percent with the option and 17.75 percent without it. Without the option the 5th percentile is −744.0.

Interpretation. Three observations follow. First, the simulated mean NPV (1,108) exceeds the deterministic-deck NPV (930) because the deck anchors the long-run *median* price at 3.75 and a lognormal price has a mean above its median (by $\exp(v/2)$, where $v = \sigma^2/(2\kappa)$ is the stationary variance of the log price; about 3.5 percent here) and because tax and cash flow are convex in price; the analyst must decide whether the deck represents the median or the mean, and use the same convention throughout. Second, the curtailment option adds only 23 (2 percent of value) here because a low-cost mine rarely needs to shut; for a high-cost mine the option would be worth far more, and ignoring it is a typical reason why DCF

understates marginal producers. Third, the distribution is wide: the interquartile range is 1,577 and the 5th-to-95th range spans more than 3,800, so a single NPV conveys false precision. A capital-allocation decision should be tested against the downside, not only the mean.

Reserves, depletion and fiscal regimes

Resource firms hold depleting assets, so valuation requires a production profile constrained by reserves (economically recoverable, classified as proven, probable or possible) rather than a perpetuity. Reserves depend on the price, since a higher price lowers the cut-off grade and converts resources into reserves. Model grade decline, sustaining capital and closure and decommissioning liabilities as negative cash flows at end of life; terminal value is generally zero. Replacing depleted reserves requires exploration spending that belongs in the forecast.

Royalties, severance and windfall taxes and state participation take a large, price-dependent share of cash flow: a royalty on revenue raises the break-even price, a profit-based tax shares the downside and a windfall tax caps the upside. Sovereign risk is handled through scenario cash flows (expropriation, changed terms) rather than the discount rate, following the principle of Chapter 3 that idiosyncratic events belong in the cash flows.

4.6 Net Asset Value for Resource Companies

The method

The NAV approach values a resource company as the sum of the net present values of its assets, less corporate costs and net debt:

```
NAV = Σ (NPV of asset i × risk weight i) + Value of exploration − PV(corporate costs) − Net debt − Other liabilities
```

NAV per share is compared to the share price through **P/NAV**. Each asset is valued by its own discounted cash flow at the common price deck, with discount rates that reflect asset-specific risk, and the sum is a break-up value rather than a going-concern estimate. The method is standard for mining and for oil and gas exploration and production, where the portfolio is a set of discrete assets at different stages.

Risk weighting by stage. Assets are not equally certain. Producing assets are valued at 100 percent of their NPV (perhaps at a lower discount rate, such as 5 percent real). Assets with a completed feasibility study and permits receive a weight for completion and construction risk (for example, 70 percent). Pre-feasibility assets receive a much lower weight (30 to 40 percent), and exploration properties are valued at a price per unit of resource, a comparable-transactions value, or at cost. The weights can be viewed as the probability of reaching production, as in the risk-adjusted NPV of pipeline assets in Chapter 3.

Example 4.7 – NAV of a diversified copper miner. Cordillera Copper has four assets and an exploration portfolio. Project NPVs, computed under the company’s price deck, are stated in millions.

Asset	Stage	Unrisked NPV	Risk weight	Risked NPV
Mine A	Producing	2,400	100%	2,400
Mine B	Producing	1,350	100%	1,350
Project C	Feasibility, permitted	1,800	70%	1,260
Project D	Pre-feasibility	900	35%	315
Exploration portfolio	Early-stage	150	n/a	150
Gross asset value				**5,475**

Corporate costs. Annual head-office costs of 60 million, tax-effected at 30 percent and capitalized at the 8 percent discount rate as a perpetuity: 60 × (1 − 0.30) / 0.08 = 525. Deduct 525.

Net debt. 1,800.

NAV. 5,475 − 525 − 1,800 = **3,150** million. With 500 million shares, NAV per share = 6.30.

P/NAV. If the share price is 5.50, market capitalization is 2,750 and P/NAV = 5.50 / 6.30 = 0.87. At a P/NAV of 0.8 the price would be 5.04; at 1.2, 7.56.

Interpretation. The share trades at a 13 percent discount to NAV. Several facts qualify this reading. First, the leverage of NAV to price is high: the equity of 3,150 is only 58 percent of gross asset value, so a 10 percent fall in the risked asset value (548) lowers NAV by 17 percent. Second, 1,725 (32 percent) of gross value rests on the two development projects and exploration, which carry execution risk that the weights only approximate. Third, premium P/NAV multiples (1.0 or above) are normally paid to companies with a growth pipeline and a record of delivering on time and budget, and discounts to those with a record of overruns; a P/NAV of 0.87 may be fair for a company with poor delivery history. Fourth, NAV measures static asset value and treats the pipeline as discrete projects; it does not capture the value of management's ability to find and develop new assets, or the cost of replacing depleted reserves, so a "going-concern" premium is arguable for good operators.

Pitfalls of NAV

NAV rests on the price deck, the discount rate and the risk weights, each a judgment. Use one consistent method across producing and development assets, avoid double counting exploration value already in a project NPV, and do not combine a spot-price NAV with a P/NAV multiple derived from historical premiums. State the deck, discount rate and weights, and show NAV at the consensus deck, at spot and at a conservative deck.

4.7 Sum-of-the-Parts Valuation

When and why

A diversified company is a portfolio of businesses with different growth, margins, risk and capital intensity, and a single multiple applied to consolidated earnings averages these differences away. **Sum-of-the-parts (SOTP)** valuation values each business on the multiple or DCF appropriate to its own peers, adds the values, deducts items that belong to the group as a whole, and compares the resulting equity value with market capitalization. The gap is the **conglomerate discount** (or premium).

The mechanics

1. **Segment** the group so that each part has an identifiable peer group and separable financials.

2. **Value each part** by EV/EBITDA, EV/EBIT or P/E of comparable pure-plays (adjusted for size, growth and margin), by DCF where the profile is unusual, or by NAV for resource segments.
3. **Sum** the segment enterprise values.
4. **Deduct corporate costs**, capitalized at an appropriate multiple. The cost of running the center belongs to no segment, and omitting it is the most common SOTP error.
5. **Deduct net debt, unfunded pensions (after tax) and minority interests**; add equity-accounted associates and non-core assets.
6. **Compare** with market capitalization and ask why a discount exists and whether it is justified.

Example 4.8 – SOTP for a conglomerate. Karst Industries has four segments. Corporate costs are 95 million a year, capitalized at 8.5 times. Net debt is 3,200 and the pension deficit is 450 before tax (tax rate 25 percent). There are 200 million shares and market capitalization is 5,400. Amounts in millions.

Segment	EBITDA	Peer EV/EBITDA	Enterprise value
Industrial Automation	420	11.0x	4,620
Specialty Chemicals	310	8.0x	2,480
Consumer Brands	260	9.5x	2,470
Legacy Distribution	150	5.5x	825
Total segments	**1,140**	**9.12x (blended)**	**10,395**

Group adjustments. Capitalized corporate costs: 95 × 8.5 = 807.5. After-tax pension deficit: 450 × 0.75 = 337.5. Net debt: 3,200.

Equity value. 10,395 − 807.5 − 3,200 − 337.5 = **6,050** million, or 30.25 per share.

Comparison with the market. Market capitalization of 5,400 is 27.00 per share. The market's discount to SOTP is 1 − 5,400 / 6,050 = 10.7 percent (SOTP is 12.0 percent above the price). The market-implied enterprise value is 5,400 + 3,200 + 337.5 = 8,937.5, or 7.84 times segment EBITDA; the SOTP enterprise value net of corporate costs is 9,587.5, or 8.41 times.

Interpretation. A 10.7 percent discount may be wholly explained by what the SOTP omits. (1) **Capital allocation inefficiency**: if cash from strong segments funds weak ones, value is destroyed, and the discount measures it. (2) **Multiples that do not transfer**: peers are larger or purer and carry liquidity and focus premiums, so 11.0x may be generous for a division with half the margin of its best peer. (3) **Separation costs**: break-up triggers tax, dis-synergies in shared services and one-time costs that can absorb 3 to 5 percent of value. (4) **Opacity**: investors discount segment results they cannot verify. The actionable question for the board or an activist is whether the discount exceeds the realizable cost of separation; if separation costs 4 percent of SOTP and executes cleanly, the net gain is about 6 percent, and hidden cross-subsidies would reduce it.

Pitfalls

The recurring errors are omitting capitalized corporate costs, using peer multiples for a structurally weaker segment, ignoring pensions, leases and minorities, and treating SOTP as achievable when separation costs, tax and loss of scale are ignored. As a check, compare the implied blended multiple (9.1x) with that of focused peers and ask why the group should be worth more in pieces.

4.8 Integrating the Methods

The approaches in this chapter share a discipline: identify what generates value, find the stable anchor, and be explicit about the cycle, price or structure that makes current accounting misleading.

Question	Banks and insurers	Cyclicals	Commodity producers	Conglomerates
What is the anchor?	Book equity and ROE	Mid-cycle margin	Long-run price from the cost curve	Segment-specific multiples
What is the key driver?	ROE minus cost of equity; capital ratio; combined ratio	Operating leverage; cycle position	Price deck; cost position; reserves	Capital allocation; separation costs
What is the main risk?	Credit/reserve error; regulation	Balance-sheet survival at the trough	Price, fiscal and execution risk	Wrong multiples; hidden cross-subsidies
Standard cross-check	P/B vs ROE regression	EV vs replacement cost	P/NAV and break-even price	Implied blended multiple

In practice, a holding such as Linden's three positions requires three different models with one common principle: the market price should be decomposed into what it implies about the unobservable driver (the fade in ROE, the position in the cycle, the long-run price, the multiple for the parts) and compared with an independent view of that driver. Reverse-engineering the market's assumption is more informative than producing another point estimate.

Case Study

Illustrative case; Meridian Financial Group is fictional.

Meridian Financial Group owns three businesses: Meridian Bank, a regional commercial lender with 62 billion of loans; Meridian Re, a property-casualty insurer; and Meridian Minerals, a 49 percent-owned copper and zinc producer that Meridian consolidates through a joint venture agreement. The board is evaluating an unsolicited approach from an investor who proposes to buy the whole group at a 20 percent premium to the current price, and an activist shareholder has argued that the group should be split into three.

Meridian Bank has equity of 5.5 billion, RWA of 46 billion, CET1 of 12.0 percent, and a return on equity that has averaged 13.2 percent over the last eight years but reached 17 percent last year because loan-loss provisions were at a cyclical low of 0.15 percent of loans, against a through-the-cycle average of 0.55 percent. Its cost of equity is 10 percent, it trades at 1.4 times book, and regulators are considering raising the CET1 requirement by one percentage point.

Meridian Re has equity of 1.6 billion and writes 2.1 billion of premiums. Its combined ratio was 93 percent last year, 96 percent the year before and 104 percent in the year of a major hurricane; the five-year average is 99 percent. It holds reserves of 3.0 billion invested at 4.2 percent. Reserve releases contributed 3 points of improvement to last year's combined ratio, and an external actuary has questioned the adequacy of reserves in a long-tail liability line.

Meridian Minerals produces 190 thousand tonnes of copper equivalent a year from two mines in the lowest-cost half of the global cost curve, with a cash cost equal to 65 percent of the long-run price assumed in the board's deck. It also holds a development project (feasibility complete, permits pending) whose unrisked NPV is 1.1 billion, and an exploration portfolio. The mines have 11 and 14 years of reserve life. Net debt of the minerals division is 0.9 billion. The royalty regime in its main jurisdiction is under review and a windfall tax has been proposed.

Group corporate costs are 120 million per year. The group has net debt (excluding the bank's funding) of 1.4 billion and an unfunded pension deficit of 380 million before tax. Group shares trade at 1.15 times consolidated book. Management says that the minerals division is "worth far more than the market gives credit for," while the activist says the bank and insurer are "being valued as if returns were at their peak."

The investor's bid price implies a premium multiple on last year's consolidated earnings, which were boosted by low provisions, a benign catastrophe year and a copper price near the top of its range. The board has asked for an independent valuation and a recommendation.

Discussion questions

1. Explain how you would value Meridian Bank using both residual income and a dividend model with a CET1 constraint. How would you normalize last year's ROE, and what would a one-point rise in CET1 do to value if the return on RWA is unchanged?
2. Evaluate Meridian Re's last three years of combined ratios. Which combined ratio would you use to normalize earnings, how would you treat the reserve releases, and what P/B would a normalized ROE justify?
3. Describe how you would build a price deck and NAV for Meridian Minerals, including how to risk-weight the development project, treat the royalty and windfall-tax risk, and reflect the 49 percent ownership.
4. Design a sum-of-the-parts valuation for the group. Which corporate costs, pension and net-debt adjustments are needed, how should the conglomerate discount be interpreted, and how would you advise the board on the bid and on the break-up proposal?

Key Takeaways

- Banks and insurers should be valued through equity, because deposits, claims reserves and other liabilities are operating inputs; enterprise-value DCF and WACC do not apply.
- The P/B-ROE relationship, $P/B = (ROE - g) / (r_e - g)$, links valuation to the return earned on equity, and residual income places most value in book plus near-term excess returns.
- A bank's distributable cash flow is net income less the equity it must retain to hold its CET1 ratio as RWA grows; capital requirements are therefore a first-order driver of value.
- Through-the-cycle provisions and combined ratios, not current ones, determine normalized ROE; reserve adequacy and catastrophe loads matter for insurers.
- Float is valuable only if its cost (underwriting loss over float) is below what the invested funds can earn; a few points of combined ratio swing ROE and justified P/B dramatically.
- For cyclical firms, value normalized (mid-cycle) earnings with mid-cycle multiples; trailing P/E is highest at the trough and lowest at the peak, and leverage can turn the trough into insolvency.
- Commodity valuation begins with a price deck anchored to futures in the near term and to the marginal cost of supply in the long run; value is highly sensitive to the price and cost position, so report break-even prices.
- Simulation of mean-reverting prices quantifies the distribution of NPV and the value of operating flexibility such as curtailment, but results depend on whether the deck represents the median or the mean.
- NAV sums risk-weighted asset values less corporate costs and net debt; P/NAV must be read against delivery record, leverage and the deck used.

- SOTP requires capitalized corporate costs, pensions and net debt; a conglomerate discount is only an opportunity if it exceeds the cost of separation and is not explained by capital misallocation or non-transferable multiples.

Review Questions

1. Why is it conceptually wrong to compute free cash flow to the firm for a bank, and what should be valued instead?
2. Explain the P/B-ROE relationship and what it implies for two banks with the same P/B but different ROE.
3. In a residual-income model, why is a terminal value less critical than in a dividend model?
4. How does the CET1 requirement constrain the dividend a bank can pay while growing, and why do higher requirements lower value?
5. What is float, how is its cost measured, and when does it create value for an insurer?
6. Why do reserve releases and catastrophe-free years distort an insurer's combined ratio, and how should an analyst normalize them?
7. Explain why a cyclical stock's P/E ratio is typically highest at the trough of the cycle.
8. List three methods of estimating normalized earnings and one situation in which normalization would be misleading.
9. Why can a price deck anchored to spot prices overvalue a mine, and what should anchor the long-run price?
10. Give two reasons why an apparent conglomerate discount may not be an arbitrage opportunity.

Problems

1. **Justified P/B.** A bank has ROE of 12 percent, growth of 3 percent and a cost of equity of 9 percent. Compute the justified P/B. If book equity is 5,000, what is the equity value, and what ROE would justify a P/B of 1.0?
2. **Residual income.** A firm has book equity of 100, expected ROE of 16 percent next year and a cost of equity of 10 percent; assume all earnings are paid out so book value stays at 100, and residual income decays with persistence $\omega = 0.5$ each year. Compute residual income for year 1 and the equity value.
3. **Capital constraint.** A bank has RWA of 50,000, earns 1.5 percent of RWA, has a cost of equity of 10 percent and grows 5 percent a year in steady state. Compute equity value (as value per unit of RWA, total, and P/B) with CET1 at 11 percent, and the value if CET1 rises to 13 percent.
4. **Insurer ROE.** An insurer has premiums of 500, equity of 400, float of 600, investment yield of 5 percent, tax of 20 percent and a combined ratio of 97 percent. Compute net income, ROE and the cost of float. What combined ratio gives a 10 percent ROE?
5. **Normalization.** A cyclical firm's EBITDA over five years was 800, 1,100, 1,500, 600 and 400. Mid-cycle EV/EBITDA is 7.0, net debt is 2,500 and there are 50 shares. Compute normalized EBITDA, enterprise value and value per share.
6. **Price deck.** A mine produces 100 units a year for three years at a cash cost of 60; prices in years 1 to 3 are 90, 80 and 70; capex is 4,000 at time zero; ignore tax; the discount rate is 10 percent. Compute NPV and the flat break-even price.
7. **NAV.** Asset NPVs are 800 (producing, 100 percent), 500 (feasibility, 60 percent) and 300 (pre-feasibility, 30 percent). Corporate costs capitalized are 150, net debt is 400 and there are 100 shares. Compute NAV per share and the share price at a P/NAV of 0.80.

8. **Sum of the parts.** Segment EBITDA of 100, 60 and 40 are valued at 10x, 7x and 6x. Corporate costs of 15 are capitalized at 8x, net debt is 500 and there are 50 shares. Compute SOTP value per share and the discount if market capitalization is 800.

Answers to Selected Problems

1. P/B = (12 − 3) / (9 − 3) = 1.5; equity value = 7,500. A P/B of 1.0 requires ROE = cost of equity = 9 percent.
2. RI_1 = (0.16 − 0.10) × 100 = 6. Value = 100 + 6 / (1.10 − 0.50) = 100 + 10 = 110.
3. Value per unit of RWA = (1.5% − 5% × 11%) / (10% − 5%) = 0.95% / 5% = 19.0 percent, i.e. 9,500; equity = 5,500, P/B = 1.73. With CET1 at 13 percent: (1.5% − 0.65%) / 5% = 17.0 percent, i.e. 8,500; equity = 6,500, P/B = 1.31.
4. Underwriting profit = 500 × 0.03 = 15; investment income = (400 + 600) × 5% = 50; pre-tax 65; net income 52; ROE 13.0 percent. Cost of float = −15 / 600 = −2.5 percent. For ROE 10 percent: net income 40, pre-tax 50, underwriting result 0, combined ratio 100 percent.
5. Average EBITDA = 4,400 / 5 = 880; EV = 6,160; equity = 3,660; per share = 73.2.
6. Cash flows 3,000, 2,000, 1,000. PV = 2,727.3 + 1,652.9 + 751.3 = 5,131.5; NPV = 1,131.5. Break-even flat price: 4,000 / (100 × 2.4869) + 60 = 76.1.
7. NAV = 800 + 300 + 90 − 150 − 400 = 640; 6.40 per share; price at P/NAV 0.80 = 5.12.
8. EV = 1,000 + 420 + 240 = 1,660; less corporate costs 120 and net debt 500 gives 1,040; 20.80 per share. Market capitalization of 800 (16.00 per share) is a 23.1 percent discount to SOTP.

Part II — Credit and Fixed Income

Chapter 5: Credit Risk Models

Learning Objectives

After studying this chapter, you should be able to:

- Decompose credit risk into probability of default, loss given default and exposure at default, and distinguish risk-neutral from real-world (physical) measures of each.
- Apply the Merton structural model to value a firm's equity and debt as options, and derive default probabilities, credit spreads and implied recovery from asset value, asset volatility and leverage.
- Calibrate asset value and asset volatility from observable equity data, and compute a KMV-style distance to default and its mapping to an expected default frequency.
- Build reduced-form (intensity) models, convert credit spreads into hazard rates and survival probabilities, and use them to price risky bonds and credit default swaps.
- Estimate and apply credit migration matrices, rating models and scorecards (Altman Z-score, logistic regression) and recognise their strengths and weaknesses.
- Compute expected and unexpected loss for loans and portfolios, and explain how model outputs feed pricing, provisioning, capital and limit-setting decisions.

Opening Vignette

(Illustrative scenario; the company and its figures are fictional.)

Marlowe Industrial Gases is a mid-sized supplier of bulk gases to steelmakers and hospitals. In the spring, its treasurer, Dana Whitcombe, meets with the credit committee of the bank that provides most of the company's revolving facility. The bank's analyst opens with three numbers and, to Dana's surprise, none of them comes from Marlowe's financial statements.

The first is a "distance to default" of 3.1, computed from Marlowe's share price, the volatility of that share price and the face value of its borrowings. The second is a five-year credit default swap spread of 210 basis points, observed in the market for Marlowe's bonds. The third is an internal rating of "BB-plus" produced by a scorecard that blends leverage, interest cover, size and sector. Three different models, three different languages, one question: how likely is it that Marlowe fails to pay, and how much would the bank lose if it did?

Dana pushes back. Marlowe's cash flows are contracted under long-term take-or-pay agreements, and its reported coverage ratios are strong. Why should an equity price, which jumped around during a recent market sell-off, move her borrowing cost? The analyst answers that the equity market is watching the same cash flows and that a falling share price with rising volatility is, mechanically, a firm moving closer to the point where equity holders would walk away. The CDS spread, she adds, is not a forecast of default at all: it embeds both the probability of default and investors' appetite for bearing it.

By the end of the meeting, Dana has learned that the facility's price is not one number but the combination of an expected-loss charge, a premium for uncertainty and a return on capital held against unexpected losses. This chapter builds the models behind each of those components, shows how they connect, and explains what each can and cannot tell a manager.

5.1 The Anatomy of Credit Risk

5.1.1 Three building blocks

Credit risk is the risk of loss arising from a counterparty's failure to meet its obligations in full and on time. For a single exposure it is conventionally decomposed into three quantities:

- **Probability of default (PD):** the likelihood that the obligor defaults within a stated horizon, typically one year, or over the life of the exposure.
- **Loss given default (LGD):** the fraction of the exposure that is lost if default occurs, after recoveries and net of workout costs. LGD = 1 − recovery rate.
- **Exposure at default (EAD):** the amount outstanding when default occurs. For a term loan this is the balance; for a revolving facility it includes the drawn amount plus the expected additional drawing before default; for a derivative it is the positive replacement value.

The expected loss on an exposure is their product:

```
EL = PD × LGD × EAD
```

Expected loss is a cost of doing business, in the same way that expected warranty claims are a cost of selling machinery. It should be priced into the loan, provisioned against, and appear in the income statement over time. The harder question, taken up at the end of the chapter, is the *unexpected* loss: the deviation from expectation that must be absorbed by capital.

5.1.2 Three families of models

The literature offers three broad ways to estimate default risk, and a practising credit professional uses all of them:

1. **Structural models** (Merton 1974 and its descendants) explain default as the economic consequence of the firm's asset value falling below its liabilities. They require a model of the firm's balance sheet and rely on option-pricing theory (Volume 1, Chapter 18).
2. **Reduced-form or intensity models** (Jarrow-Turnbull, Duffie-Singleton and others) do not explain why default happens. They treat default as the first jump of a random process with an intensity (hazard rate) that is calibrated to market prices such as bond spreads and CDS quotes.
3. **Statistical and rating-based models** estimate default probabilities from historical data: credit scoring (Altman's Z-score, logistic regression), rating agency grades, and migration matrices that describe how ratings change over time.

The first is the most economically illuminating; the second is the most useful for pricing and hedging traded credit; the third is the workhorse of lending, regulation and accounting. Understanding the links among them is the goal of this chapter.

5.1.3 Real-world versus risk-neutral probabilities

A point that causes persistent confusion is that there are two kinds of default probability.

The **real-world (physical) PD** is the actual frequency with which obligors of a given quality default. It is what you need for provisioning, capital and underwriting: if you lend to 1,000 identical borrowers, the real-world PD tells you how many will fail.

The **risk-neutral (market-implied) PD** is the probability that, when used with risk-free discounting, reproduces observed market prices. Because investors demand compensation for bearing default risk, particularly the risk that defaults cluster in recessions when wealth is scarce, risk-neutral PDs are typically larger than real-world PDs, often by a factor of two or more for investment-grade credits. This is the credit analogue of the equity risk premium.

The practical consequences are important:

- Use risk-neutral probabilities (from spreads, CDS, Merton with the risk-free rate as drift) to *value and hedge* credit instruments consistently with the market.
- Use real-world probabilities (from histories, scorecards, Merton with the true asset drift) to *measure risk*: expected loss, capital, stress testing and impairment.
- Never compare a CDS-implied PD with a historical default rate and conclude that the market is "wrong"; a gap between them is mostly the risk premium.

> **Common mistake.** Reporting a CDS-implied default probability as a forecast. A 5-year CDS spread of 210 basis points does not mean the market expects roughly a 17% chance of default; it says what number, under risk-neutral pricing and an assumed recovery, is consistent with the quoted premium.

5.1.4 Why managers should care

For a CFO, credit models are not an academic exercise. They determine the pricing of loans and bonds the firm issues, the covenants and collateral lenders request, the counterparty limits the treasury sets for banks and customers, and the provisions recognised under IFRS 9 or CECL for trade receivables and financial assets. They also shape corporate strategy: a firm that understands that its equity volatility and leverage jointly determine its credit spread will see why a leveraged recapitalisation or a risky acquisition raises debt costs even when interest cover looks comfortable.

5.2 The Merton Structural Model

5.2.1 Equity as a call option, debt as a put

Robert Merton's 1974 insight was that limited liability turns a firm's capital structure into a pair of options on its assets. Consider the simplest firm: it has asset value V today, and a single zero-coupon debt issue with face value D that matures at T. There are no dividends and no other claims.

At maturity:

- If $V_T > D$, the firm repays the debt, and equity holders keep $V_T - D$.
- If $V_T < D$, the equity holders default, the lenders take the assets, and equity is worth zero.

Equity therefore pays $\max(V_T - D, 0)$, exactly the payoff of a European call option on the assets with strike D. Debt pays $\min(V_T, D) = D - \max(D - V_T, 0)$: a risk-free bond with face D, *minus* a put option on the assets with strike D. The debt holders have, in effect, sold a put to the shareholders.

Assume that asset value follows geometric Brownian motion with volatility σ and that the risk-free rate r is constant. Black-Scholes then gives:

```
E = V·N(d1) − D·e^(−rT)·N(d2)

d1 = [ ln(V/D) + (r + σ²/2)·T ] / (σ·√T)
d2 = d1 − σ·√T
```

and the market value of debt is simply B = V − E. Several quantities of interest follow immediately:

```
Risk-neutral PD        = N(−d2)
Yield on the debt      y = −(1/T)·ln(B / D)
Credit spread          s = y − r
Expected recovery (RN) = V·N(−d1) / N(−d2)      (as a fraction of V; divide by D for % of face)
```

The risk-neutral probability of default N(−d2) is the probability that $V_T < D$ under the risk-neutral measure, in which assets grow at r. To obtain a **real-world** PD, replace r by the true expected return on assets μ in the d2 formula:

```
Real-world PD = N( −[ ln(V/D) + (μ − σ²/2)·T ] / (σ·√T) )
```

The quantity inside the parentheses is the number of standard deviations that the expected log asset value is above the default barrier: the **distance to default**.

Example 5.1 – Merton valuation of a levered firm. A firm has asset value V = 100 (in $ millions), a single zero-coupon bond with face value D = 70 maturing in T = 1 year, asset volatility σ = 25%, and a risk-free rate r = 3% (continuously compounded). Find equity value, debt value, the risk-neutral PD, the credit spread, and the real-world PD if the expected asset return is μ = 8%.

Step 1: d1 and d2.

```
ln(V/D)        = ln(100/70) = 0.35667
(r + σ²/2)·T   = 0.03 + 0.03125 = 0.06125
d1 = (0.35667 + 0.06125) / 0.25 = 1.6717
d2 = 1.6717 − 0.25 = 1.4217
```

Step 2: normal probabilities. N(d1) = N(1.6717) = 0.9527; N(d2) = N(1.4217) = 0.9224.

Step 3: equity value (call).

```
PV of face   = 70 × e^(−0.03) = 70 × 0.97045 = 67.931
E = 100 × 0.9527 − 67.931 × 0.9224 = 95.271 − 62.663 = 32.608
```

Step 4: debt value. B = V − E = 100 − 32.608 = 67.392. A risk-free zero with the same face would be worth 67.931, so the default risk costs lenders 0.539, or 0.79% of the risk-free value.

Step 5: yield and spread.

```
y = −ln(67.392 / 70) = −ln(0.96274) = 3.797%
s = 3.797% − 3.000% = 0.797%  ≈ 80 basis points
```

Step 6: default probabilities. The risk-neutral PD is N(−1.4217) = 1 − 0.9224 = 7.76%. With μ = 8%:

```
DD = [ ln(100/70) + (0.08 − 0.03125) ] / 0.25 = (0.35667 + 0.04875)/0.25 = 1.6217
Real-world PD = N(−1.6217) = 5.24%
```

Interpretation. The market-implied (risk-neutral) default probability, 7.76%, exceeds the real-world probability, 5.24%, because the asset drift is lower in the risk-neutral world. The implied loss given default is the ratio of the risk-neutral expected loss to the risk-neutral PD: expected loss is (67.931 − 67.392)/67.931 = 0.79% of the risk-free value, divided by 7.76%, which gives about 10.2%. That is surprisingly low, and it is one of the model's well-known features: because the shortfall occurs only when assets are just below D in most paths, the average recovery is high.

5.2.2 What drives the spread

The Merton model yields rich comparative statics. Three variables matter most: leverage (D relative to V), asset volatility, and maturity.

Table 5.1 – Merton credit spreads (basis points) and risk-neutral PDs. Assumptions: V = 100, σ = 25%, r = 3%.

Leverage D/V	1-year spread	3-year spread	5-year spread	10-year spread	5-year RN PD
30%	0.0	1.1	5.4	17.5	1.6%
50%	2.0	29.4	48.2	63.5	11.0%
70%	79.7	142.4	144.7	127.1	26.5%
90%	506.3	353.6	282.5	198.9	43.0%

Three lessons emerge. First, spreads rise steeply and non-linearly with leverage: a move from 50% to 70% leverage multiplies the one-year spread forty-fold. Second, the term structure of spreads is **upward-sloping for low-leverage firms** (there is time for something to go wrong, and the starting point is far from the barrier) but **downward-sloping or humped for highly levered firms** (if the firm survives the near term, volatility gives it room to recover, so the conditional probability of default falls). This shape is consistent with empirical patterns: spreads for speculative-grade issuers are often flatter or inverted relative to investment grade. Third, the model's spreads for safe firms are tiny at short maturities, much lower than observed. This **short-spread puzzle** is a major empirical weakness: real short-dated bonds of good issuers trade at material spreads because of jump risk, liquidity premia and uncertainty about the asset value itself.

Volatility matters as much as leverage, a point managers often underappreciate. Holding V = 100, D = 70, T = 1 and r = 3%, the one-year spread is about 3 basis points at σ = 15%, 80 basis points at σ = 25%, 290 basis points at σ = 35% and 780 basis points at σ = 50%. Because equity is a call option, equity holders *gain* from higher asset volatility, and lenders lose. This is the analytical root of **asset substitution** (Volume 1, Chapter 15): equity-controlled firms near distress have an incentive to take on riskier projects, and lenders protect themselves with covenants and collateral.

5.2.3 Extensions

The classic Merton model is simplistic in several ways, and the literature has tried to repair each:

- **Default only at maturity.** In practice default can happen at any time, when covenants are breached or a payment is missed. Black and Cox (1976) introduced a **first-passage** model in which default occurs the first time asset value touches a barrier. This raises PDs and spreads at short horizons.
- **Single zero-coupon debt.** Real capital structures have coupons, multiple maturities and seniorities. Geske modelled sequential compound options; Longstaff and Schwartz allowed stochastic interest rates and an exogenous recovery rate.
- **Static capital structure.** Leland's model lets the firm choose leverage by trading off the tax shield against bankruptcy costs (Volume 1, Chapter 15) and endogenises the default barrier as the point at which equity holders optimally stop servicing debt.

- **Continuous diffusion.** With purely continuous paths a firm cannot default by surprise, so short-term spreads go to zero. Jump-diffusion models (Zhou) restore short-term spreads.
- **Unobservable assets.** Asset value and volatility are not quoted; they must be inferred from equity, which is the problem solved in the next section.

5.2.4 Managerial reading of the model

The Merton framework gives managers a way to think about how decisions change credit quality. Issuing debt to repurchase shares raises D relative to V and raises the spread. Selling a volatile division and using the proceeds to repay debt lowers σ and D simultaneously and compresses the spread sharply. A longer-dated issue is not necessarily riskier for a firm that is already stressed (the downward slope in Table 5.1 reflects that survival over the first years raises recovery prospects). And **a falling share price is a credit signal**: with D constant, a decline in E is a decline in V, so the distance to default shrinks even though no covenant has yet been tested.

5.3 Distance to Default and KMV-Style Models

5.3.1 Inferring asset value and volatility

Asset value and asset volatility are not observed. What is observed is the market capitalisation of equity E, its volatility σ_E (from historical returns or option-implied volatility), and the book value of liabilities. Because equity is a call on assets, we have two equations in two unknowns:

```
(1)  E  = V·N(d1) − D·e^(−rT)·N(d2)
(2)  σ_E·E = N(d1)·σ_V·V
```

Equation (2) follows from Itô's lemma applied to E = E(V): the volatility of equity equals the delta of the call multiplied by the volatility of assets and by the leverage ratio V/E. The system is solved numerically. A convenient algorithm iterates: start with V = E + D, σ_V = σ_E·E/V, compute d1 and d2, update V from equation (1) and σ_V from equation (2), and repeat until the values converge. Convergence is typically reached within ten to fifteen iterations.

Example 5.2 – Backing out asset value and volatility from equity data. A listed firm has market capitalisation E = 40, equity volatility σ_E = 60%, debt with face value D = 60 due in T = 1 year, and r = 3%. Estimate asset value, asset volatility, the risk-neutral PD and the spread.

Step 1: initial guess. V_0 = E + D·e^(−rT) = 40 + 60 × 0.97045 = 98.227; σ_V,0 = 0.60 × 40 / 98.227 = 24.43%.

Step 2: iterate. Each iteration recomputes d1 and d2 using the current V and σ_V, then solves (1) for V and (2) for σ_V. The iterates settle after about eleven passes at:

```
V = 98.108,  σ_V = 24.78%
d1 = 2.2291,  d2 = 1.9813
N(d1) = 0.9871,  N(d2) = 0.9762
```

Step 3: check. The call value is 98.108 × 0.9871 − 60 × 0.97045 × 0.9762 = 96.842 − 56.842 = 40.000, matching E. The volatility identity gives σ_E = 0.9871 × 0.2478 × 98.108 / 40 = 60.0%, matching the input.

Step 4: credit measures.

```
Risk-neutral PD = N(−1.9813) = 2.38%
Debt value B   = V − E = 98.108 − 40 = 58.108
Yield y        = −ln(58.108/60) = 3.204%
Spread         = 3.204% − 3.000% = 20.4 bp
```

Interpretation. Equity volatility of 60% looks alarming, but the firm's asset volatility is only about 25% because leverage amplifies it: roughly 98/40 = 2.45 times, scaled down by the delta of 0.987. The implied credit spread of about 20 basis points shows the model's restraint at short maturities. A practitioner would not take that number literally; she would use the *ordering* across names and the *change over time*.

5.3.2 Distance to default

Once V and σ_V are known, the **distance to default** compresses the firm's credit state into a single number of standard deviations:

```
DD = [ ln(V / DP) + (μ − σ_V²/2)·T ] / (σ_V·√T)
```

where DP is the **default point**. Moody's KMV, which commercialised this approach, observed that firms do not default exactly when assets fall below total liabilities. Long-term debt can be rolled over and short-term debt must be paid, so the empirical default point lies between the two. KMV's rule of thumb is:

```
Default point = short-term debt + 0.5 × long-term debt
```

A simplified, widely used version treats asset value as approximately normal over the horizon and writes:

```
DD ≈ (V − DP) / (V·σ_V)
```

The distance to default is a **ranking** statistic, and this is its great strength. It combines market value (a forward-looking quantity that reflects expectations of earnings and investment), leverage and volatility into one comparable measure across firms and sectors.

5.3.3 From distance to default to expected default frequency

If asset values were normally distributed, DD would map directly to PD via N(−DD). A DD of 4 would imply a probability of roughly 0.003%, which is far lower than actual default frequencies for firms at that distance. KMV therefore maps DD to default probability **empirically**: it looks at a large historical database, groups firms by DD, and records the fraction that defaulted within the next year. The resulting **expected default frequency (EDF)** is not normal. It is fat-tailed, reflecting jumps, fraud, event risk and non-normality of asset returns.

The distinction between a theoretical (Gaussian) PD and an empirical EDF deserves emphasis. The Gaussian mapping is acceptable for ranking but badly underestimates default risk at large distances. An empirical mapping fixes the level but is only as good as the history on which it was built, and it is proprietary or calibrated by each institution.

Example 5.3 – Distance to default and its mapping. A manufacturer has asset value V = 150, asset volatility σ_V = 20%, short-term debt of 40 and long-term debt of 60. The expected asset return is 8% and the horizon is one year.

Step 1: default point.

```
DP = 40 + 0.5 × 60 = 70
```

Step 2: simple DD.

```
DD = (150 − 70) / (150 × 0.20) = 80 / 30 = 2.67
Gaussian PD = N(−2.67) = 0.38%
```

Step 3: log version with drift.

```
ln(150/70) = 0.7621
numerator  = 0.7621 + (0.08 − 0.02) = 0.8221
DD = 0.8221 / 0.20 = 4.11
Gaussian PD = N(−4.11) = 0.002%
```

Step 4: reading the difference. The two definitions give very different pictures (2.67 versus 4.11). The simple version measures distance in absolute value terms (the asset base must fall 53% of its current value to reach 70), while the log version works in return space, where falling from 150 to 70 is a log return of −76%, but with positive drift and the $-\sigma^2/2$ correction shifting the centre. Neither number is a "true" probability. In practice an institution would map the DD to an EDF using its own default history: a DD of about 2.7 to 4 might correspond to an empirical annual default frequency of perhaps a few tenths of one percent to roughly one percent, many times larger than the Gaussian figure for the upper end.

Managerial lesson. The ranking and the *direction of change* are what the figure delivers reliably. If the share price falls 30% and volatility rises, the DD shrinks; a risk manager would use the change as an early-warning trigger long before an agency downgrade.

5.3.4 Strengths and limitations of structural models

Strengths. They are forward-looking (equity prices respond within seconds); they link credit to capital structure and to equity volatility; they produce a continuous measure rather than coarse rating buckets; and they explain why credit and equity markets move together.

Limitations. They depend on equity prices and so are unavailable for private firms (although proxies using comparable firms are used); they are sensitive to market sentiment and to bubbles in equity; they assume a simple liability structure; the book value of liabilities may be a poor proxy for the default barrier (off-balance-sheet obligations, covenants, parent support); and the calibration of asset volatility is noisy. Their calculated spreads, as shown above, are typically too low for high-quality issuers over short horizons.

> **Common mistake.** Using the Merton PD in a regulatory or provisioning context without an empirical mapping. The theoretical Gaussian PD is not a calibrated default probability.

5.4 Reduced-Form Models: Hazard Rates and Survival

5.4.1 Default as a surprise

Structural models make default predictable in the sense that a continuously moving asset value approaches a known barrier. Reduced-form models take the opposite view: default is a **surprise**, the first arrival of a Poisson-type event whose intensity may vary with the economy and the firm's condition. The modeller does not need balance-sheet detail; she needs a process for the intensity and a recovery assumption, and she calibrates both to market prices.

Let τ be the (random) default time and define the **hazard rate** (default intensity) $\lambda(t)$ as the instantaneous conditional probability of default per unit time, given survival to t:

```
λ(t)·dt = P( default in [t, t+dt] | survival to t )
```

The probability of surviving to time T is then

```
S(T) = P(τ > T) = exp( − ∫₀ᵀ λ(u) du )
```

and the cumulative default probability is PD(T) = 1 − S(T). With a **constant** hazard rate λ the formula simplifies to S(T) = e^(−λT). Hazard rates are analogous to forward interest rates: the integral of the hazard is to survival probability what the integral of the short rate is to a discount factor. This analogy is not accidental. A risky zero-coupon bond can be priced by discounting at a "default-adjusted" rate, which is what makes the approach so convenient for traded credit.

Two quantities are easily confused. The **unconditional (marginal) default probability** for year k is the chance of defaulting in year k as seen from today. The **conditional default probability** is the chance of defaulting in year k given survival to its start. With a constant hazard rate, the conditional probability is the same every year, while the marginal probability declines gradually, because the firm must first survive.

5.4.2 The credit triangle

In the Duffie-Singleton framework with a fractional recovery of market value, the price of a risky zero-coupon bond is

```
P_risky(0, T) = P_riskfree(0, T) × exp( − ∫₀ᵀ λ(u)·(1 − R) du )
```

so that the credit spread is, for a constant hazard rate and recovery R,

```
s = λ × (1 − R) = λ × LGD
```

or, rearranged, **λ = s / (1 − R)**. This relation, sometimes called the **credit triangle**, is the most-used rule of thumb in credit markets. It states that the spread is the annual expected loss rate under the risk-neutral measure. A 180 basis point spread with 40% recovery implies a hazard rate of 0.0180 / 0.60 = 3.0% per annum.

Example 5.4 – Hazard rates and survival from spreads. A five-year par credit spread for a corporate issuer is 180 basis points, and the market convention for recovery is 40%.

Step 1: hazard rate.

```
λ = 0.0180 / (1 − 0.40) = 0.0180 / 0.60 = 3.00% per year
```

Step 2: survival and default probabilities.

```
S(1) = e^(−0.03×1) = 0.9704  → PD(1) = 2.96%
S(3) = e^(−0.03×3) = 0.9139  → PD(3) = 8.61%
S(5) = e^(−0.03×5) = 0.8607  → PD(5) = 13.93%
```

Step 3: marginal PDs. The one-year marginal probability is 2.96%, whereas the marginal probability in year 5 is S(4) − S(5) = e^(−0.12) − e^(−0.15) = 0.8869 − 0.8607 = 2.62%. The *conditional* probability in every year is 1 − e^(−0.03) = 2.96%.

Step 4: a term structure. Suppose instead the market quotes par spreads of 100 bp (1-year), 150 bp (3-year) and 200 bp (5-year), again with 40% recovery. Using the credit-triangle approximation, the *average* hazard to horizon T is s_T / (1 − R): 1.67% to year 1, 2.50% to year 3 and 3.33% to year 5. The cumulative hazard H(T) is the average hazard times T:

```
H(1) = 0.0167 × 1 = 0.0167
H(3) = 0.0250 × 3 = 0.0750
H(5) = 0.0333 × 5 = 0.1667
```

The implied *forward* (piecewise-constant) hazards between nodes are:

```
Years 0–1: 0.0167 / 1                 = 1.67%
Years 1–3: (0.0750 − 0.0167) / 2      = 2.92%
Years 3–5: (0.1667 − 0.0750) / 2      = 4.58%
```

and the survival probabilities are S(1) = e^(−0.0167) = 98.35%, S(3) = e^(−0.0750) = 92.77% and S(5) = e^(−0.1667) = 84.65%. The upward-sloping spread curve translates into hazard rates that *rise* over time: the market expects, in the risk-neutral sense, that default risk is back-loaded. A full bootstrap would solve the CDS pricing equation at each maturity rather than use the triangle; that is the subject of Section 5.5, and the approximation is accurate to within a few percent of the hazard for moderate spreads.

Interpretation. The hazard-rate curve plays the role for credit that the forward-rate curve plays for interest rates. It lets you value any cash flow that depends on survival (a risky bond, a loan, a trade receivable) consistently with the market.

5.4.3 Pricing a risky bond

If recovery is a fraction R of face value paid at default (the Jarrow-Turnbull convention), the price of a coupon bond is the sum of survival-weighted coupons and principal plus the discounted expected recovery. Under the simpler recovery-of-market-value assumption of Duffie and Singleton, the bond can be discounted at the *risky rate* r + λ(1 − R):

```
Price = Σ CF_t × exp( −(r + λ(1 − R))·t )
```

For a three-year zero-coupon bond with face value 100, a risk-free rate of 4%, a hazard rate of 3% and recovery of 40%, the risky rate is 0.04 + 0.03 × 0.60 = 5.8%, and the price is 100 × e^(−0.058 × 3) = 100 × e^(−0.174) = 84.03, compared with the risk-free price 100 × e^(−0.12) = 88.69. The difference of 4.66 is the market price of the expected credit loss.

5.4.4 Stochastic intensities and correlation

A constant hazard rate is a first approximation. Real credit spreads move randomly and, importantly, they move together. Models therefore make λ a stochastic process: a **Cox process** (a Poisson process with a random, non-defaultable intensity) with mean-reverting dynamics such as a CIR (square-root) process, correlated with interest rates and a market factor. The Cox structure keeps survival probabilities in closed form for affine intensities. Other extensions add jumps in the intensity (a downgrade event) or link intensity to equity prices and macro variables.

Correlation among obligors enters through the common factors in their intensities. The basic problem is that default correlation is difficult to observe, since defaults are rare. This difficulty, and the fact that tranche prices depend almost entirely on correlation assumptions, will matter greatly in Chapter 6.

5.4.5 Structural versus reduced-form: how to choose

Feature	Structural (Merton/KMV)	Reduced-form (intensity)
Source of default	Asset value falls below liabilities	Exogenous Poisson-type surprise
Main inputs	Equity price, equity vol, leverage	Bond/CDS spreads, recovery
Default predictable?	Yes (approaches a barrier)	No (a sudden event)
Short-term spreads	Too low without jumps	Naturally positive
Best use	Credit monitoring, early warning, linking to capital structure	Pricing and hedging of traded credit
Data needed	Listed equity	Liquid credit market
Main weakness	Needs unobservable asset values; model risk in barrier	No economic story; calibrates but does not explain

The two approaches converge in "hybrid" models, which make the hazard rate a function of equity value or distance to default. A practical risk manager might use structural distance to default for a private-company proxy, reduced-form hazard rates to value a CDS book, and rating-based statistics to provision a loan portfolio.

5.5 Credit Default Swaps

5.5.1 Contract mechanics

A **credit default swap (CDS)** is a bilateral contract in which the protection buyer pays a periodic premium (the CDS spread or coupon) to the protection seller until maturity or a credit event. If a credit event occurs on the reference entity (bankruptcy, failure to pay and, for some contracts, restructuring), the seller compensates the buyer for the loss: under physical settlement the buyer delivers a defaulted bond with face value equal to the notional and receives par; under cash settlement, an auction determines the market value of the defaulted debt and the seller pays notional × (1 − auction recovery).

Standard conventions in the single-name market include quarterly premium payments, fixed standard coupons of 100 or 500 basis points paid with an **upfront payment** that equates the contract's value to the quoted spread, a five-year benchmark maturity, and a 40% recovery assumption for senior unsecured debt (25% for subordinated) when converting between spread and upfront. The standard contract is an insurance-like contract with no requirement that the buyer holds the underlying debt, which means CDS can be used for hedging or for speculation and gives rise to a market whose notional can exceed the underlying debt outstanding.

5.5.2 Valuation

A CDS has two legs whose present values must be equal at inception for a par contract:

- **Premium leg (fee leg):** the buyer pays spread s on the notional each period, as long as the entity survives. There is also an *accrued premium* if default occurs between payment dates.
- **Protection leg (contingent leg):** the seller pays (1 − R) × notional at default.

Define the **risky annuity** (often called RPV01 or "risky PV01"), the present value of 1 per annum paid on the notional until default or maturity:

```
RPV01 = Σ  Δ × DF(t_i) × S(t_i)  +  accrual term
```

where Δ is the accrual fraction, DF the discount factor and S the survival probability. The protection leg is

```
Protection = (1 − R) × Σ  DF(t_i) × [ S(t_{i−1}) − S(t_i) ]
```

(an approximation assuming default occurs mid-interval, with discounting at the midpoint). The par spread equates the two legs:

```
s_par = Protection / RPV01
```

If the contract trades with a fixed coupon c rather than at par, the **upfront payment** (as a fraction of notional, from the buyer's perspective) is

```
Upfront = Protection − c × RPV01  =  (s_par − c) × RPV01
```

and the **mark-to-market** of an existing position when the spread moves from s_0 to s_1 is approximately $(s_1 - s_0) \times \text{RPV01}$ to the protection buyer.

Example 5.5 – Pricing a five-year CDS. A reference entity has a flat hazard rate λ = 3.0% per year, recovery R = 40%, and the continuously compounded risk-free rate is 3%. Premiums are quarterly. Find the par spread, the upfront payment on a contract with a 100 bp coupon, and the mark-to-market impact of spread moves.

Step 1: survival and discount factors. For quarter i with t = i/4, S(t) = e^(−0.03t) and DF(t) = e^(−0.03t). The marginal default probability in each quarter is S(t − 0.25) − S(t), which for the first quarter is 1 − e^(−0.0075) = 0.7472%.

Step 2: risky annuity. Summing the 20 quarterly terms 0.25 × DF × S plus half-period accrual for default within the period gives

```
RPV01 = 4.3036 (per unit of notional per year)
```

For comparison, a *risk-free* 5-year annuity with continuous 3% discounting and quarterly payments would be about 4.63, so survival risk reduces its value by 7%.

Step 3: protection leg.

```
Protection = (1 − 0.40) × Σ DF(t_mid) × [S(t−0.25) − S(t)] = 0.0778 (7.78% of notional)
```

Roughly: the five-year cumulative default probability is 13.93%; discounting at about 3% over an average life of 2.5 years and multiplying by LGD of 60% gives 0.1393 × 0.60 × 0.93 ≈ 0.078.

Step 4: par spread.

```
s_par = 0.07775 / 4.3036 = 1.807%  ≈ 181 bp
```

This is close to the credit-triangle value λ(1 − R) = 3.0% × 0.60 = 1.80%, confirming that the triangle is accurate for flat curves.

Step 5: upfront for a 100 bp coupon.

```
Upfront = 0.07775 − 0.0100 × 4.3036 = 0.07775 − 0.04304 = 3.47% of notional
```

The protection buyer pays 3.47 points upfront and then 100 bp per year, because the standard coupon is lower than the fair spread.

Step 6: mark-to-market. A buyer who bought protection when the spread was 100 bp earns, if the spread widens to 180 bp, approximately $(0.0180 - 0.0100) \times 4.30 = 3.44\%$ of notional. If the spread widens a further 100 bp the gain grows by roughly 100 bp × 4.3 = 4.3% of notional (a bit less in practice, because the RPV01 shrinks as survival probability falls).

Interpretation. The CDS spread's sensitivity to one basis point, the **CS01**, is approximately the RPV01 times 0.0001, i.e. about 0.043% of notional, or $4,300 on a $10 million position. That is the unit of risk for traders and for hedging a loan book.

5.5.3 Bond-CDS basis and practical issues

In an ideal world the CDS spread equals the bond's credit spread over the risk-free (swap) rate. The difference, the **CDS-bond basis**, is typically non-zero and can be large in stress. A negative basis (CDS spread below bond spread) can arise when bond liquidity is poor or funding costs for holding the bond are high; a positive basis can arise when the cheapest-to-deliver option or counterparty considerations raise the CDS price relative to the cash bond. A risk manager must also consider:

- **Counterparty risk.** The seller may default at the same time as the reference entity (wrong-way risk), which is why standardised contracts are cleared through central counterparties and margined.
- **Restructuring and definitions.** Not all restructurings trigger payment; settlement terms matter.
- **Index products.** Indices of 100 to 125 equally weighted names trade as standardised contracts; the index spread is the average (adjusted for the convexity of survival) of the constituents.
- **Recovery assumptions.** Quoting conventions fix recovery, but actual recovery is revealed only through the auction.

5.5.4 What CDS markets tell the CFO

CDS spreads on a company's own debt are a continuously updated measure of its marginal cost of unsecured funding for a given tenor. They are also a basis for counterparty valuation adjustments in derivative books, a cross-check against rating-based expectations, and, at the extremes, an early signal that bond and loan investors are repricing. A company with a CDS spread that diverges from peers with the same rating is being judged on something the rating does not capture.

5.6 Recovery and Loss Given Default

5.6.1 What determines recovery

Loss given default is as important as default probability, yet it receives less attention. A bond with PD of 2% and LGD of 80% has an expected loss of 1.6%; a secured loan to the same borrower with LGD of 25% has an expected loss of 0.5%. In practice recovery depends on:

- **Seniority and security.** Senior secured creditors recover more than senior unsecured, subordinated and equity holders, in order of the absolute priority of claims (although in actual bankruptcy proceedings that priority is sometimes bent).
- **Collateral type and coverage.** Liquid, easily valued collateral (cash, receivables, standard equipment) holds value better than specialised assets. Coverage should be measured at a stressed value, not at the origination valuation.
- **Capital structure.** The more debt ranks ahead or alongside, the less is available per dollar of claim. A firm with a thin cushion of junior debt gives senior lenders less protection.

- **Industry and the cycle.** Industries with tangible, fungible assets (utilities, shipping at a good moment, real estate) tend to show higher recoveries than asset-light services. Recovery rates decline in recessions, when many firms fail together and asset buyers are scarce.
- **Resolution process.** Time to resolution, legal system efficiency and whether the firm is reorganised or liquidated affect both the amount and the present value of the recovery.

Table 5.2 – Illustrative recovery assumptions by claim type (percent of face value).

Claim	Typical modelling assumption	Comment
First-lien senior secured loan	55–70%	Highest priority; depends on collateral coverage
Second-lien loan	30–45%	Subordinated to first-lien claims on collateral
Senior unsecured bond	35–45%	Market convention of 40% in CDS
Subordinated bond	15–25%	Convention of 25% for subordinated CDS
Junior or preferred	0–10%	Often near zero

These ranges are illustrative modelling conventions, not statistics. An institution should calibrate its own LGD estimates from its workout experience.

5.6.2 Modelling issues

Definition of recovery. Three definitions coexist: the price of the defaulted bond shortly after default (market recovery), the discounted value of cash and securities ultimately received (ultimate recovery), and the recovery rate used by the CDS auction. They can differ substantially. IFRS 9 and regulatory models focus on ultimate, discounted recovery net of workout costs.

Stochastic and correlated recovery. A model that fixes recovery at 40% ignores the fact that recovery is uncertain and, more importantly, **negatively correlated with default rates**: when more firms default, recoveries are lower. This correlation amplifies portfolio losses in recessions relative to a model with constant LGD. It is one reason regulatory frameworks require a *downturn* LGD, an estimate for adverse conditions rather than average conditions.

Sensitivity of inferred hazard to the recovery assumption. Because the spread identifies only the product $\lambda \times (1 - R)$, the implied default probability depends heavily on the recovery assumption. Take the 180 bp five-year spread of Example 5.4:

Assumed recovery	Implied hazard λ	Implied 5-year PD
20%	2.25%	10.64%
40%	3.00%	13.93%
60%	4.50%	20.15%

Moving recovery from 20% to 60% nearly doubles the implied default probability even though the spread has not changed. A CDS spread therefore measures expected loss, not default probability, and a statement about PD derived from a spread must always come with the recovery assumption attached.

5.6.3 Managerial implications

Collateral and covenants do their work through LGD. For a corporate borrower the cost of secured funding is lower than that of unsecured funding because LGD is lower, but pledging assets to one lender raises the LGD, and hence the spread, of every other creditor. This is the mechanism that makes secured debt capacity a finite resource and explains why lenders write negative-pledge clauses. When negotiating a credit facility the CFO should therefore ask not only "what is my PD?" but "what is the lender's LGD, and what in the structure changes it?"

5.7 Credit Migration and Rating Models

5.7.1 Ratings and transition matrices

Rating agencies and bank internal rating systems assign obligors to a small number of grades. Because ratings change over time, a **transition (migration) matrix** records the probability that an obligor in grade i at the start of the year is in grade j at the end. The last grade is default and is an *absorbing state*: once defaulted, always defaulted. Transition matrices serve three purposes: they give PD term structures (via matrix multiplication), they allow valuation and risk measurement of credit deterioration short of default (spread widening after a downgrade causes mark-to-market losses on bonds), and they are inputs to portfolio models such as CreditMetrics.

If P is the one-year transition matrix and migrations follow a time-homogeneous Markov chain, the k-year transition matrix is P^k. The cumulative default probability for an obligor in grade i over k years is the (i, default) element of P^k.

Example 5.6 – Multi-year default probabilities from a transition matrix. An internal rating system has five states: Strong, Good, Fair, Weak and Default. The one-year transition matrix (illustrative) is:

Table 5.3 – One-year transition matrix (illustrative; rows sum to 100%).

From To	Strong	Good	Fair	Weak	Default
Strong	92.0%	6.0%	1.5%	0.4%	0.1%
Good	5.0%	88.0%	5.0%	1.5%	0.5%
Fair	1.0%	7.0%	80.0%	7.0%	5.0%
Weak	0.0%	2.0%	10.0%	68.0%	20.0%
Default	0	0	0	0	100%

Find the cumulative default probability over one to five years for an obligor currently rated "Fair", and the marginal and conditional annual PDs.

Step 1: one-year default. Directly from the matrix, PD(1) = 5.00%.

Step 2: two-year default. The (Fair, Default) element of P^2 is the sum over intermediate states of P(Fair → k) × P(k → Default):

```
PD(2) = 0.01×0.001 + 0.07×0.005 + 0.80×0.05 + 0.07×0.20 + 0.05×1.00
      = 0.00001 + 0.00035 + 0.0400 + 0.0140 + 0.0500
      = 0.10436 = 10.44%
```

Step 3: continue the matrix powers. Computing P^3, P^4 and P^5 numerically gives the cumulative default probabilities for a Fair obligor:

Horizon (years)	Cumulative PD	Marginal PD	Conditional PD
1	5.00%	5.00%	5.00%
2	10.44%	5.44%	5.72%
3	15.85%	5.41%	6.04%
4	20.98%	5.14%	6.11%
5	25.74%	4.76%	6.02%

Marginal PD is the increment of cumulative PD; conditional PD is the marginal PD divided by the survival probability at the start of the year (for year 2: 5.44% / 95.00% = 5.72%).

Step 4: other starting grades. The five-year cumulative default probabilities from the other grades are 2.23% (Strong), 6.38% (Good) and 57.38% (Weak). A Weak obligor has a one-year PD of 20% and a five-year PD above 57%, while for a Strong obligor, a five-year PD above 2% shows that even high-quality credits can drift down and default.

Interpretation. The conditional PD for a Fair obligor *rises* in years 2 to 4 (5.00% → 6.11%) before easing. Part of the reason is that surviving Fair obligors drift toward the Weak grade, from which default is likely. Higher-quality obligors typically show a rising hazard (they migrate down), and low-quality obligors a falling hazard (survivors are the stronger ones), which echoes the Merton term-structure pattern of Table 5.1.

Limitations. The Markov assumption is only approximate: empirically, ratings show momentum (a downgraded firm is more likely to be downgraded again) and dependence on the business cycle. Annual matrices differ markedly between recessions and expansions. Matrices estimated from agency data refer to rated corporates and may not be representative of a particular bank's middle-market loans. Rare transitions, such as Strong to Default, are estimated with very few observations. Finally, ratings are "through-the-cycle" in philosophy at the agencies, so they lag the market and so differ from point-in-time measures such as distance to default.

5.7.2 Credit scoring: the Altman Z-score

Edward Altman's Z-score (1968) was one of the first multivariate credit models. Using multiple discriminant analysis on a sample of manufacturing firms that went bankrupt and a matched sample that did not, he found a linear combination of five financial ratios that best separated them:

```
Z = 1.2·X1 + 1.4·X2 + 3.3·X3 + 0.6·X4 + 1.0·X5

X1 = Working capital / Total assets
X2 = Retained earnings / Total assets
X3 = EBIT / Total assets
X4 = Market value of equity / Book value of total liabilities
X5 = Sales / Total assets
```

Conventional zones for the original public manufacturer model are Z > 2.99 ("safe"), 1.81 to 2.99 ("grey") and Z < 1.81 ("distress"). Variants exist for private firms, non-manufacturers and emerging markets with different coefficients and cut-offs.

A firm has X1 = 0.20, X2 = 0.15, X3 = 0.08, X4 = 0.90 and X5 = 1.10. Then Z = 1.2 × 0.20 + 1.4 × 0.15 + 3.3 × 0.08 + 0.6 × 0.90 + 1.0 × 1.10 = 0.240 + 0.210 + 0.264 + 0.540 + 1.100 = 2.354, in the grey zone. The score is a ranking and a screening device; the coefficients are specific to the sample, period and industry mix, and a score near a cut-off has little information.

5.7.3 Logistic regression scorecards

Modern scorecards typically use **logistic regression**, which models the log-odds of default as a linear function of explanatory variables:

```
ln( PD / (1 − PD) ) = β₀ + β₁x₁ + β₂x₂ + … + βₖxₖ
PD = 1 / (1 + e^(−z)),   z = β₀ + Σ βᵢxᵢ
```

Logistic regression is preferred to discriminant analysis because it produces a probability directly, makes weaker distributional assumptions and has interpretable coefficients: each unit of x_i multiplies the odds of default by e^{β_i}.

Example 5.7 – A logistic scorecard. A bank's scorecard for small corporates is

```
z = −3.0 + 4.0 × (Debt/Assets) − 5.0 × (EBIT/Assets) − 2.5 × (Operating cash flow / Debt)
```

Compute the PD for a borrower with Debt/Assets = 0.55, EBIT/Assets = 0.12 and OCF/Debt = 0.30, and the effect of deleveraging to 0.40.

Step 1: score.

```
z = −3.0 + 4.0×0.55 − 5.0×0.12 − 2.5×0.30
  = −3.0 + 2.20 − 0.60 − 0.75 = −2.15
```

Step 2: PD.

```
PD = 1 / (1 + e^(2.15)) = 1 / (1 + 8.585) = 10.43%
```

Step 3: deleveraging. With Debt/Assets = 0.40, z = −3.0 + 1.60 − 0.60 − 0.75 = −2.75 and PD = 1 / (1 + e^(2.75)) = 1 / (1 + 15.64) = 6.01%.

Step 4: odds interpretation. The coefficient 4.0 means that a 10 percentage-point increase in leverage adds 0.4 to the log odds, multiplying the odds by e^0.4 = 1.49. Reducing leverage by 15 points (0.55 to 0.40) cuts the log odds by 0.60, i.e. multiplies the odds of default by e^(−0.60) = 0.549: from odds of 0.1165 (10.43%) to 0.0640 (6.01%).

5.7.4 Judging a rating model

A model's quality is assessed along three dimensions:

- **Discrimination** (rank ordering): do defaulters receive worse scores than non-defaulters? Measured by the ROC curve, the area under it (AUC) and the Gini coefficient or accuracy ratio (Gini = 2 × AUC − 1). An AUC of 0.5 is no better than random; modern retail and corporate scorecards often score between 0.7 and 0.85.
- **Calibration:** does a predicted PD of 3% correspond to roughly 3% observed defaults? A model can discriminate well and be badly calibrated, because the base rate of default changes with the cycle.
- **Stability and monotonicity:** does the model behave sensibly out of sample, across the cycle and for sub-segments?

Backtesting should compare predicted and realised defaults by grade, using binomial tests with allowance for default correlation: with positively correlated defaults, a year in which realised defaults are twice the prediction is much less improbable than an independent-default test suggests. **Overfitting** is the chief enemy: a scorecard with many variables fitted to a small number of defaults will look excellent on the development sample and fail in use. Good practice includes out-of-time validation, parsimony, expert override with documented rationale, and independent model validation.

5.8 Expected Loss, Unexpected Loss and the Use of Models

5.8.1 Expected loss for a portfolio

Expected loss is additive across exposures:

```
EL_portfolio = Σ PDᵢ × LGDᵢ × EADᵢ
```

for a portfolio of obligors, regardless of correlation. Expected loss is a *cost* priced into loans and provisioned against. Under IFRS 9 it appears as the **expected credit loss (ECL)** impairment allowance. For assets whose credit risk has not increased significantly since initial recognition, the allowance equals the 12-month ECL; for assets with a significant increase in credit risk, the allowance equals the **lifetime ECL**. Lifetime ECL discounts expected losses at the effective interest rate:

```
ECL = Σₜ  marginal PD(t) × LGD(t) × EAD(t) × DF(t)
```

Example 5.8 – 12-month versus lifetime ECL. A three-year bullet loan of $10 million has a hazard rate of 2.0%, LGD of 45%, and a discount rate (effective interest rate) of 5%. Defaults are assumed to occur mid-year, and EAD remains $10 million.

Step 1: marginal default probabilities. For year t, marginal PD = e^(−0.02(t−1)) − e^(−0.02t):

```
Year 1: 1 − 0.98020      = 1.980%
Year 2: 0.98020 − 0.96079 = 1.941%
Year 3: 0.96079 − 0.94176 = 1.902%
```

Step 2: discount factors at the mid-year, e^(−0.05(t − 0.5)). Year 1: 0.9753; year 2: 0.9277; year 3: 0.8825.

Step 3: ECL by year.

```
Year 1: 0.01980 × 0.45 × 10,000,000 × 0.9753 = $86,906
Year 2: 0.01941 × 0.45 × 10,000,000 × 0.9277 = $81,031
Year 3: 0.01902 × 0.45 × 10,000,000 × 0.8825 = $75,552
```

Step 4: totals. The 12-month ECL (Stage 1) is $86,906. The lifetime ECL (Stage 2) is $86,906 + $81,031 + $75,552 = $243,489. The step-up from Stage 1 to Stage 2 is therefore a 2.8× jump in the allowance, even though the hazard rate has not changed in this example; in reality a significant increase in credit risk would also raise the hazard.

Interpretation. The "cliff" between stages creates earnings volatility when assets transfer, and an analyst should anticipate it: when a sizeable portfolio migrates, the provision charge reflects the longer lifetime horizon as well as the higher PD.

5.8.2 Unexpected loss and economic capital

Actual losses in any year differ from the expected value. The **unexpected loss (UL)** is typically measured as the standard deviation of losses, and **economic capital** is the amount needed to absorb losses at a high confidence level (say 99.9%) minus the expected loss. For a single exposure with LGD fixed, the standard deviation of loss is

```
σ_L = EAD × LGD × √( PD × (1 − PD) )
```

but portfolio losses depend critically on **default correlation**. The standard model behind regulatory credit capital is the Vasicek **asymptotic single risk factor (ASRF)** model. Each obligor's asset return is $Z_i = \sqrt{\rho} \cdot M + \sqrt{1-\rho} \cdot \varepsilon_i$, where M is a systematic factor, ε_i an idiosyncratic shock and ρ the asset correlation. Obligor i defaults if Z_i falls below the threshold $\Phi^{-1}(\text{PD})$. Conditional on the systematic factor M = m, the default probability is

```
PD(m) = Φ( [ Φ⁻¹(PD) − √ρ · m ] / √(1 − ρ) )
```

In a very large, granular portfolio the idiosyncratic risk diversifies away and the loss rate equals PD(m). The default rate that is exceeded only with probability 1 − q is obtained by setting m = $\Phi^{-1}(1-q)$, equivalently using $+\Phi^{-1}(q)$:

```
Worst-case default rate (confidence q) = Φ( [ Φ⁻¹(PD) + √ρ · Φ⁻¹(q) ] / √(1 − ρ) )
```

The unexpected loss in capital terms is (worst-case default rate − PD) × LGD × EAD.

Example 5.9 – Expected loss, capital and risk-based pricing. A bank makes a $10 million loan with PD = 2%, LGD = 45%, asset correlation ρ = 15%. Use a 99.9% confidence level. The bank's cost of equity is 12%, capital earns a risk-free benefit of 3% (it reduces funding needs), and operating cost is 0.30% of the loan.

Step 1: expected loss.

```
EL = 0.02 × 0.45 × 10,000,000 = $90,000   (0.90% of EAD)
```

Step 2: worst-case default rate. $\Phi^{-1}(0.02) = -2.0537$; $\Phi^{-1}(0.999) = 3.0902$; $\sqrt{\rho} = 0.3873$; $\sqrt{1-\rho} = 0.9220$.

```
numerator = −2.0537 + 0.3873 × 3.0902 = −2.0537 + 1.1969 = −0.8568
ratio     = −0.8568 / 0.9220 = −0.9293
Worst-case default rate = Φ(−0.9293) = 17.63%
```

Step 3: unexpected loss (capital).

```
Capital = (0.1763 − 0.0200) × 0.45 × 10,000,000 = $703,480   (7.03% of EAD)
```

Step 4: risk-based pricing. The required margin must cover expected loss, the net cost of capital and operating cost:

```
EL                   = 0.900%
Capital charge       = 7.035% × (12% − 3%) = 0.633%
Operating cost       = 0.300%
Required spread over funding ≈ 1.833%
```

Interpretation. Three-quarters of the "risk premium" in this spread is expected loss and cost; only the capital charge compensates for bearing correlated risk. If the borrower's industry has a higher asset correlation (say ρ = 25%), the worst-case default rate and the capital requirement both rise, and so does the price, even with PD and LGD unchanged. Two borrowers with identical PDs can therefore deserve different pricing if their exposures are more systematic.

5.8.3 Portfolio view

Expected loss is additive; unexpected loss is not. Adding a loan to a portfolio raises capital by less than its stand-alone UL if its default risk is not perfectly correlated with the rest. This is why diversification across industries, geographies and obligors reduces the capital per loan, and why concentration limits matter. The **marginal** contribution to portfolio risk, not the stand-alone risk, is what should drive pricing and limits. Chapter 6 develops the same logic for pools of loans held in securitisation vehicles, where the tranching of the loss distribution makes correlation the dominant driver of value.

5.8.4 Model risk and good practice

Credit models inform decisions; they do not substitute for judgement. Practitioners guard against model risk by:

- Using several models (structural, market-implied, rating-based) and investigating disagreements rather than averaging them away.
- Distinguishing risk-neutral from real-world outputs, and never mixing them in one calculation without adjustment.
- Stress testing: shocking PDs, LGDs and correlations together, since in a crisis they all move in the same adverse direction.
- Documenting assumptions (recovery, correlation, horizon, default definition), which are usually more important than the mathematics.
- Recognising that data are sparse: defaults are rare events, so estimates of PD in the safest grades and of correlation are statistically fragile.

Case Study: Redwater Components and the Refinancing Decision

(Illustrative case; Redwater Components is a fictional company and all figures are invented for teaching purposes.)

Redwater Components is a privately held automotive parts supplier with annual sales of $620 million and EBITDA of $74 million. It carries $210 million of debt: a $90 million secured term loan due in two years, a $70 million senior unsecured note due in four years and a $50 million revolving facility, fully drawn. Redwater's owners want to refinance all three into a single $230 million five-year facility and to pay a $20 million dividend from the proceeds. Their CFO, Imran Okafor, has asked the company's relationship bank for terms and has also approached two institutional lenders.

The bank's credit team compares several models. A scorecard based on leverage (debt/EBITDA of 2.8× before the transaction, 3.1× after), interest cover and size places Redwater at an internal rating equivalent to "Fair" in the matrix of Table 5.3 before the deal and "Weak-to-Fair" after. The one-year PD is 5.0% before the transaction, and the team estimates 7.5% afterward. Redwater has no listed equity, so the bank uses a structural proxy: it takes the median asset volatility of listed auto-parts suppliers (about 28%) and Redwater's estimated enterprise value of $520 million to compute a distance to default. With a default point equal to short-term debt (the revolver and the term loan maturing within two years: $140 million) plus half of long-term debt ($35 million), DP = $175 million. The simple DD is (520 − 175) / (520 × 0.28) = 2.37 before the deal. Adding the $20 million dividend funded by new debt raises the default point and reduces enterprise value only if the dividend leaks value out of the firm, so the credit team treats the DD as roughly 2.2 afterward after allowing for the larger obligation and the lower liquidity.

Market signals are available for peers but not for Redwater. Five-year CDS on a listed competitor with similar leverage trades at 310 basis points. With the market convention of 40% recovery, that implies a hazard rate of 5.17% and a five-year risk-neutral default probability of 22.8%. The bank's internal historical default experience for the "Fair" grade, however, gives a five-year cumulative PD of about 25.7% (Example 5.6), which includes migration, and a recovery on first-lien loans in the sector of about 60%.

The bank must now price the new facility. If the facility is secured by a first lien on receivables, inventory and equipment with an estimated stressed value of $180 million, coverage is 78% of the $230 million exposure, and the credit team sets LGD at 35% for the secured portion. A senior unsecured tranche would have an LGD of about 60%. The CFO is aware that the new structure puts all lenders pari passu, and the unsecured noteholders, whose consent is required for the refinancing, have already signalled that their note should be taken out at a premium because they lose their position relative to a larger secured tranche.

Imran has three proposals. The first is the full refinancing with the $20 million dividend. The second is the refinancing without the dividend, leaving the cash in the business. The third is a smaller refinancing that keeps the unsecured note in place. The bank has indicated that its minimum acceptable risk-adjusted return is a spread covering expected loss, capital charge and a cost of 0.30%, as in Example 5.9.

Discussion questions

1. Using the simple distance-to-default measure, compute DD for Redwater's proposed structure assuming enterprise value of $500 million after the dividend leaves the firm (enterprise value net of $20 million distributed) and a default point rising to $185 million. What does the change in DD suggest about the risk added by the dividend? Why might an equity-based DD be unavailable for Redwater and how does the proxy approach affect confidence in the result?
2. Compute the hazard rate and five-year risk-neutral PD implied by the peer CDS of 310 bp under recoveries of 25%, 40% and 60%. Why does the bank's historical five-year PD of 25.7% differ from the CDS-implied figure, and which figure belongs in the provision calculation and which in the pricing of a hedge?
3. For the secured facility, estimate expected loss under the post-deal PD of 7.5% and an LGD of 35% on $230 million. How would the expected loss change if the facility were unsecured with LGD 60%? How does the secured/unsecured choice affect the loan's required spread and the position of the unsecured noteholders?
4. Apply the Vasicek model with ρ = 20% and 99.9% confidence to the post-deal PD of 7.5%. Compute the capital percentage and compare the required spread to Example 5.9. Which of the three proposals would you recommend, and what covenants would you attach (consider leverage, dividend restrictions and collateral coverage tests)?

Key Takeaways

- Credit risk has three building blocks, PD, LGD and EAD, and expected loss is their product. Each component should be estimated, priced and monitored separately.
- Real-world and risk-neutral default probabilities are different objects: use the former for provisioning, capital and underwriting, and the latter for valuing and hedging traded credit.
- In the Merton model equity is a call option on the firm's assets and debt is risk-free debt minus a put; credit spreads rise steeply with leverage and asset volatility, and shareholders gain from risk that lenders bear.
- Asset value and asset volatility must be inferred from equity price and volatility by solving two simultaneous equations; distance to default compresses leverage, volatility and market value into a single ranking statistic.
- The Gaussian mapping from distance to default to PD understates tail risk; empirical EDF mappings or internal calibrations are needed for PD levels.
- In reduced-form models the hazard rate is calibrated to spreads, with spread ≈ $\lambda \times (1 - R)$; survival is $\exp(-\int\lambda)$, and the hazard curve plays the role of a forward curve for default.
- A CDS spread measures expected loss, not default probability: the same spread implies very different PDs under different recovery assumptions, and CDS pricing equates a risky-annuity premium leg to a contingent protection leg.
- Recovery depends on seniority, collateral, cycle and process, and is negatively correlated with default; downturn LGD is the appropriate basis for capital.
- Transition matrices give term structures of PD through matrix powers, show how ratings drift, and reveal that conditional PDs rise for good credits and fall for poor ones; scorecards (Z-score, logit) should be judged on discrimination, calibration and stability.

- Expected loss is additive and priced; unexpected loss depends on correlation and drives capital, so pricing should reflect marginal risk contribution. IFRS 9 ECL adds a stage-based step-up from 12-month to lifetime loss.

Review Questions

1. Define PD, LGD and EAD. Why is EAD not simply the drawn balance for a revolving credit facility?
2. Explain why a firm's equity can be viewed as a call option and its risky debt as risk-free debt less a put option. What is the economic meaning of the strike price and maturity?
3. In the Merton model, what are the effects on the credit spread of an increase in (a) leverage, (b) asset volatility, (c) the risk-free rate? Explain each in terms of option theory.
4. Why is the term structure of Merton spreads upward-sloping for low-leverage firms and humped or downward-sloping for high-leverage firms?
5. Describe the procedure for estimating asset value and asset volatility from equity data. Why can't asset volatility simply be set equal to equity volatility?
6. What is the KMV default point, and why does it lie between short-term debt and total debt? Why does KMV use an empirical mapping from DD to EDF?
7. What is a hazard rate? Explain the relationship between a hazard rate, a survival probability and a credit spread, and state the assumptions behind the credit triangle.
8. Why is the risk-neutral PD generally larger than the real-world PD? Give two uses for each.
9. Describe the two legs of a CDS and show how the par spread, upfront payment and mark-to-market are derived from them. What is the CS01?
10. What are the strengths and weaknesses of ratings transition matrices, and why is the expected-loss pricing of a loan incomplete without a capital charge?

Problems

1. **Merton model.** A firm has asset value V = 200, a zero-coupon debt with face value D = 120 due in 2 years, asset volatility 30% and a risk-free rate of 4% (continuous). Compute d1, d2, equity value, debt value, the risk-neutral PD and the credit spread (continuously compounded).
2. **Distance to default.** A firm has asset value 80, asset volatility 15%, short-term debt 20 and long-term debt 30. Using the KMV default-point rule and the simple definition DD = (V − DP)/(V × σ_V), compute the DD and the Gaussian PD.
3. **Hazard rate and recovery.** A five-year CDS spread is 240 bp. Compute the hazard rate and the five-year cumulative default probability assuming recovery of 40%, and again assuming 25%.
4. **CDS pricing.** An entity has a flat hazard rate of 2.0%, recovery of 40%, a continuous risk-free rate of 3%, and quarterly premiums over a five-year horizon. Using the legs approach, compute the risky annuity, the protection leg, the par spread, and the upfront payment on a contract with a 500 bp coupon.
5. **Migration.** A three-state system (Good, Weak, Default) has the one-year transition matrix: Good → (90%, 8%, 2%); Weak → (10%, 80%, 10%); Default absorbing. Compute the two-year transition matrix and the two-year cumulative default probability of each starting state.

6. **Expected and unexpected loss.** A $25 million exposure has PD 1.5%, LGD 40%. Compute the expected loss. Using the Vasicek model with ρ = 12% and q = 99.9%, compute the worst-case default rate and the capital (unexpected loss).
7. **Z-score.** A firm has WC/TA = 0.25, RE/TA = 0.10, EBIT/TA = 0.06, MVE/TL = 0.50 and Sales/TA = 0.90. Compute the Altman Z-score and classify it.
8. **Risky bond.** A three-year zero-coupon bond has face value 100. The risk-free rate is 4%, the hazard rate is 3% and recovery is 40% of market value. Compute the price, the risk-free price and the yield spread, using the Duffie-Singleton approximation.

Answers to Selected Problems

1. d1 = 1.6047, d2 = 1.1805; equity = 91.54, debt = 108.46; risk-neutral PD = N(−1.1805) = 11.89%; spread ≈ 1.06% (about 106 bp).
2. DP = 20 + 0.5 × 30 = 35; DD = (80 − 35)/(80 × 0.15) = 45/12 = 3.75; Gaussian PD = N(−3.75) ≈ 0.009%.
3. With R = 40%: λ = 0.0240/0.60 = 4.00%; five-year PD = 1 − e^(−0.20) = 18.13%. With R = 25%: λ = 0.0240/0.75 = 3.20%; PD = 1 − e^(−0.16) = 14.79%.
4. Risky annuity ≈ 4.407; protection leg ≈ 0.0531; par spread ≈ 1.204% (about 120 bp); upfront on a 500 bp coupon = 0.0531 − 0.05 × 4.407 = −0.167, so the protection buyer *receives* about 16.7% of notional upfront because the coupon exceeds the fair spread.
5. P^2 rows: Good → (81.8%, 13.6%, 4.6%); Weak → (17.0%, 64.8%, 18.2%). Two-year cumulative PD: Good 4.6%, Weak 18.2%.
6. EL = 0.015 × 0.40 × 25,000,000 = $150,000. Worst-case default rate = 12.06%; capital = (0.1206 − 0.015) × 0.40 × 25,000,000 ≈ $1.056 million.
7. Z = 1.2×0.25 + 1.4×0.10 + 3.3×0.06 + 0.6×0.50 + 1.0×0.90 = 0.300 + 0.140 + 0.198 + 0.300 + 0.900 = 1.838, just above the 1.81 distress cut-off, in the grey zone (very close to distress).
8. Risky rate = 0.04 + 0.03 × 0.60 = 5.8%; price = 100 × e^(−0.174) = 84.03; risk-free price = 100 × e^(−0.12) = 88.69; spread = 1.8% (λ × (1 − R)), or about 180 bp.

Chapter 6: Securitization, Structured Credit and CLOs

Learning Objectives

After studying this chapter, you should be able to:

- Explain the legal and economic structure of a securitization, including the role of the special purpose vehicle, true sale, bankruptcy remoteness and the servicer, and the motives for issuers and investors.
- Analyse the cash flows of an asset-backed security, including amortisation, prepayment (CPR and SMM) and weighted average life.
- Allocate pool losses across tranches using attachment and detachment points, and model sequential and pro-rata waterfalls with interest and principal priorities and performance triggers.
- Evaluate forms of credit enhancement (subordination, overcollateralisation, excess spread, reserve accounts) and measure how each protects senior investors.
- Explain how default correlation drives tranche value, and compute tranche expected losses using a one-factor Gaussian copula.
- Describe CLO structure and economics, compute equity returns and coverage tests, and summarise how rating agencies size credit enhancement and what lessons the structured-credit crises teach managers and investors.

Opening Vignette

(Illustrative scenario; the company and its figures are fictional.)

Alder Auto Finance lends to buyers of used cars. The company has a loan book of $900 million, a funding cost that rises whenever credit markets wobble, and a capital base that limits how fast it can grow. Its CFO, Priya Raman, is weighing a proposal from the treasury team: package $400 million of seasoned auto loans into a securitization.

The proposal is straightforward on paper. The loans would be sold to a newly created trust, which would fund the purchase by issuing three classes of notes to institutional investors. The senior class, 78% of the issue, would carry a top rating and a coupon barely above the swap rate. A middle class of 12% would be rated in the single-A to BBB range. The residual 10% of the structure, the equity or “first-loss” piece, Alder would keep.

Priya sees obvious advantages. Alder would be funding most of its loans at a rate far below its unsecured borrowing cost, because investors would be buying exposure to the loans’ cash flows rather than to Alder’s balance sheet. She would receive cash up front, lend again and earn servicing fees. But she also sees questions she cannot answer by instinct. If the senior notes are rated AAA, and the loans themselves are risky subprime-adjacent credits, where does the safety come from? Why would investors accept a 10% cushion rather than 5% or 25%? What happens to the equity piece if losses are twice what Alder’s historical data show? And what, if anything, has the market learned from securitizations that went badly wrong in the past?

The head of structured finance, Tomas Idowu, answers with a whiteboard sketch of a waterfall: cash flows pour into the top bucket first and overflow only when each bucket is full, while losses fill buckets from the bottom. “The safety is arithmetic and correlation,” he says. “The arithmetic we can compute today. The correlation is where people get hurt.” This chapter

provides the tools to do both.

6.1 The Logic of Securitization

6.1.1 From balance sheet to capital markets

Securitization transforms a pool of illiquid financial assets (loans, leases, receivables, mortgages) into tradable securities whose payments are funded by the cash flows from that pool. The basic structure has five elements:

1. The **originator** (or sponsor) creates the assets: a bank, auto lender, credit card issuer, equipment lessor or corporate lender.
2. The assets are sold to a **special purpose vehicle (SPV)**, also called a special purpose entity, trust or issuer. The SPV is legally separate from the originator and has no business except to hold the assets and issue securities.
3. The SPV issues **notes** in several classes, called tranches, with different priorities of payment and different risk and return.
4. A **servicer**, often the originator, collects payments from borrowers, pursues delinquent accounts and reports to investors. A **trustee** safeguards investor interests and distributes cash according to the documents.
5. **Credit rating agencies** assess the tranches, and **investors** (pension funds, insurers, banks, asset managers) buy them.

The central legal concept is **true sale** and **bankruptcy remoteness**. If the transfer of assets to the SPV is a true sale, the assets are not part of the originator's estate if the originator becomes insolvent, and creditors of the originator cannot reach them. Investors therefore bear the risk of the *pool*, not the risk of the originator. This is the foundation of the claim that a securitization can achieve a better credit rating than its sponsor. It is also why the analysis of the servicer (operational risk and replacement) and of legal opinions matter: if the legal separation fails, the structure's rating collapses to that of the originator.

6.1.2 Why issuers securitize

An originator securitizes for some mix of the following reasons:

- **Cheaper funding.** Because senior notes are backed by a diversified pool and protected by credit enhancement, they can be rated more highly than the originator and funded at a lower spread. This is not a free lunch, as the originator keeps the residual risk and must pay for the structure, but when the originator is lower-rated than its assets' senior tranche, the funding saving is real.
- **Diversification of funding sources.** Securitization opens a market of investors with a taste for specific risk (for example, investors who want auto-loan exposure but not the exposure to the corporate sponsor).
- **Balance sheet and capital relief.** If the transfer qualifies for derecognition under IFRS 9 (substantial transfer of risks and rewards, or loss of control) and for significant risk transfer under bank regulation, the assets leave the balance sheet and capital requirements fall. Retention of the first-loss piece often limits this relief, because the first-loss piece contains most of the expected loss.
- **Liquidity and growth.** Cash is received upfront rather than over the life of the loans, allowing faster lending.
- **Revenue recognition and servicing income.** The originator may retain servicing and excess spread.

6.1.3 Why investors buy

Investors gain access to specific risk exposures, usually with a yield pick-up relative to corporate bonds of the same rating, and a choice of tranches along a risk spectrum: senior notes for low-risk, floating-rate, high-rated assets; mezzanine for higher spread; equity for high yield and exposure to excess spread. Banks and insurers like senior tranches because, historically, regulatory capital charges for highly-rated securities were low relative to yield. This reaching for yield and regulatory arbitrage, and the dependence on ratings, are important parts of the story told in Section 6.8.

6.1.4 The balance-sheet view

Securitization does not create value by magic. What it does is allocate the cash flows and risks of a pool to the investors best placed to bear them, subject to frictions. If markets were perfect (the Modigliani-Miller setting of Volume 1, Chapter 15), securitization would be irrelevant: the total value of the securities would equal the value of the pool. Value arises from market frictions: differences in funding costs, regulatory capital rules, investor segmentation (some investors can hold only AAA paper), information asymmetries about the originator, and tax and accounting treatment. Managers should recognise the corollary that where securitization appears to create value only through ratings or regulatory treatment, the apparent value can disappear when those rules or opinions change.

6.2 ABS Mechanics: Asset Pools, Amortisation and Prepayment

6.2.1 Asset classes and cash flow patterns

Pools differ in the way they generate cash:

- **Amortising pools** (auto loans, equipment leases, mortgages, student loans) return principal and interest on a schedule. The cash flows of the notes are a function of the pool's scheduled amortisation, prepayments, defaults and recoveries.
- **Revolving pools** (credit cards, trade receivables, dealer floorplan) have short-lived assets that turn over. During a **revolving period**, principal collections are used to buy new receivables so that the pool balance stays constant; after the revolving period ends, an **amortisation period** begins and principal flows to noteholders. **Early amortisation triggers** (for example, a decline in excess spread below a threshold) end the revolving period early to protect investors.
- **Managed pools** (CLOs) are actively traded by a collateral manager within rules.

Asset-backed securities (ABS) typically describe consumer and commercial non-mortgage assets, **residential mortgage-backed securities (RMBS)** are secured by home loans, and **commercial mortgage-backed securities (CMBS)** by commercial property loans, which are generally non-amortising loans with balloon repayments.

6.2.2 Prepayment

Borrowers can usually repay early, and they do so for reasons that vary from refinancing (when interest rates fall) to sale of the asset to default-driven liquidation. Prepayment speed matters because it determines the timing of principal and thereby the **weighted average life (WAL)** of the notes and the amount of interest earned by the pool.

Prepayment speed is quoted as a **conditional prepayment rate (CPR)**: the annualised fraction of the pool's outstanding balance, after scheduled payments, that prepays. For monthly pools the **single monthly mortality (SMM)** converts CPR into a monthly rate:

```
SMM = 1 − (1 − CPR)^(1/12)
```

A CPR of 12% corresponds to SMM = 1 − 0.88^(1/12) = 1.0596% per month. A CPR of 6% gives SMM = 0.514%, and CPR 24% gives SMM = 2.261%. The standard industry benchmark for mortgages is the PSA curve, which ramps CPR from 0 to 6% over the first thirty months and holds it there; speeds are quoted as a percentage of PSA. For other asset types, prepayment is quoted in the ABS speed (ABS) convention as a percentage of the original balance.

The WAL of a tranche or pool is

```
WAL = Σ t × (principal received at t) / Σ (principal received)
```

Example 6.1 – Prepayment and weighted average life. A pool has an initial balance of 100, five annual level payments, and a loan rate of 8%. Compute the WAL with no prepayment, with a CPR of 10% and with a CPR of 25%.

Step 1: no prepayment. The level payment on a balance B with k payments remaining is B × 0.08 / (1 − 1.08^(−k)). With B = 100 and k = 5, the payment is 25.05. Interest in year 1 is 8.00, so scheduled principal is 17.05. The balance then falls to 82.95. Repeating:

Year	Beginning balance	Interest	Principal
1	100.00	8.00	17.05
2	82.95	6.64	18.41
3	64.55	5.16	19.88
4	44.66	3.57	21.47
5	23.19	1.86	23.19

Step 2: WAL, no prepayment. WAL = (1×17.05 + 2×18.41 + 3×19.88 + 4×21.47 + 5×23.19) / 100 = (17.05 + 36.82 + 59.64 + 85.88 + 115.95) / 100 = 3.154 years.

Step 3: CPR 10%. In each year, prepayment equals 10% of the balance remaining after scheduled principal, and the payment is recomputed on the new balance. Year 1: scheduled principal 17.05, prepayment 0.10 × (100 − 17.05) = 8.30, total principal 25.34, ending balance 74.66. Year 2: interest 5.97, scheduled principal 16.57, prepayment 5.81, total 22.38, ending balance 52.28. Year 3: scheduled 16.10, prepayment 3.62, total 19.72, ending 32.56. Year 4: scheduled 15.65, prepayment 1.69, total 17.34, ending 15.22. Year 5: 15.22. WAL = (25.34 + 2×22.38 + 3×19.72 + 4×17.34 + 5×15.22)/100 = (25.34 + 44.76 + 59.16 + 69.36 + 76.10)/100 = 2.747 years.

Step 4: CPR 25%. The same procedure gives a WAL of 2.247 years.

Interpretation. Faster prepayment shortens the average life from 3.15 to 2.75 and then 2.25 years, and reduces total interest earned by the pool. For an investor holding a discount note, faster prepayment accelerates the return of the discount and raises yield; for a premium note it is the opposite. Prepayment is also *correlated with interest rates* (and so with the environment in which reinvestment is least attractive). That negative convexity is the defining feature of mortgage securities, and it is why tranches are often sold on the basis of a WAL under a stated prepayment speed rather than a legal maturity alone.

6.2.3 Defaults, delinquencies and the loss timing curve

Defaults on amortising pools rise during the first years of seasoning, peak, and decline as the pool seasons and balances fall. Rating agencies and investors therefore use **cumulative net loss (CNL)** curves expressed as a percentage of the original balance, with a typical loss-timing pattern, and monitor **delinquency** (30, 60, 90 days past due), **charge-offs** and

recoveries. Performance is compared against the vintage's expected curve: if a vintage's cumulative loss is running above the expected path at a given age, the final loss will probably exceed expectations, and structural triggers are designed to react.

6.3 Tranching and the Cash Flow Waterfall

6.3.1 Attachment and detachment

Tranching allocates the pool's credit losses by seniority. Each tranche is defined by an **attachment point** A (the cumulative pool loss, as a percentage of pool balance, at which the tranche begins to lose) and a **detachment point** D (the cumulative pool loss at which the tranche is wiped out). The **thickness** of the tranche is D − A. If the cumulative pool loss is L (as a fraction of the original pool balance), the loss to a tranche, as a fraction of the tranche's size, is

```
Tranche loss fraction = min( max(L − A, 0), D − A ) / (D − A)
```

Example 6.2 – Loss allocation across tranches. A $100 million pool is divided into: a **senior** tranche with attachment 25% and detachment 100% ($75 million), a **mezzanine** tranche with attachment 10% and detachment 25% ($15 million), and an **equity** tranche with attachment 0% and detachment 10% ($10 million). Compute the tranche losses when cumulative pool losses are 4%, 12%, 20% and 30%.

Equity (0%-10%). Loss fraction = min(max(L, 0), 10%)/10%. *Mezzanine (10%-25%).* Loss fraction = min(max(L − 10%, 0), 15%)/15%. *Senior (25%-100%).* Loss fraction = min(max(L − 25%, 0), 75%)/75%.

Table 6.1 – Tranche losses by pool loss scenario.

Cumulative pool loss	Equity loss ($m / % of tranche)	Mezzanine loss ($m / %)	Senior loss ($m / %)
4% ($4m)	4.0 / 40.0%	0.0 / 0.0%	0.0 / 0.0%
12% ($12m)	10.0 / 100%	2.0 / 13.3%	0.0 / 0.0%
20% ($20m)	10.0 / 100%	10.0 / 66.7%	0.0 / 0.0%
30% ($30m)	10.0 / 100%	15.0 / 100%	5.0 / 6.7%

Interpretation. Losses always sum to the pool loss ($4m, $12m, $20m, $30m in the four rows). A pool loss of 20%, three or four times what a conservative underwriter would expect for a prime pool, still leaves the senior tranche untouched. The senior notes lose only after the 25% attachment point is breached, and then 5 on 75 is 6.7%. Equity is wiped out in any scenario with losses of 10% or more. This is the essence of tranching: the same pool produces a security that is nearly riskless in most scenarios and a security that behaves like a deep out-of-the-money put sold to the pool's investors. Like options (Volume 1, Chapter 18), each tranche is a call spread on the pool's loss: a long position in the first dollar of loss above A and a short position above D.

6.3.2 The waterfall

The **waterfall** (priority of payments) defines how collections are distributed each period. A typical structure has separate rules for interest and principal:

Interest waterfall: 1. Senior fees: trustee, servicer. 2. Interest on the Class A (senior) notes. 3. Interest on the Class B (mezzanine) notes. 4. (Optionally) replenish reserve account. 5. Remaining excess spread released to the residual holder (equity), *if performance tests are passed*; otherwise diverted to repay principal.

Principal waterfall: 1. **Sequential pay:** all principal to Class A until it is fully repaid, then to Class B, then to the equity. In **pro rata pay**, principal is shared among classes in proportion to their balance; many deals pay pro rata while performance is good and switch to sequential if triggers are breached.

Sequential pay protects the senior notes: their balance falls fastest, and their credit enhancement (the ratio of junior notes to the remaining pool) grows over time. Pro rata keeps the enhancement percentage constant and is more favourable to junior holders.

Triggers are performance tests that change the waterfall. The most common are **overcollateralisation (OC) tests**: the ratio of pool balance to the balance of the notes at or above a given class, compared with a minimum. If a class fails its test, excess interest that would have gone to the equity is diverted to repay senior principal until the test is cured. Interest-coverage tests work similarly. The effect is a self-correcting mechanism: deterioration of the pool automatically strengthens the position of the senior holders at the expense of the junior and residual holders.

Example 6.3 – A four-year waterfall with triggers. A trust holds a pool of $100 million at inception. Terms:

- The pool pays interest at 8% (WAC) on its performing balance. It amortises straight-line: scheduled principal each year equals the performing balance after defaults divided by the number of years remaining (including the current year).
- A servicing fee of 0.50% of the beginning performing balance is paid first.
- Notes: Class A $70m at 5%; Class B $20m at 7%; Equity $10m (residual).
- Defaults occur at the start of each of years 1 to 3 as a fixed percentage of the *original* pool balance. Defaulted loans pay no interest; recoveries of 50% of the defaulted balance are received in the same year as principal collections.
- Principal collections (scheduled plus recoveries) are paid *sequentially* to A, then B.
- **OC trigger:** if the performing balance after defaults divided by (A + B notes outstanding) is below 1.05, any interest remaining after paying fees and note interest is diverted to repay principal (A first, then B), rather than released to equity.

Step 1: no defaults. Year 1: interest 8.00; fee 0.50; Class A interest 3.50; Class B interest 1.40; excess spread 2.60. OC = 100/90 = 1.111, above 1.05, so 2.60 goes to equity. Scheduled principal is 100/4 = 25.00, all to A (A falls from 70 to 45). Year 2: performing balance 75, interest 6.00, fee 0.38, A interest 2.25, B interest 1.40, excess 1.97 to equity; principal 25 to A (A = 20). Year 3: performing balance 50, interest 4.00, fee 0.25, A interest 1.00, B interest 1.40, excess 1.35 to equity; principal 25: A repaid 20, B repaid 5 (B = 15). Year 4: performing balance 25; interest 2.00, fee 0.125, B interest 1.05, equity interest 0.82; principal 25: B repaid 15, the remaining 10 goes to the equity as residual principal. Equity receives 2.60, 1.97, 1.35 and 10.82: total 16.74 on an investment of 10, an IRR of 18.11%.

Step 2: 5% annual defaults (cumulative 15% of original, with 7.5 recovered in total). In year 1, defaults of 5.0 leave a performing balance of 95.0; interest is 7.60; fee 0.50; A and B interest 3.50 and 1.40; excess 2.20. OC = 95/90 = 1.056, which passes, so equity receives 2.20. Principal is 95/4 = 23.75 scheduled plus 2.5 recovery = 26.25 to A. In year 2, after further defaults the performing balance is 66.25 and the OC ratio is 66.25/(43.75 + 20) = 1.039, a **breach**: the 1.36 of excess spread is diverted to repay A rather than going to equity. In year 3 the ratio is 1.036 and 0.62 is diverted. Equity receives nothing in years 2 and 3, and only the final residual of 4.89 in year 4. Equity cash flows are 2.20, 0, 0, 4.89: total 7.09 on 10 invested, an IRR of −10.3%. Notes A and B are both repaid in full.

Step 3: summary across scenarios. Table 6.2 reports the outcomes, including the stress with 8% annual defaults both with and without the OC trigger.

Table 6.2 – Waterfall outcomes by default scenario (cash flows in $ millions).

Annual defaults (cumulative)	Equity cash flows, years 1-4	Equity total	Equity IRR	Loss to Class A	Loss to Class B
0% (0%)	2.60, 1.97, 1.35, 10.82	16.74	18.1%	0	0
2% (6%)	2.44, 1.73, 1.03, 7.60	12.80	8.5%	0	0
5% (15%)	2.20, 0, 0, 4.89	7.09	−10.3%	0	0
8% (24%), trigger on	0, 0, 0, 1.44	1.44	−38.4%	0	0
8% (24%), no trigger	1.96, 0.99, 0.08, 0	3.03	−54.8%	0	2.00
12% (36%), trigger on	0, 0, 0, 0	0	−100%	0	5.79

Interpretation. Three results are worth stressing. First, with 24% cumulative defaults and 50% recovery, the pool loses 12% of its balance, which is more than the 10% equity cushion. Both A and B nevertheless suffer no loss in the trigger-on case, because diverted excess spread and sequential principal restored the collateral cushion. Second, the same scenario *without* the trigger shows the equity receiving $3.03 million that the senior structure needed, and Class B finishing short by $2.00 million. The trigger redirected about $1.6 million of equity cash flow (3.03 versus 1.44) into early note repayment, which was enough to make Class B whole. Third, equity returns are extremely sensitive to defaults: a move from 2% to 5% per year turns an 8.5% IRR into a 10.3% loss. The equity is a leveraged claim on the pool's excess spread, and the leverage cuts both ways.

> **Common mistake.** Treating the credit rating of a senior tranche as independent of the waterfall. The rating is a statement about a specific *document*: priority rules, triggers, cure mechanics and the servicer's behaviour. Two pools with identical collateral can produce tranches of very different safety.

6.4 Credit Enhancement

6.4.1 Forms of enhancement

Credit enhancement is any feature that raises the probability that senior investors are paid in full. It can be internal (provided by the structure itself) or external (provided by a third party).

Subordination. Junior tranches absorb losses before senior tranches. The senior tranche's credit enhancement equals the combined size of the tranches below it, as a percentage of the pool. In Example 6.2, Class A has 25% subordination.

Overcollateralisation (OC). The pool balance exceeds the notes' balance. For instance, $105 million of loans supporting $100 million of notes gives 5% OC. OC absorbs losses first, ahead of any note, and it can build over time if excess spread is trapped to pay down notes faster than the pool amortises.

Excess spread. The difference between the interest earned on the pool and the sum of note interest and fees. It is the first line of defence, absorbing loss on a current basis before any principal shortfall arises. In Example 6.3 it is 2.60 in year 1 (2.6% of the pool) before defaults. A transaction with 4% excess spread and an average life of 2 years can absorb roughly 8% of cumulative losses by excess spread alone, if the spread is retained.

Reserve accounts. Cash set aside at closing or funded from excess spread, used to cover interest shortfalls or principal losses. Liquidity reserves protect against timing mismatches between collections and payments.

Triggers and early amortisation. As in Section 6.3, performance triggers redirect cash flows toward senior holders.

External enhancement. Bond insurance (monoline guarantees), letters of credit and parent guarantees. These substitute the credit of the guarantor for that of the pool and introduce counterparty exposure to the guarantor, a vulnerability that surfaced in the monoline insurers' difficulties during the credit crisis.

Hedges. Interest rate swaps convert fixed-rate collateral to floating or vice versa to match the notes, and currency swaps handle cross-currency pools. The swap counterparty adds counterparty risk, mitigated by rating triggers and collateral posting.

6.4.2 Diversification and the limits of independence

Why can a pool of risky loans support a safe senior tranche? The answer rests on diversification. If defaults are independent, the proportion of the pool that defaults concentrates tightly around its expectation, and the chance of a large loss becomes negligible.

Consider a pool of 100 equal loans, each with a default probability of 5%, independent. The number of defaults is binomial with n = 100 and p = 0.05; the mean is 5 and the standard deviation is $\sqrt{(100 \times 0.05 \times 0.95)} = 2.18$. Then:

```
P(10 or more defaults) = 2.82%
P(15 or more defaults) = 0.0136%
P(20 or more defaults) = 0.00001%
```

Under independence, the chance of a loss large enough to hit a tranche attaching at 15% (15 defaults, with zero recovery) is about one in 7,000. The senior tranche looks like a super-safe security. The weakness is the independence assumption: if defaults share common risk drivers (house prices, the business cycle, a single industry), the number of defaults is highly dispersed, and the probability of an extreme outcome is many orders of magnitude larger. The next section quantifies this.

6.5 Correlation and the Valuation of Tranches

6.5.1 The Gaussian copula

Tranche valuation requires the joint distribution of defaults, not just individual default probabilities. The market standard from the early 2000s is the **one-factor Gaussian copula**, introduced by David Li and building directly on Vasicek's loan-portfolio model (Chapter 5, Section 5.8). Each obligor i has a latent variable

```
Zᵢ = √ρ · M + √(1 − ρ) · εᵢ
```

where M is a common (systematic) factor and ε_i an independent idiosyncratic factor, both standard normal. Obligor i defaults over the horizon if $Z_i < c = \Phi^{-1}(PD)$. Conditional on M = m, defaults are independent with probability

```
PD(m) = Φ( [ c − √ρ · m ] / √(1 − ρ) )
```

In a large homogeneous pool the loss rate given m converges to LGD × PD(m). The **expected loss on a tranche** [A, D] is therefore obtained by integrating over the distribution of M:

```
E[ tranche loss ] = ∫ φ(m) · min( max(LGD · PD(m) − A, 0), D − A ) dm
```

with φ the standard normal density, and the expected loss fraction is this divided by (D − A).

The parameter ρ is the **asset correlation**. It controls how much of the risk is systematic: when ρ = 0, defaults are independent and the pool's loss rate is deterministic at LGD × PD; when ρ is high, the pool loss rate swings widely with M.

Example 6.4 – Tranche expected loss and correlation. A very large homogeneous pool has a cumulative PD of 5% over the life of the deal and LGD of 60%, so the pool's expected loss is 3.0%. Tranches are Equity (0-10%), Mezzanine (10-25%) and Senior (25-100%) as in Example 6.2. Compute each tranche's expected loss as a percentage of tranche size for ρ = 0, 10%, 30% and 50%.

Step 1: set up. c = $\Phi^{-1}(0.05) = -1.6449$. For each ρ, calculate pool loss $L(m) = 0.60 \times \Phi\left((-1.6449 - \sqrt{\rho} \cdot m) / \sqrt{(1-\rho)} \right)$ over a fine grid of m from −8 to 8 and integrate the tranche payoff with weights φ(m) dm.

Step 2: the ρ = 0 case. PD(m) = 5% for all m, so L = 3.0% always: equity loses 3.0/10 = 30% of its notional with certainty; mezzanine and senior lose nothing.

Step 3: ρ = 30% illustration. At the worst 1% of the factor (m = −2.326), $PD(m) = \Phi\left((-1.6449 + 0.5477 \times 2.326) / 0.8367 \right) = \Phi\left((-1.6449 + 1.2740) / 0.8367 \right) = \Phi(-0.4433) = 32.9\%$, so pool loss L = 0.60 × 32.9% = 19.7%, which would wipe out the equity and put 9.7 points of loss through the mezzanine tranche. At m = −3.09 the pool loss is above 25%, and the senior tranche begins to take losses.

Step 4: numerical results.

Table 6.3 – Tranche expected loss (percentage of tranche notional) as asset correlation varies. Pool PD 5%, LGD 60%.

Asset correlation ρ	Equity (0-10%)	Mezzanine (10-25%)	Senior (25-100%)
≈ 0 (independent)	30.00%	0.000%	0.000%
10%	29.80%	0.134%	0.000%
30%	26.70%	2.084%	0.023%
50%	22.24%	4.229%	0.189%

Step 5: check. The weighted sum of tranche expected losses should equal the pool's expected loss of 3.0% (in pool dollars) at every correlation. At ρ = 30%: 0.10 × 26.70% + 0.15 × 2.084% + 0.75 × 0.023% = 2.670% + 0.313% + 0.017% = 3.000%. At ρ = 50%: 2.224% + 0.634% + 0.142% = 3.000%. Correlation *redistributes* a fixed expected loss; it does not change it.

Interpretation. Rising correlation moves expected loss from the equity tranche to the mezzanine and senior tranches: the senior tranche's expected loss rises from zero to 0.19% of its notional, a tiny number in absolute terms, but a large increase relative to a rating that may be built on loss rates of a few hundredths of a percent. Equity is **long correlation** (its expected loss falls as correlation rises, because when everything defaults together there are more states in which nothing defaults), while senior tranches are **short correlation**. The mezzanine tranche is the most sensitive in relative terms and, in a thin-mezzanine structure, is exposed both to systemic shocks and to idiosyncratic losses.

6.5.2 Correlation: unobservable and unstable

The weakness of this valuation approach is that ρ is not observed directly. Its calibration relies on equity correlations, on the correlation implied by quoted index tranche prices (the "base correlation" method), or on regulatory assumptions. Practitioners soon discovered several problems:

- Implied correlation differs across tranches in the same pool (the **correlation smile**), contradicting the single-parameter model.
- Correlation is not constant: in a crisis defaults cluster, and correlations rise sharply, exactly when senior investors depend on diversification.
- The Gaussian copula has no **tail dependence**: extreme joint defaults are less likely than in the real world, where fat-tailed common shocks (a collapse in house prices, a financial panic) make simultaneous failure far more probable.
- Estimates from short histories of benign conditions systematically understate systemic risk.

A manager reading a model that reports a senior tranche as nearly riskless should ask a single question: what happens to the result if correlation doubles or if the common factor has a fat tail? The model's output is an answer to a question about structure; the *trust* placed in it is a question about assumptions.

6.6 Collateralised Loan Obligations

6.6.1 Structure

A **collateralised loan obligation (CLO)** is a securitization of a managed portfolio of senior secured leveraged loans, typically 150 to 300 obligors across many industries. Unlike a static ABS, a CLO has a **collateral manager** who buys and sells loans within rules, and the structure has a life of roughly twelve years:

- **Ramp-up period:** the manager acquires the portfolio.
- **Reinvestment period** (about four to five years): principal proceeds from repayments and sales are reinvested in new loans, subject to portfolio tests.
- **Amortisation period:** principal is paid to noteholders sequentially, from the most senior class down.
- **Non-call period:** equity holders may typically call (redeem) the deal after a non-call period, giving the equity optionality over the life of the CLO.

The liabilities consist of floating-rate notes in several rated classes (AAA, AA, A, BBB, BB) and a residual **equity** tranche (also called subordinated notes). The assets are floating-rate loans, so interest rate risk is largely absent, and the liabilities are floating-rate as well. Unlike many ABS, the CLO is **cash-flow driven** rather than market-value driven: noteholders depend on cash collections and coverage tests, not on mark-to-market triggers that force asset sales. This design is a key reason that senior CLO tranches have proved resilient in stress.

6.6.2 Portfolio tests

CLO documents impose tests on the portfolio, which the manager must comply with:

- **Collateral quality tests:** weighted average rating factor (WARF), weighted average spread (WAS), weighted average recovery rate (WARR), and a diversity score.
- **Concentration limits:** maximum proportion of CCC-rated assets (often 7.5%), second-lien loans, single-obligor and single-industry exposures, and covenant-lite loans.
- **Overcollateralisation (par coverage) tests** and **interest coverage (IC) tests** at each class level. The OC test compares the *adjusted par* of the portfolio (performing par, plus recovery value of defaulted assets, minus haircuts for excess CCC assets and discounted purchases) with the principal balance of the notes at and above each class.

A breach of OC or IC tests diverts interest cash flows away from the equity and from junior-note interest (for the most junior classes, which can have interest deferred) toward repayment of the senior notes until the test is cured. These mechanics are the generalisation of the trigger in Example 6.3.

Example 6.5 – CLO economics and equity return. A $500 million CLO portfolio of floating-rate loans has an average spread of 3.75% over the benchmark rate (SOFR, assumed 4.00%). The liability structure and spreads are:

Class	Size ($m)	Spread over SOFR	Coupon	Annual interest ($m)
AAA	310	1.40%	5.40%	16.740
AA	40	1.85%	5.85%	2.340
A	30	2.40%	6.40%	1.920
BBB	25	3.50%	7.50%	1.875
BB	25	6.50%	10.50%	2.625
Equity	70	residual		
Total	**500**			

Senior fees and manager fees total 0.45% of assets per year. Compute the equity return with no credit losses, and as the annual portfolio credit loss rises.

Step 1: asset interest. The asset coupon is 4.00% + 3.75% = 7.75%, so annual interest on $500 million is $38.75 million.

Step 2: liability interest. The sum of the interest column is 16.740 + 2.340 + 1.920 + 1.875 + 2.625 = $25.50 million on $430 million of rated notes, an average cost of 5.93%, or a spread over SOFR of 1.93%.

Step 3: fees. 0.45% × 500 = $2.25 million.

Step 4: equity excess spread with no losses. 38.75 − 25.50 − 2.25 = $11.00 million. As a percentage of the $70 million equity, the cash yield is 15.71%.

Step 5: add credit losses. Assume loss given default is 35% (65% recovery) and the annual default rate on the portfolio is CDR. Then the annual credit loss is 500 × CDR × 0.35.

Table 6.4 – Equity cash yield as annual default rate rises (unlevered loss, no reinvestment effects).

Annual default rate	Annual credit loss ($m)	Equity cash flow ($m)	Equity yield
0%	0.00	11.00	15.71%
1%	1.75	9.25	13.21%
2%	3.50	7.50	10.71%
3%	5.25	5.75	8.21%
4%	7.00	4.00	5.71%
6%	10.50	0.50	0.71%

Step 6: break-even. Equity cash flow reaches zero when loss equals 11.00, i.e. a default rate of 11.00 / (500 × 0.35) = 6.29% per year.

Step 7: sensitivities. At a 2% default rate: if the average asset spread tightens from 3.75% to 3.25% (loan repricing), equity yield falls to 7.14%; if it widens to 4.25%, it rises to 14.29%. If AAA spreads widen by 30 basis points, equity yield falls from 10.71% to 9.39%. Benchmark rates matter in a less obvious way: at SOFR of 2.00% the equity yield is 8.71%, at

4.00% it is 10.71% and at 6.00% it is 12.71%. Assets and rated notes are both floating, but $70 million of the $500 million portfolio is funded by equity, which therefore earns the benchmark rate on its own capital: each 1% rise in SOFR adds $0.70 million, or 1.0 point of equity yield.

Interpretation. The equity yield can be decomposed as the benchmark rate plus a spread margin on equity: net spread income is 500 × 3.75% − 430 × 1.93% − 2.25 = 18.75 − 8.30 − 2.25 = $8.20 million, which is 11.71% of the $70 million equity, and adding SOFR of 4.00% gives 15.71%. The margin is levered roughly 7 times (500/70) on a thin portfolio spread. Each 1% of annual default costs 2.5 points of equity yield, and each 10 basis points of asset spread is worth about 0.7 points. CLO equity is therefore sold as a high-yield instrument with embedded optionality (the call right, the ability to reinvest at wider spreads in a downturn, and non-mark-to-market financing), but with large downside in a credit cycle.

The **arbitrage** in a CLO is the gap between asset spread and the weighted cost of liabilities: it persists because the senior notes are priced as AAA-rated securities, with a much lower spread than the loans' risk would imply on a standalone basis, while the equity absorbs the risk and receives the arbitrage. The CLO arbitrage can disappear when loan spreads compress or when liability spreads widen, in which case new-issue volume dries up.

Example 6.6 – A coverage test under stress. In the same CLO, cumulative notes at each level are: AAA and AA $350m, A $380m, BBB $405m, BB $430m. The OC trigger levels are 135%, 122%, 114% and 106% respectively. Initial OC ratios are 500/350 = 142.9%, 131.6%, 123.5% and 116.3%. Test the position after the following stress: $45 million of loans have defaulted (assumed recoverable at 50%); performing par is $455 million, of which $80 million is CCC-rated; the CCC bucket limit is 7.5% of $500 million = $37.5 million; and excess CCC loans are marked at 65% of par.

Step 1: CCC excess. 80 − 37.5 = $42.5 million; haircut = 42.5 × (1 − 0.65) = $14.875 million. *Step 2: recovery on defaulted assets.* 45 × 0.50 = $22.5 million. *Step 3: adjusted par.* 455 − 14.875 + 22.5 = $462.625 million. *Step 4: tests.*

Class	Notes at and above	Adjusted OC ratio	Trigger	Result
AAA/AA	350	132.2%	135%	Fail
A	380	121.7%	122%	Fail (marginal)
BBB	405	114.2%	114%	Pass (marginal)
BB	430	107.6%	106%	Pass

Step 5: cure. Interest diverted to repay AAA principal reduces the denominators of all tests but not adjusted par. The AAA test is cured when 462.625 / (350 − x) = 1.35, which requires x = 350 − 342.685 = $7.31 million; the A test requires only x = 0.80. A diversion of about $7.3 million, 1.5% of the portfolio, cures all failures, which is two-thirds of the unstressed annual equity cash flow of $11 million.

Interpretation. The stress does not impair any note; it reroutes equity cash flow to repay AAA principal. This is the way a CLO converts credit deterioration into equity pain, and why a CLO equity investor monitors the margin by which tests are passed as closely as the default rate. The CCC haircut is a major channel: the test counts excess CCC loans at market value, so a deterioration of ratings (not just defaults) can divert cash.

6.7 How Rating Agencies Size Structures

6.7.1 Two philosophies: probability of default and expected loss

Rating agencies approach structured finance with two related philosophies. Under a **probability-of-default** (first-dollar-of-loss) approach, a tranche's rating reflects the likelihood that it suffers *any* loss. Under an **expected-loss** approach, the rating reflects the probability-weighted *size* of the loss: a tranche with a small chance of losing a large amount and a tranche with a larger chance of losing a trivial amount can have the same expected loss, and so the same rating. Different agencies emphasise one or the other, which explains why the same tranche can be rated differently and why investors should read the methodology as well as the letter grade.

Whichever philosophy is used, the analysis has the same architecture:

1. **Collateral analysis.** Estimate the pool's loss distribution: base-case default and recovery assumptions from historical vintage data, adjusted for pool composition (credit scores, loan-to-value, industry concentration, obligor concentration), and the correlation among assets. For static ABS pools the agencies use cumulative net loss expectations and stress multiples. For CLOs they use the WARF to determine a default distribution based on rating-factor tables, the diversity score to account for concentration, and recovery-rate assumptions that depend on asset seniority.
2. **Stress levels by rating category.** A higher rating requires the structure to survive a harsher scenario. The stress on the default rate and on recoveries increases with the target rating: AAA must survive the toughest scenario, BB a much milder one.
3. **Cash flow modelling.** The agency runs the actual waterfall under many scenarios of default timing (early, late, uniform), recovery lag, interest rate paths, prepayment and spread compression, to see whether each tranche is paid. This is where triggers, OC tests, and hedges are tested.
4. **Legal and operational review.** True sale and non-consolidation opinions, servicer quality and back-up servicing, swap counterparty rating triggers, commingling risk, and governing documents.
5. **Committee judgment.** Quantitative output is reviewed by a committee, which may adjust for factors the model does not capture.

6.7.2 The weighted average rating factor

For a CLO, the WARF summarises the credit quality of the pool as a single number. Each obligor's rating is mapped to a **rating factor**, approximately a proxy for the ten-year default probability in basis points, on a scale that rises steeply as ratings fall. The WARF is the weighted average of those factors. (The factors below are illustrative of the form such a scale takes.)

Example 6.7 – WARF and required enhancement. A $500 million loan portfolio has the following composition. Rating factors (illustrative): Ba2 = 940, B1 = 1,350, B2 = 1,766, B3 = 2,220, Caa1 = 2,720.

Rating	Share of portfolio	Rating factor	Contribution
Ba2	10%	940	94.0
B1	28%	1,350	378.0
B2	35%	1,766	618.1
B3	20%	2,220	444.0
Caa1	7%	2,720	190.4
WARF	100%		**1,724.5**

Step 1: WARF. The sum of the contribution column gives 1,724.5, which lies between B1 (1,350) and B2 (1,766) and is much closer to B2. The portfolio is an average single-B portfolio, typical of leveraged-loan CLOs.

Step 2: stress scenarios. Suppose the agency's *illustrative* stress for a pool of this quality, assuming the diversity score is adequate, is a cumulative default rate and a recovery assumption by target rating:

Target rating	Stress default rate	Stress recovery	Stress loss = default × (1 − recovery)	Credit enhancement in Example 6.5	Margin
AAA	55%	45%	30.25%	38%	7.75 pts
AA	45%	50%	22.50%	30%	7.50 pts
A	38%	52%	18.24%	24%	5.76 pts
BBB	30%	55%	13.50%	19%	5.50 pts
BB	22%	58%	9.24%	14%	4.76 pts

Step 3: enhancement in the structure. In Example 6.5, the equity is 70/500 = 14%, BB notes sit above it (14% to 19%), BBB from 19% to 24%, A from 24% to 30%, AA from 30% to 38% and AAA above 38%. So the credit enhancement (the attachment point) of each class is 14%, 19%, 24%, 30% and 38% as shown.

Step 4: reading the table. Every class survives its stress loss with a margin of between 4.8 and 7.8 points. In practice, agencies give some credit for excess spread (which reduces the enhancement needed) and apply cash-flow modelling that recognises timing. The point of the example is the method: the **required subordination for each rating level is the stressed pool loss for that level, and the structure is sized so that each tranche's attachment point exceeds it**.

6.7.3 Model risk in ratings

Rating methodologies compress a multi-dimensional risk into a single letter. Four sources of weakness deserve emphasis:

- **Dependence on historical data.** Default and recovery assumptions come from past experience; when underwriting standards deteriorate or an asset class is new, historical data may not be representative.
- **Correlation and systematic risk.** As Section 6.5 showed, the rating of senior and mezzanine tranches is acutely sensitive to correlation, and the correlation inputs are the least well supported by data.
- **Rating-factor mapping.** Ratings of underlying obligors measure through-the-cycle risk of default, but a CLO tranche is sensitive to the *joint* migration of the pool, and a single tranche rating does not communicate that sensitivity.
- **Conflicts of interest.** In the issuer-pays model, the agency is paid by the entity seeking the rating, creating an incentive to compete on favourable methodology and to be "shopped" by arrangers who choose the agency with the most generous assessment. Disclosure, multiple ratings and regulatory oversight are mitigants, not cures.

Rating *stability* is also different from rating *accuracy*. A structured tranche, especially a mezzanine tranche, is far more likely to experience large multi-notch downgrades than a similarly rated corporate bond because its risk is concentrated in a single common factor. A senior investor who regards a rating as a permanent label rather than a snapshot of a model is exposed to cliff risk.

6.8 Lessons from the Crises and Managerial Implications

6.8.1 What went wrong in structured credit failures

Without recounting any particular institution's history, it is possible to identify the recurring patterns that turned parts of the structured credit market into a source of systemic stress in the late 2000s, and which can recur in other forms:

1. **Weak underwriting through the originate-to-distribute chain.** When originators sell loans soon after origination and retain little risk, their incentive to screen borrowers weakens. Loan quality declined while pool-level statistics, built from the history of better underwriting, continued to look benign. This is a classic moral hazard problem: the party that can best assess credit quality is the party least exposed to its consequences.
2. **Reliance on a short, benign history.** Models calibrated to a period without a nationwide decline in house prices implied low correlation and thin tails. When house prices fell broadly, the systematic factor behind mortgage defaults exerted its full force and the assumed diversification vanished.
3. **Correlation misspecified and model risk.** The Gaussian copula's convenience (a single parameter) led to wide use and to the neglect of its limitations: no tail dependence, a correlation parameter that was not stable, and valuation of non-standard tranches by plugging in correlations observed for standard ones.
4. **Resecuritisation: leverage on leverage.** Re-packaging the mezzanine tranches of many mortgage securitizations into new "CDO" structures, whose senior tranches were rated highly, concentrated exposure to the same systematic factor. The diversification assumed in the second layer was illusory because the underlying assets shared the same risk. A tranche of tranches has enormous sensitivity to correlation: a modest increase in pool-level losses can wipe out many mezzanine tranches simultaneously, and the resecuritised senior notes, formally "protected" by subordination, are then hit.
5. **Ratings as a substitute for analysis.** Many investors, and regulatory capital rules, used ratings as the primary input, which created demand for any security that achieved a high rating regardless of the quality of the analysis behind it. Structures were engineered to just achieve a rating threshold, so that small errors in the model produced large downgrades.
6. **Maturity transformation and liquidity.** Vehicles that held long-dated structured assets and funded them with short-term paper (conduits, structured investment vehicles) were exposed when funding markets froze, even when the assets were performing. Market-value triggers forced asset sales into illiquid markets, depressing prices further. Cash-flow structures with no mark-to-market triggers, like most CLOs, fared better.
7. **Opacity and complexity.** Investors did not or could not look through multi-layer structures to the underlying loans; the complexity of waterfalls, triggers and cross-collateralisation made independent analysis costly. When doubt about asset quality arose, uncertainty about *which* securities were affected spread to all, and the market stopped functioning.
8. **Concentration of risk with the originators and insurers.** Some institutions retained the supposedly safe senior tranches on their own balance sheets, or insured them (for example through monoline guarantees or CDS) without

capital to withstand a systematic event. Risk that was apparently transferred came back through counterparty exposure and reputational support to vehicles.

A common thread is that the quantitative apparatus was not wrong in its algebra but in its *inputs and use*: models that were adequate to compare relative value were treated as reliable estimators of tail risk. That distinction between ranking and calibration, introduced for distance to default in Chapter 5, applies with equal force here.

6.8.2 The regulatory and market response

Reforms since the crisis have tended to focus on the incentive and transparency failures:

- **Risk retention** ("skin in the game"): in many jurisdictions sponsors must retain a meaningful economic interest, commonly around five percent, in the securitised exposure, and may not hedge it.
- **Higher and more risk-sensitive capital** for securitisation exposures, reducing reliance on external ratings and imposing floors on risk weights for even the most senior tranches.
- **Simple, transparent and comparable (STC)** criteria for securitisations that qualify for preferential treatment, along with standardised disclosure of loan-level data.
- **Limits on resecuritisation**, with higher capital charges for securitisations of securitisations.
- **Reforms to rating agency practice**, such as disclosure of methodologies, requirements for internal controls and rules on conflicts of interest.
- **Accounting and consolidation rules** that require an originator to consolidate vehicles it controls and to keep assets on the balance sheet where it has not transferred substantially all risks and rewards.

6.8.3 Managerial checklist

For a CFO or treasurer considering securitization as an issuer or an investor, the lessons translate into practical questions:

- **What risk have I actually transferred?** If I retain the first-loss piece, I retain most of the expected loss. Retention may be required, but it also limits capital relief and accounting derecognition. Compute the share of expected loss that stays with the sponsor: in Example 6.4, at $\rho = 30\%$ the equity tranche carries 2.67 of the 3.00 points of expected loss, or 89%.
- **Which assumptions drive the answer?** Run the waterfall under doubled losses, delayed recoveries, higher correlation, lower excess spread and prepayment shocks. Report results as ranges.
- **How do the triggers behave?** Test whether triggers cure, how much interest is diverted, and how long it takes. Examine the interaction between triggers and servicer behaviour.
- **Is the structure funded to survive illiquidity?** Prefer cash-flow structures over market-value triggers; match the term of funding to that of the assets; understand the committed liquidity lines.
- **Do I understand the counterparties?** Servicer, swap provider, guarantor, trustee. Identify wrong-way risk (the guarantor most likely to fail in the scenario in which the guarantee is needed).
- **Is the rating consistent with my own analysis?** As an investor, do not delegate underwriting to the rating; as an issuer, avoid structuring merely to a rating threshold, since any methodology change will then force a downgrade.
- **Does the structure serve a real funding or risk-transfer purpose?** If the only justification is regulatory arbitrage or rating arbitrage, the benefit is fragile.

Securitization is a durable and valuable tool: it broadens funding, matches risk to investors who want it, and can support lending to the real economy. The same arithmetic that makes a senior tranche safe under independence makes it vulnerable when defaults are correlated. Managers who understand both sides of that arithmetic can use the tool without being used by it.

Case Study: Alder Auto Finance Prices Its First Securitization

(Illustrative case; Alder Auto Finance is a fictional company and all figures are invented for teaching purposes.)

Alder Auto Finance, introduced in the opening vignette, is a regional non-bank lender with $900 million of used-car loans. The loans earn an average rate of 12.5% and have a weighted average life of about two years. Alder funds itself with a mix of bank lines and unsecured notes at an average cost of about 7.5%, and its lenders have begun to push it to diversify. CFO Priya Raman proposes securitizing a $400 million pool of seasoned loans.

Proposed structure. A newly formed trust will buy the pool and issue Class A notes of $312 million (78%) at a fixed rate of 5.20%, Class B notes of $48 million (12%) at 7.50%, and a $40 million residual (10%) retained by Alder. Alder will continue to service the loans for a fee of 1.00% per year of the pool balance. Transaction costs (legal, rating, underwriting) are 1.2% of notes sold. Rating agency analysis is expected to rate the Class A notes at a high investment-grade level and the Class B notes in the BBB category, provided the structure includes an OC trigger on the Class B notes and a non-declining reserve account of 1% of the pool. Excess spread is released to Alder only while the trigger is passed.

Historical performance. Alder's own vintage data show cumulative net losses of about 9% of original balance over the life of a pool, with the worst vintage, originated just before a regional recession, reaching 14%. The company's data cover only twelve years and one mild recession. The pool selected for sale has a slightly lower average credit score than the book as a whole, because the treasury team selected loans that were easiest to document.

Economics. With losses at 9% over a two-year average life, or roughly 4.5% per annum, the pool's annual economics look like this. The weighted average note coupon is (312 × 5.20% + 48 × 7.50%) / 400 = 4.956%. Gross interest of 12.5% less the note coupon (4.956%) less servicing (1.00%) leaves 6.544% of the pool per year as excess spread before losses, or about $26.2 million on a $400 million balance. After losses of 4.5%, 2.04% remains, about $8.2 million, which is roughly 20% on the $40 million retained piece. Alder also receives servicing income and the upfront cash, which it will lend again at 12.5%.

The debate. Tomas Idowu, head of structured finance, argues for a thinner retained piece: 5% rather than 10%, with 83% senior notes, to improve capital efficiency. Priya's chief risk officer, Elena Marsh, objects. She notes that under a recession scenario in which annual losses double to 9% of the pool, excess spread after losses becomes 6.544% − 9.0% = −2.456% per year, a shortfall of about $9.8 million annually, which the equity and reserve account would absorb within a few quarters. Cumulative losses of 18% over two years would approach, though not breach, the 22% subordination under Class A, but they would exceed the 10% equity cushion and erode the Class B notes. Elena also points out that the entire proposal assumes that auction values for repossessed cars remain stable, and that the correlation of borrower defaults with used-car prices is not captured in the historical vintages.

Finally, Priya's board asks two questions. First, whether the securitization would really lower Alder's cost of funds, given that Alder retains the first-loss piece, and second, what would happen to Alder's ability to fund itself in a market where investors lose confidence in auto ABS generally, as they did in a prior downturn. Priya has asked you to prepare a recommendation.

Discussion questions

1. Compute Alder's all-in funding cost on the securitised pool: the weighted average coupon on the $360 million of notes, the cost of the retained equity assuming a 15% required return on the $40 million, and the blended cost on $400 million. Compare it with the 7.5% unsecured funding cost, and explain why the blended number is the right comparison rather than the 4.956% note coupon.
2. Using a tranche-loss table like Table 6.1, compute losses to Class B and Class A for cumulative pool losses of 9%, 14%, 22% and 30%. How much of the expected loss does Alder retain with a 10% piece versus a 5% piece?
3. Explain the role of the OC trigger and reserve account in the proposal. Using the logic of Example 6.3, describe what happens to the equity cash flows when the losses double, and what Alder should expect to receive in years 2 and 3.
4. Critically assess the risks that the historical data may understate (correlation with used-car prices, underwriting drift, servicer concentration, funding-market liquidity). Recommend a retained percentage and any structural changes, and indicate what information you would ask Alder to produce before the deal is launched.

Key Takeaways

- Securitization moves assets to a bankruptcy-remote SPV and funds them by issuing tranched notes; its value to issuers comes from funding, capital and diversification frictions, not from any change in the total risk of the pool.
- Pool cash flows depend on amortisation, prepayment and default timing; CPR and SMM measure prepayment speed, and weighted average life, not legal maturity, is the right measure of a tranche's duration.
- Tranches are defined by attachment and detachment points, and each is economically a call spread on cumulative pool losses; losses flow up from the equity, principal flows down from the senior.
- Waterfalls with performance triggers redirect excess spread to senior holders when collateral deteriorates, and the same collateral can support tranches of very different safety depending on the document.
- Credit enhancement comes from subordination, overcollateralisation, excess spread, reserves, triggers and external support; each has costs and failure modes, and external guarantees introduce counterparty risk.
- Diversification makes senior tranches safe when defaults are nearly independent; with common factors the same tranches face much larger loss probabilities. Correlation redistributes expected loss from equity to mezzanine and senior tranches without changing the pool total.
- The Gaussian copula is convenient but has no tail dependence, a single correlation parameter that varies with tranche and time, and little empirical support in the tails; model outputs should be tested with correlation and tail stresses.
- A CLO is a cash-flow structure with a manager, reinvestment period, portfolio tests and an equity tranche whose return is a highly levered spread margin; OC and IC tests protect the senior notes by diverting equity cash.
- Rating agencies size subordination by stress scenarios (default rate and recovery by rating level), cash-flow modelling and legal review, but ratings are model outputs vulnerable to dependence on history, correlation assumptions and issuer-pays conflicts.
- Crises in structured credit came from weak underwriting, short data histories, resecuritisation, reliance on ratings, maturity mismatch and opacity; the managerial response is to keep risk where it can be monitored, stress assumptions and avoid structuring to a threshold.

Review Questions

1. Describe the roles of the originator, SPV, servicer, trustee and rating agency in a securitization. Why is true sale critical to the rating of the notes?
2. Explain why securitization can lower an originator's funding costs when it retains the first-loss tranche. In what circumstances would it fail to do so?
3. Define CPR and SMM, and explain how prepayment speed affects the WAL of a tranche and the yield of a discount or premium note.
4. Define attachment and detachment points. Why is each tranche economically a call spread on pool loss?
5. Contrast sequential and pro rata principal allocation. What are the advantages to senior and junior holders of each, and why do many deals switch from one to the other?
6. Explain how an OC trigger works, why it is "self-correcting", and why a trigger can transfer value from equity holders to the mezzanine.
7. List the main forms of credit enhancement. Which are internal and which external, and what are the principal risks of each?
8. Why does an increase in asset correlation raise the expected loss on senior tranches and reduce it on equity tranches? What is the effect on the pool's total expected loss?
9. Describe the main sources of return and risk for CLO equity. Why is the structure said to be less vulnerable to forced selling than a market-value structure?
10. Summarise the lessons of structured-credit crises for (a) originators, (b) investors and (c) rating agencies and regulators.

Problems

1. **Tranche losses.** A $200 million pool is tranched into equity (0-8%), mezzanine (8-20%) and senior (20-100%). Cumulative pool loss is 15%. Compute the loss in dollars and as a percentage of each tranche.
2. **Prepayment.** (a) Convert a CPR of 18% to an SMM. (b) A three-year level-pay loan with initial balance 100 and rate 6% has no prepayments. Compute the level payment, the principal each year and the WAL.
3. **OC test.** A trust has performing collateral of $118 million, Class A notes of $90 million and Class B notes of $20 million. The OC test for Class B requires a ratio of at least 1.10. Compute the ratio; if it fails, what happens to $2.0 million of excess interest, and does diverting it to repay Class A cure the test?
4. **Diversification.** A pool has 50 equal loans, each with a default probability of 4% and independent defaults. Compute the mean and standard deviation of the number of defaults, and the probability of 5 or more and 8 or more defaults.
5. **Conditional loss.** In a one-factor Gaussian copula with $\rho = 20\%$, PD = 4% and LGD = 50% in a large homogeneous pool, compute the conditional default rate and pool loss when the common factor is $m = -2.326$ (a one-in-a-hundred bad state).
6. **CLO equity.** A $400 million CLO has assets earning SOFR (3.00%) plus 3.50%, rated notes of $340 million with an average cost of SOFR plus 1.70%, equity of $60 million, fees of 0.40% of assets, annual default rate 2% and LGD 40%. Compute the equity cash flow and yield.
7. **CLO coverage test.** A $300 million CLO has $12 million of defaulted loans (recovery 40%) and $288 million of performing loans, of which $36 million are CCC-rated; the CCC limit is 7.5% of $300 million and excess CCC loans

are carried at 70%. Class A notes are $210 million (trigger 135%) and Class A+B notes are $250 million (trigger 118%). Compute adjusted par and each test, and how much senior principal must be repaid to cure any failure.

8. **WARF.** A portfolio has 15% Ba2 (940), 30% B1 (1,350), 30% B2 (1,766), 15% B3 (2,220) and 10% Caa1 (2,720). Compute the WARF and state which rating category it is nearest to.

Answers to Selected Problems

1. Equity loses $16.0 million, 100% of its $16 million; mezzanine loses (15% − 8%) × 200 = $14.0 million, which is 58.3% of its $24 million; senior loses nothing. Total losses are $30 million = 15% of $200 million.
2. a. SMM = 1 − (1 − 0.18)^(1/12) = 1.64% per month. (b) Payment = 100 × 0.06 / (1 − 1.06^(−3)) = 37.41; principal is 31.41, 33.30 and 35.29; WAL = (31.41 + 2 × 33.30 + 3 × 35.29) / 100 = 2.04 years.
3. Ratio = 118/110 = 1.073, below 1.10, so the test fails and the $2.0 million of excess interest is diverted to repay Class A rather than being paid to equity. After repayment, notes are $108 million and the ratio is 118/108 = 1.093, still below 1.10, so the test is not yet cured; additional diversion of about $2.7 million in total (118/1.10 = 107.27, so reduce notes by $2.73 million) is needed.
4. Mean = 50 × 0.04 = 2.0 defaults; standard deviation = √(50 × 0.04 × 0.96) = 1.386; P(≥5) = 4.90%; P(≥8) = 0.078%.
5. c = $\Phi^{-1}(0.04) = -1.7507$; $\sqrt{\rho} = 0.4472$; $\sqrt{(1-\rho)} = 0.8944$. Conditional default rate = $\Phi[(-1.7507 + 0.4472 \times 2.326) / 0.8944] = \Phi(-0.7944) = 21.35\%$; pool loss = 0.50 × 21.35% = 10.7%, over five times the unconditional expected loss of 2.0%.
6. Asset interest = 400 × 6.50% = $26.00 million; note interest = 340 × 4.70% = $15.98 million; fees = $1.60 million; credit loss = 400 × 0.02 × 0.40 = $3.20 million. Equity cash flow = 26.00 − 15.98 − 1.60 − 3.20 = $5.22 million, a yield of 8.7% on $60 million.
7. CCC limit = $22.5 million; excess = 36 − 22.5 = $13.5 million, haircut = 13.5 × 0.30 = $4.05 million; recovery on defaults = 12 × 0.40 = $4.8 million. Adjusted par = 288 − 4.05 + 4.8 = $288.75 million. Class A test = 288.75/210 = 137.5%: pass. Class A+B test = 288.75/250 = 115.5%: fail (below 118%). Cure: 288.75/(250 − x) = 1.18 gives x ≈ $5.3 million of Class A principal.
8. WARF = 0.15 × 940 + 0.30 × 1,350 + 0.30 × 1,766 + 0.15 × 2,220 + 0.10 × 2,720 = 141 + 405 + 529.8 + 333 + 272 = 1,680.8, nearest to B2 (1,766): the gap to B2 is 85 points, against 331 points above B1 (1,350).

Chapter 7: Interest-Rate Models and Rate Derivatives

Learning Objectives

After completing this chapter you should be able to:

- Construct a discount curve from deposit and par swap quotes by bootstrapping, and convert freely among discount factors, zero rates, and forward rates.
- Price caplets, caps, floorlets, floors, and European swaptions with Black-76, interpret the inputs (forward rate, strike, volatility, annuity), and verify put-call parity.
- Explain how short-rate models (Ho-Lee, Vasicek, Cox-Ingersoll-Ross) generate a term structure, and state what each model gets right and wrong, including mean reversion, negative rates, and fit to today's curve.
- Compute zero-coupon bond prices in closed form under Vasicek and CIR and check them against a Monte Carlo simulation.
- Design and evaluate a hedge for a floating-rate liability portfolio using swaps, caps, and collars, including DV01 matching, cash-flow certainty, and the accounting and liquidity consequences of each choice.
- Recognize the main model risks (volatility smile, curve construction choices, negative rates, discounting conventions) and describe what a CFO or treasurer should ask of a quant team.

Opening Vignette

Illustrative scenario: Harbor Ridge Logistics is a fictional company.

Harbor Ridge Logistics, a regional freight and warehousing operator, financed a three-year fleet and depot expansion with a $300 million five-year term loan priced at the overnight benchmark rate plus 150 basis points, reset annually. When the loan was signed, the benchmark was near 3 percent and the CFO, Maria Okonkwo, treated the floating rate as a fair price for flexibility: the company expected to prepay part of the loan from asset sales within three years.

Eighteen months later the benchmark has climbed and the forward curve slopes upward. The board's audit committee asks three questions at the quarterly meeting. First, how much would annual interest cost rise if rates moved another 100 basis points? (Answer: $3 million per year for every 100 basis points while the full loan is outstanding, which is material against free cash flow of $22 million.) Second, what would it cost to buy a ceiling on the rate, and how does that compare with locking in a fixed rate through a swap? Third, what does it mean when the bank says that "the swap is worth a few million dollars against us" after rates rose, given that no cash changed hands on day one?

The treasury team has quotes from two banks. Bank A offers a five-year pay-fixed swap at a rate that, once the CFO checks the curve, matches the average of forward rates; Bank B quotes a five-year cap at a strike of 4.5 percent for a premium of about 2.2 percent of notional. The CFO's real task is not to choose between them by instinct. It is to understand how those quotes are built from a discount curve and a volatility assumption, so that she can tell whether the premium is fair, what the hedge leaves unhedged, and how the position will behave if the curve shifts, twists, or if volatility changes.

This chapter gives you the machinery behind those quotes: how curves are built, how Black-76 values caps and swaptions, what short-rate models say about the dynamics of rates, and how to turn all of it into a hedging decision for a floating-rate liability book. Volume 1, Chapter 4 introduced bond pricing and the term structure; Chapter 19 introduced swaps and hedging. Here we go one level deeper.

7.1 The Term Structure: Discount Factors, Zero Rates, and Forwards

7.1.1 Three equivalent descriptions

Everything in fixed-income pricing starts from one object: the *discount function* P(0,T), the price today of a risk-free zero-coupon bond paying 1 at time T. Given the discount function, every other quantity is a transformation of it.

```
Discount factor:          P(0,T)
Zero rate (annual comp): z(T) = P(0,T)^(-1/T) − 1
Zero rate (continuous):  y(T) = −ln P(0,T) / T
Forward rate (annual):   F(T1,T2) = [P(0,T1) / P(0,T2) − 1] / (T2 − T1)
Instantaneous forward:   f(0,T) = −∂ ln P(0,T) / ∂T
```

The forward rate F(T1,T2) is the rate that can be locked in today for borrowing or lending between T1 and T2, because it is implied by no-arbitrage. To see this, suppose you buy one unit of the T1-maturity bond and finance it by shorting P(0,T1)/P(0,T2) units of the T2-maturity bond. Net cost today is zero. At T1 you receive 1; at T2 you pay P(0,T1)/P(0,T2). The gross return over (T1,T2) is therefore P(0,T1)/P(0,T2) per unit invested, which is exactly 1 + F·(T2−T1). The forward rate is not a forecast; it is a price. Statements such as “the forward curve predicts that rates will rise” are loose. A forward rate is the break-even rate at which a locked-in forward loan is neither a gain nor a loss, which blends expected future rates with a risk premium and convexity effects.

The yield on a coupon bond is a weighted average of zero rates; the zero rate is a geometric average of one-period forward rates:

```
(1 + z(T))^T = (1 + F(0,1)) × (1 + F(1,2)) × … × (1 + F(T−1,T))
```

Hence an upward-sloping zero curve implies forwards that lie above the zero rates, and an inverted curve implies forwards below. This identity is the most useful mental check you can apply to any quoted curve.

7.1.2 Why a single “interest rate” is never enough

Many corporate models use one discount rate for all cash flows. That is acceptable for a rough NPV (Volume 1, Chapter 6), but it fails for any instrument whose payoff depends on the timing of rates: a floating-rate loan, a callable bond, or an interest-rate swap. In a swap the fixed leg is discounted at zero rates for each payment date, and the floating leg’s expected payments are the forward rates. Using one rate throughout would price the swap incorrectly whenever the curve is not flat.

The market also uses different curves for different purposes. After the global financial crisis, practitioners separated the curve used to *project* floating payments from the curve used to *discount* collateralized derivatives. Today most derivatives are discounted on the overnight risk-free benchmark curve and projected on the benchmark curve appropriate to the floating index. For clarity this chapter uses a single-curve world in which projection and discounting use the same curve, and we note where multi-curve practice changes things (Section 7.7).

7.1.3 Bootstrapping a curve

Market quotes arrive as a mixture of instruments: overnight and short deposits, futures, and par swap rates for tenors out to thirty years or more. A *par swap rate* S(n) is the fixed rate for which an n-year swap has zero value at initiation. For an annual-pay swap, par means that the PV of the fixed leg equals the PV of the floating leg, and the floating leg of a par swap is worth 1 − P(0,n). Hence:

```
S(n) × Σ(k=1..n) P(0,k) = 1 − P(0,n)
```

Solving for the final discount factor given the earlier ones:

```
P(0,n) = [1 − S(n) × Σ(k=1..n−1) P(0,k)] / [1 + S(n)]
```

This recursion is *bootstrapping*: start with the shortest instrument, solve for its discount factor, move to the next, and so on. The result is a set of discount factors that reprices every input exactly.

Example 7.1 – Bootstrapping an annual curve from par swap rates. A treasury desk observes the following annual-pay par swap rates:

Tenor (years)	1	2	3	4	5
Par swap rate	3.00%	3.40%	3.70%	3.90%	4.05%

Step 1. With one year only, P(0,1) = 1 / (1 + 0.0300) = 0.970874.

Step 2. For n = 2: P(0,2) = [1 − 0.0340 × 0.970874] / 1.0340 = (1 − 0.033010) / 1.0340 = 0.966990 / 1.0340 = 0.935194.

Step 3. For n = 3: the sum of earlier factors is 0.970874 + 0.935194 = 1.906068. P(0,3) = [1 − 0.0370 × 1.906068] / 1.0370 = (1 − 0.070525) / 1.0370 = 0.929475 / 1.0370 = 0.896312.

Step 4. For n = 4: the sum is 1.906068 + 0.896312 = 2.802380. P(0,4) = [1 − 0.0390 × 2.802380] / 1.0390 = (1 − 0.109293) / 1.0390 = 0.890707 / 1.0390 = 0.857274.

Step 5. For n = 5: the sum is 2.802380 + 0.857274 = 3.659654. P(0,5) = [1 − 0.0405 × 3.659654] / 1.0405 = (1 − 0.148216) / 1.0405 = 0.851784 / 1.0405 = 0.818630.

Check: (1 − 0.818630) / (0.970874 + 0.935194 + 0.896312 + 0.857274 + 0.818630) = 0.181370 / 4.478284 = 4.050%, which reproduces the input.

The resulting curve:

Maturity	Discount factor	Zero rate (annual)	Zero rate (cont.)	1-year forward starting at T−1
1	0.970874	3.000%	2.956%	3.000%
2	0.935194	3.407%	3.350%	3.815%
3	0.896312	3.716%	3.649%	4.338%
4	0.857274	3.925%	3.850%	4.554%
5	0.818630	4.084%	4.003%	4.721%

Notice that the par rates rise with tenor, the zero rates rise faster than par rates, and the forwards rise faster still. The five-year par rate of 4.05 percent is a weighted average of the five forwards (3.00, 3.82, 4.34, 4.55, 4.72), weighted by the discount factors. That is the sense in which a swap “locks in the average of the forwards.”

7.1.4 Interpolation and the problem of gaps

Real markets quote swaps only at selected tenors (1, 2, 3, 4, 5, 7, 10, 15, 20, 30 years). Discount factors between nodes require an interpolation rule, and the rule matters for forwards, which are derivatives of the curve. Linear interpolation of zero rates produces forwards that jump at the nodes. Linear interpolation of the logarithm of discount factors gives piecewise constant instantaneous forwards, which is stable and arbitrage-free but not smooth. Cubic splines and monotone convex methods give smoother forwards but can overshoot between nodes and produce negative forwards when the data are noisy. Desks choose a method for stability of hedge ratios as well as for fit. The managerial implication is simple: *bucketed risk numbers depend on the curve-building convention*, so a treasury should ask the dealer which convention underlies the risk report and keep it constant over time.

7.1.5 Common mistakes

- *Mixing compounding conventions.* A 4 percent annual-compounded rate and a 4 percent continuously compounded rate differ by 8 basis points. Always state the convention.
- *Treating forwards as forecasts.* In Example 7.1, the forward rate for year five is 4.72 percent. This is the price at which the market clears a forward loan today; the realized rate may differ substantially.
- *Using a par yield as a zero rate.* A five-year par swap rate of 4.05 percent is not the five-year zero rate of 4.08 percent. The difference grows with the steepness of the curve.
- *Ignoring day counts.* An actual/360 floating leg and a 30/360 fixed leg do not have identical accrual fractions. Our examples use annual periods of exactly one year for clarity; production systems do not.

7.2 Interest-Rate Derivatives: The Product Map

Before turning to valuation it helps to organize the instruments a corporate treasury meets. All are combinations of two building blocks: *forward-rate agreements* (linear payoffs on a forward rate) and *options on forward rates* (convex payoffs).

Instrument	Payoff on each date (notional N, accrual τ)	Typical corporate use
Forward-rate agreement (FRA)	N·τ·(L − K) / (1 + L·τ), paid at start of period	Lock one future reset
Interest-rate swap (pay fixed)	N·τ·(L − K) at each date, L = floating rate	Convert floating debt to fixed
Cap (strip of caplets)	N·τ·max(L − K, 0)	Ceiling on floating cost; keep downside benefit
Floor (strip of floorlets)	N·τ·max(K − L, 0)	Protect floating-rate income, or sell to fund a cap
Collar	Long cap, short floor	Cheap corridor for cost
European payer swaption	Option to enter a pay-fixed swap at K	Hedge a *contingent* borrowing or a callable-bond position
European receiver swaption	Option to enter a receive-fixed swap at K	Hedge a prepayment or call option embedded in assets

A swap is a portfolio of FRAs, one for each reset date, all struck at the same fixed rate. A cap is a portfolio of European call options on forward rates (caplets); a floor is a portfolio of puts. Their relationship is the interest-rate analogue of put-call parity: **long cap − long floor = pay-fixed swap** (with the same strike and dates), because for each period max(L − K, 0) − max(K − L, 0) = L − K.

Two details deserve emphasis. First, in the standard convention the caplet's rate L is set at the *start* of the accrual period and the payment is made at the *end*. This "set in advance, paid in arrears" structure is what makes the Black-76 formula, discussed next, so clean. Second, the first period of a cap normally has no optionality, because the rate is already known, and is excluded from the cap; our examples follow that convention.

7.3 Pricing Caps and Floors with Black-76

7.3.1 The Black-76 model for a forward rate

In 1976 Fischer Black showed how to price options on forward and futures prices by assuming the forward is lognormally distributed. The same idea applies to forward interest rates. Fix a caplet that pays at time $T+\tau$ the amount $N \cdot \tau \cdot \max(L(T) - K, 0)$, where $L(T)$ is the rate observed at T for the period $(T, T+\tau)$. Let F be today's forward rate for that period. Under the measure in which the $T+\tau$-maturity zero-coupon bond is the numeraire (the "forward measure"), F is a martingale. Assume it is lognormal with volatility σ:

```
dF / F = σ dW   (under the T+τ forward measure)
```

Then the caplet price is the discounted expected payoff:

```
Caplet = N × τ × P(0, T+τ) × [ F·Φ(d1) − K·Φ(d2) ]
Floorlet = N × τ × P(0, T+τ) × [ K·Φ(−d2) − F·Φ(−d1) ]

d1 = [ ln(F/K) + ½σ²T ] / (σ√T)
d2 = d1 − σ√T
```

Φ is the standard normal cumulative distribution function. Several features of this formula should become reflexes:

1. **It has the Black-Scholes form** with the forward F replacing the spot grown at the risk-free rate, and the discount factor $P(0,T+\tau)$ replacing $\exp(-rT)$. There is no drift term: the forward already embeds the cost of carry. (Volume 1, Chapter 18 derived Black-Scholes; Black-76 is its forward-based sibling.)
2. **The volatility is the volatility of the forward rate, in proportional terms.** A "lognormal vol of 25 percent" at a forward of 4 percent means a standard deviation of roughly 1 percentage point per year in the rate. When rates are 0.5 percent, a 25 percent vol means 12 basis points, which is why markets quote *normal* (basis point) volatilities for low-rate currencies: Bachelier's normal model is the convention in such environments.
3. **T is the time to the rate fixing**, not to payment. Volatility accumulates until the rate is set.
4. **The payoff is paid at $T+\tau$**, so discounting uses $P(0,T+\tau)$.

7.3.2 Caps as strips, and the vol you are quoted

A cap is the sum of its caplets. Quotes in the market are expressed in terms of a single *flat vol* for the entire cap, which is the same lognormal vol applied to every caplet so that the total matches the market price. Because caplets of different expiries trade at different vols in reality, the market also infers *spot* (caplet) volatilities by stripping. Therefore a treasurer who is told "the five-year cap is quoted at 25 percent vol" should recognize that this is one summary number for a strip with different forwards and different expiries, and that the same single number applied to a different strike would not reproduce that strike's price (the *volatility smile* or *skew*; see Section 7.7).

Example 7.2 – Pricing a caplet and a floorlet. Using the curve of Example 7.1, price a caplet on a notional of 100 million that covers the year from $t = 3$ to $t = 4$, with strike $K = 4.00$ percent and flat volatility 25 percent.

Step 1. Forward rate: F = P(0,3)/P(0,4) − 1 = 0.896312 / 0.857274 − 1 = 4.5538%.

Step 2. Time to fixing T = 3, so σ√T = 0.25 × √3 = 0.25 × 1.732051 = 0.433013.

Step 3. ln(F/K) = ln(0.045538 / 0.04) = ln(1.13845) = 0.12967. Add ½σ²T = 0.5 × 0.0625 × 3 = 0.09375. Sum = 0.22342. d1 = 0.22342 / 0.433013 = 0.5160, and d2 = 0.5160 − 0.4330 = 0.0830 (0.0829 with unrounded inputs).

Step 4. Φ(d1) = Φ(0.5160) = 0.6971 and Φ(d2) = Φ(0.0829) = 0.5331.

Step 5. Caplet per unit notional = P(0,4) × τ × [F Φ(d1) − K Φ(d2)] = 0.857274 × 1 × [0.045538 × 0.6971 − 0.04 × 0.5331] = 0.857274 × [0.031744 − 0.021322] = 0.857274 × 0.010422 = 0.008933.

On 100 million notional, the caplet is worth **$893,300**.

Step 6. Floorlet = 0.857274 × [0.04 × Φ(−0.0829) − 0.045538 × Φ(−0.5160)] = 0.857274 × [0.04 × 0.4669 − 0.045538 × 0.3029] = 0.857274 × [0.018676 − 0.013794] = 0.004186 per unit, or **$418,600** on 100 million.

Step 7. Parity: caplet − floorlet = 0.008933 − 0.004186 = 0.004747. The forward swap-style value is P(0,4) × (F − K) = 0.857274 × (0.045538 − 0.04) = 0.004747. They agree.

The caplet is in the money on a forward basis (F exceeds K by 55 basis points), so its price is dominated by intrinsic value, 0.004747 per unit, plus time value of 0.004186, which equals the floorlet price. That decomposition (call = intrinsic forward value + put) is parity at work.

Example 7.3 – Pricing a four-year cap on 300 million. Harbor Ridge wants a ceiling at 4.50 percent on its loan. The first reset (t = 0 to 1) is already known, so the cap contains four caplets, fixing at t = 1, 2, 3, 4. Use the Example 7.1 curve, a flat vol of 25 percent, and notional 300 million.

Fixing T	Forward F(T,T+1)	d1	d2	Caplet (% of notional)
1	3.815%	−0.5353	−0.7853	0.1473%
2	4.338%	0.0731	−0.2805	0.4861%
3	4.554%	0.2439	−0.1891	0.6885%
4	4.721%	0.3457	−0.1543	0.8387%

To verify the third row: ln(0.045538 / 0.045) = 0.011883; ½σ²T = 0.09375; sum 0.105633; σ√T = 0.433013; d1 = 0.2439; d2 = 0.2439 − 0.4330 = −0.1891. Φ(0.2439) = 0.5963, Φ(−0.1891) = 0.4250. The caplet is 0.857274 × (0.045538 × 0.5963 − 0.045 × 0.4250) = 0.857274 × (0.027154 − 0.019125) = 0.857274 × 0.008029 = 0.006883, matching the table to rounding.

Sum of caplets: 0.1473% + 0.4861% + 0.6885% + 0.8387% = 2.1606% of notional. The cap premium is **$6.48 million** upfront. Expressed as a running cost, dividing by the sum of discount factors for the payment dates (P(0,2)+P(0,3)+P(0,4)+P(0,5) = 3.507410) gives about 61.6 basis points per year of notional over the four capped years.

Interpretation. If the cap’s holder believes forward rates are unbiased, the cap is fairly priced. The decision to buy depends on whether the company values protection against the tail (financial distress costs, covenant headroom; Volume 1, Chapter 15) more than the $6.48 million premium. A cap on a $300 million loan costs about 62 basis points per annum. If Harbor Ridge’s alternative is a swap with no upfront premium, the swap locks the cost; the cap pays only when rates exceed the strike.

7.3.3 What drives cap value

Driver	Direction	Reason
Higher forward rates	Cap value up	Caplets become more in the money
Higher strike	Cap value down	Fewer states in which it pays
Higher vol	Cap value up	Convex payoff gains from dispersion; vega is positive
Longer maturity	Cap value up	More caplets and more time for vol to accumulate
Higher discount rates	Cap value down (small)	Discounting; but also shifts the forward upward, so net effect depends on whether you hold the forward fixed

Vega deserves particular attention. For an at-the-money caplet with fixing T, vega per unit vol is approximately $P(0,T+\tau)\,\tau\,F\,\sqrt{T}\,\varphi(d1)$, where φ is the standard normal density. At ATM, $\varphi(0) = 0.3989$. This is why long-dated caps carry large vega exposure: a one-point change in vol on a five-year cap moves its price more than a one-point change on a one-year cap. A buyer of a cap is long vega; a seller of a floor to finance the collar is short vega on the floor leg. A zero-cost collar is therefore nearly vega-neutral if the floor and cap are placed symmetrically around the forward.

7.3.4 Common mistakes

- *Using the zero rate rather than the forward as F.* The caplet depends on the forward for its own accrual period, not on the cap's tenor zero rate.
- *Using the payment date for T in the volatility term.* The relevant time is the fixing date.
- *Quoting one flat vol across different strikes.* A cap at 6 percent strike and a cap at 3 percent strike trade at different implied vols because the distribution of rates is not lognormal.
- *Forgetting that the premium is paid upfront.* A cap premium is an immediate cash outflow, with implications for liquidity and for hedge accounting (Section 7.6).

7.4 Swaptions and the Swap-Rate Version of Black

7.4.1 What a swaption is

A European *payer swaption* gives the holder the right, at expiry T_0, to enter a swap in which she pays fixed at strike K and receives floating over the following n years. A *receiver swaption* is the right to receive fixed. Payers rise in value as rates rise; receivers rise as rates fall. Swaptions are priced in the form "expiry × tenor," so a "2y×3y" swaption is a two-year option on a three-year swap.

For corporates, swaptions are useful in three situations. A company that has *committed to borrow* in the future (for instance, a bond issue or acquisition financing after board approval but before market launch) can buy a payer swaption to cap the cost of the fixed rate it will lock. A company with a *callable bond* has, in effect, sold a receiver swaption to itself: the right to retire fixed-rate debt when rates fall is an option to receive fixed (the old coupon) and pay floating (new borrowing). Finally, a company with prepayable *assets* such as mortgages or leases has effectively sold a receiver swaption to its customers.

7.4.2 Black's swaption formula

Define the forward swap rate S as the fixed rate that makes the forward-starting swap worth zero, and the *annuity* (or PV01) A as the PV of receiving 1 per period during the swap's life:

```
A = Σ(i=1..n) τ_i × P(0, T_i)
S = [ P(0,T_0) − P(0,T_n) ] / A
```

Under the *annuity measure*, where the annuity is the numeraire, S is a martingale. Assuming it is lognormal with volatility σ yields:

```
Payer swaption    = N × A × [ S·Φ(d1) − K·Φ(d2) ]
Receiver swaption = N × A × [ K·Φ(−d2) − S·Φ(−d1) ]

d1 = [ ln(S/K) + ½σ²T_0 ] / (σ√T_0),   d2 = d1 − σ√T_0
```

The structure is the same as for caplets, with the annuity A taking the place of $\tau \times P(0,T+\tau)$. Parity: payer − receiver = N × A × (S − K), the value of a forward-starting swap.

Example 7.4 – A 2y×3y payer swaption. A company expects to issue three-year floating-rate debt two years from now and wants to protect against a rise in the swap rate. Use the Example 7.1 curve; the underlying swap has annual payments at t = 3, 4, 5; strike K = 4.50 percent; flat swaption vol 22 percent; notional 100 million.

Step 1. Annuity: A = P(0,3) + P(0,4) + P(0,5) = 0.896312 + 0.857274 + 0.818630 = 2.572216.

Step 2. Forward swap rate: S = [P(0,2) − P(0,5)] / A = (0.935194 − 0.818630) / 2.572216 = 0.116564 / 2.572216 = 4.5317%.

Step 3. $\sigma\sqrt{T_0} = 0.22 \times \sqrt{2} = 0.22 \times 1.414214 = 0.311127$. ln(S/K) = ln(0.045317 / 0.045) = ln(1.007044) = 0.007019. $\tfrac{1}{2}\sigma^2 T_0 = 0.5 \times 0.0484 \times 2 = 0.0484$. d1 = (0.007019 + 0.0484) / 0.311127 = 0.17810. d2 = 0.17810 − 0.31113 = −0.13303.

Step 4. Φ(0.1781) = 0.5707; Φ(−0.1330) = 0.4471 (so Φ(d2) = 0.4471, Φ(−d2) = 0.5529, Φ(−d1) = 0.4293).

Step 5. Payer = 2.572216 × [0.045317 × 0.5707 − 0.045 × 0.4471] = 2.572216 × [0.025863 − 0.020120] = 2.572216 × 0.005743 = 0.014771 per unit notional, or **$1.477 million** on 100 million.

Step 6. Receiver = 2.572216 × [0.045 × 0.5529 − 0.045317 × 0.4293] = 2.572216 × [0.024881 − 0.019456] = 0.013956, or **$1.396 million**.

Step 7. Parity: 0.014771 − 0.013956 = 0.000815, and A × (S − K) = 2.572216 × 0.000317 = 0.000815.

Interpretation. The strike is only 3 basis points below the forward swap rate, so the swaption is nearly at the money and its price, about 1.48 percent of notional, is almost entirely time value. The break-even logic is useful: the payer starts to pay off in net terms when the swap rate in two years is higher than K plus the premium amortized over the annuity. The premium of 1.477 percent of notional, divided by the annuity 2.572, adds about 57 basis points per year to the effective strike on the three-year swap, so the effective locked rate is roughly 5.07 percent (ignoring financing of the premium).

7.4.3 Relationship between caps and swaptions

Caps and swaptions respond to different aspects of the volatility surface. A cap's value depends on the *individual forward-rate* vols, while a swaption's depends on the vol of a *swap rate*, which is a weighted average of forwards. If the forward rates were perfectly correlated, the swap rate vol would equal an appropriate average of caplet vols; because forwards are

not perfectly correlated, swaption vols are lower than that average, and the gap reflects the *correlation* between forward rates. This is why a model that fits both caps and swaptions must have a rich correlation structure, and why the one- or two-factor short-rate models below can fit one market well and the other less well.

7.5 Short-Rate Models

7.5.1 Why model the short rate?

Black-76 prices one instrument at a time, taking the forward rate or swap rate and its volatility as given. It does not tell us how rates *evolve*, so it cannot price products whose payoffs depend on the path of rates or on early exercise (a Bermudan swaption, a callable bond, an amortizing swap with prepayments). For these we need a *dynamic model*. The simplest such family models only the instantaneous short rate r(t) as a stochastic process. In an arbitrage-free model, the price of a zero-coupon bond is the risk-neutral expectation of the discount factor along the path:

```
P(0,T) = E_Q[ exp( −∫₀ᵀ r(s) ds ) ]
```

Given a model for r, this expectation gives a term structure of prices, volatilities, and correlations as by-products. A good model should: (i) match the observed discount curve today; (ii) generate realistic volatility and mean reversion; (iii) avoid implausible outcomes such as large negative rates (or at least manage them); and (iv) be tractable enough to calibrate and hedge.

7.5.2 The Ho-Lee model

Thomas Ho and Sang-Bin Lee (1986) proposed the first arbitrage-free term-structure model. The short rate has a time-dependent drift and a constant volatility:

```
dr = θ(t) dt + σ dW
```

The function θ(t) is chosen so that the model reproduces *any* initial discount curve exactly. In continuous time, $\theta(t) = \partial f(0,t)/\partial t + \sigma^2 t$, where f(0,t) is the instantaneous forward curve. Two consequences follow. First, the expected short rate is $E[r(t)] = f(0,t) + \tfrac{1}{2}\sigma^2 t^2$: it follows the forward curve plus a small *convexity* term that grows with the square of time. Second, because there is no mean reversion, the variance of r(t) grows linearly in t without bound; long-dated rates can wander far from the current curve, and the model can generate large negative rates.

The upside is analytic tractability. Zero-coupon bond prices have the exact form

```
P(t,T) = [ P(0,T) / P(0,t) ] × exp( (T−t)·f(0,t) − ½σ²t(T−t)² − (T−t)·r(t) )
```

and European options on zero-coupon bonds have a Black-like closed form with bond-price volatility $\sigma_P = \sigma\,(S - T)\,\sqrt{T}$ for an option expiring at T on a bond maturing at S. Also, Ho-Lee gives the classic *futures-forward convexity adjustment*: forward rates equal futures rates minus $\tfrac{1}{2}\sigma^2 t_1 t_2$ (the difference between daily-settled futures and forward contracts caused by the correlation between rates and the margin account's reinvestment).

Example 7.5 – A Ho-Lee option on a bond and a convexity adjustment. Use the curve of Example 7.1 with σ = 1.00 percent per year. Price a one-year European call on a bond that has four years remaining at expiry (that is, a zero-coupon bond maturing at S = 5, option expiry T = 1), strike K = 0.84 per unit face.

Step 1. Inputs: P(0,1) = 0.970874 and P(0,5) = 0.818630. The forward price of the bond is P(0,5)/P(0,1) = 0.843189, slightly above the strike of 0.84.

Step 2. Bond-price volatility: σ_P = σ (S − T) √T = 0.01 × 4 × 1 = 0.0400.

Step 3. h = ln[P(0,5) / (K × P(0,1))] / σ_P + σ_P / 2. The ratio is 0.818630 / (0.84 × 0.970874) = 0.818630 / 0.815534 = 1.003796; ln = 0.003789; divided by 0.04 = 0.09472; plus 0.02 gives h = 0.11472.

Step 4. Φ(h) = Φ(0.11472) = 0.5457; Φ(h − σ_P) = Φ(0.07472) = 0.5298.

Step 5. Call = P(0,5) Φ(h) − K P(0,1) Φ(h − σ_P) = 0.818630 × 0.5457 − 0.815534 × 0.5298 = 0.446721 − 0.432071 = 0.01464 per unit face, or $1.46 million per 100 million face.

Step 6. Put = K P(0,1) Φ(σ_P − h) − P(0,5) Φ(−h) = 0.815534 × 0.4702 − 0.818630 × 0.4543 = 0.383465 − 0.371917 = 0.01155.

Step 7. Check by parity: call − put = 0.01464 − 0.01155 = 0.00310, and P(0,5) − K P(0,1) = 0.818630 − 0.815534 = 0.00310.

Convexity adjustment. For a Eurodollar-style futures contract on a rate for the period from t_1 = 2 to t_2 = 2.25 years with Ho-Lee σ = 1 percent, the adjustment is $\tfrac{1}{2}\sigma^2 t_1 t_2$ = 0.5 × 0.0001 × 2 × 2.25 = 0.000225 = 2.25 basis points. For a ten-year strip, with t_1 = 9.75 and t_2 = 10, the same formula gives 0.5 × 0.0001 × 97.5 = 48.75 basis points, which shows why ignoring the adjustment in a long futures strip creates a bias in the bootstrapped curve.

The Ho-Lee model's shortcomings are the flip side of its tractability. Since all forward rates shift by the same random amount, the model implies perfectly correlated parallel moves, and it cannot fit a term structure of volatilities.

7.5.3 The Vasicek model

Oldřich Vasicek (1977) introduced *mean reversion* into a no-arbitrage term-structure model:

```
dr = a (b − r) dt + σ dW
```

Here a > 0 is the speed of mean reversion, b is the long-run mean level of the short rate, and σ is the volatility. When r exceeds b the drift is negative; when r lies below b the drift is positive. The half-life of a deviation is ln 2 / a; for a = 0.30 it is 2.3 years.

The short rate under Vasicek is Gaussian. Its long-run distribution is normal with mean b and standard deviation σ/√(2a); with a = 0.30 and σ = 1.2 percent, the stationary standard deviation is 1.2% / √0.6 = 1.55%. Because the distribution is normal, the model gives rates a small but positive probability of being negative: with b = 5% the probability is Φ(−5/1.55) = Φ(−3.23), about 0.06 percent in the long-run distribution. At low rate levels or for high σ, this becomes a serious feature, and was historically viewed as a defect. Since 2008, when several currencies experienced negative policy rates, it has been viewed as a feature.

Zero-coupon bond prices are known in closed form:

```
P(0,T) = A(T) × exp( −B(T) × r₀ )
B(T) = [ 1 − exp(−aT) ] / a
A(T) = exp{ ( b − σ²/(2a²) ) × [ B(T) − T ] − σ² B(T)² / (4a) }
```

The long-run continuous yield (as T → ∞) is b − σ²/(2a²) rather than b, because of a Jensen's-inequality adjustment that lowers long bond yields. Volatility of the T-year yield declines as maturity increases, as B(T)/T falls: mean reversion implies that long rates are less volatile than short rates, which matches the data, and which Ho-Lee cannot replicate.

Example 7.6 – Vasicek term structure and a Monte Carlo check. Take a = 0.30, b = 5.00%, σ = 1.20%, r_0 = 3.00%. Compute P(0,5) and the continuous yields at several maturities, then verify with simulation.

Step 1. For T = 5: B = (1 − e^(−1.5)) / 0.30 = (1 − 0.223130) / 0.30 = 2.589566.

Step 2. The Jensen term: $\sigma^2/(2a^2) = 0.000144 / 0.18 = 0.0008$. So $b - \sigma^2/(2a^2) = 0.0492$. Then $(0.0492)(B - T) = 0.0492 \times (2.589566 - 5) = 0.0492 \times (-2.410434) = -0.118594$.

Step 3. $\sigma^2 B^2/(4a) = 0.000144 \times 6.706 / 1.2 = 0.000805$ ($B^2 = 6.70594$). $A = \exp(-0.118594 - 0.000805) = \exp(-0.119399) = 0.887454$.

Step 4. $P(0,5) = A \times \exp(-B\, r_0) = 0.887454 \times \exp(-2.589566 \times 0.03) = 0.887454 \times \exp(-0.077687) = 0.887454 \times 0.925257 = 0.82112$.

Step 5. Continuous yield = −ln(0.82112) / 5 = 0.19732 / 5 = 3.9417 percent.

Monte Carlo. Simulate $dr = a\,(b - r)\,dt + \sigma\,\sqrt{dt}\,Z$ using the Euler scheme with 250 steps over five years and 200,000 paths, seed 42. Accumulate $\int r\,dt$ as $\Sigma\, r_i\,dt$ and average $\exp(-\int r\,dt)$. The simulation (run in Python for this chapter) yields a mean of 0.82115 with a standard error of 0.00009, against the analytic 0.82112, a difference of about 0.3 standard errors.

Maturity T	B(T)	A(T)	P(0,T)	Continuous yield
1	0.8639	0.9932	0.96783	3.270%
2	1.5040	0.9756	0.93258	3.490%
5	2.5896	0.8875	0.82112	3.942%
10	3.1674	0.7136	0.64896	4.324%

The curve rises from the current 3 percent short rate toward the long-run 4.92 percent yield as mean reversion pulls the expected path upward. The slope of the curve is explained by the gap between r_0 and b; if r_0 exceeded b the curve would invert. This is a one-factor limitation: Vasicek's yield curve has a constrained shape set (rising, falling, or slightly humped), and the initial curve is an *output*, not an input. Fitting today's observed curve exactly requires extending the model with a time-dependent mean level, which leads to the Hull-White model.

7.5.4 The Cox-Ingersoll-Ross model

John Cox, Jonathan Ingersoll, and Stephen Ross (1985) addressed the negative-rate problem by making volatility proportional to the square root of the rate:

```
dr = a (b − r) dt + σ √r dW
```

As r approaches zero the diffusion term vanishes and the mean-reverting drift pushes r upward; if the *Feller condition* $2ab \geq \sigma^2$ holds, zero is never reached. The CIR short rate follows a non-central chi-square distribution and has a state-dependent volatility, which matches the empirical fact that rate volatility is higher when rates are higher. Zero-coupon bond prices remain closed-form:

```
γ = √(a² + 2σ²)
B(T) = 2(e^(γT) − 1) / [ 2γ + (a + γ)(e^(γT) − 1) ]
A(T) = { 2γ e^((a+γ)T/2) / [ 2γ + (a + γ)(e^(γT) − 1) ] }^(2ab/σ²)
P(0,T) = A(T) × exp( −B(T) × r₀ )
```

Example 7.7 – CIR prices and the Feller condition. Take a = 0.30, b = 5.00%, σ = 7.00%, r_0 = 3.00%. First, the Feller condition: $2ab = 2 \times 0.30 \times 0.05 = 0.0300$ and $\sigma^2 = 0.0049$. Since $0.0300 > 0.0049$, the process never reaches zero.

Step 1. $\gamma = \sqrt{(0.09 + 2 \times 0.0049)} = \sqrt{0.0998} = 0.31591$.

Step 2. For T = 5, e^(γT) = e^1.57955 = 4.8528, so e^(γT) − 1 = 3.8528. The denominator is 2γ + (a+γ)(e^(γT)−1) = 0.63182 + 0.61591 × 3.8528 = 0.63182 + 2.37299 = 3.00481.

Step 3. B(5) = 2 × 3.8528 / 3.00481 = 2.5644.

Step 4. A(5) = [2γ × e^((a+γ)×5/2) / 3.00481]^(2ab/σ²). The exponent is 0.03 / 0.0049 = 6.1224. The inside term is 0.63182 × e^(1.53978) / 3.00481 = 0.63182 × 4.6636 / 3.00481 = 0.98061, and raising to 6.1224 gives A(5) = 0.88702 (checked in Python).

Step 5. P(0,5) = 0.88702 × exp(−2.5644 × 0.03) = 0.88702 × 0.92586 = 0.82134. The continuous yield is 3.9365 percent.

Maturity T	Vasicek P(0,T)	CIR P(0,T)	Vasicek yield	CIR yield
1	0.96783	0.96783	3.270%	3.270%
5	0.82112	0.82134	3.942%	3.936%
10	0.64896	0.64998	4.324%	4.308%

The two models produce nearly identical yield curves here because their parameters were chosen so that volatility at a rate of 5 percent is similar (σ√r = 0.07 × 0.2236 = 1.57 percent against Vasicek's flat 1.20 percent). The distinction between them appears in the tails: CIR has a fatter right tail, never goes below zero, and gives a lower yield at long maturities because of the stronger convexity effect when volatility rises with the rate.

7.5.5 Comparison, extensions, and fit to the curve

Feature	Ho-Lee	Vasicek	CIR	Hull-White (extended Vasicek)
Dynamics	dr = θ(t)dt + σdW	dr = a(b−r)dt + σdW	dr = a(b−r)dt + σ√r dW	dr = [θ(t) − a r]dt + σdW
Mean reversion	No	Yes	Yes	Yes
Rates can be negative	Yes	Yes	No (Feller)	Yes
Fits today's curve exactly	Yes	No	No	Yes
Rate vol structure	Constant, absolute	Constant, absolute	Rises with level	Constant, absolute
Closed-form bond price	Yes	Yes	Yes	Yes
Typical use	Teaching, simple trees	Risk simulation, ALM	Insurers, positive-rate simulation	Bermudan swaptions, callable bonds

Practitioners generally need to fit the *current* curve exactly, because a model that misprices today's bonds creates an arbitrage against the market on day one and contaminates hedge ratios. That is why Hull-White, which adds a time-dependent θ(t) to Vasicek to fit the curve and a mean-reversion parameter to control the volatility term structure, became a market standard for Bermudan swaptions and callable-bond valuation. Beyond one-factor models, the *Heath-Jarrow-Morton* framework models the entire forward curve; the *LIBOR (or forward) market model* (Brace-Gatarek-Musiela; Jamshidian) models discrete forward rates directly and is consistent with Black-76 for caps; and the *SABR* model of Hagan and co-authors provides a smile for caplet and swaption vols. These topics are beyond this chapter, but they all share the structure we have described: a no-arbitrage drift determined by volatility, and a calibration to liquid options.

7.5.6 Risk-neutral versus real-world models

One subtlety trips up even experienced practitioners: the models above are typically specified under the *risk-neutral measure* Q, which is correct for pricing. For *risk management* (value-at-risk, economic-capital calculations, stress tests of net interest income, Chapter 8) we need the *real-world measure* P, where rates have different drifts reflecting term premia.

A Vasicek model calibrated to the cap market will not forecast the distribution of next year's rates well; a Vasicek model estimated from historical short-rate data will not price caps. A treasury or an insurer typically runs both: Q for valuation and P for scenario analysis, with a market-price-of-risk parameter linking them.

7.5.7 Common mistakes

- *Calibrating a model to the current curve using a single parameter set and then interpreting those parameters as forecasts.* A calibrated b is a risk-neutral mean, not an expected future policy rate.
- *Believing the "closed-form fit" implies the volatility structure is right.* Ho-Lee fits the curve but not the term structure of volatility.
- *Using too few Monte Carlo paths or time steps.* Euler discretization of CIR can produce negative rates unless a full-truncation or exact scheme is used.
- *Ignoring that the Feller condition is a property of calibrated parameters.* Calibrating CIR to market data often violates the condition, and many desks accept boundary hits with reflecting or absorbing conventions.

7.6 Hedging a Floating-Rate Liability Portfolio

7.6.1 Define the risk before choosing the instrument

A floating-rate liability has an unusual risk profile. Its *market value* barely moves when rates change, because the coupon resets to market (a floating-rate note resets to near par at each reset date), so a standard duration or DV01 on the liability is close to zero. Its *cash flows*, however, move one for one with the reference rate. The risk is therefore **cash-flow risk**: higher interest expense, lower coverage ratios, and in the extreme a covenant breach or a distress event. Conversely, a fixed-rate liability has market-value risk but no cash-flow risk. A treasurer must decide which of the two exposures the company actually cares about. A company with predictable operating cash flows that has board-set leverage and coverage covenants usually cares about cash-flow stability. An insurer or bank, whose assets and liabilities are both marked to economic value (Chapter 8), cares about value.

A clean sequence for the decision:

1. **Quantify the exposure.** Interest expense sensitivity: Notional × Δrate. For Harbor Ridge, 300 million × 1 percent = 3.0 million per year per 100 basis points on a flat-shift basis. Across five years, with the first reset already fixed, a persistent 100 basis point rise costs about 12 million (four years × 3 million).
2. **Link it to the objective.** Is the objective to protect interest coverage, a covenant, an earnings target, or a budget? Different objectives imply different strike levels and hedge ratios.
3. **Choose among linear and optional hedges.** A swap removes both the downside and the upside. A cap removes the downside and keeps the upside, at the cost of a premium. A collar sells part of the upside to pay for protection.
4. **Set the hedge ratio and tenor.** Many treasuries hedge 50 to 80 percent of expected debt, with a layered maturity ladder rather than a single point, to avoid locking in a single day's rate.
5. **Check accounting, liquidity, and counterparty consequences** (Section 7.6.4).

7.6.2 Swap, cap, and collar compared

Example 7.8 – Comparing a swap, a cap, and a collar for Harbor Ridge. Harbor Ridge has a 300 million five-year loan at benchmark + 1.50 percent, reset annually. The year-one rate is already set at 3.00 percent (all-in 4.50 percent, or 13.5 million). Years two through five reset at the forward rates of Example 7.1 (3.815, 4.338, 4.554, 4.721 percent for the one-

year periods starting at t = 1, 2, 3, 4, shifted in each scenario by a parallel move s).

Strategy A, unhedged. Interest cost = 13.5 + Σ 300 × (F_i + s + 1.5%) for the four resets. *Strategy B, pay-fixed swap at 4.05 percent.* The swap exchanges the benchmark for 4.05 percent, so all-in cost is 4.05% + 1.50% = 5.55 percent each year: 300 × 5.55% = 16.65 million per year, 83.25 million over five years, in every scenario. *Strategy C, cap at 4.50 percent (premium 6.48 million from Example 7.3).* Interest cost = unhedged cost − cap payoffs + premium. *Strategy D, collar:* buy the 4.50 percent cap, sell a 3.50 percent floor. Using the same flat vol of 25 percent, the four-year floor at 3.50 percent is worth 0.9373 percent of notional, or 2.81 million, so the net premium is 6.48 − 2.81 = 3.67 million. Cost = unhedged − cap payoffs + floor payments owed + 3.67 million.

For the cap and floor payoffs, we use the shifted forward rates as the realized benchmark and ignore discounting of payments for transparency (the figures below are undiscounted five-year totals in millions of dollars):

Parallel shift s	Unhedged	Swap	Cap (4.5%)	Collar (4.5% / 3.5%)
−2.00%	59.78	83.25	66.27	77.17
−1.00%	71.78	83.25	78.27	77.99
0.00%	83.78	83.25	89.44	86.63
+1.00%	95.78	83.25	91.98	89.17
+2.00%	107.78	83.25	91.98	89.17

To verify one cell, take s = +1.00 percent. Shifted forwards are 4.815, 5.338, 5.554, 5.721 percent. Cap payoffs are 300 × (0.315%, 0.838%, 1.054%, 1.221%) = 0.946, 2.514, 3.161, 3.662 million, total 10.283 million. Unhedged cost is 13.5 + 300 × (6.315% + 6.838% + 7.054% + 7.221%) = 13.5 + 300 × 0.27428 = 13.5 + 82.283 = 95.783 million. The cap strategy costs 95.783 − 10.283 + 6.482 = 91.982 million. The ceiling is visible: for years two to five the cap limits the all-in rate to 4.50% + 1.50% = 6.00 percent, which costs 300 × 6.00% × 4 = 72.0 million, plus 13.5 million in year one and 6.48 million of premium: 91.98 million.

Interpretation:

- The swap delivers certainty at a cost of 83.25 million. It is slightly cheaper than the unhedged forward-implied cost of 83.78 million in the base case (the difference of 0.53 million arises because the first period’s known rate is 3.00 percent, below the swap’s 4.05 percent fixed rate, while the swap’s par rate averages all five forwards). It loses badly if rates fall: at −2 percent it costs 23 million more than doing nothing.
- The cap is the “insurance” strategy. It costs about 5.7 million more than the unhedged base case, caps the worst outcome near 92 million, and retains most of the benefit of lower rates (66.27 million in the −2 percent case).
- The collar cuts the premium but gives up the gain from rate declines below 3.50 percent. At s = −2 percent the collar costs 77.17 million, 10.9 million more than the cap strategy.

Choosing among these is a decision about *risk preferences and distress costs*, not about which is “cheapest.” If the swap was priced fairly, it has zero expected value under the forward measure, and so does every fairly priced option; a manager adds value only to the extent that hedging reduces expected costs of financial distress, taxes, or underinvestment (the Volume 1, Chapter 19 rationale for hedging), or if the firm has a view that is not already in the curve. The honest case for a cap is that the firm’s *marginal cost of a rate shock is convex*: a 200-basis-point rise may breach a covenant, while a 100-basis-point rise would not. The honest case for a swap is that the firm values certainty for budgeting.

Mark-to-market of the swap. After entering the swap, the fixed rate is frozen at 4.05 percent. If par swap rates rise by 50 basis points in parallel, the five-year par rate becomes 4.55 percent, and the new annuity (recomputed from the shifted curve) is 4.4157. The payer swap is then worth (4.55% − 4.05%) × 4.4157 × 300 million = 0.005 × 4.4157 × 300 million = **6.62 million** in the company's favor. At inception the DV01, the change in swap value per 1 basis point, was 300 million × 4.4783 × 0.0001 = **$134,348**; after the shift, a slightly lower DV01 reflects the decline in the annuity. It is this mark-to-market, not a cash payment, that a bank means when it says the swap "is worth" a few million dollars. Such a gain is an economic offset to the higher interest expense. The swap loss in the reverse scenario would be a liquidity concern if the swap were collateralized, because the company would have to post cash (Section 7.6.4).

7.6.3 Portfolios: hedging by bucket, not by headline notional

Real floating-rate liabilities are not a single bullet loan. A portfolio typically includes revolving credit lines, term loans of different maturities, receivables-securitization funding, and floating-rate notes. The hedge must respect the *timing profile* of exposure, which is why practitioners compute *bucketed* DV01s: the sensitivity of the hedged position to a one-basis-point change in the swap rate for each tenor.

Example 7.9 – Matching a two-tranche floating-rate portfolio with swaps. Suppose that instead of one 300 million five-year loan, Harbor Ridge has two floating-rate tranches: 120 million that matures in two years and 180 million that matures in five years. The treasurer proposes to hedge 100 percent with two pay-fixed swaps at par: a two-year swap on 120 million at 3.40 percent and a five-year swap on 180 million at 4.05 percent (Example 7.1 rates). (For simplicity ignore the credit spread, which stays floating.)

Step 1. Annuities: $A_2 = P(0,1) + P(0,2) = 0.970874 + 0.935194 = 1.906068$, and $A_5 = 4.478283$ (from Example 7.1).

Step 2. DV01 (in dollars per basis point): swap 1: 120,000,000 × 1.906068 × 0.0001 = $22,873. Swap 2: 180,000,000 × 4.478283 × 0.0001 = $80,609. Total = $103,482 per basis point.

Step 3. Annual fixed payments: for years 1 and 2, 120 × 3.40% + 180 × 4.05% = 4.08 + 7.29 = 11.37 million; for years 3, 4, 5: 7.29 million. Add the credit spread of 1.50 percent on the outstanding principal (4.5 million in years 1–2 on 300 million; 2.7 million in years 3–5 on 180 million) to obtain all-in cost: 15.87 million in years 1–2 and 9.99 million in years 3–5.

Step 4. Blended fixed rate on the amortizing profile. Weighted by the PV of notional-weighted exposure, the portfolio's blended swap rate is (120 × 1.906068 × 3.40% + 180 × 4.478283 × 4.05%) / (120 × 1.906068 + 180 × 4.478283) = (7.7769 + 32.6454) / (228.728 + 806.091) = 40.4223 / 1034.819 = **3.906%**.

Step 5. Comparison with a naïve alternative: a single five-year swap on 300 million would hedge 300 million in years 3–5 even though only 180 million will still be outstanding. In years 3 to 5, Harbor Ridge would be *over-hedged* by 120 million, so a fall in rates would produce swap losses without any offsetting interest savings: a speculative position. The over-hedge costs 120 million × 1 percent = 1.2 million per year for each 100 basis points fall in rates.

The lesson: *a hedge ratio is a vector, not a scalar*. The notional profile of hedges must follow the debt amortization profile, and prepayment optionality must be considered: when the company has the right to repay debt early, a swap does not terminate automatically, and the company is left with a speculative position. In Volume 1, Chapter 19's language, the hedge is an *integral* part of the risk-management policy, not a separate trade. A *cancelable* or *extendable* swap, or a swaption, can be used when repayment timing is genuinely uncertain, but those embedded options cost money (Section 7.4).

7.6.4 Practical issues: accounting, liquidity, basis, and counterparty risk

Hedge accounting. Under IFRS 9 and US GAAP (ASC 815) a pay-fixed swap that converts floating-rate debt into synthetic fixed-rate debt can be designated a *cash flow hedge*. Fair-value changes on the effective portion go to other comprehensive income and are recycled into profit or loss as the hedged interest is incurred, which matches the swap result with the interest expense in the same period. Without designation, swap fair-value changes go through earnings and create volatility of reported profit that is the opposite of what the hedge intended. A cap's time value is treated differently from its intrinsic value under IFRS 9 (time value may be accounted for as a cost of hedging), which is why cap-based hedges may produce smoother earnings than they would under older rules. Treasury should align the hedge's critical terms (notional, reset dates, index) to the hedged debt to keep the hedge effective.

Liquidity and margin. A swap is a linear, two-sided contract. If rates fall, the swap has negative value and, if cleared or collateralized, requires margin. A cap bought for a premium has limited downside: the premium has been paid and no margin is ever posted by the buyer. A company with thin liquidity may prefer the cap for this reason alone, even if the swap is cheaper in expected terms.

Basis risk. The debt may reference a bank-specific rate, term-benchmark rate, or overnight index averaged in arrears, while the hedge references a different index or reset convention. The mismatch leaves residual cost-of-funds risk. Credit spreads on the loan also stay floating unless separately fixed.

Counterparty and credit-valuation adjustments. An uncollateralized swap that is in the company's favor creates exposure to the bank. Banks price this through a *credit valuation adjustment (CVA)* that raises the fixed rate they charge, and corporates are charged *funding valuation adjustments* in some jurisdictions. For a typical interest-rate swap the adjustment is a few basis points but can be much larger for long tenors and weak counterparties.

7.6.5 A short checklist for the CFO

Question	Why it matters
What is the cash-flow-at-risk for a 100 bp move, by year?	Sizes the hedge need; ties to coverage covenants
Is the hedge notional profile aligned with debt amortization and prepayment plans?	Avoids over-hedging and speculative residual positions
Which curve and vol surface did the bank use for the quote?	Lets you test whether the premium is reasonable
What is the all-in cost relative to the forward-implied cost?	Isolates the dealer margin and CVA
What are the margin and collateral terms?	Determines liquidity risk under adverse rate moves
Will the hedge qualify for hedge accounting?	Determines whether earnings are smoother or noisier

7.7 Model Risk and Market Practice

7.7.1 Smile, skew, and the normal model

Black-76 assumes lognormal forward rates, which implies that the implied volatility of a cap or swaption is the same for every strike. In practice, implied vols vary with strike (the *smile*). When rates were high in earlier decades, the pattern was typically a downward-sloping skew in lognormal vol: out-of-the-money floors (low strike) traded at higher lognormal vol than caps. When rates fell near zero, lognormal vols exploded because the same dollar volatility divided by a tiny forward rate gives a huge percentage vol; markets then adopted the *normal* (Bachelier) convention, which treats basis point

volatility as the parameter and allows negative forwards. The SABR model and its shifted-lognormal variants are the typical interpolation tools. For a treasurer the practical message is: the strike you choose changes the vol you pay. A deeply out-of-the-money cap can be much more or much less expensive than a flat-vol Black calculation suggests.

7.7.2 Multi-curve pricing and discounting

Since 2008 the market has recognized that the rate at which a bank can borrow for three months is not the same as the overnight risk-free rate, and that collateralized derivatives are discounted at the collateral rate. The practical result is a set of curves: a discount curve built from overnight swaps, and projection curves for each floating index. The pricing formulas in this chapter remain valid with forwards F read from the projection curve and discount factors P read from the discount curve. A common error is to compute F from the discount curve and thereby misprice caplets by a few percent.

7.7.3 Calibration and what to ask the quant team

Calibration means choosing model parameters so that model prices match market prices of selected instruments. The more parameters, the closer the fit, and the greater the risk of overfitting. A short list of questions for a quant or risk team:

- Which instruments are used in calibration, and which are priced out-of-sample?
- How stable are parameters from day to day? Unstable parameters produce unstable hedge ratios.
- How do prices change if we change the model (Hull-White versus a market model) but calibrate to the same instruments? The *model reserve* is the answer.
- Does the model permit negative rates, and is that consistent with the current market?
- How does the model price the product we are hedging (for example, a callable bond) *relative to* the hedge instrument, and what residual risk remains?

7.7.4 Managerial summary

Short-rate models and Black-76 are not forecasting tools. They are consistent frameworks for translating market prices into other prices, and for showing how different hedges respond to changes in the curve. The CFO who understands that a cap premium reflects a volatility assumption, that a swap is a weighted average of forwards, and that a hedge must follow the debt’s amortization profile can make decisions with the same rigor as an investment decision in Volume 1, Chapter 6.

Case Study

Northgate Dairy Cooperative: Refinancing in a Rising-Rate World

Illustrative case; Northgate Dairy Cooperative is fictional.

Northgate Dairy Cooperative is owned by 1,400 farms and operates six processing plants. Over the past decade it has invested heavily in automated milk-powder and cheese facilities, financed through a mix of retained surplus, member capital, and floating-rate bank debt. Today the co-operative has $540 million of borrowings: a $250 million term loan maturing in four years (benchmark + 1.75 percent, reset every six months), a $180 million revolving facility that averages $140 million drawn (benchmark + 1.40 percent, reset monthly), and $150 million of seasonal working-capital lines used heavily between March and September.

The finance director, Samuel Achterberg, presents the situation to the board. Earnings before interest and tax were $96 million last year. Interest cover, measured as EBIT divided by net interest, was 5.1 times. The loan covenants require a minimum cover of 3.0 times and a maximum net-debt-to-EBITDA of 3.5 times (currently 2.7 times). The benchmark rate

has risen 250 basis points in two years. The five-year par swap rate is 4.60 percent, the one-year benchmark rate is 4.20 percent, and the forward curve is mildly inverted beyond year three, which suggests the market expects rate cuts eventually. Implied lognormal volatility on five-year caps struck at 5.00 percent is about 30 percent.

Two board factions have emerged. One group, led by a farmer-director, wants to hedge nothing: “we have always run floating and the curve says rates will fall.” Another group wants to swap the entire term loan and revolver to fixed to protect the members’ milk payout from interest shocks. The treasury team has indicated that the revolver’s drawn amount swings seasonally, by up to $100 million, and that the cooperative plans to repay $60 million of the term loan from the sale of a distribution depot in eighteen months.

Samuel asks the treasury team to model four strategies for the next four years: no hedge; swap 70 percent of the term loan for four years; buy a four-year cap at 5.50 percent on $250 million; or a collar with cap 5.50 percent and floor 3.75 percent. The quoted cap costs 1.9 percent of notional upfront. A zero-cost collar would require a floor strike of 4.10 percent. The bank warns that the swap would require a credit support annex under which the cooperative would post cash if the mark-to-market falls below minus $8 million, and the cooperative’s cash balance is $35 million, much of it needed for the seasonal borrowing peak.

Stress analysis shows that a 200 basis point rise would raise annual interest cost by $5 million on the term loan and $2.8 million on the average revolver balance, and would cut cover to 3.6 times. A 300 basis point rise would push cover to 3.2 times. The hedging committee is also uncomfortable with the accounting: if the depot sale occurs as planned, a swap on 70 percent of the loan will exceed the remaining debt in two scenarios, and the cooperative could face an ineffective hedge.

Discussion questions

1. Using the forward curve described, explain why “the curve says rates will fall” is not a valid argument for leaving the position unhedged, and describe what the curve does and does not tell the board.
2. Compare the four strategies on cash-flow protection, expected cost, liquidity risk (note the CSA threshold), accounting treatment, and the effect of the planned $60 million prepayment. Which would you recommend, and at what hedge ratio?
3. The cap costs 1.9 percent of $250 million, which is $4.75 million. Explain how you would test whether this premium is reasonable, using Black-76 with the data given, and which inputs a dealer would need to supply.
4. How should the seasonal revolver be treated: hedge the average drawn amount, the minimum, or not at all? Propose a layered hedge that respects both the term loan’s amortization and the revolver’s seasonality.

Key Takeaways

- A discount curve is the foundation: discount factors, zero rates, and forward rates are three views of the same information, and bootstrapping from par swap rates reprices every quoted instrument exactly.
- Forward rates are break-even prices for locking in future borrowing, not forecasts; a par swap rate is a discount-factor-weighted average of forward rates.
- A cap is a strip of caplets (calls on forward rates) and a floor is a strip of floorlets; long cap minus long floor equals a pay-fixed swap, which gives a parity check on any pricing.
- Black-76 prices caplets with the forward rate as the underlying, the fixing time in the volatility term, and the payment-date discount factor; swaptions use the forward swap rate and the annuity.

- Option value is driven by volatility; a long-dated cap has large vega, and flat vols hide the smile, so the strike you choose changes the vol you pay.
- Ho-Lee fits any curve but has no mean reversion; Vasicek adds mean reversion and closed-form bonds but permits negative rates and does not fit the curve; CIR keeps rates positive via square-root volatility; Hull-White combines curve fit with mean reversion.
- Models for pricing live under the risk-neutral measure; models for risk management and stress tests need real-world dynamics, and parameters from one cannot be used in the other without adjustment.
- A floating-rate liability has cash-flow risk rather than value risk; hedging should be sized to cash-flow-at-risk and covenants, with the hedge notional profile following debt amortization and prepayment.
- A swap provides certainty but exposes the company to mark-to-market and margin; a cap provides protection at a premium with no margin; a collar trades some of the upside for a lower premium.
- Practical hedging depends on accounting designation, collateral terms, basis risk, and counterparty adjustments, not just on pricing.

Review Questions

1. Explain why a forward rate is not a forecast of the future spot rate. What other components might be embedded in it?
2. Derive the bootstrapping recursion for $P(0,n)$ from the par swap equation. Why is the floating leg of a par swap worth $1 - P(0,n)$?
3. Show using payoffs why long cap − long floor equals a pay-fixed swap with the same strike and dates.
4. In Black-76, why does volatility accumulate until the fixing date and not the payment date? Why is the discount factor taken to the payment date?
5. Why do markets quote normal (basis point) vols when rates are very low? What problem does a lognormal quote have in that environment?
6. A treasurer is quoted a 5 percent cap at 28 percent vol and a 3 percent cap at 28 percent vol. Why might both quotes be wrong or inconsistent with the smile?
7. Contrast the Ho-Lee and Vasicek models in terms of mean reversion, the long-end volatility of yields, and the ability to fit an observed discount curve.
8. What does the Feller condition guarantee in the CIR model, and why might a calibrated CIR violate it?
9. Why is the market value of a floating-rate liability nearly insensitive to rates, and why does that not make it riskless?
10. List the factors a CFO should weigh when choosing between a pay-fixed swap and a cap for a floating-rate loan with expected partial prepayment.

Problems

Use annual compounding and flat Black volatilities unless stated otherwise.

1. **Bootstrap.** A one-year deposit rate is 2.50 percent and the two-year annual-pay par swap rate is 3.00 percent. Compute $P(0,1)$ and $P(0,2)$.
2. **Forwards.** Using the discount factors from Problem 1, compute the two-year annual zero rate and the one-year forward rate starting in one year.

3. **Caplet and floorlet.** A caplet on notional 100 million covers a six-month period ($\tau = 0.5$) that fixes in two years. The forward rate is 3.80 percent, the strike is 3.50 percent, volatility is 30 percent, and the discount factor to the payment date is 0.93. Compute the caplet and floorlet prices and check parity.
4. **Swaption.** A 1y×3y payer swaption on 50 million has forward swap rate 3.20 percent, strike 3.00 percent, volatility 20 percent, and annuity 4.20. Price it.
5. **Vasicek.** With $a = 0.20$, $b = 4.00$ percent, $\sigma = 1.00$ percent, $r_0 = 2.50$ percent, compute $P(0,3)$ and the three-year continuous yield.
6. **CIR.** With $a = 0.25$, $b = 4.00$ percent, $\sigma = 6.00$ percent, $r_0 = 3.00$ percent, check the Feller condition and compute $P(0,2)$.
7. **Hedge sizing.** A company has 250 million of floating-rate debt maturing in three years. A three-year par swap has annuity 2.75. Compute the interest cost of a 40 basis point rise for one year, the DV01 of a pay-fixed swap on the full notional, and its mark-to-market gain if par rates rise by 40 basis points (ignoring the change in annuity).
8. **Convexity.** Under Ho-Lee with $\sigma = 0.90$ percent, compute the futures-forward convexity adjustment for a rate period from $t_1 = 1.5$ to $t_2 = 1.75$ years, in basis points.

Answers to Selected Problems

1. $P(0,1) = 1/1.025 = 0.975610$. $P(0,2) = (1 - 0.03 \times 0.975610) / 1.03 = 0.970732 / 1.03 = 0.942458$.
2. Two-year zero rate $= (1/0.942458)^{(1/2)} - 1 = 3.0075$ percent. Forward $= 0.975610 / 0.942458 - 1 = 3.518$ percent.
3. $\sigma\sqrt{T} = 0.30 \times 1.414214 = 0.424264$; $d1 = [\ln(3.8/3.5) + 0.5 \times 0.09 \times 2] / 0.424264 = (0.082238 + 0.09) / 0.424264 = 0.4060$; $d2 = -0.0183$. $\Phi(d1) = 0.6576$, $\Phi(d2) = 0.4927$. Caplet per unit $= 0.93 \times 0.5 \times (0.038 \times 0.6576 - 0.035 \times 0.4927) = 0.465 \times (0.024989 - 0.017245) = 0.003601$, i.e., about \$360,100. Floorlet $= 0.465 \times (0.035 \times 0.5073 - 0.038 \times 0.3424) = 0.465 \times (0.017755 - 0.013011) = 0.002206$, i.e., about \$220,600. Parity: $0.003601 - 0.002206 = 0.001395 = 0.465 \times 0.003$.
4. $d1 = [\ln(3.2/3.0) + 0.5 \times 0.04] / 0.20 = (0.064539 + 0.02) / 0.20 = 0.4227$; $d2 = 0.2227$; $\Phi(d1) = 0.6637$, $\Phi(d2) = 0.5881$. Payer $= 4.20 \times (0.032 \times 0.6637 - 0.030 \times 0.5881) = 4.20 \times (0.021238 - 0.017643) = 0.015104$ per unit, or about \$755,200.
5. $B = (1 - e^{(-0.6)}) / 0.2 = 2.2559$; $A = \exp[(0.04 - 0.0001/(0.08)) \times (2.2559 - 3) - 0.0001 \times 5.0891/0.8] = \exp[0.03875 \times (-0.7441) - 0.000636] = \exp(-0.029470) = 0.9710$; $P = 0.9710 \times e^{(-2.2559 \times 0.025)} = 0.9710 \times 0.94516 = 0.91772$. Yield $= -\ln(0.91772)/3 = 2.862$ percent.
6. $2ab = 0.0200 > \sigma^2 = 0.0036$, so Feller holds. $\gamma = 0.26401$; $B(2) = 1.5709$; $A(2) = 0.9831$; $P(0,2) = 0.9831 \times e^{(-1.5709 \times 0.03)} = 0.9831 \times 0.95397 = 0.93786$.
7. Interest rise $= 250$ million $\times 0.40\% = 1.0$ million per year. DV01 $= 250$ million $\times 2.75 \times 0.0001 = \$68{,}750$ per basis point. Gain $= 40 \times 68{,}750 = 2.75$ million.
8. $0.5 \times (0.009)^2 \times 1.5 \times 1.75 = 0.5 \times 0.000081 \times 2.625 = 0.000106 = 1.06$ basis points.

Chapter 8: Bank and Insurance Finance and Capital Regulation

Learning Objectives

After completing this chapter you should be able to:

- Read a bank balance sheet and income statement, compute net interest margin, return on assets, return on equity, and efficiency ratio, and explain how leverage, spread, cost of risk, and cost efficiency combine to produce ROE.
- Measure interest-rate risk in the banking book using repricing gap, deposit betas, and duration gap, and distinguish between earnings-based and economic-value-based views.
- Compute and interpret liquidity metrics (liquidity coverage and net stable funding ratios) and explain why maturity transformation makes banks run-prone.
- Explain, in general terms, how Basel-style regulation defines risk-weighted assets, capital tiers, buffers, and leverage requirements, and compute capital ratios, headroom, and the capital charge from the asymptotic single-risk-factor formula.
- Describe how insurers earn returns from underwriting and investment, estimate loss reserves with the chain-ladder method, and evaluate the effect of reserve error on earnings.
- Explain insurer solvency regimes, aggregate risk modules with correlations, and design a simple stress test for a bank or insurer, interpreting its results for capital planning and for risk-adjusted pricing.

Opening Vignette

Illustrative scenario: Lakeshore Financial is a fictional company.

Lakeshore Financial is a mid-sized regional bank with a small insurance subsidiary. In the early part of a rate-rising cycle, the board hears two very different stories in the same meeting. The head of retail banking reports a record quarter: net interest income has jumped as loan yields have repriced upward faster than deposit costs. Deposits have been sticky, and the bank has had little reason to raise what it pays on checking and savings accounts. Return on equity is the highest in six years.

Then the chief risk officer presents a more uncomfortable slide. The bank holds a securities portfolio that was bought when rates were low; a hundred-basis-point rise has cut its market value by several hundred million dollars, which does not appear in earnings because most of the securities are classified as held-to-maturity but would be crystallized if the bank had to sell them. A survey of large depositors suggests that nearly a third of uninsured corporate balances are held in accounts that can be moved with a click. The liquidity coverage ratio is comfortably above 100 percent, but the CRO points out that the ratio’s assumptions about the speed of outflows were calibrated to a world before digital banking.

Meanwhile, the insurance subsidiary reports that its property-and-casualty reserves have been strengthened twice in the past year because claims inflation has proved higher than assumed. The life subsidiary has sold guaranteed-rate annuities and has a duration mismatch between its assets and the promised payments. The head of the subsidiary notes that the economic solvency ratio has fallen from 160 percent to 135 percent, even though accounting earnings are positive.

The board must decide on dividends, buybacks, and growth. The chairwoman asks a simple question: “Are we more profitable or more fragile than a year ago?” The answer is that the bank is both. Profitability in financial institutions is inseparable from risk, leverage, and the regulatory capital that stands behind them. This chapter develops the tools to answer the chairwoman’s question for banks and insurers: how they earn money, how they measure and manage the risks they run, and how capital regulation constrains and shapes their decisions. It builds on Volume 1, Chapter 15 (capital structure) and Chapter 2 (financial statements), and on Chapter 7 of this volume (interest-rate models).

8.1 How Financial Institutions Make Money

8.1.1 Why banks and insurers are different from industrial companies

An industrial firm raises capital and buys real assets that produce cash flows. A bank or insurer is a *financial intermediary*: its assets are mostly financial claims, its liabilities are mostly promises to customers (deposits, insurance policies), and its equity is a thin layer that absorbs losses. Three differences matter for corporate finance.

First, *leverage is the product*. Most nonfinancial companies hold debt-to-assets ratios of 20 to 40 percent; banks hold ratios above 90 percent. The intermediary’s customers *supply* the liabilities as a by-product of using its services (payments, safekeeping, risk transfer). The Modigliani-Miller logic of Volume 1, Chapter 15 still applies, but the tax shield and distress-cost trade-offs are overlaid by *safety-net subsidies* (deposit insurance, central-bank liquidity), which blunt the discipline that market-priced debt would otherwise exert. This is the underlying reason why regulators set minimum capital: the private cost of failure to the institution understates the social cost.

Second, *the balance sheet is the income statement*. For a manufacturer, the balance sheet is a stock of equipment and the income statement shows operating profit. For a bank, interest income arises directly from the spread between asset yields and liability costs applied to the balance sheet, and for an insurer, investment income depends on the float held. Changes in balance-sheet structure change earnings immediately.

Third, *risk is hidden in the timing*. A loan portfolio can look healthy for years until losses emerge; an insurer’s profit in one year is partly determined by estimates of claims that will be paid over the next decade. Reported earnings therefore depend on provisions and reserves, which are estimates. This makes accounting quality and governance first-order issues.

8.1.2 The bank balance sheet

A stylized balance sheet shows the main components. We will use the fictional Lakeshore Bank throughout the chapter (millions of dollars).

Assets	Amount	Liabilities and equity	Amount
Cash and central-bank reserves	10,000	Retail deposits, stable	25,000
Securities	25,000	Retail deposits, less stable	25,000
Residential mortgages	25,000	Corporate operational deposits	10,000
Corporate loans	30,000	Corporate non-operational deposits	18,000
Retail unsecured loans	5,000	Wholesale funding	12,000
Other assets (including 800 of intangibles)	5,000	Other liabilities	2,000
		Shareholders’ equity	8,000
Total assets	**100,000**	**Total liabilities and equity**	**100,000**

Loans earn the highest yield but carry credit risk and are illiquid; securities provide liquidity and a modest return; reserves are the most liquid but earn the least. On the liability side, stable retail deposits are the cheapest and most reliable funding, corporate deposits are more sensitive to rates and confidence, and wholesale funding is the most expensive and most prone to withdrawal in stress. Equity is the buffer.

The core business of a bank is *maturity and liquidity transformation*: it borrows short and liquid (demand deposits) and lends long and illiquid (mortgages, business loans). This creates profit through the term premium and the liquidity premium and creates two risks: interest-rate risk (Section 8.2) and liquidity risk (Section 8.3). It also creates *credit risk* on the loan portfolio, which capital regulation (Section 8.4) is designed to address.

8.1.3 Net interest margin and the DuPont decomposition for banks

The primary profitability measure is *net interest margin* (NIM):

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NIM = (Interest income − Interest expense) / Average earning assets
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Net interest income (NII) is the dollar value of the numerator. Beyond NII, a bank earns *non-interest income* (fees, trading, payments, wealth management) and incurs *operating expenses* and *loan-loss provisions*. The efficiency ratio is operating expense divided by total revenue (NII plus non-interest income). Return on assets (ROA) is net income over average total assets, and return on equity (ROE) is net income over equity. The accounting identity ROE = ROA × (Assets / Equity) shows how leverage multiplies ROA.

Example 8.1 – Profitability decomposition for Lakeshore Bank. Loans yield 6.50 percent, securities 4.00 percent, and reserves 2.00 percent. Deposits cost 2.20 percent on average and wholesale funding 4.50 percent. Non-interest income is 900 million, operating expense 2,400 million, provisions for loan losses 600 million, and the tax rate is 25 percent. Earning assets are loans (60,000) plus securities (25,000) plus cash and reserves (10,000) = 95,000.

Step 1. Interest income = 60,000 × 6.50% + 25,000 × 4.00% + 10,000 × 2.00% = 3,900 + 1,000 + 200 = 5,100.

Step 2. Interest expense = 78,000 × 2.20% + 12,000 × 4.50% = 1,716 + 540 = 2,256.

Step 3. NII = 5,100 − 2,256 = 2,844. NIM = 2,844 / 95,000 = **2.99 percent**.

Step 4. Revenue = NII + non-interest income = 2,844 + 900 = 3,744. Pre-provision profit = 3,744 − 2,400 = 1,344. After provisions of 600, pre-tax profit = 744. Tax at 25 percent = 186. Net income = **558**.

Step 5. ROA = 558 / 100,000 = 0.558 percent. Equity multiplier = 100,000 / 8,000 = 12.5. ROE = 0.558% × 12.5 = **6.98 percent**. The efficiency ratio = 2,400 / 3,744 = **64.1 percent**.

Interpretation. A 7 percent ROE is below most banks’ cost of equity (Volume 1, Chapter 10 suggests 9 to 12 percent). Value is created only when ROE exceeds the cost of equity; the managerial question becomes which lever to pull. Raising NIM by 20 basis points adds 190 million of pre-tax profit (142.5 million after tax), which raises ROE by 1.8 points. Cutting the efficiency ratio by 5 points saves about 187 million pre-tax. Reducing provisions has the same effect but may simply postpone losses. Raising leverage multiplies ROA but raises distress risk and is capped by regulation. The ROE is a *by-product* of the interplay between spread, cost, credit quality, and leverage, and any one measure can be improved by a decision that worsens risk.

8.1.4 Other revenue and what drives it

Fee income is less rate-sensitive and capital-light, so banks seek to expand it, but it depends on franchise and customer relationships. *Trading and investment banking income* is volatile and consumes capital for market risk. *Cost of risk* (provisions as a percentage of loans) is the most cyclical component: under the expected-credit-loss approach of IFRS 9 and the current-expected-credit-loss approach in the United States, banks provision for expected losses earlier than under older "incurred loss" rules, which makes provisions more procyclical in their timing but smooths the shock of recognition. Finally, *funding costs and deposit betas* (Section 8.2) determine how much of a rise in market rates the bank retains.

8.1.5 Common mistakes

- *Reading high ROE as high quality.* A high ROE from thin capital and low provisions in a benign credit environment is fragile.
- *Comparing NIM across banks without adjusting for mix.* A bank concentrated in credit cards has a higher NIM than a mortgage lender for reasons that reflect risk.
- *Ignoring off-balance-sheet exposures.* Undrawn commitments and guarantees are claims on the bank's liquidity and capital.
- *Taking a single-year efficiency ratio as structural.* Cost-cutting that deteriorates controls or customer service has a delayed price.

8.2 Interest-Rate Risk in the Banking Book

8.2.1 Two lenses: earnings and economic value

Interest-rate risk in the banking book (IRRBB) is the exposure of a bank's earnings and capital to changes in interest rates. Two complementary views are used:

- **Earnings view (NII-at-risk).** How much does net interest income over the next one to three years change under a rate scenario? This is what management and equity analysts watch.
- **Economic-value view (EVE).** How much does the present value of the bank's assets, liabilities, and off-balance-sheet positions change under the scenario? This captures long-dated exposures that earnings ignore, and it is what the supervisor uses to judge solvency risk.

A bank can look safe on one measure and fragile on the other: a bank holding long fixed-rate securities funded by short deposits may show *rising* NII in a rate-cutting scenario and a deeply negative EVE in a rate-rising scenario, as the PV of the securities falls.

8.2.2 The repricing gap and the role of deposit betas

The simplest measure divides the balance sheet into time buckets according to when each item's interest rate resets or the item matures. The *gap* in a bucket is rate-sensitive assets (RSA) minus rate-sensitive liabilities (RSL). A change in rates of Δr changes NII over the bucket's horizon by approximately Gap $\times \Delta r$.

Example 8.2 – One-year repricing gap, with and without deposit betas. Lakeshore's balance sheet items repricing or maturing within one year: floating-rate loans 30,000 (including corporate loans and adjustable-rate mortgages), securities maturing within a year 5,000, cash and reserves 10,000. Liabilities repricing within a year: deposits 60,000 and wholesale funding 5,000.

Step 1. RSA = 30,000 + 5,000 + 10,000 = 45,000. RSL = 60,000 + 5,000 = 65,000. One-year gap = −20,000.

Step 2. A naïve application: +100 basis points changes NII by −20,000 × 1.00% = **−200 million**, and a 100 basis point fall changes it by +200 million. The bank looks *liability-sensitive*.

Step 3. But deposit rates do not move one for one with market rates. The *deposit beta* is the fraction of a change in market rates passed through to depositors. Assume a beta of 0.50 on the 60,000 of deposits (a typical value for blended checking and savings in a rising-rate episode; the actual figure varies by product and by cycle) and 1.00 on wholesale funding. The effective rate-sensitive liabilities are 60,000 × 0.50 + 5,000 × 1.00 = 35,000.

Step 4. The effective gap is 45,000 − 35,000 = +10,000, and the +100 basis point scenario changes NII by **+100 million**. The sign has flipped.

Lessons. First, the gap analysis is only as good as its behavioral assumptions about deposits (betas, and also the *duration* of non-maturity deposits, which are contractually overnight but behaviorally sticky) and about prepayment on mortgages. Second, a bank is rarely as liability-sensitive or asset-sensitive as its contractual repricing suggests. Third, deposit betas are not constant: they tend to rise as rates rise and as competition for deposits intensifies, meaning that the benefits of a rising-rate environment fade over time, an effect that crushed earnings in banks that assumed low betas indefinitely.

8.2.3 Duration gap and economic value of equity

The economic-value view uses duration. For a position with market value V and modified duration D, a parallel yield change Δy changes value by approximately −D × V × Δy (Volume 1, Chapter 4). For the bank as a whole, the change in the market value of equity ΔE is the change in assets minus the change in liabilities:

```
ΔE ≈ −(D_A × A − D_L × L) × Δy = −D_gap × A × Δy
D_gap = D_A − D_L × (L / A)
```

A positive duration gap means assets are longer than liabilities (after adjusting for leverage), so a rate rise decreases equity value.

Example 8.3 – Duration gap and equity value at risk. Lakeshore's assets have a modified duration of 2.8 years (the loan book is a mix of floating and fixed; the securities are medium-term). Liabilities (deposits and wholesale funding, recognizing the behavioral maturity of non-maturity deposits) have a modified duration of 1.1 years. Total assets are 100,000 and liabilities (including other liabilities) are 92,000.

Step 1. D_gap = 2.8 − 1.1 × (92,000 / 100,000) = 2.8 − 1.012 = **1.788 years**.

Step 2. For a +100 basis point parallel shift: ΔE ≈ −1.788 × 100,000 × 0.01 = **−1,788 million**.

Step 3. As a share of book equity (8,000), this is −22.4 percent; as a share of CET1 capital (7,200, see Section 8.4), −24.8 percent. Supervisory standards for IRRBB commonly flag a bank as an outlier when a standardized shock reduces EVE by more than 15 percent of Tier 1 capital, so this bank would attract attention.

Step 4. Hedging. A receive-fixed swap with DV01 matching the duration gap reduces exposure. The required DV01 is D_gap × A × 0.0001 = 1.788 × 100,000 × 0.0001 = 17.9 million per basis point, equivalent to a swap with notional such that notional × annuity × 0.0001 = 17.9 million; for a 5-year swap with annuity of about 4.5 (Chapter 7) the notional would be roughly 17.9 / (4.5 × 0.0001) = 39.7 billion. Even with a different mix, the hedge is large, and the bank might prefer to restructure the securities portfolio or change deposit pricing strategy. Note that this ignores optionality: prepayment on mortgages and the right of depositors to withdraw create *negative convexity* that makes the actual exposure worse for large shocks.

Notice the interplay with Chapter 7: the risk is *parallel* in this simple measure, but real exposures are *bucketed*. A bank that is long five-year assets and short two-year liabilities is hurt by a *steepening* of the curve but not necessarily by a parallel shift. This is why regulators prescribe six scenarios (parallel up, parallel down, steepener, flattener, short-rate up, short-rate down) rather than a single shift.

8.2.4 Managing IRRBB

Banks manage IRRBB through three levers: *balance-sheet structure* (the mix of fixed and floating loans, the duration of the securities portfolio), *deposit pricing* (to manage betas and deposit mix), and *derivatives* (swaps, caps, floors, swaptions). Accounting complicates the picture: securities classified as held-to-maturity do not show unrealized losses in equity or earnings, and derivatives designated as hedges receive different treatment from derivatives that are not (Chapter 7, Section 7.6). A well-run asset-liability committee (ALCO) reviews both NII and EVE under several scenarios, assesses behavioral assumptions, sets limits (typically a percentage of capital or of NII), and monitors them quarterly.

The deeper point is that a bank earns a *term premium* from being short-funded, and giving up the term premium to eliminate interest-rate risk reduces ROE. There is no free lunch: the ALCO's job is to take only the interest-rate risk that the bank is compensated for, and no more than its capital can bear.

8.2.5 Common mistakes

- *Reporting NII-at-risk over a short horizon only.* Long-dated assets can hide large EVE losses.
- *Using contractual maturities for non-maturity deposits.* Behavioral models must reflect depositor stickiness, but must also be stress-tested for rapid deposit migration.
- *Assuming deposit betas are stable through the cycle.*
- *Ignoring basis risk and optionality.* Prepayments, caps embedded in loans, and early-withdrawal options can move exposures substantially.

8.3 Liquidity Risk and Funding

8.3.1 Why banks are run-prone

A bank's liabilities are callable at par on short notice, while its assets are not. This is the Diamond-Dybvig insight: maturity transformation creates socially useful liquidity for savers but also creates a coordination problem. If depositors fear others will withdraw, it is rational to withdraw first, and a solvent bank can fail because it cannot sell illiquid assets quickly at fair value. Deposit insurance and central-bank lending reduce the problem for insured retail deposits but not for uninsured corporate balances or wholesale funding, which are the funds that flee first. Modern runs are faster than historical runs because digital channels and social-media information shorten the time to withdraw.

Two distinct concepts must be separated. *Funding liquidity risk* is the risk that the bank cannot meet its obligations as they fall due without unacceptable losses. *Market liquidity risk* is the risk that assets cannot be sold at close to fair value. The two interact: funding stress forces asset sales into weak markets, which depresses prices, reduces collateral values, and deepens the funding stress (a *liquidity spiral*).

8.3.2 The regulatory liquidity ratios

Post-crisis regulation introduced two standard ratios, described here in general terms; the exact factors vary by jurisdiction.

Liquidity coverage ratio (LCR). The LCR asks whether the bank has enough high-quality liquid assets (HQLA) to survive a severe 30-day stress:

```
LCR = Stock of HQLA / Net cash outflows over 30 days  ≥ 100%
Net outflows = Total expected outflows − min(Total expected inflows, 75% × Total expected outflows)
```

HQLA are classed by quality: Level 1 (cash, central-bank reserves, certain sovereign securities) counts at full value; Level 2 assets (certain covered bonds and corporate securities, with a haircut and a cap on their share) count at a reduced value. Outflow rates depend on funding type: stable retail deposits run off slowly (a few percent), less stable retail deposits faster, operational corporate deposits (needed for the customer's payment activities) at moderate rates, non-operational corporate deposits at rates around 40 percent in typical standards, and wholesale funding from financial institutions at 100 percent. Undrawn committed lines carry small outflow factors. Inflows from performing loans are counted at partial rates, and the 75 percent cap prevents banks from relying on inflows alone.

Net stable funding ratio (NSFR). The NSFR asks whether the bank's longer-term funding supports its longer-term assets over a one-year horizon:

```
NSFR = Available stable funding (ASF) / Required stable funding (RSF) ≥ 100%
```

Each liability is weighted by its stability (equity and long-term debt at 100 percent, stable retail deposits around 95 percent, less stable retail around 90 percent, wholesale funding shorter than six months at zero or 50 percent depending on counterparty), and each asset is weighted by its illiquidity (cash at zero, Level 1 securities at about 5 percent, mortgages at about 65 percent, other loans at about 85 percent, and assets that are encumbered or otherwise illiquid at 100 percent).

Example 8.4 – LCR and NSFR for Lakeshore Bank. Using the balance sheet of Section 8.1.2, assume the securities portfolio comprises 9,000 of Level 1 government securities, 5,000 of Level 2A assets, and 11,000 of other securities that do not qualify as HQLA. Of the wholesale funding (12,000), 8,000 matures within 30 days and 4,000 matures beyond one year. Undrawn committed facilities total 8,000. Expected inflows from performing loans within 30 days are 2,000.

LCR. Step 1. HQLA = cash and reserves 10,000 + Level 1 securities 9,000 + Level 2A 5,000 × 85% = 4,250, which is **23,250**.

Step 2. Outflows: stable retail deposits 25,000 × 5% = 1,250; less stable retail 25,000 × 10% = 2,500; operational corporate 10,000 × 25% = 2,500; non-operational corporate 18,000 × 40% = 7,200; wholesale maturing within 30 days 8,000 × 100% = 8,000; undrawn commitments 8,000 × 10% = 800. Total = **22,250**.

Step 3. Inflows of 2,000 are below the cap (75% × 22,250 = 16,688), so net outflows = 22,250 − 2,000 = **20,250**.

Step 4. LCR = 23,250 / 20,250 = **114.8 percent**. The bank meets the 100 percent minimum with a surplus of 3,000 of HQLA.

NSFR. Step 5. ASF = equity 8,000 × 100% + stable retail 25,000 × 95% + less stable retail 25,000 × 90% + operational corporate 10,000 × 50% + non-operational corporate 18,000 × 50% + long-dated wholesale 4,000 × 100% = 8,000 + 23,750 + 22,500 + 5,000 + 9,000 + 4,000 = **72,250**. The short-dated wholesale funding and other liabilities count at zero.

Step 6. RSF = Level 1 securities 9,000 × 5% = 450; Level 2A 5,000 × 15% = 750; other securities 11,000 × 50% = 5,500; mortgages 25,000 × 65% = 16,250; corporate loans 30,000 × 85% = 25,500; retail loans 5,000 × 85% = 4,250; other assets 5,000 × 100% = 5,000; undrawn commitments 8,000 × 5% = 400. Total = **58,100**.

Step 7. NSFR = 72,250 / 58,100 = **124.4 percent**.

Interpretation. Both ratios exceed 100 percent, but the cushion is thin on the LCR: if non-operational deposits were to run off at 60 percent instead of 40 percent, outflows would rise by 3,600 to 25,850 (net 23,850), and the LCR would fall to 97.5 percent. The ratio is a regulatory minimum, not a prediction; experience in recent bank failures suggests that outflows can exceed regulatory run-off assumptions. The important managerial point is that *HQLA are costly*: holding 23,250 of low-yielding assets rather than loans forgoes an earnings spread of perhaps 2 to 3 points, which at the margin is a 500-million-dollar annual cost for a bank like this. Liquidity regulation is, in effect, a tax on maturity transformation, and it explains much of the shift of lending from banks to non-banks since the crisis.

8.3.3 Funding strategy and contingency planning

A bank's funding strategy has several components. The *funding mix* trades off cost against stability: core retail deposits are cheapest and stickiest, while wholesale markets are flexible but fragile. The *maturity ladder* of wholesale funding avoids concentration of maturities in one month. *Collateral management* determines how much of the securities portfolio is unencumbered and available to pledge at the central bank. Finally, a *contingency funding plan* lists the actions available in stress, in order of speed and cost: central-bank facilities, repo of securities, asset sales, loan securitization, and, as a last resort, curtailment of lending. A good plan is *tested*: lessons from past stress episodes are that collateral eligibility, operational readiness to borrow from the central bank, and the signalling stigma of doing so are as important as the paper capacity.

A crucial linkage exists between liquidity and solvency: *market confidence determines whether solvent banks survive*. Unrealized losses on securities, even if not recognized in earnings, reduce the economic capital cushion, and uninsured depositors who see this may decide to withdraw. This is why regulators watch both the accounting capital ratio and economic measures such as EVE.

8.3.4 Common mistakes

- *Treating the LCR as the full liquidity story.* A 30-day ratio says nothing about a stress that develops over weeks, nor about intraday liquidity.
- *Counting all deposits as stable.* Concentrated, uninsured, rate-sensitive, or digitally mobile deposits behave differently from granular insured retail deposits.
- *Ignoring encumbrance.* Assets pledged to secured funders are not available to meet a new outflow.
- *Holding HQLA that lose value when rates rise without hedging.* Long government bonds are liquid but carry mark-to-market risk, and selling them realizes the loss.

8.4 Capital Regulation: The Basel Framework in General Terms

8.4.1 Why regulate capital

Regulators require banks to hold capital for three reasons. First, *externalities*: the failure of one bank can impose costs on other banks and on the real economy through counterparty exposures, fire sales, and loss of credit. Second, *moral hazard*: deposit insurance and implicit government support reduce depositors' incentives to monitor, so shareholders can gain from risk-taking at the expense of the insurer. Third, *information asymmetry*: depositors cannot assess loan portfolios. Capital reduces the probability of failure and aligns shareholders' interests with prudence, because shareholders lose first.

The international framework developed by the Basel Committee on Banking Supervision, commonly called "Basel" and implemented with local variations, rests on three pillars: *Pillar 1* sets minimum capital for credit, market, and operational risk; *Pillar 2* adds supervisory review and bank-specific add-ons for risks not fully captured, including interest-rate risk in the banking book and concentration; *Pillar 3* requires disclosure to promote market discipline. Details of calibration have been revised several times, and national regimes differ. The concepts below are stated generally and the numbers are illustrative.

8.4.2 The structure of regulatory capital

Capital is divided into tiers according to loss-absorbing quality:

- **Common Equity Tier 1 (CET1):** common shares and retained earnings, minus deductions such as goodwill and other intangibles, deferred tax assets from loss carryforwards, and certain holdings in other financial institutions. This is the highest-quality, going-concern capital.
- **Additional Tier 1 (AT1):** perpetual instruments, such as contingent convertible bonds, that can absorb losses by write-down or conversion before the bank fails.
- **Tier 2:** subordinated debt that absorbs loss in resolution (gone-concern capital).

Requirements are expressed as ratios to *risk-weighted assets* (RWA), and typically include a minimum (for example, CET1 of 4.5 percent of RWA), a *capital conservation buffer* (a further 2.5 percent), and additional buffers: a *countercyclical buffer* set by national authorities in a credit boom, and surcharges for globally or domestically systemically important banks. Falling into the buffer does not cause failure but restricts dividends, buybacks, and bonuses, a powerful practical constraint. A separate *leverage ratio* (Tier 1 capital divided by a non-risk-based measure of total exposure, with a minimum around 3 percent plus surcharges for the largest banks) acts as a backstop to the risk-weighted measure, which can be gamed or mis-calibrated. Finally, large banks must hold *total loss-absorbing capacity* (TLAC) or comparable debt that can be bailed in during resolution.

8.4.3 Risk-weighted assets

Under the *standardized approach*, each exposure is multiplied by a risk weight set by regulation according to asset class and, for some classes, external ratings or collateral: sovereign exposures in the domestic currency at 0 percent, high-quality covered bonds and agency paper at about 20 percent, residential mortgages with moderate loan-to-value at about 35 percent, retail exposures at 75 percent, corporates at 100 percent (with lower weights for highly rated or small-business exposures in some regimes), and equity and high-risk items at higher weights. Off-balance-sheet items are converted into credit-equivalent amounts with credit conversion factors. RWA also includes charges for *market risk* (the trading book) and *operational risk*, expressed as RWA equivalents by multiplying the capital charge by 12.5 (the reciprocal of the 8 percent original minimum). Under the *internal ratings-based (IRB)* approach, approved banks estimate probability of default (PD), loss given default (LGD), exposure at default (EAD), and maturity, and a regulatory formula converts them to RWA (Section 8.4.5).

Example 8.5 – RWA, CET1 ratio, and headroom for Lakeshore Bank. The balance sheet of Section 8.1.2 with the securities split of Example 8.4. Risk weights: cash and reserves 0 percent; Level 1 government securities 0 percent; Level 2A 20 percent; other securities 50 percent; residential mortgages 35 percent; corporate loans 100 percent; retail loans 75 percent; other assets (excluding the 800 of intangibles, which are deducted from capital) 100 percent. Operational and market risk RWA total 6,800.

Step 1. Credit RWA: Level 2A 5,000 × 20% = 1,000; other securities 11,000 × 50% = 5,500; mortgages 25,000 × 35% = 8,750; corporate loans 30,000 × 100% = 30,000; retail loans 5,000 × 75% = 3,750; other assets (5,000 − 800) × 100% = 4,200. Total = 1,000 + 5,500 + 8,750 + 30,000 + 3,750 + 4,200 = **53,200**.

Step 2. Add operational and market RWA 6,800: total RWA = **60,000**.

Step 3. CET1 = equity 8,000 − intangibles 800 = **7,200**. CET1 ratio = 7,200 / 60,000 = **12.0 percent**.

Step 4. Assume the bank's overall CET1 requirement including buffers is 4.5% (minimum) + 2.5% (conservation) + 1.0% (systemic surcharge) = **8.0 percent**. Required CET1 = 0.08 × 60,000 = 4,800. Excess capital = 7,200 − 4,800 = **2,400**, or 4.0 points of RWA.

Step 5. Growth capacity: with 7,200 of CET1 and an 8.0 percent requirement, maximum RWA = 7,200 / 0.08 = 90,000. The bank could add **30,000 of RWA**, for example, 30,000 of new corporate loans at 100 percent weight, without raising capital, though supervisors usually expect a management buffer above the requirement.

Step 6. Leverage ratio: Tier 1 capital of 7,200 over an exposure measure of approximately 100,000 (assets of 99,200 after deductions plus 800 of credit-converted off-balance-sheet items) = **7.2 percent**, comfortably above a 3 percent minimum.

Step 7. Dividend capacity. Net income of 558 (Example 8.1) retained entirely adds 558 to CET1 and 0.93 percentage points to the ratio before RWA growth. If the bank pays out 60 percent, retention is 223 and the ratio rises by 0.37 points; sustainable growth in CET1, 223 / 7,200 = 3.1 percent, supports RWA growth of about the same rate while holding the ratio constant.

Interpretation. The 12.0 percent CET1 ratio is attractive, but note how heavily the answer depends on risk weights. If the bank reclassified 10,000 of corporate lending from 100 percent to 75 percent weights through better ratings or collateral, RWA would fall by 2,500 and the ratio would rise to 12.5 percent, an improvement achieved without raising a dollar of capital. This is both the usefulness and the weakness of risk-based capital: it aligns capital with risk, and it creates incentives to *optimize* RWA. The leverage ratio is the check against excess optimization.

8.4.4 The economic logic of capital: expected loss, unexpected loss, and the capital charge

Credit risk has two parts. *Expected loss (EL)* = PD × LGD × EAD is the average annual loss and is a cost of doing business, priced into loan spreads and covered by provisions. *Unexpected loss (UL)* is the variability around the average. Capital is held against UL: the loss at a high confidence level (for instance, 99.9 percent over one year) minus the expected loss. The portfolio loss distribution is highly skewed because defaults are correlated through common economic factors. Diversification reduces idiosyncratic risk but cannot eliminate the systematic factor.

8.4.5 The asymptotic single-risk-factor formula

The IRB risk-weight function is based on the model of Oldřich Vasicek (applied to loan portfolios), the same intuition as the Merton structural model of default (Volume 1, Chapter 18's discussion of equity as a call on assets). Assume each borrower's asset return has a systematic component (one common factor, such as the economy) and an idiosyncratic component, with asset-return correlation ρ. For a large, granular portfolio, the default rate conditional on a bad draw of the common factor at confidence level q is:

```
Conditional PD = Φ [ ( Φ⁻¹(PD) + √ρ × Φ⁻¹(q) ) / √(1 − ρ) ]
Capital requirement K = LGD × ( Conditional PD − PD )
Risk-weighted assets = K × 12.5 × EAD
```

Φ is the standard normal cumulative distribution function and Φ^{-1} is its inverse. The "−PD" term subtracts expected loss, since expected loss is covered by provisions and pricing. The regulatory formula adds a maturity adjustment and sets ρ as a decreasing function of PD (between 12 and 24 percent for corporate exposures); a fixed $\rho = 0.15$ is used below for clarity.

Example 8.6 – The ASRF capital charge for a corporate loan. PD = 1.00 percent, LGD = 45 percent, $\rho = 0.15$, q = 99.9 percent. $\Phi^{-1}(0.01) = -2.3263$ and $\Phi^{-1}(0.999) = 3.0902$.

Step 1. $\sqrt{\rho} = 0.38730$; $\sqrt{(1-\rho)} = 0.92195$.

Step 2. Numerator: $-2.3263 + 0.38730 \times 3.0902 = -2.3263 + 1.1969 = -1.1294$.

Step 3. Ratio = $-1.1294 / 0.92195 =$ **−1.2251**. Conditional PD = $\Phi(-1.2251) =$ **11.03 percent**.

Step 4. K = $0.45 \times (0.1103 - 0.0100) = 0.45 \times 0.1003 =$ **4.51 percent** of EAD.

Step 5. Risk weight = $12.5 \times 4.51\% =$ **56.4 percent**.

The same calculation for other PDs, using the same LGD and correlation:

PD	Conditional PD at 99.9%	K (% of EAD)	Risk weight
0.50%	6.74%	2.81%	35.1%
1.00%	11.03%	4.51%	56.4%
2.00%	17.63%	7.03%	87.9%
5.00%	31.35%	11.86%	148.2%

Observe that the capital charge is *concave* in PD: doubling PD from 1 to 2 percent raises K by about 56 percent, not 100 percent, because expected loss is excluded and because the conditional PD increases less than proportionally. At 5 percent PD, the risk weight exceeds 100 percent and a corporate loan consumes substantially more capital than a loan at 100 percent standardized weight.

Example 8.7 – Risk-adjusted return on capital (RAROC) and the minimum margin. Lakeshore considers a 100 million corporate loan with the risk parameters of Example 8.6. The funding-adjusted margin (loan rate less cost of funds) is 2.20 percent, operating costs are 0.40 percent of the loan, and the tax rate is 25 percent. The capital held is 11 percent of RWA (the bank's target, which exceeds the 8.0 percent requirement), and capital replaces debt funding that would cost 4.0 percent.

Step 1. EL = $1.00\% \times 45\% \times 100 =$ **0.45 million**.

Step 2. RWA = $56.4\% \times 100 = 56.4$ million. Capital = $11\% \times 56.4 =$ **6.20 million**.

Step 3. Pre-tax risk-adjusted profit = margin 2.20 − costs 0.40 − EL 0.45 = **1.35 million**; after tax at 25 percent = 1.0125 million.

Step 4. Capital benefit: since 6.20 million of the loan is funded by equity rather than debt costing 4.0 percent, the bank saves $6.20 \times 4.0\% = 0.248$ million pre-tax, or 0.186 million after tax.

Step 5. RAROC = $(1.0125 + 0.186) / 6.204 =$ **19.3 percent**.

Step 6. With a hurdle rate equal to a cost of equity of 12 percent, the loan creates value: the economic profit is (19.3% − 12%) × 6.20 = 0.45 million. The minimum margin that earns exactly 12 percent solves $(m - 0.85) \times 0.75 + 0.186 = 0.12 \times 6.204 = 0.7445$, so $m - 0.85 = 0.7445$ and **m = 1.59 percent**. A borrower with a higher PD would need a higher margin, as it consumes more capital.

Interpretation. RAROC links regulation to pricing: because capital is a scarce resource, the bank should allocate it to uses with the highest risk-adjusted return, and *price loans to the capital they consume*. A bank that prices all corporate loans at the same margin regardless of PD cross-subsidizes risky borrowers. The same logic underlies Volume 1, Chapter 7's capital rationing discussion: when capital is constrained, the criterion is return per unit of constrained resource.

8.4.6 Criticisms and the capital debate

Three debates are worth knowing. First, *procyclicality*: risk-sensitive requirements rise in downturns when PDs increase, forcing banks to deleverage when the economy needs credit; countercyclical buffers and through-the-cycle PD estimation are partial responses. Second, *model risk and RWA variability*: different banks can assign materially different RWA to the same portfolio, which led to output floors that bound IRB results to a fraction of the standardized approach. Third, the *cost of equity debate*, a Modigliani-Miller argument: if bank funding were frictionless, raising capital would simply lower the required return on equity and leave the weighted cost of funding unchanged. Critics note that this argument fails when deposit insurance subsidies, debt tax shields, and informational frictions matter, which is why banks resist higher requirements. Advocates of higher capital point out that, even allowing for these frictions, the social benefit of lower crisis probability probably exceeds the modest increase in lending cost. As an MBA leader you should hold both arguments: capital is expensive for a single bank's shareholders and cheap for the economy as a whole.

8.4.7 Common mistakes

- *Equating regulatory capital with economic capital.* The regulatory ratio is a floor; economic capital reflects the bank's own risk profile and concentrations.
- *Treating the CET1 ratio as the only constraint.* Leverage ratio, TLAC, liquidity ratios, and stress-test buffers can each bind.
- *Optimizing RWA without regard to risk.* Reducing RWA by reclassification does not reduce the underlying loss potential.
- *Ignoring that provisions already cover expected loss.* Capital is for unexpected loss; double counting EL and UL overstates the capital need.

8.5 Stress Testing

8.5.1 Purpose and design

Capital ratios computed from current balance sheets answer "are we compliant today?" Stress tests answer "would we remain adequately capitalized after a severe but plausible shock?" Supervisors run *top-down* tests, imposing scenarios on all large banks and comparing results against hurdle rates, and banks run *bottom-up* internal tests tailored to their portfolios. After the 2008 crisis, stress tests became a central instrument of prudential supervision and now directly influence permitted dividends and buybacks.

A stress test has five components:

1. **Scenario.** A path for macroeconomic and financial variables over, typically, three years: GDP, unemployment, house prices, interest rates, equity prices, credit spreads. A *severely adverse* scenario is generally calibrated to be at least as severe as a deep post-war recession. *Reverse stress tests* ask the opposite question: what scenario would make the bank fail?

2. **Translation to risk parameters.** Models map macro variables to PD, LGD, prepayment, deposit outflows, and fee income. This step is where most model risk lies, because the relationships are estimated from limited historical data.
3. **Projection of pre-provision net revenue (PPNR).** Revenue and expenses under stress: NII falls if margins compress, fees fall with activity, costs are sticky.
4. **Balance-sheet dynamics.** RWA typically *rise* in stress because rating migration raises risk weights and because drawn credit lines increase; assumptions about whether the bank can shrink its balance sheet matter.
5. **Capital roll-forward and comparison to hurdle.** Start CET1 + stressed net income (PPNR − losses − taxes) − dividends = ending CET1; ratio = ending CET1 / stressed RWA.

8.5.2 A worked stress test

Example 8.8 – A three-year stress test for Lakeshore Bank. Starting point: CET1 of 7,200 and RWA of 60,000 (Example 8.5), ratio 12.0 percent. Baseline pre-provision net revenue is NII 2,844 + non-interest income 900 − operating expense 2,400 = 1,344 per year. The severely adverse scenario implies:

- PPNR falls by 30 percent to 940.8 per year (lower fees, margin compression, sticky costs).
- Default and loss parameters rise as follows (annual PD × LGD × EAD, applied to the loan book of Section 8.1.2):

Segment	EAD	Baseline PD / LGD	Stressed PD / LGD	Stressed annual loss
Corporate loans	30,000	1.2% / 50%	8.0% / 55%	1,320.0
Residential mortgages	25,000	0.3% / 15%	3.0% / 35%	262.5
Retail unsecured	5,000	2.0% / 80%	10.0% / 85%	425.0
Total	**60,000**	Baseline loss 271.3		**2,007.5**

- Stressed loss intensity is 100 percent of the annual figure in years 1 and 2, and 60 percent in year 3 as the scenario eases.
- RWA rises 7, 12, and 15 percent over the baseline in years 1, 2, and 3 because of rating migration.
- The tax rate is 25 percent, losses generate a tax credit (assume it is recognized), and dividends are suspended.

Step 1. Corporate loss: 30,000 × 8.0% × 55% = 1,320.0. Mortgage loss: 25,000 × 3.0% × 35% = 262.5. Retail loss: 5,000 × 10.0% × 85% = 425.0. Total 2,007.5.

Step 2. Year 1: Pre-tax result = 940.8 − 2,007.5 = −1,066.7; tax credit 266.7; net result = −800.0. CET1 = 7,200 − 800.0 = 6,400.0. RWA = 60,000 × 1.07 = 64,200. Ratio = **9.97 percent**.

Step 3. Year 2: same loss rate; result = −800.0; CET1 = 5,600.0; RWA = 60,000 × 1.12 = 67,200; ratio = **8.33 percent**.

Step 4. Year 3: loss = 0.60 × 2,007.5 = 1,204.5; pre-tax = 940.8 − 1,204.5 = −263.7; net = −197.8; CET1 = 5,402.2; RWA = 69,000; ratio = **7.83 percent**.

Year	PPNR	Credit losses	Net result	CET1	RWA	CET1 ratio
Start				7,200.0	60,000	12.00%
1	940.8	2,007.5	−800.0	6,400.0	64,200	9.97%
2	940.8	2,007.5	−800.0	5,600.0	67,200	8.33%
3	940.8	1,204.5	−197.8	5,402.2	69,000	7.83%

Interpretation. The CET1 ratio falls 4.2 percentage points, to 7.83 percent. This remains above the 4.5 percent regulatory minimum, so the bank does not fail. But it falls *below* the 8.0 percent combined requirement including buffers, which in practice means distribution restrictions and a supervisory dialogue about a capital plan. The stress consumes 1,798 of capital in absolute terms, but RWA inflation accounts for a large part of the ratio decline: with RWA held constant at 60,000, ending CET1 would give 9.0 percent. A bank can therefore improve its stress performance both by raising CET1 and by reducing RWA sensitivity, for example by shortening loan maturities or by holding collateralized exposures.

How should management respond? Three options: build capital by retaining earnings now (cutting buybacks), hold more capital against the most loss-intensive segment (corporate loans contribute 66 percent of the stressed loss), or purchase credit protection or securitize part of the portfolio. Each has a cost, and the stress test makes the cost-benefit trade-off explicit.

8.5.3 Limits and good practice

Stress tests are as good as their scenarios and models. Weaknesses include: reliance on historical correlations that break down in unprecedented episodes; omission of liquidity feedback loops (a bank that passes the capital test can still fail on liquidity); static assumptions about management actions; and the risk that all banks use similar scenarios and thus converge on similar mitigations, which makes the system more homogeneous and therefore more vulnerable. Good practice includes reverse stress tests, multiple scenario narratives (rates up with inflation, rates down with deflation), integrating liquidity, and explicit “management action” assumptions that are documented and credible.

8.6 Insurance Company Finance

8.6.1 The insurance business model

An insurer sells promises: if an insured event occurs, it will pay a claim. Premiums are collected up front; claims are paid later, sometimes decades later. In the interval the insurer invests the premium, a pool of funds known as the *float*. Insurers earn money in two ways: *underwriting profit* (premiums less claims and expenses) and *investment income* on the float and on capital. The distinctive feature is that the *cost of goods sold is unknown at the time of sale*: the insurer must estimate the claims it will pay.

The two main lines are different in financial structure:

Feature	Property and casualty (non-life)	Life and annuity
Liability horizon	Short to medium (1 to 10+ years; long tail for liability lines)	Long (10 to 50+ years)
Main liability risk	Claims frequency, severity, inflation, catastrophes	Mortality, longevity, lapse, guaranteed rates
Typical profit sources	Underwriting margin plus investment income	Investment spread, loadings, fees
Key estimate	Loss reserves	Actuarial reserves and discount rates
Principal financial risk	Reserve inadequacy, catastrophe, reinsurance credit	Interest-rate mismatch, longevity, lapse, credit

Reinsurance transfers part of the risk to other insurers, trading premium for protection and reducing capital need, but introducing counterparty credit risk.

8.6.2 Underwriting performance: the combined ratio

For non-life insurers, the central measure is the *combined ratio*:

```
Combined ratio = Loss ratio + Expense ratio
Loss ratio     = Incurred claims (incl. claim-handling costs) / Earned premium
Expense ratio  = Acquisition costs and operating expenses / Earned premium
```

A combined ratio below 100 percent means an underwriting profit. Many insurers deliberately underwrite near or above 100 percent because investment income on the float makes the total business profitable. The return on equity is the sum of underwriting and investment contributions after tax.

Example 8.9 – Property-and-casualty ROE and the effect of reserve error. An insurer has earned premium of 1,000 million, a loss ratio of 66 percent, an expense ratio of 29 percent, an investment portfolio of 3,000 million yielding 4.0 percent, a tax rate of 21 percent, and equity of 900 million.

Step 1. Combined ratio = 66% + 29% = **95 percent**. Underwriting profit = 1,000 × (1 − 0.95) = **50 million**.

Step 2. Investment income = 3,000 × 4.0% = **120 million**.

Step 3. Pre-tax income = 50 + 120 = 170. Tax at 21 percent = 35.7. Net income = **134.3 million**. ROE = 134.3 / 900 = **14.9 percent**.

Step 4. Suppose the company later discovers adverse development of prior-year claims totalling 75 million (the reserves were inadequate). This is charged to current-year underwriting: underwriting result = 50 − 75 = −25. Pre-tax income = −25 + 120 = 95; after-tax income = 95 × 0.79 = **75.05 million**; ROE = **8.3 percent**.

Step 5. The reserve error of 75 million is only 3 percent of loss reserves of 2,500 million, yet it cuts ROE by nearly half. The leverage in the business is reserve-to-equity: reserves of 2,500 against equity of 900 means a 1 percent error in reserves equals 25 million, or 2.8 percent of equity.

Interpretation. Reported underwriting results depend on reserve estimates, and small percentage errors translate into large equity impacts because reserves are a multiple of equity. Investors therefore scrutinize *prior-year reserve development* and the *quality of reserves* before trusting a low combined ratio. Moreover, investment income depends on interest rates: a decline of 100 basis points on the 3,000 million portfolio reduces income by 30 million pre-tax, as much as 3 points of loss ratio would.

8.6.3 Loss reserving: the chain-ladder method

Reserves are estimated from *loss development triangles*: cumulative paid (or incurred) claims by accident year (rows) and development year (columns). The *chain-ladder method* assumes that future development in each year follows the historical pattern: the ratio of cumulative claims at development year j+1 to those at year j, the *age-to-age factor*, is stable. Multiplying the latest observed cumulative claim by the product of the remaining factors gives the estimated *ultimate* loss; the *reserve* is ultimate minus paid to date.

Example 8.10 – Chain-ladder reserve estimate. Cumulative paid claims by accident year (millions) after one to four years of development:

Accident year	Dev 1	Dev 2	Dev 3	Dev 4
1	1,000	1,800	2,200	2,350
2	1,100	2,000	2,450	
3	1,250	2,250		
4	1,400			

Step 1. Volume-weighted age-to-age factors: f_1 = (1,800 + 2,000 + 2,250) / (1,000 + 1,100 + 1,250) = 6,050 / 3,350 = **1.8060**; f_2 = (2,200 + 2,450) / (1,800 + 2,000) = 4,650 / 3,800 = **1.2237**; f_3 = 2,350 / 2,200 = **1.0682**.

Step 2. Cumulative development factors to ultimate (assuming no development beyond year 4, a tail factor of 1.00): from Dev 3: 1.0682; from Dev 2: 1.2237 × 1.0682 = 1.3071; from Dev 1: 1.8060 × 1.2237 × 1.0682 = 2.3606.

Step 3. Ultimates: accident year 1: 2,350 (fully developed). Year 2: 2,450 × 1.0682 = 2,617.0. Year 3: 2,250 × 1.3071 = 2,941.0. Year 4: 1,400 × 2.3606 = 3,304.9.

Step 4. Reserves = ultimate − paid to date: year 1: 0; year 2: 2,617.0 − 2,450 = 167.0; year 3: 2,941.0 − 2,250 = 691.0; year 4: 3,304.9 − 1,400 = 1,904.9. **Total reserve = 2,762.9 million**.

Step 5. Sensitivity to the tail. If claims continue to develop by 3 percent beyond year 4 (a tail factor of 1.03, plausible for a liability line with long-tailed claims), every ultimate rises by 3 percent: the sum of ultimates is 11,213, so reserves increase by 336 million to **3,099 million**, a 12 percent increase from the base estimate, for what looks like a small change in an assumption.

Interpretation. Chain-ladder is simple and transparent, but it has weaknesses: (1) it assumes the future resembles the past, so changes in claims handling, inflation, or mix distort it; (2) it is highly leveraged on the most recent accident year, whose factor is the product of all remaining factors and so has the largest estimation error; (3) the tail factor is a judgment, not data. Actuaries cross-check with alternative methods (Bornhuetter-Ferguson, which blends the chain-ladder ultimate with an a priori loss ratio; frequency-severity models) and report a *range*, with management choosing a point within it. This is exactly the kind of choice where governance matters: an insurer under earnings pressure has an incentive to select the low end.

8.6.4 Catastrophe risk, reinsurance, and capital

Property insurers face *catastrophe* losses (hurricanes, earthquakes, floods) that are highly correlated across policies in a region. These cannot be diversified by writing more policies in the same area. Insurers use probabilistic *catastrophe models* to estimate the distribution of annual losses and their tail, then buy *reinsurance*: quota-share (proportional), excess-of-loss (non-proportional, paying losses above an attachment point), and *catastrophe bonds* (securitization that transfers peak risk to capital markets). The reinsurance decision is a capital-structure decision: it reduces required capital and earnings volatility at the cost of ceded premium and counterparty risk. A rule of thumb is that insurers buy more reinsurance when capital is scarce and less when it is abundant, so reinsurance pricing is cyclical, hardening after large losses.

8.7 Insurer Asset-Liability Management and Solvency Regimes

8.7.1 Life insurers and the duration mismatch

A life insurer with guaranteed long-term payouts faces a *reinvestment and discounting* problem similar to the bank's, but with longer horizons and less ability to reprice liabilities. If assets are shorter than liabilities, falling rates raise the present value of liabilities more than they raise assets, which reduces surplus. The problem became acute in low-rate environments

for insurers that sold long-dated guarantees.

Example 8.11 – Duration mismatch in a life insurer. A life insurer owes a single payment of 100 million in 10 years and discounts at 3.00 percent. It holds assets with value equal to the liability today, in a six-year zero-coupon bond.

Step 1. Liability PV = $100 / 1.03^{10} = 100 / 1.34392 =$ **74.409 million**. Asset: a six-year zero with face value F such that $F / 1.03^{6} = 74.409$; $F = 74.409 \times 1.19405 = 88.849$ million.

Step 2. Modified durations: liability 10 / 1.03 = **9.709** years; assets 6 / 1.03 = **5.825** years. Duration gap = 5.825 − 9.709 = −3.884 years: assets are shorter than liabilities.

Step 3. Rate scenarios (a parallel shift of all yields):

Yield	Asset value	Liability value	Surplus change
2.00%	78.895	82.035	−3.140
3.00%	74.409	74.409	0.000
4.00%	70.218	67.556	+2.662

A fall of one percentage point in yields creates a **3.14 million** deficit (4.2 percent of the initial liability value) from a position that was exactly matched in value.

Step 4. Hedge with a receive-fixed swap. DV01 of the liability = $74.409 \times 9.709 \times 0.0001 = 0.07224$ million; DV01 of assets = $74.409 \times 5.825 \times 0.0001 = 0.04335$ million; the gap is 0.02889 million per basis point. A 10-year receive-fixed swap at 3.00 percent has an annuity of $(1 - 1.03^{-10}) / 0.03 = 8.530$, so its DV01 per unit notional is 8.530×0.0001. Required notional = $0.02889 / (8.530 \times 0.0001) =$ **33.9 million**. This closes the first-order gap; convexity differences remain, and the swap introduces collateral requirements, which are a liquidity risk if rates rise.

Alternatively, the insurer could extend the duration of its assets by purchasing long government bonds or infrastructure debt, or lessen its liability exposure by changing guarantee terms in new products. The point matches Chapter 7's treatment: *hedge ratios are first-order; convexity, basis risk, and collateral are the residual risks that actually hurt*. In a prolonged episode of falling rates, the insurer's real problem is *reinvestment risk* on the shorter assets, as the cash flows from maturing bonds must be reinvested at lower yields to meet fixed guarantees.

8.7.2 Solvency regimes in general terms

Insurers are regulated to protect policyholders. Two broad approaches exist. *Factor-based* regimes (for example, risk-based capital in the United States) compute required capital by applying factors to asset and liability categories. *Economic* or *market-consistent* regimes (Solvency II in the European Union, the Swiss Solvency Test, and similar frameworks elsewhere) value assets and liabilities at market-consistent values, define *own funds* as the excess, and require own funds to exceed a *solvency capital requirement (SCR)* calibrated to a 99.5 percent one-year value-at-risk. A smaller *minimum capital requirement* triggers harsher intervention. The principles are similar to Basel's: risk-sensitive requirements, supervisory review, and disclosure, with an emphasis on technical provisions and long-term guarantees.

The SCR is built bottom-up from *risk modules*: market risk (interest rates, equity, credit spread, property, currency), underwriting risk (non-life, life, health: premium, reserve, catastrophe, mortality, longevity, lapse), counterparty default risk, and operational risk. Module capital charges are aggregated with a *correlation matrix* to recognize diversification:

```
BSCR = √( Σᵢ Σⱼ ρᵢⱼ × SCRᵢ × SCRⱼ )
SCR  = BSCR + Operational risk charge (+/− adjustments)
Solvency ratio = Own funds / SCR
```

Example 8.12 – Aggregating an insurer's SCR. Module charges are market 400, insurance 300, and counterparty credit 120 (millions); each pair has a correlation of 0.25; operational risk is 60 (added without diversification). Own funds are 900.

Step 1. Sum of squares: $400^2 + 300^2 + 120^2 = 160{,}000 + 90{,}000 + 14{,}400 = 264{,}400$.

Step 2. Cross terms: $2 \times 0.25 \times (400 \times 300 + 400 \times 120 + 300 \times 120) = 0.5 \times (120{,}000 + 48{,}000 + 36{,}000) = 0.5 \times 204{,}000 = 102{,}000$.

Step 3. BSCR = $\sqrt{(264{,}400 + 102{,}000)} = \sqrt{366{,}400} =$ **605.3**. The simple sum of modules is 820, so diversification benefit is 214.7 (26.2 percent).

Step 4. SCR = 605.3 + 60 = **665.3**. Solvency ratio = 900 / 665.3 = **135.3 percent**. Surplus above the requirement = 234.7.

Step 5. Sensitivity: a 100 basis point fall in rates that reduces own funds by 150 (the life book of this insurer is larger than the stylized single payment of Example 8.11) leaves 750 of own funds, and the ratio falls to 750 / 665.3 = **112.7 percent**; if it also raises the market-risk module because of lower rates, say to 440, the SCR rises and the ratio falls further. Insurers often set an internal target ratio well above 100 percent, such as 150 to 200 percent, because the regulatory minimum triggers supervisory intervention and market confidence erodes before it is reached.

Interpretation. The square-root aggregation says that diversification is powerful when correlations are low and vanishes when they approach 1. Correlation assumptions are set by regulators and are heavily debated, since in crises correlations between market and insurance risks may be higher than the standard formula assumes. The *own funds* side also matters: it depends on the discount curve (including the treatment of any "volatility adjustment" or matching adjustment, which allows some credit-spread movements to be excluded from liabilities), which means regulatory design affects measured solvency and, indirectly, investment strategy.

8.7.3 What banks and insurers share, and where they differ

Dimension	Banks	Insurers
Main liability	Deposits and wholesale funding, callable	Policy liabilities, contingent on events
Run risk	High (liquidity-driven)	Lower (except lapse in life products; cat events)
Dominant balance-sheet risk	Credit and interest-rate	Underwriting, reserving, interest-rate (life)
Key capital constraint	CET1/RWA, leverage ratio, liquidity ratios	SCR or RBC ratio, own funds quality
Profit driver	Net interest margin, fees	Underwriting margin, investment spread
Key estimate in earnings	Loan-loss provisions	Loss reserves, actuarial assumptions
Typical hedging tools	Swaps, caps, floors, securitization	Swaps, reinsurance, cat bonds

A financial conglomerate (like Lakeshore Financial) must also manage *double leverage* (debt raised at the holding company and injected as equity into subsidiaries) and *contagion* between subsidiaries, and regulators often impose group-level supervision for that reason.

8.8 Managerial Implications: Capital Allocation, Payout, and Governance

8.8.1 Capital as the scarce resource

For a financial institution the binding constraint on growth is typically capital, not cash. This changes the toolkit of Volume 1. Capital budgeting (Chapters 6 and 7) becomes *capital allocation*: each business line, product, and client relationship is evaluated by risk-adjusted return on the regulatory and economic capital it consumes (Example 8.7). Cost of capital (Chapter 11) becomes a hurdle rate applied to allocated equity. Payout policy (Chapter 16) becomes a *capital plan*: dividends and buybacks are permitted only if projected capital ratios under stress remain above requirements (Example 8.8). Finally, mergers (Chapter 22) among financial institutions are shaped by regulatory approval and by the capital implications of combining balance sheets.

A sound capital-allocation framework has four elements: (i) a consistent measure of risk-based capital for each business; (ii) transfer pricing of funding and liquidity costs to businesses so that cheap deposits and costly wholesale funding are credited and charged appropriately; (iii) a hurdle rate that reflects the cost of equity; and (iv) incentives that reward risk-adjusted rather than raw profit. Without these, managers are paid for growth in high-spread, high-risk lines, and the bank accumulates risk in its tail.

8.8.2 Governance, incentives, and the real lessons of failures

Bank and insurer failures typically share a short list of causes: concentration in one asset class or funding source; growth outpacing controls; duration or liquidity mismatches justified by accounting classification; reserves or provisions set optimistically; and incentive systems that rewarded short-term earnings. The common thread is that *the accounting result and the economic result diverged and the board did not ask why*. For an MBA leader the practical checklist is: demand both earnings-based and economic-value views of rate risk; ask what depositors or policyholders could do in the worst week; check the quality of reserves and provisions against independent estimates; insist that limits be expressed as a percentage of capital and be enforced; and tie compensation to risk-adjusted outcomes with deferral and clawback.

8.8.3 Summary

Financial institutions earn money by bearing risks that customers wish to shed: maturity transformation, credit risk, and insurance risk. Regulation constrains these risks through capital and liquidity requirements that are broadly similar in principle across banks and insurers, and different in detail. Good management consists of understanding the economics behind the regulatory ratios, measuring risks in more than one way, pricing for the capital consumed, and holding buffers above the minimum that reflect the bank's own judgment about tail events.

Case Study

Tidewater Mutual Group: Growth, Duration, and Capital

Illustrative case; Tidewater Mutual Group is fictional.

Tidewater Mutual Group is a regional financial group consisting of Tidewater Bank, a commercial bank with $30 billion of assets, and Tidewater Life, a life insurer with $8 billion of invested assets. The group is listed, and its chief executive, Priya Raman, has promised shareholders a return on equity of 12 percent within two years. Current group ROE is 8.5 percent.

The bank's growth strategy centres on commercial real estate and small-business lending, where margins are 150 basis points higher than on residential mortgages. Commercial real estate loans carry a risk weight of 100 percent under the standardized approach, against 35 percent for mortgages, so growth consumes capital quickly. The bank's CET1 ratio is

11.2 percent against a requirement, including buffers, of 8.5 percent. The treasury department notes that 65 percent of the bank's deposits are retail, 25 percent are corporate operational deposits, and 10 percent are non-operational deposits from a small number of large corporate and municipal clients, which have been growing as the bank offered above-market rates to fund loan growth. The bank's LCR is 112 percent. The securities portfolio, 20 percent of assets, has an average duration of 5.5 years and unrealized losses equal to 18 percent of CET1; most securities are classified as held to maturity.

Tidewater Life sells fixed annuities with a guaranteed crediting rate of 3.25 percent for five years, followed by renewal rates. It has grown annuity sales rapidly by offering a rate 40 basis points above competitors. The investment portfolio has a duration of 6.0 years against a liability duration of 8.5 years, and the CIO reaches for yield by buying longer-dated private credit, which is illiquid. The economic solvency ratio is 148 percent against an internal target of 150 percent and a regulatory trigger of 100 percent. Lapse rates on annuities have risen from 4 to 7 percent as customers shop for higher rates. The company's actuary reports that if lapse rates reach 12 percent, the insurer would be forced to sell illiquid assets at a discount to meet withdrawals.

The group CFO, Marcus Delgado, has prepared a plan: raise commercial real estate lending by 25 percent over two years, increase annuity sales by 30 percent, buy back $150 million of stock, and finance the buyback with a holding-company debt issue, which would downstream proceeds as equity to subsidiaries. A stress test conducted by the risk team under a scenario of rising rates, falling property prices and a 15 percent rise in unemployment shows the bank's CET1 ratio falling to 7.9 percent and the insurer's solvency ratio falling to 104 percent; the test does not model the interaction between deposit outflows and securities losses, nor between lapses and asset sales.

The board's risk committee chair, Elena Voss, is uneasy. "Every number in this plan passes a regulatory test today," she says. "My question is what happens in a bad week, not a bad year."

Discussion questions

1. Compute, using the data given, the approximate additional RWA and CET1 needed for a 25 percent increase in commercial real estate lending if commercial real estate is currently 20 percent of the bank's assets (assume 100 percent risk weight). Does the plan still leave the bank above its 8.5 percent requirement after the buyback?
2. Identify the liquidity and interest-rate vulnerabilities in the bank's deposit and securities positions. Which metrics beyond the LCR would you ask for, and what would reverse stress tests look like?
3. Explain the duration mismatch at Tidewater Life, the interaction between lapses and asset illiquidity, and propose a hedging and product-design response. How would you test whether growth in annuities is value-creating after the cost of capital?
4. Evaluate the holding-company debt issue and buyback (double leverage). Which stakeholders are exposed, and what would you recommend to the board?

Key Takeaways

- Banks and insurers are leveraged intermediaries whose balance sheets generate their income; ROE equals ROA times the equity multiplier, and a high ROE may reflect low capital or low provisions rather than quality.
- Net interest margin depends on asset yields, funding costs, and deposit betas; naive repricing gaps can have the wrong sign once deposit behavior is modelled.
- Earnings-based and economic-value measures of interest-rate risk answer different questions; a bank can show rising NII and falling economic value at the same time.

- Maturity transformation creates run risk; the LCR and NSFR set minimum buffers and structural funding ratios but are not predictions, and uninsured, concentrated deposits behave differently from granular retail balances.
- Basel-style capital is built from tiers (CET1, AT1, Tier 2), risk-weighted assets, buffers, and a leverage-ratio backstop; falling into the buffer restricts distributions even before the minimum is breached.
- The IRB formula derives from a single-factor model: capital is the conditional default rate at a high confidence level, less expected loss, scaled by LGD; the capital charge is concave in PD, and capital should be priced into loans through RAROC.
- Stress tests project losses, revenue, and RWA under adverse scenarios; RWA inflation and the omission of liquidity feedback are common blind spots.
- Insurers earn underwriting profit and investment income on the float; because reserves are a multiple of equity, small reserve errors have large effects on earnings, and the tail factor is a major judgment.
- Life insurers face duration mismatch, lapse, and longevity risk; solvency regimes aggregate risk modules with correlations to a 99.5 percent one-year standard, and diversification vanishes as correlation rises.
- Capital is the scarce resource of financial institutions: allocate it by risk-adjusted return, plan payouts from stressed capital rather than current ratios, and keep buffers above regulatory minima.

Review Questions

1. Why is leverage a product feature of banks rather than only a financing choice? How does deposit insurance change the Modigliani-Miller logic?
2. Decompose ROE for a bank into spread, non-interest income, cost, provisions, tax, and leverage. Which components are most cyclical?
3. Explain why a bank with a negative contractual one-year repricing gap might benefit from rising rates. What assumptions are crucial?
4. Distinguish NII-at-risk from economic value of equity. Give an example of a balance sheet for which they give opposite signals.
5. Describe the Diamond-Dybvig argument for bank runs. How do deposit insurance, the LCR, and central-bank facilities each address it, and what do they leave open?
6. What is the difference between the LCR and the NSFR? Why are both needed?
7. Why does the IRB capital charge subtract expected loss? What role does the asset correlation play?
8. Give two reasons the leverage ratio is retained alongside risk-weighted requirements.
9. Why do insurers' combined ratios understate or overstate profitability? What should an analyst check about reserves?
10. Explain how square-root aggregation of risk modules works and why diversification credit is controversial.

Problems

1. **Profitability.** A bank has earning assets of 70,000 yielding 6.0 percent and interest-bearing liabilities costing 1,960 in total. Non-interest income is 500, operating expense 1,500, provisions 250, tax rate 20 percent, total assets 80,000, equity 6,000. Compute NIM, net income, ROA, and ROE.
2. **Capital ratio.** Exposures: corporate loans 40,000 (100 percent weight), mortgages 30,000 (35 percent), sovereign bonds 20,000 (0 percent), interbank deposits 10,000 (20 percent), other assets 5,000 (100 percent). CET1 capital is

6,900 and the requirement including buffers is 8.0 percent. Compute RWA, CET1 ratio, excess capital, and the additional 100-percent-weight lending capacity.

3. **LCR.** HQLA consist of 12,000 of Level 1 assets and 4,000 of Level 2A assets (85 percent recognition). Expected outflows are 18,000 and expected inflows are 6,000 (subject to a 75 percent cap on outflows). Compute the LCR.
4. **Repricing gap.** Rate-sensitive assets within one year are 30,000 and rate-sensitive liabilities 38,000. Compute the change in NII for +150 basis points and for −50 basis points.
5. **Duration gap.** Assets are 50,000 with modified duration 4.0; liabilities are 46,000 with modified duration 1.5; equity is 4,000. Compute the duration gap and the change in the market value of equity for a +100 basis point shift, and express it as a percentage of equity.
6. **ASRF.** For PD = 2.0 percent, LGD = 45 percent, ρ = 0.15, and confidence 99.9 percent, compute the capital charge K and risk weight. What RWA does a 200 million exposure generate?
7. **Insurer ROE.** Earned premium is 800, loss ratio 68 percent, expense ratio 28 percent, investments 2,400 at 3.5 percent, tax rate 21 percent, equity 700. Compute the combined ratio and ROE. Then compute ROE if there is 40 of adverse prior-year reserve development.
8. **Solvency aggregation.** An insurer has market risk SCR of 300 and insurance risk SCR of 200 with correlation 0.25 and no other charges. Own funds are 520. Compute the SCR, the solvency ratio, and the ratio if the correlation were 1.0.

Answers to Selected Problems

1. Interest income = 70,000 × 6.0% = 4,200; NII = 4,200 − 1,960 = 2,240; NIM = 2,240 / 70,000 = 3.20 percent. Pre-tax profit = 2,240 + 500 − 1,500 − 250 = 990; tax = 198; net income = 792. ROA = 792 / 80,000 = 0.99 percent; ROE = 792 / 6,000 = 13.2 percent.
2. RWA = 40,000 + 30,000 × 0.35 + 0 + 10,000 × 0.20 + 5,000 = 40,000 + 10,500 + 2,000 + 5,000 = 57,500. CET1 ratio = 6,900 / 57,500 = 12.0 percent. Required = 0.08 × 57,500 = 4,600; excess = 2,300. Maximum RWA = 6,900 / 0.08 = 86,250; additional capacity = 28,750.
3. HQLA = 12,000 + 0.85 × 4,000 = 15,400. Inflows are below 75 percent × 18,000 = 13,500, so net outflows = 12,000. LCR = 15,400 / 12,000 = 128.3 percent.
4. Gap = −8,000. For +150 bp: −8,000 × 1.5% = −120. For −50 bp: +40.
5. $D_{gap} = 4.0 − 1.5 \times (46{,}000 / 50{,}000) = 4.0 − 1.38 = 2.62$ years. $\Delta E \approx −2.62 \times 50{,}000 \times 0.01 = −1{,}310$, which is 32.75 percent of equity.
6. $\Phi^{-1}(0.02) = −2.0537$; numerator = −2.0537 + 0.3873 × 3.0902 = −0.8568; divided by 0.92195 = −0.9294; Φ = 0.1763. K = 0.45 × (0.1763 − 0.02) = 0.0703 (7.03 percent). Risk weight = 12.5 × 7.03% = 87.9 percent. RWA = 0.879 × 200 = 175.9 million.
7. Combined ratio = 96 percent; underwriting profit = 32; investment income = 84; pre-tax = 116; net = 116 × 0.79 = 91.6; ROE = 13.1 percent. With 40 adverse development: underwriting = −8; pre-tax = 76; net = 60.0; ROE = 8.6 percent.
8. $SCR = \sqrt{300^2 + 200^2 + 2 \times 0.25 \times 300 \times 200} = \sqrt{90{,}000 + 40{,}000 + 30{,}000} = \sqrt{160{,}000} = 400$. Ratio = 520 / 400 = 130 percent. With correlation 1.0, SCR = 500 and the ratio is 104 percent.

Part III — Governance, Compensation and Behavior

Chapter 9: Corporate Governance and Boards

Learning Objectives

After studying this chapter you should be able to:

- Explain corporate governance as the system of rights, institutions, and incentives that addresses two distinct agency problems: manager versus shareholders, and controlling shareholder versus minority shareholders.
- Describe board structure, fiduciary duties, independence criteria, and the work of the audit, compensation, nominating/governance, and risk committees, and assess how well a given board is designed to monitor and advise.
- Compute voting power, cash-flow rights, and control wedges under cumulative voting, dual-class shares, and pyramid structures, and evaluate how much minority investors lose when a controller diverts value.
- Analyze takeover defenses, proxy contests, and activism using the economics of free-riding, and judge when a shareholder's stake justifies the cost of intervening.
- Compare governance systems across countries (outsider versus insider, one-tier versus two-tier boards, stakeholder orientation) and explain why one model cannot simply be transplanted to another legal and ownership setting.
- Design and interpret a board evaluation, and read the empirical evidence on governance and firm value critically, distinguishing correlation, selection, and causation.

Opening Vignette

Illustrative scenario; the company and people are fictional.

Harborline Marine Systems makes navigation electronics and ship-board control software. It went public in 2009 and, seventeen years later, has a market capitalization of about $4.2 billion. Its founder and chair, Dagmar Ellison, is 71. She holds 100% of a special Class B share with ten votes per share and about 9% of the cash-flow rights; the public holds Class A shares with one vote each. The result is that Ellison controls roughly 50% of the votes, though she owns well under a tenth of the equity.

The board has eleven directors. Nine are labelled independent, but three of them have served for more than fifteen years, one is a retired partner of the firm's outside law firm, and one sits on the board of a company whose chief executive sits on Harborline's compensation committee. The chief executive, Marcus Oyelaran, was hired six years ago and is widely regarded as capable. Ellison, however, has begun to push for the acquisition of a yacht-electronics business owned by her late husband's family trust, at a price that Harborline's banker calls "full but defensible."

Institutional investors holding 38% of Class A shares have written to the lead independent director. They ask four questions. Who negotiates for Harborline against the family trust? Does the audit committee have a process for related-party transactions? What is the plan for succession when Ellison steps down? And will the board agree to a sunset clause that converts Class B shares into Class A shares after ten more years or on her death?

The lead director, Priya Raman, now faces a problem that no valuation model solves. The transaction might be good for Harborline. It might be a transfer of value to the family at the expense of Class A holders. The board's structure, independence, processes, and the incentives of the people in the room will determine which it is, and whether outside investors can ever tell the difference. This chapter develops the tools for thinking about exactly that situation.

9.1 Governance as an Agency Problem

9.1.1 Why governance exists

Volume 1 treated the firm as a nexus of decisions in which management maximizes value for shareholders. Corporate governance asks the harder question: *what makes managers actually do that?* Investors who supply capital cannot watch every decision, cannot write a contract that anticipates every contingency, and often cannot easily exit. The managers and controlling owners who run the firm have information, discretion, and interests of their own. Governance is the collection of legal rules, ownership structures, boards, contracts, markets, and norms that constrains the use of that discretion and channels it toward value creation.

The modern framing begins with Adolf Berle and Gardiner Means, who observed in the 1930s that dispersed ownership in large American corporations separated ownership from control. It was formalized by Michael Jensen and William Meckling, who defined agency costs as the sum of monitoring expenditures by the principal, bonding costs borne by the agent, and the residual loss that remains because interests are never perfectly aligned. Shleifer and Vishny's survey of corporate governance emphasized the other side of the story: the way in which suppliers of finance assure themselves of getting a return on their investment. Without such assurance, capital becomes expensive or unavailable, a point we return to when discussing cross-country evidence.

A useful decomposition of agency costs is the following.

```
Agency cost  =  monitoring cost  +  bonding cost  +  residual loss
Residual loss = value of the best feasible policy  -  value of the policy actually chosen
```

The residual loss is the economically important part. It includes overpriced acquisitions pursued for empire-building, capital projects maintained because they preserve jobs, dividends withheld to keep the firm's cash under management's control, and related-party transactions that move value to insiders. None of these appears as a line item in the financial statements.

9.1.2 Two agency problems, not one

Textbook discussions often assume dispersed shareholders and powerful managers: the *Type I* agency problem of Berle and Means. In much of the world, and in a sizeable minority of large U.S. firms, the dominant conflict is different. A controlling shareholder, such as a family, a state, or a parent company, has the votes to appoint the board and direct management. The conflict is between the controller and the *minority* shareholders: the *Type II* problem.

Feature	Type I: Manager vs. dispersed owners	Type II: Controller vs. minority
Typical setting	Widely held U.S. or U.K. public firm	Family, state, or group-controlled firm; dual-class firm
Who has power	Management, often with a friendly board	Controlling shareholder, appointing board and management
Classic misbehavior	Empire building, perquisites, entrenchment, risk avoidance	Tunneling, related-party deals, dilutive issues, low payouts to minorities
Main discipline	Board, takeover market, pay design, activists	Minority-protection law, independent directors, courts, exit via price
Monitoring free-rider problem	Severe (small stakes)	Concentrated on the minority side
Takeover market helps?	Often	Rarely; the controller can block bids

The distinction matters because governance prescriptions that fix one problem can worsen the other. Independent directors help in a widely held firm only if they are not captured by the CEO; in a controlled firm they are chosen, directly or indirectly, by the very party whose conduct they are supposed to police. Powerful shareholders reduce Type I conflicts but create Type II conflicts. The right design question is always *whose interests might diverge from whose, and what does the monitor depend on?*

9.1.3 The toolkit of governance mechanisms

Governance mechanisms fall into internal and external categories, and in each category it helps to identify which agency problem the mechanism addresses.

Internal mechanisms include the board of directors, shareholder voting rights, the structure of executive pay (Chapter 10), debt covenants and leverage that commit free cash flow to creditors (Jensen's free-cash-flow argument, Volume 1, Chapter 15), ownership concentration, and information systems such as audit and internal control.

External mechanisms include the market for corporate control (hostile takeovers and the threat of them), the product and labor markets, the legal system and its enforcement, regulation of disclosure, securities analysts, the financial press, credit rating agencies, and institutional and activist investors.

These mechanisms are partly substitutes and partly complements. A country with weak courts may rely on concentrated ownership as a substitute for legal protection. A firm with a staggered board and a poison pill (Section 9.4) has switched off the takeover market and therefore leans harder on the board. When reading any governance system, ask which mechanisms are doing the work and what happens if one fails.

9.1.4 Fiduciary duties and legal standards

Directors in common-law jurisdictions owe two principal duties. The **duty of care** requires informed, deliberate decision-making. The **duty of loyalty** forbids self-dealing and requires that directors place the firm's interests ahead of their own. In Delaware, where most large U.S. companies are incorporated, courts apply the **business judgment rule**, a presumption that disinterested, informed directors acted in good faith for the company. The presumption is rebutted when directors are conflicted, uninformed, or act in bad faith; then courts review the transaction under tougher standards such as *entire fairness*, which examines both fair dealing (process) and fair price. Special standards apply in contests for control, in the tradition of the *Unocal* and *Revlon* decisions: directors facing a hostile bid must show a reasonable threat and a proportional response, and once a sale of the company becomes inevitable they must seek the best price reasonably available.

For a finance leader the practical consequence is process. In a transaction with a controller or a conflicted insider, the combination of an independent special committee with its own advisers and real authority to say no, and approval by a majority of the minority, shifts both the legal standard and the economics. It does so because it replicates, imperfectly, the arm's-length negotiation that a controller's presence would otherwise eliminate. We apply this to Harborline later in the chapter.

9.2 Board Structure, Independence, and Duties

9.2.1 What a board does

A board has four functions that fit into one sentence: it *selects, monitors, rewards, and replaces* the chief executive; it *approves strategy and major capital decisions*; it *oversees risk, compliance, and financial reporting*; and it *serves as a source of advice and external connections*. The first function is the one the other three depend on. A board that cannot

remove a failing CEO has little leverage over anything else.

Boards do not manage. They meet perhaps six to ten times a year for a few hours each, depend on management for information, and consist of part-time directors with other commitments. The value a board adds is therefore concentrated in a few high-stakes judgments: whom to hire and fire, whether to authorize a bet-the-company acquisition, how to respond to a crisis, how to structure pay. Good governance design maximizes the quality of these few decisions.

9.2.2 Size, composition, and leadership

Board size. Empirical work, beginning with Yermack's study of large U.S. firms, found that smaller boards are associated with higher valuations, in a pattern consistent with coordination and free-riding costs rising in group size. The finding is correlational. Complex, diversified, or heavily regulated firms need larger boards to staff committees and span expertise, and size is a choice made by the firm. A defensible practical range for a large listed company is eight to twelve directors, with larger numbers requiring a clear reason.

Insiders, outsiders, and independents. Inside directors are executives; outside directors are non-executives; *independent* directors are outside directors with no material relationship to the firm or its management. Listing standards in the United States require boards to have a majority of independent directors, and require fully independent audit, compensation, and nominating committees for listed firms, with some exemptions for controlled companies. Independence tests typically look at employment within the last three to five years, material commercial relationships, family ties, receipt of consulting fees, and interlocking directorships.

CEO duality. Should the CEO also be chair? Combining the roles concentrates power and makes the chair, who sets the agenda and controls information flow to the board, a subordinate of herself in her role as CEO. Splitting the roles is the norm in the United Kingdom and common in continental Europe. In the United States combined roles remain common, often paired with a **lead independent director** with authority to call executive sessions and set agendas. The evidence that separation improves performance is weak, and good outcomes appear to depend on the quality of the individuals and processes more than on the label. Some firms deliberately keep a former CEO as chair during a transition, which can help or entrench, depending on whether the former CEO lets go.

Tenure, refreshment, and diversity. Long tenure builds firm-specific knowledge but may erode independence of mind. Board refreshment tools include term limits, retirement ages, and annual performance reviews. Diversity of background, experience, and perspective has been shown in some studies to improve decision quality and monitoring, but the evidence on narrow demographic measures and firm performance is mixed and sensitive to the sample and method. The managerial argument rests on a simpler point: homogeneous boards tend to share blind spots, which is costly where a firm faces uncertain strategic choices.

9.2.3 What independence does and does not buy

Formal independence is a necessary but not sufficient condition for effective monitoring. Three weaknesses deserve emphasis.

First, *formal independence is a checklist; independence of mind is a disposition*. A director who owes the CEO a board seat, who earns a high fee, who would lose prestige by dissent, or who values collegiality may satisfy every listing criterion and still defer.

Second, *independent directors are chosen by someone*. In practice, nominating committees often take CEO suggestions, and in controlled firms the controller chooses. Genuine independence needs a nomination process that the CEO and controller do not dominate.

Third, *independence is not expertise*. A director with no ties to the firm but no understanding of its technology, financial instruments, or competitive environment cannot detect problems. Boards need both independence and competence; the two are traded off in director selection.

The research literature offers a nuanced picture. Boards with more independent directors are associated with better CEO turnover decisions following poor performance, with better monitoring in takeover bids (lower bid premiums paid by acquirers on average where boards are independent), and with fewer accounting irregularities when the audit committee has financial expertise. But across studies, the simple relationship between the *fraction* of independent directors and firm performance is weak, partly because boards adapt to circumstances (struggling firms add outsiders, so adding outsiders correlates with poor performance) and partly because independence in form often differs from independence in substance. We take up this identification problem in Section 9.8.

9.2.4 Director incentives, time, and liability

Directors are paid a fee, usually part cash and part stock, and are exposed to reputational and legal risk but not to large financial loss. Their time commitment is constrained. A director who serves on four boards and holds a full-time executive job may have little capacity to probe a complex acquisition. Proxy advisors and large institutions commonly set limits, such as no more than four or five public company boards for non-executives and fewer for sitting CEOs. Stock-based director pay aligns incentives with shareholders but excessive equity pay or pay structured like executive pay may compromise independence from the CEO's agenda.

Directors are typically protected by indemnification in the charter and bylaws, by directors-and-officers (D&O) insurance, and in many jurisdictions by exculpation clauses for breaches of the duty of care (but not loyalty or bad faith). This protection makes service feasible and encourages reasonable risk-taking but it also means that legal liability is not the principal discipline; reputation and the possibility of losing board seats are.

Common mistake. Treating the percentage of independent directors as a governance score. A board of ten independents who see only what the CEO chooses to show, who have never met in an executive session without management, and who hold no committee authority is not meaningfully independent in function.

9.2.5 Information, agenda, and the role of the chair

The most underrated determinant of board effectiveness is information. Directors receive board packs written by management. If the CEO and CFO control the content, timing, and form of information, the board is a spectator. Good practice includes direct access to the heads of internal audit, compliance, and risk; meetings with managers below the CEO; independent advisers for committees; periodic site visits; and a board calendar that allocates time to strategy, risk, talent, and capital allocation rather than only to approval of management's agenda. The chair or lead director shapes all of this.

A practical test of the board's grip on strategy is whether directors can answer, without help, four questions: What are the two or three most important sources of value in this business? What would cause us to abandon the current strategy? Who are our three most plausible successors for the CEO, and what would each need to learn? And what is the biggest risk we are not discussing? Boards that cannot answer are in no position to evaluate management's proposals.

9.3 Board Committees and Board Process

9.3.1 Why committees

Committees allow the board to divide labor, develop expertise, and handle conflicts of interest. Their power is substantial: the audit committee decides who audits the company and how to respond to accounting disputes; the compensation committee sets the pay of the CEO; the nominating committee decides whom to put forward for election. For this reason, listing standards and codes require committee independence, and shareholders examine committee composition closely.

9.3.2 The audit committee

The audit committee oversees financial reporting, internal control, the internal audit function, and the external auditor. After the accounting scandals of the early 2000s, the Sarbanes-Oxley Act of 2002 required that listed-company audit committees consist solely of independent directors, that they appoint, compensate, and oversee the external auditor, that they establish procedures for complaints about accounting and auditing matters, and that at least one member be a financial expert (or the company explain). It also required CEO and CFO certification of financial statements and management's assessment of internal control over financial reporting.

The committee's critical tasks are often prosaic: reading the auditor's report on critical accounting estimates, challenging management's judgments on revenue recognition, impairments, and reserves (the accounting judgments behind the ratios in Volume 1, Chapter 2), meeting privately with the auditor and the head of internal audit, and reviewing related-party transactions. The effectiveness of the committee depends on whether its members have the time, skepticism, and expertise to ask follow-up questions.

9.3.3 The compensation committee

The compensation committee sets CEO and senior executive pay, approves incentive plan design, evaluates performance against goals, and oversees disclosure. Independence rules require that members be independent and that the committee have authority over its own consultants. Chapter 10 develops pay design in detail; here the governance point is that the committee sits at the center of a recognized conflict. The CEO influences who sits on the committee, the peer group used to benchmark pay, and the information on which performance is judged. Benchmarking against peer groups tends to ratchet pay upward, because most boards want to pay at or above the median.

9.3.4 The nominating and governance committee

This committee identifies director candidates, recommends committee assignments, manages board evaluation, oversees corporate governance guidelines, and often leads CEO succession planning (in other firms, the full board does). Its most valuable role is to maintain a **skills matrix**: a map of the competencies the board needs (industry, finance, technology, cybersecurity, international operations, regulation, human capital) against the competencies it has, which is updated as strategy and risks change.

9.3.5 The risk committee and other committees

Banks and other financial institutions in many regimes are required to have board-level risk committees. In non-financial firms, risk oversight is handled variously by the audit committee, a dedicated risk committee, or the full board. After the global financial crisis, regulators placed more emphasis on the role of the board in setting risk appetite, approving risk limits, and ensuring that incentives do not reward excessive risk taking (Chapter 10 revisits this). Other committees address special topics: technology and cybersecurity, sustainability, executive or finance committees, and ad hoc **special committees** formed to evaluate conflicted transactions such as a controller buyout or related-party acquisition.

9.3.6 Committee design principles

Committee	Core task	Typical independence requirement	Key risk if weak
Audit	Financial reporting, controls, auditor oversight	Fully independent; financial expertise	Misstatement, fraud, weak controls
Compensation	Executive pay, incentive design	Fully independent in most listing regimes	Pay unlinked to performance, ratcheting
Nominating/governance	Director selection, evaluation, succession	Fully independent in most listing regimes	Entrenched board, thin skills
Risk	Risk appetite, limits, stress testing	Mostly independent; mandatory for large banks in many regimes	Unseen risk accumulation
Special committee	Conflicted transactions	Disinterested directors, own advisers, authority to say no	Value transfer to insiders

A committee is only as strong as its mandate, its resources, and the willingness of the board to defer to it. Good practice gives committees budgets for independent advisers, requires regular executive sessions without management, and rotates chairs to prevent capture.

9.3.7 Board process: executive sessions, minutes, and decision records

Processes are mundane and decisive. Regular executive sessions allow directors to speak candidly without management present. Concise, accurate minutes that record the questions asked and the alternatives considered create the evidence that directors were informed, which matters if their decisions are later challenged. An annual agenda that includes strategy offsites, a deep-dive on each major business, and a rolling review of capital allocation against the cost of capital and returns (Volume 1, Chapters 6 and 11) keeps the board from drifting into reaction mode.

9.4 Shareholder Rights, Voting, and Takeover Defenses

9.4.1 The voting machinery

Shareholders do not run the firm, but they hold three powers that shape governance: they elect directors, they approve fundamental changes (mergers, charter amendments, sales of substantially all assets), and they may sue. The mechanics of the first power vary and matter more than most managers appreciate.

Straight (plurality) voting. Each share casts one vote per open seat, and the nominees with the most votes win. A holder of a bare majority can elect every director. Minorities get nothing.

Cumulative voting. Each share carries as many votes as there are open seats, and the holder may allocate them all to one candidate. This gives minority holders a path to representation. The minimum number of shares needed to elect `k` of `D` directors from `S` voting shares is:

```
Shares needed = S × k / (D + 1) + 1
```

Majority voting. In an uncontested election a director must receive more votes for than against; a director who fails typically must tender a resignation that the board may accept or reject. This replaced plurality voting at most large U.S. firms after campaigns by institutional investors, though the resignation tender still leaves the board with discretion.

Staggered (classified) boards. Directors serve overlapping three-year terms, one-third elected each year. A hostile bidder who wins a proxy contest in year one controls only a third of the board and cannot take control until the following year's meeting. Staggered boards are among the strongest anti-takeover devices, and institutional pressure has led most large firms to de-stagger.

Proxy access. A shareholder or group holding a minimum stake for a minimum time (often 3% for three years in the U.S. practice that emerged) may nominate directors on the company's own proxy card, avoiding the cost of a separate solicitation.

Example 9.1 – Cumulative voting and minority representation. Grumman Basin Instruments has 1,000,000 voting shares and nine open board seats. A dissident group wants to know how many shares it needs to elect two directors under (a) straight voting and (b) cumulative voting.

Step 1: Straight voting. Each of the nine seats is decided separately. A holder of 500,001 shares wins every seat. A holder of 499,999 shares wins none, because the opponent holds a majority on each ballot. The dissident group's minimum for two seats is therefore 500,001 shares, the same as for all nine: the system is winner-take-all.

Step 2: Cumulative voting threshold. Shares needed = 1,000,000 × 2 / (9 + 1) + 1 = 200,000 + 1 = 200,001.

Step 3: Verify. The dissidents hold 200,001 shares and cast 200,001 × 9 = 1,800,009 votes. They split them between two candidates: 900,004 and 900,005. The incumbents hold 799,999 shares, giving 799,999 × 9 = 7,199,991 votes. If incumbents run nine candidates and split evenly they get 799,999 votes each, below the dissidents' 900,004. The nine winners are the top nine vote-getters: both dissident nominees and seven incumbents. If incumbents instead run only seven candidates, each gets 1,028,570 votes, and the same result holds. Incumbents cannot shut the dissidents out.

Step 4: Three seats. Shares needed = 1,000,000 × 3 / 10 + 1 = 300,001. A bloc of just over 30% elects a third of the board, versus nothing under straight voting.

Interpretation. Cumulative voting converts a minority stake of 20% into 22% of the board (two of nine). It protects minorities in controlled firms but can also give a disruptive faction a seat that raises board conflict. Many U.S. states make it optional; many large firms exclude it in their charters.

9.4.2 Takeover defenses and their economics

The market for corporate control is the ultimate external governance mechanism: a poorly run firm trades below its potential value, and a bidder can profit by buying it and changing management (Volume 1, Chapter 22). Takeover defenses raise the cost of that process. They are defended on the grounds that they give boards bargaining power to extract higher premiums, protect long-term investment from short-term pressure, and prevent coercive bids. They are criticized on the grounds that they entrench managers who prefer a quiet life.

Defense	Mechanism	Governance effect
Poison pill (shareholder rights plan)	If an investor crosses a threshold (often 10-20%), other holders may buy shares at a deep discount, diluting the acquirer	Forces bidder to negotiate with the board; a board can remove the pill, shareholders cannot unless they replace the board
Staggered board	Only part of the board is elected each year	Delays control change by a year or more
Supermajority provisions	Mergers need 66-80% approval	Lets a large blockholder or insiders veto
Dual-class structure	Insiders hold super-voting shares	Complete bid-proofing while insiders remain
Golden parachutes	Large payments to executives on a change of control	Reduces managers' resistance to bids, but costs the bidder
Limits on special meetings and written consent	Shareholders cannot call meetings between annual meetings	Slows activist and bidder campaigns

Pills can be defended economically as bargaining tools: a board that can say no can negotiate a better price for shareholders. But a pill that is never redeemable, or a board that is also staggered, is no longer a negotiating instrument; it is a wall. The evidence from event studies finds that adoption of defenses tends to be accompanied by declines in the value of firms that have weak governance elsewhere, while firms with strong boards and large outside holdings can use them with less harm. As with all governance evidence, causality is hard to separate from selection.

9.4.3 Proxy contests, activism, and the free-rider problem

An activist investor buys a stake in a company, argues for strategic or governance changes, and seeks board seats through negotiation or a proxy fight. The economics of activism are dominated by one problem identified by Grossman and Hart: *free-riding*. If the activist spends resources to raise firm value, every shareholder benefits in proportion to ownership, but only the activist bears the cost. A small shareholder therefore rationally refuses to bear the cost of intervention, and nobody intervenes unless a large holder's share of the gain exceeds the cost.

Example 9.2 – Activist breakeven stake. A fund considers a campaign at Tidewell Chemicals, which has a market capitalization of $2,000 million. The fund believes that a break-up, board change, and capital-allocation reform would raise equity value by 10%. The campaign, including legal, advisory, and communication expenses, would cost $8 million. It estimates a 50% probability that the campaign succeeds.

Step 1: Value created if successful. $\Delta V = 10\% \times \$2{,}000$ million = $200 million.

Step 2: Fund's gain at a 3% stake, assuming certain success. 3% × $200 million = $6.0 million. Cost is $8.0 million, so net = –$2.0 million. A 3% stake does not pay even if success is certain.

Step 3: Breakeven stake with certain success. $s^* = \text{Cost} / \Delta V = 8 / 200 = 4.0\%$.

Step 4: Include the 50% success probability. Expected gain = $p \times s \times \Delta V - \text{Cost}$, so breakeven $s^* = 8 / (0.5 \times 200) = 8.0\%$. At 3% the expected net payoff is $0.5 \times 6.0 - 8.0 = -\$5.0$ million.

Step 5: Interpretation. Only large stakes justify intervention, which is why activists seek coalitions ("wolf packs") that share costs, build stakes cheaply before disclosure, use derivatives, or pursue campaigns that offer secondary private gains. At a 5% stake and 80% success probability, expected gain = $0.8 \times 0.05 \times 200 - 8 = 0.0$ (breakeven). The model highlights that governance activism concentrates on large companies with big potential gains and on firms with reachable boards, and that passive index funds, with tiny stakes in each firm, rely on stewardship teams and voting rather than campaigns.

The same logic explains the structure of institutional stewardship: large asset managers hold stakes of a few percent across thousands of companies, so intervention at any one firm is rarely worth the cost, but the aggregate value of voting policies, engagement, and the reputational weight of a large holder can be substantial.

9.4.4 Managerial implications

For a CEO or CFO, shareholder rights are not an abstraction. They determine how easily activists can reach your board, how much a bidder must pay, and what investors will pay for governance-sensitive stocks. A rational approach is to ask for each provision: *what problem does this solve, and is there a cheaper way to solve it?* A pill adopted in response to a specific threat and subject to shareholder ratification costs little; a standing set of entrenching provisions raises the cost of capital over time if investors regard management as unaccountable.

9.5 Ownership Structure and Controlling Shareholders

9.5.1 Dispersed and concentrated ownership

In the United States and United Kingdom, large listed companies tend to be widely held, though institutional holdings are concentrated among a few large asset managers. In much of continental Europe, Asia, and Latin America, most listed firms have a controlling shareholder. Faccio and Lang's study of Western European firms documented the prevalence of family control; La Porta, Lopez-de-Silanes, and Shleifer documented similar patterns across countries, with family and state control common outside the Anglo-American world.

Concentrated ownership is not inherently bad. A large owner has an incentive to monitor, which solves the free-rider problem. Family owners often have long horizons, deep firm knowledge, and reputations to protect. Evidence from some markets finds that founder-controlled and family firms perform well, particularly where founders remain in a leadership role. The problem is that a controller can also divert value, and the more the controller's voting power exceeds her share of cash flows, the stronger the temptation.

9.5.2 Separating control from cash-flow rights

Three devices let a controller hold votes disproportionate to money: dual-class shares, pyramids, and cross-holdings.

Dual-class shares. The company issues shares with unequal voting power. Founders hold high-vote shares and sell low-vote shares to the public. Supporters argue that this protects long-term vision from short-term market pressure and lets entrepreneurs raise equity without losing control. Critics argue that it entrenches founders whose abilities may decline, shields them from takeovers and activists, and allows them to extract private benefits with little financial exposure.

Pyramids. A holding company controls a second company, which controls a third. At each level the controller holds only a share of the equity but has control of the whole chain. Cash-flow rights multiply down the chain while control follows the weakest link.

Cross-holdings. Companies in a group own shares in each other, creating a web in which management of each firm is protected from outside shareholders. Japanese keiretsu and some Korean chaebol structures historically relied on this.

The central measure is the **wedge** between control and cash-flow rights.

```
Cash-flow rights  =  product of ownership stakes along the chain (pyramid)
                  =  ownership of total equity (dual-class: shares held / total shares)
Control rights    =  weakest link in the chain (pyramid)
                  =  votes held / total votes (dual-class)
Wedge             =  Control rights  -  Cash-flow rights
Ratio             =  Cash-flow rights  /  Control rights
```

Example 9.3 – Dual-class voting power. Sundial Media has 80,000,000 Class A shares (1 vote each) and 20,000,000 Class B shares (10 votes each). The founder owns all 20,000,000 Class B shares and 10% of Class A shares.

Step 1: Total votes. Class A votes = 80,000,000 × 1 = 80,000,000. Class B votes = 20,000,000 × 10 = 200,000,000. Total = 280,000,000.

Step 2: Founder's votes. Class A held = 10% × 80,000,000 = 8,000,000 votes. Class B held = 200,000,000 votes. Total = 208,000,000.

Step 3: Voting power. 208,000,000 / 280,000,000 = 74.3%.

Step 4: Cash-flow rights. Shares held = 8,000,000 + 20,000,000 = 28,000,000 out of 100,000,000 = 28.0%.

Step 5: Wedge and ratio. Wedge = 74.3% – 28.0% = 46.3 percentage points. Ratio = 28.0 / 74.3 = 0.377.

Step 6: Minimum stake for control. How little equity could the founder hold and keep a majority of votes? Suppose the founder holds only x Class B shares (10 votes) and no Class A, with 80,000,000 Class A outstanding. A majority requires 10x / (80,000,000 + 10x) > 0.5, which simplifies to 10x > 80,000,000, so x > 8,000,000. With 8,000,000 Class B shares plus 80,000,000 Class A, the founder holds 8,000,000 / 88,000,000 = 9.1% of the equity and exactly 50% of the votes.

Interpretation. With roughly 9% of the economics the founder can dominate every shareholder vote. Each $1 of private benefit the founder extracts costs her only $0.09 of her own wealth through the equity claim; if the benefit is worth more than 9 cents to her, she gains. This is the Harborline pattern from the vignette. Each additional vote per share raises the control-to-cash-flow ratio and therefore the incentive to extract.

Example 9.4 – A pyramid. The Okonkwo family owns 40% of Holdco. Holdco owns 50% of Midco. Midco owns 60% of OpCo, a listed operating firm.

Step 1: Control. At each link the controlling stake is a majority of the votes (or enough for effective control): 40% in Holdco, 50% in Midco, 60% in OpCo. Under the weakest-link rule, the family's control rights equal the smallest stake in the chain, which is its 40% in Holdco (a 40% stake is normally enough for effective control of a widely held firm).

Step 2: Cash-flow rights. 0.40 × 0.50 × 0.60 = 0.12, or 12%.

Step 3: Wedge and ratio. Wedge = 40% – 12% = 28 percentage points. Cash-flow/control ratio = 12 / 40 = 0.30.

Step 4: Capital invested. If OpCo has equity worth $1,000 million, the family's economic stake is 12% × $1,000 million = $120 million, yet it directs the whole $1,000 million asset base. Its control per dollar invested is 1,000/120 = 8.3 times.

Step 5: Incentive to divert. If OpCo sells an asset to a family-owned company at $50 million below fair value, the family loses 12% × 50 = $6 million as an OpCo shareholder and gains $50 million as the buyer, a net of +$44 million. Outside OpCo shareholders (88% of the equity, including Holdco and Midco's other owners) bear $44 million.

Interpretation. Pyramids make it possible to control large assets with small capital, allowing financing of expansion without diluting control, but they also magnify the incentive to move value between layers, a practice discussed next.

9.5.3 Tunneling and private benefits of control

Tunneling, in the terminology of Johnson, La Porta, Lopez-de-Silanes, and Shleifer, is the transfer of resources out of a firm to its controlling shareholder. It takes many forms: below-market asset sales to the controller, above-market purchases from controller-owned firms, excessive pay to family members, loans on favorable terms, transfer pricing, dilutive equity issues, and mergers between firms in the same group in which one side is favored. A subtler form is **propping**: the controller injects funds into a failing group firm to protect its reputation or access to credit, which may favor the controller's other interests.

Minority shareholders cannot easily observe tunneling, and, even if they can, remedies are costly. The result is a discount in the price of minority shares. The size of that discount reflects the probability and size of expected extraction, the transparency of the firm, and the strength of legal protection.

Example 9.5 – When does tunneling pay, and what do minorities lose? A controller owns 55% of the cash-flow rights of a firm worth $1,000 million before any diversion. Diversion is costly: for each $1 diverted, the controller receives only $0.75 because of legal fees, disguise costs, and inefficiency (a deadweight cost of 25%). Suppose the controller diverts 10% of firm value, $100 million.

Step 1: Firm value after diversion. The firm's value falls by the whole amount diverted: $1,000 – $100 = $900 million.

Step 2: Controller's position. Share of remaining value = 55% × 900 = $495 million. Receipts from diversion = 0.75 × 100 = $75 million. Total = $570 million. Before diversion the controller held 55% × 1,000 = $550 million. Net gain = $20 million.

Step 3: Minority's position. Before: 45% × 1,000 = $450 million. After: 45% × 900 = $405 million. Loss = $45 million, 10% of their value.

Step 4: Breakeven ownership. The controller gains per dollar diverted if the receipts per dollar, 0.75, exceed her cost per dollar, her cash-flow stake θ. So diversion pays if $\theta < 0.75$. Here $\theta = 0.55 < 0.75$, so diversion pays. At $\theta = 0.80$ the controller would not divert. If the deadweight cost were 50% (receipts $0.50 per dollar) diversion would pay only if $\theta < 0.50$.

Step 5: Net social loss. Total value after diversion = 900 + 75 = $975 million (firm's remaining value plus what the controller actually receives); the loss of $25 million is the deadweight cost. The $45 million minority loss is therefore larger than the controller's $20 million gain: a transfer plus a destruction of value.

Interpretation. The lower the controller's cash-flow stake and the cheaper diversion is (weak law, opaque accounting), the greater the incentive. This is why legal protection of minorities, disclosure, and the cash-flow stake of controllers act as substitutes: high controller ownership can partially replace weak law. It is also why a wedge, as in Examples 9.3 and 9.4, raises concern.

9.5.4 The value of control and the minority discount

When a controlling block is sold, the price per share often exceeds the price of minority shares. The **control premium** reflects both the ability to improve the firm (value that would be shared by all shareholders) and the private benefits of control (value not shared). In the cross-section, Dyck and Zingales found that control premiums vary widely across countries and are larger where investor protection is weaker, consistent with larger private benefits.

Example 9.6 – Interpreting a control block price. Karst Industrial has 100 million shares trading at $10 each ($1,000 million market capitalization). A holder of 30 million shares (30%) sells the block to a strategic buyer for $14 per share, and the buyer gains control of the board.

Step 1: Premium. (14 – 10) / 10 = 40%.

Step 2: Total paid over market. 30 million × $4 = $120 million.

Step 3: As a fraction of firm value. 120 / 1,000 = 12% of market capitalization.

Step 4: Interpretation. The buyer pays $120 million for the combination of (a) value from changing strategy (shared with other shareholders if it increases the share price) and (b) private benefits (not shared). If the market expects the strategic improvements to raise the share price to $11.50, the shared component is $1.50 per share and the block's private-benefit component is $2.50 per block share, worth $75 million. Whether minorities are better or worse off depends on how the buyer behaves after taking control: if the premium reflects shared gains, minorities benefit through the price; if it reflects private benefits to be extracted, minorities lose. This is why many jurisdictions have mandatory bid rules requiring a bidder who crosses a threshold (often 30%) to offer the same price to all shareholders, equalizing the sale of control. The United States takes a different approach, relying on fiduciary duties and the market, with fewer mandatory bid rules.

9.5.5 Managing controlled-company governance

For firms with controllers, practical safeguards exist that reduce the minority discount and lower the cost of capital.

Safeguard	How it protects minorities
Majority-of-minority approval for related-party deals	Gives outside holders a veto
Independent special committee with own advisers	Replicates arm's-length negotiation
Time-based sunset on super-voting rights	Limits duration of entrenchment
Ownership-based sunset (e.g., conversion when controller's cash stake falls below a threshold)	Links votes to economic exposure
Minimum free float and independent majority on key committees	Preserves some outside influence
Full related-party disclosure and audit committee review	Reduces opacity
Tag-along and mandatory bid rights	Ensures minorities share in a sale of control
Independent lead director with authority	Provides a recognized counterweight

For Harborline, the sensible response to the institutional investors' letter is a package: a special committee of directors with no tie to the Ellison family or the executives, empowered to negotiate with or reject the yacht-electronics purchase and to hire independent advisers; approval by a majority of the votes cast by Class A holders; full disclosure of the fairness analysis; and a sunset clause converting Class B shares on Ellison's death or incapacity or after a fixed period. These measures do not guarantee a fair price but they make an unfair one costly to impose, and they signal that the board expects to be held to account.

9.6 Governance Across Countries

9.6.1 Outsider and insider systems

Governance systems are commonly grouped into two ideal types. In **outsider** (market-based) systems, such as the United States and United Kingdom, ownership is dispersed, equity markets are deep, boards are mostly independent, and disclosure and shareholder litigation are central. In **insider** (relationship-based) systems, such as much of continental Europe, Asia, and Latin America, ownership is concentrated in families, banks, groups, or the state, and control operates through blocks, board seats, and relationships. Real systems are hybrids, and they are evolving.

Dimension	Outsider systems (U.S., U.K.)	Insider systems (e.g., continental Europe, East Asia, Latin America)
Typical ownership	Dispersed; large institutional holders	Concentrated: family, bank, group, or state
Dominant agency problem	Manager vs. shareholders	Controller vs. minority
Board model	Mostly one-tier; independent majority	Two-tier in Germany and the Netherlands; mixed elsewhere
Takeover market	Active	Limited by blocks, dual-class, cross-holdings
Minority protection	Securities law, litigation, disclosure	Varies widely; mandatory bid rules, related-party rules
Stakeholder role	Primarily via contracts and law	Often formal (co-determination) or informal (bank and employee ties)
Financing	Equity and bond markets	Bank loans, retained earnings, group financing

9.6.2 Legal origin and investor protection

La Porta, Lopez-de-Silanes, Shleifer, and Vishny argued that legal origin (common law, French civil law, German civil law, Scandinavian) shapes the level of investor protection, and that stronger protection is associated with larger equity markets, more dispersed ownership, and lower private benefits of control. The "law and finance" research program has been challenged on measurement, on the role of political economy and history, and on convergence in the law, but its core insight remains influential: *where outside investors are poorly protected, ownership concentrates, and where it is concentrated, the marginal share held by outsiders is worth less.* Enforcement matters at least as much as the text of the law: a protective statute that courts do not apply provides little shelter.

9.6.3 Board models: one tier and two tier

In the **one-tier** model the board contains executives and non-executives and meets together; in the **two-tier** model, found in Germany, the Netherlands, and Austria, a **management board** (Vorstand) runs the company and a **supervisory board** (Aufsichtsrat) appoints, oversees, and may dismiss the management board. Executives cannot sit on the supervisory board, which separates management from supervision structurally. In Germany, large companies are subject to co-determination: employee representatives hold up to half the seats on the supervisory board of the largest firms, though the chair, who is generally a shareholder representative, has a casting vote. Proponents argue that co-determination builds trust and long-term firm-specific investment. Critics argue that it complicates restructuring and can distort capital allocation. The empirical evidence on firm performance is mixed.

9.6.4 Codes, comply-or-explain, and stewardship

The United Kingdom's Cadbury Committee (1992), formed in the wake of corporate failures, introduced the approach of a code of best practice with a "comply or explain" rule: listed companies must either follow the code's provisions or disclose and justify deviations. This soft-law approach has been widely adopted in Europe and elsewhere. It places the burden of judgment on investors, who must engage with explanations; it works best where investors are active and credible.

The U.K. Stewardship Code (first issued in 2010) and similar codes in Japan (2014), and other markets encourage institutional investors to monitor, engage, and vote responsibly. Japan's Corporate Governance Code (2015) introduced independent directors, comply-or-explain, and later pressure to unwind cross-shareholdings and improve capital efficiency.

Over time, boards in Japan have added independent directors, share buybacks have grown, and cross-holdings have declined, though the process is gradual.

9.6.5 State-owned enterprises, family firms, and business groups

State-owned enterprises (SOEs) carry a different mix of objectives: commercial returns, employment, industrial policy, and political goals. The state is both owner and regulator, creating conflicts. The OECD's guidance for SOE governance recommends separating the ownership function from regulation, requiring professional boards, and applying the same transparency standards as private firms. Listed SOEs raise a particular minority problem: outside shareholders may find that company decisions serve non-commercial aims.

Family firms and business groups (chaebol, Indian and Latin American groups, Italian pyramids) can be efficient where capital markets are underdeveloped, because internal capital markets, reputation, and trust substitute for weak institutions. Their risks are succession (a competent founder may be followed by a less able heir), pyramid-style tunneling, and entrenchment. A central empirical finding in the literature is that family firms managed by professional CEOs, or by founders, perform better than those run by heirs, other things equal.

9.6.6 Convergence or persistence?

Hansmann and Kraakman argued that global competition would drive convergence on a shareholder-focused model; others, such as Bebchuk and Roe, emphasized path dependence: existing ownership structures, rents, and complementarities make change costly. In practice there has been partial convergence in form (more independent directors, audit committee rules, disclosure standards) with persistence in substance (ownership concentration, board culture, labor relations). A manager operating internationally should expect that board practices and shareholder expectations in each jurisdiction differ, and that importing another country's rule without its supporting institutions may fail. A U.S.-style independent audit committee in a family-controlled Asian company will face a different set of pressures than in a dispersed-ownership firm.

9.7 Board Evaluation and Effectiveness

9.7.1 Why evaluate

A board that never evaluates itself relies on hope. Evaluation identifies gaps in skills, process, and dynamics; it signals to investors that the board takes its role seriously; and it supports refreshment decisions that are otherwise awkward. U.K. codes require annual evaluation with an external facilitator at least every three years for larger companies, and many other regimes expect annual evaluation. Evaluation should cover the board, its committees, and, in the best practice, each director.

9.7.2 What to evaluate

A comprehensive evaluation addresses five areas:

1. **Strategy oversight.** Does the board understand the sources of value, challenge strategic assumptions, and monitor execution?
2. **Risk, audit, and controls.** Are risks identified and discussed, and do audit and risk committees receive reliable, timely information?
3. **CEO evaluation, compensation, and succession.** Is the CEO evaluated against clear goals, are incentives aligned (Chapter 10), and is there a ready pipeline for emergency and planned succession?

4. **Information and process.** Are board materials timely and decision-useful, do meetings leave time for debate, are executive sessions held?
5. **Board dynamics and composition.** Do directors challenge each other constructively, does the board hear dissent, and is the skills mix right for the strategy?

Methods include questionnaires, structured interviews with each director and senior executives (best practice includes interviews by an external adviser), observation of meetings, review of minutes and board papers, and benchmarking against peers. Self-assessment is cheap but tends toward generosity; external review is costlier but more candid.

9.7.3 Turning evaluation into action

An evaluation is only worth its follow-up. A good output is a short list of actions with owners and dates: replace a director whose skills no longer fit, reallocate meeting time from reports to debate, set up a committee of the board to examine succession, or clarify decision rights between the board and management. A common failure is to produce a report that is filed and forgotten, or one that sanitizes problems with generalities.

Example 9.7 – A weighted board scorecard. Corvane Logistics' board used a five-dimension scorecard on a 1-5 scale. Directors rated themselves; an external adviser's interviews with directors and senior managers gave an independent rating. Weights reflect the importance the nominating committee attached to each area.

Dimension	Weight	Directors' self-rating	External rating	Gap
Strategy oversight	25%	4.2	3.4	0.8
Risk, audit, controls	20%	4.0	3.8	0.2
CEO evaluation and succession	20%	3.8	2.9	0.9
Information and process	20%	4.1	3.6	0.5
Dynamics and composition	15%	4.4	3.5	0.9

Step 1: Weighted self-rating. 0.25 × 4.2 + 0.20 × 4.0 + 0.20 × 3.8 + 0.20 × 4.1 + 0.15 × 4.4 = 1.050 + 0.800 + 0.760 + 0.820 + 0.660 = 4.09.

Step 2: Weighted external rating. 0.25 × 3.4 + 0.20 × 3.8 + 0.20 × 2.9 + 0.20 × 3.6 + 0.15 × 3.5 = 0.850 + 0.760 + 0.580 + 0.720 + 0.525 = 3.435.

Step 3: Overall gap. 4.09 – 3.435 = 0.655. Directors rate themselves about 0.66 points more favorably than the external review does.

Step 4: Weighted gaps (gap × weight). Strategy 0.20; risk 0.04; succession 0.18; information 0.10; dynamics 0.135. Total = 0.655, matching Step 3.

Step 5: Priorities. Strategy oversight has the largest weighted gap (0.20), followed by succession (0.18) and dynamics (0.135). Succession and dynamics have the largest raw gaps (0.9). The audit and risk area is where self and external views agree.

Interpretation. The pattern is typical: boards believe they are strong where they feel comfortable and rate themselves generously on dynamics, while outsiders see a lack of candid challenge and unprepared succession. The action list should target the three largest weighted gaps, beginning with a strategy session at which directors, not management, set the agenda.

9.7.4 CEO succession as the board's central test

Succession planning is the clearest test of a board. A sound process maintains a standing list of internal candidates, assesses them with outside help, exposes them to the board through presentations and operating responsibilities, and defines emergency succession (if the CEO is incapacitated tomorrow, who steps in and what happens next). Firms with weak processes face value-destroying outcomes: delayed decisions, outside hires chosen under pressure, and predecessors who linger as chair and undermine successors. Research on CEO turnover finds that boards are more likely to dismiss CEOs after poor performance when boards are more independent and when blockholders are present, consistent with the view that the removal decision is the core test of monitoring.

9.8 Governance and Firm Value: Evidence, Interpretation, and Limits

9.8.1 What the evidence claims

A large literature relates governance to firm value. Selected threads, each described cautiously:

- **Governance indices.** Gompers, Ishii, and Metrick built an index counting 24 governance provisions (mostly takeover defenses and shareholder-rights restrictions) for U.S. firms and found that, in the 1990s, firms with strong shareholder rights had higher valuations (Tobin's Q) and a portfolio of such firms earned large abnormal returns relative to firms with weak rights. Subsequent work by Bebchuk, Cohen, and Ferrell reduced the index to six "entrenching" provisions (the E-index) with similar associations. But later studies, including by Bebchuk and coauthors themselves, found that the abnormal-return pattern largely vanished in the 2000s, consistent with either market learning (investors priced the provisions), the index capturing something else that changed, or a sample-specific phenomenon.
- **Board size and composition.** Yermack's finding of a negative relationship between board size and Tobin's Q; mixed results for independence; evidence that independent boards are better at CEO dismissal and acquisition bid decisions.
- **Cross-country studies.** Firms in countries with stronger investor protection are valued higher; within countries, firms that adopt stronger governance (for example by cross-listing in the U.S.) show valuation premiums that some interpret as "bonding" (Coffee) to a stronger legal regime and others attribute to the selection of better firms into cross-listing.
- **Dual-class firms.** Gompers, Ishii, and Metrick, and Masulis, Wang, and Xie, find that valuation falls as the wedge between votes and cash-flow rights widens, a finding consistent with agency costs. At the same time, some high-profile dual-class firms perform very well, particularly in early years when the founder's vision creates value. Several recent studies suggest that the penalty rises with time since IPO, motivating sunset clauses.
- **Activism.** Studies of hedge fund activism (Brav, Jiang, Partnoy, Thomas, and others) find positive abnormal returns around the announcement of activist stakes and improvements in operating performance in several measures, though critics question the durability of gains and the effects on long-term investment.
- **Takeover defenses.** Mixed results: staggered boards are associated with lower valuations in some studies, but identification is difficult because firms choose the defense.

9.8.2 The identification problem

Almost all governance variables are *chosen*. A firm's governance structure reflects its history, ownership, industry, and growth prospects. If governance is correlated with unobserved firm quality, then regressions of value on governance yield biased estimates.

Consider the standard pitfalls.

Omitted variables. Better-run firms (higher "quality") may both adopt good governance and have high value. Controlling for observable characteristics helps only to the extent that quality is observable.

Reverse causality. Poorly performing firms attract activists and adopt reforms in response, so good governance might follow weak performance, producing a negative correlation even if governance helps.

Simultaneity and endogenous choice. The theory of optimal governance (Himmelberg, Hubbard, and Palia; Demsetz and Lehn) predicts that firms choose governance structures appropriate to their circumstances, so that differences across firms should not be expected to correlate with value even if governance matters.

Sample and period dependence. The G-index result (strong in the 1990s, absent later) illustrates that associations may not be stable.

Credible identification uses natural experiments: staggered adoption of state anti-takeover laws, regulatory changes such as the Sarbanes-Oxley independence requirements affecting firms differentially depending on their pre-existing board composition, instrumental variables, regression discontinuities at close shareholder votes (Cuñat, Giné, and Guadalupe examine governance proposals passing or failing by narrow margins), and event studies of sudden shocks. These approaches reduce, without eliminating, concerns.

Example 9.8 – How omitted quality produces a spurious governance premium. A researcher studies 1,500 firms. Each firm has an unobserved *quality* score (mean 0, standard deviation 1) and a governance score. Good firms tend to have good governance: governance = 0.6 × quality + noise, where noise has standard deviation 0.8, so that the governance score has standard deviation about 1.0 and its correlation with quality is about 0.6. Tobin's Q is generated as Q = 1.50 + 0.30 × quality + random noise (standard deviation 0.5). By construction, governance has *no causal effect* on Q. The data below come from a Python simulation of this process (random seed 42).

Step 1: Naive regression of Q on governance. Estimated slope = 0.190, standard error = 0.0138, t-statistic = 13.8. A researcher would announce that a one-standard-deviation improvement in governance is associated with a Q higher by 0.19, which is 13% of the mean Q of 1.5, and a highly "significant" result.

Step 2: The theoretical bias. The slope from regressing Q on governance equals 0.30 × Cov(quality, governance) / Var(governance) = 0.30 × 0.60 / 1.0 = 0.18. The estimate 0.190 matches this calculation closely.

Step 3: Controlling for quality (if it could be observed). Estimated slope on governance = 0.025 (standard error 0.0156, t = 1.62), not different from zero. The coefficient on quality = 0.281 (close to the true 0.30).

Step 4: Add a small true effect. If the true governance effect were 0.05 and quality were still unobserved, the naive slope in the same simulation would be 0.234, overstating the true effect by a factor of nearly five.

Interpretation. Whenever governance is correlated with an unobserved quality, a regression will attribute quality's contribution to governance. Evidence that survives sensitivity to omitted variables, such as natural experiments and sharp discontinuities, is much stronger than a cross-sectional correlation. This is not an argument that governance does not matter. It is an argument for humility about the size of the effect and for scrutiny of how it was estimated.

9.8.3 What can be concluded

A balanced reading:

1. *Governance matters most where agency problems are severe and external discipline is weak.* The cross-country and dual-class evidence is consistent with this.
2. *No single structural feature (board size, independence percentage, CEO duality) reliably predicts performance.* Substance and process matter more than form.
3. *Boards matter most in a few decisions.* Evidence is strongest for CEO turnover, takeover bids, and related-party transactions.
4. *Provisions are priced.* In the 2000s the abnormal return to "good governance" portfolios disappeared, consistent with markets learning to price known provisions.
5. *Governance reforms can change behavior without raising value*, if they mandate form (for example, independence tests) without changing incentives.

For the manager, the practical message is that governance is a risk-management and cost-of-capital issue before it is a source of excess returns. Weak governance raises the probability of fraud, tunneling, and value-destroying acquisitions; investors price this through the discount rate and the minority discount. Strong governance will not rescue a bad strategy but it limits the downside and maintains the credibility that lowers financing costs (Volume 1, Chapter 11).

9.8.4 Common mistakes in governance analysis

- Equating *compliance* with *effectiveness*: ticking boxes on independence, committees, and policies.
- Reading a governance score as a causal driver of returns without considering selection.
- Ignoring the identity of the controlling shareholder; a "highly independent" board in a controlled company may be independent of management but not of the controller.
- Treating all dual-class firms alike: a ten-year sunset and an independent committee are very different from perpetual unequal voting with no independent oversight.
- Assuming that more governance is always better: excessive process can slow decisions, make boards risk-averse, and crowd out entrepreneurship.

Case Study

Illustrative case; Altamar Foods and all individuals are fictional.

Altamar Foods is a listed packaged-foods company with revenue of $3.1 billion and a market capitalization of $2.4 billion. The Vasquez family holds 100% of the Series V shares (five votes each) and 22% of total equity, giving it 59% of the votes. Public shareholders, among them two large asset managers holding 11% and 7% of the equity, own the remainder. The board has ten members: the chair and CEO, Rafael Vasquez (third generation); his sister, a non-executive director; the CFO; and seven directors classified as independent. The independent directors have a median tenure of 13 years, and four were originally recommended by the family.

Altamar's return on invested capital has averaged 6.8% over five years against a weighted average cost of capital of about 8.5% (Volume 1, Chapter 11). The group has made three acquisitions in adjacent categories, each at a premium above 35% and none yet earning its cost of capital. The CEO's cash bonus is tied to revenue growth and adjusted EBITDA.

Last quarter the board received a proposal from the CEO that Altamar acquire Sol Verde Beverages for $640 million. Sol Verde is 70% owned by a company controlled by the Vasquez family's cousins, who also hold 6% of Altamar directly. Management's fairness analysis, prepared with the company's usual investment bank, values Sol Verde at $610 to $690

million. Sol Verde's EBITDA multiple at the price is 13.5 times, versus a sector median of about 10.5 times. The CEO describes the acquisition as "keeping a heritage brand in the family of companies."

The audit committee chair, Helen Mbeki, notes that the committee has no written procedure for related-party transactions and that the proposal reached the full board with only a two-week review period. The two asset managers have written to the board demanding a special committee, a majority-of-the-minority vote, and a reconsideration of the company's dual-class structure. One manager has said that it will vote against the re-election of the nominating committee chair and the lead independent director if no steps are taken.

In the week before the board meeting, two of the independent directors privately tell Mbeki that they believe the price is too high but "do not want to cause a family rupture." A third says that the independent directors "owe the family their positions but must be seen to act independently." Mbeki must recommend a process by the end of the week.

Altamar's CFO has given her one more fact: financing the purchase with $400 million of new debt and $240 million of cash would raise net leverage from 2.1 times EBITDA to 2.9 times, close to the covenant limit of 3.0 times.

Discussion Questions

1. Identify the agency problems at Altamar (Type I and Type II) and explain which features of the ownership and board structure aggravate each. Compute the wedge between control and cash-flow rights, and estimate how much of a $50 million overpayment to the cousins' company would be borne by the Vasquez family itself.
2. Design the process by which Altamar's board should evaluate the Sol Verde proposal: who should sit on the special committee, what authority and advisers it needs, which approvals should be required, and what information the committee should demand beyond the fairness range.
3. Evaluate the independence of the seven "independent" directors. Which facts weaken the claim of independence in substance, and what changes to nomination, tenure, and executive sessions would help? What would you do about the two directors who privately doubt the price but fear a "family rupture"?
4. The asset managers propose a sunset clause for the Series V shares. Assess the arguments for and against, propose specific sunset terms, and discuss how the existence of the clause might change the valuation investors place on Altamar's minority shares. Use the identification issues from Section 9.8 to explain why you would not rely on a cross-sectional regression to argue the point.

Key Takeaways

- Governance addresses two agency problems: managers versus dispersed shareholders, and controllers versus minority shareholders. Prescriptions that fix one may aggravate the other.
- Boards add most of their value in a few decisions: hiring, evaluating, and replacing the CEO; approving large capital commitments; overseeing risk and reporting; and handling conflicted transactions.
- Formal independence is necessary but not sufficient. Substance depends on who nominates directors, what information they get, how long they serve, and whether they will say no.
- Committees concentrate power over auditing, pay, and nominations, so their independence, resources, and authority matter more than their number.
- Voting rules determine whether minorities can obtain representation. Cumulative voting needs $S \times k / (D + 1) + 1$ shares to elect k of D directors; plurality voting gives a majority holder every seat.

- Activism is governed by free-riding: a stake of size s pays only if $s \times p \times \Delta V$ exceeds the cost, so campaigns need large stakes, coalitions, or very large potential gains.
- Dual-class shares and pyramids separate control from cash-flow rights; the wedge raises the controller's incentive to divert, because diversion pays whenever the controller's cash-flow stake is below the receipts per dollar diverted.
- Tunneling transfers more from minorities than it gains for the controller, because diversion is wasteful. Independent special committees, majority-of-the-minority approvals, disclosure, and sunsets reduce the minority discount.
- Governance systems differ across countries because ownership, law, and enforcement differ; copying one country's rules without its supporting institutions often fails.
- Evidence on governance and value is vulnerable to omitted variables and reverse causality. Natural experiments are more credible than cross-sectional correlations, and effects found in one period (such as the 1990s governance-index results) may not persist.

Review Questions

1. Distinguish the Type I and Type II agency problems and give one governance mechanism that helps with each and one that could worsen the other.
2. Explain the duty of care, the duty of loyalty, and the business judgment rule. When does a court apply entire fairness instead?
3. Why might the percentage of independent directors be a poor measure of board quality? Give three reasons.
4. What are the main arguments for and against combining the roles of CEO and chair? What role does a lead independent director play?
5. Compare straight and cumulative voting. Why is cumulative voting valuable to minorities, and why do many companies opt out?
6. Using the free-rider logic of Grossman and Hart, explain why passive index funds rarely launch proxy contests, and why activists try to assemble coalitions.
7. Define the wedge between control and cash-flow rights. How do dual-class shares and pyramids create it, and why does it matter for tunneling incentives?
8. Compare the governance systems of the United States, Germany, and Japan with respect to ownership, board model, and the role of stakeholders.
9. Why is it hard to establish that governance causes higher firm value? Describe a research design that could improve identification.
10. Describe what a good board evaluation covers, how external facilitation changes the outcome, and why evaluation without follow-up is ineffective.

Problems

Problem 9.1. A company has 5,000,000 voting shares and 12 open board seats. Under cumulative voting, what is the minimum number of shares needed to elect 3 directors? Compare this with the shares needed to elect any director under straight voting where the rest of the vote is cast as a single bloc.

Problem 9.2. A firm has 60,000,000 Class A shares (1 vote) and 15,000,000 Class B shares (10 votes). The founder holds all Class B shares and 5% of Class A shares. Compute the founder's voting power, cash-flow rights, and wedge.

Problem 9.3. A family owns 35% of Holdco; Holdco owns 55% of Midco; Midco owns 70% of OpCo. (a) Compute the family's control rights and cash-flow rights in OpCo. (b) OpCo sells an asset to a family-owned company for $80 million below fair value. Compute the family's net gain and the amount borne by other OpCo shareholders (treat the family's cash-flow share as its OpCo stake).

Problem 9.4. A controller owns 60% of a firm worth $2,000 million. She diverts 5% of firm value ($100 million) and receives $0.80 per dollar diverted. Compute the controller's gain, the minority's loss, the deadweight loss, and the breakeven cash-flow stake above which she would not divert.

Problem 9.5. An activist considers a campaign at a company with market capitalization $5,000 million, expecting a 6% increase in equity value if successful. The campaign costs $12 million and the probability of success is 60%. Compute the breakeven stake and the expected net payoff at a 4% stake.

Problem 9.6. A holder sells a 20 million-share block (of 50 million shares outstanding) at $26 per share when the market price is $20. Compute the premium, the total premium paid, and the premium as a percentage of market capitalization. If the market price rises to $22 on the announcement, split the premium per block share into a shared component and a private component and compute the private component in total.

Problem 9.7. A board scorecard weights Strategy 30%, Risk 30%, and Succession 40%. Directors' self-ratings are 4.5, 4.0, and 3.5; external ratings are 3.8, 3.9, and 2.6. Compute the weighted self-rating, the weighted external rating, the overall gap, and the weighted gap for each dimension.

Problem 9.8. Quality has a variance of 1.0, governance has a variance of 1.25, and the covariance between quality and governance is 0.50. Tobin's Q rises 0.40 per unit of quality (a quality effect) and 0.02 per unit of governance (a true governance effect). If quality is unobserved, what slope would a regression of Q on governance estimate? How many times larger is it than the true effect?

Answers to Selected Problems

- **9.1.** Shares needed = 5,000,000 × 3 / 13 + 1: 5,000,000 × 3 / 13 = 1,153,846.15, so the minimum is 1,153,847 shares (about 23.1%). Under straight voting a holder needs 2,500,001 shares to win any seat against a unified majority bloc.
- **9.2.** Votes: Class A held = 3,000,000; Class B votes = 150,000,000; total founder votes = 153,000,000 of 60,000,000 + 150,000,000 = 210,000,000, so 72.9%. Cash-flow rights = (3,000,000 + 15,000,000) / 75,000,000 = 24.0%. Wedge = 48.9 percentage points.
- **9.3.** (a) Control = 35% (weakest link); cash-flow = 0.35 × 0.55 × 0.70 = 13.475%. (b) Family loses 13.475% × 80 = $10.78 million as OpCo shareholder and gains $80 million, a net gain of $69.22 million. Other shareholders bear 80 – 10.78 = $69.22 million.
- **9.4.** Gain: before 0.60 × 2,000 = 1,200; after 0.60 × 1,900 + 0.80 × 100 = 1,140 + 80 = 1,220; gain $20 million. Minority: before 800, after 760; loss $40 million. Deadweight loss = $20 million (diverted 100, received 80). Breakeven stake = 80%.
- **9.5.** Breakeven stake = 12 / (0.60 × 300) = 6.67% (ΔV = 6% × 5,000 = $300 million). At 4%: 0.60 × 0.04 × 300 – 12 = 7.2 – 12 = –$4.8 million.
- **9.6.** Premium = 26/20 – 1 = 30%. Total premium = 20 million × $6 = $120 million = 12% of market capitalization ($1,000 million). Shared component = $2 per share; private component = $4 per block share; total private = $80 million.

- **9.7.** Self = 0.3 × 4.5 + 0.3 × 4.0 + 0.4 × 3.5 = 3.95. External = 0.3 × 3.8 + 0.3 × 3.9 + 0.4 × 2.6 = 3.35. Gap = 0.60. Weighted gaps: strategy 0.3 × 0.7 = 0.21; risk 0.3 × 0.1 = 0.03; succession 0.4 × 0.9 = 0.36; total 0.60.
- **9.8.** Naive slope = 0.02 + 0.40 × 0.50 / 1.25 = 0.02 + 0.16 = 0.18, which is nine times the true effect of 0.02.

Chapter 10: Executive Compensation and Incentives

Learning Objectives

After studying this chapter you should be able to:

- Contrast the optimal-contracting and managerial-power views of executive pay, and use the informativeness principle to judge whether a pay design rewards what executives actually control.
- Value stock options, restricted stock, and performance share units, compute their grant-date fair value and deltas with the Black-Scholes model, and explain why executives value them at less than they cost the firm.
- Calculate pay-for-performance sensitivity and compare realized with granted pay, and interpret the “dollars per $1,000 of shareholder value” measure.
- Compare total shareholder return (TSR) and return on invested capital (ROIC) as performance metrics, show how each can mislead, and design a metric set that links pay to economic profit.
- Analyze how convex payoffs (options), leverage, and inside debt shape executives’ appetite for risk, and relate the analysis to financial-institution regulation.
- Apply clawback and malus mechanics, evaluate say-on-pay votes and pay-ratio disclosures, and recommend a defensible incentive plan for a given strategic situation.

Opening Vignette

Illustrative scenario; the company and individuals are fictional.

Northgate Components, a $10 billion supplier of precision parts to automotive and industrial customers, is preparing for its annual meeting. Last year’s say-on-pay proposal passed with only 62% of votes cast in favor, a shock to a board accustomed to 95%. The compensation committee chair, Naomi Falkner, has promised investors a redesign.

The existing plan has three pieces. Chief executive Julian Braddock earns a base salary of $1.4 million. His annual bonus pays from 0% to 200% of a $2.1 million target, based on adjusted earnings per share. His long-term award, with a grant-date value of about $9 million, is half stock options and half restricted stock units that vest on time. Over the last five years the stock rose 6% a year, in line with the sector. Braddock’s realized pay nonetheless averaged $12 million, mostly because the bonus paid out above target in every year as adjusted EPS grew through buybacks while operating profit stayed flat.

Investors have raised four complaints. Adjusted EPS excludes restructuring charges that recur every year. The time-vested restricted units pay out whether or not the stock performs. The peer group used to benchmark pay includes firms two to three times larger. And Northgate has earned a return on invested capital of about 7% against a cost of capital near 9% for five years, while the CEO has championed growth through acquisitions at a series of premiums.

Falkner’s advisor proposes shifting half the long-term award to three-year performance shares based on relative TSR, adding ROIC as a second metric, replacing adjusted EPS in the bonus with operating profit after capital charge, and adopting a stronger clawback. Braddock argues that relative TSR will reward luck and punish him for the sector’s cycle,

and that a capital-return metric will make him reject growth projects. Both objections have merit. This chapter develops the tools for deciding how much weight each deserves: valuation of the instruments, the math of sensitivity and risk, the strengths and traps of each metric, and the governance of the process.

10.1 Why Pay Design Matters: Two Views of the Contract

10.1.1 Pay as a solution to an agency problem

Chapter 9 framed governance as a response to agency problems. Executive compensation is the most direct instrument available to the board for aligning the executive's interests with those of shareholders, and it is the most visible: it is publicly disclosed, personally salient, and politically charged. Three questions organize the subject. *How much* should an executive be paid in total? *In what form*, and with what mix of fixed and variable pay? And *for what*: which performance measures, over what horizon?

Volume 1 equipped you to value firms and projects. Here the object being valued is a contract, and the "return" the board seeks is better decisions by the executive. A pay plan is best judged by the behavior it induces under the executive's own private objectives, not by its stated intent.

10.1.2 The optimal-contracting view

In the standard principal-agent framework, the board designs a contract that maximizes shareholder value subject to the executive's participation constraint (she must earn at least her outside option) and incentive-compatibility constraint (she chooses the action that maximizes her own expected utility given the contract). The risk-averse executive must be compensated for bearing risk, so more pay-for-performance means higher average pay, the classic trade-off between incentives and insurance.

Two results from this literature guide design. The **informativeness principle** (Holmstrom, 1979) says that pay should depend on any measure that provides information about the executive's effort, controlling for the information in other measures. Its corollary is *relative performance evaluation*: if peer returns reveal the common shock that the executive cannot control, filtering it out should improve efficiency by reducing the risk imposed on her. The **multitask problem** (Holmstrom and Milgrom, 1991) warns that when an executive has several tasks and only some are measurable, strong incentives on the measurable tasks pull effort away from the unmeasured ones. A CEO paid on current EPS will cut research spending and maintenance, which are hard to measure and whose benefits come later.

10.1.3 The managerial-power view

Bebchuk and Fried argued, in their critique of executive pay practice, that pay is not set at arm's length. Executives influence the board that sets their pay: they can affect director nominations, supply the information used to evaluate performance, and benefit from peer groups and consultants that reward higher pay. Boards are reluctant to defy executives; shareholders are dispersed and their ability to object is limited by "outrage costs": directors will tolerate generous packages unless disclosure makes them publicly embarrassing. The view predicts several patterns: stealth compensation through pensions and perquisites, options that reward luck (such as stock-price gains driven by a rising market), pay that rises with firm size irrespective of performance, windfalls on departure, and weaker pay-performance links where governance is weak.

Both views contain truth. Most observed features, such as heavy reliance on equity, can be consistent with either story. What distinguishes them is how practices vary with governance quality, and how plan terms respond to scrutiny. The managerial-power lens is a useful audit tool: for every design feature, ask whether it would survive if the executive had no

influence on the committee.

10.1.4 Levels of pay and the market for talent

Pay levels have risen far faster than median worker pay in many countries over recent decades. Explanations include growth in firm size and scale of responsibility, a competitive market for managerial talent and rising demand for general (portable) skills, changes in equity use, and weaker governance. Gabaix and Landier showed that in a model where talent is differentiated only slightly and the best CEO is assigned to the biggest firm, small differences in talent produce large differences in pay, and pay rises in proportion to firm size when firm sizes rise. The model rationalizes the cross-sectional correlation between pay and firm size and its time-series growth, but it does not itself explain why pay should be insensitive to performance. A practical consequence is that **benchmarking** relative to peer groups pushes pay up when most boards seek to pay at or above the median ("Lake Wobegon" effect). Careful peer selection (similar size, industry, complexity) and a documented rationale are therefore more important than the percentile target.

10.1.5 Regulatory and accounting context

A few institutional features recur in the design problem. Equity awards are expensed at **grant-date fair value** under U.S. GAAP (ASC 718) and IFRS 2; the expense is spread over the vesting period, so the accounting cost of an option grant is its estimated fair value, not what the executive eventually receives. U.S. tax law limits the deductibility of compensation above $1 million for certain top officers (the earlier exception for performance-based pay was eliminated in 2017), reducing one historical tax reason for options and performance shares. Disclosure rules in the U.S. require a compensation discussion and analysis, summary compensation tables, a pay-versus-performance table, and the ratio of CEO to median employee pay; the U.K., Europe, and other markets have analogous requirements, often with binding shareholder votes on the pay policy. We return to these in Section 10.7.

10.2 The Anatomy of an Executive Pay Package

10.2.1 Components

Component	Purpose	Typical form	Main weakness
Base salary	Fixed pay, retention	Cash, set against peers	Not tied to performance; floor on cost
Annual bonus	Reward near-term operating results	Cash, 0-200% of target against financial and strategic goals	Short horizon; target-setting games; earnings management
Stock options	Reward share-price appreciation	Right to buy shares at grant-date price	Rewards market beta; convex; encourages risk
Restricted stock/RSUs	Retention and alignment	Shares vesting over time	Pay for presence; value even when stock falls
Performance shares (PSUs)	Reward multi-year goals	Shares earned on TSR, ROIC, EPS or other metrics	Metric design; complexity
Pensions and deferred pay	Retention; inside debt	Defined benefit, SERP, deferred comp	Opaque; debt-like
Benefits and perquisites	Convenience, security	Aircraft, security, club memberships	Visibility; stealth pay
Severance and change-in-control	Retention through transition; insurance	Multiple of salary and bonus; accelerated vesting	Pay for failure; conflicts in takeovers

10.2.2 Fixed versus variable pay and mix

The proportion of pay that is at risk varies with seniority, industry, and firm risk. For large U.S. companies, long-term equity commonly makes up well over half of the CEO's target pay opportunity, with salary typically a small fraction. In practice, the right mix depends on (1) how much of the outcome the executive can influence, (2) how risk-averse and wealth-constrained she is, (3) how noisy performance measures are, and (4) the horizon over which effort pays off. A CEO of a highly cyclical business gets more noise from stock price and a smaller fraction of that variation reflects her actions, so the optimal weight on stock-based pay is lower than for a growth company whose value depends on discretionary choices.

10.2.3 Target pay and the performance range

Annual and long-term plans use schedules: a *threshold* (below which nothing is paid), a *target* (payout of 100%), and a *maximum* (cap, commonly 150-200%). The shape of this schedule affects behavior. A cliff at threshold encourages gaming near the boundary; a cap above which extra performance earns nothing encourages the executive to defer revenue; goals set too easily deliver payouts without effort. Goal-setting should be tied to the board-approved plan and to external expectations, and the board should specify in advance the treatment of adjustments (acquisitions, divestitures, accounting changes, restructuring) rather than make discretionary adjustments after the fact. The vignette's complaint that "adjusted EPS excludes recurring restructuring" is exactly this issue.

10.2.4 Ownership guidelines and holding requirements

Most large firms require executives to hold shares equal to a multiple of salary (commonly five to six times for the CEO) and often to retain a portion of net shares from vesting until retirement or beyond. Their purpose is to prevent executives from cashing out in a way that decouples their wealth from long-term value. The choice of holding period matters. Edmans, Gabaix, Sadzik, and Sannikov show in a dynamic model that deferring vesting and linking it to outcomes after the executive leaves can reduce incentives for short-term manipulation. Yet very long holding periods increase the executive's risk, so require a higher average pay, and can encourage a leaving executive to maintain a sub-optimal strategy if she still holds equity.

10.2.5 Common mistakes in pay design

- *Paying for what is easy to measure.* Revenue growth and adjusted EPS are easy to compute and easy to inflate.
- *Rewarding market movements.* Options with no hurdle pay out in rising markets; at-the-money options reward the sector tide.
- *Using the same metric for annual and long-term plans.* Double-counting makes incentives overweight a single measure.
- *Setting targets by last year plus a few percent.* Without reference to the board-approved strategy, targets drift.
- *Ignoring the fall-back.* When equity falls out of the money, boards re-price or add supplementary awards, which destroys the incentive's credibility.

10.3 Equity Instruments: Options, Restricted Stock, and Performance Shares

10.3.1 Stock options

An executive stock option gives the right to buy shares at a fixed exercise price K (usually the grant-date price) for a period of up to ten years, normally subject to vesting. It is a call option, so Volume 1, Chapter 18 supplies the tools. Because the option is non-transferable, the executive cannot hedge it with the underlying shares and must exercise early or forfeit on

departure. Therefore its expected term is shorter than the contractual term and its value to the executive is lower than market value.

Why cost exceeds value. Hall and Murphy showed that a risk-averse, undiversified executive who cannot trade the option values it below the Black-Scholes cost to the firm, because she bears the idiosyncratic risk the market would diversify away. This "cost-value wedge" means that options are an expensive way to deliver a given amount of perceived pay, but it also means that executives who value options at a deep discount to cost should respond more to explicit cash and stock than to the headline value of an option grant. Boards should therefore think of option grants in terms of the *incentive they provide* (delta and vega) and not of the dollar value they report.

Example 10.1 – Grant-date value and leverage of an option award. Northgate's share price is $50, there are 200 million shares outstanding (market capitalization $10 billion), and the compensation committee considers granting the CEO 400,000 at-the-money options with a ten-year term. Using an expected term of 6.5 years, annualized volatility of 30%, a risk-free rate of 4%, and a continuous dividend yield of 1.5%, we compute the Black-Scholes value (Volume 1, Chapter 18).

Step 1: Inputs. S = 50, K = 50, T = 6.5, σ = 0.30, r = 0.04, q = 0.015.

Step 2: d_1 and d_2.

```
d₁ = [ln(S/K) + (r − q + σ²/2) T] / (σ √T)
   = [0 + (0.04 − 0.015 + 0.045) × 6.5] / (0.30 × 2.5495)
   = 0.455 / 0.7649 = 0.5949
d₂ = d₁ − σ √T = 0.5949 − 0.7649 = −0.1700
```

Step 3: Cumulative normal values. $N(d_1) = 0.7240$; $N(d_2) = 0.4325$.

Step 4: Call value.

```
c = S e^(−qT) N(d₁) − K e^(−rT) N(d₂)
  = 50 × e^(−0.0975) × 0.7240 − 50 × e^(−0.26) × 0.4325
  = 50 × 0.90712 × 0.7240 − 50 × 0.77105 × 0.4325
  = 32.84 − 16.68 = $16.16
```

The value is 32.3% of the share price.

Step 5: Grant-date fair value. 400,000 × $16.16 = $6.47 million. This is the amount expensed over the vesting period and typically reported as the award's value.

Step 6: Delta and share-equivalents. Option delta = e^(–qT) $N(d_1)$ = 0.90712 × 0.7240 = 0.6568. The grant behaves like 400,000 × 0.6568 = 262,711 shares, with a dollar delta of 262,711 × $50 = $13.14 million.

Step 7: Leverage compared with restricted stock. An RSU award of the same $6.47 million grant-date value buys 129,314 shares ($6.47 million / $50) with a dollar delta of $6.47 million. The option award therefore delivers 13.14 / 6.47 = 2.03 times as much dollar sensitivity to the share price per dollar of reported cost. (This ignores the executive's discount for risk and illiquidity.)

Step 8: Dilution. 400,000 options are 0.2% of 200 million shares. The company's expense is $6.47 million, but the economic cost to shareholders is the value transferred, which depends on where the share price ends up.

Step 9: Granted versus realizable pay. Three years later, with 3.5 years remaining, the option's Black-Scholes value at three alternative share prices is (same r, q, σ): S = $38 gives $5.55 per option, $2.22 million in total with zero intrinsic value; S = $50 gives $12.15, $4.86 million; S = $70 gives $26.78, $10.71 million, of which intrinsic value is $20 × 400,000 = $8.00 million. The reported grant value of $6.47 million is therefore a poor guide to what the executive actually realizes, and it makes the case for showing *realizable* pay alongside grant-date pay when judging alignment.

10.3.2 Restricted stock and restricted stock units

RSUs deliver shares on vesting, so they have positive value even if the stock falls, and they retain executives (unvested units are forfeited on departure). From the viewpoint of incentives they are less leveraged than options: dollar delta equals the value of shares. Their weakness is that time-vested RSUs reward presence, not performance. Many investors consider them acceptable only as a minority of the package, and in start-up or turnaround settings where a share-price hurdle would not be credible.

10.3.3 Performance share units

PSUs deliver a number of shares that depends on the achievement of performance goals over a multi-year period, usually three years. A typical design specifies a target number of units and a payout curve of 0% to 200% of target. Goals may be market-based (relative TSR), financial (ROIC, EPS, free cash flow), or a mix. Valuing PSUs requires distinguishing two kinds of condition. For *market conditions*, such as relative TSR, accounting rules require the expected payout to be built into the grant-date fair value through a Monte Carlo simulation, and the expense is not reversed if the condition fails. For *performance conditions*, such as ROIC, the expense is recognized according to the probable outcome and trued up. This difference explains part of why firms choose one or the other.

PSUs can include modifiers (such as capping the payout when absolute TSR is negative, which stops the plan paying above target in a falling market) and can be settled in shares so that executive wealth remains tied to the share price after vesting.

10.3.4 Measuring pay-for-performance sensitivity

The incentive to raise firm value depends on how much the executive's wealth changes when shareholder wealth changes. Jensen and Murphy (1990) proposed the **pay-performance sensitivity**: the change in the executive's wealth per $1,000 change in shareholder wealth, including the effects of the value of shares and options held, bonus and salary revisions, and the risk of dismissal. Their central empirical finding was that the sensitivity for U.S. CEOs was low, on the order of a few dollars per $1,000, and that most of it came from stock ownership, not from the pay-contract design. Subsequent work, notably by Hall and Liebman, found that the growth of option grants in the 1990s raised the sensitivity substantially, though with high dispersion across firms. For our purposes, the exact estimate matters less than the calculation.

```
Dollar delta   =  Σ (shares-equivalent held × share price)
PPS (per $1,000 of shareholder value)  =  1,000 × Dollar delta / Market capitalization
```

Example 10.2 – Pay-performance sensitivity of a CEO's holdings. Julian Braddock holds 300,000 shares of Northgate, the 400,000 newly granted options of Example 10.1, and 600,000 older options with an exercise price of $30 and 3 years remaining (volatility 30%, r = 4%, q = 1.5%). Share price is $50 and market capitalization is $10 billion.

Step 1: Older options. Black-Scholes value = $22.37; delta = 0.8769.

Step 2: Dollar delta by component.

```
Shares:            300,000 × $50                    = $15.00 million
New options:       400,000 × 0.6568 × $50           = $13.14 million
Old options:       600,000 × 0.8769 × $50           = $26.31 million
Total dollar delta                                  = $54.44 million
```

Step 3: Sensitivity. A 1% rise in the share price changes Braddock's wealth by about 1% × $54.44 million = $0.544 million.

Step 4: PPS. PPS = 1,000 × $54.44 million / $10,000 million = $5.44 per $1,000 of shareholder value. Braddock's wealth moves by about half a percent of any change in firm value.

Step 5: Interpretation. Half a percent is meaningful in dollars but small relative to the stake of an owner-manager. A $200 million project with a 10% chance of failure that costs shareholders $20 million in expected value alters Braddock's expected wealth by only about $0.11 million, small compared with the personal costs, such as effort, reputational risk, and career risk, that the project may impose. Even large dollar deltas may leave the executive indifferent to many marginal decisions.

Step 6: A caution. Dollar delta from existing holdings is a *wealth-sensitivity*, not the strength of *new* incentives. Executives with huge accumulated wealth in company stock may be over-exposed (and so risk-averse), while those with little equity may have low alignment regardless of the size of a new grant.

10.4 Choosing Performance Metrics: TSR, ROIC, and Their Alternatives

10.4.1 What a metric must do

A performance metric should satisfy five conditions: it should be *aligned* with long-run shareholder value, *controllable* enough that the executive's actions move it, *measurable* with limited manipulation, *timely* enough to provide feedback, and *understandable*. No metric satisfies all five. The practical response is a small portfolio of complementary measures (typically two or three) with different weaknesses, a long measurement window, and board discretion bounded by pre-specified rules.

Metric	Strength	Weakness	Manipulation / distortion
Absolute TSR	Directly shareholder-aligned; hard to manipulate; no accounting	Dominated by market and sector moves; sensitive to start and end dates	Timing of news, buybacks, guidance
Relative TSR	Filters common shocks (informativeness principle)	Peer group choice; payoffs depend on rank among close competitors; rewards “least bad”	Strategic peer selection; sabotage incentive in theory
EPS growth	Familiar; short feedback	Ignores capital employed; boosted by buybacks and leverage; adjustments	Accruals, buybacks, cutting investment
Operating profit / EBITDA	Controllable; less leverage-sensitive	Ignores capital charge	Underinvestment in maintenance; capital-light accounting choices
ROIC (average)	Links profit to capital; discourages value-destroying growth	Rewards shrinking the base; underinvestment when ROIC high; accounting-based	Asset write-offs, leasing, sale-leasebacks, depreciation policy
Economic profit (EVA-type)	Charges for capital; consistent with NPV	Complexity; needs a capital-cost estimate	Capital definition
Free cash flow	Hard to inflate by accruals	Lumpy; can be raised by cutting investment	Working capital timing, capex deferral
Strategic/ESG scorecards	Capture non-financial value drivers	Subjective; small weights are noise	Goal-setting leniency

10.4.2 TSR: strengths and limits

Total shareholder return is the return an investor earns from price change plus reinvested dividends over the period:

```
TSR (annualized) = [ (P_end + reinvested dividends) / P_start ]^(1/n) − 1
```

Its appeal is that it is the shareholder’s own outcome, with no accounting discretion. Its problems are well known. First, TSR confuses what the executive controls with what she doesn’t: sector multiples, interest rates, and market sentiment dominate the short-term variance of returns, which is why **relative TSR** compares performance to a peer group or index to remove common shocks. Second, TSR is *expectations-adjusted*: a firm that starts the period with a high valuation must deliver extraordinary results merely to earn a normal return, whereas a firm starting from depressed prices earns TSR from re-rating. The start-date effect is one of the main reasons that three-year TSR goals can reward or punish for a decision made before the executive arrived or before the plan began. Third, TSR is vulnerable to value-neutral financial engineering that boosts short-term returns, and to disclosure timing. Fourth, relative TSR payouts depend on rank, so a peer group that is small or has idiosyncratic firms generates noisy payouts.

Relative TSR plans commonly pay at target for median performance, vest 0% below the 25th-30th percentile, and pay 200% at the 75th-90th percentile, with a cap on payout when absolute TSR is negative. A key design feature is the smoothness of the payout curve. A cliff creates a large discontinuity in the payoff that, in valuation terms, resembles a digital option and encourages gaming.

Example 10.3 – Valuing and stress-testing a relative-TSR performance share. Northgate considers granting performance shares that pay on three-year TSR rank against ten peers. The payout schedule, as a percentage of target units, is: 0% if the rank percentile is below the 25th; 50% at the 25th percentile; 100% at the 50th; 150% at the 75th; 200% at the 90th and above; linear in between. Percentile = (number of firms ranked below Northgate) / 10 (so rank 1 of 11 is the 100th

percentile and rank 11 is the 0th). Assume all eleven firms have a volatility of 30% per year and a pairwise correlation of 0.5 in annual returns, the risk-free rate is 4%, and Northgate's share price is $50. For simplicity assume total returns are reinvested (so TSR equals the price return on the total-return index) and that the executive receives shares worth S_3 at the end.

Step 1: Rank probabilities. Because the firms are statistically identical, Northgate's rank is equally likely to be any of 1 through 11, each with probability 1/11 = 9.09%.

Step 2: Payout at each rank.

Rank (of 11)	Percentile	Payout (% of target)
1	100%	200%
2	90%	200%
3	80%	166.7%
4	70%	140%
5	60%	120%
6	50%	100%
7	40%	80%
8	30%	60%
9	20%	0%
10	10%	0%
11	0%	0%

For example, at the 80th percentile: 150% + (0.80 – 0.75) / (0.90 – 0.75) × 50% = 150% + 16.7% = 166.7%. At the 40th percentile: 50% + (0.40 – 0.25) / 0.25 × 50% = 80%.

Step 3: Expected payout. Sum of payouts = 2.00 + 2.00 + 1.667 + 1.40 + 1.20 + 1.00 + 0.80 + 0.60 = 10.667; divided by 11 = 0.970. The expected payout is 97.0% of target, slightly below 100% because of the 25% cliff to zero and the cap at 200%. The probability of paying nothing is 3/11 = 27.3%; the probability of the maximum is 2/11 = 18.2%.

Step 4: Fair value by Monte Carlo. The fair value must reflect that the shares delivered are worth more when Northgate ranks high, because high rank is correlated with a high share price. We simulated 200,000 three-year paths (seed 7) for eleven correlated lognormal total-return indices, with return drift $(r - \sigma^2/2)\,T$ and common-factor correlation 0.5, and valued the payoff payout × S_3 discounted at the risk-free rate. Results: the simulated mean payout is 0.970 (matching the exact calculation), the simulated probability of zero payout is 27.3%, and the probability of maximum is 18.1%. The unconditional discounted expected share value is 1.001 × $50 = $50.05. The fair value per target unit is 1.202 × $50 = $60.1.

Step 5: Interpret the gap. If payout and share value were independent, fair value would be 0.970 × $50.05 = $48.5 per target unit. The Monte Carlo value of $60.1 is about 24% higher, because the payout is paid in more shares exactly when each share is worth more. The grant-date fair value used for accounting is therefore *greater than* the price of a share at target. A board that sets the number of target units by dividing a dollar value by the share price will under-count the cost, and a board that sets it from the simulated fair value will grant about 17% fewer units for the same intended value ($50.0 / $60.1 = 0.83).

Step 6: Design lesson. The 27% chance of zero and the 18% chance of a maximum payoff illustrate the cliff and cap. Smoothing the schedule near the 25th percentile and extending the cap would reduce the incentive to take excess risk near thresholds.

10.4.3 ROIC and economic profit

Volume 1 and Chapter 21 develop the value-driver formula for a firm growing at g with return on new invested capital equal to ROIC:

```
Value of operations  =  NOPAT × (1 − g / ROIC) / (WACC − g)
Economic profit (EP)  =  (ROIC − WACC) × Invested capital
NPV of a project      =  PV of the economic profits it generates
```

The formula shows that growth creates value only when ROIC exceeds WACC. A firm with NOPAT of $1,000 million, g = 3%, ROIC = 20%, and WACC = 9% is worth 1,000 × (1 – 0.03/0.20) / (0.09 – 0.03) = 1,000 × 0.85 / 0.06 = $14,167 million; with no growth it would be 1,000 / 0.09 = $11,111 million. The $3,056 million difference is the value created by growing at a spread of 11 points. The same growth at ROIC = WACC creates nothing.

ROIC is therefore an attractive pay metric: it links profit to capital and discourages growth at poor returns, a direct remedy for the empire-building behavior that Chapter 9 identified. But it introduces two opposite distortions, because an *average* ratio is not a *marginal* decision rule.

1. *High-ROIC firms underinvest.* If bonuses depend on the ROIC level, a manager of a high-ROIC business will reject projects that earn more than the cost of capital but less than the current average.
2. *Low-ROIC firms overinvest in poor projects that raise the average.* A project that earns less than the cost of capital but more than the firm's average ROIC raises the ratio but destroys value.

Additional distortions arise from accounting: ROIC rises when assets are depreciated, written off, or leased rather than owned, and when acquisition goodwill is excluded from invested capital (which flatters acquirers). Using a capital definition that includes goodwill and capitalized leases, measuring over several years, and using the *spread* (ROIC – WACC) times capital, i.e., economic profit, fixes the first-order problems.

Example 10.4 – The ROIC trap, and why economic profit fixes it. *Firm A* (a high-return business) has invested capital of $5,000 million, NOPAT of $1,000 million (ROIC 20%), and WACC of 9%. It has a project requiring $100 million that would earn a perpetual NOPAT of $12 million (ROIC 12%).

Step 1: NPV of the project. NPV = 12 / 0.09 – 100 = 133.33 – 100 = +$33.33 million. The project should be accepted.

Step 2: Effect on average ROIC. New ROIC = (1,000 + 12) / (5,000 + 100) = 1,012 / 5,100 = 19.84%. ROIC falls by 0.16 points. A manager paid on ROIC would reject the project.

Step 3: Effect on economic profit. EP before = (0.20 – 0.09) × 5,000 = $550 million. EP after = 1,012 – 0.09 × 5,100 = 1,012 – 459 = $553 million. EP rises by $3 million per year; its perpetuity value 3 / 0.09 = $33.33 million equals the NPV. A manager paid on EP accepts the project.

Firm B (a low-return business) has invested capital of $5,000 million, NOPAT of $300 million (ROIC 6%), and WACC of 9%. A project requiring $100 million would earn a perpetual NOPAT of $8 million (ROIC 8%).

Step 4: NPV. 8 / 0.09 – 100 = 88.89 – 100 = –$11.11 million. The project destroys value.

Step 5: Effect on ROIC. (300 + 8) / 5,100 = 6.04%, an increase from 6.00%. A manager paid on ROIC accepts a value-destroying project.

Step 6: Effect on EP. EP before = (0.06 – 0.09) × 5,000 = –$150 million. EP after = 308 – 0.09 × 5,100 = 308 – 459 = –$151 million. EP falls by $1 million, whose perpetuity value –1 / 0.09 = –$11.11 million equals the NPV. Under EP the manager rejects.

Interpretation. Level-of-ROIC bonuses induce the wrong decision in both directions. A spread-times-capital measure (economic profit, ideally with a multi-year window and a bank-style carry-forward of negative amounts) aligns incentives with NPV (Volume 1, Chapter 6).

10.4.4 TSR versus fundamentals: a decomposition

Because TSR reflects both operating results and changes in the multiple investors will pay, it is helpful to decompose it for the compensation committee.

```
Price return  =  (1 + EPS growth) × (1 + change in P/E) − 1
TSR           =  price return + dividend contribution (internal rate of return on price path and
dividends)
```

Example 10.5 – How much of TSR is management's doing? Northgate begins the period with a share price of $40 and EPS of $2.50 (P/E = 16.0). Over three years EPS grows 5% annually, to $2.50 \times 1.05^3 = \$2.894$. Dividends are $1.00, $1.05, and $1.10 at the ends of years 1 to 3.

Step 1: Scenario 1, the multiple holds at 16.0. Ending price = 2.894 × 16.0 = $46.31. The annualized return r solves $-40 + 1.00/(1+r) + 1.05/(1+r)^2 + 1.10/(1+r)^3 + 46.31/(1+r)^3 = 0$, giving r = 7.50% per year.

Step 2: Scenario 2, the multiple falls to 14.0 (a sector de-rating). Ending price = 2.894 × 14.0 = $40.52. The same equation gives r = 3.04% per year.

Step 3: Isolate the multiple. The operating performance, EPS growth of 5% a year and the same dividends, is identical in both scenarios. The de-rating from 16.0 to 14.0 (a 12.5% fall in the multiple, or 1/1.1429 = 0.875 times) reduces annualized TSR by 7.50 – 3.04 = 4.46 percentage points.

Step 4: Interpretation. If the CEO's three-year PSU is paid on absolute TSR against a hurdle of 6%, she receives nothing in Scenario 2 even though she has delivered exactly what the plan promised, and she receives a large payout in Scenario 1 if the sector re-rates upward even if EPS growth is poor. This is the limitation that motivates *relative* TSR (the sector de-rating hits peers too) and *operating-metric* measures such as ROIC and economic profit that are less sensitive to multiples. Equally, relative TSR does not remove firm-specific re-rating, which can reflect real changes in expected performance.

10.4.5 A recommended metric architecture

A defensible architecture for the vignette's firm would include:

- An annual bonus on *operating profit after capital charge* (economic profit) and a small number of strategic goals with measurable outcomes, with adjustments constrained by a pre-approved list.
- A long-term PSU based on a three-year cumulative measure, split between *relative TSR* (weight 50%, with a negative-absolute-TSR cap at target) and *average ROIC or economic profit* (weight 50%), measured with goodwill and capitalized leases in invested capital.

- Time-vested restricted stock as a minority of the long-term award, used mainly for retention, and options only where the board wants convexity (for example for a high-growth firm).
- Stock ownership guidelines and holding periods that extend beyond vesting.

The combination balances an external benchmark (relative TSR) with an internal capital-discipline measure (ROIC or EP). Neither measure alone is sufficient, and the resulting complexity is justified only if directors can explain each element.

10.5 Risk-Taking Incentives

10.5.1 Convexity, vega, and risk appetite

An executive's wealth is a function of firm value. If that function is linear (shares), she is exposed to downside as well as upside, and a risk-averse executive will tend to choose less risk than diversified shareholders would. If the function is convex (options), the executive gains from variance: the value of an option rises with volatility, and the sensitivity of option value to volatility is its **vega**. The executive can raise the value of her options by taking on projects that increase the volatility of firm value, even without raising its mean.

```
Vega (per 1.00 of σ)  =  S e^(−qT) φ(d₁) √T
Executive vega        =  number of options × option vega
```

Smith and Stulz explained how convex payoffs can offset risk aversion and induce hedging, and Core and Guay showed that portfolio vega and delta vary widely across firms. Empirical research finds that higher vega is associated with higher stock-return volatility, more R&D and leverage, and fewer diversifying acquisitions. Whether this is good or bad depends on the firm. For a firm whose managers underinvest in risky positive-NPV growth options, convexity may be exactly what the board wants. For a highly leveraged bank, it is dangerous.

Example 10.6 – Vega, risk shifting, and inside debt. Using the Northgate inputs from Example 10.1, compute the value of the 400,000-option grant at different levels of volatility.

Step 1: Option values. Black-Scholes values per option for S = K = $50, T = 6.5, r = 4%, q = 1.5%: at σ = 20%, $12.26; at σ = 30%, $16.16; at σ = 45%, $21.76. Grant values: $4.90 million, $6.47 million, and $8.71 million.

Step 2: Vega. Vega per option at σ = 30% = 50 × e^(–0.0975) × φ(0.5949) × √6.5 = 45.355 × 0.3341 × 2.5495 = $38.65 per unit of σ, or $0.3865 per volatility point (1% change in σ). For the whole grant, 400,000 × $0.3865 = $154,600 per volatility point.

Step 3: A value-neutral risk shift. Suppose management can change strategy so that the volatility of equity returns increases from 30% to 45% with no change in expected firm value. Share price stays at $50 (for simplicity). The CEO's new grant increases in value by $8.71 million – $6.47 million = $2.24 million. Shareholders see no change in expected value, although they bear additional risk (and, if the firm has debt, creditors bear more of the downside).

Step 4: Inside debt and neutrality. Suppose the firm has D/E = 0.8 (market values). The CEO's pension and deferred compensation, which are unsecured claims on the firm and resemble debt, total $12 million, and her equity-linked holdings (shares and options, measured by dollar delta) are $30 million. Edmans and Liu showed that, with the CEO's inside debt-to-equity ratio equal to the firm's debt-to-equity ratio, she is indifferent to risk shifts that transfer value between claimholders. Neutral inside debt = 0.8 × $30 million = $24 million. Her actual ratio is 12 / 30 = 0.40, half the firm's ratio of 0.80, so she is biased toward shareholders and equity-like risk. A $12 million increase in inside debt would remove the bias.

Step 5: Interpretation. A CEO whose wealth is almost entirely equity-linked prefers risk shifts that favor equity over debt, increasing leverage, volatility, and payouts. Creditors price this behavior at issuance (Volume 1, Chapter 15), and regulators of banks, for whom the creditors include depositors and the public, address it directly.

10.5.2 Banks and the post-crisis response

The global financial crisis focused attention on pay in financial firms. A bank's debt-holders and depositors, backed by an implicit or explicit public guarantee, are not well placed to price risk-taking by management; a pay plan that rewards short-term revenue or returns on equity with large equity-based bonuses may induce risk accumulation with fat-tail losses borne by others. The Financial Stability Board's principles for sound compensation practices and the Basel supervisory expectations, together with national rules such as the EU capital-requirements directive's bonus cap and deferral rules, push toward: deferral of a large share of variable pay (often three to five years), payment of a significant portion in shares or contingent capital instruments, **malus** (reduction of unvested pay) and clawback for misconduct or material risk failures, risk-adjusted performance measures, and board risk committee involvement in setting pay. Studies of bank CEO pay before the crisis find mixed relationships between equity incentives and risk-taking and subsequent performance; the evidence that equity-heavy pay caused the crisis is not definitive, but the logic of convexity with limited liability is robust. A related idea is to tie deferred pay to the firm's debt (such as bail-in bonds) so that executives' payoffs include the downside faced by creditors.

10.5.3 Short-termism and horizon

A related concern is that executives manage reported results or stock price over short horizons at the expense of long-term value: cutting research and advertising, deferring maintenance, timing news around vesting dates, or using buybacks to meet EPS goals. Stein's model of myopic management shows how an executive whose compensation depends on the near-term stock price may choose projects that raise current market expectations. Bolton, Scheinkman, and Xiong show how option-heavy, short-vesting pay can encourage deliberate pursuit of short-term price run-ups. Remedies include longer vesting, holding periods after vesting, multi-year metrics, and incentives tied to measures that look ahead (strategic milestones, customer retention, product pipeline), though the last are harder to verify.

10.6 Clawbacks, Malus, and Holding Requirements

10.6.1 The logic of recovery

Incentive pay is paid on measured results. If the measurement later proves wrong because of a restatement, misconduct, or a hidden risk that materializes, the executive has been paid for performance that did not occur. Recovery provisions restore the link between pay and the true outcome. **Clawback** refers to recovering pay already received; **malus** to reducing or cancelling unvested or deferred pay before it is delivered. Both rely on the board's willingness to invoke them, which makes the design (triggers, scope, and discretion) as important as the policy statement.

In the United States, Section 304 of the Sarbanes-Oxley Act required reimbursement of bonuses and equity gains by the CEO and CFO after a restatement caused by misconduct, but it was seldom enforced and left no private right of action. Under Rule 10D-1, adopted by the SEC in 2022 and implemented through exchange listing standards from late 2023, listed companies must adopt policies to recover incentive-based compensation erroneously awarded to current and former executive officers over the three completed fiscal years before an accounting restatement. The recovery is *no-fault* (it applies whether or not the executive was responsible), is computed on a pre-tax basis, and the company cannot indemnify

the executive. Boards may decline recovery only in narrow circumstances, such as when the cost of recovery would exceed the amount recovered. Many companies go beyond the rule with *discretionary* clawbacks for misconduct or reputational harm, and for non-financial triggers such as risk-management failures.

10.6.2 Design questions

Design question	Options	Considerations
Trigger	Restatement only; misconduct; risk failures; reputational harm	Narrow triggers are certain but leave gaps; broad triggers need judgment
Fault	No-fault vs. fault-based	No-fault is simple and stops disputes about intent; fault-based can be fairer
Who	Named executive officers vs. all participants	Wider coverage deters but is costlier to administer
Pay covered	Cash bonus, performance equity, time-vested equity	Rule 10D-1 covers incentive-based pay only; stock-price-based pay needs reasonable estimation
Look-back	Three years (U.S. rule); longer for banks	Longer periods improve deterrence but require reliable record-keeping
Net or gross of tax	Gross (pre-tax) under U.S. rule	Executive bears the tax cost unless it can be reclaimed

Example 10.7 – Clawback after a restatement. Marlow Industrial's annual bonus has a target of $1.5 million. Payout is 0% if EPS is below $2.70; 50% at $2.70; 100% at $3.00; 200% at $3.30 or more; linear in between. Reported EPS was $3.20. A later restatement of revenue recognition reduces EPS for that year to $2.85.

Step 1: Payout on reported EPS. $3.20 lies between $3.00 and $3.30: payout % = 100% + (3.20 – 3.00) / (3.30 – 3.00) × 100% = 100% + 66.7% = 166.7%. Bonus paid = 1.667 × $1.5 million = $2.50 million.

Step 2: Payout on restated EPS. $2.85 lies between $2.70 and $3.00: payout % = 50% + (2.85 – 2.70) / 0.30 × 50% = 75%. Bonus that should have been paid = 0.75 × $1.5 million = $1.125 million.

Step 3: Erroneously awarded amount. $2.50 million – $1.125 million = $1.375 million, recoverable on a pre-tax basis. If the executive's combined tax rate was 40%, she retained $1.50 million after tax on the original bonus, so the after-tax cost of repayment to her is $1.375 million less any tax relief she can obtain (recovering a tax deduction in a later year is possible in some jurisdictions but is not guaranteed). Her net after-tax bonus on a correct calculation would have been 0.6 × $1.125 million = $0.675 million.

Step 4: Interpretation. A restatement that lowers EPS by about 11% ($0.35 on $3.20) removes 55% of the bonus ($2.50 million to $1.125 million) because of the steepness of the payout curve. Steep payout schedules amplify the incentive to manage earnings near thresholds and make the consequences of restatement large. A board should expect the effects when reviewing the plan: gradual curves and multi-year averaging reduce both the temptation and the cost of recovery.

10.6.3 Limits of clawbacks

Clawbacks work after the fact and only for events that are discovered, can be quantified, and fall within the look-back period. They do not repair the damage from a bad acquisition or excess risk that never involves a restatement. They can also induce executives to demand higher fixed pay to compensate for the risk of recovery. For that reason clawbacks are a complement to, not a substitute for, good metric design, strong controls (Chapter 9), and long vesting.

10.7 Governance of Pay: Say-on-Pay, Disclosure, and the Process

10.7.1 Say-on-pay

The United Kingdom introduced an advisory vote on the directors' remuneration report in 2002, and in 2013 added a *binding* vote on the forward-looking pay policy at least every three years. In the United States, the Dodd-Frank Act of 2010 requires listed companies to hold an advisory say-on-pay vote at least once every three years, with a vote every six years on the frequency; most large companies hold it annually. Other jurisdictions use binding or advisory votes of various kinds. Because U.S. votes are non-binding, their force comes from reputational and governance consequences: proxy advisors and institutional investors treat low support, commonly below 70% or 80%, as a trigger for engagement, a required board response, and possibly votes against compensation committee members. Companies that receive low support commonly respond by changing plan design, increasing disclosure, or adding performance conditions.

Research on say-on-pay finds that support is strongly related to simple measures of pay-performance alignment (pay relative to TSR and to peers), the use of problematic practices (such as excise tax gross-ups or repricing of options), and the recommendation of the proxy advisor. The evidence on whether say-on-pay reduces pay levels is mixed; there is better evidence that it reduces the incidence of the most criticized practices and that it increases boards' responsiveness to investors.

Example 10.8 – Reading the vote and the pay ratio. At Northgate's last meeting, 200 million shares were outstanding and 88% voted. The say-on-pay proposal received 62% of votes cast. The CEO's reported total compensation was $14.2 million and the median employee's annual pay was $62,000.

Step 1: Votes cast. 200,000,000 × 0.88 = 176,000,000.

Step 2: Votes for and against. For = 0.62 × 176,000,000 = 109,120,000. Against = 176,000,000 – 109,120,000 = 66,880,000.

Step 3: Shift required to reach 70% support. The target is 0.70 × 176,000,000 = 123,200,000 votes for. Required shift = 123,200,000 – 109,120,000 = 14,080,000 votes, about 21% of the "against" votes (14.08 / 66.88 = 21.1%) or 8% of all votes cast. Because institutional holders cast most of the "against" votes, engagement with a small number of holders could achieve this.

Step 4: Pay ratio. $14,200,000 / $62,000 = 229 to 1.

Step 5: Interpretation. The pay ratio by itself is a weak indicator: it depends on the workforce composition (a firm with many outsourced or part-time workers has a different median), the volatility of equity awards in the numerator, and the geographic distribution of employees. The ratio's value is as a prompt to examine internal pay structure and to compare year to year within the same firm. The more useful analysis for the committee is the *pay-for-performance* comparison: realizable pay over three to five years against TSR relative to peers, which in Northgate's case shows high payouts despite sector-average returns.

10.7.2 Disclosure and the pay-versus-performance table

The U.S. rule requiring a table of "compensation actually paid" against TSR, peer-group TSR, net income, and a company-selected measure helps shareholders compare. Its weakness is the methodology for "actually paid" (which counts changes in the fair value of equity awards) and the lack of a single standard, but it moves the debate from grant-date value to realizable pay. In other jurisdictions, regulators ask for a similar link and for clearer discussion of the performance outcomes achieved.

10.7.3 Compensation committee process

A strong process has six elements: an independent compensation committee with the authority to hire its own consultant (and with a consultant free from other conflicting work for management); a defined peer group and philosophy approved at the start of the cycle; goal-setting that ties targets to the board-approved strategic plan and to external expectations; scenario analysis showing payouts under low, target, and high outcomes (including the realizable value of equity at several share prices, as in Example 10.1); a pre-specified policy on adjustments; and a record of discretion exercised, with reasons. Shareholder engagement before the proxy season, rather than after a failed vote, is the most reliable way to avoid surprises.

10.7.4 Severance and change in control

Severance and change-in-control provisions affect incentives at the critical points of a CEO's tenure. Double-trigger vesting (accelerated only if the executive is terminated after a change in control) is regarded as better practice than single-trigger vesting (accelerated on the transaction itself), which can induce executives to favor a sale. Large cash payments at termination for poor performance undermine the credibility of the performance link. Excise-tax gross-ups, retirement benefits that exceed those of other employees, and "evergreen" agreements are widely disfavored.

10.7.5 Putting it together: a design checklist

Question	What good looks like
What decisions do we want the CEO to make differently?	Named, specific, tied to the strategic plan
Does the metric reflect what she controls?	Relative TSR for market-exposed outcomes; ROIC/EP for capital allocation; operating metrics for execution
Is the metric hard to game?	Pre-agreed adjustment policy; multi-year measurement; capital includes goodwill and leases
Is the horizon long enough?	Three-year performance period plus holding period beyond vesting
Is the downside real?	Meaningful equity ownership, malus and clawback, no repricing
Are risk incentives balanced?	Check vega and inside debt against leverage and strategy
Will the process survive scrutiny?	Independent committee, disclosed rationale, investor engagement

Case Study

Illustrative case; Brightwell Energy Services and all individuals are fictional.

Brightwell Energy Services supplies drilling and well-servicing equipment to oil and gas producers. Its market capitalization is $6 billion, its net debt is $2.4 billion (net debt to EBITDA of 2.2 times), and its returns are highly cyclical: ROIC ranged from –2% in the last downturn to 16% at the peak, averaging 8% against a WACC of 9.5%. CEO Dana Whitfield joined five years ago from a larger competitor.

Her pay for last year included a $1.3 million salary, a cash bonus of $3.9 million (150% of target; the metric was EBITDA growth against budget), and long-term awards with a grant-date value of $10.5 million: 60% stock options with a seven-year term, 40% time-vested restricted stock units. Whitfield holds shares and RSUs worth $18 million and options with a dollar delta of $22 million. Her pension and deferred compensation total $5 million.

During the last cycle upswing, Brightwell's stock rose 85% in 18 months, as did those of its peers. Whitfield's options, granted near the trough, grew from a grant-date value of $6 million to a realizable intrinsic value of $31 million. The same year the company committed $900 million to a fleet-expansion program and acquired a competitor at 11.5 times EBITDA,

against a historical sector average of about 8 times. When the cycle turned, EBITDA fell 40%, leverage rose to 3.4 times, the covenant limit was 3.75, and the stock lost 55% of its value. The board agreed to re-price Whitfield's underwater options and to grant a one-time retention award of $4 million in RSUs "to ensure continuity".

At the annual meeting, 41% of votes were cast against say-on-pay. Two large holders have written to the compensation committee chair, Arturo Delgado. They ask the committee to explain why options struck near the trough, without a performance hurdle, are an appropriate reward for a cyclical firm; why the EBITDA metric ignores the capital deployed to produce it; why option re-pricing was needed when peers' stock also fell; and why debt-holders' interests were not considered when the capital program was approved. A third holder, a credit-oriented fund, points out that Brightwell's bonds fell more than the equity of its better-capitalized peers.

Delgado has asked the committee's consultant for a redesigned plan. The consultant proposes: replacing options with performance shares measured on a three-year average of ROIC and relative TSR against a peer group of eight oilfield services firms; changing the bonus metric from EBITDA growth to economic profit with a cumulative multi-year bank; requiring Whitfield to hold 50% of net vested shares for two years after retirement; adding a debt-like element by paying 20% of deferred pay in a long-dated, unsecured deferred cash plan; and adopting a clawback covering both restatements and "material risk management failures". Whitfield replies that relative TSR may punish her for events she cannot control and that economic profit will make her reluctant to invest at the bottom of the cycle, when assets are cheap.

Discussion Questions

1. Compute the Black-Scholes value (inputs of your choosing: $S = K = \$40$, $T = 5$, $\sigma = 40\%$, $r = 4\%$, $q = 0$) and delta of an at-the-money option, then explain why Whitfield's trough-priced options produced a large payout in an industry-wide upswing. What does the informativeness principle suggest about indexing exercise prices to peers or the sector?
2. Evaluate the proposed metric architecture. For each of ROIC, relative TSR, and economic profit, explain how it could mislead in a cyclical business and what design adjustments (cyclical averaging, carry-forward banks, capital definition) would improve it. Address Whitfield's two objections.
3. Using the ideas of vega and inside debt, assess the risk-taking incentives of the current plan given leverage of 2.2 to 3.4 times EBITDA and the covenant constraint. Estimate roughly the inside-debt-to-equity ratio of Whitfield's package and compare it with the firm's debt-to-equity ratio. What change in pay design would reduce the bias toward risk?
4. Assess the re-pricing and the retention award from the managerial-power and optimal-contracting perspectives. What process should the committee follow in responding to the say-on-pay result, and how should it communicate with the three investor groups whose concerns differ?

Key Takeaways

- Executive pay can be understood as optimal contracting under agency problems and as an outcome of managerial influence over the board. Use both lenses and test each design feature against the question: would it survive if the executive had no influence on the committee?
- The informativeness principle says to pay on measures that carry information about the executive's actions and to filter out common shocks; the multitask problem says that strong incentives on measurable tasks starve unmeasured ones.

- Options cost the firm more than they are worth to an undiversified executive, yet they deliver about twice the dollar sensitivity per dollar of reported value as stock, as Example 10.1 shows (2.03 times). Judge awards by delta and vega, not by headline value.
- Realizable pay can differ widely from grant-date pay; show both, at several share prices, before judging alignment.
- Pay-performance sensitivity per $1,000 of shareholder value (Example 10.2: $5.44) measures alignment; large dollar amounts can still correspond to small fractions of firm value.
- TSR is a clean shareholder measure but is dominated by market, sector, and multiple changes (Example 10.5: a de-rating cut annualized TSR by 4.5 points). Relative TSR helps but depends on peer choice, rank-based cliffs, and caps.
- ROIC levels mislead in both directions (Example 10.4): high-ROIC managers reject good projects and low-ROIC managers accept bad ones. Economic profit, with a capital charge and multi-year measurement, aligns incentives with NPV.
- Convex payoffs reward volatility (vega), and inside debt offsets the equity bias; neutrality arises when the executive's inside debt-to-equity ratio matches the firm's. Banks require extra safeguards: deferral, malus, clawback, and risk-adjusted metrics.
- Clawbacks restore the link between measured and true performance, but steep payout curves magnify both the incentive to manage earnings and the cost of restatement (Example 10.7), and clawbacks cannot substitute for sound metrics and strong controls.
- Say-on-pay, disclosure, and a disciplined committee process, including engagement with investors before a vote, turn pay design from a one-time decision into a governed system.

Review Questions

1. Contrast the optimal-contracting and managerial-power views of executive pay. Give two observed practices that each view explains better.
2. State the informativeness principle and the multitask problem. How does each bear on the choice between EPS growth and relative TSR?
3. Why do executives value options at less than Black-Scholes cost? What follows for how boards should size option grants?
4. Compare stock options, RSUs, and performance share units in terms of retention, pay-performance sensitivity, risk incentives, and accounting treatment.
5. Explain pay-performance sensitivity. Why might a CEO with a high dollar delta still be indifferent to a project that has a modest negative effect on shareholder value?
6. List three weaknesses of absolute TSR as a performance metric and explain how relative TSR addresses some of them and creates others.
7. Why can a ROIC-level bonus lead to under-investment in some firms and over-investment in others? How does economic profit change the result?
8. Define vega and explain how an executive's option portfolio may encourage risk shifting. What does the Edmans-Liu neutrality condition for inside debt require?
9. Distinguish clawback from malus. What are the main features of the U.S. Rule 10D-1 regime, and what are its limits?

10. A say-on-pay proposal receives 62% support. Describe the process the compensation committee should follow, and what changes in design or disclosure you would expect.

Problems

Problem 10.1. A company with share price $40 plans to grant 250,000 at-the-money options. Use S = K = 40, T = 5 years, σ = 35%, r = 3%, q = 1%. Compute the Black-Scholes value per option, the grant-date fair value, the option delta, and the dollar delta of the grant.

Problem 10.2. A CEO holds 100,000 shares and 200,000 options with a delta of 0.55. The share price is $80 and the market capitalization is $8 billion. Compute the dollar delta and the pay-performance sensitivity per $1,000 of shareholder value. By how much does her wealth change if the share price rises 5%, assuming delta is constant?

Problem 10.3. A relative-TSR plan has five firms in the comparison (the company and four peers). Percentile = (firms ranked below) / 4. Payout: 0% below the 25th percentile; 50% at the 25th; 100% at the 50th; 150% at the 75th; 200% at the 100th (linear in between). Assuming each rank is equally likely, compute (a) the expected payout and the probability of zero payout, and (b) the expected payout if the plan instead pays 0% below the 50th percentile, 100% at the 50th, and 200% at the 100th (linear in between).

Problem 10.4. A division has invested capital of $2,000 million, NOPAT of $360 million, and a cost of capital of 10%. A project requiring $50 million would add perpetual NOPAT of $6 million. Compute the project's NPV, the division's ROIC before and after, and the change in economic profit. Would a manager paid on ROIC level accept it? On economic profit?

Problem 10.5. A stock begins at $30 with EPS of $2.00 (P/E = 15). EPS grows 6% a year for two years. Dividends are $0.80 at the end of each year. Compute annualized TSR over two years if the P/E at the end of year 2 is (a) 15 and (b) 12.

Problem 10.6. An executive holds 500,000 at-the-money options on a stock with S = K = $60, T = 5, r = 4%, q = 0. Compute the Black-Scholes value per option and the position value at σ = 25% and at σ = 40%. What is the gain to the executive if a value-neutral strategy raises volatility from 25% to 40%?

Problem 10.7. An annual bonus has a target of $2.0 million. Payout is 50% at EPS $4.00, 100% at $4.50, 200% at $5.00 or above, and 0% below $4.00 (linear in between). Reported EPS was $4.90, later restated to $4.40. Compute the bonus paid, the bonus that should have been paid, and the amount recoverable.

Problem 10.8. A company has 150 million shares and 80% turnout. The say-on-pay proposal received 58% of votes cast. (a) How many votes were cast for and against? (b) How many votes must shift from against to for to reach 70% support? (c) The CEO's reported pay is $9.6 million and the median worker earns $48,000; compute the pay ratio.

Answers to Selected Problems

- **10.1.** Value per option = $12.93; grant value = 250,000 × 12.93 = $3.23 million; delta = e^(–0.01×5) × N(d_1) = 0.6641; dollar delta = 250,000 × 0.6641 × $40 = $6.64 million.
- **10.2.** Dollar delta = 100,000 × $80 + 200,000 × 0.55 × $80 = $8.0 million + $8.8 million = $16.8 million. PPS = 1,000 × 16.8 / 8,000 = $2.10 per $1,000. A 5% rise changes wealth by about 5% × $16.8 million = $0.84 million.
- **10.3.** (a) Percentiles are 100%, 75%, 50%, 25%, 0% with payouts 200%, 150%, 100%, 50%, 0%; the mean is (2.0 + 1.5 + 1.0 + 0.5 + 0) / 5 = 1.00 (100% of target); probability of zero = 1/5 = 20%. (b) Payouts: 200%, 150%, 100%, 0%, 0%; the mean is 450% / 5 = 90%; probability of zero = 40%.

- **10.4.** NPV = 6 / 0.10 – 50 = +$10 million. ROIC before = 360 / 2,000 = 18.0%; after = 366 / 2,050 = 17.85%. EP before = 360 – 200 = $160 million; after = 366 – 205 = $161 million, +$1 million per year (PV = 1 / 0.10 = $10 million, equal to the NPV). A ROIC-level manager would reject (ROIC falls); an EP manager would accept.
- **10.5.** EPS at year 2 = 2.00 × 1.06^2 = $2.2472. (a) Price = 2.2472 × 15 = $33.71; TSR solves $-30 + 0.8 / (1 + r) + (0.8 + 33.71) / (1 + r)^2 = 0$, so r = 8.59%. (b) Price = 2.2472 × 12 = $26.97; r = –2.45%.
- **10.6.** At σ = 25%: $18.19 per option, $9.09 million. At σ = 40%: $24.71 per option, $12.36 million. Gain = $3.26 million.
- **10.7.** Paid: 100% + (4.90 – 4.50) / 0.50 × 100% = 180%, so $3.6 million. Should have been paid: 50% + (4.40 – 4.00) / 0.50 × 50% = 90%, so $1.8 million. Recoverable = $1.8 million (pre-tax).
- **10.8.** (a) Votes cast = 120,000,000; for = 69,600,000; against = 50,400,000. (b) 70% needs 84,000,000 for, so 14,400,000 votes must shift. (c) 9,600,000 / 48,000 = 200 to 1.

Chapter 11: Shareholder Activism, Proxy Contests and Shareholder Rights

Learning Objectives

- Explain why shareholder activism exists in terms of agency costs, the free-rider problem and the economics of concentrated blocks, and compute the break-even stake and success probability for an activist campaign.
- Describe the stages of the modern activist playbook, the main categories of activist demands, and the evidence on how markets react to activist involvement.
- Work through the mechanics of voting: record dates, plurality versus majority standards, cumulative voting, staggered boards, the universal proxy card, and the arithmetic of winning a proxy contest.
- Analyze takeover defenses, in particular the poison pill, classified boards and dual-class structures, quantify their dilution and control effects, and explain how Delaware and UK law constrain them differently.
- Evaluate the role of institutional investors, index funds and proxy advisors in determining contest outcomes, and explain how stewardship codes and engagement practices have changed the balance of power.
- Design a practical preparation and response plan for a board facing an activist approach, including settlement terms and the trade-offs between engagement and resistance.

Opening Vignette

Illustrative scenario. Harlow Brands is a fictional consumer-products company.

On a Tuesday morning in March, the chair of Harlow Brands received a courteous four-page letter. Tidewater Capital, a fictional activist fund, disclosed that it owned 6.1 percent of Harlow’s shares and wished to “share its perspective on unlocking value.” The letter was polite. The attached presentation was not. It showed that Harlow’s operating margin trailed its closest peers by six points, that the company carried a household-appliance division that had consumed capital for a decade without earning its cost of capital, and that three of the nine directors had served for more than fifteen years. It proposed a sale or spin-off of the appliance division, a margin plan, an additional $400 million of buybacks, and two new independent directors with operating experience.

Harlow’s CEO was initially inclined to dismiss the letter. The share price was only 8 percent below its five-year high, the board was staggered, and the company had a poison pill with a 15 percent trigger. But the chief financial officer ran a quick exercise: if Tidewater’s numbers were right, the stock was worth 25 percent more under the activist plan. Harlow’s top ten holders owned 48 percent of the shares, and two of them were index managers whose published voting guidelines favored board refreshment after twelve years of tenure. Within a week the stock rose 7 percent on volume, the proxy advisors had asked for meetings, and three large holders had called the investor relations desk.

The board now faced a set of decisions that no single chapter of finance theory fully answers. Should it engage privately or fight publicly? Is the pill a shield or a provocation? What will institutional holders actually do when the ballot arrives? And underneath every tactical question lay a more basic one, which is whether Tidewater had found genuine value or merely a good story.

This chapter provides the tools to answer such questions: the economics of the activist's position, the mechanics of voting, the legal architecture of takeover defense, and the incentives of the institutions that decide the outcome.

11.1 The Governance Problem Behind Activism

11.1.1 Agency costs and the bundle of shareholder rights

Volume 1 (Chapter 15) treated the separation of ownership and control as a source of agency costs: managers may over-invest, retain cash, entrench themselves, or take perquisites, because they bear only a small part of the value they destroy. Corporate law responds with a bundle of shareholder rights whose purpose is to make the agency problem tolerable. The core rights are: to vote on the election and removal of directors; to vote on fundamental transactions such as mergers, charter amendments and asset sales; to propose resolutions under defined conditions; to receive information; to sue directors for breach of fiduciary duty; and to sell shares freely. The economic content of these rights is simple. Shareholders are the residual claimants, so they have the strongest incentive to maximize firm value, and the law gives them the power to replace the people who run the firm.

The difficulty is that the power is rarely used. A shareholder owning 0.01 percent of a company captures 0.01 percent of any value improvement, yet pays 100 percent of the cost of analyzing the firm, mobilizing other investors and voting an informed ballot. This is the rational apathy and free-rider logic formalized by Grossman and Hart in their 1980 analysis of takeovers: each small holder prefers that someone else pay the cost of change, because the gain accrues to all holders in proportion to their shares regardless of who incurred the cost. Attempted takeovers and activist campaigns are public goods for the shareholder base, and public goods are under-supplied.

11.1.2 Voice, exit and the value of a block

Albert Hirschman distinguished between exit (selling) and voice (arguing for change). For a diversified investor with a small position, exit is almost always cheaper than voice. For an investor with a large block, exit is costly because selling depresses the price and surrenders the potential gain, while voice has a payoff that scales with the block. Shleifer and Vishny's 1986 analysis of large shareholders makes this point precisely: a block holder internalizes enough of the benefit of monitoring to justify the cost, and so concentrated ownership can partly solve the free-rider problem.

Activist funds are professional block holders designed to exploit this logic. They accept concentration risk, typically holding a handful of positions of 3 to 10 percent, and they build specialized capabilities in valuation, legal strategy, communications and director recruitment that make the cost of voice low and the probability of success higher than for a passive investor. Their compensation, a management fee plus a share of profits (commonly 20 percent) above a hurdle, rewards them for converting a share-price gap into realized gains.

Gilson and Gordon have emphasized that activists therefore occupy a particular institutional niche. Diversified institutions, such as index funds and many mutual funds, are "rationally reticent": they do not originate campaigns, but they hold the votes. Activists originate and finance campaigns, but hold few votes. The outcome of a contest is often determined by whether the reticent majority is persuaded that the activist's plan beats management's. This division of labor, rather than any ideology, explains most of the behavior discussed in this chapter.

11.1.3 What activism is trying to fix

Activists tend to target a recognizable set of governance and strategic failures:

- **Capital allocation.** Persistent investment in divisions earning less than the cost of capital, acquisitions at high premiums, and cash hoards. In the language of Volume 1 (Chapter 6 and Chapter 11), the firm accepts projects with negative NPV or fails to return capital whose best use is outside the firm.
- **Portfolio structure.** Conglomerate discounts, where the sum of the parts is worth more than the whole, and divisions that would be worth more to another owner.
- **Operating performance.** Margin gaps against peers that management cannot explain.
- **Governance and incentives.** Long-tenured boards, pay that is weakly linked to returns on capital, and entrenchment devices.
- **Balance sheet.** Under-leveraged companies whose capital structure leaves tax shields unused (Volume 1, Chapter 15), or over-leveraged ones that need deleveraging.

A recurring analytical task for boards is to separate these claims into those that are verifiable, those that are arguable, and those that are wrong. Activism is not self-validating: a campaign can be well-organized and still be mistaken about value. The activist's incentive is to profit, not to be right, though over a repeated game the two are correlated.

11.2 The Activist Playbook

11.2.1 Stages of a campaign

Although each campaign is different, the structure is recognizable.

1. Target selection. The activist screens for firms with identifiable gaps: low valuation relative to peers, a diversified portfolio, poor returns on invested capital, a weak or entrenched board, and a share register in which institutions hold a high percentage of the votes. A company with a staggered board and a majority-vote vulnerability may be less attractive than one with annual elections.

2. Accumulation. The fund accumulates shares up to the disclosure threshold. In the United States, a person who acquires beneficial ownership of more than 5 percent of a class of voting equity must file a Schedule 13D, and under the amended rules this filing is due within five business days of crossing the threshold. Between the first purchase and the filing date, the activist can keep buying quietly, which allows it to build a stake before the market learns of its interest. Derivatives, including cash-settled swaps and options, can provide economic exposure without counting toward the threshold in some jurisdictions, though this practice is controversial and subject to evolving regulation. Regulatory merger-control filings (the Hart-Scott-Rodino rules in the United States) can also apply to acquisitions above specified sizes, even for investors with no intention of control.

3. Private engagement. The activist typically approaches the board or CEO privately to present its thesis. Many campaigns end here, with management adopting some or all of the proposals. The activist's credibility depends on a readiness to go public.

4. Public escalation. If private talks fail, the activist publishes an open letter and a detailed presentation, often with a website. It lobbies other shareholders, sell-side analysts and the press. The public stage sets the narrative that proxy advisors and large holders will evaluate.

5. Formal action. The activist can nominate directors, submit shareholder proposals, call a special meeting where charter and bylaws permit, or solicit written consents. Nomination is subject to advance-notice bylaws that fix a window (commonly 90 to 120 days before the anniversary of the previous annual meeting). Missing the window is fatal, which is why boards draft bylaw provisions carefully and why courts scrutinize them.

6. Settlement or vote. Most campaigns that reach the formal stage settle. A typical settlement adds one to three directors (some of them activist nominees, others mutually agreed independents), establishes a board committee to review strategy, and includes a standstill: the activist agrees not to exceed a specified ownership level or run another campaign for a period such as twelve months, and to vote with the board on certain matters. Fewer campaigns go to a vote, and those that do are expensive.

7. Exit. After the strategic review or buyback, the activist sells shares into the price recovery. A holding period of one to three years is typical, though there are many exceptions.

11.2.2 Types of activists

Activist type	Typical stake	Primary tools	Time horizon	Principal constraint
Hedge fund activist	3-10%	Letters, board nominations, proxy fights, settlement	1-3 years	Concentration and liquidity risk; fund redemptions
Wolf pack (loosely coordinated funds)	Combined 8-15%, each below thresholds	Parallel purchases, shared thesis	Months to 2 years	Group-filing rules; legal risk if deemed coordinated
Public pension fund	Small (under 1%)	Votes, proposals, engagement, letters	Long	Political pressures; diversified portfolio
Strategic or corporate activist	5-20%	Board seats, bid threat	Long	Antitrust and regulatory approval
Retail or “gadfly” proponent	Minimal	Shareholder proposals, questions at meetings	Variable	Eligibility thresholds and low vote shares
Thematic or ESG-oriented investor	Small to moderate	Proposals, engagement, vote-no campaigns	Long	Depends on the support of large institutions

The key distinction is between investors who can credibly threaten to take control of the board and those who can only signal views. The second group depends almost entirely on persuading others.

A “wolf pack” is a loosely coordinated set of funds that each buy below the 5 percent threshold, or file separately, yet together hold a large block. Legal treatment depends on whether the investors have formed a “group” under securities law: coordination is permitted only up to the point at which it amounts to an agreement to act together in acquiring, holding or voting shares. This is a gray area that regulators and litigants dispute.

11.2.3 Types of demands and the value mechanism

Demand	Mechanism of value	Primary risk
Divest or spin off a division	Removes conglomerate discount; redeploys capital	Stranded costs; tax leakage; the “sum of parts” may be illusory
Return capital (buyback, special dividend)	Reduces agency cost of free cash flow; raises ROE	Leverage rises; buybacks near highs destroy value
Cost and margin program	Closes the gap to peer margins	Underinvestment; short-term focus
Sale of the company	Control premium captured by shareholders	Limited buyers; deal risk; undervaluation if timing is poor
Board change	Better oversight and capital discipline	Activist directors may serve the activist’s horizon
Leveraged recapitalization	Tax shield and discipline (Volume 1, Chapter 15)	Financial distress costs

The managerial lesson is that the demand set maps onto classical corporate finance questions: Is the firm earning its WACC on each business? Is it over- or under-leveraged? Is there a buyer who can use the assets better? An activist who asks these questions is not inventing a new theory. The more serious debate is about horizon and execution.

11.2.4 The economics of the activist’s position

The activist’s calculation can be written down. Let the market capitalization be V, the stake fraction s, the proportional value uplift if the campaign succeeds be u, the probability of success be p, the price impact of accumulating the stake (the premium paid over the undisturbed price) be π, and the fixed cost of the campaign be C. Then:

```
Expected profit = p × u × V × s  −  π × V × s  −  C
```

The first term is the stake’s share of the value created. Note what is *not* in the formula: the activist captures only fraction s of the uplift, while the other 1 − s of the shareholders obtain the rest for free. That is the free-rider problem, and it sets a floor on the stake the activist must hold.

Example 11.1 – Break-even economics of an activist campaign.

Harlow Brands has a market capitalization of V = $5,000 million. Tidewater buys s = 6% ($300 million at the undisturbed price) and, because its buying lifts the price, pays an average premium of π = 4% above the undisturbed price. It estimates the campaign (advisers, solicitation, legal) will cost C = $12 million. If the plan succeeds, it believes the equity value rises by u = 15%.

Step 1: cost of the stake over and above undisturbed value.

```
Price-impact cost = π × V × s = 0.04 × 5,000 × 0.06 = $12.0 million
```

Step 2: gain on the stake if successful.

```
Uplift on whole firm = 0.15 × 5,000 = $750 million
Tidewater's share      = 750 × 0.06  = $45 million
Other shareholders     = 750 × 0.94  = $705 million (free ride)
```

Step 3: expected profit as a function of p.

```
E[profit] = p × 45 − 12 − 12 = 45p − 24
p = 0.3:  13.5 − 24 = −$10.5 million
p = 0.5:  22.5 − 24 = −$1.5 million
p = 0.7:  31.5 − 24 = +$7.5 million
```

Step 4: break-even success probability.

```
p* = 24 / 45 = 53.3%
```

Step 5: break-even stake at p = 0.5. Profit per unit of stake is 0.5 × 0.15 × 5,000 − 0.04 × 5,000 = 375 − 200 = 175 per unit of s, so

```
E[profit] = 175 × s − 12  =>  s* = 12 / 175 = 6.86%
```

At s = 9% the profit would be 175 × 0.09 − 12 = +$3.75 million; at 6% it is −$1.5 million.

Interpretation: At 6% and a 50 percent chance, Tidewater loses money in expectation, even though the campaign creates $750 million of value for shareholders in total if it succeeds. The campaign is socially valuable but privately marginal. This arithmetic explains why activists seek larger stakes, lower accumulation costs (derivatives, quiet buying before the 13D filing), cooperative co-investors that share the campaign cost, and settlements that make success probable. It also explains why the market often assigns a premium merely on the news of an activist stake: investors are pricing the probability p, not the outcome.

11.2.5 Common mistakes in analyzing activist claims

Three errors recur.

1. **Confusing the headline gap with achievable value.** A comparison of the firm’s multiple to the average peer multiple ignores the reasons for the gap. Boards should decompose any claimed uplift into assumptions about margin, growth, capital intensity and risk, and test each.
2. **Ignoring execution cost.** Spin-offs create stranded costs and separation expenses; buybacks reduce financial slack. The cost of a transaction shows up in the denominators of the value case.
3. **Treating the share price reaction as proof.** A positive abnormal return on announcement shows that the market assigns some probability to value creation. It does not show that the plan is correct, and it tells you little about the long-term outcome.

11.2.6 What the evidence says

A sizeable empirical literature, including work by Brav, Jiang, Partnoy and Thomas and by Bebchuk, Brav and Jiang, finds that the announcement of a hedge fund activist stake is associated with positive abnormal returns of several percent on average, and that targets tend to show improved operating performance and payouts over the following years. Critics argue that some of the improvement reflects selection (activists choose firms about to improve), that effects on long-term investment and employment are contested, and that the average masks a wide dispersion of outcomes. A balanced reading is that activism is, on average, associated with positive abnormal returns without long-term reversals in the typical sample, that it varies greatly across campaigns, and that its causal effect is hard to isolate. Boards should therefore neither dismiss activism as short-termism by default nor accept it as invariably value-creating.

The following example shows how the standard tool, the event study, is applied.

Example 11.2 – Measuring the market reaction to an activist disclosure.

Tidewater's 13D filing becomes public on day 0. Using a market model estimated on prior data, Harlow's expected return is $\alpha + \beta R_m$ with $\alpha = 0.03\%$ per day and $\beta = 1.1$. The daily standard deviation of Harlow's abnormal returns in the estimation window is 1.5%. Observed data:

Day	Harlow return	Market return	Expected return (0.0003 + 1.1 × market)	Abnormal return
−1	0.4%	0.2%	0.03% + 0.22% = 0.25%	0.15%
0	6.2%	0.1%	0.03% + 0.11% = 0.14%	6.06%
+1	1.1%	−0.3%	0.03% − 0.33% = −0.30%	1.40%

Step 1: cumulative abnormal return over the window [−1, +1].

```
CAR = 0.15% + 6.06% + 1.40% = 7.61%
```

Step 2: test statistic. Under independence the standard deviation of the three-day CAR is $\sigma\sqrt{n}$.

```
SE = 1.5% × √3 = 2.598%
t  = 7.61 / 2.598 = 2.93
```

A t-statistic of 2.93 is significant at the 1 percent level.

Step 3: dollar value.

```
7.61% × $5,000 million = $380.5 million
```

Interpretation: The market added about $380 million to Harlow's equity value. Compare this with Example 11.1: the 6 percent stake is worth 6% × 380.5 = $22.8 million immediately, against $24 million of price impact and campaign costs. The announcement return alone does not cover Tidewater's costs, so it still needs the plan to succeed. Note the small positive abnormal return on day −1: leakage or pre-announcement buying by others shows up before the formal disclosure, a pattern commonly observed.

11.3 Voting Mechanics and the Proxy Contest

11.3.1 From record date to certified result

Shareholders vote on matters at annual or special meetings, almost always by proxy. The process has a precise sequence, and activists and boards both win or lose on details.

1. **Record date.** The board fixes the date on which ownership determines who may vote. Shares bought after the record date do not carry votes for that meeting. This matters for activists accumulating late and for the phenomenon of "empty voting," discussed below.
2. **Solicitation.** Each side distributes proxy materials and a proxy card. A dissident running its own slate files a proxy statement with the securities regulator and mails or e-delivers it to holders. The cost is large: legal fees, proxy solicitors, printing and communication can run into many millions for a large company, and the incumbent board usually spends company money while the dissident spends its own unless it wins and seeks reimbursement.
3. **Quorum.** The meeting needs a minimum percentage of shares represented (often a majority). Retail apathy makes quorum a real issue for some companies, which is why they hire solicitors to chase turnout.

4. **Voting standard.** The company's charter or bylaws determine how much support wins. The key variants are in the table below.
5. **Tabulation.** An independent inspector of elections counts the proxies, resolves disputes over the validity of cards, and certifies the result. When two proxy cards are submitted by one holder, the latest dated card controls.

11.3.2 Voting standards

Standard	How it works	Implication for contests
Plurality	Nominees with the most votes win, up to the number of seats. "Withhold" votes do not defeat a nominee in an uncontested election.	In a contested election, the dissident needs more votes than the incumbent nominee on each seat.
Majority voting	In an uncontested election, a nominee must receive more "for" than "against" or "withheld." Resignation policies often apply.	Gives shareholders a real vote-no weapon. Contested elections generally revert to plurality.
Cumulative voting	Each share gets as many votes as there are seats; the holder can concentrate them on fewer candidates.	Allows a minority to elect representatives.
Staggered (classified) board	Directors serve multi-year terms, with only a fraction elected each year.	Slows any change of control of the board to two annual meetings or more.
Supermajority	A higher threshold (for example two-thirds) for certain actions.	Raises the cost of removal, merger approval or charter change.

Broker non-votes are another detail. In the United States, brokers may vote uninstructed shares only on "routine" matters. Director elections are not routine, so uninstructed retail shares do not count as votes cast for director elections. In an uncontested election, this reduces turnout; in a contested election, retail participation is driven by solicitation and by the incumbent's access to the retail list.

11.3.3 The universal proxy card

For decades, a dissident and the incumbent each had their own card, and a holder voting on one card could not pick individual nominees from both slates. This was a significant advantage for incumbents, who could attack the dissident's whole slate without worrying that a few of their own directors would be dropped. The United States adopted a universal proxy rule, applicable to contested meetings after August 2022, that requires both sides to list all nominees on a single card so that each holder can choose among the whole field. Several other jurisdictions already allowed split voting, since in the UK, for example, voting is on individual resolutions at a meeting.

The consequence is a competition on the merits of individual candidates. Activists tend to nominate fewer candidates, aimed at a minority of seats, and put forward highly qualified people, because a universal card allows holders to vote for the strongest of both sides. Boards can no longer argue that a vote for the activist's nominee means a vote against the entire incumbent team. The result has been more settlements, since the incumbents' risk of losing individual seats rose.

11.3.4 The arithmetic of a contest

Contest outcomes follow from three quantities: the number of shares that actually vote, the support within each constituency, and the influence of advisors.

Example 11.3 – Counting votes in a proxy contest.

Harlow has 200 million shares outstanding. Tidewater holds 16 million (8%) and will vote them for its nominees. The remaining shares are held by other institutions (124 million) and retail investors (60 million). Turnout is expected to be 92% for institutions and 45% for retail. Retail holders tend to favor incumbents; assume that only 20% of retail votes cast support the dissident.

Step 1: votes cast.

```
Institutions voting = 124 × 0.92 = 114.08 million
Retail voting       =  60 × 0.45 =  27.00 million
Tidewater           =               16.00 million
Total votes cast    =              157.08 million
Majority threshold  = 157.08 / 2  =  78.54 million
```

Step 2: retail support for the dissident.

```
Retail for Tidewater = 27.00 × 0.20 = 5.40 million
```

Step 3: institutional support Tidewater needs.

```
Needed = 78.54 − 16.00 − 5.40 = 57.14 million
As a share of institutional votes cast = 57.14 / 114.08 = 50.09%
```

So the institutions are the swing vote, and Tidewater needs just over half of them.

Step 4: two scenarios driven by the proxy advisor recommendations. Suppose that when the leading advisors recommend the dissident slate, 65% of institutional votes follow; when they recommend against, only 25% do.

```
Advisor supports:  16 + 0.65×114.08 + 5.40 = 16 + 74.15 + 5.40 = 95.55 million → 60.8% (win)
Advisor opposes:   16 + 0.25×114.08 + 5.40 = 16 + 28.52 + 5.40 = 49.92 million → 31.8% (lose)
```

Interpretation: A swing of 40 percentage points in the institutional vote, from the advisor recommendation, flips the result, and the activist's own 8 percent and the retail vote are tiny by comparison. This is why both sides spend the campaign courting a few dozen institutional holders and two advisory firms, and why index managers' voting policies matter. It is also why the sensible unit of analysis is the share register, not the shareholder count.

11.3.5 Cumulative voting and minority representation

With cumulative voting, a holder multiplies the number of shares by the number of seats up for election and may allocate that total across candidates as desired. The minimum share of voting stock needed to guarantee k of N seats is:

```
Threshold = k / (N + 1)    (plus one share)
```

Example 11.4 – How little it takes under cumulative voting.

A company has 60 million voting shares and elects N = 9 directors in a single election. A dissident wants k = 2 seats.

Step 1: threshold.

```
Shares needed = 60,000,000 × 2 / (9 + 1) + 1 = 12,000,001 shares  (20% of votes cast)
```

Step 2: verification. The dissident casts 12,000,001 × 9 ≈ 108.0 million votes, split over two candidates at about 54.0 million each. The other holders have 48,000,000 × 9 = 432.0 million votes. To elect seven candidates they can allocate about 61.7 million to each (432 ÷ 7), above the dissident's 54.0 million, so they win seven seats. To elect an eighth

candidate they would need over 54.0 million votes for it and for each of the other seven, a total of 8 × 54.0 = 432 million plus one vote, which they do not have. The dissident therefore guarantees two seats.

Step 3: compare with other regimes.

```
Straight voting, annual election of all 9: the majority holder (50%+1) elects all 9 seats.
Cumulative, N = 9, k = 1: 10% threshold.   k = 3: 30% threshold.
Cumulative with staggered board, N = 3 seats up, k = 1: 25% threshold.
```

Interpretation: Cumulative voting converts a 20 percent block into two board seats, where straight voting gives it none. It is rare among large public companies, in part because management and boards dislike factional boards; staggering the board raises the threshold since fewer seats are up in any year (the denominator N + 1 shrinks). Boards that want to dilute minority power therefore combine a staggered board with a small number of seats per class.

11.3.6 Votes without economic interest

Votes and economic interest can be separated. A holder who borrows shares just before the record date and returns them just after, or who hedges the economic exposure with derivatives while keeping the votes, has “empty voting” power. Hu and Black described this phenomenon, which can lead to votes cast in ways that reduce the value of the shares being voted. Securities lending by index funds also removes votes from the register unless the lender recalls shares before the record date. Regulators and large institutions now often recall loans before important votes. For the board, the point is practical: reconcile the record-date register with the actual economic holders before a contest.

11.4 Takeover Defenses

11.4.1 The conflict of interest

A hostile bidder offers shareholders a premium, usually 25 to 40 percent over the undisturbed price, to sell directly to the bidder, bypassing the board. Directors have an obvious conflict: they may lose their positions. The legal and economic question is whether defenses protect shareholders by allowing the board to bargain for a higher price and resist coercive or opportunistic bids, or protect managers from discipline. Both effects exist. The empirical literature is mixed and depends on the defense: staggered boards, in particular, have been associated in several studies with lower valuations, while the role of poison pills in raising bid premia is a classic subject of debate.

11.4.2 The poison pill

The shareholder rights plan, or poison pill, was designed in the 1980s by the law firm Wachtell, Lipton. The board issues one right per share as a dividend. The rights are initially worthless and cannot trade separately. If a person or group acquires more than a threshold percentage (commonly 10 to 20 percent) without board approval, the rights become exercisable:

- **Flip-in.** All holders other than the acquirer may buy additional shares of the target at a deep discount, typically half the market price. The acquirer’s stake is diluted and its economic loss is large.
- **Flip-over.** If the target is merged into the acquirer afterward, holders may buy shares of the acquirer at a discount.

The board can redeem the rights for a nominal amount (such as $0.001 each) at any time before they are triggered. So the pill does not bar a deal; it bars an *unapproved* acquisition and forces the bidder to negotiate with the board, which can then redeem the pill for a bidder that offers an acceptable price.

Example 11.5 – How a flip-in right dilutes an acquirer.

Harlow has 100 million shares at $40, so equity value is $4,000 million. A rights plan sets a 15% trigger. A hostile acquirer crosses the trigger and holds 15 million shares ($600 million at market). Each share carries one right. Upon triggering, each right (except the acquirer's) allows its holder to pay an exercise price of $80 and receive shares with a market value of twice the exercise price, $160 — that is, 4 shares at the $40 pre-trigger price.

Step 1: shares issued if all other holders exercise.

```
Rights held by others = 100 − 15 = 85 million
New shares issued     = 85 × 4 = 340 million
Cash received         = 85 × $80 = $6,800 million
```

Step 2: post-dilution price (assuming the firm's operating value is unchanged and the cash is added to its assets).

```
Total equity value = 4,000 + 6,800 = $10,800 million
Total shares       = 100 + 340 = 440 million
Price per share    = 10,800 / 440 = $24.55
```

Step 3: the acquirer.

```
Acquirer's stake     = 15 / 440 = 3.41% (down from 15%)
Value of its holding = 15 × 24.55 = $368.2 million (down from $600 million)
Loss                 = $231.8 million, or 38.6%
```

Step 4: the other holders. Each exercising holder pays $80 and receives 4 shares worth 4 × 24.55 = $98.18, a gain of $18.18 per right. Holders who do not exercise would suffer the same dilution as the acquirer, which is why the exercise right is crucial.

Interpretation: The pill does not make the acquisition impossible, it makes it ruinous without the board's consent. In practice the rights are never exercised, because no rational bidder crosses the trigger. The pill's real function is to give the board time and bargaining power, and to deter creeping accumulation of control. Observe that an activist who stays below the trigger (say 9.9 percent) is not affected. For this reason, some companies set a low trigger, even 10 percent or less, against activists, and courts have scrutinized whether such pills are a proportionate response to a threat. The Delaware Chancery Court in the Sotheby's litigation upheld a pill with a lower threshold for active investors than for passive ones, on the facts presented to it, but boards should not infer that every such pill would survive.

11.4.3 Staggered boards and other structural defenses

Defense	Mechanism	Effect on a bidder or activist	Typical critique
Staggered (classified) board	Three classes, one elected per year	Control needs two annual meetings; proxy fight takes at least 24 months to win majority	Entrenchment; associated with lower valuation in several studies
Poison pill	Dilution on unapproved crossing of trigger	Forces negotiation with the board	Can be used to block premium bids
Supermajority provisions	Two-thirds or more to approve merger or remove directors	Raises vote hurdle	Gives a small block veto power
Limits on written consent and special meetings	Shareholders cannot act between annual meetings	Slows an activist	Reduces accountability
Advance notice bylaws	Strict deadlines and information requirements for nominations	Procedural obstacles for dissidents	Can be used to disqualify late or imperfect nominations
Dual-class shares	Founders hold super-voting stock	Control cannot be acquired without the controlling holder	Wedge between votes and cash flow (see Section 11.5)
Golden parachutes	Large payments to executives on change of control	Raises the cost of acquisition	Rewards failure if poorly designed; can align executives with a sale if well designed
Anti-takeover statutes	For example Delaware Section 203, which bars certain business combinations for three years after a holder crosses 15% without board approval	Deters creeping control	Limits market for control

Staggering is especially potent combined with a pill. A bidder who wins a tender offer but cannot replace the board for two years cannot redeem the pill, and so cannot complete a merger. This was the structure tested in the Airgas litigation, where the Delaware courts permitted a board with a staggered structure and a pill to decline a bid it considered inadequate until the shareholders elected a new majority, a ruling that confirmed the strength of the combination.

11.4.4 Law: Delaware and the United Kingdom

Feature	Delaware (United States)	United Kingdom
Board's power to defend	Allowed, subject to enhanced scrutiny (*Unocal*): the board must show a reasonable threat and a proportionate response	Board may not take frustrating action without shareholder approval once an offer is imminent (Takeover Code, General Principle and Rule 21)
Poison pills	Permitted; upheld in *Moran v. Household International* (1985) and subsequent cases	Effectively unavailable for the same reason
Duty when selling the company	*Revlon* duties: maximize price when a sale or break-up is inevitable	Board must give independent advice and cannot prefer a bidder without shareholder consent
Mandatory bid	No general rule	A holder reaching 30% must make an offer to all shareholders (Rule 9)
Timetable	Flexible	Tight timetable; bidder must "put up or shut up" within 28 days of a public approach (Rule 2.6)
Nomination procedures	Governed by bylaws and charter	Shareholders holding 5% or more may requisition a general meeting; directors may be removed by ordinary resolution
Voting on directors	Typically annual or staggered	Annual re-election is standard under the corporate governance code for large companies

The contrast reflects different philosophies: the Delaware approach trusts the board, subject to judicial review of its motives and proportionality, while the UK approach reserves the decision to shareholders. Delaware law also recognizes that where a board's actions interfere with shareholder voting, the board must show a compelling justification, as in *Blasius Industries v. Atlas Corp.* (1988). A manager of an international firm should therefore treat "takeover defense" as a jurisdiction-specific topic.

11.4.5 Valuing the decision to defend

Defenses cost shareholders something when the bidder withdraws and the price falls back, but they may increase the price when they allow the board to negotiate. The decision is an expected-value problem with a time dimension, which the next example makes explicit.

Example 11.6 – Should the board use the pill to negotiate?

Harlow's unaffected price is $50. A hostile bidder offers $62 in cash. The board's cost of equity is 10% per year. Two paths are available.

Path A (no defense). The offer closes with probability 0.85 in three months (0.25 year). If it fails, the price returns to $50.

Path B (use the pill and negotiate). The board negotiates for nine months (0.75 year). Outcomes: with probability 0.60 the bidder raises to $68; with probability 0.15 a white knight pays $65; with probability 0.25 the bidder walks away and the price returns to $50 (taken as received immediately, since the share is simply held).

Step 1: value of A.

```
PV of deal   = 0.85 × 62 / 1.10^0.25 = 52.70 / 1.0241 = $51.46
Failure      = 0.15 × 50 = $7.50
Value A      = 51.46 + 7.50 = $58.96
(Undiscounted expectation: 0.85 × 62 + 0.15 × 50 = $60.20)
```

Step 2: value of B.

```
Deals: (0.60 × 68 + 0.15 × 65) = 40.80 + 9.75 = $50.55 received in 0.75 year
PV   = 50.55 / 1.10^0.75 = 50.55 / 1.0741 = $47.06
Walk-away = 0.25 × 50 = $12.50
Value B   = 47.06 + 12.50 = $59.56
(Undiscounted expectation: $63.05)
```

Step 3: break-even raised price. The value of B equals A when the discounted deal term equals 58.96 − 12.50 = 46.46, i.e. undiscounted deal receipts of 46.46 × 1.0741 = 49.90, so that 0.60 × X + 9.75 = 49.90 and X = $66.92.

Interpretation: The undiscounted expected values (60.20 versus 63.05) overstate the advantage of resistance by about $2.85 per share, because negotiation takes longer and has a failure risk. After discounting, B beats A by only $0.60, and the board needs to extract at least $66.92 in the “raised bid” outcome to justify resistance. If the walk-away probability were higher than 25 percent, or if the price following a failed bid would fall below $50 (a typical outcome when the market was pricing a premium), resistance would not pay. Boards should run this calculation explicitly rather than assuming that “we got a better price” justifies the delay, and should recognize that their own sense of the probabilities is subject to the optimism discussed in Chapter 12.

11.5 Dual-Class Shares and the Control Wedge

11.5.1 Votes without cash flow rights

A dual-class structure gives one class of shares (often held by founders) multiple votes per share and the other class one vote or none. Several large technology and media companies have used such structures, and some issue non-voting shares to the public. The effect is a wedge between control rights and cash-flow rights. The founder controls the firm with a small fraction of the economic claim, and so bears only a small part of the cost of a value-reducing decision.

Grossman and Hart’s analysis of the private benefits of control, and later work by Bebchuk, Kraakman and Triantis on control-enhancing mechanisms, show why the wedge matters. A controlling holder will take an action when its private benefit exceeds its share of the loss in firm value. With a cash-flow stake c, the founder takes any action with private benefit B and value loss L when:

```
B > c × L        i.e.  L < B / c
```

The lower c is, the larger the value-destroying actions the founder will tolerate.

Supporters of dual-class structures respond that founders may have a unique vision, that short-term pressure from the market can harm long-run innovation, and that investors know the structure at the time of purchase and price it accordingly. These arguments are more persuasive for a young firm than for a mature one, which is why many index providers, institutional investors and governance advocates favor *sunset* provisions that convert the super-voting shares to ordinary shares after a fixed period or when the founder’s stake falls below a threshold, and why some index providers have restricted the inclusion of companies with unequal voting rights.

Example 11.7 – The wedge in numbers.

A company has 100 million Class A shares carrying 10 votes each and 900 million Class B shares carrying one vote each. The founder holds all of the Class A shares and 45 million Class B shares.

Step 1: cash-flow rights.

```
Founder shares   = 100 + 45 = 145 million
Total shares     = 1,000 million
Cash-flow stake c = 145 / 1,000 = 14.5%
```

Step 2: voting rights.

```
Founder votes = 100 × 10 + 45 × 1 = 1,045 million
Total votes   = 100 × 10 + 900 × 1 = 1,900 million
Voting share  = 1,045 / 1,900 = 55.0%
```

Step 3: wedge ratio.

```
Wedge = 55.0% / 14.5% = 3.79
```

Step 4: incentives. Suppose the founder can pursue an acquisition that benefits the founder privately by B = $20 million but reduces firm value by L = $100 million.

```
Founder's share of the loss = 0.145 × 100 = $14.5 million < $20 million benefit → the founder proceeds
With one-share-one-vote and 55% ownership: 0.55 × 100 = $55 million > $20 million → the founder declines
Break-even loss the founder will tolerate: L* = 20 / 0.145 = $137.9 million
```

Interpretation: Under dual-class, the founder will accept a loss of up to $137.9 million to gain a private benefit of $20 million; under one-share-one-vote at the same control, the limit is $20 / 0.55 = $36.4 million. The wedge thus multiplies the loss that the controller will impose on minority holders by about 3.8. Investors price this risk through a lower multiple, and governance structures such as independent directors, related-party transaction approvals by minority holders, and sunset clauses are the tools that reduce the discount.

11.6 Institutional Investors and Proxy Advisors

11.6.1 Who decides

In most large public companies, the shareholder register is dominated by institutions: index and passive managers, active mutual funds, pension funds, sovereign wealth funds, insurers and hedge funds. Their incentives differ in ways that affect voting.

Investor type	Typical incentive	Approach to voting and engagement
Passive index manager	Low fees; no ability to exit a company without leaving the index	Policy-driven voting guided by published guidelines; engagement through stewardship teams; rarely originates campaigns
Active mutual fund	Outperform benchmark; can sell	Analysis-driven; may support activists if the thesis is credible; sometimes sells instead of voting
Public pension fund	Long-term liabilities; political oversight	Principled positions on governance; sometimes initiates proposals
Sovereign wealth fund	Long horizon; reputational concerns	Often low-profile; votes by guidelines
Hedge fund (event-driven)	Short-term profit	May support or oppose depending on the payoff structure
Insurers	Liability matching	Engagement, but limited activism

The largest index managers collectively hold a substantial share of most large-cap registers, which gives them great influence on outcomes. Bebchuk and Hirst, in "The Specter of the Giant Three," argue that their incentives to invest in stewardship are weak because they earn a small fee per dollar and their funds compete on cost. Defenders argue that large index managers have expanded their stewardship teams and that their long-term horizon and inability to sell make engagement rational. The truth is probably between: large investors do engage, but their capacity is thin relative to the number of companies they hold, and they depend on policy-based voting.

11.6.2 Incentives of passive holders

A passive manager cannot sell a poorly performing company without tracking error. This gives it a stake in company improvement, but the return from improving any single company is small relative to the cost of engagement because the manager holds thousands of positions and its fee does not depend on relative performance. Passive managers therefore adopt standardized policies, such as board diversity targets, limits on director over-boarding, or voting against directors at companies with unequal voting rights, and apply them at scale. For a board, these policies become de facto governance standards.

When an activist campaign arises, passive managers analyze the issues using a standard framework: Is the company's performance poor relative to peers? Are the activist's proposed changes plausible? Does the board's response appear credible? Many have indicated that they prefer settlement to a vote, and they often ask for a face-to-face meeting with the director nominees. In the vignette, Harlow's index holders had already published a policy favoring board refreshment after long tenures, which put the incumbent board on the defensive before the campaign began.

11.6.3 Proxy advisors

Institutions that vote thousands of ballots every spring outsource part of the analysis to proxy advisory firms, the best known of which are ISS and Glass Lewis. The advisors publish voting policies, analyze each contest, and issue recommendations. Because votes follow recommendations to a significant degree, an advisor's call can influence a contest, as Example 11.3 illustrated. The influence is a subject of controversy: companies complain of errors, conflicts of interest and one-size-fits-all policies, while institutions emphasize that they make their own decisions and use advisors for research and operational support. Many large managers have reduced their reliance on advisory recommendations, devoting internal analysis to the biggest holdings. Regulators have considered rules covering proxy advisor conflicts and disclosure; the rules and their enforcement have changed over time, so managers should consult current requirements.

For boards, the lesson is practical: communicate directly with the largest holders and advisors well before the contest, and respond to published policies by explaining departures rather than assuming they will be ignored.

11.6.4 Say-on-pay and engagement

Dodd-Frank (2010) required advisory shareholder votes on executive compensation in the United States, and the EU's revised Shareholder Rights Directive (2017) created similar rights across member states. Say-on-pay votes are non-binding but have a strong signaling effect: a failed vote prompts a board response, often changes to pay design, and is a leading indicator of weak institutional confidence. In an activist campaign, compensation is a common line of attack, because it is easy to illustrate a gap between pay and performance. A compensation committee that has already engaged with holders and aligned pay with returns on capital is harder to attack.

11.7 Stewardship, Engagement and the Evolving Rights Agenda

11.7.1 Stewardship codes

Stewardship codes set expectations for how institutional investors monitor, engage with and vote at investee companies. The UK Stewardship Code, first issued by the Financial Reporting Council in 2010 and substantially revised in 2020, asks signatories to report on how they integrate stewardship and investment, how they engage, how they escalate, and what outcomes they achieve. Japan introduced a stewardship code in 2014, followed by a corporate governance code in 2015, as part of an effort to raise returns on capital in listed companies. The EU's Shareholder Rights Directive, as revised in 2017, requires institutional investors and asset managers to disclose engagement and voting policies on a comply-or-explain basis. These codes are generally voluntary or comply-or-explain, and they work by disclosure and reputation, not by legal compulsion.

The economic logic is to reduce the cost to institutions of acting as monitors and to encourage "voice" over "exit." Criticism has focused on box-ticking: reports can describe process more than substance. Still, the codes have changed the market by normalizing engagement, making voting records public, and creating an expectation that large holders will escalate when private dialogue fails.

11.7.2 Rights that change the contest

Several shareholder rights shift the balance between boards and holders, and boards should know where they stand on each.

- **Majority voting for directors.** Replaces plurality in uncontested elections so that a director who fails to win majority support is expected to offer a resignation.
- **Proxy access.** Allows long-term holders meeting an ownership threshold to include their own nominees on the company's proxy statement. A common market design in the United States requires holdings of 3 percent for three years and permits nominees up to 20 percent of the board, with a cap on the number of holders who may aggregate their shares into a group.
- **Declassification.** Moves the board to annual elections. Many large companies have done so following shareholder pressure.
- **Right to call a special meeting and to act by written consent.** Reduces the time an activist must wait for the annual meeting.
- **Shareholder proposals.** In the United States, Rule 14a-8 allows eligible shareholders to include a proposal in the company's proxy statement, subject to ownership thresholds and subject matter exclusions. Such proposals are

generally precatory, meaning they ask but do not bind, but high votes pressure boards to respond.

- **Removal rights.** Shareholders may remove directors by simple majority in many jurisdictions; in others, removal is allowed only for cause when the board is classified.

11.7.3 Environmental, social and governance proposals

Shareholder proposals on climate, human capital and political spending have become more common. Their effect depends on the support of large institutions, which vary in how they treat them: some evaluate each proposal for materiality to long-term value and the company's current disclosures, and others have become more cautious in recent years. The corporate finance question is the same as for any activist demand: does the proposal identify a risk or opportunity that affects cash flows or discount rates, and is the board's response proportionate? Treating ESG proposals as an "activism" category separate from valuation, rather than as inputs to the cost of capital and growth assumptions (Volume 1, Chapter 11 and Chapter 21), is a mistake in both directions.

11.7.4 The short-termism debate

A central objection to activism is that it shortens corporate horizons: activists press for buybacks and cost cuts, investment suffers, and long-run value is sacrificed for quick returns. Advocates reply that managers also have short horizons (pay, tenure, reputational concerns), that capital returned to investors is not destroyed but reallocated, and that the evidence for long-term harm is weak. Both can be partly right: activism is beneficial when it removes empire-building and discipline failure and harmful when it pressures a firm with genuine long-horizon investment opportunities to under-invest. The relevant test is the one applied throughout this book: does the proposed action raise the present value of expected cash flows after accounting for risk? A board that can show that its capital allocation clears the cost of capital, with explicit hurdle rates and post-audit results, has the best defense against the short-termism charge being turned against it.

11.8 Preparing for and Responding to an Activist

11.8.1 Think like the activist

The most effective preparation is a self-assessment that adopts the activist's lens, often done with outside advisors. It examines:

1. **Valuation gap.** Compare the firm's multiples and returns on capital with peers and with a sum-of-the-parts valuation. Identify the divisions with the lowest returns.
2. **Capital allocation record.** Review acquisitions and capital projects against the promised returns. Incorporate post-audit evidence.
3. **Governance profile.** Board tenure, skills, independence, committee structure, and the link between pay and performance.
4. **Share register.** Who owns the stock, how they vote, how they have voted in other contests, and which holders are swing voters.
5. **Structural defenses.** Which provisions exist, how a court or the holders would view them, and which would provoke more opposition than they prevent.
6. **Communication.** Whether the investor narrative is credible and consistent, and whether the board speaks to holders directly.

11.8.2 Engagement choices

Response	When it makes sense	Risk
Private engagement and partial adoption	The activist's analysis has merit; the demands are compatible with the board's strategy	Appearing weak; the activist returns for more
Settlement with board seats	The vote is uncertain; the cost of a contest is large; the nominees are credible	Dilutes board cohesion; sets a precedent
Public rebuttal with data	The thesis rests on errors that can be demonstrated	Escalation; focuses on personalities
Strategic review	Value of a sale or separation is genuinely uncertain	Signals that the company is for sale
Structural defense (pill, bylaw changes)	A threat to control without a premium; creeping accumulation	Governance backlash; legal exposure if disproportionate
Full contest	Board is confident of the value case and the vote	Cost; distraction; risk of losing seats

11.8.3 Settlement terms

Settlements usually specify the number and identity of new directors, board committee assignments, a standstill period during which the activist may not increase its stake beyond a stated percentage or launch a campaign, a voting commitment (to support the board's slate at the next meeting), non-disparagement, and expense reimbursement. Each term shifts risk. A standstill protects the board but may reduce the activist's incentive to cooperate, and a short standstill invites a second campaign. Boards should evaluate the settlement the same way they would evaluate any contract: what is the cost of the concession, what do they gain, and what is the alternative if they refuse?

11.8.4 Common mistakes by boards

- **Dismissing the thesis because of the messenger.** The activist's motives do not change the validity of its analysis.
- **Adopting defenses after the campaign begins.** Late defensive moves attract litigation and criticism and may be judged in a harsher light than defenses adopted in calm.
- **Neglecting the swing holders.** Spending resources on rebuttal rather than on talking to the ten investors whose votes determine the outcome.
- **Over-promising.** A board that responds with aggressive targets it cannot meet has made the activist's case at the next earnings call.
- **Inconsistent messages.** Management says the strategy is working; the board agrees to review it.

The managerial lesson is that the best defense against activism is a business that earns its cost of capital, a board that holds management to account, and a share register that understands the strategy. Structural defenses can buy time, but time is useful only if it is used to address the gap that attracted the activist.

Case Study: Harlow Brands and Tidewater Capital

Illustrative case. All names and figures are fictional.

Harlow Brands (market capitalization $5.0 billion; 200 million shares; price $25) is a consumer-products company with two segments. Branded household goods (about 70 percent of operating profit) earn a return on invested capital of 13 percent and have a WACC of 8 percent. The appliance division, with $900 million of invested capital, earns 4.5 percent against a WACC of 8.5 percent. Group operating margin is 11 percent against peer median 17 percent. Over the past decade, Harlow has made three appliance acquisitions at average premiums of 35 percent, each described as “transformational.” The board has nine directors in three classes; the average tenure is eleven years, and two directors have served for eighteen. The CEO has been in office for twelve years, and long-term incentive pay is based on adjusted earnings per share growth. A rights plan with a 15 percent trigger was adopted two years ago in response to a rumored approach.

Tidewater Capital discloses a 6.1 percent stake (acquired at an average of $24.20) and proposes: (1) sale or spin-off of the appliance division, which it values at 0.9 times invested capital, or $810 million, against the current implied carrying value of $900 million; (2) a margin plan to reach 14 percent in three years; (3) a $400 million buyback; (4) two new directors; and (5) incentive pay tied to return on invested capital. Tidewater estimates that these steps would raise the equity value by 25 percent over two years.

The CEO’s initial reaction is to refuse to meet. The chair, a director for fifteen years, notes the staggered board and says that it would take Tidewater two years to win control. The lead independent director argues that the appliance numbers cannot be dismissed and that the board’s own reputation depends on its response. The CFO prepares an analysis showing that a spin-off would leave $55 million of stranded corporate costs and trigger tax leakage of around $60 million, partly offsetting the benefit. She also notes that the $400 million buyback would raise net debt to 2.3 times EBITDA, from 1.5 times, which is within the covenant but would reduce the room for the next acquisition. Tidewater’s analysts respond that stranded costs can be eliminated over eighteen months and that the divisions’ tax basis is higher than the company assumes.

The share register shows 48 percent held by the top ten institutions, of which three are index managers with published policies on board tenure, two are active managers with a history of supporting engagement, and one is a pension fund that has voted against directors at companies with persistent underperformance. After Tidewater’s letter, the stock rises to $26.70. The proxy advisors request meetings. An analyst note raises the possibility that a strategic buyer might be interested in the appliance division at about one times invested capital.

Three weeks later Tidewater signals that it will nominate two directors at the next annual meeting, within the advance-notice window, and publishes a presentation titled “Harlow: Closing the Gap.” The board must decide whether to engage, negotiate or contest. It also must consider whether to bring forward a declassification proposal and whether the pill should be amended.

Discussion Questions

1. Using the structure of Example 11.1, estimate Tidewater’s break-even probability of success if it expects a 25 percent uplift on a 6.1 percent stake, pays an average 3 percent premium on accumulation and spends $15 million on the campaign. How does the answer change if it can build the stake to 9 percent?
2. Evaluate Tidewater’s claim about the appliance division. Compute the economic profit on the division’s invested capital and discuss whether a sale at 0.9 times invested capital creates or destroys value after stranded costs and tax leakage.
3. Map the share register to the vote count in the style of Example 11.3. Which holders are swing voters, and what would the board need to show them in order to prevail? What is the effect of the universal proxy card on Tidewater’s choice of nominees?

4. Should the board declassify and keep the pill? Discuss the effect of each on Tidewater's bargaining power, on the credibility of the board, and on the likelihood of a settlement. What settlement terms would you propose?

Key Takeaways

- Activism exists because diversified shareholders face a free-rider problem. Concentrated, specialized holders can internalize enough of the gain to pay for monitoring and for a campaign.
- An activist captures only its pro rata share of any value created. The break-even stake and probability of success (Example 11.1) explain accumulation tactics, co-investment and the preference for settlement.
- The activist playbook has defined stages: accumulation to the disclosure threshold, private engagement, public escalation, formal nomination, and settlement or vote. Most campaigns settle.
- Announcement returns are positive on average, but they price the probability of success and do not prove that the plan is correct. Evidence on long-term effects is contested and varies widely across campaigns.
- Voting rules decide the contest: record date, plurality versus majority voting, cumulative voting, staggered boards and the universal proxy card. The arithmetic of turnout and swing holders determines the result.
- Institutional holders and proxy advisors are the swing vote. Passive managers vote by policy, so published guidelines are de facto governance standards.
- The poison pill raises the cost of unapproved accumulation by diluting the acquirer. It does not prevent a deal approved by the board, but combined with a staggered board it is a strong barrier.
- The value of defense is an expected-value, time-dependent question. Resistance pays only if the extra price exceeds the cost of delay and the risk of losing the bid (Example 11.6).
- Dual-class structures create a wedge between control and cash-flow rights, increasing the loss a controller will tolerate for a given private benefit. Sunset provisions and minority protections reduce the discount.
- Delaware and UK law give opposite answers on board power to defend. International firms must treat takeover defense as jurisdiction-specific.
- The best protection against activism is performance: returns above the cost of capital, a credible board and direct engagement with the share register before a campaign begins.

Review Questions

1. Explain why a diversified shareholder has little incentive to monitor management, and describe how a block holder alters the calculation.
2. Why does an activist capture only part of the value it creates? List three tactics by which activists try to improve their economics.
3. Describe the stages of an activist campaign, and explain the significance of the 13D disclosure and the advance-notice window.
4. What is a wolf pack, and why is the legal treatment of coordination unsettled?
5. Distinguish plurality, majority and cumulative voting. How does each affect a minority shareholder's ability to elect directors?
6. How did the universal proxy card change the tactics of dissidents and incumbents?

7. Explain how a flip-in poison pill works and why it deters unapproved acquisitions without preventing a negotiated deal.
8. Why is a staggered board more powerful when combined with a poison pill?
9. Contrast the legal approaches of Delaware and the United Kingdom to board defensive action, and explain what each approach assumes about who should decide.
10. Discuss the role of passive index managers and proxy advisors in determining contest outcomes. What are the principal criticisms of each?

Problems

Problem 11.1. An activist buys 4% of a company with a market capitalization of $8,000 million, paying an average premium of 3% over the undisturbed price because of its own buying. It expects a 12% rise in equity value if the campaign succeeds, with probability 0.5, and the campaign costs $20 million. (a) Compute the expected profit. (b) Compute the break-even probability of success.

Problem 11.2. A company has 50 million shares at $30. A rights plan has a 20% trigger. An acquirer crosses it, holding 10 million shares. Each right (other than the acquirer's) allows the purchase of shares worth twice the $90 exercise price, at the $30 pre-trigger price. Assuming all other holders exercise, compute (a) the number of new shares, (b) the post-dilution share price, (c) the acquirer's ownership percentage, and (d) the acquirer's loss.

Problem 11.3. A company has 150 million shares: an activist holds 13.5 million, other institutions 84 million, and retail 52.5 million. Turnout is 90% for institutions and 40% for retail; 20% of retail votes cast support the activist. (a) Compute the total votes cast. (b) What percentage of institutional votes cast must the activist win to obtain a majority?

Problem 11.4. A company has 80 million voting shares and uses cumulative voting. (a) If 7 seats are up and a dissident wants 2, what is the minimum number of shares it needs? (b) If only 5 seats are up (staggered) and it wants 1, what is the minimum?

Problem 11.5. A target has an unaffected price of $30. A hostile bidder offers $39 cash. The cost of equity is 12%. Path A (no defense): the bid closes with probability 0.8 in six months; otherwise the price is $30. Path B (negotiate): in one year, probability 0.5 of a $44 deal, probability 0.2 of a $41 deal with a white knight, probability 0.3 the price is $30 (received now). Compute the present value per share of each path and state which is better.

Problem 11.6. A company has 50 million Class A shares (20 votes) and 450 million Class B shares (1 vote). The founder holds all Class A and 20 million Class B. (a) Compute the founder's cash-flow stake and voting share. (b) The founder can take an action with a private benefit of $30 million. What is the maximum firm-value loss the founder would tolerate?

Problem 11.7. On an activist announcement, a stock returns 1.0%, 5.5% and 0.8% on days −1, 0 and +1. The market returned 0.5%, −0.2% and 0.3%. The market model has $\alpha = 0.02\%$ and $\beta = 0.9$; daily abnormal return standard deviation is 1.8%. Market capitalization is $3,000 million. Compute (a) the CAR, (b) the t-statistic, and (c) the dollar value.

Problem 11.8. A holder with a $2 million position believes that a proposal would raise the firm's value by 6%. The probability that its vote is decisive is 2%, and the cost of researching and voting is $3,000. (a) Compute the expected benefit. (b) Is voting rational? (c) What probability of being decisive makes the holder indifferent?

Answers to Selected Problems

11.1. (a) Stake = 0.04 × 8,000 = $320 million; price impact = 0.03 × 320 = $9.6 million; gain if successful = 0.12 × 8,000 × 0.04 = $38.4 million. E[profit] = 0.5 × 38.4 − 9.6 − 20 = −$10.4 million. (b) p* = (9.6 + 20) / 38.4 = 77.1%.

11.2. (a) Others hold 40 million rights; each yields 2 × 90 / 30 = 6 shares, so 240 million new shares. (b) Cash = 40 × 90 = $3,600 million; equity value = 1,500 + 3,600 = $5,100 million; shares = 290 million; price = $17.59. (c) 10 / 290 = 3.45%. (d) Holding falls from $300 million to 10 × 17.59 = $175.9 million, a loss of $124.1 million.

11.3. (a) Institutions 84 × 0.9 = 75.6; retail 52.5 × 0.4 = 21.0; activist 13.5; total 110.1 million. (b) Majority = 55.05 million; retail for activist = 4.2; needed from institutions = 55.05 − 13.5 − 4.2 = 37.35 million, or 49.4% of institutional votes cast.

11.4. (a) 80,000,000 × 2 / 8 + 1 = 20,000,001 shares (25%). (b) 80,000,000 / 6 + 1 ≈ 13,333,334 shares (16.7%).

11.5. A: 0.8 × 39 / 1.12^0.5 + 0.2 × 30 = 31.2 / 1.0583 + 6.0 = 29.48 + 6.0 = $35.48. B: (0.5 × 44 + 0.2 × 41) / 1.12 + 0.3 × 30 = 30.2 / 1.12 + 9.0 = 26.96 + 9.0 = $35.96. Path B is better by about $0.48.

11.6. (a) Cash-flow stake = (50 + 20) / 500 = 14%; votes = (50 × 20 + 20) / (50 × 20 + 450) = 1,020 / 1,450 = 70.3%. (b) L* = 30 / 0.14 = $214.3 million.

11.7. (a) Expected returns: 0.02 + 0.45 = 0.47%; 0.02 − 0.18 = −0.16%; 0.02 + 0.27 = 0.29%. Abnormal: 0.53%, 5.66%, 0.51%. CAR = 6.70%. (b) SE = 1.8 × √3 = 3.118%; t = 2.15. (c) 6.70% × 3,000 = $201 million.

11.8. (a) 2,000,000 × 0.06 × 0.02 = $2,400. (b) No; the expected benefit is below the $3,000 cost, so abstaining is rational (though for the group of holders as a whole, voting is valuable, which is the free-rider problem). (c) 3,000 / 120,000 = 2.5%.

Chapter 12: Behavioral Corporate Finance and Managerial Biases

Learning Objectives

- Distinguish the two branches of behavioral corporate finance (irrational investors facing rational managers, and rational markets facing biased managers) and explain which implications each has for financing and investment decisions.
- Identify the main managerial biases (overconfidence, optimism, anchoring, framing, loss aversion, sunk-cost escalation and myopia) and trace how each distorts capital budgeting, acquisitions and financing.
- Quantify bias effects with numerical models: shrinkage and the winner's curse, anchoring-and-adjustment, prospect-theory evaluation of escalation, narrow framing, market-timing transfers and quasi-hyperbolic discounting.
- Build reference-class ("outside view") forecasts and adjust a capital budget for the planning fallacy.
- Design debiasing techniques and decision processes, including premortems, red teams, pre-commitment, calibration training and stage-gated funding.
- Evaluate organizational design, incentives and governance choices that make good decisions robust to individual bias.

Opening Vignette

Illustrative scenario. Brightwater Logistics is a fictional company.

Brightwater Logistics, a regional freight and warehousing firm, had a record of steady returns and a CEO, Marta Quillon, who was widely admired for a string of successful acquisitions. In the year in question, her team proposed the largest project in the firm's history: a $200 million automated distribution hub. The capital committee's model showed an NPV of $76 million at the company's 10 percent hurdle rate. The projections assumed construction in three years, a ramp to $52 million of annual cash flow, and a ten-year operating life. Everyone on the committee had seen the model. Nobody had seen the history of similar projects.

Over the following three years, the project cost $260 million, a year late. Customer volumes ramped slowly. At the first post-completion review, the hub's cash flow was $41 million a year, not $52 million. The team's response was predictable: the delay was due to supply chain problems that no one could have foreseen, the cost overrun reflected inflation, and the lower volumes were temporary. A further $35 million was approved to expand capacity so that the hub would "reach scale." When the board eventually asked why the original forecast had been so favorable, the project sponsor pointed to the model's internal consistency. It had been audited three times.

Brightwater's finance team later reviewed the firm's last twenty major projects. Seventeen had finished late, fourteen had overrun their budgets, and the average realized cash flow was 15 percent below the plan. Every one of those projects had been approved on a positive NPV. The firm did not have a valuation problem. It had a human-judgment problem: optimistic forecasts, anchoring on the budget, escalation of commitment after sunk costs, and a decision process that rewarded the sponsor for advocacy rather than accuracy.

This chapter examines these biases systematically, shows how to measure their financial effects, and describes how organizations can build decision processes that do not depend on any individual being unbiased.

12.1 Two Branches of Behavioral Corporate Finance

12.1.1 The standard model and its behavioral amendments

Volume 1 developed corporate finance on the assumption that managers maximize shareholder value, forecast without systematic error, and discount cash flows at a rate that reflects risk (Volume 1, Chapters 6, 10 and 11). The standard framework does not require markets to be perfectly efficient, but it does require that managers act as value-maximizing agents. Behavioral corporate finance asks what changes when one or both assumptions fail.

Baker, Ruback and Wurgler, in their survey of the field, organized the literature into two approaches. They are worth keeping distinct, because they lead to different advice.

Approach 1: Irrational investors, rational managers. Markets can misprice securities, because of limits to arbitrage (Shleifer and Vishny) and investor sentiment. Managers are assumed to understand this and to respond: issuing equity when it is overvalued, repurchasing when it is undervalued, paying dividends when investors place a premium on them (the "catering" theory of Baker and Wurgler), and timing mergers. The firm is run for the benefit of long-term shareholders, but market prices are noisy signals.

Approach 2: Rational investors, irrational managers. Markets price assets correctly, but managers are subject to the psychological biases documented in the decision-making literature, such as overconfidence, anchoring, loss aversion, and escalation. The firm's decisions deviate from value maximization because of managerial cognition, not because of market mispricing.

Real life combines both. This chapter treats Approach 2 as the main subject, since it directly addresses the quality of the decisions that finance leaders make, and treats Approach 1 mainly as the problem of market timing in Section 12.5.

12.1.2 Why bias is predictable and costly

Psychologists distinguish errors that are random from errors that are systematic. Random error averages out across many decisions. Bias does not: it pushes in the same direction every time. Kahneman, Tversky and many successors documented that people use heuristics (rules of thumb) such as availability, representativeness, and anchoring and adjustment, and that these produce predictable departures from the rational benchmark. Financial decisions are fertile ground for such errors because they are made under uncertainty, on the basis of forecasts, with ambiguous feedback arriving years later.

The managerial importance is that biased decisions are not noise around a correct answer. Because they are systematic, they can be modeled, measured and, to some degree, corrected. Most of this chapter is about doing exactly that.

12.1.3 A taxonomy of biases for finance leaders

Bias	Core idea	Typical corporate manifestation	Corrective principle
Overconfidence	Overestimating the precision of one's knowledge or the probability of success	Overpaying for acquisitions; narrow forecast ranges; excess investment	Calibration; wider ranges; outside view
Optimism and the planning fallacy	Expecting better-than-typical outcomes; ignoring base rates	Overruns, delays, optimistic synergies	Reference-class forecasting
Anchoring	Insufficient adjustment from an initial number	Bids anchored on asking price; budgets anchored on last year	Independent estimates before exposure to anchors
Framing and narrow framing	Evaluating decisions in isolation or by their description	Rejecting positive-NPV risky projects individually	Aggregate decisions at the portfolio level
Loss aversion	Losses weigh more than equal gains	Hurdle rates inflated above risk-adjusted rates; avoidance of write-offs	Evaluate on incremental value; pre-set rules
Sunk-cost fallacy and escalation	Past costs influence future decisions	Throwing good money after bad	Incremental analysis; kill criteria; independent review
Confirmation bias	Seeking evidence that supports the existing view	One-sided due diligence	Red teams; disconfirming tests
Hindsight bias	Believing outcomes were predictable after the fact	Weak learning from post-audits	Decision journals recorded before outcomes
Myopia and present bias	Overweighting near-term outcomes	Cutting R&D to meet earnings	Longer-horizon metrics; governance
Groupthink	Suppression of dissent for cohesion	Unchallenged proposals in committees	Structured dissent; anonymous voting

12.2 Overconfidence and Optimism

12.2.1 Three kinds of overconfidence

Moore and Healy distinguish three forms. *Overestimation* means thinking one's performance is better than it is. *Overplacement* means believing oneself better than others. *Overprecision* means excessive certainty about the accuracy of one's beliefs, shown in confidence intervals that are too narrow. Overprecision is the most persistent and the most relevant to finance, since valuation is a forecast with a distribution of outcomes, and a point estimate with too narrow an interval leads to excessive commitment.

Optimism is distinct from overconfidence, though they interact. Optimism is a belief that outcomes will be better than they are on average; overconfidence is a belief that one's estimates are more accurate than they are. A manager can be both: confident that a project will deliver $52 million a year, and wrong about both the level and the uncertainty.

12.2.2 Evidence from corporate decisions

Richard Roll's 1986 hubris hypothesis explained takeovers as the result of acquirers overestimating their ability to find value, so that they pay too much and the winners' premia are in part transfers from acquirer to target shareholders. Heaton (2002) analyzed how managerial optimism can lead to overinvestment when managers believe the market undervalues the firm, and to a preference for internal funds over external financing. Malmendier and Tate measured overconfidence in a practical way: they identified chief executives who held in-the-money stock options until expiry, which is inconsistent with

prudent diversification, and found that such executives made more acquisitions, especially diversifying deals, and relied more on internal financing, with investment more sensitive to cash flow. Their approach illustrates a key point for finance: overconfidence is revealed by behavior, and its consequences are observable in capital allocation.

Surveys of finance executives, such as those that ask for 80 percent confidence intervals of the following year's market return, have found that realized values fall inside the stated intervals far less often than 80 percent. The overprecision finding is robust across domains: people's "90 percent" ranges contain the truth perhaps half the time.

12.2.3 Why overconfidence is systematically selected

Overconfident managers may be rewarded by promotion processes. They advocate more convincingly, take on ambitious projects, and are more likely to be in positions of authority. Goel and Thakor have modeled how competition for promotion can favor overconfident managers. The result is that the people who make large capital decisions are not a random sample: they may be more optimistic than average. At the firm level, boards that admire "decisive" CEOs and projects that must be presented as winners to be funded compound the problem.

12.2.4 The winner's curse in capital budgeting

Overconfidence interacts with a statistical effect that does not depend on psychology at all. When a firm chooses projects by selecting those with the highest estimated NPV, it systematically selects projects whose estimates contain positive error. This is the winner's curse (a term used by Thaler in the context of auctions), also known as selection bias or regression to the mean. A manager who understands it shrinks the estimates toward the mean; an overconfident manager treats the selected estimates as unbiased.

Suppose true project NPVs are normally distributed across opportunities with mean zero and standard deviation σ_T, and the estimate is the true NPV plus independent error with standard deviation σ_E. The expected true NPV given an estimate x is:

```
E[true NPV | estimate = x] = x × σ_T² / (σ_T² + σ_E²)
```

The factor $\sigma_T^2 / (\sigma_T^2 + \sigma_E^2)$ is the shrinkage coefficient. The larger the forecast error relative to the dispersion of true values, the less the estimate should be trusted. An overconfident forecaster, who believes σ_E is smaller than it is, uses too large a coefficient.

Example 12.1 – Winner's curse and shrinkage in project selection.

A firm evaluates many projects. True NPVs across opportunities are $N(0, 20^2)$ in $ millions (a prior that most projects are near zero NPV, which is realistic in competitive markets). Forecast errors are $N(0, 20^2)$, independent of the true value.

Step 1: shrinkage coefficient with the correct error.

```
k = 20² / (20² + 20²) = 400 / 800 = 0.50
A project with estimated NPV +$30m has expected true NPV = 0.50 × 30 = $15m
```

Step 2: an overconfident team believes its error SD is only 10.

```
k_believed = 400 / (400 + 100) = 0.80
Believed expected true NPV = 0.80 × 30 = $24m
Overstatement = 24 − 15 = $9m, or 60% above the correct value
```

Step 3: simulation. To see the effect on a whole selected portfolio, we simulate 100,000 projects with these distributions (seed 42) and select by estimated NPV.

Selection rule	Share of projects selected	Mean estimated NPV	Mean true NPV	Share with true NPV < 0
Accept if estimate > 0	50.0%	$22.5m	$11.2m	25.6%
Top 20% by estimate	20.0%	$39.6m	$19.8m	10.3%
Estimate > $30m	14.4%	$44.6m	$22.2m	7.4%

Analytic check for the first row: with $\sigma_T = 20$ and total SD of estimate $\sqrt{800} = 28.28$, $E[\text{estimate} \mid \text{estimate} > 0] = 28.28 \times \sqrt{(2/\pi)} = \$22.6\text{m}$, and $E[\text{true NPV} \mid \text{estimate} > 0] = 0.5 \times 22.6 = \$11.3\text{m}$, in line with the simulation.

Interpretation: In every case, the realized NPV is about half the estimate. The firm that treats the sum of estimated NPVs as its expected value gains has overstated it by a factor of two even if no one is dishonest. The effect gets worse the more projects the firm screens and the higher the selection bar, since the highest estimates are the most contaminated by error. The practical remedies are to shrink estimates before ranking, to evaluate forecaster track records (so that σ_E is estimated rather than assumed), and to require independent validation for the largest estimates, since those are the ones most likely to be wrong.

12.2.5 Managerial implications

- Treat a forecast as a distribution, not a number. Ask for ranges and test whether the proposer’s past ranges contained the outcomes.
- Estimate forecast error from the firm’s own post-audit data and use it to shrink.
- Be aware that the more selective the pipeline, the larger the winner’s curse. A firm with a hundred proposals and a hurdle of 25 percent IRR will likely accept the most error-contaminated numbers.
- Separate the person who prepares the forecast from the person who will be evaluated on the result when possible.

12.3 Anchoring, Adjustment and Framing

12.3.1 How anchors work

Tversky and Kahneman showed that people estimating an uncertain quantity start from an initial value, the anchor, and adjust, but adjust insufficiently. Even irrelevant anchors, such as a random number from a wheel, shift subsequent estimates. Anchors in business are rarely random: last year’s budget, the asking price of the seller, the current share price, the 52-week high, a competitor’s bid, or the first number said in a negotiation.

A simple model describes the effect. The final estimate is a weighted average of the anchor A and the independent estimate I:

```
Final = w × A + (1 − w) × I
```

where w is the anchoring weight, between 0 and 1. If the decision-maker were fully rational, w would be zero when the anchor carries no information. Experiments suggest w is positive and sometimes large, especially when the decision-maker is uncertain or time-pressured.

12.3.2 Anchoring in acquisitions

Takeover premia illustrate anchoring. Baker, Pan and Wurgler found that offer prices cluster around the target's recent peak price, and that bids at or above that reference point are more likely to be accepted, even though a past peak is not a fundamental value. In deal negotiations, the seller's opening ask, the investment banker's "football field" range and the prior transaction multiples in the sector all serve as anchors.

Example 12.2 – Anchoring and the walk-away price.

Brightwater is considering acquiring a smaller carrier. Its own DCF valuation of the target's standalone equity is $65 per share (Volume 1, Chapter 21). It estimates synergies worth $8 per share. The maximum rational price, the walk-away, is therefore 65 + 8 = $73 per share. There are 40 million target shares. The seller's advisers publicly signal a price of $95 per share. Suppose that the decision-maker's bid is formed by anchoring and adjustment with weight w.

Step 1: bid as a function of w.

```
Bid = w × 95 + (1 − w) × 65
```

Anchoring weight w	Bid ($/share)	Bid minus walk-away ($73)
0.00	65.00	−8.00
0.10	68.00	−5.00
0.20	71.00	−2.00
0.35	75.50	+2.50
0.50	80.00	+7.00

Step 2: the weight at which the bid equals the walk-away.

```
w* = (73 − 65) / (95 − 65) = 8 / 30 = 0.267
```

Step 3: dollar cost of anchoring at w = 0.35.

```
Overpayment = (75.50 − 73.00) × 40 million = $100 million
```

Interpretation: An anchoring weight of only about 27 percent is enough to consume the entire synergy value and begin transferring wealth to the seller. At 35 percent, the buyer overpays by $100 million. The result is a plain dollar number that reflects a modest cognitive tendency. This example explains why disciplined acquirers set their walk-away price before contact with the seller or banker, record it in writing, require the board's approval to exceed it, and examine independently whether any claimed synergy is real. It also shows why the first number in a negotiation has value: a seller who opens at $95 has pulled the final price up even if it is never accepted.

12.3.3 Anchoring in budgeting and targets

Budgets anchor on the previous year's number. Managers who build next year's plan as "last year plus 5 percent" are anchoring, and divisions with high anchors keep them. Zero-based budgeting, which requires each item to be justified from scratch, reduces the anchor at the cost of effort. Targets anchored on aspiration, such as "15 percent growth," change the way the downside is judged: a result of 8 percent is "a miss" and invites risky efforts to close the gap.

12.3.4 Framing effects

Framing refers to the sensitivity of choices to the description of the same facts. A restructuring described as “an 85 percent chance of keeping the company intact” attracts support that “a 15 percent chance of failure” does not. In finance, framing influences hurdle rates, mental accounting (treating a windfall differently from regular earnings), and the choice between presenting a proposal as a gain relative to the status quo or as a loss avoided. The best discipline is to restate each proposal in at least two frames and check whether the decision changes. If it does, the decision is being driven by presentation.

12.4 Loss Aversion, Sunk Costs and Escalation of Commitment

12.4.1 Prospect theory in brief

Kahneman and Tversky’s prospect theory replaces expected utility with a value function defined over gains and losses relative to a reference point. The function is concave for gains, convex for losses, and steeper for losses than for gains (loss aversion). A common parameterization is:

```
v(x) = x^α              for x ≥ 0
v(x) = −λ × (−x)^α      for x < 0
with α ≈ 0.88 and λ ≈ 2.25
```

Three consequences matter for finance. First, people are risk-averse in gains but risk-seeking in losses: a sure loss is less attractive than a gamble with the same expected loss. Second, losses loom larger than gains, so a 50-50 gamble must offer roughly twice as much upside as downside to be accepted. Third, the reference point matters: the same outcome is treated as a gain or loss depending on the starting point, and managers’ reference points are often the budget, the previous result, or the break-even of the original investment.

12.4.2 The sunk-cost fallacy

A sunk cost is a cost that has been incurred and cannot be recovered whatever the firm does now. Rational decisions consider only incremental cash flows (Volume 1, Chapter 6): continue if the expected present value of future benefits exceeds the future costs of obtaining them. Yet experiments by Arkes and Blumer showed that people who have invested more are more likely to continue with a failing course of action, and Staw’s work on escalation of commitment documented how decision-makers who are personally responsible for a failing project often invest further to justify their earlier choice. The causes include loss aversion (stopping makes the loss “real”), self-justification (admitting error damages reputation), project completion bias, and agency problems (a manager’s career is attached to the project).

Example 12.3 – Escalation: incremental analysis versus the loss frame.

Brightwater has spent $120 million on an automated hub that is behind schedule. Completing it requires another $40 million. Management’s updated view is that there is a 25 percent chance the hub will succeed, in which case it will be worth $150 million in present value at completion, and a 75 percent chance of failure, in which case it will be worth zero.

Step 1: incremental (rational) analysis. The $120 million is sunk.

```
E[benefit of completion] = 0.25 × 150 = $37.5 million
Incremental cost         = $40 million
Incremental NPV          = 37.5 − 40 = −$2.5 million → stop
```

Step 2: the loss frame. The manager has a mental account that includes the $120 million. If the project is stopped, the account closes at −$120 million. If it continues, the account closes at 120 + 40 = −$160 million with probability 0.75 (failure) or −160 + 150 = −$10 million with probability 0.25 (success).

Expected final position: stop −$120 million; continue 0.25 × (−10) + 0.75 × (−160) = −2.5 − 120 = −$122.5 million. In dollars, stopping is better by $2.5 million, which matches Step 1.

Step 3: prospect-theory evaluation of the account (α = 0.88, λ = 2.25).

```
v(−120) = −2.25 × 120^0.88 = −152.01
v(−10)  = −2.25 × 10^0.88  = −17.07
v(−160) = −2.25 × 160^0.88 = −195.80
V(stop)     = −152.01
V(continue) = 0.25 × (−17.07) + 0.75 × (−195.80) = −4.27 − 146.85 = −151.12
```

The prospect-theory value of continuing (−151.12) exceeds that of stopping (−152.01).

Interpretation: Even though continuing destroys $2.5 million in expected value, a decision-maker who frames the choice as a loss account and has the standard curvature prefers to gamble. The margin in this illustration is small because the parameters are conservative, but real escalation is typically reinforced by other forces that this value function omits: reputational cost of admitting failure, accounting write-offs that the manager wants to defer, and sponsors' career incentives. The numerical lesson is that the pull toward continuing is greatest exactly when the odds are poor but there is some chance of recovering the account, which is the profile of many failing projects. The remedy is procedural: review go/no-go decisions on incremental analysis only, preferably by someone who did not approve the original investment, and agree on explicit kill criteria in advance.

12.4.3 Narrow framing and hurdle-rate inflation

Loss aversion also shows up when managers evaluate each project separately, a pattern Kahneman and Lovallo called *narrow framing* or "bracketing." A manager responsible for a divisional budget who fears a visible loss on any single project may reject positive-NPV projects with a risk of loss, even though the firm holds a portfolio in which independent risks diversify (Volume 1, Chapters 8 and 10). Graham and Harvey's surveys of corporate managers suggest firms often use hurdle rates well above their cost of capital; some of the gap is justified by real options and estimation error, but some probably reflects such behavioral caution.

Example 12.4 – The cost of narrow framing.

A division head can undertake up to ten independent identical projects. Each yields +$30 million with probability 0.6 and −$20 million with probability 0.4. The head is loss-averse with λ = 2.5 and evaluates each project on its own.

Step 1: expected value and loss-averse value of one project.

```
E[NPV]    = 0.6 × 30 + 0.4 × (−20) = 18 − 8 = +$10 million
Perceived = 0.6 × 30 − 2.5 × 0.4 × 20 = 18 − 20 = −$2 million → reject
```

Step 2: the portfolio view. With k successes out of ten, the total is 30k − 20(10 − k) = 50k − 200, which is positive when k ≥ 5.

```
Expected total = 10 × 10 = $100 million
Standard deviation of one project = √(0.6×30² + 0.4×20² − 10²) = √(540 + 160 − 100) = √600 = $24.5 million
Standard deviation of ten = 24.5 × √10 = $77.5 million
P(total > 0) = P(k ≥ 5) = 83.4%
```

Step 3: loss-averse value of the portfolio. Computing the distribution of the total exactly gives E[gains] = $103.44 million and E[losses] = $3.44 million.

```
Perceived value of portfolio = 103.44 − 2.5 × 3.44 = $94.84 million → accept
```

Interpretation: The same manager who rejects each project in isolation would accept the ten as a bundle. Narrow framing costs the firm $100 million of expected value. The practical implication is to require that risky positive-NPV projects be evaluated against the firm's total portfolio (or at least a division-wide portfolio), to give division heads a hurdle rate based on systematic risk rather than total project variance, and to reward decisions by the quality of the process, not by the outcome of individual bets. A firm that punishes every failed project will end up with a pipeline of safe projects.

12.5 Market Timing, Mispricing and Managerial Myopia

12.5.1 Financing as a response to mispricing

Under the first behavioral approach, managers are rational and markets are not. If the share price differs from intrinsic value, the firm can transfer wealth between groups of shareholders through its financing decisions. Stein's 1996 analysis, "Rational Capital Budgeting in an Irrational World," showed that a manager acting for long-term shareholders should base the investment decision on fundamental value (discounting at a rate that reflects risk, as in Volume 1, Chapter 11), but base the financing decision on the market price: issue shares when overpriced, and repurchase when underpriced. The two decisions are separable in principle.

Baker and Wurgler (2002) documented that firms issue equity when their market-to-book ratios are high, and that the effects on capital structure are persistent: a firm's leverage today is correlated with the weighted average of its past market-to-book ratios. They interpreted capital structure as the cumulative result of past market timing, not the outcome of a target leverage ratio, which differs from the trade-off theory in Volume 1 (Chapter 15). Subsequent work has debated the persistence and the interpretation, but the central observation that equity issuance is concentrated after periods of high valuation is well established. Shleifer and Vishny (2003) extended the logic to mergers: overvalued acquirers use their stock as currency to buy less overvalued targets, and gain, at the expense of the long-term holders of the combined firm.

12.5.2 Who wins and who loses

The interests of three groups diverge when a firm times the market. Long-term shareholders (those who hold through the correction) gain when the firm issues overpriced shares. New shareholders who buy at the inflated price lose. Short-term holders who sell into the inflated market gain. Hence the meaning of "acting in the shareholders' interest" depends on whose interest is intended, a question the standard model avoids by assuming that price equals value.

Example 12.5 – Wealth transfers from issuing overvalued shares and repurchasing undervalued ones.

Brightwater has 100 million shares. Management's analysis puts the intrinsic value at $40 per share (equity value $4,000 million), but the market price is $60.

Issuance. The firm issues 10 million new shares at $60, raising $600 million, and holds the cash. Assume intrinsic value of the existing business is unchanged.

Step 1: new intrinsic value per share.

```
Intrinsic equity value after = 4,000 + 600 = $4,600 million
Shares                       = 110 million
Intrinsic value per share    = 4,600 / 110 = $41.82 [243]
```

Step 2: gain to existing holders and loss to new holders.

```
Gain per old share = 41.82 − 40.00 = $1.82
Total gain         = 1.82 × 100 = $181.8 million
New holders: paid 10 × 60 = $600 million for a claim worth 10 × 41.82 = $418.2 million → loss of $181.8
million
```

The gain to long-term holders exactly equals the loss to the new buyers.

Repurchase. Alternatively, suppose the market price is $30 against an intrinsic value of $40, and the firm buys back 10 million shares.

Step 3: value after repurchase.

```
Intrinsic equity value after = 4,000 − 300 = $3,700 million
Shares                       = 90 million
Intrinsic value per share    = 3,700 / 90 = $41.11
Gain per remaining share     = 1.11;  total = 1.11 × 90 = $100.0 million
Selling holders: received 10 × 30 = $300 million for a claim worth 10 × 40 = $400 million → loss of $100.0
million
```

Interpretation: Timing creates no value for the firm as a whole; it redistributes between continuing and exiting or entering shareholders. The gain depends entirely on the manager's judgment that the market is wrong, and managers are prone to overconfidence about mispricing. If the manager believes the stock is overvalued when it is fairly valued, the issuance is merely neutral; but if the manager believes the stock is undervalued and repurchases when it is fairly valued, or worse overvalued, the firm hands value to sellers and destroys liquidity. Managerial discipline requires an explicit valuation range, an estimate of the probability that the manager's valuation is wrong (borrowing the calibration approach in Section 12.7), and a comparison of the transfer against the cost of signaling (issuance usually lowers the price, as Volume 1, Chapter 15 explained under pecking-order theory).

12.5.3 Catering and the reaction to sentiment

Baker and Wurgler's catering theory of dividends (2004) proposes that managers initiate or omit dividends depending on whether investors currently put a premium on dividend payers. Similar logic applies to share splits, stock-based acquisitions, names (companies that adopted "dot-com" names around 1999) and, more recently, the framing of strategy in language that is currently fashionable. Catering is not irrational for a manager who maximizes the current price, but it is a form of short-horizon behavior if the price premium is transitory.

12.5.4 Myopia and present bias

Managers can have short horizons for rational reasons (compensation, job risk, takeover risk) and for behavioral ones. Stein's 1989 analysis of "efficient capital markets, inefficient firms" showed how managers who care about the current stock price may sacrifice long-term value if investors cannot observe the quality of investment. Graham, Harvey and Rajgopal's survey of chief financial officers found that a large majority were willing to give up economic value, for example by delaying a positive-NPV project, to meet an earnings target.

Behavioral economics adds *present bias*. People weight the near future more heavily than exponential discounting would imply. A convenient model is quasi-hyperbolic (beta-delta) discounting: a payoff at time $t \geq 1$ is valued at $\beta \times \delta^t \times$ payoff, where δ is the long-run discount factor and $\beta < 1$ an additional penalty on everything that is not immediate.

Example 12.6 – Present bias and the R&D that is never funded.

A division can spend $10 million on R&D now, with a payoff of $14 million in two years. The firm's cost of capital is 10%, so $\delta = 1/1.10 = 0.9091$. The divisional manager's bonus is paid on this year's earnings, and the manager discounts with $\beta = 0.6$.

Step 1: value to the firm.

```
PV = 14 / 1.10² = 14 / 1.21 = $11.57 million
NPV = 11.57 − 10 = +$1.57 million → fund
```

Step 2: perceived value to the present-biased manager.

```
Perceived PV = β × δ² × 14 = 0.6 × 0.8264 × 14 = $6.94 million
Perceived NPV = 6.94 − 10 = −$3.06 million → reject
```

Step 3: the critical β.

```
β* = 10 / (0.8264 × 14) = 10 / 11.57 = 0.864
```

Only a manager with β above 0.864 would fund the project.

Interpretation: A small positive NPV is fragile against even a moderate present bias. With $\beta = 0.6$ the manager rejects the project, and a project with NPV three times larger would also fail if its payoffs were far enough in the future. Bonus structures tied to current accounting earnings reinforce the bias, because R&D is expensed. Companies reduce the problem with multi-year incentive plans, deferred and clawed-back compensation, stock holding requirements, and metrics that reflect long-term investment such as returns on capital and pipeline progress. A board that sees a pattern of delayed, small-NPV, long-payoff projects should suspect present bias before it concludes that opportunities have dried up.

12.6 Behavioral Capital Budgeting

12.6.1 The planning fallacy and the inside view

Kahneman and Tversky introduced the *planning fallacy*: people underestimate the time, cost and risk of their own projects, even when they know that similar projects have overrun. Their explanation is that planners adopt an *inside view*, constructing a scenario from the specifics of the case, and ignore the *outside view*, the statistical distribution of outcomes in a reference class of similar projects. Lovallo and Kahneman argued that the inside view is a major cause of failure in large undertakings, and Flyvbjerg developed *reference-class forecasting* for infrastructure projects, which adjusts a forecast by the empirical distribution of overruns for comparable projects.

Why does the inside view fail so reliably? Each project has a plan, and a plan describes a path in which things go right. Possible delays are numerous, specific and unforeseeable; individually each is improbable, so the planner does not include any. But the probability that *some* delay occurs is high. Moreover, the person who prepares the plan is typically the advocate who needs approval, and organizations rarely reward the planner who says, "this is probably a bad idea."

12.6.2 The outside view in practice

Reference-class forecasting follows three steps:

1. **Identify the reference class.** Choose a set of past projects similar enough in type and scale. For a hub, this might be the firm's own last twenty projects plus industry data.

2. **Obtain the distribution of the outcome of interest.** For example, the ratio of actual to budgeted cost, delay in months, and actual to forecast cash flow.
3. **Position the current project within the distribution.** Start from the reference class median and adjust only for specific, documented differences. Set the budget at a chosen percentile (the 70th or 80th) according to the firm's tolerance for overruns.

The result is typically a higher cost, a longer schedule and lower benefits than in the plan, and thus a lower NPV. Whether the project remains attractive after adjustment is exactly the information the board needs.

12.6.3 Quantifying the planning fallacy

Example 12.7 – Inside versus outside view for the distribution hub.

Brightwater's plan for the hub (a simplified version of the vignette) has the following cash flows, in $ millions, at a 10% discount rate. Costs: $100 at t = 0, $60 at t = 1, $40 at t = 2. Inflows: $52 per year for ten years starting at t = 3 (t = 3 to t = 12).

Step 1: inside-view NPV.

```
PV of costs   = 100 + 60/1.10 + 40/1.10² = 100 + 54.55 + 33.06 = $187.60 million
Annuity factor (10 years, 10%) = 6.1446
PV at t = 2 of inflows = 52 × 6.1446 = 319.52
PV at t = 0           = 319.52 / 1.10² = 264.06
NPV = 264.06 − 187.60 = +$76.46 million
```

Step 2: outside-view adjustment from the reference class: cost 30% higher in each period, a one-year delay in the start of inflows (first inflow at t = 4), and realized inflows 10% lower ($46.8 million a year, t = 4 to t = 13).

```
PV of costs   = 130 + 78/1.10 + 52/1.10² = 130 + 70.91 + 42.98 = $243.88 million
PV of inflows = 46.8 × 6.1446 / 1.10³ = 287.57 / 1.3310 = $216.05 million
NPV = 216.05 − 243.88 = −$27.83 million
```

Step 3: which adjustment matters most? Apply each alone.

Adjustment applied alone	NPV ($ million)	Change from +76.46
None (inside view)	+76.46	–
30% cost overrun only	+20.18	−56.28
One-year delay only	+52.45	−24.01
10% lower inflows only	+50.05	−26.41
All three (outside view)	−27.83	−104.29

Step 4: break-even inflows in the outside-view case. The annual inflow at which the adjusted NPV is zero is PV of costs divided by the PV factor of a ten-year annuity starting at t = 4:

```
Required inflow = 243.88 / (6.1446 / 1.10³) = 243.88 / 4.6165 = $52.83 million per year
```

That is more than the $52 million in the original plan.

Interpretation: A project with a comfortable $76 million NPV on the inside view loses $28 million when the base rates of overruns, delays and shortfalls are applied. The cost overrun alone removes about three-quarters of the original NPV. In the adjusted case the project would need $52.83 million of annual inflows, above even the original forecast, merely to break even. The difference between +$76 million and −$28 million is not about the discount rate or the model's mathematics; it comes entirely from assumptions about execution. The cost of this adjustment is small compared with the cost of building a project that destroys value, and the technique forces the sponsor to explain why the project will beat its reference class.

12.6.4 Hurdle rates, risk adjustment and optimism

Some firms respond to optimism by raising the hurdle rate. This is a blunt tool. A uniform premium penalizes long-lived projects more than short ones (a point that follows from compounding), penalizes the honest sponsor as much as the optimistic one, and distorts the choice toward quick-payback projects. It also confuses two kinds of risk: systematic risk, which belongs in the discount rate (Volume 1, Chapters 10 and 11), and forecast bias, which belongs in the cash flows. The principled approach is to correct the cash flows for bias, using the outside view, and to keep the discount rate for risk. A related error is to discount optimistic expected cash flows at a high rate and call the result "conservative": it would be better to estimate honest expected cash flows and discount at the cost of capital, using scenario analysis or simulation to display the spread.

12.6.5 Post-audits and learning

Hindsight bias undermines learning. After the fact, people overestimate how predictable an outcome was and misremember their earlier forecasts. A post-audit that compares outcomes to the original plan is therefore only useful if the plan was recorded in a form that cannot be revised. Good practice involves a decision record at approval stating the key assumptions, the range of outcomes, the expected value and the criteria that would cause the project to be stopped. Reviews then compare the outcome to the record and identify the causes of variance as skill, luck, or bias.

12.7 Debiasing and Decision Processes

12.7.1 Individual and organizational debiasing

Fischhoff and subsequent researchers have found that "debiasing" individuals by warning them is only modestly effective. Knowing about overconfidence does not make people less confident. More effective techniques change the decision *process*, so that the environment, not willpower, corrects the bias. Kahneman, Lovallo and Sibony have argued that organizations can review decisions for biases by checklists and structured questions, because it is easier to see bias in others' proposals than in one's own.

12.7.2 The premortem

Gary Klein's *premortem* is a simple technique. Before the decision is final, the team imagines that the project has failed completely, one or two years in the future, and each member writes down the reasons for the failure. The shift from "what might go wrong?" to "it did go wrong; why?" uses *prospective hindsight*, which research has shown to increase the number and specificity of reasons generated. It also licenses dissent: in a normal meeting, expressing doubts is disloyal; in a premortem, it is the task.

A well-run premortem has these steps: (1) brief the team on the plan; (2) announce that the project has failed; (3) each person writes failure reasons independently (to avoid conformity); (4) collect and cluster the reasons; (5) assign each major risk an owner, a mitigation and, where possible, a measurable early-warning indicator; (6) revise the plan, or its funding,

accordingly. For Brightwater's hub, a premortem might surface reasons such as "automation vendor delivers late," "anchor customer renegotiates volume," and "labor costs for installation rise," each of which could be priced into the base-rate adjustment of Example 12.7 and monitored.

12.7.3 Other techniques

- **Red teams and devil's advocates.** Assign a team to argue against the proposal, with access to the same data. The red team's incentives must be genuine; a token dissenter is ineffective, and an assigned dissenter can be dismissed as "just playing a role."
- **Independent estimates before discussion.** Collect forecasts privately and then discuss. This reduces anchoring on the first speaker or the most senior person and exploits the average of independent estimates, which is usually more accurate than any individual (the wisdom-of-crowds effect, which depends on independence). The Delphi method formalizes this in rounds.
- **Pre-commitment and kill criteria.** Decide in advance what evidence would cause the project to stop. Ulysses contracts, such as pre-agreed stage gates, remove the temptation to rationalize after the fact.
- **Decision journals.** Record the decision, the reasoning, the probabilities and the expected outcomes in writing at the time, to defeat hindsight bias.
- **Separate roles.** Separate the advocate, the evaluator and the decision-maker. The group that proposes the project should not evaluate it.
- **Base-rate checklists.** Require every proposal to state the reference class, the base rate of success, and why this case is different.
- **Anonymous voting and rotating dissent.** Reduces the effect of rank and conformity (groupthink, in Janis's terminology).
- **Stage-gated funding.** Release money in tranches tied to milestones, which converts a single large commitment into a sequence of smaller decisions with options to abandon (Volume 1, Chapter 18, real options). Stage gates directly attack escalation, as each gate is a fresh decision on incremental value.

12.7.4 Calibration training

Overconfidence is one of the biases that responds to feedback. Forecasters who receive repeated, quantified feedback on their probability judgments, as in weather forecasting or tournaments of probabilistic forecasts studied by Tetlock, become better calibrated. The measures are simple. For *interval* forecasts, the hit rate (the proportion of outcomes inside the stated interval) should match the stated confidence. For *probability* forecasts, the Brier score, the mean squared difference between the forecast probability and the 0/1 outcome, measures accuracy, with lower being better.

Example 12.8 – Measuring calibration and correcting intervals.

Brightwater's treasury team has made twenty forecasts of quarterly cash balances, each with an 80% confidence interval. Eleven of the twenty actuals fell inside the interval.

Step 1: the hit rate.

```
Hit rate = 11 / 20 = 55%   (target: 80%)
```

Step 2: statistical significance. If the intervals were truly 80%, the number of hits would be Binomial(20, 0.8).

```
P(X ≤ 11) = Σ(k = 0 to 11) C(20,k) × 0.8^k × 0.2^(20−k) = 0.0100  (about 1%)
```

The team's intervals are too narrow, and the shortfall is unlikely to be chance.

Step 3: implied correction. For a normal error with the team's stated standard deviation σ_s, an 80% interval is ± 1.2816 σ_s. Suppose the actual standard deviation is σ_a, and the observed hit rate of 55% implies the interval covers ± 0.7554 σ_a (the z-value for a central 55% region).

```
1.2816 σ_s = 0.7554 σ_a   →   σ_a / σ_s = 1.2816 / 0.7554 = 1.70
```

The team should widen its intervals by about 70%.

Step 4: a probability-forecast example. A manager gives probabilities that eight initiatives will meet their milestones:

Initiative	1	2	3	4	5	6	7	8
Forecast probability	0.90	0.80	0.70	0.90	0.60	0.95	0.85	0.75
Outcome (1 = met)	1	0	1	0	1	1	0	1

```
Brier = [(0.1)² + (0.8)² + (0.3)² + (0.9)² + (0.4)² + (0.05)² + (0.85)² + (0.25)²] / 8
      = (0.01 + 0.64 + 0.09 + 0.81 + 0.16 + 0.0025 + 0.7225 + 0.0625) / 8 = 2.4975 / 8 = 0.312
Average forecast = 0.806;  realized frequency = 5/8 = 0.625
Benchmarks: always forecasting 0.5 gives 0.250; always forecasting the base rate 0.625 gives 0.234
```

Interpretation: The manager's Brier score of 0.312 is worse than simply forecasting 50-50 or the base rate, and the average forecast of 80.6 percent exceeds the realized frequency of 62.5 percent. This is the signature of overconfidence. With only eight observations the evidence is weak, which is itself a lesson: calibration requires many forecasts and sustained tracking. A firm that records probabilities for every major milestone and scores them every quarter builds the data to correct itself. The interval example shows the remedy when the data are sufficient: widen the ranges and size contingency, hedges and capital buffers to the wider, honest distribution.

12.8 Designing the Organization for Better Decisions

12.8.1 Incentives and the interaction with bias

Behavioral and agency problems interact. A manager with a career incentive to see a project through will rationalize escalation more readily. A compensation scheme with a payoff only for success encourages optimism, and one with a penalty for any failure encourages loss avoidance. The firm should reward the quality of forecasts as well as results: for example, recognizing managers whose forecasts are accurate, whose projects reach their milestones or are stopped early when evidence warrants. Gains from decision quality are typically larger than gains from marginal changes in incentive strength.

12.8.2 Governance and the role of the board

The board is an important debiasing mechanism because its members are not committed to the proposal. Effective boards require a standard decision template (reference class, base rates, premortem results, kill criteria) for major capital commitments, review post-audits with the same seriousness as approvals, and ensure that at least one director with relevant expertise is empowered to challenge the CEO. They also recognize that a charismatic and successful CEO is a risk factor

for unchallenged proposals, because past success increases confidence in future judgment, a variant of the overconfidence finding in Section 12.2. A related caution is about overcorrection: a board that rejects every proposal with a positive probability of failure is displaying the same loss aversion it is trying to avoid.

12.8.3 A decision-quality checklist

Question	Bias targeted
Who prepared the forecast, and what is their incentive?	Optimism; advocacy
What is the reference class, and where does this project fall in its distribution?	Planning fallacy; inside view
What range of outcomes did we estimate, and how often have our past ranges contained the outcome?	Overprecision
Was the walk-away value set before seeing the counterparty's number?	Anchoring
What would the decision be if the money already spent were zero?	Sunk cost
Is this project evaluated alone or as part of a portfolio?	Narrow framing
What did the premortem identify, and who owns each risk?	Confirmation; groupthink
What evidence would cause us to stop, and who decides?	Escalation
Are we financing based on a view of market mispricing, and how confident are we?	Overconfidence in timing
Is the incentive horizon consistent with the investment horizon?	Myopia

12.8.4 Limits of debiasing

Debiasing has costs. Procedures take time; a premortem for every minor decision would paralyze the firm. They should be proportional to the stakes and the irreversibility of the decision. Debiasing may also produce false confidence: a firm that has a checklist may believe it is immune. The disciplined view is that bias is a persistent feature of human judgment and the right response is to build processes, track the record, and keep measuring.

There is a further caution about the evidence. Behavioral finance is sometimes criticized for providing a catalog of biases without a unified theory and for fitting each anomaly with a different explanation. Some findings, especially those from laboratory studies, do not always replicate in professional settings with feedback and incentives, and some documented patterns in market behavior have weakened over time. The conclusions of this chapter should be read in that light: the biases reviewed here are supported by multiple lines of evidence, and the procedures are inexpensive relative to the stakes, but the magnitudes in any given firm should be measured and not assumed.

Case Study: The Meridian Fuel-Cell Program

Illustrative case. All names and numbers are fictional.

Meridian Industrial is a fictional manufacturer of heavy equipment with $4.0 billion of revenue and a reputation for engineering excellence. Four years ago, under the leadership of its CEO, Daniel Okafor, Meridian launched a program to develop hydrogen fuel-cell power units for industrial vehicles. The original business case called for $180 million of development cost over three years, a commercial launch in year 4, and an NPV of $210 million at a 10 percent discount

rate. The program was championed by Okafor himself, who had presented it to the board as “the next Meridian.” Okafor’s confidence was supported by a record: two earlier product launches had succeeded, and the company’s stock had risen 70 percent during his tenure.

Approval was swift. The board’s capital committee reviewed a forty-page model; it asked about the discount rate and the terminal value but did not ask how long comparable programs had taken or what share of them reached commercial viability. A consultant’s market study, commissioned by the program sponsor, projected volumes about double those in a competing analyst’s note. Management’s explanation was that analysts “did not understand the technology.”

The program has now spent $310 million, the launch has slipped two years, and cell durability remains below customer requirements. At the latest review, the program team asked for another $60 million to complete a “final” redesign. They estimated a 30 percent chance that the redesign would meet durability targets, in which case the program’s value at launch would be about $190 million in present value. The team also noted that cancellation would require a $310 million write-off and might trigger a covenant discussion with lenders. Okafor argued that “we have come too far to stop” and that the engineering team would lose its best people if the program was canceled. Two directors privately doubted the program but did not speak, since the CEO was respected and the previous review had passed without objection.

A newly appointed director with a background in private equity asked the finance team to compile a reference class. The finance team found that of eleven similar internal and industry development programs, three reached commercial viability within the original budget and schedule, five reached viability with budgets 50 to 120 percent higher and delays of two or more years, and three were abandoned. The median overrun in those that completed was 80 percent. Meridian’s own last three major programs had finished with an average cost overrun of 45 percent.

Meridian’s CFO has been asked to prepare a recommendation. She is aware that approving the additional $60 million would be consistent with the sponsor’s wishes, that rejecting it would be seen as a rebuke of the CEO, and that either decision will be examined in the light of what happens next. She has proposed a premortem, a review by a team that had no role in the original approval, and a stage-gated plan in which the $60 million would be released in three tranches, each tied to a specific durability test. The CEO has objected that this would “signal a lack of confidence” and delay the redesign by two months.

Discussion Questions

1. Compute the incremental NPV of the requested $60 million, treating the $310 million as sunk. What probability of success would make the request break even? How might the stage-gated structure change the economics (describe qualitatively, referring to real options)?
2. Identify the biases visible in the original approval and in the current request. For each, state the evidence in the case and the process change that would address it.
3. How should the reference-class data be used? Suggest how a board could translate the data into an adjusted forecast for the original business case, and what that adjustment implies about the quality of the original approval.
4. The CEO objects to the premortem and the staged release. What are the organizational and incentive issues? How would you design the decision rights and incentives so that a director who doubts the program can say so, and so that the CEO is not penalized for stopping?

Key Takeaways

- Behavioral corporate finance has two branches: rational managers exploiting irrational markets (timing, catering) and biased managers operating in rational markets (overconfidence, anchoring, escalation). Advice differs by branch.
- Biases are systematic, not random. Because they push in a consistent direction, they can be measured, modeled and corrected, which is the purpose of the quantitative examples in this chapter.
- Selecting the projects with the highest estimated NPV selects the largest forecast errors. Estimates should be shrunk toward the mean by a coefficient based on the firm's measured forecast error (Example 12.1).
- Overprecision is the most robust form of overconfidence: stated ranges are too narrow. Calibration tracking and interval widening are practical corrections (Example 12.8).
- Anchors, including the seller's ask, the budget, or last year's number, pull estimates toward themselves. A walk-away price set in advance is the best defense in negotiations (Example 12.2).
- Sunk costs are irrelevant to the decision; the incremental NPV decides. Loss framing and self-justification pull managers toward escalation, so decision rights and kill criteria should be set in advance (Example 12.3).
- Evaluating risky positive-NPV projects one at a time under loss aversion rejects value that the portfolio would accept. Aggregate decisions and judge managers on process as well as outcomes (Example 12.4).
- Market timing transfers wealth between shareholder groups without creating firm value, and it depends on the manager's judgment that the market is wrong, which is itself subject to overconfidence (Example 12.5).
- Present bias and earnings-based incentives make small, long-payoff projects fragile. Multi-year incentives and long-horizon metrics reduce myopia (Example 12.6).
- The planning fallacy is corrected by the outside view: apply the reference class distribution of cost, schedule and benefit outcomes to the project, and put risk in the discount rate and bias in the cash flows (Example 12.7).
- Process beats exhortation. Premortems, red teams, independent estimates, decision journals, stage gates and separation of advocate and decision-maker make decisions robust to individual bias, at proportionate cost.

Review Questions

1. Distinguish the two approaches in behavioral corporate finance. Give one financing decision and one investment decision that each approach would analyze.
2. Define overestimation, overplacement and overprecision. Which is most relevant to valuation, and why?
3. Explain why selecting the highest-NPV projects produces a winner's curse even when forecasters are unbiased. What does the shrinkage coefficient depend on?
4. Describe Malmendier and Tate's measure of CEO overconfidence and what it implies for acquisitions and financing.
5. Explain the anchoring-and-adjustment model. How would you design a negotiation process to limit the effect of the counterparty's opening number?
6. Why is the sunk-cost fallacy a fallacy? Give two psychological reasons that managers continue to fund failing projects.
7. What is narrow framing, and how does it cause a loss-averse manager to reject positive-NPV projects? How would you fix it?
8. Explain Stein's argument that investment decisions should be based on fundamental value while financing decisions respond to market price. What are the risks of acting on a perceived mispricing?

9. What is the planning fallacy, and how does reference-class forecasting address it? Why is a uniformly higher hurdle rate an inferior fix?
10. Describe a premortem, why prospective hindsight helps, and how to ensure the exercise yields action rather than ritual.

Problems

Problem 12.1. True project NPVs are distributed $N(0, 15^2)$ and forecast errors $N(0, 25^2)$, independent. (a) Compute the shrinkage coefficient and the expected true NPV for a project estimated at +$36 million. (b) A team believes its error SD is only 10. What expected true NPV does it infer, and by how much does it overstate?

Problem 12.2. A bidder's DCF value for a target is $120 per share, synergies are $15 per share, and the seller anchors at $160. The bidder's bid is $0.4 \times 160 + 0.6 \times 120$. (a) Compute the bid. (b) Compare with the walk-away price and compute the overpayment on 50 million shares. (c) At what anchoring weight does the bid equal the walk-away price?

Problem 12.3. A project has $80 million already spent. Completing it costs $25 million more, and there is a 40% chance of a $55 million payoff (present value) and a 60% chance of zero. (a) Compute the incremental NPV. (b) What probability of success breaks even? (c) What is the incremental NPV if the payoff is $70 million?

Problem 12.4. A firm has 200 million shares with an intrinsic value of $25 per share and a market price of $35. It issues 20 million shares at $35 and keeps the cash. (a) Compute the new intrinsic value per share. (b) Compute the gain to existing holders and the loss to new holders. Then, in a different scenario with price $20 and intrinsic value $25, the firm repurchases 15 million shares. (c) Compute the intrinsic value per share afterward and the gain to the remaining holders.

Problem 12.5. A project costs $50 million at $t = 0$ and $30 million at $t = 1$ and returns $28 million per year for seven years starting at $t = 2$. The discount rate is 12%. (a) Compute the inside-view NPV. (b) Apply a 40% cost overrun, a one-year delay and 15% lower inflows ($23.8 million per year, $t = 3$ to $t = 9$). Compute the outside-view NPV. (c) Compute the annual inflow required to break even in the adjusted case.

Problem 12.6. A project pays +$40 million with probability 0.5 and −$25 million with probability 0.5. A manager has loss aversion $\lambda = 2$. (a) Compute the expected value and the manager's perceived value. (b) If eight independent copies are available, compute the number of successes needed for a positive total and the probability of this. (c) Compute the expected total.

Problem 12.7. An R&D project costs $8 million now and returns $12 million in three years. The cost of capital is 9%. A manager has $\beta = 0.7$ and $\delta = 1/1.09$. (a) Compute the NPV to the firm. (b) Compute the perceived value to the manager. (c) Compute the critical β.

Problem 12.8. A manager makes 25 forecasts of costs, each with a 90% confidence interval. Fifteen actuals fall inside the intervals. (a) Compute the hit rate. (b) Compute the probability of observing 15 or fewer hits if the intervals were truly 90%. (c) By what factor should the intervals be widened, assuming normal errors?

Answers to Selected Problems

12.1. (a) $k = 225 / (225 + 625) = 0.2647$; expected true NPV $= 0.2647 \times 36 = \$9.53$ million. (b) $k = 225 / (225 + 100) = 0.6923$; inferred = $24.92 million; overstatement = $15.39 million, about 2.6 times the correct value.

12.2. (a) $0.4 \times 160 + 0.6 \times 120 = 64 + 72 = \$136$. (b) Walk-away $= 120 + 15 = \$135$; overpayment $\$1 \times 50$ million $= \$50$ million. (c) $w^* = (135 - 120) / (160 - 120) = 0.375$.

12.3. (a) 0.4 × 55 − 25 = 22 − 25 = −$3.0 million (stop). (b) p* = 25 / 55 = 45.5%. (c) 0.4 × 70 − 25 = +$3.0 million (continue).

12.4. (a) (200 × 25 + 20 × 35) / 220 = 5,700 / 220 = $25.91. (b) Gain per share 0.909 × 200 = $181.8 million; new holders paid $700 million for a claim worth 20 × 25.909 = $518.2 million, a loss of $181.8 million. (c) (5,000 − 15 × 20) / 185 = 4,700 / 185 = $25.41; gain = 0.405 × 185 = $75.0 million, which equals the sellers' loss of 15 × 5 = $75 million.

12.5. (a) PV of costs = 50 + 30/1.12 = 76.79; inflows: 28 × annuity factor for 7 years at 12% (4.5638) = 127.79 at t = 1, so PV = 127.79 / 1.12 = 114.10; NPV ≈ +$37.31 million. (b) Costs: 70 + 42/1.12 = 107.50; inflows 23.8 × 4.5638 = 108.62 at t = 2, PV = 108.62 / 1.12^2 = 86.59; NPV ≈ −$20.91 million. (c) Required inflow = 107.50 / (4.5638 / 1.2544) = 107.50 / 3.6382 = $29.55 million per year, above the original $28 million.

12.6. (a) E = 0.5 × 40 − 0.5 × 25 = +$7.5 million; perceived = 20 − 2 × 12.5 = −$5.0 million (reject). (b) Total = 40k − 25(8 − k) = 65k − 200 > 0 requires k ≥ 4; P(k ≥ 4) = 163/256 = 63.7%. (c) 8 × 7.5 = $60 million.

12.7. (a) PV = 12 / 1.09^3 = 12 / 1.2950 = $9.27 million; NPV = +$1.27 million. (b) 0.7 × 9.27 = $6.49 million, below the $8 million cost (reject). (c) β* = 8 / 9.27 = 0.863.

12.8. (a) 15 / 25 = 60%. (b) P(X ≤ 15 | n = 25, p = 0.9) ≈ 0.00008 (about 8 in 100,000). (c) The 90% interval is ± 1.645σ_s; a 60% hit rate corresponds to ± 0.8416σ_a; ratio 1.645 / 0.8416 = 1.95, so widen intervals by about 95%.

Part IV — Private Markets and Real Assets

Chapter 13: Venture Capital in Depth

Learning Objectives

After studying this chapter you should be able to:

- Describe the venture capital fund lifecycle (fundraising, investment period, harvest, wind-down) and explain how fund structure shapes investor behavior and the founders who take the money.
- Explain why venture returns follow a power-law distribution, simulate a portfolio, and derive the implications for fund size, concentration, and reserves.
- Apply the VC method of valuation, adjust it for dilution and failure risk, and recognize why headline discount rates of 30-60% are not "required returns" in the CAPM sense.
- Calculate payoffs under non-participating, participating, and capped-participating liquidation preferences, and value preferred stock as a bundle of options using breakpoint analysis.
- Build and update a capitalization table through seed, Series A, and Series B rounds, including the option-pool shuffle, and compute the effect of full-ratchet and weighted-average anti-dilution protection.
- Evaluate venture debt (cost, warrants, covenants) and secondary transactions (pricing, discounts, tender offers) from both the company's and the investor's perspective.

Opening Vignette

(Illustrative company; all names and figures are fictional.)

Maya Okafor and Dev Ranganathan founded Lumen Freight, a software company that matches under-used truck capacity with small shippers. After eighteen months of living on savings and a small angel check, they sat down in a conference room with three term sheets for a $12 million Series A.

The first, from a large, prestigious fund, offered a $48 million pre-money valuation, a 1x non-participating preferred, and a board of two investors, two founders, and one independent. The second, from a newer fund, offered $60 million pre-money but with a 1.5x participating preferred, a full-ratchet anti-dilution clause, and a redemption right after five years. The third, from a corporate venture arm of a logistics conglomerate, offered $55 million but wanted a right of first refusal on any sale of the company.

Dev wanted the $60 million. "Higher valuation means less dilution, and dilution is the whole game," he argued. Maya had spent the weekend building a spreadsheet. At an exit of $150 million, she showed, the second term sheet left the founders and employees with less money than the first. At $400 million the two were close. Only above roughly $450 million did the higher headline valuation start to help the founders, and Lumen's own plan put the most likely outcomes well below that. Worse, if the next round happened at a lower price, the full ratchet would hand the second investor a large share of the company.

"So the price per share is not the price," Dev said slowly.

"The price is the whole bundle," Maya replied. "A venture term sheet is not a number. It is a set of options, and we are the ones writing them."

This chapter explains how to read that bundle: how venture funds make money, why outcomes are so skewed, how preferred stock is built and valued, how cap tables evolve, and what other forms of capital (venture debt, secondaries) do to the equation. It builds on option pricing (Volume 1, Chapter 18) and company valuation (Volume 1, Chapter 21).

13.1 The Venture Capital Model: Funds, Lifecycle, and Incentives

13.1.1 What venture capital is and is not

Venture capital (VC) is equity (and equity-linked) financing of young, high-growth companies whose value lies mainly in future growth options rather than in current cash flows. Such companies are typically unprofitable, have few tangible assets to pledge, and face outcomes that range from total loss to extraordinary success. Debt is a poor fit because lenders cap their upside at the interest rate while still bearing downside; the Modigliani-Miller logic of Volume 1, Chapter 15 reminds us that when bankruptcy and agency costs are high and cash flows are nonexistent, equity is the only sensible instrument. Public markets are a poor fit too because disclosure, compliance, and short-term scrutiny are costly for a company still searching for its business model.

VC is therefore a specialized solution to three problems at once: *information asymmetry* (founders know far more than investors), *uncertainty* (even the founders do not know whether the product will work), and *agency* (founders can take actions that benefit themselves at investors' expense, such as burning cash on a vanity project or stalling a profitable sale). Almost every feature of venture contracting, from staged financing to board control to liquidation preferences, is a response to one of these three problems.

It helps to distinguish the stages, though the labels are conventions rather than rules:

Stage	Typical purpose	Typical investors	Main risk
Pre-seed / seed	Build a prototype, find early users	Angels, micro-funds, accelerators	Does anyone want this?
Series A	Prove repeatable demand, build the core team	Early-stage VC funds	Can the model be repeated and scaled?
Series B / C	Scale sales, expand markets	Growth-oriented VC funds	Unit economics and execution
Late stage / growth	Prepare for IPO or acquisition	Growth funds, crossover investors, sovereign funds	Market conditions, exit timing

13.1.2 The limited partnership structure

A venture fund is usually organized as a limited partnership with two kinds of partners. *Limited partners* (LPs) are institutions such as pension funds, endowments, foundations, sovereign wealth funds, and wealthy families. They supply nearly all the capital but have limited liability and no say in individual investments. The *general partner* (GP) is the management company that sources, selects, governs, and exits the investments. The GP typically commits a small fraction of the fund (often 1-3%) from its own pocket.

The LP does not hand over the money on day one. Instead, LPs make a *commitment* and the GP issues *capital calls* (drawdowns) as investments are made and fees are due. This matters for returns: the LP has uncalled commitments that must be kept liquid, but is credited with returns only on capital actually drawn (Chapter 14 returns to the consequences for IRR). Distributions flow back when portfolio companies are sold or go public.

13.1.3 The fund lifecycle

A typical venture fund lives about ten years, with extensions of one or two years common. The life divides into four phases.

1. **Fundraising (months 0-12+).** The GP markets the fund to LPs using the track record of previous funds. A first-time fund has no such record and relies on the team's personal history. Closes occur in stages (first close, final close).
2. **Investment period (years 1-5).** The GP identifies and funds new companies, usually 20-40 for a diversified early-stage fund, and holds back *reserves* for follow-on rounds in the winners. A common rule of thumb is to keep 40-60% of the fund for follow-ons, though this varies widely by strategy.
3. **Harvest (years 4-10).** Portfolio companies grow, raise further rounds, and are eventually acquired, go public, or fail. Distributions begin, usually from the early winners.
4. **Wind-down (years 10-12+).** Remaining stakes are sold, sometimes into the secondary market (Section 13.7), and the partnership is dissolved.

Because the GP raises a new fund every two to four years once the old one is substantially invested, a firm is typically managing several overlapping funds. The new fund's fundraising depends on the *marks* (reported valuations) of the old one, which creates an incentive problem we discuss in Chapter 14.

13.1.4 Why the structure creates specific behaviors

The structure has consequences that every founder and every investor should understand:

- **Time horizon.** A fund with a ten-year life needs an exit for each investment within roughly seven to nine years. A founder who would happily run a profitable $30 million business for 20 years is a poor fit for a fund that needs a $500 million exit to matter.
- **Return threshold.** A fund investor in a $300 million fund needs, at a minimum, to return the fund through a handful of outcomes. A $50 million sale that is wonderful for the founder may be economically irrelevant to the fund.
- **Option mentality.** The GP shares in gains (carried interest) but not in losses beyond its small commitment. This asymmetry encourages risk-taking, which is precisely what LPs want from the venture allocation, but it also encourages "swinging for the fences" in situations where a sure modest outcome would be better for founders and employees.
- **Staged commitment.** Money is delivered in rounds tied to milestones. Each round is a real option for the investor to continue or stop (see Volume 1, Chapter 8 on real options), and also a moment of renegotiation power.

Common mistake. Treating all investors as interchangeable "money." Two investors who pay the same price per share can have very different fund sizes, ownership targets, reserve policies, and timelines, and therefore very different preferences about whether to accept an acquisition offer in year four.

13.1.5 What GPs actually get paid

LPs typically pay an annual management fee (commonly around 2% of commitments during the investment period, stepping down afterward) and *carried interest* (commonly 20% of profits after returning capital, sometimes with a preferred return). In venture, carry is frequently paid without a hurdle rate on a whole-fund basis, since expected returns are skewed enough that a hurdle rarely binds. The details, and their effect on LP net returns, are the subject of Chapter 14. For now, notice that the management fee supports the firm regardless of performance, whereas carry is a call option on the fund's profits, with strike equal to the capital contributed.

13.2 Power-Law Returns and Portfolio Construction

13.2.1 The shape of venture outcomes

If you invest in a diversified portfolio of large listed stocks, returns are approximately symmetric and bounded below by -100%. Venture outcomes are different. A seed-stage company is a bet on a distribution with a very large mass at or near zero and a thin right tail of enormous outcomes. Empirically, in most venture portfolios a small fraction of companies generate the large majority of total value. A distribution of this kind is often described as a *power law*: the probability of an outcome of size x declines roughly as x raised to a negative power, so extremely large values are far more likely than a normal or lognormal curve with the same mean would suggest.

You do not need the technical definition to see the practical point. Define the **multiple of invested capital (MOIC)** for company *i* as exit proceeds attributable to the fund divided by dollars invested:

```
MOIC_i = Proceeds_i / Invested_i
Fund MOIC = Σ Proceeds_i / Σ Invested_i
```

If a fund invests equal amounts in 25 companies, the fund MOIC is the simple average of the 25 company MOICs. The average of a skewed distribution is dominated by a few terms, and, crucially, the *typical* fund draws a sample of the distribution that may not contain any of the rare extreme draws at all.

13.2.2 A simulated portfolio

Example 13.1 – Simulating a power-law portfolio. Suppose an illustrative early-stage fund invests an equal amount in each of *n* companies. For each company, the gross multiple (proceeds divided by cost) is drawn independently from this distribution:

Outcome bucket	Multiple on cost	Probability	Contribution to expected multiple	Share of expected value
Write-off	0.0x	45.0%	0.000	0.0%
Partial loss	0.5x	30.0%	0.150	9.0%
Modest	2x	15.0%	0.300	18.0%
Good	6x	7.0%	0.420	25.1%
Great	20x	2.5%	0.500	29.9%
Outlier	60x	0.5%	0.300	18.0%
Expected multiple per company		**100%**	**1.670**	**100%**

Step 1: expected multiple per company. 0.45(0) + 0.30(0.5) + 0.15(2) + 0.07(6) + 0.025(20) + 0.005(60) = 0 + 0.15 + 0.30 + 0.42 + 0.50 + 0.30 = **1.67x**.

Step 2: what drives it. Three percent of companies (the 20x and 60x buckets) supply 0.50 + 0.30 = 0.80 of the 1.67, or **47.9%** of the expected value. Three-quarters of companies (the write-off and partial-loss buckets) return less than cost.

Step 3: simulate funds. Using a Monte Carlo simulation (a fixed random seed, 100,000 simulated funds for each portfolio size), we obtain:

Companies in fund	10th percentile fund MOIC	Median	90th percentile	Mean	P(MOIC < 1.0x)	P(MOIC ≥ 3.0x)
10	0.35x	1.10x	3.30x	1.67x	44.7%	13.4%
25	0.64x	1.40x	3.22x	1.67x	30.2%	12.6%
50	0.85x	1.51x	2.70x	1.67x	17.3%	6.0%
100	1.05x	1.60x	2.40x	1.67x	7.6%	1.8%

Step 4: read the table. The mean is 1.67x for every portfolio size, as it must be: diversification does not change expected return. What changes is the spread. A 10-company fund has a 45% chance of failing to return capital even though the underlying strategy has a 1.67x expected multiple. The *median* outcome is below the mean in every row, and for small portfolios it is well below. A 100-company fund has almost no chance of a spectacular result, because the extreme tail is averaged out.

Step 5: the lottery-ticket probability. The chance that a given company is a 20x-or-better outcome is 3.0%. In a 25-company fund, the probability of holding **no** such company is 0.97^25 = **46.7%**. Nearly half of such funds will miss the right tail entirely, and those funds will likely return less than about 1.0-1.2x.

Conclusion. The strategy has a decent expected return, a mediocre median, and a severe dependence on whether the GP's sample happens to contain an outlier. The simulation assumes independence and that the GP cannot predict winners; skill is the claim that a particular GP can shift these probabilities, for instance by raising the chance of a 20x outcome from 3.0% to 4.0%. That one-point shift raises the expected multiple by roughly 0.2x, which is the whole difference between a good and a mediocre fund. ∎

13.2.3 Implications of the power law

Several practical rules follow from the shape of the distribution.

(1) "Return the fund" math. A fund of size F that wants a 3x gross multiple must generate proceeds of 3F. If the fund owns 10% of a company at exit, the exit equity value needed for that single position to return the entire fund is F / 0.10 = 10F. A $300 million fund with 10% ownership at exit needs a $3 billion outcome from one company just to give investors their money back. This arithmetic explains why large funds pass on "good" $200 million exits, why ownership targets exist (a fund wants 10-20% at entry so that dilution still leaves a meaningful stake), and why a fund's size constrains what it can profitably do.

(2) Concentration versus diversification. Standard portfolio theory (Volume 1, Chapter 9) says to diversify idiosyncratic risk. In venture the right tail is the *source* of return, so diversification beyond a point sacrifices the very thing that makes the strategy work. The table in Example 13.1 shows the tradeoff: diversifying from 25 to 100 companies cuts the failure probability from 30% to under 8% but also cuts the chance of a 3x fund from 12.6% to 1.8%. The economically relevant question for the GP is how many "shots" are needed so that, given its sourcing and selection skill, at least a few land in the right tail.

(3) Reserves. Because winners become identifiable only after a few rounds, funds hold reserves to double down. The value of reserves is the value of an option to buy more of companies that have revealed themselves as winners at a price that still leaves attractive returns. But the follow-on price is higher, so the follow-on return is lower than the initial return; funds that over-reserve to mediocre companies, or under-reserve to the best, both lose.

(4) Fees bite harder on small outcomes. If the median fund returns about 1.4x gross, fees and carry consume a large fraction of the profit (Chapter 14 quantifies this). That is why LPs focus on top-quartile managers: the dispersion between top and bottom quartile VC funds is far wider than in most other asset classes, so manager selection matters more than asset-class allocation.

(5) Founders' view. A founder should understand that the investor's rational behavior is to *encourage* risk. Rejecting a $60 million acquisition offer in order to try for $600 million makes sense to a fund holding 10% of an otherwise-valuable portfolio of 25 companies; it may make less sense to a founder who owns 20% and has no other portfolio.

Common mistake. Judging a venture fund by its failures. In a power-law world, a fund with half its companies returning zero can be spectacularly successful; and a fund with a near-perfect "hit rate" of modest 2x exits can be a failure because it has no right tail.

13.3 Valuing a Startup: The VC Method and Its Limits

13.3.1 Why standard DCF struggles

Volume 1, Chapter 21 valued enterprises by forecasting free cash flows and discounting at WACC. For a startup, near-term free cash flows are negative, terminal value is almost all of the value, the growth path is unknowable, and the beta needed for CAPM does not exist for a company with no traded equity and no stable business. A DCF can be built, but its output is a function of assumptions the analyst cannot verify. Practitioners therefore use more transparent frameworks, of which the best known is the *venture capital method*, usually attributed in its modern form to William Sahlman and colleagues at Harvard Business School.

13.3.2 The VC method step by step

The method works backward from an estimated exit:

1. Forecast the company's value (enterprise or equity) at the exit date T, typically using a revenue or earnings multiple of comparable companies.
2. Choose a *target rate of return* r and compute the required multiple on investment: (1 + r)^T.
3. Compute the **post-money valuation today** that would deliver that multiple: PostMoney = ExitValue / (1 + r)^T, ignoring dilution.
4. Compute the required **ownership at exit**: Own_exit = Investment × (1 + r)^T / ExitValue.
5. Adjust for future dilution: if the investor will retain a fraction *k* of its stake after subsequent financings and option grants, the required ownership *today* is Own_today = Own_exit / k.
6. The *pre-money valuation* is the post-money minus the investment.

Recall the definitions that matter throughout this chapter:

```
Post-money = Pre-money + New investment
Investor ownership (immediately after) = Investment / Post-money
Price per share = Pre-money / Fully diluted shares outstanding before the round
```

where "fully diluted" counts all shares, options (granted and ungranted pool), warrants, and convertible securities treated as converted.

13.3.3 A worked example

Example 13.2 – The VC method with dilution. An investor considers a $15 million Series A in a software company. The investor's base case: in five years the company has revenue of $120 million, and a comparable-company multiple of 5.0x revenue gives an exit value of $600 million. The fund's target return is 40% per year. The investor expects to retain 70% of its ownership after future rounds and option-pool refreshes.

Step 1: exit value. 120 × 5.0 = **$600 million**.

Step 2: required multiple. (1.40)^5 = **5.378x**.

Step 3: required value of the stake at exit. $15M × 5.378 = **$80.67 million**.

Step 4: required ownership at exit. 80.67 / 600 = **13.45%**.

Step 5: adjust for dilution. Own_today = 13.45% / 0.70 = **19.21%**.

Step 6: valuations. Post-money = 15 / 0.1921 = **$78.09 million**; pre-money = 78.09 − 15 = **$63.09 million**.

Comparison with the no-dilution answer. Ignoring dilution, Post-money = 600 / 5.378 = $111.56 million and pre-money = $96.56 million. The dilution adjustment lowers the supportable pre-money by about **$33.5 million (35%)**. A founder who ignores future dilution will systematically overestimate what a disciplined investor can pay.

Conclusion. Under these assumptions the investor can afford a pre-money of about $63 million, not $97 million. The negotiation will be between roughly these bounds, with the founder arguing for higher exit values or lower retention and the investor arguing the opposite. ∎

13.3.4 What the 40% really means

A 40% target return is far above any CAPM-style cost of capital for a company, even with a high beta. Does this mean venture investors are earning extraordinary risk-adjusted returns, or that they are miscalibrated? Neither, usually. The high target is a rule-of-thumb *correction for three things*:

1. **Survival bias in the forecast.** The base-case exit of $600 million is the outcome conditional on success; the unconditional expected value is lower. The target return partly compensates for the probability that the company fails.
2. **Illiquidity and effort.** The investor locks up capital for years and spends significant time on board and portfolio support.
3. **Optimism in management projections.** Founders' forecasts are typically optimistic, and discounting at a high rate is a crude haircut.

The method is more defensible if these effects are separated. A scenario-based approach does so explicitly. Continue the previous example with four scenarios for the exit value in year 5:

Scenario	Probability	Exit value ($M)	Probability × value ($M)
Breakout	15%	900	135.0
Solid success	25%	400	100.0
Modest sale	20%	100	20.0
Failure	40%	0	0.0
Expected exit value	**100%**		**255.0**

The expected exit value is **$255 million**, not $600 million. Discount at a more moderate rate reflecting risk and illiquidity, say 25%, with the same 70% retention:

```
Value of 100% of company, post-money = 255 × 0.70 / (1.25)^5
                                     = 178.5 / 3.0518
                                     = $58.5 million
```

So the scenario approach supports a **post-money of about $58.5 million** (pre-money about $43.5 million), considerably lower than the $78.1 million from the single-scenario method at 40%. The gap shows that the "40%" in the simple method was not conservative enough here: the base case of $600 million was a 15%-probability-style outcome rather than the mean, and the 40% hurdle only partly compensated. Different assumptions can move the answer in either direction, which is exactly the point: the method is a framework for organizing disagreement, not a source of precise values.

Managerial implications. Founders should come to a negotiation with their own scenario table. Investors respond more to credible downside analysis than to a hockey stick. The valuation conversation is also a conversation about dilution, so the founder should model the entire future financing path (Section 13.5), not just the current round.

Common mistakes. (i) Treating the target return as a cost of capital. (ii) Forgetting that the exit multiple itself is uncertain and cyclical: a company worth 5x revenue in a hot market may be worth 2x in a cold one. (iii) Applying the retention factor to the wrong base: retention should reflect *expected* future rounds, which are themselves correlated with success (successful companies raise at higher prices and dilute less).

13.4 Preferred Stock: A Bundle of Options

13.4.1 Why venture investors buy preferred, not common

Founders and employees hold *common stock*. Venture investors almost always buy *convertible preferred stock*, a security that gives them (a) a senior claim on proceeds up to some amount, and (b) the right to convert into common if that is worth more. The rationale is the combination of problems from Section 13.1: preferred protects investors in the downside, where most outcomes live, while retaining the upside through conversion. It also provides a natural valuation wedge: *the price per share that the investor pays is the price of the preferred, which is worth more than the common held by founders*. When a startup announces a "$100 million valuation," that figure is the preferred price multiplied by all shares, including common. The common stock is worth less than the headline; we quantify how much in Section 13.4.4.

The key economic terms are:

- **Liquidation preference multiple** (1x, 1.5x, 2x…): the amount per share paid before common receives anything, in a liquidation or sale.
- **Participation:** whether, after receiving the preference, the preferred also shares in the remainder as if converted ("participating"), or must choose between the preference and conversion ("non-participating").
- **Participation cap:** a limit on total proceeds under participation, often a multiple of the original investment (for example 3x), above which the holder converts to common.
- **Seniority:** *standard* (later series is paid before earlier series), *pari passu* (all series share pro rata in proportion to their preferences), or occasionally *stacked* in some other order.
- **Dividends:** cumulative or non-cumulative accrual that increases the preference over time, or dividends payable only if declared.

- **Redemption rights:** the right to require the company to buy back the shares after a stated period; seldom exercised, but a source of leverage.

13.4.2 Liquidation preference mathematics

Define for a single series of preferred: investment I, as-converted ownership s (the fraction of fully diluted shares the preferred would own if converted), and exit equity proceeds X after debt and costs. The payoffs to the preferred holder are:

```
Non-participating, multiple m:   P(X) = max( min(X, m·I), s·X )
Participating (uncapped):        P(X) = min(X, m·I) + s·max(X − m·I, 0)
Participating, capped at C:      P(X) = max( min( min(X, m·I) + s·max(X − m·I, 0), C ), s·X )
```

The common stockholders receive X − P(X). Notice the structure: the non-participating preferred is a debt-like claim of m·I plus a conversion option; the participating preferred keeps the debt-like claim *and* takes its pro-rata share of what is left, so it never needs to convert unless a cap binds.

Example 13.3 – One investment, three preference structures. A venture fund invests I = $10 million for s = 25% of a company (post-money $40 million). The common stockholders (founders, employees) hold the remaining 75%. The multiple is m = 1. Compare three structures at exit equity values X from $5 million to $200 million: (A) 1x non-participating; (B) 1x participating, uncapped; (C) 1x participating with a 3x cap ($30 million total).

Step 1: breakpoints.

- A. The investor converts when s·X > I, i.e. 0.25X > 10, so X > **$40 million**.
- B. No conversion is ever optimal; the investor always receives $10 million plus 25% of the excess.
- C. The participating payoff reaches the cap when 10 + 0.25(X − 10) = 30, i.e. 0.25(X − 10) = 20, X = **$90 million**. Above $90 million the investor stays at $30 million until conversion yields more: 0.25X > 30, X > **$120 million**.

Step 2: payoffs.

Exit value X ($M)	A: investor	A: common	B: investor	B: common	C: investor	C: common
5	5.0	0.0	5.0	0.0	5.0	0.0
10	10.0	0.0	10.0	0.0	10.0	0.0
20	10.0	10.0	12.5	7.5	12.5	7.5
40	10.0	30.0	17.5	22.5	17.5	22.5
60	15.0	45.0	22.5	37.5	22.5	37.5
100	25.0	75.0	32.5	67.5	30.0	70.0
120	30.0	90.0	37.5	82.5	30.0	90.0
200	50.0	150.0	57.5	142.5	50.0	150.0

Step 3: sample calculation, X = $60 million. (A): 0.25 × 60 = 15 > 10, so converts: investor $15.0M, common $45.0M. (B): 10 + 0.25(60 − 10) = 10 + 12.5 = $22.5M; common 60 − 22.5 = $37.5M. (C): 22.5 is below the cap of 30 and above 0.25 × 60 = 15, so $22.5M.

Step 4: cost of participation to the common. At X = \$100 million the common loses \$7.5 million under (B) relative to (A), exactly 0.75 × \$10 million: participation transfers to the investor the preference amount times the common's share. This is a constant transfer for all X above the conversion point, which is why participation is "expensive" at every exit size. At X = \$200 million the difference is still \$7.5 million (142.5 vs 150).

Conclusion. The preference matters most at *moderate* exits, where most real-world outcomes cluster. A cap on participation (C) removes the extra transfer at high values: at X ≥ \$120 million, (C) and (A) are identical. A founder should read a "1x participating" term sheet as an extra \$7.5 million tax on exit in this deal, however attractive the headline valuation. ∎

13.4.3 Stacked preferences and the multi-round problem

With multiple rounds, seniority and the aggregate preference stack matter. Suppose total preferences across all series equal \$60 million. For exits below \$60 million the common receives nothing under *any* structure, and the order of payment between series determines who bears the loss. Under *standard seniority* the latest investors are repaid first, which protects the newest, highest-priced money; under *pari passu* all series share losses proportionally. Founders typically prefer pari passu because it reduces the incentive of late investors to demand a sale at a moderately high price that pays their own preference but wipes out the earlier holders and common. This is the central conflict: with a large preference stack, different classes want different exit thresholds, and the *effective* alignment among shareholders is lower than the cap table percentages suggest.

A helpful diagnostic is to build an *exit waterfall chart*: for each exit value from zero to a high upper bound, list who gets what. The kinks in the chart (the breakpoints, such as \$40 million and \$90-120 million in Example 13.3) are the values at which incentives change. Management teams should know these breakpoints cold, because they determine which acquisition offers are worth taking from whose point of view.

Management carve-outs. In a sale below the preference stack, common stockholders, including the executives who must run the transition, may receive nothing, creating an incentive to sabotage or abandon a deal. A *carve-out plan* reserves a share of proceeds (often 5-15%) for management and employees before the preferred is paid. It is a classic fix for misaligned incentives.

13.4.4 Valuing preferred and common with option-pricing

The liquidation payoffs are piecewise-linear functions of firm equity value X, which is the signature of an option. Following Black-Scholes (Volume 1, Chapter 18) and, in practice, the *option pricing method* (OPM) used in 409A and purchase-price-allocation valuations, we can value each class by treating each as a combination of call options on total equity value, where the strike prices are the *breakpoints* at which payoff slopes change.

Take Example 13.3, structure (A). The payoffs are:

- Preferred: min(X, 10) for X ≤ 40, then 0.25X above 40. Slope 1 on [0, 10], slope 0 on [10, 40], slope 0.25 beyond 40.
- Common: 0 on [0, 10], slope 1 on [10, 40], slope 0.75 beyond 40.

Writing C(K) for the value of a European call on total equity with strike K:

```
Preferred = [C(0) − C(10)] + 0.25 · C(40)
Common    = C(10) − 0.25 · C(40)
```

The first bracket is a dollar-for-dollar claim on the first \$10 million (a long position in the underlying and a short call at 10); the second term is the conversion option, giving the preferred 25% of the value above \$40 million.

Example 13.4 – OPM valuation of preferred versus common. Equity value today S = $40 million, which is the post-money implied by the investor's price. Assume the exit horizon T = 4 years, a risk-free rate r = 4%, and volatility σ = 70% (typical of an early-stage company). Use the Black-Scholes call formula without dividends:

```
C(K) = S·N(d1) − K·e^(−rT)·N(d2)
d1 = [ln(S/K) + (r + σ²/2)·T] / (σ√T),  d2 = d1 − σ√T
```

Step 1: C(0) = S = **$40.000M**.

Step 2: C(10). ln(40/10) = 1.3863; (r + σ²/2)T = (0.04 + 0.245) × 4 = 1.14; σ√T = 0.70 × 2 = 1.40. d1 = (1.3863 + 1.14)/1.40 = 1.8045; d2 = 0.4045. N(d1) = 0.9644, N(d2) = 0.6571. C(10) = 40(0.9644) − 10 e^(−0.16)(0.6571) = 38.58 − 10(0.8521)(0.6571) = 38.58 − 5.60 = **$32.98M**.

Step 3: C(40). ln(1) = 0; d1 = 1.14/1.40 = 0.8143; d2 = −0.5857. N(d1) = 0.7923, N(d2) = 0.2790. C(40) = 40(0.7923) − 40(0.8521)(0.2790) = 31.69 − 9.51 = **$22.18M**.

Step 4: values.

- Preferred = (40.00 − 32.98) + 0.25(22.18) = 7.02 + 5.55 = **$12.57M**, or **31.4%** of equity value.
- Common = 32.98 − 0.25(22.18) = 32.98 − 5.55 = **$27.43M**, or **68.6%** of equity value.

Step 5: per-share comparison. The preferred represents 25% of the shares but has value of 31.4% of equity. The common is 75% of the shares and is worth 68.6%. Value per 1% of shares: preferred 12.57/25 = $0.503M; common 27.43/75 = $0.366M. The ratio is **0.728**: each common share is worth about **73%** of a preferred share, a 27% *discount*, even though the two classes have the same voting and cash-flow rights above the $40 million breakpoint.

Step 6: sensitivity to volatility. Repeating the calculation:

σ	Preferred value ($M)	Common/preferred per-share ratio
40%	12.05	0.773
70%	12.57	0.728
100%	12.10	0.769

Conclusion. The preference gives the preferred roughly $2.6 million of extra value relative to its 25% pro-rata share of $40 million, paid for by the common. The value is not monotonic in volatility: preferred holders are short a call at 10 (which hurts them when volatility rises) and long a call at 40 (which helps them). This is an important lesson: the headline "post-money valuation" overstates the value of the founders' stock, and the discount depends on the capital structure rather than on any judgment about the company. In practice this is why a company's 409A common valuation, used to set option strike prices, comes in well below the preferred round price. ∎

Common mistakes. (i) Equating the preferred price with the common price when valuing employee stock options. (ii) Ignoring that option-based values are sensitive to the assumed time to exit and volatility, both of which are judgments. (iii) Believing a higher valuation always benefits founders: an inflated price paired with aggressive terms can reduce what the common receives at plausible exit values.

13.4.5 Control and governance terms

Economic terms are only half of the bundle. Investors also negotiate control rights that function as additional options:

- **Board composition.** The seat count and who chooses the independent director. A board majority controls hiring and firing of the CEO, the budget, and the decision to sell.
- **Protective provisions (veto rights).** Investors must approve certain actions: issuing senior securities, selling the company, changing the charter, taking on debt, changing the board size. A veto over a sale is a particularly powerful option: it can hold up an acquisition until the investor's preference is satisfied.
- **Drag-along rights** require minority holders to approve a sale that the board and a required majority approve. **Tag-along** (co-sale) rights allow investors to sell alongside a founder who is selling stock.
- **Pro rata (pre-emptive) rights** let existing investors buy their proportional share of future rounds, preserving ownership and giving them a free option to follow winners.
- **Information rights, registration rights, and founder vesting.** Founder shares usually vest over four years with a one-year cliff, so that a founder who leaves early does not walk away with a large unvested stake.
- **Pay-to-play provisions.** Investors who do not participate in a down round lose some preferred rights (anti-dilution protection or conversion into common). This protects the company from passive investors who benefit from protective terms without providing capital.

Managerial implication. Negotiate in order of *economic impact at plausible exit values*, not in order of how scary the legal language sounds. A full-ratchet provision might never be triggered; a 1.5x participating preference will be triggered in almost every outcome that matters.

13.5 Cap Table Modeling and Anti-Dilution

13.5.1 The capitalization table

A *capitalization table* ("cap table") lists every holder of equity or equity-linked securities, the number of shares or options held, and the percentage of the fully diluted company each represents. It is the central bookkeeping device of venture finance and the first thing a diligent investor will check. Errors in cap tables, such as unrecorded option grants or convertibles that convert on unexpected terms, are among the most common causes of closing delays and post-closing disputes.

Four modeling points deserve emphasis.

(1) Fully diluted shares include the pool. The *option pool* is a block of shares reserved for current and future employees. In pricing, investors count the *entire* pool (granted and ungranted) in the fully diluted share count, because they assume it will be issued.

(2) The option-pool shuffle. Investors commonly insist that the pool be sized to cover hiring for the next 12-24 months and that any top-up be created *before* the round, i.e. included in the pre-money. This is called the "option-pool shuffle" or "pre-money pool". It lowers the *effective* pre-money valuation to the existing holders because the new pool shares are counted in the pre-money share count but belong to future hires, not to the existing holders.

(3) Pre-money versus post-money mechanics. Price per share = Pre-money / pre-round fully diluted shares (including any top-up). New shares = Investment / Price. Post-round ownership = new shares / total post-round shares = Investment / Post-money.

(4) Convertible instruments. Early funding often comes via *convertible notes* (debt that converts into equity at the next priced round at a discount or cap, plus interest) or *SAFEs* (simple agreements for future equity, which convert without interest or maturity). Each includes a *valuation cap* and/or *discount*. In a post-money SAFE, for example, an investment of $500,000 with a $5 million post-money cap converts into 10% of the company *before* the new round's shares and before the new option pool top-up; the conversion price is the lower of the cap price and the discounted round price. The practical lesson: the dilution from convertibles is not apparent until the priced round, so model them before they surprise you.

13.5.2 A three-round cap table

Example 13.5 – From seed to Series B. Two founders hold 8,000,000 shares of common stock. We model three financings.

Seed. A seed investor puts in $1.0 million at a $4.0 million pre-money (so post-money is $5.0 million and the investor owns $1.0M / $5.0M = **20%**). The round also creates an option pool equal to **10%** of the post-money fully diluted shares. The founders therefore own 100% − 20% − 10% = **70%**.

Step 1: total seed-round shares. 8,000,000 / 0.70 = **11,428,571** shares. Seed investor: 0.20 × 11,428,571 = **2,285,714**; pool: 0.10 × 11,428,571 = **1,142,857**. Price per share = $1,000,000 / 2,285,714 = **$0.4375**.

Series A. $8.0 million invested at a $24.0 million pre-money; post-money $32.0 million; Series A investor owns 8/32 = **25%**. The investor requires the pool be topped up to **10% of post-money** fully diluted shares and that the top-up be included in the pre-money.

Step 2: total post-A shares. The founders and seed investor hold 8,000,000 + 2,285,714 = 10,285,714 shares, which must equal 100% − 25% − 10% = 65% of the post-A total. Total = 10,285,714 / 0.65 = **15,824,176** shares. Series A shares = 0.25 × 15,824,176 = **3,956,044**; total pool = 0.10 × 15,824,176 = **1,582,418**, an increase of 1,582,418 − 1,142,857 = **439,561** new pool shares.

Step 3: price. $8,000,000 / 3,956,044 = **$2.0222** per share. Check: pre-round FD shares = 15,824,176 − 3,956,044 = 11,868,132; × $2.0222 = $24.0 million. ✓

Step 4: effect of the pool shuffle. Excluding the new pool shares, the existing holders' shares (founders, seed, and the original pool) number 11,428,571, valued at $2.0222 each = **$23.11 million**. The headline $24.0 million pre-money is effectively **$23.11 million** to existing holders: the pool top-up costs them 439,561 × $2.0222 = **$0.89 million** of headline value (3.7%).

Series B. $20.0 million at an $80.0 million pre-money; post-money $100.0 million; Series B owns **20%**. The pool is not topped up.

Step 5: total post-B shares. Pre-B shares of 15,824,176 represent 80% of the post-B total, so total = 15,824,176 / 0.80 = **19,780,220**. Series B shares = 0.20 × 19,780,220 = **3,956,044**; price = $20,000,000 / 3,956,044 = **$5.0556**.

Holder	After seed: shares	%	After Series A: shares	%	After Series B: shares	%
Founders	8,000,000	70.0%	8,000,000	50.6%	8,000,000	40.4%
Seed investor	2,285,714	20.0%	2,285,714	14.4%	2,285,714	11.6%
Series A			3,956,044	25.0%	3,956,044	20.0%
Series B					3,956,044	20.0%
Option pool	1,142,857	10.0%	1,582,418	10.0%	1,582,418	8.0%
Total	**11,428,571**	**100%**	**15,824,176**	**100%**	**19,780,220**	**100%**
Price per share	$0.4375		$2.0222		$5.0556	

Step 6: values and returns. At the Series B post-money of $100 million, the founders' stake is 40.44% × $100M = **$40.4 million**; the seed investor's stake is 11.56% × $100M = **$11.6 million**, a **11.6x** paper multiple on $1.0 million. The share price has risen from $0.4375 to $5.0556, a factor of 11.56, matching the seed multiple.

Conclusion. The founders sold no shares yet went from 70% to 40% ownership. That is not necessarily bad: the value of their stake rose in this scenario from $3.5 million ($5.0M post × 70%) to $40.4 million. Dilution matters in relation to the price at which it occurs. The cap table also reveals that the pool, which had to be topped up at Series A, was effectively paid for by the founders and seed investor, and that its size drifts downward as later rounds dilute it. ∎

13.5.3 Anti-dilution protection

Because preferred is priced at a given valuation, investors worry about *down rounds*: a later financing at a lower price per share than they paid. **Anti-dilution provisions** adjust the conversion price of earlier preferred downward when new shares are sold below it, so that the earlier investors convert into more common shares. There are three standard forms:

1. **Full ratchet.** The conversion price drops to the new, lower issue price, regardless of how many shares are issued at that price. Harsh: even a tiny down-round issuance resets the entire series.
2. **Broad-based weighted average.** The new conversion price is a weighted average of the old price and the new price, with weights reflecting the amount of new money relative to all outstanding shares (including options and convertibles):

```
CP2 = CP1 × (A + B) / (A + C)
```

where CP1 is the old conversion price; A is the fully diluted shares outstanding before the issue ("broad-based" counts options, warrants, and pool); B is the number of shares that the new money would have bought at CP1 (= new money / CP1); and C is the number of new shares actually issued. 3. **Narrow-based weighted average.** The same formula but A counts only outstanding preferred and common, excluding options and the pool, which produces a larger adjustment.

The adjustment is a transfer, not a creation of value: shares issued to the protected investor reduce the percentages of everyone else, *including the new investor* unless the new investor re-prices to account for it. In practice, new investors insist on a price that reflects post-adjustment dilution, which makes the real cost fall on founders, employees, and unprotected holders.

Example 13.6 – Full ratchet versus weighted average. A company's Series A investors bought 4,000,000 preferred shares at CP1 = **$2.00** (an $8.0 million investment). There are 12,000,000 other fully diluted shares (founders, seed, and the pool), so A = 16,000,000. A new Series B invests **$5.0 million at $1.00** per share, issuing C = 5,000,000 new shares: a 50% down round.

Case 1: no protection. Series A converts 1:1 into 4,000,000 common. Post-round total = 12,000,000 + 4,000,000 + 5,000,000 = 21,000,000. Series A owns 19.05%; Series B 23.81%; other holders 57.14%.

Case 2: full ratchet. CP drops to $1.00. Series A converts into $8,000,000 / $1.00 = **8,000,000** shares. Total = 12,000,000 + 8,000,000 + 5,000,000 = 25,000,000. Series A owns **32.0%**; Series B 20.0%; others **48.0%**.

Case 3: broad-based weighted average. B = $5,000,000 / $2.00 = 2,500,000. CP2 = 2.00 × (16,000,000 + 2,500,000) / (16,000,000 + 5,000,000) = 2.00 × 18,500,000 / 21,000,000 = **$1.7619**. Series A converts into $8,000,000 / 1.7619 = **4,540,541** shares. Total = 12,000,000 + 4,540,541 + 5,000,000 = 21,540,541. Series A owns **21.08%**; Series B 23.21%; others **55.71%**.

Case 4: narrow-based weighted average, with A counting only 14,000,000 shares (excluding a 2,000,000-share pool): CP2 = 2.00 × (14,000,000 + 2,500,000) / (14,000,000 + 5,000,000) = **$1.7368**; Series A converts into 4,606,061 shares and owns **21.32%**.

	Series A converts into	Series A %	Series B %	Other holders %	Founders' % (8M shares)
No protection	4,000,000	19.05%	23.81%	57.14%	38.10%
Broad-based WA	4,540,541	21.08%	23.21%	55.71%	37.14%
Narrow-based WA	4,606,061	21.32%	23.14%	55.54%	37.04%
Full ratchet	8,000,000	32.00%	20.00%	48.00%	32.00%

Conclusion. The weighted-average clauses transfer about 2 points of ownership to Series A (21.1% versus 19.0%), and the full ratchet transfers about 13 points. Founders with 8,000,000 shares fall from 38.1% to 37.1% under the broad-based formula and to 32.0% under full ratchet, a difference of about 5 points, equivalent to roughly 14% of their stake. Note too that Series B's own stake drops from 23.8% to 20.0% under the ratchet; a rational Series B would demand a lower price or more shares to compensate, making the real cost even larger. Full ratchets are rare in healthy markets and become more common when capital is scarce. ∎

Common mistake. Reading "anti-dilution" as protection from *all* dilution. It protects only against *price* dilution in a down round, not against ordinary dilution from new rounds at higher prices or from option grants. Another mistake is ignoring the *carve-outs* (exempt issuances such as option-pool grants) that define when the clause is triggered.

13.6 Venture Debt

13.6.1 What it is and who uses it

Venture debt is a term loan, usually extended by specialized lenders or banks to venture-backed companies that have raised an equity round from institutional investors. The lender looks not to current cash flow or hard collateral but to the *quality of the equity sponsors*, the cash runway, and the likelihood of the next round. Typical features are a three-to-four-year term, an initial interest-only period followed by amortization, a floating or fixed interest rate well above bank rates, an upfront fee, an end-of-term (final payment) fee, and *warrants* to purchase a small amount of equity. Warrant coverage is expressed as a percentage of the loan: 8% coverage on a $5 million loan means warrants with an aggregate strike price of $400,000, usually priced at the latest round's price.

Companies use venture debt to extend runway between equity rounds (so that the next round can be raised at a better valuation), to finance equipment or receivables, to fund an acquisition, or to avoid dilution. The theory is straightforward: if the company will use the money to hit a milestone that raises the valuation of the next round by more than the cost of the

loan, the loan is cheaper than equity. Equity is the most expensive form of capital for a company that expects to succeed, since it surrenders a share of all future value; debt is cheaper when failure is unlikely before the loan is repaid.

13.6.2 The cost of venture debt

The stated interest rate understates the all-in cost. A correct analysis computes the internal rate of return on the lender's cash flows including fees and warrants.

Example 13.7 – All-in cost of a venture loan. A company borrows $5.0 million for 36 months. Terms: interest 11% per year (paid monthly on the outstanding balance), interest-only for the first 12 months and then equal monthly principal payments over the remaining 24 months, a 1% upfront fee ($50,000, deducted from proceeds), and a 4% end-of-term fee ($200,000, paid at month 36). Warrants cover 8% of the loan ($400,000 at a strike of $2.00 per share, so 200,000 shares).

Step 1: cash flows. Month 0: the company receives 5,000,000 − 50,000 = **$4,950,000**. Months 1-12: interest of 5,000,000 × 0.11/12 = **$45,833** per month. Months 13-36: principal of 5,000,000/24 = $208,333 per month plus interest on the declining balance (month 13 interest 45,833, payment = **$254,167**); the month-36 payment is $208,333 + interest of $1,910 + end-of-term fee $200,000 = **$410,243**.

Step 2: totals. Total payments = **$6,322,917**: principal $5,000,000, end-of-term fee $200,000, and interest of **$1,122,917**.

Step 3: cash-only cost. Solving for the monthly rate that equates $4,950,000 to the payment stream gives an annualized (compounded) rate of **13.9%**. The upfront fee and final payment lift the cost from the 11% coupon to nearly 14%.

Step 4: warrants. Assume the lender exercises at month 36 at a price per share P (simplification: exercise exactly at maturity, net of strike).

Share price at month 36	Warrant gain ($)	Lender's/borrower's all-in cost
$0 to $2.00 (at or below strike)	0	13.9%
$4.00	200,000 × (4 − 2) = 400,000	17.2%
$8.00	200,000 × (8 − 2) = 1,200,000	23.0%

Step 5: dilution. With, say, 20,000,000 fully diluted shares, 200,000 warrant shares are 1.0% of the company (200,000/20,200,000 = 0.99%).

Conclusion. The all-in cost is 13.9% if the company does not outperform and rises toward 17-23% in success cases. This is still typically cheaper than equity, whose expected cost for a successful company may exceed 30%, but it is a fixed obligation that must be paid in the failure scenarios in which equity absorbs the loss. The right comparison is between the dilution avoided and the risk of default if the milestone is missed. ∎

13.6.3 Covenants and risks

Venture loans carry *covenants* tailored to early-stage firms: a minimum cash balance, a restriction on additional debt and on dividends, sometimes a revenue or "material adverse change" clause, and often a lien on all assets other than intellectual property (with a negative pledge on IP). The most dangerous clause for a startup is the *material adverse change* default: a vague standard that a lender may invoke when the company's prospects look weak, precisely when the borrower most needs patience.

Use venture debt when (a) the company has strong, recently raised equity backing; (b) the use of proceeds creates value faster than interest accrues; and (c) there is a credible plan to repay or refinance. Avoid it when runway after the loan still ends before a plausible milestone, since you will then face a refinancing crisis with a senior secured creditor holding the

keys.

13.7 Secondaries, Liquidity, and Exits

13.7.1 Why the secondary market exists

The ten-year fund life and the multi-year gap between founding and exit leave both founders and investors with illiquid holdings. A **secondary transaction** is a sale of an existing security by an existing holder to a new buyer, with no new capital going into the company. There are several forms:

- **Founder or employee secondary:** a sale of common shares by employees to a new or existing investor, often in conjunction with a primary round, which lets people diversify, pay taxes, or buy a house without a full exit.
- **Company tender offer:** the company, or a designated investor, offers to buy shares from many holders at a set price.
- **LP-stake secondary:** a limited partner sells its interest in a fund (all remaining NAV plus unfunded commitments) to a specialized secondary buyer.
- **GP-led secondary:** the GP moves one or several portfolio companies from an old fund into a new vehicle (Chapter 14 covers continuation funds in detail), offering existing LPs the choice to sell or roll.
- **Direct secondary of preferred stock:** a venture fund sells a position in a particular company to a new investor at a negotiated price.

13.7.2 Pricing a secondary

Secondaries almost always trade at a *discount to net asset value (NAV)* when the buyer is paying cash up front for a pool of illiquid assets with uncertain timing. Pricing depends on the quality of the underlying holdings, the pace of expected distributions, the amount of unfunded commitments (which the buyer must fund), and market conditions. The buyer's return rises as the discount deepens; the seller trades return for liquidity.

Example 13.8 – Pricing an LP-stake secondary. An LP holds a fund interest with a reported NAV of **$50 million**. A secondary buyer projects the following *net* cash flows to the holder (distributions less future capital calls), in $ millions, at the end of years 1-6:

Year	1	2	3	4	5	6
Net cash flow	1	5	13	15	14	12

The total is **$60 million**, or 1.2x of NAV. We compute the buyer's IRR at several purchase prices (expressed as a percentage of NAV):

Price (% of NAV)	Purchase price ($M)	Total cash back / price	Buyer's IRR
100%	50.0	1.20x	4.5%
90%	45.0	1.33x	7.2%
85%	42.5	1.41x	8.7%
80%	40.0	1.50x	10.4%

Step 1: reading the table. If the buyer requires a 10% return, it will bid about 80% of NAV; a buyer content with 7% will pay 90%.

Step 2: seller's side. The seller who accepts 85% gives up $7.5 million of reported NAV in exchange for immediate liquidity and the removal of unfunded-commitment risk. This is attractive if the seller faces a liquidity need, has an overweight in private assets (the "denominator effect," when public holdings fall and private holdings stay marked), or believes the projected cash flows are optimistic.

Conclusion. The discount is not a statement that NAV is wrong. It is the price of immediacy and information risk. A buyer that expects stronger flows than the seller will bid higher; if the cash flows slip by a year, each IRR in the table falls by roughly a percentage point or more. ∎

13.7.3 Exits: IPO, acquisition, and the choice between them

The principal exit routes are an *acquisition* (strategic or financial buyer), an *IPO* (Volume 1, Chapter 13 on raising equity), or, occasionally, a recapitalization or sale to a secondary investor. Each has implications that follow from the terms in this chapter:

- In an acquisition, the liquidation waterfall governs payments directly. Escrows and earn-outs defer part of the price, and employee retention packages may be negotiated separately.
- In an IPO, preferred converts automatically (a *qualified IPO* clause usually requires a minimum price or proceeds), so liquidation preferences disappear. This is why participating preferred creates a pro-sale bias: it pays more at modest acquisition prices than in an IPO.
- Lock-up periods keep insiders from selling for typically 90-180 days after the IPO, and post-IPO share price volatility becomes the holder's risk.

Managerial implication. Founders considering a secondary sale should weigh the signaling effect (investors may interpret heavy founder selling as a lack of confidence) against the benefit of reduced personal risk, which can actually make a founder *more* willing to pursue high-variance, high-return strategies that the fund wants.

Case Study: Tessera Analytics Weighs Its Series B

(Illustrative company; all figures are fictional.)

Tessera Analytics sells subscription software that predicts equipment failures for regional hospitals. Founded four years ago by Dr. Imani Brooks (CEO) and Tomas Reyes (CTO), the company has $9 million in annual recurring revenue (ARR), growing 80% per year, and burns about $1.1 million per month. Its balance sheet holds $6.6 million of cash, about six months of runway.

Tessera's capitalization is as follows. The founders hold 8.0 million common shares. A Series A investor, Cedar Ridge Ventures, bought 3.0 million preferred shares two years ago at $3.33 per share ($10 million for 25% of the company, 1x non-participating, standard seniority, broad-based weighted-average anti-dilution). An option pool of 1.0 million shares remains, of which 0.4 million are ungranted. Total fully diluted shares are 12.0 million.

The board has three offers on the table.

Offer 1 (Summit Growth): $20 million Series B at a $60 million pre-money, 1x non-participating preferred, pari passu with Series A. A condition: the option pool must be topped up to 12% of post-money shares, with the top-up included in the pre-money.

Offer 2 (Harbor Capital): $20 million at a $75 million pre-money, but with a 1.5x participating preferred (uncapped), seniority over Series A, and a redemption right after five years. Harbor would also take two of the seven board seats and a veto over any sale below $300 million.

Offer 3 (venture debt plus bridge): Cedar Ridge offers a $4 million convertible bridge note that converts at a 20% discount to the Series B price with a $50 million cap, and a specialized lender offers $5 million of venture debt at 12% interest, 1% fee, 4% end-of-term fee, and 8% warrant coverage at the Series A price. Together these would extend runway to about 11 months, enough, management believes, to reach $15 million of ARR and raise a larger Series B in the autumn.

Dr. Brooks leans toward Harbor because of its $75 million valuation. Tomas worries about the participating preference and the veto. The Cedar Ridge partner favors the bridge because it avoids setting a price and preserves the Series A's pro rata right. The venture lender wants a first-priority lien and a minimum-cash covenant of $2 million.

The management plan shows three exit scenarios in five years: a $600 million outcome (probability 20%), a $250 million sale (35%), and a $60 million sale (25%), with a 20% chance of failure. The numbers in the plan assume 55% annual growth falling to 35%.

Cedar Ridge's partner cautions: "With a $250 million exit, the difference between these terms is real money; with a failure, nothing matters; with a $600 million exit, everyone is happy, but the founders may regret the veto."

Discussion questions

1. For each of Offers 1 and 2, compute the post-money valuation, the new investor's ownership, the price per share, and the effective pre-money after the pool shuffle in Offer 1. Which offer has the higher *effective* valuation to existing holders?
2. Build the exit waterfall for the three non-failure scenarios under Offers 1 and 2 (include the Series A's 1x non-participating preference). At which exit value do the common stockholders do equally well under the two offers? Use the scenario probabilities to compute the expected proceeds to the common for each offer.
3. Evaluate Offer 3: compute the all-in cost of the venture debt, estimate the dilution from the bridge note at a $50 million cap, and discuss the risk that the debt covenant (minimum cash) is breached before the autumn round. How would you stress-test the plan's 55% growth assumption?
4. Which offer would you recommend, and what term would you negotiate to improve it? Consider the incentives of each investor in the $60 million exit scenario and the role of the Series A's anti-dilution protection if the autumn round is a down round.

Key Takeaways

- A venture fund is a ten-year limited partnership whose structure (commitments, capital calls, fixed life, carry) shapes investor time horizons and appetite for risk; founders should evaluate investors' fund size, reserves, and timeline, not just price.
- Venture outcomes follow a power law: a few percent of companies deliver roughly half of all value, so fund returns depend on whether the portfolio contains an outlier, and mean returns say little about the typical fund.
- "Return the fund" arithmetic (fund size divided by ownership at exit) explains why funds pass on modest exits and insist on meaningful ownership.

- The VC method works backward from an exit value using a target return and a retention factor; it is a framework for negotiation, not a precise valuation, and the high target rate substitutes for explicit failure probabilities.
- Preferred stock is a bundle of options: a senior claim plus a conversion right. Participation transfers a roughly constant amount from the common to the investor at all exit values above conversion, and caps limit that transfer.
- Option-pricing methods show that common stock is worth materially less than the preferred price (about 27% less in our example); the headline post-money overstates what the founders own.
- Cap table modeling must include the option-pool shuffle, convertible instruments, and each round's effect on ownership; the pre-money pool top-up lowers the effective valuation to existing holders.
- Full-ratchet anti-dilution can shift double-digit ownership points in a down round, while weighted-average protection shifts only a couple of points; both transfer cost to founders and unprotected holders.
- Venture debt costs more than its coupon: fees and warrants lift the all-in cost to 14-23% in our example, and its fixed obligations and covenants can matter most when the company is weakest.
- Secondaries trade at discounts to NAV that compensate buyers for illiquidity and information risk; they provide liquidity for LPs, employees, and founders, but pricing, signaling, and conflicts of interest need to be managed.

Review Questions

1. Why is equity, rather than debt, the dominant form of financing for early-stage ventures? Which three problems do staged financing, board control, and liquidation preferences address?
2. Explain the difference between the fund's commitments and its drawn capital. Why does the timing of capital calls matter to LPs?
3. Why does diversification across more companies not raise a venture fund's expected return, and why can it still reduce the probability of a very high return?
4. A fund of $400 million owns 8% of a company at exit. What exit value is required for that holding alone to return the fund? What does this imply for the kind of company the fund can back?
5. Why is the 40% target return used in the VC method not a cost of capital? What does it implicitly contain?
6. Contrast 1x non-participating, 1x participating, and capped participating preferred. At what exit values do they differ, and who bears the cost?
7. Explain how a convertible preferred share can be decomposed into option positions and why this implies that common stock is worth less than the preferred price.
8. What is the option-pool shuffle? Who bears its cost, and how can a founder negotiate it?
9. Compare full-ratchet, broad-based weighted-average, and narrow-based weighted-average anti-dilution. Why does a new investor care about the existing investors' anti-dilution terms?
10. When is venture debt a sensible choice and when is it dangerous? Why is a warrant part of the compensation?

Problems

1. **Pre/post-money.** A startup has 9,000,000 fully diluted shares (including pool) and raises $6.0 million at an $18.0 million pre-money. Find the price per share, the number of new shares, the post-money valuation, and the new investor's ownership.

2. **VC method.** An investor puts in $5.0 million. In four years the company is expected to have an exit value of $200 million. The target return is 35% per year and the investor expects to retain 80% of its ownership after later rounds. Find the required multiple, the required ownership at exit, the required ownership today, and the implied post-money and pre-money valuations.
3. **Liquidation preferences.** A fund invests $8.0 million for 20% of a company. Compute the fund's proceeds and the common's proceeds at an exit value of $50 million under (a) 1x non-participating preferred, and (b) 1.5x participating preferred (uncapped). At what exit value does the non-participating preferred convert?
4. **Option-pool shuffle.** A company has 6,000,000 shares (no pool). A new investor will invest $5.0 million at a $15.0 million pre-money (headline), and requires a post-money pool of 12% of fully diluted shares, created pre-money. Find total post-money shares, the price per share, the new pool size, and the effective pre-money valuation for existing holders.
5. **Anti-dilution.** Series A holds 1,000,000 preferred shares bought at $5.00 (convertible 1:1). There are 10,000,000 fully diluted shares before a down round that issues new shares for $3.0 million at $2.50 per share. Compute the new conversion price and the number of common shares into which the preferred converts under (a) broad-based weighted average and (b) full ratchet.
6. **Venture debt.** A company borrows $4.0 million for two years; it pays 12% annual interest at the end of each year and repays the principal at maturity. The lender takes a 1% upfront fee, and receives warrants on 200,000 shares with a $1.00 strike. At maturity the shares are worth $3.00. Compute the lender's IRR (a) ignoring the warrants and (b) including the warrants' gain at maturity. Treat the fee as deducted from proceeds.
7. **Secondary pricing.** A secondary buyer expects net cash flows of 2, 4, 6, 8, 8 ($ millions) at the end of years 1 through 5 from an LP interest with NAV $20 million. Compute the buyer's IRR if the price is (a) 100% of NAV and (b) 80% of NAV. What is the cash-on-cash multiple at 80%?
8. **Power law.** A fund of $250 million invests in 40 companies, each with a 2% chance of being an outlier (a 50x or better outcome), independently. (a) What is the probability that the fund contains no outlier? (b) If the fund owns 12% of a company at exit, what exit value is needed for that company alone to return the fund?

Answers to Selected Problems

1. Price = $18.0M / 9,000,000 = **$2.00**; new shares = 6.0M / 2.00 = **3,000,000**; post-money = **$24.0M**; ownership = 6/24 = **25%**.
2. Required multiple = 1.35^4 = **3.3215x**; stake value at exit = 5 × 3.3215 = $16.61M; ownership at exit = 16.61/200 = **8.30%**; today = 8.30%/0.80 = **10.38%**; post-money = 5/0.1038 = **$48.2M**; pre-money = **$43.2M**.
3. a. Converts if 0.2X > 8, i.e. X > $40M. At X = 50, 0.2 × 50 = 10 > 8, so converts: fund **$10.0M**, common **$40.0M**. (b) 12 + 0.2(50 − 12) = 12 + 7.6 = **$19.6M**; common **$30.4M**. Participation costs the common $9.6M here.
4. Existing holders are 100% − 25% − 12% = 63%, so total = 6,000,000 / 0.63 = **9,523,810** shares. Investor shares = 0.25 × 9,523,810 = 2,380,952; price = $5.0M / 2,380,952 = **$2.10**. Pool = 0.12 × 9,523,810 = **1,142,857** shares. Effective pre-money = 6,000,000 × $2.10 = **$12.6M** (the headline $15.0M includes the $2.4M value of the pool).
5. B = new money / CP1 = $3.0M / $5.00 = 600,000; C = 3.0M / 2.50 = 1,200,000. (a) CP2 = 5.00 × (10,000,000 + 600,000) / (10,000,000 + 1,200,000) = 5.00 × 10.6/11.2 = **$4.7321**; conversion shares = 5,000,000 / 4.7321 = **1,056,604**. (b) CP = $2.50, shares = 5,000,000 / 2.50 = **2,000,000**.

6. Proceeds = 4,000,000 − 40,000 = $3.96M. Interest = $480,000 per year. (a) Cash flows −3.96, +0.48, +4.48: IRR = **12.6%**. (b) Warrant gain = 200,000 × (3 − 1) = $0.4M; final flow 4.88: IRR = **17.2%**.

7. a. Flows −20, 2, 4, 6, 8, 8: IRR = **10.1%**. (b) Price $16M: IRR = **17.6%**; multiple = 28/16 = **1.75x**.

8. a. 0.98^40 = **44.6%**. (b) 250 / 0.12 = **$2,083 million** (about $2.1 billion).

Chapter 14: Private Equity Fund Economics

Learning Objectives

After studying this chapter you should be able to:

- Describe the structure of a private equity fund (commitments, capital calls, fee base, life) and compute management fees and total fee drag under common fee schedules.
- Build a distribution waterfall with return of capital, a compounding preferred return (hurdle), a GP catch-up, and carried interest, under both European (whole-fund) and American (deal-by-deal) conventions, including clawbacks.
- Explain the J-curve and calculate IRR, MOIC, DPI, RVPI, and TVPI, recognizing what each measure rewards and hides.
- Compute a Kaplan-Schoar public market equivalent and interpret it alongside IRR.
- Analyze GP/LP alignment mechanisms and evaluate a GP-led continuation fund from the perspective of selling LPs, rolling LPs, new investors, and the GP.
- Identify the major pitfalls in measuring private equity performance: subscription-line distortions, stale valuations, vintage effects, survivorship bias, and fee layering.

Opening Vignette

(Illustrative company; all names and figures are fictional.)

Rosalind Park is the chief investment officer of the Millbrook Teachers' Retirement System, a pension fund with $14 billion in assets and a 12% allocation to private markets. The investment committee meets on Thursday to consider a $150 million commitment to Granite Peak Partners Fund IV, a $2 billion buyout fund.

Granite Peak's marketing deck is impressive. Its three previous funds report net IRRs of 24%, 21%, and 19%, "top-quartile" status, and a gross multiple of 2.4x on realized investments. Rosalind's analyst, Tomas, has done his homework and returns with uncomfortable questions. The 21% IRR for Fund III includes a two-year period in which investments were bridged with a credit line, so that LPs' contributions were delayed. The reported "net" multiple of 1.9x includes 35% of value still held as unrealized NAV, which the GP itself marks. The fund's fee base steps down only at the end of the investment period, and transaction fees paid by portfolio companies are offset only 80% against management fees. Meanwhile, the S&P-based public benchmark over the same period returned strongly, so the true "excess" return is far less than 21% minus a hurdle rate.

"None of this is fraud," Tomas says. "Every number is accurate. They are just not the numbers that answer your question."

Rosalind's question is simple: *if we commit $150 million, how much of the profit will we actually keep, and how does that compare with owning index funds, adjusted for illiquidity and risk?* To answer it she must reconstruct the fund from first principles: what the LPs pay, in what order the cash is divided, how timing inflates or deflates the IRR, and where the GP's incentives agree or conflict with her own. This chapter provides that toolkit. It builds on the buyout analysis in Volume 1, Chapter 23 and on the venture fund structure in Chapter 13.

14.1 The Private Equity Fund: Structure and Terms

14.1.1 The partnership and its documents

A private equity (PE) fund is, like a venture fund, a finite-life limited partnership. Limited partners commit capital; the general partner (GP) manages the fund through a separate management company; and the relationship is governed by the *limited partnership agreement* (LPA). Strategies are conventionally grouped as *buyout* (control acquisitions of mature companies, often with leverage; see Volume 1, Chapter 23), *growth equity* (minority stakes in profitable growing companies), *venture capital* (Chapter 13), *distressed and special situations*, *private credit*, and *real assets* (infrastructure, real estate). The economic mechanics discussed here (fees, hurdle, carry, waterfalls, performance measurement) are common to all, though the parameters differ.

The LPA is a risk-allocation contract. The terms that matter most to economics are:

Term	Typical feature	Why it matters
Fund size / commitments	Total capital pledged by LPs	Basis for fees; limits deal size
Investment period	4-6 years from first close	Fees on commitments during this period
Term	10 years plus 1-2 extensions	Sets the horizon for exits and for IRR
Management fee	~1.5-2.0% of commitments, stepping down	Pays the firm regardless of performance
Carried interest	~20% of profits	The GP's performance incentive
Preferred return (hurdle)	~8% compounded annually	LPs get this before the GP shares profit
GP catch-up	50-100% to GP after hurdle	Restores the GP's full 20% share
Waterfall type	Whole-fund (European) or deal-by-deal (American)	Determines timing and risk of carry
GP commitment	1-5% of commitments	The GP's "skin in the game"
Clawback	GP must return over-distributed carry	Protects LPs under deal-by-deal waterfalls
Key-person clause	Suspends investing if named partners leave	Protects against loss of the team

The numbers are illustrative of market practice and vary widely across strategies and over time; particular funds may differ substantially.

14.1.2 The cash flow pattern

LPs commit, GPs call capital, and cash flows back over time. From the LP's viewpoint:

- **Paid-in capital (contributions):** cash actually transferred to the fund, to pay for investments, fees, and expenses.
- **Unfunded commitment:** the portion not yet called; LPs must hold it ready and cannot invest it in illiquid assets.
- **Distributions:** cash returned, from asset sales, dividends, or recapitalizations.
- **Net asset value (NAV):** the GP's estimate of the fair value of the remaining holdings.

The LP's return is computed on contributions and distributions plus the residual NAV, which will be central in Sections 14.4 and 14.7.

14.2 Management Fees and Gross-to-Net Returns

14.2.1 Fee base and schedule

The management fee is normally set as a percentage of a *fee base*. During the investment period the base is usually *committed capital*; afterward it commonly steps down to *invested capital* (cost of unrealized investments) and/or the percentage falls. The choice of base matters because committed capital is fixed whereas invested capital depends on pacing and realization.

Total fees as a percentage of commitments can be surprisingly large. Two features amplify the effect. First, fees on committed capital mean that the LP pays for money that has not yet been invested. Second, fees are *paid out of LP capital*, so a part of what LPs contribute never reaches a company: the fund must earn more than the fee load on the capital actually invested just to give back the LP's contributions.

Other charges, each a subject of LP negotiation, include:

- **Fund expenses** (legal, audit, administration, and often a portion of the GP's operating costs).
- **Transaction and monitoring fees** paid by portfolio companies to the GP; LPs usually negotiate an *offset* of 80-100% against the management fee.
- **Organizational costs** capped by the LPA.
- **Placement fees** to intermediaries, usually borne by the GP or netted against fees.

Common mistake. Comparing funds on "headline" 2% and 20% only. The fee base, step-down, offsets, expense allocation, and hurdle design can shift the net LP return by more than the headline numbers.

14.2.2 A complete fund model

To make the economics concrete we build one example fund and follow it through the chapter.

Example 14.1 – Gross-to-net bridge for a $1,000 million fund. A buyout fund has $1,000 million of commitments and a ten-year life. Terms: management fee 2.0% of commitments in years 1-5, then 1.5% of the cost basis of unrealized investments (measured at the start of each year) in years 6-10; 20% carry on the whole fund after an 8% compounded preferred return, with a 100% GP catch-up. To simplify, all cash flows occur at year-end (which understates the pref slightly in years of heavy calls) and the GP commitment is ignored.

The fund invests $800 million over five years (the other portion of capital funds fees) and generates gross sales proceeds of $1,700 million, a **gross multiple of 2.125x**:

Year	Investments	Gross proceeds	Management fee
1	200	0	20.0
2	200	0	20.0
3	200	40	20.0
4	150	100	20.0
5	50	200	20.0
6	0	300	9.6
7	0	350	7.5
8	0	300	5.0
9	0	250	2.9
10	0	160	1.1
Total	**800**	**1,700**	**126.1**

Step 1: cost basis for the fee. Each exit realizes cost equal to proceeds divided by the gross multiple of 2.125. The unrealized cost basis at the end of years 1 to 10 is, in $ millions: 200.0; 400.0; 581.2; 684.1; 640.0; 498.8; 334.1; 192.9; 75.3; 0.0.

Step 2: fees. Years 1-5: 2% × 1,000 = **$20.0M** per year, total $100.0M. Year 6: 1.5% × 640.0 = **$9.6M**; year 7: 1.5% × 498.8 = **$7.5M**; year 8: 1.5% × 334.1 = **$5.0M**; year 9: 1.5% × 192.9 = **$2.9M**; year 10: 1.5% × 75.3 = **$1.1M**. Total fees = **$126.1M**, or **12.6% of commitments**.

Step 3: LP contributions. Contributions equal investments plus fees: 800 + 126.1 = **$926.1M** (the remaining $73.9M of commitments is never called). Per year: 220.0; 220.0; 220.0; 170.0; 70.0; 9.6; 7.5; 5.0; 2.9; 1.1.

Step 4: gross IRR. Treating investment outflows and gross proceeds as the cash flow stream (before fees and carry), the **gross IRR is 18.9%**.

Step 5: IRR after fees, before carry. LP contribution and gross proceeds as cash flows give **15.4%**.

Step 6: net IRR after carry. Carry (computed in Example 14.4) is $154.8M, leaving LPs with $1,545.2M of distributions on $926.1M contributed: **net TVPI of 1.67x** and a **net IRR of 13.3%**.

Stage	Multiple	IRR
Gross (investments to proceeds)	2.125x	18.9%
After fees (LP contributions to proceeds)	1,700 / 926.1 = 1.84x	15.4%
After carry (net to LP)	1,545.2 / 926.1 = 1.67x	13.3%

Step 7: total drag. The gross profit is 1,700 − 800 = **$900M**. Fees take $126.1M (14.0%), carry takes $154.8M (17.2%), together **$280.9M or 31.2%** of the gross profit. The IRR spread is **5.6 percentage points** (18.9% less 13.3%).

Conclusion. A fund with a respectable 2.1x gross multiple delivers a 1.67x net multiple. The fee drag of 3.5 IRR points exists even if the GP adds no value; the further 2.1 points are carry on profits. LPs should always evaluate net-of-fee, net-of-carry cash flows, since gross numbers (often presented in marketing materials for individual deals) are not what the LP receives. ∎

14.3 Carry, Hurdles, Catch-Up, and the Waterfall

14.3.1 The four tiers

A *distribution waterfall* specifies the order in which cash is divided between the LPs and the GP. The standard "European" (whole-fund) waterfall has four tiers:

1. **Return of capital.** 100% to the LPs until they have received all contributed capital (including capital used for fees and expenses).
2. **Preferred return (hurdle).** 100% to the LPs until they have received a compounded annual return (commonly 8%) on contributed capital, measured from the date of each contribution.
3. **Catch-up.** A share, often 100%, to the GP until the GP has received 20% of the cumulative profit distributed so far (that is, until carry equals 20% of pref plus catch-up).
4. **Carried-interest split.** 80% to the LPs and 20% to the GP thereafter.

With a full catch-up and a profitable fund, the GP ends up with exactly 20% of *total* profits: the hurdle matters for *timing* and for marginal funds, but not for the final split in a successful one. With **no** catch-up (a "hard hurdle"), the GP receives carry only on profit above the hurdle, so its share is below 20% of total profit. The catch-up percentage is therefore a significant economic negotiation.

The hurdle is *not* the same as a minimum return guarantee. It is a threshold that must be cleared before carry begins, and it makes carry resemble a call option on fund profits: strike = contributed capital grown at the hurdle rate.

14.3.2 A single-period waterfall

Example 14.2 – Waterfall with hurdle and catch-up. An LP contributes **$100 million** which the fund holds for five years before a single distribution. The terms: 8% compounded preferred return, 100% GP catch-up, 20% carry. Compute the division of proceeds of (a) $250 million, (b) $150 million, (c) $120 million.

Step 1: hurdle amount. 100 × 1.08^5 = **$146.93M**; the preferred return itself is 146.93 − 100 = **$46.93M**.

Step 2: catch-up target. The GP catches up until it holds 20% of profit distributed, i.e. carry = 0.20 × (pref + catch-up), so catch-up = 0.25 × pref = 0.25 × 46.93 = **$11.73M**.

(a) Proceeds $250M. Tier 1+2: $146.93M to the LP. Tier 3: $11.73M to the GP. Remaining = 250 − 146.93 − 11.73 = **$91.33M**; LP 80% = $73.07M, GP 20% = $18.27M. Totals: LP = 146.93 + 73.07 = **$220.0M** (2.20x), GP = 11.73 + 18.27 = **$30.0M**, which is exactly 20% of the $150M profit. ✓

(b) Proceeds $150M. LP receives the hurdle amount of 146.93. Only 150 − 146.93 = $3.07M is left, which goes to the GP as part of the catch-up (not yet complete). The GP receives **$3.07M**, only **6.1%** of the $50M profit. The LP receives $146.93M (1.47x).

(c) Proceeds $120M. Below the hurdle: the LP receives all $120M; the GP receives nothing. The LP has a profit of $20M (3.7% annual return) but the GP gets no carry.

Step 3: effect of removing the catch-up. In (a) without a catch-up, the GP receives 20% × (250 − 146.93) = **$20.61M** (13.7% of profit) rather than $30.0M. The catch-up is worth $9.39M to the GP.

Conclusion. The hurdle protects LPs from paying carry on mediocre outcomes, but the catch-up largely cancels the hurdle's effect once returns clear it. Note the sharp "cliff": between proceeds of $146.93M and $158.66M (= 146.93 + 11.73) every extra dollar goes entirely to the GP. Such a region creates an incentive for the GP to push exit values just above the hurdle,

and for LPs to prefer partial catch-ups (e.g., 50%). ∎

14.3.3 European versus American waterfalls and the clawback

In an **American (deal-by-deal)** waterfall, carry is calculated and paid on each realized investment, typically after the LPs have received capital and pref on that investment *and sometimes on previous realized losses*. The GP therefore receives carry earlier, but can be paid carry on early winners that is later not warranted when other investments lose money. To protect LPs, the LPA contains a **clawback**: at the end of the fund, if the GP has received more than 20% of cumulative profit (or more than the fund's hurdle supports), it must return the excess, sometimes limited to after-tax amounts and secured by an escrow or personal guarantees of the partners.

Example 14.3 – Deal-by-deal carry and clawback. A small fund invests \$50M in Deal A and \$50M in Deal B. To isolate the mechanics we use no hurdle and 20% carry. Deal A is sold in year 3 for \$150M (profit \$100M). Deal B is sold in year 5 for \$20M (loss \$30M).

Step 1: American waterfall at year 3. Profit on A is \$100M; carry = 20% × 100 = **\$20M** paid to the GP in year 3.

Step 2: year 5. Deal B loses \$30M. Cumulative profit = 100 − 30 = \$70M; total carry earned = 20% × 70 = **\$14M**.

Step 3: clawback. The GP received \$20M but is entitled to only \$14M, so it must return **\$6M**. If the GP has spent the money or the partners are not creditworthy, LPs bear the loss unless an escrow exists.

Step 4: compare with a whole-fund (European) waterfall. Under a whole-fund test the LPs must first receive all \$100M contributed for both deals. At year 3 the \$150M of proceeds returns that \$100M and leaves \$50M of profit, so carry is 20% × 50 = **\$10M** (not \$20M). At year 5 the loss on B reduces cumulative profit to \$70M and total carry to \$14M, so the GP receives a further **\$4M**. No clawback is needed. In year-5 dollars at an 8% discount rate, the American GP's position is 20 × 1.08^2 − 6 = 23.33 − 6 = **\$17.3M**, versus 10 × 1.08^2 + 4 = 11.66 + 4 = **\$15.7M** for the European GP: a **\$1.7M** timing advantage, plus the credit risk that the LPs bear if the clawback is not paid.

Conclusion. The American waterfall shifts the risk of subsequent losses from the GP to the LP until the clawback is settled, and gives the GP an incentive for early exits of winners. LPs prefer the European convention for exactly this reason; GPs prefer the American. Many modern funds use a European waterfall, and deal-by-deal variants often include interim clawbacks, escrows of 20-30% of carry distributions, and a requirement to return all capital on realized deals plus allocable fees and any write-downs before carry is paid. ∎

14.3.4 The whole-fund waterfall in time

For a multi-year fund we must handle the compounding hurdle on the *outstanding balance* of unreturned capital and unpaid preference. The procedure at each distribution date is:

1. Accrue the 8% hurdle on the opening balance of (unreturned capital + unpaid pref).
2. Add any new contributions.
3. Pay the distribution first to return unreturned capital, then to pay accrued pref (Tier 1/2).
4. If both are fully paid, pay 100% to the GP until it holds 20% of the profit (pref paid plus catch-up), then split 80/20.

Example 14.4 – The whole-fund waterfall for the \$1,000M fund. Continue Example 14.1. All cash flows are at year-end; the hurdle accrues annually on the opening balance. Using the contributions (220, 220, 220, 170, 70, 9.6, 7.5, 5.0, 2.9, 1.1) and the gross proceeds (0, 0, 40, 100, 200, 300, 350, 300, 250, 160):

Year	Contribution	Gross proceeds	To LP	To GP (carry)	LP net cash flow	Cumulative LP net cash flow
1	220.0	0	0.0	0.0	-220.0	-220.0
2	220.0	0	0.0	0.0	-220.0	-440.0
3	220.0	40	40.0	0.0	-180.0	-620.0
4	170.0	100	100.0	0.0	-70.0	-690.0
5	70.0	200	200.0	0.0	+130.0	-560.0
6	9.6	300	300.0	0.0	+290.4	-269.6
7	7.5	350	350.0	0.0	+342.5	+72.9
8	5.0	300	226.4	73.6	+221.4	+294.3
9	2.9	250	200.6	49.4	+197.7	+492.0
10	1.1	160	128.2	31.8	+127.1	+619.1
Total	**926.1**	**1,700**	**1,545.2**	**154.8**		

Step 1: years 1-6, return of capital. In years 3-6 the entire proceeds go to the LPs as a return of capital. By the end of year 6 the LPs have received 40 + 100 + 200 + 300 = 640 against contributions of 220 × 3 + 170 + 70 + 9.6 = 909.6, leaving **$269.6M** of unreturned capital.

Step 2: year 7. The pref accrues 8% on the opening balance (unreturned capital of $269.6M plus unpaid pref of $230.6M), so the year-7 pref balance reaches $270.6M. Of the $350M distribution, the first $277.1M returns the remaining capital (269.6 plus the year-7 contribution of 7.5) and **$72.9M** is applied to preferred return. A pref balance of **$197.7M** (270.6 − 72.9) remains unpaid.

Step 3: year 8 (the hurdle clears). The pref balance grows by 8% to $213.5M. Of the $300M distribution: $5.0M returns the new capital, $213.5M pays the remaining pref (cumulative pref paid: 72.9 + 213.5 = **$286.4M**). The leftover is 300 − 218.5 = **$81.5M**. The catch-up needs 0.25 × 286.4 = **$71.6M** to the GP; the last $9.9M is split 80/20: LP $7.9M, GP $2.0M. GP total in year 8 = 71.6 + 2.0 = **$73.6M**; LPs receive 218.5 + 7.9 = **$226.4M**.

Step 4: years 9-10. The catch-up is complete, so after returning each year's small contribution, every dollar is split 80/20. Year 9: 250 − 2.9 = 247.1; GP gets 0.2 × 247.1 = **$49.4M**; LP gets 200.6. Year 10: 160 − 1.1 = 158.9; GP gets **$31.8M**; LP gets 128.2.

Step 5: check. Total carry = 73.6 + 49.4 + 31.8 = **$154.8M**. Profit after fees = 1,700 − 926.1 = $773.9M; 20% of that is **$154.8M**. ✓ The catch-up was completed, so the GP earns exactly 20% of total profit net of fees.

Step 6: the carry timing. Despite cumulative gross proceeds of $990M by the end of year 7 (more than the $800M invested), the GP received no carry until year 8. Fee-funded capital counts as contributed capital, so the fund had to pay back all fees and pref before the GP shared profit.

Conclusion. The European waterfall defers carry until the LPs have a clear economic gain. The LP's cumulative net cash flow turns positive in year 7: that point is the *breakeven* of the J-curve, discussed next. ∎

14.4 The J-Curve and the Measures: IRR, MOIC, DPI, TVPI

14.4.1 Why the J-curve exists

Plot the cumulative net cash flow to an LP over the life of a typical fund and you obtain a **J-curve**: the line dips below zero in the early years and rises later. There are three causes, and a good analyst distinguishes them.

1. **Fees are paid from day one, but gains arrive late.** Management fees begin at the first close; investment gains require years of operational improvement and a sale.
2. **Conservative or cost-based interim marks.** Many GPs hold investments at cost for the first year or so (valuation guidelines often start from the transaction price), so the fund's reported NAV lags its economic value in the early years and then catches up. Early reported IRRs are therefore biased downward.
3. **Capital is drawn before it is returned.** Contributions cluster in years 1-4 and distributions in years 4-10, so cumulative cash flow must fall before it rises.

In Example 14.4, the LP's cumulative net cash flow bottoms at **−$690M in year 4** (the maximum net cash exposure, 69% of commitments) and breaks even in **year 7**. Even a fund with a 1.67x final multiple is cash-negative for six years.

Practical implications. (i) An LP's *commitment pacing* must account for the fact that net exposure peaks at far less than 100% of commitments but takes years to return; most institutions commit to several funds each year (vintage diversification) so that new J-curves are always being laid on top of maturing ones. (ii) Interim IRRs, particularly in the first four years, are not meaningful performance measures: they reflect the stage of the fund more than its quality. (iii) The J-curve is deeper when fees are charged on committed capital, and shallower for funds that buy mature, cash-generating companies and pay early dividends.

14.4.2 The standard LP-side metrics

The industry uses a family of ratios defined on the LP's paid-in capital:

```
DPI  (distributed to paid-in)  = Cumulative distributions / Paid-in capital
RVPI (residual value to paid-in) = Current NAV / Paid-in capital
TVPI (total value to paid-in)  = DPI + RVPI
MOIC (gross multiple)          = (Realized proceeds + Unrealized value) / Invested capital
Net IRR                        = IRR of (−contributions, +distributions, + terminal NAV)
```

All of these are normally reported *net* of fees and carry when computed on LP cash flows. DPI is realized, "cash-on-cash" performance; RVPI depends on the GP's valuations; TVPI combines them. For the example fund (Example 14.4):

End of year	Cumulative paid-in	Cumulative LP distributions	DPI
3	660.0	40.0	0.06x
5	900.0	340.0	0.38x
6	909.6	640.0	0.70x
7	917.1	990.0	1.08x
8	922.1	1,216.4	1.32x
10	926.1	1,545.2	1.67x

DPI crosses 1.0x in year 7, at the same point as the break-even of cumulative net cash flow. At the end, with no NAV left, DPI = TVPI = 1.67x. An LP who sees TVPI of 1.5x in year 4 with a DPI of 0.1x is observing mostly the GP's own marks, not cash. The rule among sophisticated LPs is: *DPI is the only number that cannot be revised.*

14.4.3 IRR versus MOIC

IRR and MOIC answer different questions. The **multiple** tells you how many dollars came back for each dollar invested, ignoring time. The **IRR** tells you the annualized, compounded rate at which money grew, which favors quick exits. Each measure can mislead alone.

Example 14.5 – Two deals, two rankings. Deal A: invest $100M, receive $150M after 2 years. Deal B: invest $100M, receive $250M after 8 years.

Step 1: multiples. A: 150/100 = **1.50x**; B: 250/100 = **2.50x**.

Step 2: IRRs. A: (1.5)^(1/2) − 1 = **22.5%**; B: (2.5)^(1/8) − 1 = **12.1%**.

Step 3: reconcile with reinvestment. IRR implicitly assumes that interim proceeds are reinvested at the IRR itself. If instead the LP can reinvest proceeds only at 8% (roughly the cost of capital for such an allocation), Deal A's $150M grows over the remaining six years to 150 × 1.08^6 = **$238.0M** at year 8. The *modified IRR* (MIRR) over the full eight years is (238.0/100)^(1/8) − 1 = **11.4%**, which is below B's 12.1%. If the reinvestment rate were only 3%, A's MIRR would drop to 7.6%.

Step 4: dollar profit. A creates $50M of profit and B creates $150M. If the LP had a fixed budget of $100M for eight years, B is more valuable unless the LP can reinvest A's cash at above about 12.1%.

Conclusion. A quick exit lifts IRR and lowers the multiple, and vice versa. GPs know this: a fund can raise its IRR by selling winners early (or by recapitalizing to return cash quickly), even when holding would produce more total dollars. LPs should look at IRR *and* multiple *and* DPI, and ask what the reinvestment alternative actually is. ∎

14.4.4 Subscription lines and the IRR illusion

Many funds use a **subscription credit facility** (a "sub line"): the fund borrows against LPs' uncalled commitments to make investments, then calls capital from LPs weeks, months, or even a year later to repay the loan. The line smooths capital calls and speeds closings, and it has legitimate uses. It also *mechanically raises reported IRR* by shortening the time between the LP's contribution and the distribution, while *lowering the multiple* by the interest paid.

Example 14.6 – The sub-line effect on IRR. The fund buys a company for $100M and sells it four years later for $180M. We compare two ways of financing the purchase.

Case 1: no sub line. The LPs fund the $100M at time 0. Multiple = 180/100 = **1.80x**; IRR = (1.8)^(1/4) − 1 = **15.8%**.

Case 2: sub line. The fund borrows $100M at time 0 at 6% and calls the LPs' capital at the end of year 1, when the loan plus interest equals 100 × 1.06 = $106M. LPs contribute **$106M at year 1** and receive $180M at year 4 (three years later). Multiple = 180/106 = **1.70x**; IRR = (180/106)^(1/3) − 1 = **19.3%**.

Step 3: compare. The reported IRR rises by **3.5 percentage points** (15.8% to 19.3%) even though the LPs' profit has *fallen* by $6M (from $80M to $74M). The LP's dollar gain is lower, but its measured IRR is higher, because the LP's money was "invested" for three years rather than four. The effect grows with the length of the borrowing: a two-year line would give (180/112.4)^(1/2) − 1 = 26.6% (using 100 × 1.06^2 = 112.36).

Step 4: interpret. The sub line is a form of leverage at the fund level supplied by the LPs' balance sheets (the commitments secure the loan), and the benefit accrues to the GP's headline IRR and, through the IRR-based carry hurdle, can accelerate carry. But the IRR "improvement" is not alpha. The right adjustment is to compute IRR from the *date capital would have been called without the line* (many LPs now request "unlevered" or "as-if-called" returns) and to look at the multiple, which is penalized by the interest.

Conclusion. In Rosalind's due diligence (Opening Vignette), the 21% IRR for Fund III includes a period during which credit lines delayed contributions. If the average delay were a year on all deals, the unlevered IRR would be materially lower while the multiple would remain unchanged: the reported rank of the fund relative to its peers may be an artifact of financing choices. ∎

14.5 Public Market Equivalents

14.5.1 Why IRR cannot be compared directly with an index return

An index return, such as the annual return on a broad equity index, is a *time-weighted* return: it does not depend on when an investor puts money in or takes it out. A private equity IRR is *money-weighted*: it depends on the timing of cash flows controlled by the GP (through calls and distributions). Comparing an IRR with an index return is like comparing a mile time with a lap time. The **public market equivalent (PME)** family solves the problem by asking: *what would the investor have had if the same dollars, at the same dates, had been invested in the index?*

14.5.2 The Kaplan-Schoar PME

Kaplan and Schoar proposed the simple and widely used ratio: compound every contribution and every distribution forward to a common date at the index return and take the ratio.

```
KS-PME = [ Σ D_t × (I_T / I_t) + NAV_T ] / [ Σ C_t × (I_T / I_t) ]
```

where D_t is distributions at time t, C_t is contributions, I_t is the index level at time t, and NAV_T is the terminal value at the end date (zero for a liquidated fund). A KS-PME above 1.0 means the fund returned more than the index would have on the same cash flows; below 1.0, less. The measure is a *ratio of future values*, so it is not an annualized rate, and it has the great virtue of being well-defined even for funds with negative IRRs.

Example 14.7 – A KS-PME. An LP's fund has the following net cash flows at the end of years 0 through 6 (in $ millions, contributions negative), and the index levels are:

Year	0	1	2	3	4	5	6
Fund cash flow	−100	−50	−50	0	+30	+80	+150
Index level	100	108	104	121	130	138	160
Factor I_6 / I_t	1.600	1.481	1.538	1.322	1.231	1.159	1.000

Step 1: future value of contributions. 100 × 1.600 = 160.0; 50 × 1.481 = 74.07; 50 × 1.538 = 76.92. Sum = **$310.99M**.

Step 2: future value of distributions. 30 × 1.231 = 36.92; 80 × 1.159 = 92.75; 150 × 1.000 = 150.0. Sum = **$279.68M**. (The fund is fully liquidated at year 6, so NAV_6 = 0.)

Step 3: ratio. KS-PME = 279.68 / 310.99 = **0.899**.

Step 4: other metrics for the same fund. Multiple (TVPI) = 260/200 = **1.30x**. The fund's IRR is **5.7%**. The index rose from 100 to 160 over six years, or 8.1% per year (time-weighted).

Step 5: Long-Nickels equivalent. Suppose the same contributions had gone into the index and the distributions had been withdrawn at the same dates. The index-mimicking account would still hold **$31.3M** at year 6 (= 310.99 − 279.68, the shortfall). Using that residual as the terminal value, the "index-equivalent IRR" is **8.2%**, compared with the fund's **5.7%**: an underperformance of about **2.5 percentage points** per year.

Conclusion. The fund returned a positive 1.30x multiple and a positive IRR of 5.7%, yet it destroyed value relative to simply buying the index on the same dates: it ended with about 10% fewer dollars (in year-6 terms) than the index alternative. A pension fund should also remember that PME ignores risk: private equity portfolios often carry higher leverage and a higher equity beta than the index, so the hurdle for "outperformance" should be above 1.0 (for example the index scaled by a beta greater than one). ∎

14.5.3 Interpreting PME and its variants

- **KS-PME > 1** is evidence of excess return relative to the index, *before* adjusting for risk and illiquidity. It is a benchmark for performance, not a measure of alpha.
- **Direct alpha** (Gredil, Griffiths, and Stucke) converts the same idea into an annualized excess return by finding the constant rate at which the benchmark-compounded cash flows net to zero; it is easier to compare with other asset classes and interpretable as an annual spread.
- **PME+ and Long-Nickels** variants adjust the scaling of distributions so that the benchmark account never goes negative.
- **The choice of index matters.** Small-cap equity or sector-matched indices are more appropriate for buyout and growth strategies than a large-cap index; for leveraged buyouts, a levered benchmark may be needed.
- **PME and time.** A single PME for a fund of a given vintage describes the whole life of the fund; it does not say whether performance was generated early or late, or whether it was luck.

The measurement conclusion is that no single metric is sufficient. A prudent LP monitors DPI (realized cash), TVPI (marked value), net IRR (timing), and PME (market-relative performance), and interprets them in the context of vintage and peer group (Section 14.7).

14.6 GP/LP Alignment and Continuation Funds

14.6.1 Where interests align and where they diverge

The 2-and-20 structure is meant to align the GP with LPs, but the alignment is partial. Consider the GP's payoff. It receives fees regardless of performance, plus carry, which is a call option on fund profits with a strike (the hurdle) and no downside. A GP's expected economics therefore depend on risk: because the option value rises with volatility, the GP has an incentive to take more risk than LPs, who bear the downside, would choose. Specific conflicts include:

Conflict	Source	Typical mitigation
Excess risk-taking	Carry is a call option; LPs bear the losses	GP commitment; hurdle; clawback
Fee-driven fund growth	Fees on commitments are large and certain	Fee step-downs; fee on invested capital; size caps
Early exits / IRR gaming	Carry and fundraising favor high IRR	Multiples and DPI reporting; whole-fund waterfall
Overvaluation of NAV	Marks drive fundraising for the next fund	Independent valuation, audit, ILPA-style reporting
Fee leakage	Transaction, monitoring, and director fees paid to GP	80-100% fee offset
Cross-fund conflicts	Several funds competing for deals	Allocation policies; advisory committee approval
Holding too long / too short	Fund-life constraints and extension fees	LP-approved extensions; term limits

The GP commitment. A GP investing 2-5% of the fund's capital alongside the LPs bears real losses. For a $2 billion fund, a 2% commitment is $40 million, an amount that is large relative to the personal wealth of partners and can be financed in part by *waiving* management fees (a "fee waiver") in exchange for an investment of the same amount in the fund: the partners swap pre-tax fee income for an equity position. This increases alignment in form, but LPs should distinguish a GP commitment funded from personal wealth from one funded by fee waivers, since the latter reduces the cash the GP actually puts at risk.

Carry as option value. If the fund's expected return is low, carry may be worth little; if volatile, its option value is higher. As an illustration, consider a fund with $100 of NAV today, a hurdle that requires $146.9 at the end of five years, a catch-up and a 20% carry. A simple Black-Scholes-style treatment of the GP's share as a call struck at the hurdle value gives a carry value that rises with fund volatility, which explains why risk-seeking is a persistent concern even in well-designed funds; the mitigation lies in LP governance (advisory committees, diversification limits, and investment guidelines in the LPA), not in the waterfall alone.

Key-person and no-fault provisions. An LPA typically allows a supermajority of LPs to remove the GP without cause ("no-fault removal") or to suspend the investment period on a key-person event. These rights are rarely exercised but affect bargaining power during disputes.

Co-investment. LPs are increasingly offered the right to invest directly in individual deals alongside the fund, usually with reduced or zero fees and carry. For the LP, co-investments are a cost-saving device and a way to deploy more capital with a manager it trusts; for the GP, they enable larger deals without a larger fund but create allocation questions (which LPs get the best deals?) and adverse selection risk (the GP may offer co-investments in deals it finds hard to fill).

14.6.2 Continuation funds

A **continuation fund** (or continuation vehicle, CV) is a GP-led secondary transaction: the GP of an aging fund moves one or several trophy assets into a new fund that it also manages, funded by new investors (typically specialist secondary buyers) and by existing LPs who elect to roll. Existing LPs are offered a choice: take cash at the transaction price or roll their interest into the new vehicle on similar or new terms. The GP typically *crystallizes* the carry earned to date on the assets, often by rolling that carry into the new vehicle as GP capital.

The rationale is genuine: the best companies in a fund may need more time to compound than the fund's term allows; the sale to a CV provides immediate liquidity to LPs who want it and continued exposure to those who want it. But the structure is also a *conflict of interest* in its purest form: the GP is on both sides of the transaction, setting the price at which the old fund sells and the new fund buys, and it generates fees and carry crystallization from the transaction. Governance safeguards therefore include an independent fairness opinion, a competitive bidding process led by an outside advisor, LP advisory committee approval, an election period of 30 days or more with full information, a roll option on equal-or-better terms, and a significant GP rollover.

Example 14.8 – Anatomy of a continuation fund. Fund I owns a company with a cost basis of **$250M** and a reported NAV of **$400M**. The GP proposes to sell it to a new continuation vehicle at **$380M**, a 5% discount to NAV, with an independent valuation supporting the price. Terms of the CV: management fee ignored for simplicity; carry of 10% on profits up to a 1.5x multiple on the $380M price (that is, on profit up to $190M), and 20% above that; no hurdle. The GP crystallizes its Fund I carry on the sale and rolls that amount into the CV. Fund I's carry rate is 20% and the fund has cleared its hurdle.

Step 1: crystallized carry. Profit at the sale price = 380 − 250 = **$130M**; carry = 20% × 130 = **$26M**. If the GP rolls the carry into the CV, it takes no cash from this part of the transaction, and of the $380M price, $26M is "paid" in carry (rolled) and **$354M** is new capital from buyers and rolling LPs.

Step 2: selling LPs' outcome. An LP that sells takes its share of the net proceeds: relative to the $400M NAV, it gets 5% less (and has paid carry on the price, not on the NAV). It trades 5% of value for immediate liquidity and certainty.

Step 3: CV investors' outcome. Suppose the asset is sold three years later for $520M. Profit above the $380M entry = **$140M**, which is within the first tier (up to $190M), so carry = 10% × 140 = **$14M**. Investors receive 520 − 14 = **$506M** on $380M: net multiple **1.33x**, net IRR = (506/380)^(1/3) − 1 = **10.0%**.

Step 4: sensitivity to exit value.

Exit value ($M)	Profit above $380M	Carry ($M)	Net multiple to CV investors	Net IRR (3 years)
380	0	0.0	1.00x	0.0%
450	70	7.0	1.17x	5.2%
520	140	14.0	1.33x	10.0%
650	270	35.0	1.62x	17.4%
800	420	65.0	1.93x	24.6%

(At $650M: carry = 10% × 190 + 20% × 80 = 19 + 16 = $35M. At $800M: 19 + 20% × 230 = 19 + 46 = $65M.)

Step 5: the GP's economics. The GP's rolled carry is $26M of the $380M, or 6.8% of the CV capital. At the $520M exit, the GP receives 26/380 × 506 = **$34.6M** on its rolled capital plus $14M of new carry, **$48.6M** in total, on $26M of capital that was previously only a contingent claim in Fund I. The GP has converted a paper claim into a tradable stake, restarted its fee stream on the asset, and secured a new carry option on further appreciation. The selling LPs have forgone further appreciation above $380M (they would have received it in Fund I with an 80/20 split), and rolling LPs accept a 10-20% carry on new profit.

Conclusion. The CV can be good for everyone: LPs gain a choice, the asset gets more time, and buyers earn an attractive return if the entry price is sound (10% net in the base case). But each party's outcome depends on the fairness of the $380M price. A price 10% too low transfers value from selling LPs to the new investors and the GP; a price too high transfers it the

other way. This is why independent valuation, competitive price discovery, and meaningful LP choice matter more than the headline terms. ∎

Common mistake. Treating a CV as simply "extending the fund." It is a sale and a repurchase of the same asset by two vehicles under one manager, and each party's decision depends on information the GP has and the LPs may not.

14.7 Performance Measurement Pitfalls

14.7.1 Stale and smoothed valuations

NAVs of private assets are *appraisals*, not prices. They are updated quarterly or less often, based on comparable-company multiples, recent financings, and GP judgment. The result is **smoothing**: reported returns lag true economic returns, show lower volatility, and have positive serial correlation. Smoothing makes private equity look safer and less correlated with public markets than it is, which flatters its Sharpe ratio and affects its role in portfolio allocation (see Volume 1, Chapter 10 on the CAPM and portfolio theory).

A simple model of smoothing (Geltner) is that the reported return is a weighted average of the true return and last period's reported return:

```
r_reported(t) = (1 − a) · r_true(t) + a · r_reported(t−1)
```

where a is the smoothing parameter ($0 \le a < 1$) and also the first-order autocorrelation of reported returns when true returns are independent. The variance relation is $\text{Var(reported)} = \text{Var(true)} \times (1 - a)/(1 + a)$, so the "true" volatility is the reported volatility multiplied by $\sqrt{[(1 + a)/(1 - a)]}$. The true return series can be recovered by "unsmoothing": $r_{true}(t) = [r_{reported}(t) - a \cdot r_{reported}(t-1)] / (1 - a)$.

Example 14.9 – Unsmoothing volatility and the Sharpe ratio. A buyout index reports an annualized volatility of **9.0%** and an average excess return over cash of **6.0%**. The first-order autocorrelation of the reported returns is **0.50**.

Step 1: volatility multiplier. $\sqrt{[(1 + 0.5)/(1 - 0.5)]} = \sqrt{3} =$ **1.732**.

Step 2: unsmoothed volatility. 9.0% × 1.732 = **15.6%**.

Step 3: Sharpe ratios. Reported: 6.0/9.0 = **0.67**. Unsmoothed: 6.0/15.6 = **0.38**.

Step 4: sensitivity. With autocorrelation 0.30 the multiplier is $\sqrt{(1.3/0.7)} = 1.36$, giving volatility 12.3% and Sharpe 0.49; with 0.60 the multiplier is $\sqrt{(1.6/0.4)} = 2.00$, giving volatility 18.0% and Sharpe 0.33.

Conclusion. The reported Sharpe ratio of 0.67 is nearly twice the economically relevant 0.38. Allocation models that use reported volatilities will overweight private equity, and risk measures (value at risk, stress tests) will understate its drawdown risk, particularly in a crisis when public markets fall immediately but private marks decline with a delay, producing the "denominator effect" at the plan level. ∎

14.7.2 Other major pitfalls

(1) Interim IRR is not final IRR. Funds in years 1-4 sit in the J-curve, and funds in years 5-8 carry unrealized NAV that is subject to revision. Comparing funds of different ages on net IRR is meaningless. Compare funds of the same *vintage year* only.

(2) Vintage dispersion. Returns depend heavily on the entry year: a fund that deploys in a period of low valuations can earn double the return of one that deploys at the peak, with identical skill. Quartile rankings must be done within vintage, and an LP's own commitment pacing diversifies across vintages.

(3) Survivorship and self-reporting bias. Databases of fund performance are built from voluntary reporting by GPs; poor performers and defunct funds may be underrepresented. Averages in data sets therefore tend to overstate the typical outcome, and "top-quartile" claims by GPs are based on selected benchmark universes and selected measures (gross vs net, IRR vs multiple, a flattering subset of funds).

(4) Persistence of performance. Early academic work found that top-quartile managers tended to remain top-quartile, which made manager selection the central skill; more recent evidence suggests persistence has weakened as the industry matured and as the number of funds grew. LPs should not assume that a track record of the *prior* fund predicts the *next* one, especially when fund size has risen sharply. Fund size growth dilutes returns because the opportunity set for a given strategy is limited.

(5) Gross versus net, and fee layering. A deal-level "2.4x gross multiple" says nothing about the LP's return after fees, carry, and expenses; Example 14.1 turned 2.125x into 1.67x. Fund-of-funds and feeder structures add a second layer of fees (commonly around 0.5-1% plus 5-10% carry), which can push net returns below public alternatives. Co-investment and direct programs reduce the layers but require internal expertise.

(6) Realized versus unrealized. NAV-heavy funds depend on the GP's marks. Check the *mix* of realized and unrealized value; look at the realized track record separately; and compare valuation multiples to what the same GP actually achieved on previous exits ("markup versus mark-out" analysis).

(7) Cash-flow timing choices. Besides sub lines, GPs can delay calls, accelerate distributions through dividend recapitalizations (which return cash quickly and raise the IRR even if the remaining company is more leveraged and riskier), and time exits to the fund's fundraising calendar.

(8) Benchmark choice and illiquidity. Comparing private equity to a large-cap index understates the risk of small leveraged companies, and ignoring the illiquidity premium that LPs rightly require means that an excess return of a few points may not compensate for the costs of illiquidity, fees, and the need to hold uncalled commitments.

Managerial implications. For a pension CIO such as Rosalind Park, the practical protocol is: (i) require net-of-all-fees cash flows by date for every fund in the track record; (ii) compute DPI, TVPI, net IRR with and without the sub line, and KS-PME for a size- and strategy-matched benchmark; (iii) compare within vintage; (iv) examine realized versus unrealized value; (v) negotiate the fee base, offsets, waterfall type, clawback and GP commitment; and (vi) compare the entire package with a lower-cost liquid alternative, such as a levered small-cap index. For a corporate leader whose company is owned by a PE fund, understanding fund economics explains why the sponsor presses for a particular exit horizon, a dividend recapitalization, or a sale into a continuation vehicle.

Case Study: Alder Point Partners and the Fund II Continuation Vehicle

(Illustrative company; all figures are fictional.)

Alder Point Partners is a mid-market buyout firm with three funds under management. Fund II ($900 million, vintage 2014) is in its twelfth year after two one-year extensions. It has distributed 1.6x of paid-in capital and holds one remaining asset: Brightwater Logistics, a regional freight-forwarding company bought for $250 million (equity) in 2017. Brightwater has grown earnings at 14% per year and is carried at $600 million of NAV. Fund II's terms: 2% fee on commitments (stepping down to 1.5% on invested capital after year 5), 20% carry with an 8% compounded hurdle and 100% catch-up, whole-fund

waterfall. Fund II has already cleared its hurdle and paid carry on prior distributions. The GP's advisory committee has been told that continuing to hold Brightwater beyond the extension would require a third extension that the LPs are unlikely to grant.

Alder Point proposes to move Brightwater into a continuation vehicle, "Alder Point Continuation Fund I", at a price of $552 million, a 8% discount to NAV, supported by a fairness opinion from a boutique advisor and a competitive process in which three secondary buyers submitted indications between $520 million and $565 million. The lead investor, a secondary fund, would commit $330 million. Fund II's LPs may either sell for cash at the transaction price or roll into the CV. Alder Point would roll its crystallized carry from Fund II (20% of the gain on the Brightwater sale) and commit an additional $15 million of new capital. The CV would charge a 1.25% management fee on invested capital and a tiered carry: 10% of profits up to 1.5x, 20% above 1.5x, no hurdle. The business plan projects a sale in four years at $850 million.

Opinions in the LP community differ. A large endowment, which would like to realize cash, supports the sale. A pension plan with a 12% private markets allocation, led by its CIO, is concerned that the process was run by the GP, that the winning bidder is a long-standing co-investor of Alder Point, and that the GP's crystallized carry is large compared with the GP's new cash commitment. A smaller family office points out that Fund II has been the firm's best performer, and Alder Point's fundraising for Fund IV is under way, so a strong "realized" track record for Fund II (through a sale rather than an extension) would help the fundraise. One LP asks whether the 8% discount is too large, given that Brightwater's earnings have not declined, and whether LPs would get a better outcome by simply approving a one-year extension.

The advisory committee must decide whether to approve the transaction. It has asked its analyst to estimate the economics for each party.

Discussion questions

1. Compute the GP's crystallized carry on Brightwater at the $552 million price (cost $250 million, carry 20%, hurdle cleared). Compare it with the new cash capital of $15 million. What does this say about alignment in the CV? What additional term would you request?
2. Compute the net multiple and 4-year IRR to CV investors if Brightwater is sold for $850 million, using the tiered carry (ignore management fees), and for sale prices of $552 million and $650 million. Compute also the GP's total proceeds in the $850 million case, assuming its rolled carry is invested pro rata with the CV capital.
3. Evaluate the 8% discount from the viewpoint of a selling LP: at what discount does a seller break even versus a one-year extension, assuming the asset's value grows 14% in that year and the LP has a cost of capital of 10%? What information would you want about the competitive process?
4. Identify the conflicts of interest in this transaction and the governance safeguards that would make you comfortable voting to approve it. How should an LP decide between rolling and selling?

Key Takeaways

- A PE fund is a finite-life partnership whose economics are set in the LPA: the fee base and step-down, the hurdle, the catch-up, the waterfall type, the clawback, and the GP commitment collectively determine what LPs keep.
- Fees take a large, certain share of profit: in the worked fund, fees consumed 14% and carry 17% of gross profit, turning a 2.1x gross multiple and 18.9% gross IRR into a 1.67x net multiple and 13.3% net IRR.
- The European waterfall has four tiers (return of capital, preferred return, catch-up, split); with a full catch-up the GP ends with 20% of total profit if the fund is successful, so the hurdle matters mostly for timing and for marginal funds.

- Deal-by-deal (American) waterfalls pay carry earlier and shift the risk of later losses to LPs until a clawback is settled; the clawback's value depends on escrow and the creditworthiness of GP partners.
- The J-curve arises from fees, conservative early marks, and the timing of calls and distributions; interim IRRs in the first years say more about fund age than quality, and DPI is the metric that cannot be revised.
- IRR rewards speed and implicitly assumes reinvestment at the IRR; multiples ignore time; neither alone is sufficient, and MIRR makes the reinvestment assumption explicit.
- Subscription lines raise reported IRR and lower the multiple by shortening the invested period; one-year delays can add several points of IRR without creating any additional dollars of profit.
- A Kaplan-Schoar PME compares a fund to the index on the same cash flow dates; a fund can have a positive IRR and a multiple above 1.0 and still have a PME below 1.0.
- Carry is a call option, so GPs have an incentive to take more risk than LPs; the GP commitment, advisory committees, and fund guidelines address this imperfectly.
- Continuation funds can provide liquidity and more time for the best assets but place the GP on both sides of a transaction; independent valuation, competitive price discovery, genuine LP choice, and a significant GP roll are the key safeguards.
- Reported private-equity risk is understated by smoothed appraisals, and performance comparisons must be done within vintage, net of all fees, and alongside a public benchmark.

Review Questions

1. Describe the sequence of cash flows between an LP and a PE fund from commitment to wind-down. Why is the unfunded commitment a cost to the LP?
2. Why do fees on committed capital make the J-curve deeper? How does a step-down to invested capital change the economics for the LP?
3. Explain each tier of a European waterfall. What happens to the GP's share of total profit if the catch-up is 50% rather than 100%, or if there is no catch-up?
4. Compare European and American waterfalls from the LP's and the GP's viewpoints. What is a clawback and what weaknesses does it have?
5. Explain why carried interest is analogous to a call option. What behavior does this encourage, and how might an LP mitigate it?
6. Why might a fund with a lower multiple have a higher IRR than another fund? Give a numerical example.
7. Define DPI, RVPI, and TVPI. Why do experienced LPs emphasize DPI?
8. How do subscription lines affect IRR and multiple? How would you adjust performance for them?
9. What does a KS-PME of 0.90 mean? Why can a fund with a positive IRR have a PME below 1?
10. List four pitfalls in measuring PE performance and explain how an LP can guard against each.

Problems

1. **Fee drag.** A $500M fund charges 2% of commitments for 5 years and 1.5% of $250M of invested capital for the next 5 years. (a) Compute total fees and total fees as a percentage of commitments. (b) If the fund invests only the capital net of fees (that is, commitments less fees), what is the amount invested?

2. **Single-period waterfall.** An LP contributes $100M; four years later the fund distributes $180M. Hurdle 8% compounded, 100% catch-up, 20% carry. Compute the hurdle amount, the catch-up, the LP's and GP's totals, and the GP's share of profit. Compare the GP's take with no catch-up.
3. **Clawback.** Deal A ($50M invested) is sold for $150M in year 3; Deal B ($50M invested) is sold for $20M in year 5. Carry is 20% with no hurdle on a deal-by-deal basis. Compute the carry paid after A, the carry owed after B, and the clawback.
4. **IRR versus multiple.** Deal A turns $100M into $160M after 2 years; Deal B turns $100M into $300M after 7 years. Compute the multiples and IRRs. Which is better if the LP can reinvest at 8%?
5. **Subscription line.** A fund buys a company for $100M and sells it after 3 years for $150M. Compute the multiple and IRR with no sub line. Then suppose the fund borrows for the first year at 5%, with LPs funding the loan payoff at the end of year 1. Recompute the multiple and IRR and comment.
6. **PME.** An LP's fund has cash flows (years 0-4): −100, −50, 0, +90, +100. The index levels are 100, 110, 120, 130, 150. Compute the KS-PME, the multiple, and the fund's IRR. Does the fund beat the index?
7. **Continuation fund.** An asset has cost $100M and is sold into a continuation vehicle at $180M. The old fund's carry is 20% and has cleared its hurdle. In the CV, carry is 10% of profit above $180M. The asset is sold for $270M three years later. Compute the crystallized carry, the CV carry, and the CV investors' net multiple and IRR (ignore fees).
8. **Total fees.** A $300M fund charges 2% of commitments for 5 years, then 1.5% on $150M of invested capital for 5 years. Compute total fees and total fees as a percent of commitments. If gross profit on the invested capital is $270M, what share of gross profit do fees take?

Answers to Selected Problems

1. a. Years 1-5: 0.02 × 500 × 5 = $50.0M; years 6-10: 0.015 × 250 × 5 = $18.75M; total **$68.75M = 13.75%** of commitments. (b) Invested = 500 − 68.75 = **$431.25M**.
2. Hurdle = 100 × 1.08^4 = **$136.05M**. Profit above hurdle = 180 − 136.05 = 43.95; the pref is 136.05 − 100 = 36.05, so catch-up = 0.25 × 36.05 = **$9.01M**. Remainder = 43.95 − 9.01 = $34.94M; LP 80% = 27.95; GP 20% = 6.99. LP total = 136.05 + 27.95 = **$164.0M**; GP total = 9.01 + 6.99 = **$16.0M** = 20% of the $80M profit. Without a catch-up, GP = 0.2 × 43.95 = **$8.79M**.
3. Carry after A = 0.2 × 100 = **$20M**. After B, cumulative profit = 100 − 30 = 70, carry owed = **$14M**; clawback = **$6M**.
4. Multiples: **1.6x** and **3.0x**. IRRs: 1.6^(1/2) − 1 = **26.5%**; 3^(1/7) − 1 = **17.0%**. With 8% reinvestment A's $160M grows to 160 × 1.08^5 = $235.1M at year 7, a multiple of 2.35x and MIRR 13.0%, versus B at 17.0%: **B is better**.
5. No sub line: **1.50x**, IRR = 1.5^(1/3) − 1 = **14.5%**. With the line: LPs contribute $105M at year 1 and receive $150M at year 3: multiple = 150/105 = **1.43x**; IRR = (150/105)^(1/2) − 1 = **19.5%**. IRR rises 5 points while the multiple falls and profit falls from $50M to $45M.
6. FV of contributions: 100 × 1.5 + 50 × (150/110) = 150 + 68.18 = 218.18. FV of distributions: 90 × (150/130) + 100 × (150/150) = 103.85 + 100 = 203.85. KS-PME = 203.85/218.18 = **0.934**. Multiple = 190/150 = **1.27x**. IRR = **7.7%**. The fund does **not** beat the index (PME < 1).
7. Crystallized carry = 0.2 × 80 = **$16M**. CV profit = 270 − 180 = 90; CV carry = 0.1 × 90 = **$9M**. Net to investors = 261; multiple = 261/180 = **1.45x**; IRR = 1.45^(1/3) − 1 = **13.2%**.

8. Fees = 0.02 × 300 × 5 + 0.015 × 150 × 5 = 30 + 11.25 = **$41.25M = 13.75%** of commitments. As a share of gross profit: 41.25/270 = **15.3%**.

Chapter 15: Real Assets, Infrastructure and Project Finance

Learning Objectives

After studying this chapter you should be able to:

- Explain why sponsors use a special-purpose vehicle (SPV) and non-recourse debt, and identify the project characteristics that make project finance preferable to corporate finance.
- Map the contractual web around a project (EPC, O&M, offtake, supply, financing and security documents) and explain how each contract moves a specific risk away from lenders.
- Apply the principle of risk allocation to construction, resource, price, operating, financial and political risks, and evaluate a risk matrix critically.
- Compute and interpret DSCR, LLCR, PLCR and gearing; size debt by sculpting repayments to a target DSCR; and explain lock-up and default covenants.
- Describe the structure of a project finance model (timing, inputs, construction, operations, debt, tax, waterfall, outputs) and the audit and sensitivity discipline that makes it reliable.
- Evaluate renewable-energy project economics (energy yield, P50/P90, degradation, LCOE, equity IRR) and assess a public-private partnership (PPP) on value for money and risk transfer.

Opening Vignette

Illustrative scenario; the company and all figures are fictional.

Marlow Power Partners is a mid-sized developer that has spent three years assembling a 100 MW solar farm in a sunny, grid-connected region. It holds the land lease, the grid connection agreement and a 20-year power purchase agreement (PPA) with a regional utility at a fixed price per megawatt-hour. Construction will cost about 120 million, including financing fees and interest during construction. Marlow's own balance sheet is modest: total equity of about 60 million, and its existing corporate lenders will not lend another 90 million against a company this size.

Marlow's chief financial officer, Imani Okafor, proposes a different route. Rather than borrowing at the corporate level, Marlow will create a new company, Marlow Solar One Ltd, that owns only this farm. Banks will lend to that company against the project's contracted cash flows, with no claim on Marlow's other assets. Marlow and a pension-fund co-investor will provide the equity.

The board is sceptical. "You are telling me lenders will put in three-quarters of the money and we can walk away if it fails?" asks the chair. Imani's reply frames the chapter: the lenders are not betting on Marlow, they are betting on a contract web. The engineering, procurement and construction (EPC) contractor guarantees the price and completion date; the PPA fixes the revenue; an operator guarantees availability; and the debt is sized so that, even in a bad solar year, cash flow covers debt service with a margin. The lenders will charge for the risks they keep, and they will police the structure with covenants.

Over the next hour the board asks the questions this chapter answers. Why a separate company? What exactly does "non-recourse" protect? How is the debt sized, and what does a debt service coverage ratio of 1.30 mean? What happens if the sun is weaker than forecast, or if construction runs six months late? And, ultimately: does a project that clears all of the lenders' tests still earn the sponsors an acceptable return?

15.1 Why Project Finance Exists

15.1.1 Corporate finance versus project finance

In the corporate finance of Volume 1 (Chapters 6, 11 and 15), a firm raises capital against the whole of its balance sheet. Lenders look at the firm's diversified earnings, assets and management, and they have recourse to everything the firm owns. The investment decision (the project's NPV) is made inside the firm, and financing is largely a secondary decision governed by capital-structure trade-offs.

Project finance inverts this. The project is financed as a stand-alone economic unit: a legally separate company is created to build and operate one asset (or a closely related group of assets), and lenders look to that company's cash flows, and to the assets it owns, as the primary and sometimes sole source of repayment. The financing is therefore inseparable from the project's contracts, risk allocation and cash-flow forecast. In effect, the lenders underwrite a contract structure rather than a corporate credit.

The distinction is one of degree, not kind. Few project loans are *completely* non-recourse; during construction, sponsors often provide completion guarantees, equity commitments or standby facilities. The term "limited recourse" is more accurate for most real transactions. What matters is that recourse to the sponsor is bounded, defined by contract, and usually falls away once the asset is built and proven.

15.1.2 Which projects suit project finance

Project finance is not suited to every investment. It works best when several conditions hold:

1. **Large, discrete, long-lived assets** with high up-front capital cost and low operating cost: power plants, wind and solar farms, transmission lines, toll roads, pipelines, ports, LNG terminals, hospitals, water treatment plants, data-center campuses.
2. **Predictable, contractable revenue.** The more of the revenue stream is fixed by a long-term contract with a creditworthy counterparty (take-or-pay offtake, availability payment, regulated tariff), the more debt the project can support.
3. **Transparent, ring-fenceable cash flows.** The asset must be separable from the sponsor's other businesses. Shared assets, cross-subsidies and intra-group pricing make ring-fencing difficult.
4. **Controllable risks.** Construction, operating and technology risks need to be understood and allocatable by contract. A first-of-a-kind technology with no performance guarantee is generally unfinanceable on a non-recourse basis.
5. **A finite life.** Lenders prefer an asset whose cash flows will pay off the debt well before the end of the contract or asset life, leaving a tail.

15.1.3 Why sponsors choose it

Sponsors accept the cost and complexity for four reasons. **Higher leverage:** because debt is sized to project cash flows rather than the sponsor's credit, a project may carry 70 to 85 percent debt even if the sponsor's own gearing target is far lower. **Limited balance-sheet impact:** depending on accounting standards, ownership and control, the debt may be off

balance sheet or at least labelled non-recourse, though rating agencies often look through to guarantees. **Risk isolation:** a project failure is confined to the equity invested and contractual guarantees. **Partnership and discipline:** a joint-venture SPV lets sponsors, offtakers and infrastructure funds each hold a stake that suits their risk appetite, and lenders' advisers scrutinise the project more thoroughly than many boards would.

15.1.4 The costs of project finance

Documentation is extensive, with transaction costs commonly a few percent of project cost; lenders charge margins above corporate debt and impose covenants and reserve accounts that delay equity distributions; and negotiation takes many months. As the Modigliani-Miller logic of Volume 1, Chapter 15 reminds us, leverage does not create value by itself; value comes from tax shields, risk-sharing, reduced agency costs, or access to capital that would otherwise be unavailable.

> **Common mistake.** Treating project finance as "free leverage." The risk the sponsor does not bear through recourse is borne by lenders, who price it through higher margins, tighter covenants and lower leverage. Cheaper-looking debt is paid for elsewhere in the structure.

15.2 The SPV and the Contractual Web

15.2.1 The special-purpose vehicle

The **special-purpose vehicle** (SPV, sometimes "project company" or "ProjectCo") is a newly formed company whose only business is the project. Its constitution typically prohibits other activities, other debt and other subsidiaries. The SPV signs all the project contracts, holds the permits and the assets, borrows from the lenders, and pays dividends to its shareholders (the sponsors) only after the lenders' cash waterfall has been satisfied.

The SPV's purpose is to make a defined set of cash flows legally and economically separate. This protects lenders against a sponsor's insolvency (the project's assets are not part of the sponsor's estate) and protects the sponsor against the project's insolvency (the project's creditors cannot claim against the sponsor except under agreed guarantees). It also ensures that project cash flow cannot be diverted to the sponsor's other needs.

15.2.2 The parties

The sponsors (developers, utilities, infrastructure and pension funds) contribute equity. Senior lenders (commercial banks, export-credit agencies, development banks, institutional bond investors) hold first-ranking security, with mezzanine lenders behind them if any. Around the SPV sit the EPC contractor, the operator, the offtaker, input suppliers, insurers, public authorities and a team of advisers (independent engineer, market and insurance consultants, model auditor, lawyers).

15.2.3 The contract map

The project's value is carried by its contracts. Table 15.1 shows the core contracts and the principal risk each is designed to address.

Table 15.1 – Core project contracts and the risks they address

Contract	Counterparty	Risk addressed	Typical lender protection
EPC contract	Contractor (often a consortium)	Cost overrun, delay, defects, performance shortfall	Fixed price, fixed date, liquidated damages (LDs), performance bond, parent guarantee
Offtake / PPA	Utility, government agency, corporate buyer	Price and volume of revenue	Long-term fixed or indexed price, take-or-pay or availability payment, creditworthy counterparty
O&M agreement	Operator	Availability, maintenance cost	Fixed fee, availability guarantee, bonus-malus
Supply agreement	Fuel, feedstock or equipment supplier	Input price and availability	Pass-through or fixed price, supply-or-pay
Concession / licence	Public authority	Right to operate, regulation, change in law	Compensation for termination, change-in-law protection
Insurance	Insurers	Physical damage, delay in start-up, business interruption	Lenders named as loss payees, minimum cover
Loan agreement and security	Lenders	Repayment	Security over assets, accounts, contracts, shares; covenants; step-in rights
Direct agreements	Each key counterparty and the lenders	Termination, counterparty default	Lender right to cure and step in before contract termination

Two features of this map deserve emphasis. First, the **direct agreements** (also called consents or tripartite agreements) are crucial but easy to overlook: they give the lenders the right to be notified of defaults and to cure them, or to step in and replace the SPV, before the offtaker or contractor terminates. Without them, a contract that looks strong on paper can be terminated at the moment the lenders most need it. Second, every contract has a *counterparty credit risk* of its own: an LD of 10 percent of the EPC price is only as good as the contractor's balance sheet or parent guarantee.

15.2.4 The security package and cash waterfall

Lenders take security over everything the SPV owns: the assets, the bank accounts, the receivables, the rights under the project contracts and the sponsors' shares in the SPV. Practically, the lenders' strongest tool is **control over the cash**. All project revenues are paid into controlled accounts, and a contractual waterfall determines the order in which cash is released:

1. Operating costs, taxes and statutory payments.
2. Senior debt fees and interest.
3. Scheduled senior principal.
4. Top-up of the **debt service reserve account** (DSRA), commonly sized at six months of the next debt service payments.
5. Top-up of the **major maintenance reserve account**, where applicable.
6. Mezzanine or subordinated debt service.
7. Any mandatory prepayments (cash sweep).
8. Distributions to equity, only if the lock-up and default tests are passed (Section 15.4.4).

The waterfall explains why project equity is the residual claim in the strictest sense: its return arrives last, and only if the project is healthy. It also explains why equity IRR in project finance is sensitive to relatively small changes in revenue, as leveraged cash flows magnify operating variation (Example 15.5).

15.3 Risk Allocation

15.3.1 The governing principle

The central rule of project finance risk allocation is simple to state and difficult to apply: **each risk should be borne by the party best able to control it, or, if it cannot be controlled, best able to price and diversify it, at least cost.** An EPC contractor controls construction methods and procurement, so it should bear cost-overrun and delay risk (and be paid a premium for doing so). A government controls regulation and tariffs, so it should bear change-in-law risk. Nobody controls weather, so resource risk is shared between insurance, sponsor equity and lenders' debt cushion, and priced through the DSCR.

When the parties allocate risk to the wrong party, the project becomes more expensive, not cheaper. A contractor asked to accept an unpriceable risk will either add a large contingency or walk away. A government that retains a risk it cannot manage will bear the consequences later, often at greater cost.

15.3.2 The risk matrix

Table 15.2 shows a stylised risk matrix for a contracted renewable project. Actual matrices are far more detailed, but the pattern is representative.

Table 15.2 – Stylised risk allocation matrix for a contracted solar project

Risk	Description	Primary holder	Mitigants
Construction cost	Capex exceeds budget	EPC contractor	Fixed-price, date-certain turnkey contract; contingency; parent guarantee
Completion / delay	COD (commercial operations date) later than planned	EPC contractor, then sponsor	Delay LDs sized to cover interest and fixed costs; delay-in-start-up insurance; sponsor completion support
Technology / performance	Panels or inverters underperform	EPC contractor, equipment supplier	Performance tests, warranties, performance LDs
Resource (irradiation)	Solar yield below forecast	Equity and lenders	Independent yield assessment (P50/P90), debt sized on P90 or with DSCR margin, DSRA
Offtake / price	Counterparty fails to pay; price is lower	Offtaker (price), lenders (credit)	Long-term PPA, creditworthy offtaker, termination payments
Operating cost	O&M costs rise	O&M contractor, then SPV	Fixed-price O&M with indexation caps
Interest rate	Floating rates rise	SPV	Fixed-rate debt or swaps covering 75-100 percent
Foreign exchange	Revenue and debt in different currencies	SPV or offtaker	Currency-matched debt, indexed tariffs
Political / regulatory	Expropriation, retroactive tariff change, change in law	Government, political-risk insurers	Concession compensation, MIGA or ECA cover, international arbitration
Force majeure	Events beyond all parties' control	Shared	Insurance, termination compensation formulas
Merchant / residual	Revenue after PPA expires	Equity	Debt fully repaid before PPA end; no reliance on merchant revenue

15.3.3 Construction and completion risk

Construction is usually the riskiest phase: cash is committed, nothing is yet earning, and the collateral is a partly built asset worth far less than its cost. Lenders therefore insist on **fixed-price, date-certain, turnkey** contracts, **performance bonds** or parent guarantees, and **delay and performance LDs**. A well-structured delay LD covers at least the extra interest during construction and the SPV's fixed costs, so lenders' ratios are not impaired; but LDs are usually capped (often 10 to 20 percent of contract price). Lenders also require budget contingency and sponsor equity undertakings, sometimes backed by letters of credit. Example 15.1 quantifies the gap between delay LDs and the true cost of delay.

15.3.4 Operating, market and political risk

During operations, technical risk shrinks and volume, price and counterparty risks dominate. For contracted assets the offtaker's credit quality is critical: a PPA with a weak utility is worth much less than one with a strong buyer, and lenders reflect this in margin, DSCR and tenor. Where revenue is partly merchant, lenders demand higher DSCRs, shorter tenors, cash sweeps and sometimes hedges. Political risk includes expropriation and inconvertibility but, most commonly in practice, retroactive changes to tariffs, subsidies or taxes; mitigants include political-risk insurance, multilateral participation, treaty protection and stabilisation clauses, though none is perfect.

15.3.5 Quantifying risk: a delay example

Example 15.1 – Delay liquidated damages versus the actual cost of delay. The Marlow solar project (Section 15.6) has 90 million of senior debt at 5.5 percent, fully drawn at the completion date. The EPC contract provides delay LDs of 12,000 per day (that is, 0.012 million). Construction finishes 180 days late. The SPV continues to incur 0.30 million of fixed costs over the delay period (insurance, site and administration costs not covered by the EPC price).

Step 1: Extra interest during construction. Interest over 180 days (taken as half a year) on 90 million at 5.5 percent: 90 × 0.055 × 0.5 = 2.475 million.

Step 2: Additional fixed costs. 0.30 million.

Step 3: Total delay cost to the SPV. 2.475 + 0.30 = 2.775 million.

Step 4: LD recovery. 0.012 million per day × 180 days = 2.160 million.

Step 5: Shortfall. 2.775 − 2.160 = 0.615 million. This is a direct cost to equity (or must be funded by a contingency or standby equity).

Step 6: Broader consequence. The PPA term and tariff typically start at COD, so the entire cash-flow profile shifts six months later; the debt grace period and the equity cash flows both move. The equity's loss of time value is additional to the shortfall and, because equity bears the residual, can easily exceed it.

Interpretation. LDs rarely make the SPV whole. They are sized to protect lenders' coverage ratios, not to compensate sponsors for lost time value. Sponsors should negotiate LD rates on the basis of the project's real daily cost of delay, not a standard percentage of contract value.

15.4 Non-Recourse Debt: Sizing, Coverage Ratios and Covenants

15.4.1 Cash flow available for debt service

All project coverage ratios start from **cash flow available for debt service (CFADS)**: the cash the project generates in a period that is available to pay interest and scheduled principal. It is defined contractually in the loan agreement, but in essence:

```
CFADS = Revenue − Operating costs − Taxes paid ± Working-capital movements
        (± Capex or maintenance reserve movements, as defined in the loan agreement)
```

Debt service itself is not deducted, and neither is any distribution to equity. Defining CFADS carefully is itself a negotiation: sponsors prefer definitions that include insurance proceeds, reserve releases and interest income; lenders prefer narrow ones. Two models with the same cash flows but different CFADS definitions can produce different headline ratios, so always read the definition before comparing ratios across deals.

15.4.2 The coverage ratios

Debt service coverage ratio (DSCR). For each period t:

```
DSCR_t = CFADS_t / (Interest_t + Scheduled principal_t)
```

It measures how many times the period's debt service is covered by the cash flow. A DSCR of 1.30 means CFADS can fall by about 23 percent (1 − 1/1.30) before debt service is no longer fully covered. Lenders focus on the **minimum** DSCR over the loan life, and the **average** DSCR, which indicates the typical cushion.

Interest cover ratio (ICR). CFADS divided by interest alone; used mainly during early operations or for bullet loans.

Loan life coverage ratio (LLCR). At any date, LLCR compares the present value of CFADS over the remaining loan life to the outstanding debt:

```
LLCR = [ Σ (t=1..N) CFADS_t / (1 + r)^t  +  DSRA balance ] / Debt outstanding
```

where r is the weighted average cost of the senior debt (the discount rate for the loan) and N is the remaining loan life. LLCR is a "solvency" style measure: it asks whether the cash flows over the loan's remaining life are sufficient, in present-value terms, to repay the debt. A project whose DSCR is dipping in one bad year might still have a comfortable LLCR if later years are strong.

Project life coverage ratio (PLCR). The same calculation, but with CFADS over the whole project (or contract) life rather than the loan life. The difference between PLCR and LLCR measures the "tail": the cash flow after the loan is repaid, which acts as an additional cushion. A tail of two years on a 20-year project is typical for contracted renewables.

Gearing (debt-to-total-capital). Debt / (Debt + Equity) at financial close; often capped (for example, at 75 percent) regardless of what DSCR sizing would allow.

Table 15.3 compares the ratios.

Table 15.3 – Coverage ratios compared

Ratio	Formula	Horizon	What it tells lenders	Typical use
DSCR	CFADS / debt service	One period	Liquidity cushion each year	Sizing, distribution lock-up, default trigger
ICR	CFADS / interest	One period	Ability to pay interest	Early operations, bullet debt
LLCR	PV(CFADS over loan life) / debt	Loan life	Solvency over remaining loan	Refinancing tests, default triggers
PLCR	PV(CFADS over project life) / debt	Project life	Tail cushion beyond loan	Tenor decisions, refinancing capacity
Gearing	Debt / (Debt + Equity)	At close	Equity skin in the game	Debt-capacity cap, equity commitment

Typical DSCR targets vary by revenue quality: contracted infrastructure with strong offtakers may carry minimum DSCRs of around 1.20 to 1.35 on a P50 case; assets with merchant exposure often require 1.7 or higher or conservative P90-based sizing. These ranges are illustrative; actual terms depend on the market, jurisdiction and offtaker rating.

Example 15.2 – Computing DSCR, average DSCR and LLCR. A small project has the following annual CFADS and scheduled debt service over a six-year loan (million). The loan's cost is 6 percent.

Year	1	2	3	4	5	6
CFADS	14	16	18	18	16	14
Debt service	10	11	12	12	11	9

Step 1: Annual DSCRs. 14/10 = 1.400; 16/11 = 1.455; 18/12 = 1.500; 18/12 = 1.500; 16/11 = 1.455; 14/9 = 1.556.

Step 2: Minimum and average. The minimum DSCR is 1.400 (year 1); the average of the six ratios is (1.400 + 1.455 + 1.500 + 1.500 + 1.455 + 1.556)/6 = 8.865/6 = 1.477.

Step 3: Debt outstanding at the start. Since debt service repays the loan, the debt outstanding equals the PV of debt service at 6 percent: $10/1.06 + 11/1.06^2 + 12/1.06^3 + 12/1.06^4 + 11/1.06^5 + 9/1.06^6 = 9.434 + 9.790 + 10.075 + 9.505 + 8.220 + 6.345 = 53.369$ million.

Step 4: PV of CFADS over the loan life at 6 percent. $14/1.06 + 16/1.06^2 + 18/1.06^3 + 18/1.06^4 + 16/1.06^5 + 14/1.06^6 = 13.208 + 14.240 + 15.113 + 14.258 + 11.956 + 9.869 = 78.644$ million.

Step 5: LLCR. 78.644 / 53.369 = 1.474 (ignoring any DSRA balance).

Interpretation. The LLCR is close to the average DSCR because debt service is roughly proportional to CFADS. Where debt service is front-loaded relative to CFADS (as with a level annuity on a declining cash-flow profile), the minimum DSCR falls well below the LLCR, which is why lenders watch both.

15.4.3 Sizing debt: sculpting to a target DSCR

Rather than offering a loan and testing the ratios afterwards, lenders usually **size the debt from the cash flows**. The sponsor and lenders agree a target DSCR, a tenor and an interest rate; the model then computes the maximum debt the project can service. Two repayment profiles are common.

Annuity or level repayment. Debt service is constant (or a standard mortgage-style profile). The binding DSCR is the lowest, which occurs in the year when CFADS is lowest. If CFADS declines over time (as with degrading solar output), the minimum DSCR occurs late in the loan, and debt capacity is limited by that year.

Sculpted repayment. Debt service in each period is set equal to CFADS divided by the target DSCR. Repayments thereby track the cash-flow profile, and the DSCR is constant over the loan life. Debt capacity is the present value of the sculpted debt service at the loan's interest rate:

```
Debt service_t = CFADS_t / target DSCR
Debt capacity  = Σ Debt service_t / (1 + r)^t
```

Sculpting maximises debt for a given minimum DSCR, and with a constant DSCR the LLCR at financial close equals the target DSCR as well. The practical complication is **circularity**: interest depends on the opening balance, which depends on earlier repayments, which depend on debt service sizing, which depends on CFADS, which (through taxes and interest income) can depend on interest. Models solve this with iteration, macros or closed-form formulas.

In addition to the DSCR test, lenders apply a **gearing cap**, so debt capacity is the lesser of the DSCR-sized amount and the maximum percentage of total project cost.

Example 15.3 – Sculpted debt sizing for the Marlow solar project, with a binding DSCR. The project has the following assumptions (pre-tax, for clarity; real models include tax):

- Capacity 100 MW at a 25 percent capacity factor: P50 energy = 100 × 0.25 × 8,760 h = 219,000 MWh = 219 GWh in year 1.
- Degradation 0.5 percent per year.
- PPA price: 62 per MWh, flat nominal for 20 years.
- Operating costs: 2.6 million in year 1, escalating at 2 percent per year.
- Total project cost: 120 million.
- Debt: 18-year tenor, 5.5 percent fixed, target DSCR 1.30 (sculpted), gearing cap 75 percent.

Step 1: Revenue and CFADS, year 1. Revenue = 219,000 MWh × 62 = 13.578 million. CFADS = 13.578 − 2.600 = 10.978 million.

Step 2: CFADS, later years. Energy in year t is 219 × 0.995^(t−1) GWh; costs rise 2 percent a year. Year 2: revenue 13.510, costs 2.652, CFADS 10.858. Year 18: revenue 12.469 ($219 \times 0.995^{17} \times 62/1{,}000$), costs 3.641, CFADS 8.828. Year 20: CFADS 8.557 million. The profile declines steadily, because both degradation and cost inflation erode CFADS while the price is flat.

Step 3: Sculpted debt service. Year 1: 10.978 / 1.30 = 8.445 million. Year 2: 10.858 / 1.30 = 8.352. Year 18: 8.828 / 1.30 = 6.791.

Step 4: Debt capacity. The PV of these 18 amounts at 5.5 percent is 87.392 million.

Step 5: Gearing test. 87.392 / 120 = 72.8 percent, below the 75 percent cap (90 million). **The DSCR constraint is binding**; debt is 87.4 million.

Step 6: Repayment schedule. Year 1 interest = 87.392 × 5.5% = 4.807; principal = 8.445 − 4.807 = 3.638; closing balance 83.754. Year 2 interest = 4.606; principal = 3.746; balance 80.008. The final payment in year 18 has interest of 0.354 and principal of 6.437, leaving a zero balance.

Step 7: LLCR and PLCR at close. The LLCR equals 1.30 by construction. The PLCR, which includes years 19 and 20, is 1.370; the extra 0.07 is the tail.

Step 8: Compare with a level annuity. A level 18-year annuity at 5.5 percent for the same 87.392 million has annual debt service of $87.392 \times 0.055 / (1 - 1.055^{-18}) = 7.771$ million. The DSCR is then 10.978/7.771 = 1.413 in year 1, 1.287 in year 9 and 8.828/7.771 = **1.136 in year 18**. A lender with a 1.30 minimum DSCR would have to cut the loan substantially under an annuity profile. Sculpting therefore supports considerably more debt for the same risk metric.

Interpretation. The sculpted profile front-loads less principal than an annuity, because CFADS falls over time. The project's equity gets cash earlier too, which matters for IRR.

15.4.4 Covenants: lock-up, default and reserves

Coverage ratios become operational through covenants, usually two thresholds:

- **Distribution lock-up** (for example, historic and projected DSCR below 1.15): equity distributions are blocked and cash is trapped in the project account. If the ratio recovers within a period, the trapped cash is released; if not, it may be applied to prepay debt.
- **Event of default** (for example, DSCR below 1.05): lenders may accelerate, enforce security or step in.

The gap between the base case and the lock-up ratio is the cushion for operating variation; the gap between lock-up and default is the cure period, which gives the sponsor time to react. Lock-ups are a gentle but effective form of control: they protect lenders, but also cost sponsors, because trapped cash defers equity returns and the IRR penalty is immediate.

Reserve accounts provide liquidity. A **DSRA** of six months' debt service allows the project to meet payments through a temporary shortfall. A **major maintenance reserve account** accumulates cash for large future repairs (for example, inverter replacement or turbine overhaul). Sponsors often replace cash-funded reserves with letters of credit, which free cash but add a contingent liability.

Other covenants restrict additional debt, change of control and amendments to project contracts, and require hedging, insurance and reporting.

15.4.5 Debt instruments and tenor

Bank term loans dominate, with miniperm structures (shorter tenor, balloon, cash sweep to incentivise refinancing) common where bank appetite is limited to seven or ten years. **Project bonds** attract institutional investors seeking long-dated stable cash flows. **Export-credit and development-finance** loans bring long tenors and political-risk comfort. **Mezzanine** debt fills the gap between senior debt and equity at higher cost. Longer tenor raises debt capacity but shrinks the tail and raises refinancing and merchant risk, so loans are usually repaid a year or two before the PPA or concession ends.

15.5 The Structure of a Project Finance Model

15.5.1 Why the model is the centrepiece

In a project financing the model is a contractual document: lenders size debt from it, test covenants against it, and attach it to the loan agreement as the base case, usually after an independent model audit.

15.5.2 Standard architecture

A good model separates inputs, calculations and outputs and flows information one way: inputs and scenarios; a timing module with period flags; construction (capex, IDC, sources and uses); operations (volumes, degradation, prices, costs); tax and accounting; debt (drawdown, sculpting, reserves, hedging); ratios and covenant tests; the cash waterfall; returns (project IRR, equity IRR, break-even price); and a checks and sensitivities module.

15.5.3 Good modelling practice

Several principles distinguish reliable models from fragile ones.

Consistent timing. Use one time axis (monthly, quarterly or semi-annual) with flags to control which formulas operate in which period. Mismatched periods between revenue and debt service lead to errors, particularly with semi-annual repayment and annual PPAs.

One calculation per row, no hard-coded numbers in formulas. Every assumption lives in the inputs sheet and is traceable. A single constant embedded in a formula is a classic source of undiscovered error.

Integrity checks. The balance sheet balances in every period; sources equal uses; debt closes to zero at maturity; cash waterfall outflows equal inflows. A master check cell aggregates all of them.

Circularity control. IDC, fees capitalised in the loan, sculpting and tax can all create circular references. Sound models either resolve them algebraically, use a macro with convergence checks, or break the loop with a copy-paste approach; circularity left uncontrolled produces unstable results.

Nominal and real consistency. Indexation, cost inflation and the discount rate must all be in the same terms; mixing real cash flows with nominal discount rates is a persistent infrastructure-model error (see Volume 1, Chapter 6).

15.5.4 Sensitivity and scenario analysis

The standard sensitivity set covers energy yield (P50, P90, P99), price or volume, operating costs, capex, unhedged interest rates, inflation, tax and COD delay. Lenders run combined downside cases in which the minimum DSCR must stay above 1.0, and often above the default covenant. **Break-even analysis** turns “cushion” into a number: the fall in yield that takes the minimum DSCR to the lock-up level, the default level and 1.00 (Example 15.5). Monte Carlo analysis complements deterministic cases for resource and merchant price risk, but its output is only as good as the assumed correlations and tails.

15.6 Renewable Project Economics

15.6.1 From resource to revenue

A utility-scale solar farm is, economically, a long-dated annuity with an unusually transparent structure: almost all cost is incurred up front, operating cost is small, fuel is free, and revenue depends on two numbers, the quantity of electricity produced and the price at which it is sold.

Energy yield. The first-year energy is capacity times capacity factor times hours in a year:

```
Energy (MWh) = Capacity (MW) × Capacity factor × 8,760
```

For 100 MW at a 25 percent capacity factor this is 219,000 MWh. Capacity factors differ widely by technology and location, and a point of capacity factor matters: in Section 15.6.5 a fall from 25 to 22 percent adds more than 9 per MWh to the cost of energy.

Degradation. Panels lose a small fraction of output each year; we use 0.5 percent a year, so output in year t is the first-year output times 0.995^(t−1).

P50 and P90. An independent engineer estimates the long-term average annual yield (the **P50**, the level exceeded with 50 percent probability) and its uncertainty. The **P90** is the yield level that is exceeded 90 percent of the time; the **P99** is exceeded 99 percent of the time. If annual yield is approximately normal with standard deviation σ, then P90 ≈ P50 − 1.28σ and P99 ≈ P50 − 2.33σ. These statistics are reported for a one-year horizon and a ten-year horizon; because inter-annual variation partly averages out, the ten-year P90 lies closer to the P50 than the one-year P90. Lenders look at the P90 (and sometimes P99) to test debt service in a bad year, and sponsors use the P50 for expected returns. When the P90 is, say, 8 percent below the P50, the difference is the “resource uncertainty haircut.”

Price. A fixed-price PPA gives certainty; an indexed PPA transfers inflation risk to the offtaker; a merchant tail exposes the project to wholesale prices. For merchant exposure the key concept is the **capture price**, the average price the asset actually receives, which for solar is usually below the baseload average because solar plants all produce at the same hours and depress midday prices as penetration rises.

Costs. Operating costs (O&M, insurance, land lease, grid and asset-management fees, property taxes) are often fixed in real terms and indexed; major maintenance such as inverter replacement can appear as a lumpy mid-life cost.

15.6.2 The cost of electricity: LCOE

The **levelised cost of energy (LCOE)** is the constant price per MWh at which the discounted revenues equal the discounted costs over the asset’s life:

```
LCOE = [ Capex + Σ Opex_t / (1 + r)^t ]  /  [ Σ Energy_t / (1 + r)^t ]
```

The discount rate r is the project’s cost of capital. LCOE is the average cost, and the price at which the project exactly earns r. It is useful for comparing technologies and for judging whether a PPA price is plausible, but it has limits: it ignores timing and location of generation (a MWh at noon is not worth a MWh at 7 p.m.), integration and grid costs, the effect of taxes and subsidies unless explicitly modelled, and the financing structure beyond the single discount rate.

Example 15.6 – LCOE and its sensitivity. Using the Marlow project’s data (capex 120, opex 2.6 growing 2 percent, energy 219 GWh falling 0.5 percent a year, 20 years), compute the LCOE at a 7 percent discount rate.

Step 1: PV of operating costs. Σ 2.6 × 1.02^(t−1) / 1.07^t for t = 1 to 20 = 32.032 million.

Step 2: PV of capital and operating cost. 120 + 32.032 = 152.032 million.

Step 3: PV of energy. Σ 219,000 × 0.995^(t−1) / 1.07^t = 2,237,396 MWh (discounted).

Step 4: LCOE. 152.032 million / 2,237,396 MWh = **67.95 per MWh**.

Step 5: Sensitivity. At a 6.375 percent discount rate (a 75/25 blend of 5.5 percent debt and 9 percent equity, ignoring tax) the LCOE falls to 65.48. At 5 percent it is 60.26; at 9 percent, 76.19. Changing capex to 100 or 140 (at 7 percent) gives 59.01 or 76.89. Reducing the capacity factor to 22 percent raises the LCOE to 77.22.

Interpretation. The LCOE moves by roughly 8 to 9 per MWh for each 20 million of capex, and by about 16 per MWh between 5 and 9 percent cost of capital. Renewable economics are capital-cost and financing-cost economics: a solar farm is a bet on the cost of capital as much as on the sun. This is why the sponsors' ability to access cheap, long-dated, non-recourse debt is a competitive advantage in renewables.

15.6.3 Equity returns and the leverage trap

Lenders' tests tell the sponsors how much debt is available. They do not tell them whether the project is worth doing. For that, sponsors evaluate **project IRR** (unlevered cash flow against total cost) and **equity IRR** (cash distributions against equity invested), and compare equity IRR to the required return on equity for the risk class (Volume 1, Chapters 10 and 11).

Leverage raises equity IRR only when the project's return exceeds the cost of debt. If project IRR is below the debt cost, leverage reduces equity IRR; if it is equal, leverage is neutral. This is the simple algebra behind positive and negative leverage.

Example 15.4 – Equity IRR, the gearing cap and the break-even PPA price. Continue with the Marlow data, but first compare two PPA prices.

Case A: PPA price 62 per MWh. From Example 15.3, debt is 87.392 million, equity is 120 − 87.392 = 32.608 million, and the sculpted DSCR is 1.30 each year. Equity cash flow each year is CFADS minus debt service. Year 1: 10.978 − 8.445 = 2.533; year 2: 10.858 − 8.352 = 2.506; years 19 and 20, after the loan is repaid, receive full CFADS of about 8.69 and 8.56. The equity IRR is **5.40 percent**. The unlevered project IRR (cash flows of −120 followed by 20 CFADS values) is 5.47 percent, essentially equal to the 5.5 percent cost of debt, so leverage adds nothing. The project does not clear a 9 percent equity hurdle.

Case B: PPA price 66 per MWh. Revenue in year 1 is 219,000 × 66 = 14.454 million; CFADS is 14.454 − 2.600 = 11.854; year 2: 11.730; year 3: 11.605; year 18: 9.633; year 20: 9.353.

Step 1: DSCR-based capacity. Sculpting at 1.30 gives debt capacity of 94.708 million.

Step 2: Gearing cap. 75 percent of 120 is 90.0 million; the cap binds. Debt is 90.0 million, equity 30.0 million.

Step 3: Scale the debt service. The sculpted debt service is multiplied by 90 / 94.708 = 0.95029. Year 1 debt service is (11.854 / 1.30) × 0.95029 = 8.665 million; year 2: 8.574; year 3: 8.483; year 18: 7.041. The DSCR is now 1.30 / 0.95029 = **1.368** every year (and LLCR is 1.368).

Step 4: Equity cash flows. Year 1: 11.854 − 8.665 = 3.189; year 2: 3.155; year 18: 2.591; year 19: 9.494; year 20: 9.353. Total distributions over 20 years are 70.96 million on 30.0 invested.

Step 5: Equity IRR. Solving −30.0 + Σ CF_t/(1+IRR)^t = 0 gives **8.81 percent**. The unlevered project IRR is 6.51 percent versus a 5.5 percent debt cost, so leverage is positive.

Step 6: Break-even price. Searching over price, the equity IRR reaches 9.00 percent at a PPA price of about **66.23 per MWh**. At that price the minimum DSCR is 1.374.

Interpretation. Between 62 and 66, the binding constraint switches from DSCR (cash flow limits debt) to gearing (the lender caps equity thinness). Above 66 additional price is a pure equity windfall: each additional 1 per MWh of price above 66 lifts equity IRR by roughly 0.8 percentage points (about 8.8 percent at 66, 10.5 percent at 68, 12.1 percent at 70). That is the operating leverage of a capital-heavy asset magnified by financial leverage.

> **Common mistake.** Reading a high equity IRR as evidence of a good project. Equity IRR conflates asset quality with financing. Always check the project IRR against the cost of debt, and the equity IRR against the risk-appropriate cost of equity. A 75 percent geared project with a 9 percent equity IRR is a thinly profitable asset with thick leverage.

15.6.4 Resource risk, covenants and the cushion

The sculpted debt of Example 15.4 (Case B) fixes debt service in nominal terms. If energy output falls short of the P50, CFADS falls proportionally more than revenue because operating costs are largely fixed, and the DSCR falls in an amplified way. This is the operating leverage that makes resource risk the main driver of equity risk.

Example 15.5 – Stress test: P90 production, lock-up and default thresholds. Keep the 66 per MWh case with debt of 90.0 million and the debt service schedule of Example 15.4. Now test lower production, expressed as a fraction of P50 in every year, with other inputs unchanged.

Step 1: Production at 95 percent of P50. Year 1 revenue is 14.454 × 0.95 = 13.731; CFADS = 13.731 − 2.600 = 11.131; DSCR = 11.131 / 8.665 = 1.285. In year 18 revenue at full P50 is 13.272, so at 95 percent it is 12.608; CFADS = 12.608 − 3.641 = 8.967 and DSCR = 8.967 / 7.041 = **1.274**, the minimum. Equity IRR falls to 6.00 percent.

Step 2: Production at 92 percent of P50 (a plausible illustration of a one-year-style P90 haircut). Year 1 DSCR = 1.235; the minimum is 1.217 in year 18. Equity IRR is **4.25 percent**.

Step 3: Production at 90 percent of P50. Minimum DSCR = **1.180**; equity IRR = **3.05 percent**.

Step 4: Break-even production levels. The production factor at which the minimum DSCR falls to the 1.15 lock-up level is **88.4 percent of P50**; to the 1.05 default level, **83.1 percent**; to 1.00, **80.5 percent**.

Interpretation. A drop of 8 percent in production (a P90-like year if applied permanently) lowers equity IRR from 8.8 to 4.25 percent, a fall of more than half, while the lenders' minimum DSCR remains well above the lock-up. The structure protects lenders far more than it protects equity. Lenders would face distress only after production falls roughly 17 to 20 percent below P50 permanently, an event far outside the P90 range for a single well-studied solar site, so a project with a 1.30 sculpted DSCR is safe from a creditor's perspective, while the equity can lose most of its return in a poor but not extreme year.

15.6.5 Merchant tail and hedging

A 20-year PPA may be shorter than the asset's 30-year life. Lenders give the merchant tail little or no value in sizing; sponsors should separate "contracted equity value" from "merchant tail value", since a large share of valuation resting on uncertain tail prices is a warning sign. Floating-rate debt is normally swapped to fixed for 75 to 100 percent of the exposure. For the Example 15.3 loan of 87.4 million, a 100 bp rise on the same principal schedule would lift year-1 debt service by about 0.87 million and cut the minimum DSCR from 1.30 to about 1.18.

15.7 Infrastructure, PPPs and Value for Money

15.7.1 Public-private partnerships

A **public-private partnership (PPP)** is a long-term contract under which a private consortium designs, builds, finances, operates and maintains an asset (a DBFOM structure) and is paid over 20 to 30 years. The SPV is the contracting entity, the structure of this chapter applies, and the public authority is the offtaker.

The central choice is the **payment mechanism**:

- **Availability payment.** The authority pays a fixed periodic amount as long as the asset is available at the specified quality (a hospital with working wards, a road open to traffic), with deductions for unavailability or poor performance. Demand risk stays with the public sector. This produces stable, bond-like cash flows and high leverage.
- **Demand-based (user-pay) revenue.** Tolls or user charges depend on traffic or usage. The private party bears demand risk, so debt sizing is more conservative, coverage ratios are higher, and equity returns are volatile. Projects with demand risk historically have a higher incidence of renegotiation and distress, because traffic forecasts are prone to optimism.
- **Hybrids.** The authority pays per user, or a base payment plus a share of upside, with caps and collars.

15.7.2 Value for money

Public authorities justify PPPs through **value for money (VfM)**: the claim that delivery through the private consortium is cheaper, on a risk-adjusted whole-life basis, than conventional public procurement. The standard tool is the **public sector comparator (PSC)**, a hypothetical cost of public delivery including a quantified value of the risks the public sector would retain.

The core difficulty is financing cost. Governments borrow more cheaply than private consortia, so the private financing premium must be earned back through risk transfer (construction cost and delay risk, whole-life maintenance risk) and efficiency. Whether the premium is justified depends on the discount rate used to compare the options, a long-running debate: discounting at the government's low borrowing rate makes PPPs look expensive; discounting at a risk-adjusted rate makes public delivery look cheaper than it really is because the risks that public delivery retains are not priced. A balanced approach values the transferred risks explicitly.

Example 15.7 – The financing premium in an availability PPP. A hospital costs 200 million to build. The consortium finances it with 180 million of senior debt (90 percent gearing) at 5.5 percent over 25 years and 20 million of equity. The annual operating and lifecycle cost is 6.0 million. Lenders require a minimum DSCR of 1.20. The government can borrow at 3.5 percent.

Step 1: Annual debt service. A 25-year annuity on 180 million at 5.5 percent: $180 \times 0.055 / (1 - 1.055^{-25}) = 13.419$ million.

Step 2: CFADS needed. $1.20 \times 13.419 = 16.103$ million.

Step 3: Availability payment. CFADS plus operating cost: $16.103 + 6.000 =$ **22.103 million a year**.

Step 4: Equity return. Equity receives CFADS minus debt service = $16.103 - 13.419 = 2.684$ million a year for 25 years on 20 million invested: IRR = **12.75 percent** (before tax and before any risk events).

Step 5: Public comparator financing. If the government borrows 200 million at 3.5 percent over 25 years: $200 \times 0.035 / (1 - 1.035^{-25}) = 12.135$ million, so with operating cost the annual cost is 18.135 million.

Step 6: The premium. 22.103 − 18.135 = 3.968 million a year. Its present value at 3.5 percent over 25 years is 65.4 million, which is 21.9 percent of the PV of the public comparator's cost (298.9 million versus 364.3 million).

Interpretation. For the PPP to represent value for money, the transferred risks (construction overruns, delay, lifecycle failure, poor-quality operations) must be worth more than about 65 million in PV on a 200 million project, or the consortium must deliver measurable cost savings of that order. Both are plausible on a project with a history of public-sector overruns, but not automatic; the calculation shows why VfM claims require evidence rather than assertion. It also illustrates why the project's debt financing is only part of the story: the equity margin (12.75 percent on a 90 percent geared structure) is a significant component of the premium.

15.8 Implications for Corporate Leaders

For corporate managers, whether as sponsors, offtakers, contractors or lenders, four lessons stand out.

Treat financing as part of the investment case. The same asset has different equity returns under different structures. Compare the project's unlevered return with the corporate WACC (Volume 1, Chapter 11) to see whether the asset creates value, and use the financing to optimise, not to rescue a weak investment.

Treat guarantees and equity-contribution agreements as contingent debt. A parent guarantee of 30 million affects the sponsor's rating and covenant headroom even if the project debt is non-recourse on paper.

Value the tail and the options separately. Contracted cash flows deserve a low discount rate; merchant tails and expansion or repowering options deserve a higher rate or an option-pricing treatment (Volume 1, Chapter 18).

Mind the rate environment. Infrastructure returns are sensitive to financing costs, as the LCOE sensitivity showed. Projects approved at low rates may become uneconomic before financial close, so consider pre-close rate hedges and indexation in offtake pricing. Infrastructure financings that fail typically do so through optimistic demand forecasts, weak counterparties, short-tenor debt assumed to be refinanced, or thin coverage ratios with little room for error; lender-driven downside cases and independent audits exist to catch exactly these.

Case Study

Harbourline Toll Road: When the Traffic Forecast Meets the Lenders

Illustrative case; the firm, the project and all figures are fictional.

Aurelia Infrastructure Partners, a private infrastructure fund, has won a 30-year concession to build and operate Harbourline, a 42-kilometre toll road linking a port to an industrial region. The construction cost is 800 million, to be funded with 560 million of senior bank and bond debt (70 percent gearing) and 240 million of equity from Aurelia and two pension-fund co-investors. Unlike a typical renewable project, Harbourline has no availability payment: revenue is tolls, which depend on traffic.

The traffic forecast, prepared by the sponsors' consultant, projects steady growth as the industrial region develops. The lenders' own traffic adviser is more cautious: its base case is about 12 percent lower in the first five years, and its downside case, which assumes a competing rail freight link opens in year 8, is about 25 percent below the sponsors' forecast by year 10. The concession agreement allows toll increases in line with inflation but caps real increases, and gives the authority a right to terminate for default with compensation limited to senior debt outstanding after year 5.

On the construction side, the road is built under a fixed-price, date-certain EPC contract with a consortium of two contractors, each jointly and severally liable, backed by performance bonds of 10 percent of contract price. Delay LDs are set at a daily rate sized to cover interest and fixed costs, capped at 15 percent of the contract price. Geotechnical risk is excluded from the fixed price for a 3-kilometre section crossing marshland, with a provisional sum and a shared-overrun mechanism.

The debt is sized by the lenders on their traffic adviser's base case with a minimum DSCR of 1.60, an average DSCR of 1.80, and a sculpted repayment over 24 years. A DSRA covers six months of debt service. Lock-up is set at a DSCR of 1.30 and default at 1.10. The interest rate is hedged for 80 percent of the debt with a swap whose counterparty is a highly rated bank. The remaining 20 percent floats.

Under the sponsors' own traffic forecast, equity IRR is about 11 percent. Under the lenders' base case it is about 8 percent. Under the lenders' downside case, the minimum DSCR falls to 1.18 in year 9, below the lock-up threshold for three consecutive years, with equity distributions deferred until traffic recovers. The equity IRR in that case falls to about 3 percent.

Aurelia's investment committee is divided. One partner argues that the sponsors' forecast reflects the best available information about regional growth and that the lenders' downside is overly conservative. Another points out that the equity's 240 million is exposed to a risk, traffic, that no counterparty will take: the concession authority will not guarantee volumes, and the EPC and O&M contracts cannot hedge demand. A third argues for a revenue-sharing mechanism with the authority that would raise the DSCR in bad outcomes in exchange for giving up some upside, and for bringing in a third pension co-investor to reduce Aurelia's own exposure.

The committee must decide whether to proceed at the proposed capital structure, renegotiate the concession terms, restructure the debt (for instance with a longer tenor and lower gearing), or walk away.

Discussion Questions

1. Using the principle that risks belong with the party best able to control or diversify them, identify which Harbourline risks are well allocated and which are not. Which risk is retained by equity, and why can it not be transferred by contract?
2. Explain how the lenders arrived at a minimum DSCR of 1.60 when the Marlow solar project (Example 15.3) was sized at 1.30. What features of revenue risk justify the difference, and how would the debt capacity change if the lenders had sized at 1.30?
3. The sponsors' forecast gives an equity IRR of about 11 percent, the lenders' base case about 8 percent. If the equity's required return is 10 percent, what should the committee conclude, and what additional analysis (for example, a probability weighting of cases) would you request?
4. Evaluate the proposal for revenue sharing with the authority, and the proposal to add a third co-investor. How does each affect the risk, return and control of Aurelia's equity, and what would you negotiate?

Key Takeaways

- Project finance lends against a stand-alone SPV's contracted cash flows rather than the sponsor's balance sheet; "non-recourse" is in practice limited recourse, bounded by guarantees, equity undertakings and completion support.
- The value of the structure sits in the contract web (EPC, offtake, O&M, direct agreements) and in the lenders' control over cash through the waterfall, reserves and covenants.

- Allocate each risk to the party best able to control or price it; mis-allocation raises cost rather than lowering it.
- DSCR measures period liquidity cushion, LLCR and PLCR measure solvency over loan and project life; the lender's minimum DSCR and the gearing cap jointly determine debt capacity.
- Sculpting debt service to CFADS maximises debt for a given DSCR and equalises the cushion over time; a level annuity on a declining cash-flow profile concentrates risk in late years.
- Lock-up and default covenants convert coverage ratios into action; the gap between base case and lock-up is the operating cushion, and trapped cash costs equity IRR immediately.
- Project models are structural documents: consistent timing, no hard-codes, integrity checks, scenario switches, controlled circularity and consistent real/nominal treatment.
- Renewable economics are capital-cost economics: LCOE is highly sensitive to capex, capacity factor and the cost of capital; resource risk is amplified by operating leverage.
- Leverage raises equity IRR only when project IRR exceeds the cost of debt; high equity IRR can reflect financing rather than asset quality, and the lenders are protected far better than equity by a 1.30 DSCR.
- PPP value for money depends on whether transferred risk and efficiency exceed the private financing premium over public borrowing, which requires evidence, not assertion.

Review Questions

1. Why does a sponsor create an SPV instead of borrowing against the corporate balance sheet? What are the benefits and costs for the sponsor and for the lender?
2. Explain the difference between non-recourse and limited-recourse financing, and give three examples of sponsor support that appear in real deals.
3. What makes a project suitable for project finance? Give an example of an investment that is not suitable and explain why.
4. Why are direct agreements critical for lenders, and what could happen without them?
5. Describe the standard cash waterfall and explain why equity is the residual claim.
6. Define DSCR, LLCR and PLCR, and explain what each captures that the others do not.
7. What is debt sculpting, and why does it increase debt capacity compared with an annuity profile when CFADS declines over time?
8. Explain the roles of distribution lock-up and the default covenant. Why might a sponsor prefer a lower lock-up threshold, and why might lenders resist?
9. Why can a project with high equity IRR be a poor investment? What comparisons would you make?
10. What are the main reasons a PPP may or may not provide value for money relative to conventional public procurement?

Problems

1. A project has CFADS of 30, 32, 34, 34 and 30 million over five years and scheduled debt service of 22, 24, 25, 25 and 22 million. Compute each annual DSCR, the minimum and average DSCR, and the LLCR at a 5 percent discount rate.

2. A project has flat CFADS of 20 million a year for 10 years. Lenders require a DSCR of 1.25 on a sculpted profile at 6 percent. (a) What is the debt capacity? (b) If total project cost is 150 million and the gearing cap is 70 percent, what is the maximum debt and which constraint binds?
3. A wind farm costs 80 million, produces 150 GWh a year (flat) for 20 years, and has flat operating costs of 2 million a year. Compute the LCOE at an 8 percent discount rate (per MWh).
4. An asset costs 100 million and generates 9 million a year for 20 years. It is financed with 70 million of debt at 6 percent repaid by a level annuity over 20 years. Compute the unlevered IRR, annual debt service, the levered equity IRR, and the average DSCR.
5. A project has 150 million of debt at 6 percent. Completion is 90 days late (use a 360-day year). Delay LDs are 20,000 per day and extra fixed costs are 0.4 million. Compute the delay cost, the LD recovery and the shortfall.
6. A solar project has P50 energy of 300 GWh, an annual yield standard deviation of 6 percent of P50, a price of 60 per MWh, operating costs of 3 million, and debt service of 14 million. Compute P90 energy (using 1.2816σ), CFADS and DSCR at P50 and at P90.
7. A hospital PPP costs 100 million, funded 85 percent with debt at 6 percent over 20 years (annuity) and the rest with equity. Lenders require DSCR of 1.15 and operating costs are 3 million a year. Compute the annual availability payment and the equity IRR. Comment on the result.
8. A toll road earns 20 million a year of revenue with operating costs of 6 million and debt service of 10 million. What percentage fall in revenue brings DSCR to the lock-up level of 1.15 and to the default level of 1.05? (Costs are unchanged.)

Answers to Selected Problems

1. DSCRs: 1.364, 1.333, 1.360, 1.360, 1.364; minimum 1.333, average 1.356. PV of CFADS at 5 percent is 138.44; PV of debt service (the outstanding debt) is 102.12; LLCR = 1.356.
2. a. Annuity factor for 10 years at 6 percent is 7.3601; debt service 20 / 1.25 = 16.0; capacity = 16.0 × 7.3601 = 117.76 million. (b) The 70 percent cap gives 105 million; the gearing cap binds (the DSCR-sized debt of 117.8 would be 78.5 percent gearing).
3. Annuity factor (20 years, 8 percent) = 9.8181. PV of cost = 80 + 2 × 9.8181 = 99.636 million; PV of energy = 150,000 × 9.8181 = 1,472,722 MWh; LCOE ≈ 67.65 per MWh.
4. Unlevered IRR 6.39 percent. Debt service = 70 × 0.06 / $(1 - 1.06^{-20})$ = 6.103 million. Equity cash flow 9 − 6.103 = 2.897 million a year on 30 million: equity IRR 7.30 percent. DSCR = 9 / 6.103 = 1.475 each year. (Leverage is positive, but only marginally, since 6.39 percent is barely above 6 percent.)
5. Interest during delay = 150 × 0.06 × 90/360 = 2.25; plus 0.4 gives 2.65 million. LD recovery = 20,000 × 90 = 1.8 million. Shortfall = 0.85 million.
6. P90 = 300 × (1 − 1.2816 × 0.06) = 276.93 GWh. At P50: revenue 18.0, CFADS 15.0, DSCR 1.071. At P90: revenue 16.616, CFADS 13.616, DSCR 0.973. The debt service is too high even at P50 for a 1.30-type sizing and does not cover in a P90 year.
7. Debt 85 million: debt service = 85 × 0.06 / $(1 - 1.06^{-20})$ = 7.411 million; CFADS = 1.15 × 7.411 = 8.522; availability payment = 8.522 + 3 = 11.522 million. Equity cash flow = 8.522 − 7.411 = 1.111 million a year on 15 million invested: IRR ≈ 4.08 percent, below any plausible equity hurdle. The authority would need to pay more (higher DSCR/equity margin), or gearing would be raised, or the term lengthened.

8. DSCR at lock-up: CFADS = 11.5 so revenue = 17.5, a fall of 12.5 percent. At default: CFADS = 10.5 so revenue = 16.5, a fall of 17.5 percent.

Chapter 16: Pensions, Endowments and Institutional Investors

Learning Objectives

After studying this chapter you should be able to:

- Distinguish defined benefit (DB) and defined contribution (DC) pension arrangements by who bears investment, longevity and inflation risk, and compute a projected benefit obligation and a DC replacement ratio.
- Measure a plan's funding ratio, explain how discount-rate choice drives reported solvency, and evaluate the economic and accounting arguments in the discount-rate debate.
- Explain liability-driven investing (LDI): duration and DV01 matching, hedge ratios, swap overlays and collateral risk, and size a hedge.
- Frame asset allocation for a liability-bearing investor as a surplus problem rather than an asset-only problem, and interpret simulated funding-ratio distributions.
- Describe the endowment model, spending rules that smooth distributions, and the main types and governance issues of sovereign wealth funds.
- Assess how pension obligations affect a corporate sponsor's leverage, valuation, cash flows and strategy, and compare options for managing them.

Opening Vignette

Illustrative scenario; the company and all figures are fictional.

Greyfield Industries is a mid-cap manufacturer with operating earnings (EBITDA) of 300 million, net financial debt of 600 million and an equity market capitalisation that analysts value at roughly 1.4 billion. Its chief financial officer, Tomasz Reyes, has just finished a bruising meeting with the company's pension trustees and its lead lender.

The legacy final-salary pension plan, closed to new members a decade ago, has obligations of about 1.1 billion measured at a high-quality corporate bond yield and assets of 800 million. The plan is 73 percent funded. Half a percentage point of fall in the discount rate last year added almost 80 million to the deficit; strong equity markets partly offset it, but only partly. The trustees want higher cash contributions and a de-risking plan. The lender, for its part, has begun to describe Greyfield's "adjusted leverage" as the sum of financial debt and the after-tax pension deficit, a ratio that is close to breaching a covenant-style threshold the bank uses internally.

Tomasz sees the underlying tension. The plan's assets are 60 percent in equities and growth assets; its liabilities behave like a very long-dated bond portfolio. A drop in rates and a drop in equities together would be the worst of both worlds. The trustees favour liability matching; the finance team worries about locking in a deficit and tying up cash that could fund growth. An insurer has offered an indicative buy-in price for part of the pensioner liabilities. Meanwhile, the board is asking a deceptively simple question: how much is the pension plan really costing the shareholders, and is the deficit a debt?

Answering requires several tools: the economics of DB and DC promises, the mechanics of funding ratios and discount rates, liability-driven investing, asset allocation under liabilities, and the valuation implications for the sponsor. The same toolkit also explains how other long-horizon institutions, endowments and sovereign funds, decide how much risk to take and how much to spend.

16.1 The Institutional Investor Landscape

16.1.1 Liabilities define the investor

An **institutional investor** manages pooled capital on behalf of beneficiaries, with a mandate shaped by obligations or objectives rather than by the personal preferences of an individual. The most important lesson of this chapter is that **the liability, or spending objective, defines what is "risky" for the investor**. For an individual saving for retirement, volatility of wealth is the risk. For a pension fund with long-dated, inflation-linked promises, the risk is that assets fall short of liabilities, which can happen even when assets rise, if liabilities rise faster. For an endowment, the risk is failure to maintain purchasing power while sustaining a spending stream. For a sovereign fund the objective may be stabilising a budget, saving resource revenue for future generations, or funding a pension reserve.

The consequence is that standard portfolio theory (Markowitz, Volume 1) must be adapted: the relevant optimisation is over **surplus** (assets minus liabilities) or the funding ratio, and the relevant "risk-free asset" is not Treasury bills but the asset that matches the liabilities.

16.1.2 Main types

Table 16.1 – Principal institutional investor types

Type	Liability or objective	Horizon	Liquidity need	Distinctive features
Corporate DB pension	Contractual, inflation-linked benefits	Very long, but with maturing cash outflows	Moderate, rising as plan matures	Sponsor stands behind plan; regulation; accounting impact on sponsor
Public-sector DB pension	Benefit promises to public employees	Very long	Moderate	Contributions set politically; often discounted at expected return; high equity and alternatives
DC plan (provider)	None at plan level; member bears outcome	Individual-specific	Member-driven	Default funds, glide paths, fees matter
University endowment	Perpetual spending, intergenerational equity	Perpetual	Low, but needs to cover spending and commitments	Large alternatives share; spending rule
Charitable foundation	Grants and mission; often a minimum payout	Perpetual	Low	Mission-related investing
Insurance company	Policy claims, annuities	Long, matched	Regulated	Capital rules, credit-heavy portfolios
Sovereign wealth fund	Stabilisation, savings, development or pension reserve	Varies by mandate	Varies	Governance, transparency and political economy issues

16.1.3 Why they matter for corporate finance

Institutional investors are the dominant owners of public equity and corporate debt, the main providers of long-dated infrastructure and private-market capital (Chapter 15), and, for the many firms that sponsor DB plans, a source of balance-sheet risk. Understanding their constraints explains demand for long-duration bonds, the use of derivatives overlays, the pursuit of illiquidity premia, and shareholder-engagement behaviour. It also explains why a pension deficit shows up in credit analysis, in M&A due diligence (Volume 1, Chapters 22 and 23), and in rating agency methodology.

16.2 Defined Benefit versus Defined Contribution

16.2.1 The two promises

A **defined benefit** plan promises a specified retirement income, typically a fraction of final or career-average salary for each year of service (for example, 1/60th of final salary per year), often with inflation indexation. The sponsor, directly or through a trust, bears the investment risk, the longevity risk (members living longer than assumed), the inflation risk and the risk that salaries rise faster than expected. A **defined contribution** plan promises only contributions: the member accumulates an account, invests it among available options, and at retirement converts it to income (or draws it down). The member bears the investment, longevity and inflation risks, and the sponsor's obligation ends with the contribution.

Between these poles are hybrids: cash-balance plans (a notional account credited with a fixed or index-linked return), collective DC (risk pooled across members), and DB plans with risk-sharing features such as conditional indexation.

Table 16.2 – DB and DC compared

Dimension	Defined benefit	Defined contribution
Promise	Retirement income formula	Contribution rate
Investment risk	Sponsor (or trust surplus/deficit)	Member
Longevity risk	Sponsor	Member (unless annuitised)
Inflation risk	Sponsor if benefits indexed	Member
Sponsor cost predictability	Low: depends on markets and rates	High: fixed contribution
Member outcome predictability	High (replacement ratio defined)	Low
Balance-sheet effect on sponsor	Net obligation or surplus recognised	None beyond current contributions
Portability	Low	High
Pooling benefits	Yes: longevity pooled, long horizon, professional investment, lower fees	Limited, unless collective
Key governance question	Funding and de-risking	Defaults, fees, decumulation

The global drift from DB to DC reflects employer unwillingness to bear financial risk, regulation, accounting rules that made DB deficits visible on balance sheets, and greater workforce mobility. It does not mean that risk has vanished; it has moved to individuals, who are less able to diversify longevity risk and less informed on investment.

16.2.2 Measuring a DB obligation

The measures used by actuaries and accountants differ in how they treat future salary growth:

- The **accumulated benefit obligation (ABO)** is the present value of benefits earned to date, based on current salaries.
- The **projected benefit obligation (PBO)** is the present value of benefits earned to date, based on **projected final salaries**. Under the projected unit credit method, each year of service earns a unit of benefit, and the PBO values the

units accrued so far.

- The **service cost** is the present value of the additional benefit earned in the current year; the **interest cost** is the unwinding of the discount on the opening obligation. Accounting standards (such as IAS 19 in IFRS, and ASC 715 in US GAAP) require these components in the sponsor's financial statements, with actuarial gains and losses recognised in other comprehensive income under IFRS.

Example 16.1 – A member's PBO, service cost and interest cost. A 45-year-old member has 15 years of service, a current salary of 80,000, and salaries are expected to grow 3 percent a year. The plan pays 1/60th of final salary for each year of service, starting at age 65, payable for 20 years with 2 percent annual indexation. The discount rate is 5 percent. (Treat the member as certain to survive and retire at 65, for clarity; real valuations apply mortality, turnover and other decrements.)

Step 1: Projected final salary. Retirement is in 20 years: $80{,}000 \times 1.03^{20} = 80{,}000 \times 1.8061 = 144{,}489$.

Step 2: Accrued pension at 65. 15/60 × 144,489 = 36,122 a year (first payment, indexed thereafter).

Step 3: Value at 65 of a pension of 1 in its first year, indexed at 2 percent for 20 years, discounted at 5 percent. Annuity factor = Σ (t = 1..20) 1.02^(t−1) / 1.05^t = 14.665 (payments at year-end).

Step 4: PV at 65 of the accrued pension. 36,122 × 14.665 = 529,730.

Step 5: PBO today. Discount 20 years at 5 percent: $529{,}730 / 1.05^{20} = 529{,}730 / 2.6533 =$ **199,656**.

Step 6: ABO for comparison (no salary growth: pension 15/60 × 80,000 = 20,000): 20,000 × 14.665 / 2.6533 = 110,545. The PBO is 80 percent larger than the ABO, because the plan promises pensions linked to final salary.

Step 7: Service cost. One more year of service adds 1/60 of final salary: 144,489/60 = 2,408 a year of pension, worth 2,408 × 14.665 / 2.6533 = **13,310**.

Step 8: Interest cost. 5% × 199,656 = **9,983**.

Step 9: Roll-forward. The PBO a year later (member now 46, 16 years of service, with 19 years to retirement) is (199,656 + 13,310) × 1.05 = 223,615, which equals $16/60 \times 144{,}489 \times 14.665 / 1.05^{19}$. (Here the service cost is valued at the start of the year; conventions differ on timing.)

Interpretation. The PBO grows faster than the interest cost alone because of service accrual; and because the discount rate is applied over 20 years, a half-point change in the rate moves the member's PBO by roughly 8 to 10 percent. The sensitivity of the obligation to the discount rate is the origin of most pension-risk management.

16.2.3 The DC outcome: contributions, returns and fees

For a DC member the outcome is a random accumulation. Three variables dominate: the **contribution rate** (and its persistence), the **net return** (gross return minus charges), and the **time horizon**. Because early contributions compound for decades, delays and gaps are costly, and so are small recurring fees.

Example 16.2 – DC replacement ratio and the cost of fees. A member starts work at salary 60,000, receives annual salary growth of 3 percent for 35 years, and contributes 12 percent of salary (employee and employer combined) at the end of each year. The account earns 5 percent a year net of fees. At retirement the balance buys a 20-year level annuity at 4.5 percent.

Step 1: Final salary. $60{,}000 \times 1.03^{34} = 163{,}914$ (salary in year 35).

Step 2: Accumulated balance. Balance_t = Balance_(t−1) × 1.05 + 0.12 × Salary_t. Summing over 35 years gives **972,775**.

Step 3: Annuity factor. 20 years at 4.5 percent: $(1 - 1.045^{-20})/0.045 = 13.008$.

Step 4: Annual retirement income. 972,775 / 13.008 = 74,783.

Step 5: Replacement ratio. 74,783 / 163,914 = **45.6 percent** of final salary.

Step 6: Add a 1 percent annual fee (net return 4 percent): balance **815,203**, income 62,670, replacement ratio **38.2 percent**. The fee reduces the retirement balance by 16.2 percent, and the replacement ratio by 7.4 percentage points.

Step 7: Lower contribution (8 percent, no extra fee): balance 648,517, replacement ratio 30.4 percent.

Interpretation. A DC member needs a high sustained contribution rate and low costs to achieve a replacement ratio comparable to a typical DB promise. The 4.5 percent annuity rate is itself a function of interest rates at retirement, so DC members bear interest-rate risk on conversion to income. A well-designed default fund therefore **de-risks as retirement approaches**, shifting from growth assets toward assets that resemble annuity pricing, which is the DC analogue of liability-driven investing.

16.2.4 Managerial implications of the DB/DC choice

For a sponsor, the DB/DC decision is a **risk-allocation** decision. A DB plan transfers risk from employees to shareholders; shareholders bear a risk that is correlated with their own (the company's earnings often weaken in the same recessions in which equities fall and pension deficits widen) and that is hard to diversify. A sponsor with strong cash flows and a stable workforce may prefer the retention benefits of DB; a cyclical or levered sponsor may find the volatility intolerable. Closing a DB plan to new entrants or future accrual stops the growth of the risk but not the legacy exposure, which remains for decades.

16.3 Funding Ratios and the Discount-Rate Debate

16.3.1 The funding ratio

The **funding ratio** is the ratio of plan assets to the value of liabilities:

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Funding ratio = Assets / Liabilities
Surplus (deficit) = Assets − Liabilities
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A funding ratio above 100 percent means a surplus on the chosen measure; below 100 percent, a deficit. The ratio is a ratio of two random variables. Assets move with markets; liabilities move with interest rates, inflation expectations, mortality assumptions and plan changes. Because liabilities are long-dated, they behave like a long-duration, often inflation-linked bond: when yields fall, liabilities rise, and unless the assets rise by a similar amount, the funding ratio falls.

Two numbers are therefore needed to understand a plan's risk: the current funding ratio, and the **sensitivity of the ratio to market moves**. A plan that is 100 percent funded but with assets of duration 3 years against liabilities of duration 14 is in a weak position, since a fall in rates immediately creates a large deficit.

16.3.2 Valuing liabilities: which discount rate?

The liability is a stream of expected benefit payments. The present value depends on the discount rate, and the choice is one of the most contested questions in pension finance. Four positions are common.

1. The expected return on plan assets (public-plan convention). Many public-sector plans historically discounted at the assumed long-run return on their (equity-heavy) portfolio, say 7 percent. This produces a low liability and a high funding ratio, and it implicitly assumes the investment premium will be earned. It also makes a riskier asset mix look cheaper: raise

the equity share, raise the assumed return, reduce the reported liability and the required contributions.

2. A high-quality corporate bond yield (accounting convention). IAS 19 and US GAAP require corporate sponsors to discount at a market yield on high-quality (typically AA) corporate bonds with maturities matching the obligations. The reasoning is that the benefit promise has low default risk and that the liability should reflect market-consistent prices.

3. A government bond or swap yield (regulatory or "risk-free" convention). Some regulators and many actuaries use sovereign or swap curves, sometimes with a spread. The rate is lower than the AA yield, producing a larger liability.

4. The sponsor-credit-adjusted rate. Some argue the liability's value should reflect the possibility that the sponsor cannot pay, which would argue for a higher discount rate for weak sponsors. Regulators generally reject this, since the benefits are protected by contract and law.

16.3.3 The economic argument

The argument for a low, market-consistent rate rests on financial economics. A benefit that is fixed (or indexed) and for which payment is safe is economically a debt-like claim. The value of a claim depends on the **riskiness of the cash flows, not on the riskiness of the assets chosen to fund it**. This is the logic of Modigliani-Miller applied to pensions: the asset allocation cannot change the value of the liability. If a plan invests in equities and expects a 7 percent return, that does not reduce the amount owed to members; it means the sponsor has taken on a risky position, with a possible gain and possible loss, relative to the matching asset.

The theoretical literature supports this. Sharpe (1976) modelled the pension guarantee as a put option written on the plan's assets, so that the sponsor (and any insurer, such as a pension guarantee fund) holds a short put with strike equal to the liability; riskier assets raise the put's value and so transfer value from beneficiaries or insurers to shareholders. Treynor (1977) and Black (1980) argued that sponsors should invest in bonds and exploit the tax advantages of the arrangement, since equity risk in the plan is not valued at its expected return. Tepper (1981) and others discussed the tax arbitrage of funding a plan with bonds financed by corporate debt. The empirical record of corporate behaviour is mixed, though, and many plans hold substantial equities.

16.3.4 The counter-arguments

The case for the expected-return rate has some sound elements. The plan is a **going concern** with a long horizon; contributions are made over time; benefits are paid over many decades; and a diversified portfolio can plausibly earn a premium over bonds, so if the plan and sponsor are both long-lived, expected funding costs may be lower than a market-consistent valuation implies. Public plans may also have taxing power behind them: the "sponsor" is a government that can raise taxes, and the covenant is stronger than any corporate's.

The rebuttal is that **expected returns are not guaranteed**. If the premium is earned, the plan will outperform a bond strategy; if it is not, someone, usually taxpayers or future employees, will pay. Discounting at the expected return implicitly assumes the plan is risk-free to fund at the equity premium, and it discourages recognition of risk. A well-managed plan can disclose both: the liability at a market-consistent rate for risk measurement, and an expected-cost estimate at an expected return for budgeting.

Table 16.3 – Discount-rate choices and their consequences

Convention	Typical rate relation	Liability	Reported funding ratio	Incentive effect
Expected asset return	Highest	Lowest	Highest	Rewards risk-taking (raising the assumed return lowers the liability)
High-quality corporate bond yield	Middle	Middle	Middle	Market-consistent; sponsor deficits visible
Government or swap curve	Lowest	Highest	Lowest	Conservative; matches regulatory and buy-out pricing

Example 16.3 – Discount-rate choice and the funding ratio. A plan will pay 40 million at the end of each year for 25 years. Plan assets are 520 million.

Step 1: PV of liabilities at different rates. PV = $40 \times [1 - (1 + r)^{-25}] / r$.

Discount rate	PV of liabilities	Funding ratio (520 / PV)
4.0%	624.88	83.2%
4.5%	593.13	87.7%
5.0%	563.76	92.2%
5.75%	523.71	99.3%
6.0%	511.33	101.7%
7.0%	466.14	111.6%

Step 2: Interpretation of the gap. The same plan, with the same assets and the same promised payments, is 83 percent funded at a 4 percent risk-free-style rate and 112 percent funded at a 7 percent expected-return rate. Reported solvency depends on a convention, not on any change in what is owed.

Step 3: Duration and convexity at 5 percent. The Macaulay duration of the payment stream is 10.52 years, so modified duration = 10.52 / 1.05 = 10.02. Convexity is 153.8. A 100 bp fall in rate predicts a change of $-(-0.01) \times 10.02 + 0.5 \times 153.8 \times 0.01^2 = +10.02\% + 0.77\% = +10.79\%$, against an exact change of +10.84% (from 563.76 to 624.88). A 100 bp rise predicts $-10.02\% + 0.77\% = -9.25\%$, against an exact −9.30% (to 511.33).

Step 4: Funding-ratio sensitivity. With assets fixed at 520, a 100 bp fall in the rate turns a 92.2 percent funding ratio into 83.2 percent; a 100 bp rise lifts it to 101.7 percent. Because liabilities are convex, a fall in rates raises liabilities by more than a rise lowers them: an asymmetry that LDI strategies need to manage.

Interpretation. One percentage point on the discount rate moves the liability by about 10 percent (the duration), and therefore the funding ratio by about 9 points here. Anyone assessing a pension plan should ask first: *what rate, and why?* And second: *how would the ratio change if that rate fell by 100 bp?*

16.3.5 Common mistakes

> **Common mistake 1.** Comparing funding ratios across plans without checking the discount-rate convention. A public plan at “80 percent funded” on an expected-return basis may be far worse than a corporate plan at 80 percent on an AA-bond basis.

> **Common mistake 2.** Treating a funding ratio above 100 percent as safe. A plan at 105 percent with a large unhedged duration gap can slip below 90 percent in a bad year for rates. The right question is the *distribution* of the ratio, not its point estimate.
>
> **Common mistake 3.** Assuming a high expected return reduces the cost of pensions. It reduces the *expected* cost, at the price of greater variance and a possible shift of the burden to other stakeholders.

16.4 Liability-Driven Investing

16.4.1 The idea

Liability-driven investing (LDI) is the practice of structuring the plan's assets so that their value moves with the liabilities, thereby stabilising the funding ratio. It splits the portfolio into two parts. A **liability-hedging portfolio** (long-duration government and high-grade corporate bonds, inflation-linked bonds, interest-rate and inflation swaps) is designed to replicate the liability's sensitivity to rates and inflation. A **return-seeking portfolio** (equities, private markets, credit, real assets) is designed to earn the excess return needed to close the deficit and pay for future accruals. The plan chooses how much risk to run in the second part, and then removes as much as it wishes of the unrewarded interest-rate and inflation risk in the first.

The logic follows from Section 16.3: the liability has a duration, so interest-rate exposure is an unrewarded risk unless the plan is paid to bear it. Hedging it is not market timing; it is the removal of a mismatch.

16.4.2 Duration, DV01 and hedge ratios

The key sensitivities are:

```
Modified duration D = (1/P) × Σ t × CF_t / (1 + y)^t  /  (1 + y)
DV01 = P × D × 0.0001      (value change for a 1 bp fall in yield)
Hedge ratio = DV01 of assets / DV01 of liabilities
```

DV01 is the money change for one basis point and is the practical way to compare assets and liabilities, because it accounts for size. A plan with assets of 520 and liabilities of 564 needs the **dollar** duration of its assets to approach the liability's, not simply the same percentage duration. This is why an underfunded plan with a duration-matched asset portfolio still has an uncovered liability exposure: assets are smaller than liabilities, so a 1 percent move in the liability is larger in money terms than a 1 percent move in the assets.

Typical hedging policies set a **target hedge ratio** (for example 50 percent today, rising toward 100 percent as the funding ratio improves) and may hedge interest rates and inflation separately. Many plans use **triggers** or glide paths: as funding improves, risk assets are sold and hedges added, locking in gains.

16.4.3 Instruments

Physical bonds provide matching exposure but require capital that could otherwise be in return-seeking assets. **Interest-rate swaps** (paying floating, receiving fixed) give duration exposure with little cash outlay, and **inflation swaps** give inflation exposure. **Repurchase agreements** let a plan lever a gilt or Treasury portfolio. The efficient design, with a sizeable overlay on top of a modest bond portfolio, frees capital for return-seeking assets, but at the price of **collateral and liquidity risk**. A fall in yields creates swap gains, a rise in yields creates swap losses that must be met with cash collateral. If yields jump, the plan needs liquid assets immediately, and may have to sell other holdings into a falling market. The UK gilt market in September 2022 provided a vivid illustration: rapid yield increases triggered large collateral calls on

leveraged LDI portfolios, forcing asset sales that pushed yields still higher until the central bank intervened. The lesson, accepted since by regulators, is to hold a **liquidity buffer** sized to withstand a large yield shock and to stress-test the combined position.

Example 16.4 – Sizing an LDI swap overlay. Using Example 16.3 at a 5 percent discount rate: liabilities 563.76, assets 520 (funding ratio 92.2 percent). The assets are 60 percent equities (no interest-rate sensitivity assumed) and 40 percent bonds (208) with modified duration 8. A 20-year receiver swap has modified duration 12.46 (same as a 20-year par bond at 5 percent).

Step 1: Liability DV01. Liability modified duration = 10.02, so DV01 = 563.76 × 10.02 × 0.0001 = **0.565**. (A 1 bp fall in yield raises liabilities by 0.565 million.)

Step 2: Asset DV01. 208 × 8.0 × 0.0001 = **0.166**.

Step 3: Hedge ratio. 0.166 / 0.565 = **29.4 percent**. The plan is 70 percent unhedged on rates.

Step 4: DV01 needed to reach 100 percent. 0.565 − 0.166 = 0.399.

Step 5: Swap notional. 0.399 / (12.46 × 0.0001) = **319.9 million** of receive-fixed swaps (about 61 percent of assets as notional, with minimal initial cash).

Step 6: Outcomes for a 100 bp move (equities held constant; asset effects approximated linearly):

Scenario	Liabilities	Assets, unhedged	Funding ratio, unhedged	Assets with swaps	Funding ratio, hedged
Start	563.76	520.00	92.2%	520.00	92.2%
Rates −100 bp	624.88	536.64	85.9%	576.50	92.3%
Rates +100 bp	511.33	503.36	98.4%	463.50	90.6%

The swap gains 39.86 when rates fall (319.9 × 12.46 × 1%) and loses 39.86 when they rise.

Step 7: Collateral requirement. In the +100 bp scenario the swaps lose 39.86 million, or 7.7 percent of assets, which must be funded in cash (collateral) over a short horizon.

Interpretation. The hedged funding ratio stays within about 90.6 to 92.3 percent, against 85.9 to 98.4 percent unhedged. The hedge **removes upside as well as downside** (the plan gives up the gain from a rate rise) and demands liquid collateral of nearly 8 percent of assets for a 100 bp shock. The small residual drift in the hedged case (92.3 percent at −100 bp vs 90.6 at +100 bp) arises from convexity: the liability's convexity exceeds a linear swap position's. A real design would add liquidity buffers and rebalance the hedge as yields move.

16.4.4 Longevity and inflation

Interest rates are not the only liability risk. **Longevity risk**: if members live longer than assumed, liabilities increase and no market asset hedges the exposure cheaply. Instruments include **buy-ins** (the plan buys an insurance policy that pays the benefits of specified members), **buy-outs** (the insurer takes on liabilities and the sponsor exits), and **longevity swaps**. **Inflation risk** arises where benefits are indexed; inflation-linked bonds and inflation swaps hedge much of it, though caps and floors in the benefit rules complicate the match.

16.5 Asset Allocation for Liability-Bearing Investors

16.5.1 From asset-only to surplus optimisation

The classical mean-variance problem (Markowitz) chooses weights to maximise expected return for a given portfolio variance. For a plan, the right objective is the **surplus** (or the funding ratio), whose variance includes the covariance between assets and liabilities. Sharpe and Tint's surplus optimisation recasts the problem in terms of a portfolio return and a liability return; the key formula for the variance of the funding ratio's change is approximately:

```
Var(FR change) ≈ FR² × [ σ_A² + σ_L² − 2 ρ σ_A σ_L ]
```

where σ_A and σ_L are the volatilities of asset and liability returns and ρ is their correlation. For a plan with a modest deficit, **the minimum-risk portfolio is the liability-matching portfolio**, not a cash or short-bond portfolio. Equities, which have a low correlation with liabilities, add a lot of surplus risk for the same portfolio volatility. The planner's problem is to decide how much surplus risk (the "risk budget") the sponsor can tolerate, then to maximise expected surplus return per unit of risk.

16.5.2 Setting the risk budget

The risk budget depends on three things. The **funding ratio and the sponsor's covenant**: a poorly funded plan with a weak sponsor has little capacity for risk, while a well-funded plan with a strong sponsor can take risk because the sponsor can fund a shortfall. The **plan's maturity** (the ratio of retirees to active members and the proportion of cash outflow to assets) which determines liquidity needs and the horizon over which risk can be recovered. And **stakeholder preferences**: trustees, sponsor, regulator, members.

Practical frameworks include **funding-ratio glide paths** (reduce risk as the funding ratio rises toward a target, such as full funding on a buy-out basis), **surplus-at-risk or funding-ratio-at-risk limits** (for example, the 95th percentile one-year loss in the funding ratio must not exceed a set amount), and **journey-plan** models that simulate paths to a long-term target.

16.5.3 Simulation: how much risk is worth taking?

Closed-form formulas can only go so far, because liabilities, rates and equities interact through time, contributions arrive, and decisions depend on state variables. Most plans therefore use stochastic simulation.

Example 16.5 – Funding-ratio simulation under three policies. Consider a stylised plan starting with assets 0.90 per unit of liabilities, with liability duration 10. Interest rates follow a mean-reverting process (starting at 4.5 percent, long-run level 4.5 percent, speed 0.15 a year, volatility 0.8 percentage points a year). Liabilities earn the prevailing yield less duration times the change in rates. Equity returns have mean 7.5 percent and volatility 16 percent, with a correlation of −0.2 with the rate shock (equities tend to fall when rates rise). Conventional bonds have duration 8; the liability-matching portfolio has duration 10 and earns 0.2 percent above the liability return. The sponsor contributes 2 percent of initial liabilities each year. We simulate 100,000 paths over 10 years (computed with Python; seed 7).

Table 16.4 – Simulated funding ratio after 10 years

Policy	Mean	Median	5th pct	95th pct	P(FR < 1.0)	P(FR < 0.8)	Std dev
A: 60% equity, 40% conventional bonds, unhedged	1.274	1.230	0.784	1.913	22.4%	5.8%	0.350
B: 40% equity, 60% bonds, 80% of the bond sleeve liability-matched	1.209	1.190	0.883	1.600	16.7%	1.4%	0.219
C: 25% equity, 75% liability-matched	1.160	1.153	0.958	1.388	10.2%	0.1%	0.131

Reading the results. Policy A has the highest expected funding ratio (1.274) but also the widest distribution: its 5th percentile is 78 percent funded, and there is a 5.8 percent chance of ending below 80 percent funded. Policy C gives up about 11 points of expected funding ratio (1.274 to 1.160), but cuts the standard deviation from 0.350 to 0.131 and the probability of ending below 80 percent from 5.8 percent to 0.1 percent. Policy B sits in between. Note that even Policy C has a 10 percent chance of a funding ratio below 100 percent after 10 years, because the starting deficit is 10 percent, the contribution is modest, and the liability-matched assets earn only a small premium over liabilities.

Interpretation. There is no "right" policy without a view of the sponsor's capacity for contributions and the cost of failure. If the sponsor can afford deficit contributions and values the upside, A may be justified; if a bad outcome would breach covenants or threaten the firm, B or C is more appropriate. The simulation's value is that it makes the trade-off between expected funding and tail risk explicit. Its limitation is that the answers depend entirely on the assumed correlations and premia: in a model where equities earn a lower premium or correlate more with rate falls, the ordering shifts.

16.5.4 Alternatives, illiquidity and diversification

Large plans also hold private equity, real estate, infrastructure (Chapter 15), private credit and hedge funds, for diversification and a possible illiquidity premium. The concerns are fees, valuation lags that understate true risk, and illiquidity when collateral or benefit payments fall due. A useful discipline is to classify assets by liquidity under stress and ensure the liquid portion covers collateral and benefit needs for a defined period in a severe scenario.

16.6 Endowments and Foundations

16.6.1 The objective: intergenerational equity

An endowment exists in perpetuity and must serve current and future beneficiaries equally. Its objective is to **maximise the real, sustainable spending stream** while preserving the real value of the corpus. The fundamental tension is between today's spending (the university's budget depends on distributions) and tomorrow's capital. There are no external liabilities in the pension sense, but there are **implicit liabilities**: the budget share the endowment funds, and the donors' expectation that the capital will last.

The required real return is approximately the spending rate plus the cost of investment management plus the excess of higher-education cost inflation over general inflation. With a 5 percent spending rate, 0.5 percent investment costs and inflation in the institution's costs of 2.5 percent, the required nominal return is about 8 percent: a demanding hurdle that drives the search for equity and alternative exposure.

16.6.2 The endowment model

The **endowment model**, associated with David Swensen's management of the Yale endowment and popularised through the 1990s and 2000s, rests on several claims. First, a **long horizon and absence of near-term liabilities** allow the investor to bear illiquidity and volatility, and so to earn the associated premia. Second, a heavy **equity orientation**, since equities have provided the highest long-run real returns. Third, **diversification away from traditional marketable assets** into private equity, venture capital, real assets and absolute-return strategies, where skilled manager selection might add value. Fourth, a **governance structure** with a small, expert investment committee and a long-tenured staff able to make unconventional decisions and to access top managers.

The critiques are also instructive. Manager selection, not asset-class exposure, accounts for much of the model's performance, and access to top managers is scarce, so the model is hard to replicate at smaller institutions. Valuation smoothing in private assets understates risk. The 2008 financial crisis exposed **liquidity risk**: endowments with large unfunded private-market commitments and falling public holdings faced a "denominator effect," with illiquid assets forming an inflated share of a shrunken portfolio, while still owing capital calls and budget distributions. Some endowments sold assets at poor prices or cut budgets; the lesson is to size illiquid commitments against stressed liquidity, not just average liquidity. A simpler alternative, often called a **reference-portfolio** approach (used publicly by some sovereign funds), benchmarks the actual portfolio against a low-cost mix of equities and bonds and requires active risk to be justified against it.

16.6.3 Spending rules

A naive rule, spending a fixed percentage of current market value each year, makes distributions as volatile as markets; a rule that spends a fixed real amount, regardless of market value, risks depleting the fund. Most institutions therefore use a **smoothing rule**. The best-known is a weighted average of last year's spending (grown by inflation) and a target percentage of current market value:

```
Spending_t = w × Spending_(t−1) × (1 + inflation) + (1 − w) × s × MarketValue_(t−1)
```

where w is the smoothing weight (for example 0.8) and s the long-run spending rate (for example 5 percent). Variants use a moving average of market value over three to five years.

Example 16.6 – A smoothing spending rule versus a fixed 5 percent rule. An endowment starts with a market value of 1,000. Year-1 spending is 45. Inflation is 2.5 percent. Smoothing weight w = 0.8, spending rate s = 5 percent. Returns over five years are +12%, −20%, +8%, +10%, +6%. Spending occurs at the start of each year; the balance then earns the return.

Step 1: Smoothing rule, year 1. Spending 45; market value after = (1000 − 45) × 1.12 = 1,069.60.

Step 2: Year 2. Spending = 0.8 × 45 × 1.025 + 0.2 × 0.05 × 1,069.60 = 36.90 + 10.70 = **47.60**. Market value = (1,069.60 − 47.60) × 0.80 = 817.60.

Step 3: Year 3. Spending = 0.8 × 47.60 × 1.025 + 0.01 × 817.60 = 39.03 + 8.18 = **47.21**. Market value = (817.60 − 47.21) × 1.08 = 832.03.

Step 4: Years 4 and 5. Spending 47.03 and 47.20; market values 863.50 and 865.28.

Step 5: Fixed 5 percent rule. Spending is 50.0, 53.2 (5% of 1,064.0), 40.4 (5% of 808.6), 41.5 and 43.4; ending market value 873.07.

Year	Return	Smoothing rule spending	Fixed 5% rule spending
1	+12%	45.00	50.00
2	−20%	47.60	53.20
3	+8%	47.21	40.43
4	+10%	47.03	41.48
5	+6%	47.20	43.35

Interpretation. After the 20 percent crash, the fixed rule forces spending to fall by 24 percent (53.2 to 40.4), exactly when the university's needs for support are greatest. The smoothing rule changes spending by less than 1 percent. The price is that spending runs ahead of the market for a time (it is above 5 percent of market value in years 3 to 5), which depletes capital slightly: the smoothing rule's ending value (865.28) is below the fixed rule's (873.07). The smoothing rule thus trades capital preservation in the short run for budget stability, which is usually the right trade for an institution that must run a payroll.

16.7 Sovereign Wealth Funds

16.7.1 Types and purposes

A **sovereign wealth fund (SWF)** is a state-owned investment fund, typically funded from commodity revenue, fiscal surpluses or foreign-exchange reserves. Classification by purpose is more useful than by name:

- **Stabilisation funds** smooth budgets against commodity-price swings; they need liquidity and low risk.
- **Savings or intergenerational funds** convert finite resource wealth into a permanent financial asset; they have a long horizon and can take equity and illiquid risk.
- **Development funds** invest in domestic economic development, which blurs commercial and policy objectives.
- **Pension reserve funds** pre-fund future public pension liabilities; they resemble large pension plans.
- **Reserve investment corporations** manage excess foreign-exchange reserves for higher return.

16.7.2 Investment and governance issues

SWFs combine the long horizon of endowments with unusual size, which creates market-impact and liquidity constraints and often leads to indexing-style core holdings with active satellites. Their distinctive questions are governance questions: who decides the spending rule (the finance ministry, the fund board), how withdrawals are linked to fiscal rules, how transparent the fund is, whether its investments are commercial or political, and how conflicts are managed. The **Santiago Principles** (2008), a voluntary set of generally accepted principles and practices agreed by SWFs, address legal framework, governance and accountability, and investment and risk-management practices. Funds that tie withdrawals to a fiscal rule based on the expected real return of the fund, and that publish their investment reference portfolio and results, are widely regarded as having strong governance; funds without such rules face the risk that the capital is drawn down for short-term fiscal needs.

For corporate leaders, SWFs matter as large, patient co-investors and acquirers. They often invest alongside private equity and infrastructure sponsors (Chapter 15), buy minority stakes in listed companies, and are active in real-estate and private-market co-investment. Because of national-security and foreign-investment screening, their approach to governance and

disclosure can affect a transaction's regulatory path.

16.8 Implications for Corporate Sponsors

16.8.1 The pension plan as part of the corporate balance sheet

For the sponsor, a DB plan is economically a **debt-like liability coupled with a risky asset portfolio**. The net deficit is a claim ahead of shareholders (the sponsor must eventually fund it, often with tax deductibility), and the volatility of the deficit is a source of earnings and credit risk. Analysts and rating agencies therefore adjust: they add the deficit (usually after tax) to net debt, and they adjust EBITDA or cash flow for the service cost and deficit repair payments.

Example 16.7 – Adjusting leverage and equity value for a pension deficit. Return to Greyfield (Opening Vignette): EBITDA 300; net financial debt 600; operating enterprise value (before pension) 2,000; pension liabilities 1,100; plan assets 800; tax rate 25 percent; liability duration 14.

Step 1: Pre-tax and after-tax deficit. Deficit = 1,100 − 800 = 300; after tax (contributions are deductible) = 300 × (1 − 0.25) = **225**.

Step 2: Funding ratio. 800 / 1,100 = 72.7 percent.

Step 3: Adjusted leverage. Reported net debt / EBITDA = 600 / 300 = 2.0x. Adjusted (with after-tax deficit) = (600 + 225) / 300 = **2.75x**. Using the pre-tax deficit gives 3.0x.

Step 4: Equity value. Ignoring the deficit: 2,000 − 600 = 1,400. Deducting the after-tax deficit: 2,000 − 600 − 225 = **1,175**, an effective 16 percent reduction in equity value, even though the pension liability does not appear in financial debt.

Step 5: Deficit repair cash cost. Spreading 300 over 10 years at a 5 percent rate gives a level payment of 300 × 0.05 / (1 − 1.05^{-10}) = 38.85 a year pre-tax, or 29.14 after tax: about 10 percent of EBITDA for a decade.

Step 6: Rate shock. If the discount rate falls 50 bp, the liability rises by roughly 1,100 × 14 × 0.5% = 77, so the deficit increases from 300 to about 377 (and the after-tax deficit to about 283), adjusted leverage from 2.75x to about 2.94x, with assets unchanged. If half the interest-rate risk were hedged, the increase would be about half as large.

Interpretation. The deficit consumes 225 of equity value and lifts leverage by three-quarters of a turn; a half-point fall in discount rates, the sort of move that occurs in a normal year, adds almost another fifth of a turn. That is why lenders, rating agencies and acquirers treat the plan as debt, and why the board should evaluate pension decisions alongside capital structure.

16.8.2 Strategic options for the sponsor

A sponsor facing a large DB exposure has a menu of options, whose costs and benefits differ.

- **Fund the deficit faster**, with cash or with assets such as property or a contingent asset arrangement. This reduces risk but uses cash. It may be tax-efficient, since contributions are generally deductible. Sponsors sometimes borrow to fund the plan (a debt-for-pension swap), which is value-neutral before tax effects but lowers risk for members.
- **Reduce the risk of the plan assets** through LDI and de-risking, trading expected return for lower variance (Section 16.5). This is the cheapest way to cut deficit volatility.
- **Close the plan** to new entrants or to future accrual, replacing it with a DC arrangement. This halts growth in exposure but does not remove legacy liabilities.

- **Transfer risk to an insurer.** A buy-in covers a portion of liabilities with an insurance policy; a buy-out removes them from the sponsor. The price is typically higher than the accounting liability (the insurer discounts at a lower rate and loads for capital and profit), so completing a buy-out often requires a cash contribution above the reported deficit, but leaves the sponsor free of longevity and investment risk.
- **Liability management exercises**, such as enhanced transfer values or pension increase exchanges, which offer members a capital sum in lieu of future benefits, must be handled with care for fairness and regulatory reasons.
- **Negotiate the contribution schedule and security**, balancing the trustees' demand for contributions against the sponsor's capital needs. The relationship between the strength of the sponsor's covenant and the plan's risk is two-way: a stronger sponsor can support a riskier plan.

16.8.3 Valuation and M&A implications

In valuation (Volume 1, Chapter 21) the pension deficit should be treated as a debt-like item in the bridge from enterprise value to equity value, after tax, with careful attention to the discount rate used and to whether the deficit is calculated on a market-consistent basis. In M&A and LBO transactions (Volume 1, Chapters 22 and 23), the pension plan can be a deal issue: buyers will demand a price adjustment, or a funding commitment, and will examine the trustees' power to demand additional contributions on a change of control. Because leveraged buyers cannot support highly volatile liabilities, an LBO often leads to a rapid de-risking or insurance transaction. Finally, **ratings and covenants**: sponsors should check how the lender treats pension deficits in covenant definitions, since an unexpected fall in rates can trigger a technical breach.

Case Study

Norwood Rail Pension Plan: De-Risk Now or Wait for Rates?

Illustrative case; the firm, the plan and all figures are fictional.

Norwood Rail Holdings is a regional rail and freight operator with annual EBITDA of 420 million and net financial debt of 1.05 billion. It sponsors the Norwood Staff Pension Scheme, a final-salary DB plan closed to new members eight years ago and to future accrual two years ago, when management replaced it with a DC plan for active employees. The legacy plan has 14,000 members: about 40 percent pensioners, 45 percent deferred members and 15 percent still working for the firm with frozen benefits.

On the trustees' latest valuation, using a discount rate based on high-quality corporate bond yields, liabilities are 2.4 billion and assets are 2.0 billion, a funding ratio of 83 percent and a deficit of 400 million. The liability has a modified duration of about 13 years and is largely inflation-linked. The asset portfolio is 55 percent equities and private markets, 15 percent credit, and 30 percent government bonds and cash, with a modest swap overlay that hedges about 35 percent of the interest-rate sensitivity of the liabilities. The sponsor currently contributes 45 million a year toward the deficit under an agreed recovery plan that runs ten years.

The chief financial officer, Amara Bello, faces three pressures. First, the company's main lender has told her that it adds 100 percent of the pension deficit, after tax, to net debt in its leverage covenant, which has a ceiling of 3.5 times. At a 25 percent tax rate, the adjusted leverage is already above 3.2 times, and the lender has noted that a further fall in discount rates could breach the covenant. Second, the trustees want a "journey plan" to full funding on a conservative basis within ten years, and want a hedge ratio of 80 percent as soon as possible. They also want the sponsor to give security over a property portfolio as a contingent asset. Third, the board wants to resume dividends and to fund a fleet replacement programme of 600 million, and sees the pension contributions as competing with those priorities.

An insurer has offered an indicative quote to buy in the pensioner liabilities, about 960 million on a reported basis, at a price of roughly 1.07 billion. The price exceeds the accounting liability because the insurer prices on a lower discount rate and includes a capital and profit margin. The trustees are interested but note that using plan assets to pay the premium would leave the remaining plan with a higher proportion of deferred members, whose liabilities are longer and more uncertain, and with less liquid assets.

Amara has also been approached by the chair of the trustees about an alternative: the sponsor could issue a 300 million bond and make a one-off contribution to the plan, in return for a reduction in the annual recovery contributions and a more flexible investment strategy. Her treasury team notes that the bond would raise reported financial debt, but would reduce the after-tax pension deficit by the same amount (before any tax effects), so adjusted leverage would be unchanged on day one. It would, however, replace an uncertain, contractually soft obligation with hard debt that carries covenants and has to be refinanced.

Her analysts have run a simulation. With the current allocation and a 35 percent hedge, the one-year 5th-percentile fall in the funding ratio is about 11 points, and the 5th percentile of the deficit in five years is about 800 million. At an 80 percent hedge ratio with 40 percent in return-seeking assets, the one-year 5th-percentile fall narrows to about 5 points, but the expected funding ratio in five years falls by about 4 points and the plan’s liquidity needs for swap collateral rise to roughly 12 percent of assets in a 100 basis point shock.

Discussion Questions

1. Estimate the after-tax pension deficit and Norwood’s adjusted leverage using the figures given. Explain why the lender’s treatment is economically defensible, and what the sponsor can and cannot do to influence it.
2. Evaluate the trustees’ proposal to raise the hedge ratio to 80 percent. What does it do to the distribution of the deficit, to expected funding and to liquidity? What preparations (collateral buffer, rebalancing rules) would you require before agreeing?
3. Compare the insurer buy-in at 1.07 billion with the alternative of keeping the pensioner liabilities. Under what circumstances is the premium over the accounting liability justified, and what happens to the residual plan’s risk?
4. Assess the bond-financed contribution proposal. Is it value-neutral? Consider tax, covenants, the sponsor’s credit rating, the trustees’ security demands, and the competing demands of the dividend and fleet programme, and recommend a sequenced strategy.

Key Takeaways

- For any institution with liabilities or a spending objective, the relevant risk is the risk to the surplus or funding ratio, not the volatility of assets in isolation.
- DB plans place investment, longevity and inflation risk on the sponsor; DC plans place them on members; the shift to DC moves risk rather than removing it, and fees and contribution persistence dominate DC outcomes.
- The PBO depends heavily on salary projection and the discount rate; the half-point to one-point discount-rate sensitivity of a long-duration plan is typically 5 to 10 percent of the liability.
- The funding ratio depends on the discount-rate convention: the same plan can appear 83 percent or 112 percent funded; a market-consistent rate separates the value of the promise from the risk of the assets chosen to fund it.
- Liability-driven investing reduces unrewarded interest-rate and inflation risk by matching DV01, freeing the risk budget for return-seeking assets, but it requires liquidity to meet collateral calls and removes upside from rate rises.

- A hedge ratio is a ratio of DV01s, so an underfunded plan with a percentage duration match is still unhedged in money terms.
- Asset allocation for plans is a trade-off between expected funding and tail risk; simulations make this explicit but depend on assumed correlations and premia.
- The endowment model exploits a long horizon through equities and illiquid assets, but its success depends on manager access and liquidity discipline, and smoothing spending rules protect budgets at some cost to capital in downturns.
- Sovereign wealth funds differ by purpose (stabilisation, savings, development, pension reserve); governance, spending rules and transparency matter as much as asset allocation.
- For sponsors, an after-tax pension deficit is a debt-like claim that reduces equity value and raises adjusted leverage; the options (faster funding, de-risking, closure, risk transfer) should be evaluated together with capital structure and dividend policy.

Review Questions

1. Why does the liability or spending objective, rather than the asset volatility, define the risk of an institutional investor? Illustrate with a pension fund and an endowment.
2. Compare DB and DC plans in terms of who bears investment, longevity and inflation risk, and explain why a sponsor might close a DB plan but still have large obligations for decades.
3. Distinguish the ABO, PBO, service cost and interest cost. Why is the PBO larger than the ABO for a final-salary plan?
4. Explain why funding ratios computed with different discount rates are not comparable. What are the main arguments for and against discounting at the expected return on assets?
5. Using the Sharpe pension-put idea, explain why increasing the riskiness of plan assets can transfer value to shareholders at the expense of members or a pension guarantee fund.
6. What is DV01, and why is the hedge ratio expressed as a ratio of DV01s rather than durations? Why can an underfunded plan with matched durations still be exposed to interest-rate changes?
7. Describe the benefits and risks of using swaps for LDI, with particular attention to collateral and liquidity risk.
8. Why is the minimum-risk portfolio for a pension plan the liability-matching portfolio, and how should a plan decide how much surplus risk to take?
9. Describe the main features and the main criticisms of the endowment model. What did the 2008 crisis reveal about illiquid commitments?
10. Why do corporate lenders and rating agencies add the pension deficit to debt, and what options does a sponsor have for managing the exposure?

Problems

1. A plan will pay 30 million at the end of each year for 20 years. Assets are 330 million. Compute the funding ratio when liabilities are discounted at (a) 4 percent and (b) 6 percent.
2. A plan has liabilities of 800 million with modified duration 12, and assets of 700 million with modified duration 6. (a) Compute the DV01 of each, and the hedge ratio. (b) Estimate the funding ratio after a 50 bp fall in rates, using

linear duration approximations.

3. A 40-year-old member has 10 years of service and a salary of 50,000, growing 2.5 percent a year. The plan pays 1/60th of final salary per year of service from age 65, level for 18 years. The discount rate is 4.5 percent. Compute the projected final salary, the accrued pension and the PBO (ignore mortality and indexation).
4. A DC member earns 50,000, receives 2 percent annual raises for 30 years, and contributes 10 percent of salary at each year end. The account earns 6 percent net. The balance at retirement buys a 20-year level annuity at 5 percent. Compute the balance, annual income and replacement ratio against final-year salary.
5. An endowment's previous-year spending was 22 million, inflation is 3 percent, the opening market value is 480 million, the smoothing weight is 0.7 and the long-run spending rate is 5 percent. Compute this year's spending and its percentage of market value.
6. A plan has liability DV01 of 0.80 million and asset DV01 of 0.35 million. A receiver swap has modified duration 15. What swap notional hedges the remaining DV01? What notional would achieve a 60 percent hedge ratio?
7. A sponsor has EBITDA 120 million, net debt 360 million, pension deficit 150 million before tax, tax rate 21 percent and operating EV 1,000 million. Compute adjusted leverage with the after-tax deficit and the equity value after deducting net debt and the after-tax deficit.
8. A plan with funding ratio 0.90 has asset volatility 10 percent and liability volatility 9 percent. Compute the standard deviation of the one-year change in the funding ratio (approximate formula $FR \times \sqrt{(\sigma_A^2 + \sigma_L^2 - 2\rho\sigma_A\sigma_L)}$) when $\rho = 0.60$ and when $\rho = 0.95$. Interpret.

Answers to Selected Problems

1. Annuity factors: 13.5903 at 4 percent and 11.4699 at 6 percent. PV of liabilities: 407.71 and 344.10. Funding ratios: (a) 80.9 percent; (b) 95.9 percent.
2. a. Liability DV01 = 800 × 12 × 0.0001 = 0.96; asset DV01 = 700 × 6 × 0.0001 = 0.42; hedge ratio = 0.42 / 0.96 = 43.75 percent. (b) Liabilities rise by about 800 × 12 × 0.005 = 48 to 848; assets rise by 700 × 6 × 0.005 = 21 to 721; funding ratio falls from 87.5 percent to 85.0 percent.
3. Final salary = $50{,}000 \times 1.025^{25} = 92{,}697$. Accrued pension = 10/60 × 92,697 = 15,450. Annuity factor (18 years, 4.5 percent) = 12.160, so value at 65 = 187,874; PBO = $187{,}874 / 1.045^{25} \approx 62{,}509$.
4. Final salary = $50{,}000 \times 1.02^{29} = 88{,}792$. Balance = 491,516. Annuity factor (20 years, 5 percent) = 12.4622; income = 39,441; replacement ratio = 44.4 percent.
5. Spending = 0.7 × 22 × 1.03 + 0.3 × 0.05 × 480 = 15.862 + 7.200 = 23.06 million, which is 4.80 percent of market value.
6. Remaining DV01 = 0.45; notional = 0.45 / (15 × 0.0001) = 300 million. For a 60 percent hedge ratio, total DV01 needed is 0.48, so swap DV01 = 0.13 and notional = 86.7 million.
7. After-tax deficit = 150 × 0.79 = 118.5. Adjusted leverage = (360 + 118.5) / 120 = 3.99x (versus 3.0x reported). Equity value = 1,000 − 360 − 118.5 = 521.5 million (versus 640 before the deficit).
8. For $\rho = 0.60$: $0.90 \times \sqrt{(0.0100 + 0.0081 - 0.0108)} = 0.90 \times \sqrt{0.0073} = 7.69$ percent. For $\rho = 0.95$: $0.90 \times \sqrt{(0.0100 + 0.0081 - 0.0171)} = 0.90 \times \sqrt{0.0010} = 2.85$ percent. The higher the correlation between assets and liabilities (that is, the better the hedge), the lower the surplus volatility.

Part V — Advanced Transactions

Chapter 17: Advanced M&A: Structuring, Tax and Accounting

Learning Objectives

After completing this chapter, you should be able to:

- Compare asset purchases, stock purchases and merger structures, and explain how each allocates tax, liability and contract risk between buyer and seller.
- Quantify the value of a tax basis step-up, and decide when a gross-up payment to the seller is justified.
- Apply the logic of purchase price accounting under IFRS 3 and ASC 805, including goodwill, identifiable intangibles, deferred taxes and contingent consideration, and explain what these entries do and do not say about economic value.
- Value an earnout as an option on post-closing performance, and design earnout terms that limit gaming and disputes.
- Analyze fixed-exchange-ratio, fixed-value and collared stock deals, and value a contingent value right (CVR) as a contingent claim.
- Evaluate risk allocation tools (representations and warranties, escrows, indemnities, insurance) and the mechanics of financing commitments, termination fees and deal certainty.

Opening Vignette

Illustrative scenario; all names and figures are fictional.

Harlow Instruments is a mid-sized industrial technology company with a market capitalization of about $4.2 billion. Its chief executive, Priya Ramanathan, has spent eighteen months courting Verity Sensors, a privately held maker of precision measurement equipment with $400 million of proposed enterprise value. Verity is owned by a founder family (62 percent), a growth equity fund (30 percent) and management (8 percent). The strategic logic is clean. The structure is not.

The founder family wants cash and, above all, wants to defer tax; the fund wants a clean exit before its vintage year ends; management wants to keep working and to be paid for the growth they believe in. Verity's two largest customers have change-of-control clauses. One of Verity's product lines is under a patent dispute whose outcome is unknowable for three years. Harlow's board is nervous about its own share price, which has been volatile, and about a rival bidder who could appear with an all-cash offer. Harlow's lenders are willing to provide a bridge facility but want to understand what happens if antitrust review drags on.

Over six weeks, the deal team discovers that "price" is the smallest part of the negotiation. Buying Verity's assets would give Harlow a tax basis step-up worth millions but would trigger a second layer of tax for some sellers and require consents from every counterparty. A stock purchase preserves contracts but inherits unknown liabilities. A stock-for-stock merger defers seller tax but exposes the sellers to Harlow's share price and Harlow to dilution. An earnout could bridge the valuation gap on the growth forecasts, but it could also poison the working relationship with the managers who must earn it. A contingent value right could address the patent dispute. A representations-and-warranties insurance policy could replace the founders' fear of a long escrow.

This chapter equips you to work through exactly these trade-offs. The central lesson is that structure is value: the same $400 million headline can be worth very different amounts to buyer and seller, and the skill of a deal-maker lies in splitting that difference intelligently rather than arguing about the headline.

17.1 Deal Structure as a Value Lever

Volume 1, Chapter 22 treated mergers and acquisitions mainly as a valuation exercise: estimate standalone value, add synergies, decide how much to pay, and compare cash with stock as the means of payment. That framing is correct but incomplete. Two deals with the same headline price can transfer very different amounts of wealth, because price is only one of at least five terms that determine who gets what:

1. **Form of transaction** (asset purchase, stock purchase, merger), which determines tax treatment, liability transfer and the consents required.
2. **Form of consideration** (cash, stock, notes, contingent claims), which determines who bears the risk that value changes between signing and closing and after closing.
3. **Timing and conditionality** (financing conditions, regulatory approvals, termination rights), which determines the probability that the deal closes at all.
4. **Risk allocation** (representations, warranties, indemnities, escrows, insurance), which determines who pays when something undisclosed turns up.
5. **Accounting and reporting**, which does not change cash flows but changes reported earnings, covenants, management incentives and sometimes the stock price.

A useful way to hold these together is to treat the deal as the sale of cash flows plus a bundle of risks and options. The headline price pays for the cash flows; every other term prices a risk or option, and each can be shifted between the parties at a cost. The ideal allocation puts each risk with the party that can bear it most cheaply, and avoids giving one party risk over an outcome that the other controls.

The Three Basic Forms

Stock purchase. The buyer acquires the target's shares directly from its shareholders. The legal entity survives unchanged: its contracts, permits, employees and liabilities (including unknown ones) stay inside it. For the buyer, the tax basis of the target's assets generally carries over; the buyer takes a basis in the shares equal to the price paid, but that "outside" basis is realized only on a future sale of the shares. Each selling shareholder must agree, so a stock purchase from a dispersed shareholder base is impractical without a statutory squeeze-out. For private targets with a handful of holders, it is standard.

Asset purchase. The buyer acquires selected assets and assumes selected liabilities, leaving the rest behind in the seller's legal shell. The buyer takes a fair-value ("cost") tax basis in each acquired asset, which is the source of the step-up discussed below. Excluded liabilities, such as historical product liability or unpaid taxes, stay with the seller (subject to successor liability doctrines that can reach buyers in some situations). The cost is administrative: contracts, leases and permits often need third-party consent for assignment, employees must be transferred, and title to each asset must be conveyed.

Merger. A statutory merger combines two entities by operation of law, with shareholder votes rather than individual sales. In a *forward merger* the target merges into the acquirer and ceases to exist. In a *reverse triangular merger* the acquirer forms a subsidiary that merges into the target, so the target survives as a subsidiary of the acquirer; contracts and permits

normally remain with the surviving entity (though some contracts treat a merger as an assignment or change of control, which must be checked). The reverse triangular form is the workhorse of public-company deals precisely because it combines the legal continuity of a stock purchase with the ability to bind all shareholders through a vote.

Tax-Free Reorganizations and the Continuity Requirement

In many jurisdictions, a share-based acquisition can qualify for tax deferral at the shareholder level if enough of the consideration is acquirer stock and certain continuity-of-interest and business-purpose tests are met. Under such a reorganization the target shareholders do not recognize gain on the stock they receive; their tax basis in the target shares carries over to the new shares, and the gain is deferred until they sell. The buyer does not step up asset basis. This creates a fundamental trade-off that recurs throughout this chapter:

- **Taxable deal:** sellers pay tax now; the buyer may obtain a step-up in asset basis (in an asset deal or a deemed asset deal) that creates future tax deductions.
- **Tax-deferred deal:** sellers defer tax; the buyer inherits the old basis and loses the deductions.

The size of each side of the trade-off is estimable, and it is what the negotiation should be about. Specific rules differ by country and change over time, so the numbers in this chapter are illustrative; always have tax counsel confirm current treatment before relying on any structure.

Table 17.1 – Comparison of Basic Deal Forms

Feature	Stock purchase	Asset purchase	Reverse triangular merger
Who sells	Each shareholder	The target entity	Shareholders vote; dissenters may have appraisal rights
Target legal entity	Survives	Remains with seller (assets move)	Survives as subsidiary
Liabilities	All inherited	Only those assumed	All inherited (survivor entity)
Contract consents	Only change-of-control clauses	Assignment consents for each contract	Change-of-control and anti-assignment clauses must be reviewed
Buyer tax basis in assets	Carryover (unless election)	Fair value (step-up)	Carryover (unless election)
Seller tax	Single level at shareholder	Possible two levels if corporate target	Single level, possibly deferred if stock consideration qualifies
Typical use	Private targets with few holders	Carve-outs, distressed sellers, asset-light buyers	Public targets, many holders

Notice that the right column’s legal continuity makes it the default for acquirers who value a clean transfer of the operating entity, while the middle column is preferred where the buyer wants to cherry-pick or needs the tax basis.

17.2 Tax Basis Step-Ups

What a Step-Up Is and Why It Has Value

When a buyer pays a price above the target’s tax basis in its assets, it would ideally be able to depreciate or amortize the full price, not the old, lower basis. A *step-up* resets the tax basis of the assets to fair market value (or the allocated purchase price). The incremental depreciation and amortization deductions reduce taxable income in future years. The tax saving is a

real cash flow; its present value is the value of the step-up.

The step-up deduction applies to allocable categories. Under the simplified approach used in the US and many other systems, the purchase price is allocated across asset classes in a residual method: cash and receivables first, then inventory, then tangible fixed assets, then identifiable intangibles, and finally goodwill. Fixed assets are depreciated over their tax lives; acquired intangibles including goodwill are in the US amortized straight-line over 15 years for tax purposes (the "Section 197" rule). Because goodwill and intangibles dominate in knowledge-based and technology businesses, the step-up is often a 15-year deduction stream.

The present value is straightforward:

```
Annual tax shield = (Step-up ÷ Amortization years) × Tax rate
PV of shield      = Annual shield × Annuity factor(r, n)
Annuity factor    = [1 − (1 + r)^−n] ÷ r
```

The discount rate deserves thought. The shield is a claim on future taxable income, so it is only as safe as the buyer's ability to use deductions. Most practitioners discount at a rate near the cost of debt or at the WACC. Using the cost of debt treats the shield as relatively safe (the standard Modigliani-Miller treatment in Volume 1, Chapter 15, where interest tax shields are discounted at the cost of debt when the debt is fixed); using the WACC treats it as risky because it depends on future profitability. We use the buyer's WACC of 9 percent below. Both choices are defensible, and a sensitivity range is better than false precision.

Example 17.1 – Value of a step-up. Harlow buys Verity's assets for $400 million. Verity's tax basis in those assets is $150 million, so the step-up is $400 − $150 = $250 million. Assume the entire step-up is amortizable straight-line over 15 years, Harlow's tax rate is 25 percent, and the discount rate is 9 percent.

Step 1 – Annual deduction: $250 ÷ 15 = $16.667 million.

Step 2 – Annual tax saving: $16.667 × 0.25 = $4.1667 million.

Step 3 – Annuity factor for 15 years at 9 percent:

```
[1 − 1.09^−15] ÷ 0.09 = [1 − 0.27454] ÷ 0.09 = 8.0607
```

Step 4 – PV of the shield: $4.1667 × 8.0607 = **$33.59 million**.

That is 8.4 percent of the $400 million price. Sensitivity: at a 6 percent discount rate (a debt-like rate), the annuity factor is 9.7122 and the PV is $40.47 million; at a 12 percent rate it is 6.8109 and $28.38 million. The step-up is therefore worth roughly $28 to $40 million depending on the risk assumed, and any structural cost that the sellers demand in return needs to be compared with that range.

Managerial implication: the headline of "$33.6 million of free value" is the maximum amount the buyer should be prepared to share with the sellers for access to the step-up. Whether any is shared depends on how much of the cost falls on the sellers, as the next example shows.

Who Bears the Cost: Single and Double Taxation

A step-up requires the seller to be taxed as if the assets were sold, not the shares. If the seller is a corporation whose shareholders are the ultimate owners, an asset sale triggers corporate-level tax on the asset gain, and shareholders are taxed again when the proceeds are distributed. This double tax typically exceeds the value of the step-up to the buyer, so for C-

corporation targets a plain asset purchase rarely makes economic sense. (There are exceptions: if the target has large net operating losses to absorb the gain, or if it is a subsidiary of a consolidated group whose parent has a low stock basis.)

For pass-through targets, the picture reverses. A partnership, LLC taxed as a partnership, or S corporation is taxed once at the owner level whether the buyer purchases assets or shares. Jurisdictions often allow an *election* that treats a stock purchase as an asset purchase for tax purposes (in the US, Section 338(h)(10) for S corporations and subsidiaries of consolidated groups, Section 336(e) in related circumstances, and an "F reorganization" followed by an LLC interest sale for S corporations). The buyer gets a step-up; the legal form is a stock purchase, preserving contracts and avoiding the consent problem; and the seller bears one layer of tax, though the *character* of income can change.

The character point matters. A step-up deal can convert part of the seller's gain from low-rate capital gain on shares to higher-rate ordinary income, typically on depreciation recapture and inventory or receivables. That incremental seller tax is the real price of the buyer's step-up and the number to negotiate.

Example 17.2 – Sharing the step-up surplus in a pass-through deal. Verity is an S corporation. Under a 338(h)(10)-style election, $60 million of the $250 million total gain is recharacterized as ordinary income (recapture) taxed at 37 percent instead of the 20 percent capital gain rate on the shares. The buyer's benefit is $33.59 million from Example 17.1.

Step 1 – Seller's incremental tax: $60 × (0.37 − 0.20) = $10.20 million.

Step 2 – Gross-up payment X needed to leave the sellers whole. The additional price is itself taxed at 20 percent as additional capital gain, so we need X × (1 − 0.20) = $10.20 million, giving X = $10.20 ÷ 0.80 = **$12.75 million**.

Step 3 – Buyer's value of the extra price. The extra $12.75 million is also allocated to amortizable assets and creates its own shield: ($12.75 ÷ 15) × 0.25 × 8.0607 = **$1.71 million**.

Step 4 – Buyer's net gain from the election: $33.59 + $1.71 − $12.75 = **$22.55 million**.

The sellers are indifferent (they get whole), and the buyer is $22.55 million better off. Total surplus created is $22.55 million, which could be split in a negotiation: if the sellers hold out for half of the buyer's net gain, an additional $11.3 million would be paid, and the final price would be about $424 million. The mistake to avoid is treating the election as a pure cost to sellers who ought to give it away: it is a positive-sum structure whose surplus should be shared, and a buyer who insists on all of it risks losing the seller's cooperation on the deal's other terms.

Contrast: the C-corporation case. Suppose instead Verity were a C-corporation with an inside asset basis of $150 million and shareholder stock basis of $100 million, corporate tax of 25 percent and shareholder capital gain tax of 20 percent. In a stock sale at $400 million, shareholders net $400 − 0.20 × (400 − 100) = $340 million. In an asset sale at $400 million, the corporation pays 0.25 × (400 − 150) = $62.5 million of tax, distributes $337.5 million, and shareholders net $337.5 − 0.20 × (337.5 − 100) = $290.0 million. To leave the shareholders with $340 million, the corporation must receive $400 million after corporate tax and the pre-tax price must be $(400 − 0.25 × 150) ÷ 0.75 = $483.33 million, an $83.33 million gross-up. The buyer's step-up on the larger price is $333.33 million; its PV shield at 9 percent is ($333.33 ÷ 15) × 0.25 × 8.0607 = $44.78 million. The buyer pays $83.33 million to receive $44.78 million, a net loss of $38.55 million. This is why asset deals are rarely done for ordinary C-corporation targets.

17.3 Purchase Price Accounting

What the Standards Require

Under IFRS 3 *Business Combinations* and the US equivalent, ASC 805, every acquisition of a business is accounted for by the **acquisition method**. The steps are the same under both frameworks:

1. Identify the acquirer.
2. Determine the acquisition date (the date control passes).
3. Recognize and measure, at fair value, the identifiable assets acquired and the liabilities assumed, including intangible assets that the target never recorded on its own balance sheet (customer relationships, developed technology, trade names, order backlog), with limited exceptions (deferred taxes, employee benefits and a few others follow their own standards).
4. Measure the consideration transferred at fair value, including the acquisition-date fair value of any contingent consideration.
5. Recognize **goodwill** as the residual: consideration plus any non-controlling interest (and previously held interest) minus the net identifiable assets at fair value. If the residual is negative, a bargain purchase gain is recognized after reassessing the measurement.

```
Goodwill = Consideration transferred (incl. contingent consideration at fair value)
         + Fair value of non-controlling interest (or its proportionate share of net assets under IFRS)
         + Fair value of previously held equity interest
         − Fair value of identifiable net assets acquired
```

The core point for finance professionals is that purchase price accounting is *an allocation of the price you paid*, not a valuation of what you got. The residual called goodwill is by construction whatever is left: it may reflect synergies, an assembled workforce, overpayment, or a mistake in valuing the identifiable assets. Treating the goodwill figure as the "value of the synergies" is a common error. A bidder who overpays in a hot market ends up with more goodwill, not less; the accounting never tells you the price was too high until an impairment test fails years later.

Differences Between IFRS and US GAAP That Matter in Practice

- **Goodwill amortization.** Neither IFRS nor US GAAP (for public companies) amortizes goodwill; both require annual impairment testing at the level of the cash-generating unit (IFRS) or reporting unit (US GAAP). US private companies may elect to amortize goodwill over up to ten years as an alternative. The impairment tests differ in detail (IFRS compares carrying amount with recoverable amount at the cash-generating-unit level; US GAAP compares carrying amount with fair value at the reporting-unit level), and IFRS permits no reversal of goodwill impairment.
- **Non-controlling interest.** IFRS allows a choice, acquisition by acquisition, between measuring non-controlling interest at fair value ("full goodwill") or at its proportionate share of the net identifiable assets ("partial goodwill"). US GAAP requires fair value.
- **Contingent consideration.** Both require recognition at acquisition-date fair value. Liability-classified contingent consideration is remeasured at fair value every period, with changes going through earnings. Equity-classified contingent consideration is not remeasured. This is why the structure of an earnout (cash versus shares) affects reported volatility.

Deferred Taxes in Purchase Accounting

In a stock deal without a tax election, the buyer steps up the *book* basis of the target's assets to fair value but inherits the old *tax* basis. The difference is a temporary difference, and the standards require a **deferred tax liability (DTL)** on the book-tax difference (with exceptions for goodwill itself). The DTL increases the net liabilities assumed and therefore *increases* goodwill. As the intangible assets amortize, the DTL unwinds, offsetting the tax effect of the book amortization so that the after-tax earnings charge reflects the true cash tax position.

Example 17.3 – Purchase price allocation with deferred taxes. Harlow acquires 100 percent of Verity's shares for $400 million in cash, with no tax election. Verity's identifiable net assets at fair value, and their tax bases, are:

Item	Fair value ($m)	Tax basis ($m)	Book-tax difference ($m)
Net working capital	40	40	0
Property, plant and equipment	120	100	20
Customer relationships	90	0	90
Developed technology	60	0	60
Trade name (indefinite life)	20	0	20
Total	**330**	**140**	**190**

Tax rate: 25 percent. Customer relationships have a ten-year life, the technology five years, and the trade name is not amortized.

Step 1 – DTL: 190 × 0.25 = **$47.5 million**.

Step 2 – Net identifiable assets: 330 − 47.5 = $282.5 million.

Step 3 – Goodwill: 400 − 282.5 = **$117.5 million**, which equals 29.4 percent of the price.

Step 4 – Annual intangible amortization: customer relationships 90 ÷ 10 = $9 million; technology 60 ÷ 5 = $12 million; total $21 million per year in the first five years.

Step 5 – Earnings effect: pre-tax charge $21 million. The DTL unwinds by 21 × 0.25 = $5.25 million (a deferred tax benefit), so the after-tax reduction to net income is 21 − 5.25 = **$15.75 million**. After year 5, the technology is fully amortized and the charge falls to $9 million pre-tax ($6.75 million after tax).

Step 6 – Compare with a step-up structure. If the same deal were structured as a 338(h)(10)-type asset deal, the tax basis would equal fair value and goodwill, as defined for tax, would be amortizable. In book terms there would be no DTL on the identifiable assets, so goodwill would be $400 − $330 = **$70 million**, and the tax deductions would reduce cash taxes (as in Example 17.1) rather than being "booked" through deferred tax. The book goodwill is lower by exactly the DTL: $117.5 − $70 = $47.5 million.

Managerial implication: the DTL "grosses up" goodwill, and the after-tax book amortization of the finite-lived intangibles will depress reported EPS for years. Management that cares about reported EPS accretion, whether because of bonus metrics, covenants or investor communication, should model this explicitly, and should communicate "cash EPS" (excluding acquisition amortization) alongside it. The cash flows are the same either way; the controversy is entirely over how to describe them.

Contingent Consideration and Post-Closing Remeasurement

Because contingent consideration is a liability recognized at fair value, a good outcome after closing produces an *expense*. If Verity's managers achieve their targets and the earnout liability rises from $8 million to $30 million, Harlow reports a $22 million charge, even though the business is doing better than expected. The reverse (poor performance reduces the liability) produces a gain. Analysts usually strip these items from "adjusted" earnings, but covenant definitions, bonus plans and credit metrics may not. A treasurer should therefore check how earnout remeasurement flows through the definition of EBITDA in debt agreements before agreeing to the structure.

17.4 Earnouts: Bridging Valuation Gaps

Why Earnouts Exist

An **earnout** is a deferred, contingent payment to the seller that depends on the target's post-closing performance (revenue, EBITDA, product milestones, customer retention). Earnouts are used when buyer and seller disagree on the business's future: the seller believes the forecasts and wants to be paid for them, the buyer doubts them and does not want to pay for growth that never materializes. By making part of the price contingent, the earnout lets each party "put its money where its beliefs are." It also helps in three other situations: when management will stay on and incentives matter; when the seller has superior information (adverse selection), because a seller who accepts a large contingent component signals confidence; and when the buyer is cash-constrained and prefers to defer payments to the period when the business generates the cash.

The costs are equally real. Earnouts create **agency conflicts**: the buyer controls the business after closing and can reduce measured performance (shifting costs, pricing for market share, integrating in ways that move revenue); the seller-managers can inflate measured performance (pulling forward sales, cutting investment, discounting aggressively) at the expense of long-term value. Many earnout disputes end in litigation, and a surprisingly large share of the legal conflicts in M&A involve earnout definitions.

Earnouts as Options

Economically, a typical earnout is a **call option (or a call spread) on a performance metric**. A payment of 60 percent of revenue above a threshold, capped at a maximum, is a long call at the threshold and a short call at the cap, with the metric as the underlying. This framing delivers several valuable insights:

- The value of the earnout rises with **volatility** of the metric, not only with its expected level. A buyer who thinks only about the expected case misses the value of the upside tail.
- A threshold *above* the expected outcome still has substantial value because of the option-like payoff. Valuing the earnout at the expected-case payoff (a deterministic forecast) systematically understates its value, as the example below shows.
- The cap reduces the cost to the buyer and reduces the seller's incentive to push beyond the cap.
- Because payments are made years later, discounting matters, and the right rate reflects the risk of the metric and the credit risk of the buyer.

In practice, contingent consideration is valued with Monte Carlo simulation or option pricing models, using risk-adjusted projections (the metric's drift reduced so that discounting at a near-risk-free rate is appropriate) or real-world projections discounted at a higher rate. Auditors require this for acquisition accounting, and the discount rate must be consistent with the projection's risk treatment.

Example 17.4 – Valuing a revenue earnout. Verity's current annual revenue is $100 million. The sellers will receive 60 percent of year-3 revenue above $120 million, capped at $40 million. The cap is reached when $0.6 \times (R_3 - 120) = 40$, i.e. $R_3 = 120 + 66.67 = \$186.67$ million. Assume year-3 revenue is lognormal with a risk-adjusted expected growth rate of 7 percent per year (so the expected R_3 is $100 \times 1.07^3 = \$122.50$ million) and annual volatility of 20 percent. The payment is made at the end of year 3, and we discount at 5 percent (a risk-adjusted rate reflecting the buyer's credit risk).

Step 1 – Naive deterministic value: at the expected revenue of $122.50 million the payout is $0.6 \times (122.50 - 120) = \$1.50$ million. The PV at 5 percent would be $1.30 million, which looks trivial.

Step 2 – Option value. The payoff is $0.6 \times [\text{Call}(K=120) - \text{Call}(K=186.67)]$ on revenue with forward F = 122.50. For a lognormal variable, the expected value of a call payoff is:

```
E[(R − K)+] = F·N(d1) − K·N(d2)
d1 = [ln(F/K) + ½σ²T] ÷ (σ√T),   d2 = d1 − σ√T
```

With $\sigma\sqrt{T} = 0.20 \times 1.7321 = 0.3464$:

- For K = 120: d1 = [ln(1.02087) + 0.06] ÷ 0.3464 = (0.02065 + 0.06) ÷ 0.3464 = 0.2328; d2 = −0.1136. N(d1) = 0.5920, N(d2) = 0.4548. E = 122.50 × 0.5921 − 120 × 0.4548 = 72.53 − 54.58 = $17.95 million (rounded).
- For K = 186.67: d1 = [ln(0.65627) + 0.06] ÷ 0.3464 = (−0.42115 + 0.06) ÷ 0.3464 = −1.0436; d2 = −1.3900. N(d1) = 0.1483, N(d2) = 0.0824. E = 122.50 × 0.1486 − 186.67 × 0.0824 = 18.20 − 15.38 = $2.82 million.

Step 3 – Expected payoff: 0.6 × (17.95 − 2.82) ≈ $9.08 million (the unrounded calculation gives $9.083 million).

Step 4 – Present value at 5 percent over three years: 9.083 ÷ 1.157625 = **$7.85 million**.

Step 5 – Monte Carlo check. A simulation of 1,000,000 paths gave an expected payoff of $9.077 million (PV $7.84 million); the probability that revenue exceeds $120 million is 45.5 percent, and the probability the cap binds is 8.2 percent. The analytic and simulated values agree.

Step 6 – Sensitivity to volatility (PV at 5 percent): with σ = 10 percent, the earnout is worth $4.98 million; with σ = 20 percent, $7.85 million; with σ = 30 percent, $8.83 million.

Interpretation: the deterministic view says $1.30 million; the option view says $7.85 million, six times higher. If the parties negotiate as if the earnout is worth little, the sellers are giving up value, or the buyer is under-accrued. For Harlow's accounting, the acquisition-date fair value of the contingent consideration is $7.85 million, which is part of the consideration transferred and therefore adds to goodwill. Each subsequent period, the liability is remeasured as time passes and revenue evolves.

Designing Earnouts That Work

Experience suggests a set of design rules:

1. **Choose a metric the seller can influence but the buyer cannot easily manipulate.** Revenue is harder to game than EBITDA (which depends on allocated overheads and integration costs) but invites unprofitable growth. Gross profit or contribution margin is a compromise. For biotech and technology, milestone metrics (regulatory approvals, product launches) are clean and observable.
2. **Define accounting terms precisely.** State which accounting standards apply, forbid changes in policies, and specify treatment of intercompany transactions, shared costs and purchase accounting adjustments.

3. **Impose operating covenants, carefully.** Buyers promise not to take actions with the primary purpose of reducing the earnout (a "good faith" covenant is common); sellers want stronger protections such as separate books and minimum investment levels. The stronger the covenants, the less the buyer can integrate, which dilutes the strategic logic of the deal.
4. **Keep the earnout period short.** One to three years is typical. A long earnout locks the seller and buyer into a relationship neither controls, and an extended performance period lets the metric be affected by factors unrelated to the seller's effort.
5. **Cap and floor.** Caps limit the buyer's exposure; floors (minimum guaranteed payments) are really deferred purchase price and should be treated as debt-like.

Set-off rights deserve particular attention. If the buyer can deduct indemnity claims from the earnout, the earnout becomes a form of security for the sellers' warranties; sellers then bear more of the risk that the buyer will assert claims to reduce the payment. The combination of a contested earnout and a contested indemnity claim is the single most reliable recipe for post-closing litigation.

17.5 Share Consideration, Collars and Contingent Value Rights

Who Bears Price Risk Between Signing and Closing

When part or all of the consideration is acquirer stock, the value the target's shareholders receive floats with the acquirer's share price between signing and closing, a period that can run from weeks to over a year if regulatory approvals are slow. The exchange mechanism determines who bears the risk. Three designs dominate.

Fixed exchange ratio. Each target share receives a fixed number of acquirer shares (say 0.5). The target shareholders bear the risk that the acquirer's stock falls and enjoy the gains if it rises. The acquirer issues a fixed number of shares: its *dilution* is certain, but the *value* it gives up fluctuates. Because the target holders effectively own a fixed percentage of the combined company, a fixed ratio makes the deal a true merger of equals in risk-sharing terms: both sets of shareholders share in the combined entity's fortunes, including the market's reaction to the deal.

Fixed value (floating exchange ratio). Each target share receives acquirer shares worth a fixed dollar amount (say $50) at closing, so the number of shares issued adjusts. The target is protected against acquirer price declines; the acquirer bears the price risk and may have to issue many more shares if its price falls (an unbounded dilution risk). This looks like a cash deal funded by stock issuance at the closing date: the target holders behave like creditors with a claim of fixed value.

Collars blend these designs inside a price band. Two common varieties:

- *Fixed-ratio collar (floating outside the band).* The exchange ratio is fixed within the band (acquirer price from $90 to $110, say), so the value floats within the band; outside it, the ratio floats to hold the value at the band's edge ($45 at the bottom, $55 at the top). The target is protected from large declines; the acquirer is protected from paying more than a maximum value.
- *Fixed-value collar (floating within the band).* The exchange ratio floats within the band to deliver a fixed $50, and is fixed outside the band, so that value is allowed to float beyond the edges. Within the band the target is fully protected; outside the band, risk reverts to the target.

In option language, a collar is a position in a *call spread and a put spread* on the acquirer's price, held by one side and written by the other. Each collar variety assigns the options differently. A cap and floor on the value delivered are an option strategy (long put at the floor, short call at the cap, from the target's perspective). The target thus gives away upside in exchange for downside protection, and the acquirer does the reverse. Because these options are valuable (the price volatility of a stock over several months is large), the choice of collar should be priced rather than negotiated by intuition. The negotiation currency is the band width and the exchange ratio at the center; wider bands transfer more risk to the target.

Example 17.5 – Fixed ratio, fixed value, and collared structures. Harlow offers an exchange for a public target with 100 million shares. At signing Harlow trades at $100 and the headline price is $50 per target share, equivalent to 0.5 Harlow shares. Compare three structures as Harlow's price at closing varies. (Fixed-value collar band $90–$110; fixed-ratio collar value limits $45 and $55 with the same band.)

Harlow price at close ($)	Fixed ratio: ER / value per target share ($)	Fixed-value collar: ER / value ($)	Fixed-ratio collar: ER / value ($)
70	0.5000 / 35.00	0.5556 / 38.89	0.6429 / 45.00
80	0.5000 / 40.00	0.5556 / 44.44	0.5625 / 45.00
90	0.5000 / 45.00	0.5556 / 50.00	0.5000 / 45.00
100	0.5000 / 50.00	0.5000 / 50.00	0.5000 / 50.00
110	0.5000 / 55.00	0.4545 / 50.00	0.5000 / 55.00
120	0.5000 / 60.00	0.4545 / 54.55	0.4583 / 55.00
130	0.5000 / 65.00	0.4545 / 59.09	0.4231 / 55.00

Step 1 – Fixed-value collar: within $90–$110 the ratio is $50 ÷ price (0.5556 at $90, 0.5 at $100, 0.4545 at $110). Outside the band the ratio freezes at the edge: 50 ÷ 90 = 0.5556 below $90, 50 ÷ 110 = 0.4545 above $110. Value at $70 = 0.5556 × 70 = $38.89; at $130 = 0.4545 × 130 = $59.09.

Step 2 – Fixed-ratio collar: ratio 0.5 within the band; outside, the ratio floats to hold value at $45 (below $90): at $70 the ratio is 45 ÷ 70 = 0.6429; at $80, 45 ÷ 80 = 0.5625; and at $55 above $110: at $120 the ratio is 55 ÷ 120 = 0.4583; at $130, 55 ÷ 130 = 0.4231.

Step 3 – Cost to Harlow (value delivered for 100 million target shares). At a Harlow price of $70: fixed ratio $3,500 million (50.0 million shares); fixed-value collar $3,889 million (55.56 million shares); fixed-ratio collar $4,500 million (64.29 million shares). At $130: fixed ratio $6,500 million (50.0 million shares); fixed-value collar $5,909 million (45.45 million shares); fixed-ratio collar $5,500 million (42.31 million shares).

Step 4 – Interpretation. Harlow's cost ranges from $3.5 to $6.5 billion under a fixed ratio, $3.9 to $5.9 billion under the fixed-value collar, and $4.5 to $5.5 billion under the fixed-ratio collar. The fixed-ratio collar gives the target value insurance across the entire range and the narrowest range of costs to Harlow, but it forces Harlow to issue up to 64.3 million shares if its price collapses (a 29 percent increase in the shares issued versus the 50 million base). The simplest fixed ratio costs Harlow the most dilution protection but offers the target the least value protection.

Managerial implication: the right design depends on the relative risk aversion and information of the two parties. A target whose shareholders are diversified and that wants to share in synergies prefers a fixed ratio; a target with concentrated holders who need certainty of value prefers a collar; an acquirer worried about its stock's volatility and willing to pay for deal completion will concede a collar but may insist on a cap on shares issued.

Contingent Value Rights

A **contingent value right (CVR)** is a security or contractual right issued to target shareholders in addition to the upfront consideration, entitling them to a payment if a specified future event occurs: regulatory approval of a drug, achievement of sales milestones, resolution of litigation, or a minimum price of the acquirer's stock in a future period. CVRs are the commodity form of the earnout, issued to many holders, sometimes transferable and listed, and often used in pharmaceutical and biotech acquisitions where the target's value depends on binary clinical or regulatory outcomes that the parties value differently.

There are two main varieties. *Event-driven* CVRs pay a fixed amount if a defined event occurs by a deadline (a digital, or binary, payoff). *Price-protection* CVRs pay the difference, if positive, between a reference price and the acquirer's stock price on a future date, which is a put option on the acquirer's shares and effectively a value guarantee for stock-consideration holders; they give target holders the downside protection of a collar without a closing-date adjustment.

Example 17.6 – Valuing an event-driven CVR. Harlow acquires Brightline Therapeutics (60 million shares outstanding) for $30 in cash per share plus one CVR per share that pays $5 if regulatory approval for a lead compound is obtained within three years. Assume a risk-neutral-equivalent probability of approval of 35 percent (clinical-stage probabilities are largely idiosyncratic, so a near-risk-free-plus-credit rate is appropriate) and a discount rate of 5 percent.

Step 1 – Expected payoff per CVR: $0.35 \times \$5 = \$1.75$.

Step 2 – Present value, assuming payment at the end of year 3: $\$1.75 \div 1.05^3 = \$1.75 \div 1.157625 =$ **$1.512 per CVR**.

Step 3 – Total fair value of the liability: 60 million × $1.512 = **$90.7 million**. The maximum payout is 60 × $5 = $300 million, so the CVR's expected value is about 30 percent of its face amount.

Step 4 – Sensitivity to the discount rate: at 12 percent the PV per CVR is $1.246, giving a liability of $74.7 million. A simple sensitivity to probability: at 20 percent probability the PV is $0.864 per CVR ($51.8 million); at 50 percent it is $2.160 ($129.6 million).

Step 5 – Accounting. The $90.7 million is recognized at the acquisition date as contingent consideration (a liability measured at fair value), added to the $1,800 million in cash ($30 × 60 million) to give total consideration of $1,890.7 million for goodwill purposes. If approval is subsequently obtained, the liability steps up to $300 million and a $209 million charge (before the time-value adjustment) flows through earnings. If approval is rejected, the liability is eliminated and a gain of the full carrying amount is booked.

Managerial implication: the headline price for the target is "$30 plus a CVR worth up to $5", but its economic price is $31.51, 5 percent higher than the cash-only price; the board should compare offers on the basis of fair value, not maximum value. A bidder who offers a large nominal CVR with a low probability of triggering is offering less than it seems, and a target board that accepts such a CVR should demand covenants on efforts and, ideally, a minimum milestone payment.

17.6 Representations, Warranties and Risk Allocation

The Architecture of Risk Allocation

The acquisition agreement is a risk allocation device. The seller makes **representations and warranties** (reps) about the business: financial statements are accurate, no undisclosed liabilities exist, the company owns its intellectual property, it complies with laws, taxes have been paid, contracts are in force. The buyer relies on the reps in setting its price. If a rep is false, the buyer has a remedy: **indemnification** (the seller pays for losses arising from the breach), typically subject to limits.

Several terms define the economics:

- **Basket (deductible or tipping).** A threshold of losses the buyer must absorb before the seller owes anything. A *deductible* basket means the seller pays only losses above the threshold; a *tipping* (first-dollar) basket means that once the threshold is crossed, the seller pays all losses from the first dollar. Baskets, commonly around 0.5 to 1 percent of the transaction value, screen out trivial claims.
- **Cap.** The maximum amount the seller must pay for general rep breaches, frequently 10 percent of price in traditional deals (lower with insurance). Fundamental reps (title, authority, capitalization) and fraud usually sit outside the cap, up to the purchase price.
- **Survival period.** How long after closing a claim can be made: commonly 12 to 24 months for general reps, longer for tax and fundamental reps.
- **Escrow or holdback.** Part of the purchase price kept in an account (or withheld) as security for indemnity claims.
- **Materiality scrape.** A clause that ignores materiality qualifiers when calculating losses or determining whether a breach occurred, expanding the seller's exposure.
- **Sandbagging.** Whether the buyer may claim for a breach it knew about before closing. A pro-sandbagging clause lets the buyer close and still recover; an anti-sandbagging clause bars the claim.
- **Disclosure schedules.** Exceptions to the reps; the more the seller discloses, the less the reps protect the buyer. Disclosure is how the real negotiation over risk takes place.

Rep-and-warranty provisions determine where the *unknown unknowns* fall. The principle that the party best able to know or control a risk should bear it suggests that reps about the business's historical operations belong with the sellers, while risks created by the buyer's own post-closing actions belong with the buyer.

Representations and Warranties Insurance

Representations and warranties insurance (RWI) has become common in private-company deals. The buyer purchases a policy (buy-side policy) that pays for losses from breaches of the seller's reps, subject to a **retention** (like a deductible), a **limit** (usually 10 percent of enterprise value), and **exclusions** (known matters, forward-looking statements, certain categories such as underfunded pensions, asbestos, or specific issues identified in due diligence). The premium, often a few percent of the limit paid as a one-time cost, and the underwriting fee are borne by the buyer, the sellers, or split. The insurer relies on the buyer's due diligence reports to decide what to exclude. As a result, due diligence quality directly determines the scope of coverage, and the insurer's exclusions list is a useful independent diligence output.

For the seller, RWI converts a large indemnity and escrow into a small one (say 0.5 percent of price) plus any residual exposure for fraud and fundamental reps. For the buyer, the policy provides a creditworthy counterparty and a longer survival period (often three years for general reps and six for fundamental), at the cost of the retention and exclusions. For both, RWI shifts the negotiation: the buyer is willing to accept a smaller seller indemnity, and the seller is willing to improve the price, because it keeps more certain proceeds.

Example 17.7 – Escrow versus insurance. Verity's enterprise value is $400 million. A traditional package: sellers provide a 10 percent ($40 million) escrow for 18 months. The alternative is a buy-side RWI policy with a $40 million limit, a retention of $2 million (0.5 percent), and a premium of 3 percent of the limit. Assume post-closing losses from rep breaches (in $ million) have the following distribution:

Loss ($m)	0	2	5	10	25	60
Probability	0.86	0.06	0.04	0.025	0.01	0.005

Step 1 – Premium: 3% × $40 = **$1.2 million**, or 0.30 percent of enterprise value.

Step 2 – Expected total loss: 2 × 0.06 + 5 × 0.04 + 10 × 0.025 + 25 × 0.01 + 60 × 0.005 = 0.12 + 0.20 + 0.25 + 0.25 + 0.30 = **$1.12 million**.

Step 3 – Insurance recovery for each loss level: min(max(L − 2, 0), 40) = 0, 0, 3, 8, 23, 40. Expected recovery: 3 × 0.04 + 8 × 0.025 + 23 × 0.01 + 40 × 0.005 = 0.12 + 0.20 + 0.23 + 0.20 = **$0.75 million**. This is 62.5 percent of the premium, a plausible loss ratio once the insurer's expenses and profit and the loading for correlated, severe tail events are considered.

Step 4 – Buyer's uncovered expected loss under RWI: $1.12 − 0.75 = **$0.37 million** (the retention on each loss plus the $20 million excess in the 1-in-200 case where loss is $60 million: 2 × (0.06 + 0.04 + 0.025 + 0.01 + 0.005) + 18 × 0.005 = 0.28 + 0.09 = 0.37).

Step 5 – Traditional escrow: the buyer recovers min(L, 40): expected recovery $1.02 million (the tail of $60 million is truncated at $40 million). The buyer's uncovered expected loss is $0.10 million, apparently better than RWI. But the escrow's recovery is conditional on the seller's willingness to pay and the buyer's ability to prove the claim, and the escrow's cost to the sellers is substantial.

Step 6 – Cost to sellers of the escrow versus RWI. If sellers value capital at 8 percent per year, trapping $40 million for 18 months costs 40 × (1.08^1.5 − 1) = **$4.89 million** in time value, against $2 million of escrow (the residual retention-related escrow, for which the opportunity cost is 2 × 0.1236 = $0.25 million) under RWI, so they save about $4.6 million. Whoever pays the $1.2 million premium, the total surplus from RWI is: sellers save $4.64 million in time value, the buyer pays $1.2 million for the policy and bears slightly more expected loss ($0.37 million versus $0.10 million); net joint saving about $4.64 − 1.2 − 0.27 = $3.2 million. A buyer may reasonably demand a price reduction or a split of the premium in return for accepting the substitution.

Interpretation: RWI is not "free coverage". It reallocates costs from sellers to insurers and buyers and it trades the certainty of exclusions for a creditworthy counterparty. The deal team should model the joint surplus to find where the negotiation should land.

17.7 Financing Commitments and Deal Certainty

The Certainty Problem

Between signing and closing, a deal can fail for reasons unrelated to the price: financing falls through, regulators object, a shareholder vote fails, a rival bids, a material adverse change occurs, or the market moves. The target bears the cost of a failed deal (employee attrition, customer defection, a damaged auction, a signal that something is wrong), and in a deal that fails the target is typically worth less than before the process began. Sellers therefore pay for **certainty**, usually by accepting a lower headline price from a bidder with fewer outs, and buyers choose how much certainty to offer by selecting the conditions, commitments and fees in the agreement.

Financing Commitments

A cash acquisition requires committed funds. The buyer typically obtains a **commitment letter** from its banks to provide bridge or term loans on specified terms, with a commitment fee and a **ticking fee** that accrues if closing is delayed. In competitive deals the sellers demand "certain funds": the buyer's obligation to close is not conditional on obtaining

financing. Lenders respond with *limited conditionality*, often described as "SunGard" provisions after the type of leveraged buyout in which they first became standard in the US. Under those provisions, the conditions to funding are cut back to a short list: accuracy of a few specified representations, no material adverse effect on the target, delivery of required documents. Lenders accept the reduced conditionality for fees, but they require that the buyer's equity contribution, collateral and documentation be effectively final at signing, and they hold the buyer to rigorous standards.

Conditions, MAC Clauses and Regulatory Risk

The **material adverse change (MAC) clause** allows the buyer to refuse to close if the target suffers a change that is materially adverse to its business, but courts interpret it narrowly: the change must be durable (measured in years, not months), significant relative to the target's overall condition, and not within the carve-outs (general economic conditions, industry-wide changes, changes in law or accounting, pandemics, failure to meet projections) that negotiators routinely include. In practice, the MAC clause is a weak exit, and a buyer that wants out of a deal is more likely to rely on other conditions. A wise buyer therefore negotiates specific closing conditions (key customer retention, no injunction, no specified litigation outcome) and treats the MAC as a backstop.

Regulatory approval is usually the most material risk in strategic deals. The agreement allocates it through three terms:

- **Efforts covenants.** "Reasonable best efforts" or "hell or high water" provisions, the latter obliging the buyer to accept whatever divestitures or remedies the regulator requires. Sellers push for the stronger form; buyers for a cap on the burdens they must accept.
- **Outside date.** The date after which either party may terminate if the deal has not closed, often extendable if only regulatory approval is outstanding.
- **Reverse termination fee (RTF).** A fee payable by the buyer to the seller if the deal fails for specified reasons (regulatory failure or financing failure), compensating the seller and creating an incentive for the buyer to perform. RTFs are usually set higher than the seller's own "break-up" fee payable on a competing bid: typical ranges are a few percent of equity value for the target's fee and higher for regulatory RTFs in sensitive sectors.

Once a deal is announced, the target trades at a discount (the spread) to the offer price, which can be decoded into an implied probability of completion. If P is the market price, O the offer, D the price without the deal, T the time to close and r the risk-free rate:

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Implied probability p = (P − D) ÷ (O ÷ (1 + r)^T − D)
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The simple version ignores time value and sets the offer's discounting term to zero: p = (P − D) ÷ (O − D). The merger arbitrageur earns the spread if the deal completes and suffers the fall to D if it does not, a payoff like writing an out-of-the-money put.

Example 17.8 – Implied probability of completion. Brandt Systems, a public target, has agreed to be acquired for $50 per share in cash, expected to close in six months. Brandt traded at $38 before the announcement and trades at $48.20 now. The risk-free rate is 4 percent.

Step 1 – Naive implied probability (no time value): (48.20 − 38) ÷ (50 − 38) = 10.20 ÷ 12 = 85.0 percent.

Step 2 – Present value of the offer: 50 ÷ 1.04^0.5 = 50 ÷ 1.0198 = $49.03.

Step 3 – Time-adjusted implied probability: (48.20 − 38) ÷ (49.03 − 38) = 10.20 ÷ 11.03 = **92.5 percent**.

Step 4 – Gross spread: (50 − 48.20) ÷ 48.20 = 3.73 percent over six months, equivalent to $(1.0373)^2 - 1$ = **7.6 percent annualized**.

Interpretation: once the time value is taken into account, the market implies a probability of completion of about 92.5 percent. The naive method understates it by 7.5 points, because it attributes the entire spread to risk. If Harlow, in a similar position as acquirer, sees its target trading at a price implying a 70 percent probability, the market is signaling concern about financing, regulatory approval or a competing bid, and management should take it as a prompt to identify and address the issue. A rising spread with no news is informative as well.

Valuing Certainty

For the seller, certainty has a price that can be calculated. A deal with a conditional financing structure is a probability-weighted lottery between the deal price and the failed-deal value.

Example 17.9 – What a seller will pay for a financing commitment. A buyer offers $400 million. If the deal fails, the target's value falls to $340 million (distraction, attrition, and market stigma). Bidder A offers $400 million with a financing condition, and the seller estimates an 80 percent probability of closing. Bidder B offers $400 million with committed financing, a reverse termination fee of $12 million, and a 88 percent estimated probability of closing.

Step 1 – Bidder A expected value: 0.80 × 400 + 0.20 × 340 = 320 + 68 = **$388.0 million**.

Step 2 – Bidder B expected value without the RTF: 0.88 × 400 + 0.12 × 340 = 352 + 40.8 = **$392.8 million**.

Step 3 – RTF contribution (assuming the fee is payable on failure for financing or regulatory reasons): 0.12 × 12 = **$1.44 million**, so the total expected value is $394.24 million.

Step 4 – Value of certainty to the seller: 394.24 − 388.0 = **$6.24 million**. The seller would therefore accept a price for Bidder B up to $6.24 million ÷ 0.88 ≈ $7.1 million below $400 million (about 1.8 percent less) and be no worse off, ignoring risk aversion. Because sellers are not risk-neutral (a private founder cannot diversify the deal), the true willingness to pay is higher.

Managerial implication: the buyer who can offer certainty cheaply, through a strong balance sheet, committed financing, and a credible regulatory path, has a competitive advantage in auctions that is worth several percent of the price. Conversely, a buyer offering certainty it cannot honor has simply sold the sellers an unhedged put.

Case Study: Meridian Orthopedics and Calder Biologics

Illustrative case; the companies and figures are fictional.

Meridian Orthopedics, a listed medical device company with revenue of $2.8 billion and net debt of 1.6 times EBITDA, wishes to acquire Calder Biologics, a privately held maker of bone-graft substitutes. Calder's trailing revenue is $180 million, EBITDA is $36 million, and it is growing at 22 percent. The sellers are the founder, Dr. Imani Okafor (55 percent), a venture fund (35 percent) and employees (10 percent). Calder is an S corporation (the fund holds through a pass-through vehicle), so a tax election is available. Its tax basis in assets is $60 million.

Meridian values Calder's standalone business at $420 million on a conservative forecast and sees $50 million of present-value synergies. The sellers, citing a pipeline product (CB-7, a next-generation graft) that is awaiting regulatory clearance, are asking $560 million. The parties agree to explore a structure that bridges the gap.

Proposals on the table.

1. **Upfront price of $450 million in cash**, financed with a bridge loan taken out by bonds. Meridian's lenders will give committed funds, with a ticking fee.
2. **338(h)(10)-type election** treating the purchase as an asset purchase. Meridian estimates that $330 million of the price would be allocated to amortizable intangibles and goodwill, creating a PV step-up shield of about $38 million at its discount rate. The sellers' incremental tax from recapture and ordinary income treatment is $9 million.
3. **A two-year earnout** of up to $60 million, paid at 50 cents per dollar of revenue above $260 million in year two. The sellers' forecast has year-two revenue at $290 million; Meridian's has $250 million. Meridian wants EBITDA as the metric, since it will integrate Calder into its sales force; Dr. Okafor wants revenue, and is concerned that Meridian will redirect sales representatives to its own products.
4. **A CVR-style milestone** of $40 million payable if CB-7 is cleared by the regulator within 30 months. Meridian estimates a 45 percent probability. The sellers estimate 75 percent.
5. **Insurance.** The sellers have refused any indemnity above 1 percent of price, citing the fund's wind-down, and propose RWI with a limit of $45 million. Insurers will exclude a pending dispute over a competitor's patent that Calder has discussed in due diligence. Meridian's counsel believes the patent claim could cost up to $25 million.
6. **Regulatory and certainty terms.** Calder's largest hospital customer can terminate on a change of control. Antitrust review is expected to take four to six months. The sellers want a reverse termination fee of 4 percent of the price.

Meridian's board concerns. The board worries about goodwill and amortization of acquired intangibles diluting reported EPS in the first two years, about the earnout liability volatility, and about distraction. The CFO, Samuel Reyes, believes a fixed price of $560 million is too high, but that a structure that matches the sellers' beliefs with contingent payments could be acceptable and might also retain Dr. Okafor as head of the division for the earnout period.

Task. The board has asked you, as the deal team, to recommend a structure, a price composition, and a negotiating strategy.

Discussion Questions

1. Using the numbers in the case, estimate the joint surplus from the tax election, and recommend how to split it. What price would you offer on that basis, and how would the allocation affect Meridian's reported goodwill and EPS compared with a straight stock deal?
2. Value the earnout and the CVR under both Meridian's and the sellers' assumptions. Treat the earnout as a call spread on year-two revenue with volatility of 25 percent; discuss why the valuations differ and how a contract could make each side comfortable. Which metric should the earnout use, and what covenants are needed to limit gaming?
3. Design the risk-allocation package. How would you deal with the patent claim (which RWI will exclude), the change-of-control right of the key customer, and the sellers' unwillingness to provide an escrow? What would a specific indemnity or price-adjustment mechanism look like?
4. Design the certainty package: what financing, conditions and termination fee would you offer, and how much should you be willing to pay for certainty if an alternative bidder is believed to have a 20 percent chance of emerging? Justify with expected-value reasoning.

Key Takeaways

- Structure is value: form of transaction, consideration, conditions, risk allocation and accounting each carry economic consequences beyond the headline price.

- A tax step-up is a stream of future deductions whose present value (often 6 to 10 percent of price at 15-year amortization) is the maximum the buyer should share with the sellers; the sellers' incremental tax is the cost to be compensated.
- For C-corporation targets, double taxation usually makes plain asset purchases uneconomic, while for pass-through targets elections can create value for both sides.
- Purchase price accounting allocates the price paid; goodwill is a residual that includes any overpayment, and deferred taxes increase goodwill when the buyer cannot step up tax basis.
- Contingent consideration is a liability measured at fair value and remeasured through earnings, so good performance after closing can produce reported charges.
- Earnouts and CVRs are options on performance; they should be valued with volatility, not at the expected-case payoff, and their design must anticipate manipulation by both sides.
- In stock deals, the exchange mechanism (fixed ratio, fixed value, collars) allocates acquirer price risk between signing and closing; each variety is an option position that should be priced.
- Representations and warranties allocate unknown risks; RWI can replace large escrows and founder indemnities, at the cost of retention, exclusions and a premium, and known risks need specific solutions.
- Deal certainty (committed financing, clean conditions, reverse termination fees, regulatory commitments) has a calculable value to sellers, and spreads in target stock prices reveal the market's estimate of completion probability.
- The best structure assigns every risk to the party that can bear it most cheaply and every value-creating term to joint negotiation.

Review Questions

1. Why can two acquisitions with the same headline price differ in value to the buyer and the seller? List at least four deal terms that cause this.
2. Explain the difference between a stock purchase, an asset purchase and a reverse triangular merger in terms of liability, consents and tax basis.
3. Why does an asset sale by a C-corporation usually destroy value relative to a stock sale, and under what circumstances might it not?
4. What discount rate would you apply to the tax shield from a step-up, and why? What are the consequences of using the cost of debt versus the WACC?
5. Explain why goodwill is not a measure of synergy value. What does an impairment tell the reader of financial statements?
6. Why does a deferred tax liability increase goodwill in a stock acquisition without a tax election?
7. Explain why an earnout is an option and how volatility affects its value. What happens to the value of the earnout if the cap is removed?
8. Compare fixed-ratio and fixed-value stock deals. Which side bears the price risk in each, and how does a collar change the allocation?
9. How do RWI policies change the negotiating positions of buyers and sellers? Which risks remain with the seller and which with the buyer?

10. How can a merger arbitrage spread be used to infer the market's view of deal completion? What would cause the implied probability to be a biased estimate?

Problems

1. **Step-up value.** A buyer pays $600 million for assets with a tax basis of $200 million. The step-up is amortized straight-line over 15 years; the tax rate is 21 percent and the discount rate 8 percent. Compute the annual tax shield and its present value.
2. **C-corporation gross-up.** A C-corporation target is to be sold for $300 million in a stock deal. Inside asset basis is $100 million; shareholders' stock basis is $80 million. Corporate tax is 21 percent and shareholder capital gains tax is 20 percent. (a) What do the shareholders net in a stock sale? (b) What pre-tax asset sale price leaves them equally well off? (c) If the buyer's tax rate is 25 percent and discount rate 10 percent, what is the PV of the buyer's step-up shield at that price, and does the buyer gain?
3. **Purchase accounting.** A buyer pays $500 million in cash for a business whose identifiable net assets have a fair value of $380 million and a tax basis of $200 million. The tax rate is 30 percent and there is no tax election. Compute the deferred tax liability, the goodwill, and goodwill as a percentage of price. If $100 million of the fair value is a finite-lived intangible with a 10-year life, compute the annual pre-tax amortization, deferred tax benefit and after-tax earnings reduction.
4. **Earnout.** Revenue is $200 million today, with an expected growth rate of 6 percent per year and volatility of 25 percent. The sellers receive 50 percent of year-2 revenue above $230 million, capped at $30 million. Compute the expected payoff and PV at 6 percent. (Use the lognormal call formula with forward $F = 200 \times 1.06^2$.) Compare it with the deterministic estimate.
5. **Collar.** An acquirer trades at $80 and offers $48 per target share (exchange ratio 0.6). Under a fixed-value collar with band $72 to $88, compute the exchange ratio and value per target share for acquirer prices of $60, $72, $80, $88 and $100.
6. **CVR.** A buyer issues 80 million CVRs, each paying $4 if a milestone occurs within two years (probability 40 percent). Discount rate is 6 percent. Compute the fair value per CVR and in total, and compare with the maximum payout.
7. **RWI.** Post-closing losses have the distribution: $0 (0.82), $1 million (0.08), $4 million (0.05), $12 million (0.03), $30 million (0.015) and $70 million (0.005). The RWI policy has a retention of $1.5 million and a limit of $25 million and costs 3 percent of the limit. Compute the expected insurer payout and compare it with the premium.
8. **Merger arbitrage.** A target offered $32 in cash, closing in nine months, trades at $29.40 and had traded at $24 before the announcement. The risk-free rate is 5 percent. Compute the naive and time-adjusted implied probabilities, and the annualized spread.

Answers to Selected Problems

1. Annual shield = (400 ÷ 15) × 0.21 = $5.60 million. Annuity factor (8%, 15) = 8.5595. PV = **$47.93 million**.
2. a. 300 − 0.20 × (300 − 80) = **$256.0 million**. (b) Needed corporate distribution D: D − 0.20 × (D − 80) = 256, so D = (256 − 16) ÷ 0.8 = $300 million; price P with P − 0.21 × (P − 100) = 300 gives P = (300 − 21) ÷ 0.79 = **$353.16 million** (a $53.16 million gross-up). (c) Step-up = 253.16; PV shield = (253.16 ÷ 15) × 0.25 × 7.6061 = **$32.09 million**. The buyer pays $53.16 million for $32.09 million of value: a net loss of about $21.1 million.

3. DTL = (380 − 200) × 0.30 = **$54 million**; goodwill = 500 − 380 + 54 = **$174 million** (34.8 percent of price). Amortization = $10 million per year; deferred tax benefit = $3 million; after-tax earnings reduction = **$7 million**.
4. F = 224.72; cap strike = 230 + 30 ÷ 0.5 = 290; σ√T = 0.3536. Expected call at 230 = $29.33 million; at 290 = $12.33 million; expected payoff = 0.5 × (29.33 − 12.33) = **$8.50 million**; PV = **$7.57 million**. The deterministic payoff is zero since the expected revenue ($224.72 million) is below the $230 million threshold.
5. $60: ratio 0.6667, value $40.00; $72: 0.6667, $48.00; $80: 0.6000, $48.00; $88: 0.5455, $48.00; $100: 0.5455, $54.55.
6. Expected payoff per CVR = 0.40 × 4 = $1.60; PV = 1.60 ÷ 1.1236 = **$1.424**; total **$113.9 million** versus maximum **$320 million** (35.6 percent).
7. Insurer payouts: 0, 0, 2.5, 10.5, 25, 25. Expected payout = 2.5 × 0.05 + 10.5 × 0.03 + 25 × 0.015 + 25 × 0.005 = 0.125 + 0.315 + 0.375 + 0.125 = **$0.94 million**. Premium = 0.03 × 25 = $0.75 million. The expected payout exceeds the premium, so this policy would not be offered; the insurer would raise the premium or the retention. (Expected total loss = $1.44 million.)
8. Naive probability = (29.40 − 24) ÷ (32 − 24) = **67.5 percent**. PV of offer = 32 ÷ 1.05^0.75 = $30.85; time-adjusted = 5.4 ÷ 6.85 = **78.8 percent**. Gross spread = 2.6 ÷ 29.4 = 8.84 percent over nine months, or **11.96 percent annualized**.

Chapter 18: Cross-Border Deals and Emerging Markets

Learning Objectives

After completing this chapter, you should be able to:

- Decompose the risks of an emerging-market investment into country, currency, political, and operating components, and decide which belong in the discount rate and which in the cash flows.
- Estimate a country risk premium using sovereign spreads and relative volatility, and compare the main ways of applying it to the cost of equity and WACC.
- Value cross-border cash flows consistently in local or hard currency, using inflation and interest-rate differentials, and avoid the mixed-currency errors that distort valuations.
- Quantify political risk through scenario and probability analysis, and evaluate mitigation through insurance, treaty protection and deal structure.
- Analyze the effect of capital controls, withholding taxes and trapped cash on the value of a subsidiary, and compare repatriation routes.
- Evaluate joint-venture structures, local-partner dynamics and capital structure choices in emerging markets, including the currency of debt.

Opening Vignette

Illustrative scenario; all names and figures are fictional.

Solano Beverages is a mid-sized consumer company based in a high-income country, with $1.8 billion of revenue and a cost of capital of 8 percent. Its chief strategy officer, Tomas Villareal, has found what looks like a once-in-a-decade opportunity: a regional bottler in the fictional republic of Aurelia, a market of 90 million people with rising incomes and a thinly served beverage market. The bottler, Aurelia Refrescos, earns the local equivalent of $60 million of EBITDA and has a 38 percent market share. Its owners, a prominent local family, would sell 60 percent and keep 40 percent.

The financial model looks spectacular. At Solano's 8 percent cost of capital, the value of the target is $620 million. Then the questions begin. Aurelia's sovereign bonds yield 6.5 percentage points more than comparable US Treasuries. Its currency has lost about 45 percent against the dollar over five years. Inflation is running at 9 percent. A former finance minister tells Tomas that dividend remittances were restricted for eight months during the last crisis. The family partner has a cousin who is a senior regulator. The local banking system will lend only in local currency, at 15 percent, while a dollar loan from an international bank costs 7 percent.

Tomas's team must now answer questions that the single-country valuation models of Volume 1 did not prepare them for. What cost of capital should apply? In which currency should the cash flows be projected? How much should political risk reduce value, and can it be insured? How will cash reach the parent? How should the joint venture be governed so that Solano can protect its investment without a local partner's cooperation? And how should the venture be financed?

This chapter develops the tools to answer those questions. The unifying theme is *consistency and explicitness*: every risk should be identified, assigned to either the cash flows or the discount rate (never both, never neither), and priced.

18.1 The Anatomy of Cross-Border Risk

Volume 1 introduced valuation (Chapter 21), the cost of capital (Chapter 11) and foreign exchange risk management (Chapter 20) for a single-country corporation. When the target or project is in another country, the valuation framework is the same but the inputs are harder. The sources of additional risk can be classified:

- **Country (sovereign) risk.** The risk that macroeconomic conditions, sovereign default, banking crises or recession in the host country affect all firms there. It manifests in higher sovereign bond yields, and in correlated declines in earnings and asset values.
- **Currency risk.** The local currency may depreciate relative to the investor's currency, eroding the home-currency value of local cash flows. Currency risk is closely related to inflation: high-inflation countries typically see depreciation.
- **Political risk.** Government actions that reduce the value of the investment: expropriation or nationalization, creeping expropriation (progressively adverse regulation and taxation), contract repudiation, price controls, local content rules, and breaches of licenses.
- **Transfer and convertibility risk.** The inability to convert local currency into hard currency or to move funds out of the country because of capital controls.
- **Legal and governance risk.** Weak enforcement of contracts and property rights, opaque accounting, related-party dealing, and corruption exposure (including legal risk at home under anti-bribery statutes).
- **Operating and market risk.** Infrastructure, supply-chain reliability, consumer-income volatility, and competitor behavior.
- **Liquidity and exit risk.** Illiquid local equity markets, restrictions on foreign sales, and limited buyers make exit uncertain and costly.

Two principles organize the rest of the chapter.

Principle 1: Diversifiable risk does not earn a premium, but cross-border risk is not always diversifiable. In the textbook CAPM (Volume 1, Chapter 10), only systematic risk is priced. If international investors hold the global market portfolio, the relevant risk of an emerging-market investment is its covariance with the world market, not its total volatility. Emerging-market returns correlate with global returns far less than their volatility alone would suggest, so a pure global-CAPM approach would conclude that a country risk premium is small. But several considerations push the other way: emerging-market returns are more correlated with the global market in crises (when diversification is most needed); many investors, especially corporates with concentrated exposure, are not globally diversified; and events such as expropriation or controls are *not* priced by a beta but are real losses that must go somewhere in the valuation. Practice has therefore moved to explicit adjustments, but one should know that the debate is unresolved and the *size* of the premium is a judgment.

Principle 2: Treat each risk once. The discount rate and the cash flows are two places to put risk. A risk that has been reflected in an expected-cash-flow scenario (a 5 percent chance of expropriation reducing flows) should not also be loaded into the discount rate, or the same risk is double-counted. A rule that works in practice: *systematic* and persistent country risk (macro volatility, general sovereign risk) enters through the discount rate; *discrete, diversifiable or insurable* events (expropriation, blocked funds, a specific currency devaluation) enter through scenarios, expected cash flows or contractual protection.

Table 18.1 – Where to Put Each Risk

Risk	Preferred treatment	Why
General sovereign / macro risk	Country risk premium in discount rate	Persistent and correlated across firms
Expected inflation and depreciation	Cash-flow currency and consistent discount rate	Predictable differential, not a risk premium
Currency volatility	Hedging cost or scenarios; small premium only if systematic	Mostly diversifiable; hedgeable at forward prices
Expropriation, nationalization	Probability-weighted scenarios or insurance cost	Discrete event with known loss
Capital controls	Cash-flow timing scenarios or structure (offshore accounts, royalties)	Affects the timing and amount of repatriation
Corruption, governance	Reduce cash-flow estimates, require control rights, use as walk-away criterion	Not a pricing problem alone
Illiquidity at exit	Lower terminal multiple or illiquidity discount	Affects the exit value

The weakness of the classic approach, "add a few points to the discount rate for risk," is that it is not tied to specific risks, so it penalizes long-duration cash flows disproportionately, ignores mitigation, and cannot be debated. The remainder of this chapter replaces the lump-sum premium with explicit analysis.

18.2 Country Risk Premiums

The Sovereign Spread as a Starting Point

The yield spread on the host government's hard-currency (typically US dollar) bonds over the risk-free rate of the same currency measures the market's price of default risk on the country. A sovereign that borrows in dollars at 3.5 percentage points over Treasuries is perceived as riskier than one that pays 1 point, and the spread captures both the probability of default and the loss given default (the bond pricing and default logic of Volume 1, Chapter 4). Other measures include sovereign credit default swap (CDS) spreads and agency ratings that can be mapped to spreads.

Spreads are the premium paid by *bondholders* of the government. The equity market is riskier than the bond market in the same country, because equity is a residual claim. The standard adjustment scales the bond spread by the ratio of equity to bond volatility, for example

```
Country risk premium (CRP) = Sovereign spread × (σ_equity ÷ σ_bond)
```

where the ratio is estimated from the local equity market's and the sovereign bonds' volatilities (typical estimates for emerging markets lie between 1.2 and 2.0, but the figure should be estimated from data for the country, not assumed). The logic is intuitive: if equity is 1.4 times as volatile as the sovereign's bonds, its premium for country risk should be proportionately higher.

An alternative is to calculate the equity risk premium directly from the country's equity market history (long-run local equity returns over the risk-free rate); this is unstable in many countries because their equity markets are short-lived, volatile and subject to survivorship effects.

Applying the CRP

There are three common ways of using the CRP in the cost of equity of a project in the host country, each with a different economic interpretation:

1. **Additive (bond-spread view).** Treat country risk as a separate premium that applies equally to all investments in the country, regardless of beta:

```
k_e = r_f + β × ERP_mature + CRP
```

2. **Beta-scaled.** Treat country risk as an extra component of the market risk premium, which scales with the project's beta:

```
k_e = r_f + β × (ERP_mature + CRP)
```

This implies that a low-beta firm is less exposed to country risk, which is realistic for a utility serving a domestic market but dubious for an exporter, whose cash flows may be *less* tied to the country's economy than its beta suggests.

3. **Lambda (exposure) approach.** Measure each firm's exposure to country risk with a factor λ, typically estimated as the share of the firm's revenue (or operating profit) derived from the country relative to the average firm:

```
k_e = r_f + β × ERP_mature + λ × CRP
```

A firm that sells globally and operates a single local plant has λ well below 1; a firm entirely dependent on local demand has λ close to or above 1. This approach respects the fact that risk differs among firms in the same country.

The cost of debt follows the same logic: the firm's borrowing cost in hard currency is typically the risk-free rate plus the sovereign spread plus a company-specific credit spread (the sovereign is usually the ceiling or near-ceiling for local corporates, though some multinationals borrow below their sovereign's rate through parent guarantees and offshore structures).

Example 18.1 – Cost of equity and WACC for an emerging-market bottler. Inputs (US dollar terms): risk-free rate 4.0 percent, mature-market equity risk premium 5.0 percent, beta 1.1, Aurelia's sovereign spread 3.5 percent, equity-to-bond volatility ratio 1.4, a company credit spread of 2.0 percent above the sovereign, a target capital structure of 35 percent debt and 65 percent equity, and a tax rate of 25 percent. For the lambda approach assume λ = 0.8.

Step 1 – Country risk premium: 3.5% × 1.4 = **4.9%**.

Step 2 – Mature-market cost of equity: 4.0% + 1.1 × 5.0% = 9.5%.

Step 3 – Cost of equity under each approach:

- Additive: 9.5% + 4.9% = **14.4%**.
- Beta-scaled: 4.0% + 1.1 × (5.0% + 4.9%) = 4.0% + 10.89% = **14.89%**.
- Lambda (λ = 0.8): 9.5% + 0.8 × 4.9% = 9.5% + 3.92% = **13.42%**.

Step 4 – Cost of debt in dollars: 4.0% + 3.5% + 2.0% = **9.5%**; after tax: 9.5% × 0.75 = 7.125%.

Step 5 – WACC (dollar terms):

- Additive: 0.65 × 14.4% + 0.35 × 7.125% = 9.36% + 2.494% = **11.85%**.
- Beta-scaled: 0.65 × 14.89% + 0.35 × 7.125% = 9.679% + 2.494% = **12.17%**.
- Lambda: 0.65 × 13.42% + 0.35 × 7.125% = 8.723% + 2.494% = **11.22%**.

Step 6 – Impact on value. Compared with Solano's home WACC of 8 percent, the cost of capital is 3.2 to 4.2 percentage points higher. For a business with a perpetuity-like profile and growth of 3 percent, value is proportional to 1 ÷ (WACC − g): at 8 percent the multiple is 1 ÷ 0.05 = 20.0; at 11.85 percent it is 1 ÷ 0.0885 = 11.3; at 12.17 percent, 10.9; at 11.22 percent, 12.2. The valuation at the higher rates is roughly 40 to 45 percent lower than at the home-market rate.

Interpretation: the choice among the three approaches changes WACC by about 0.95 percentage points, which is less than the difference between any of them and ignoring country risk altogether. That makes the *existence* of the premium more important than its application method. But the dispersion is not trivial: a deal team should present the valuation range rather than a point estimate, and should choose λ using the firm's actual exposure, not by default.

Cautions on Country Risk Premiums

- The CRP is a *market-based* number: it already reflects the sovereign's market-priced default risk and tends to rise in crises and fall in booms. Using a one-time snapshot embeds a market mood into a long-run valuation. Many practitioners average over a cycle or use a normalized premium.
- The CRP is *double-counted* if the cash flows already contain a pessimistic scenario for the same risk.
- The CRP does not capture risks specific to the investment that a sovereign spread does not price, such as expropriation of a single industry, and does not capture currency risk. It captures the cost of the sovereign's default, which correlates with these other risks but is not the same.
- For multinational investors holding portfolios of international assets, part of the premium may be diversified. This is why λ tends to be below 1 for exporters.
- Spreads in *hard currency* are used, not local-currency government yields, because local-currency yields include the expected depreciation, which would be double-counted if cash flows are also converted at forecast exchange rates.

18.3 Currency in Valuation

The Consistency Rule

The most common error in cross-border valuation is mixing currencies: projecting cash flows in local currency and discounting at a hard-currency rate, or the reverse. The rule is:

> **The currency and inflation assumptions embedded in the cash flows must match those embedded in the discount rate.**

There are two equivalent methods, which give the same value if executed consistently:

Method A – Value in local currency, convert at spot. Project cash flows in local currency (including local inflation), discount at a local-currency cost of capital, and convert the resulting present value at today's spot exchange rate.

Method B – Value in hard currency. Convert each projected local cash flow into the hard currency at a forecast exchange rate, and discount at a hard-currency cost of capital.

The link between the methods is a pair of parity conditions. *Purchasing power parity* (PPP) says that the expected change in the exchange rate equals the inflation differential; *interest rate parity* says that the forward rate reflects the interest-rate differential. In the "relative" form used for forecasting:

```
E[S_t] = S_0 × [(1 + i_local) ÷ (1 + i_hard)]^t
(1 + k_local) = (1 + k_hard) × (1 + i_local) ÷ (1 + i_hard)
```

where S is the price of one unit of hard currency in local currency, i the expected inflation rates, and k the cost of capital in each currency. Under these relationships, the PV is identical in both methods.

PPP holds poorly over months and years, since real exchange rates move for long periods; but as a valuation device it has an attractive feature: it removes any assumed *real* appreciation or depreciation, keeping the valuation neutral. A forecast of real depreciation, or of overvaluation corrected later, should be an explicit scenario, not a hidden assumption.

Example 18.2 – Valuation in local and hard currency. Aurelia Refrescos's free cash flows (in millions of local currency, LC) for years 1 to 5 are 120, 135, 150, 165 and 180, and a terminal value at the end of year 5 of 1,800. Spot is LC 20 per US dollar. Expected inflation is 9 percent in Aurelia and 2.5 percent in the United States. The dollar WACC is 11.85 percent (additive approach from Example 18.1).

Step 1 – Local-currency WACC: 1.118538 × (1.09 ÷ 1.025) − 1 = 1.118538 × 1.063415 − 1 = 0.18947, so **18.95 percent**.

Step 2 – Method A. Discount the local cash flows (year-5 flow including terminal value is 1,980):

Year	Cash flow (LC m)	Discount factor at 18.95%	PV (LC m)
1	120	0.8408	100.89
2	135	0.7068	95.42
3	150	0.5942	89.13
4	165	0.4996	82.43
5	1,980	0.4200	831.57
Total			**1,199.43**

Convert at spot: 1,199.43 ÷ 20 = **$59.97 million**.

Step 3 – Method B. Forecast exchange rates S_t = 20 × (1.09 ÷ 1.025)^t: 21.27, 22.62, 24.05, 25.58 and 27.20 LC per dollar. Cash flows in dollars: 120 ÷ 21.27 = 5.64; 135 ÷ 22.62 = 5.97; 150 ÷ 24.05 = 6.24; 165 ÷ 25.58 = 6.45; 1,980 ÷ 27.20 = 72.80 (all $ million). Discount at 11.85 percent: the PV is **$59.97 million**, identical to Method A.

Step 4 – The common error: discount the local-currency cash flows at the dollar WACC and convert at spot. PV = Σ LC CF ÷ 1.1185^t ÷ 20 = **$77.93 million**, a 30 percent overvaluation. The error arises because the local cash flows contain nine percent inflation growth (and, implicitly, depreciation) that the dollar discount rate does not charge for.

Interpretation: the 30 percent error is not exotic; it arises in practice whenever a spreadsheet-based model "converts at the spot rate" to avoid forecasting the exchange rate. The correct response to uncertainty about exchange rates is to hold the parity-consistent forecast as a base case and then run scenarios of real depreciation, not to ignore depreciation.

Currency Risk: Hedging or Pricing?

A forecast exchange rate is an *expectation*; the *risk* around it is separate. Corporate hedging of long-dated emerging-market currency exposure is costly and sometimes impossible. The forward rate implied by interest differentials reflects the expected depreciation, but liquid forwards seldom extend beyond one to three years in many emerging markets, and cross-currency swaps may be available only for a few tenors. Strategic exposures (the value of the business in dollars) are rarely hedged financially; they are managed through *natural hedges*: pricing in or indexing prices to hard currency where possible, sourcing inputs locally, matching the currency of debt to the currency of cash flows, and diversifying across countries.

The expected cost of hedging is the forward premium: if the forward rate implies 6.3 percent annual depreciation and the firm expects only 4 percent, hedging costs 2.3 percent a year in expectation, in exchange for eliminating the variance. Whether the variance reduction is worth the cost depends on the firm's costs of financial distress, as discussed in Volume 1, Chapter 20.

18.4 Political Risk

Forms of Political Risk

Political risk is the possibility that government actions or political events change the value of an investment. A useful taxonomy ranges from the dramatic to the gradual.

- **Expropriation and nationalization.** Outright seizure of assets, with or without compensation. Direct expropriation of foreign-owned industrial assets has become rarer than in the mid-twentieth century, but resource sectors, utilities and strategic industries remain exposed.
- **Creeping expropriation.** A series of measures, such as discriminatory taxes, forced price cuts, local-ownership requirements, license revocation, and tightening of operating conditions, that together destroy value without any single seizure. This is the more common form in practice and the harder to insure.
- **Contract repudiation and renegotiation.** The government revokes or renegotiates concessions, power-purchase agreements, tariffs or tax holidays after the investor's capital is sunk. The "obsolescing bargain" is the classic dynamic: before the investment, the investor holds bargaining power because the country wants the capital; after the plant is built, the capital is hostage to the government.
- **Currency inconvertibility and transfer restrictions** (covered in Section 18.5).
- **Political violence.** War, civil disturbance, terrorism and sabotage.
- **Regulatory and tax change.** Windfall taxes, retroactive tax claims, export restrictions.

Political risk is not unique to emerging markets: elections, trade wars and sanctions affect developed economies as well, but the distribution of outcomes differs, with weaker institutions leaving a wider gap between the contract and its enforcement.

Quantifying Political Risk

Because the events are discrete and of uncertain probability, expected-value analysis is more informative than a discount-rate adjustment. For a stream of cash flows C_t, an annual probability of loss p of the investment's remaining value, and recovery fraction ρ of that remaining value on loss, the value is:

```
V = Σ_t [ S_{t−1} × p × ρ × RV_t ÷ (1 + r)^(t−1) ] + Σ_t [ S_t × C_t ÷ (1 + r)^t ]
S_t = (1 − p)^t (survival probability), RV_t = PV at t−1 of the remaining cash flows
```

A shortcut for the zero-recovery case: discounting survival-weighted cash flows is equivalent to discounting the unweighted cash flows at a higher rate k ≈ (1 + r) ÷ (1 − p) − 1. With r = 12 percent and p = 4 percent, k = 16.67 percent. This provides a useful rule of thumb: *an annual expropriation probability p adds roughly p ÷ (1 − p) times (1 + r) to the discount rate*, but it overstates the effect on near-term cash flows and understates it for the terminal value. It is a convenient equivalence for communication, not a replacement for scenario analysis.

Where do the probabilities come from? Sources include sovereign risk ratings and country risk services; the history of expropriation or contract renegotiation in the industry and country; the investor's own relationship with the government; and market evidence such as the yield spread on bonds of sectors vulnerable to expropriation. Estimates are inevitably subjective; the discipline of making them explicit and testing the sensitivity of the decision to them is the benefit.

Example 18.3 – Expected value of an investment exposed to expropriation. Solano's Aurelian project generates US dollar cash flows of $30 million per year for years 1 to 4, and $30 million plus a $300 million terminal value in year 5. The cost of capital without political risk is 12 percent. Assume a 4 percent chance at the start of each year that the government seizes the business for that year onward.

Step 1 – Base value with no political risk: $30 \div 1.12 + 30 \div 1.12^2 + 30 \div 1.12^3 + 30 \div 1.12^4 + 330 \div 1.12^5 = 26.79 + 23.92 + 21.35 + 19.07 + 187.25 =$ **$278.37 million**.

Step 2 – Zero compensation. Weight each cash flow by its survival probability 0.96^t: $30 \times 0.96 = 28.80$; $30 \times 0.9216 = 27.65$; $30 \times 0.8847 = 26.54$; $30 \times 0.8493 = 25.48$; $330 \times 0.8154 = 269.07$. Discounting at 12 percent gives 25.71 + 22.04 + 18.89 + 16.19 + 152.68 = **$235.52 million**, 15.4 percent below the base and equivalent to discounting the unweighted flows at 16.67 percent.

Step 3 – Partial compensation. If the government pays compensation equal to 25 percent of the remaining value at the time of seizure, the expected value rises to **$246.23 million** (11.5 percent below base; equivalent discount rate about 15.4 percent). If compensation is 50 percent, the value is **$256.95 million** (7.7 percent below base).

Step 4 – Insurance comparison. Political risk insurance (available from multilateral agencies and private insurers) might cost about 1.2 percent per year of the insured amount. On an insured amount of $330 million the premium is $3.96 million per year, with a present value over five years at 12 percent of $3.96 \times 3.6048 =$ **$14.3 million**.

Interpretation: the modeled expected loss from the expropriation risk without recovery is $42.85 million, and the PV of the insurance premium is $14.3 million. The difference tells us either that the 4 percent probability is too pessimistic for what the insurers believe, or that the policy's coverage is narrower than the modeled loss (waiting periods, exclusions for creeping expropriation and for events caused by the investor, valuation of the insured amount at book value, limits on tenor, and the requirement that the investor exhaust local remedies). The investor should read what the policy actually covers before treating it as a substitute for the scenario analysis.

Mitigating Political Risk

Mitigation falls into four categories:

1. **Contractual and legal protection.** Bilateral investment treaties (BITs) and multilateral investment frameworks offer foreign investors the right to arbitrate disputes with host governments before international tribunals, with protections against uncompensated expropriation and discriminatory treatment. Investors often structure holdings through a subsidiary in a country that has a favorable treaty with the host, a practice that is legal but increasingly scrutinized by governments and tribunals when done after a dispute is foreseeable. Stabilization clauses in concession contracts, which freeze the tax and regulatory regime, give some protection but their enforceability depends on the jurisdiction.
2. **Insurance and guarantees.** Political risk insurance from national export-credit and investment-insurance agencies, multilateral guarantee programs and private insurers covers expropriation, inconvertibility and political violence. Coverage is typically available for equity and debt, for terms up to 15 years or more, and prices vary by country and sector.

3. **Structural and financial.** Financing the venture with local debt, or with loans from multilateral development institutions (whose presence deters expropriation, since the government wishes to preserve its relationship with the institution), reduces net exposure. Staging investment so that the capital at risk at any time is small, and recovering capital quickly (high-dividend payout and fees), shortens the exposure period. Keeping critical technology, brands or intellectual property under the parent's control (so that the local operation cannot function without ongoing supply) reduces the value of seizure to the government.
4. **Relationship and operational.** Partnerships with local firms, local hiring and sourcing, visible contribution to the national economy and engagement with the government and community raise the cost of hostile action. The risk of this approach is that relationships with specific individuals may not survive political change and can carry legal exposure under anti-corruption laws.

A caution applies to reliance on relationships: a partner with political connections is both an asset (access, speed) and a liability (dependence on particular officials, reputational and legal exposure if connections shade into improper payments). Compliance with home-country anti-corruption law, which can apply to the actions of local partners and agents, is a condition of the investment, not an optional extra. Any investment case that depends on an improper advantage should be rejected.

18.5 Capital Controls, Repatriation and Trapped Cash

Why Cash Flows in the Model Differ from Cash Received by the Parent

A discounted cash flow valuation of a foreign subsidiary values the free cash flow generated in the host country. But the investor values *cash that reaches the parent* (or at least cash that can be freely redeployed). Between the two lie taxes, legal restrictions and costs.

- **Local corporate income tax.** Paid in the host country on profits.
- **Withholding taxes** on dividends, interest and royalties paid across borders, often reduced by tax treaties.
- **Home-country tax** on foreign income: a worldwide system taxes foreign earnings with a credit for foreign taxes paid; territorial and participation-exemption systems exempt most foreign dividends. The details are specific to each country and change frequently; the structure of the analysis stays the same.
- **Capital controls.** Limits on converting local currency to hard currency or on transferring it out of the country: approval requirements, quotas, mandatory waiting periods, minimum local reserves, or multiple exchange rates (an official rate for some transactions and a worse "parallel" rate for others).
- **Legal restrictions on distributions.** Statutory reserves, minimum capital requirements, and earnings-based limits on dividends.
- **Exchange conversion costs.** Spreads, official fees, and in some cases the cost of obtaining hard currency at the parallel market rate.

Valuation must therefore model **repatriation** explicitly: how much and when cash reaches the parent, by which routes, and with what leakage. Where cash can be freely repatriated at low cost, one can discount the free cash flow to the firm. Where it cannot, the value to the parent can be much lower than the value to a local investor.

Repatriation Channels

Parent companies move cash out of a foreign subsidiary through several channels, each with different tax and legal characteristics:

Channel	Direction	Typical tax features	Constraints
Dividends	Sub to parent	Local tax paid first, then withholding tax; no deduction locally	Distributable reserves, central bank approval in some countries
Royalties / license fees	Sub to parent	Deductible locally, withholding tax, taxable at home (with credit)	Transfer-pricing arm's-length limits, local approval requirements
Management / service fees	Sub to parent	Deductible locally, withholding tax possible	Need documented services, arm's-length
Interest on intercompany loans	Sub to parent	Deductible locally subject to thin-capitalization rules; withholding tax	Limits on interest deduction, repayment of principal is not taxed
Repayment of shareholder loans	Sub to parent	Generally not taxed as principal	Approval and registration requirements
Transfer pricing on goods	Both	Affects profit location	Anti-avoidance rules, customs, tax audits
Capital reductions / liquidation	Sub to parent	Possibly taxed as dividend or capital gain	Creditor protection, approvals

Because the channels have different tax treatments, the parent can often reduce leakage by using a mix. But aggressive use of transfer pricing to move profits out of a country (particularly when controls limit dividends) creates tax-authority and legal risk; the structure must satisfy the arm's-length principle and withstand audit.

Blocked and Trapped Cash

When controls bind, cash accumulates locally. The value of trapped cash is less than its face value, because of the time it will take to release it, the exchange rate at which it is converted (possibly a worse official or parallel rate), the inflation that erodes its local value while waiting, and the political risk that the rules will tighten. A standard approach is to value the cash as the PV of its expected release schedule at a rate reflecting these costs.

Example 18.4 – Leakage along the repatriation route and the value of blocked funds. Part 1: Dividend versus royalty route. The Aurelian subsidiary has $100 million of pre-royalty pre-tax profit. The local tax rate is 25 percent, dividend withholding tax is 10 percent, royalty withholding tax is 15 percent, and the parent's home tax rate is 21 percent with credit for foreign taxes. The parent could extract cash purely by dividends, or by a royalty of $50 million (arm's length) and dividends on the remainder.

Dividend only. Local tax = 0.25 × 100 = 25.0. Net income = 75.0, distributed. Dividend withholding = 0.10 × 75.0 = 7.5. Cash to parent = 67.5. Home tax: gross foreign income 100, home tax at 21 percent = 21.0, credit for local tax and withholding of 25.0 + 7.5 = 32.5 exceeds the 21.0 liability (credit limited to 21.0), so there is no residual home tax. Total taxes = 32.5; **parent receives 67.5** (67.5 percent of pre-tax profit).

Royalty plus dividend. Royalty 50.0, deductible. Local pre-tax profit becomes 50.0; tax = 12.5; net income = 37.5; dividend withholding = 3.75; dividend to parent after withholding = 33.75. Royalty withholding = 0.15 × 50 = 7.5; royalty received = 42.5. Home tax: royalty income 50 × 21% = 10.5, less credit for royalty withholding 7.5, residual **3.0**. Dividend: no

residual (credit). Total taxes = 12.5 + 3.75 + 7.5 + 3.0 = 26.75. Cash to parent after home tax = 33.75 + 42.5 − 3.0 = **73.25** (73.25 percent).

The royalty route saves 5.75 (5.75 percent of the pre-tax profit) because the royalty is deductible against a 25 percent local rate and bears a lower combined rate. But this applies only up to the arm's-length royalty rate, and the tax authority may challenge excess. In some jurisdictions, royalties are further restricted by exchange control.

Part 2: Blocked funds. The subsidiary holds the equivalent of $100 million in local currency that the central bank will release in four equal annual installments of $25 million. The appropriate discount rate reflecting the sovereign risk, the opportunity cost of capital and the risk of further delay is 15 percent.

PV = 25 × [1 ÷ 1.15 + 1 ÷ 1.15^2 + 1 ÷ 1.15^3 + 1 ÷ 1.15^4] = 25 × 2.8550 = **$71.37 million**, a 28.6 percent haircut to face value.

Interpretation: cash and subsidiary cash flows should not be valued at face value when controls apply. The haircut is the combined cost of delay and risk, and it grows if the release schedule is uncertain. When a target holds large trapped cash balances, the buyer should price them at the haircut value, not the balance sheet value (a common difference between the headline "net cash" and the economic one).

In practice, managers limit idle local cash, repatriate frequently through every legitimate channel, use trapped balances to repay local debt or fund sound local investments (not merely because the cash is there), and recognize that controls rarely appear alone: price caps, forced conversion of export proceeds and discriminatory taxes often arrive together, so scenarios should be consistent.

18.6 Joint Ventures and Local Partners

Why Joint Ventures

Foreign investors enter emerging markets through wholly owned greenfield investment, acquisition of a local firm, a minority stake, a contractual alliance (licensing, franchising, distribution), or a **joint venture (JV)** with a local partner. JVs are chosen for four reasons: *regulatory* (many countries cap foreign ownership in sectors such as telecoms, banking, media, retail, mining), *access* (the partner brings customers, distribution, licenses, land, and political relationships), *risk sharing* (sharing capital at risk in an uncertain market), and *learning* (the foreign firm gets market knowledge). The costs are shared control, conflicts over strategy and dividends, risk of technology leakage, and the possibility that the partner becomes a competitor once it has learned the business.

The Economics and Governance of a JV

A JV is incomplete contracting in its most stressful form: the partners must agree ex ante on a business whose outcomes and the partners' interests change over time. The main design questions are:

- **Ownership and control.** Economic ownership (share of cash flows) and control (board seats, veto rights, management appointments) need not coincide. A minority investor can hold *reserved matters* (veto rights over budget, capital increases, related-party transactions, changes of strategy, new debt, dividend policy, sale of assets, and appointment of the CEO and CFO) that protect its investment without majority ownership. Consolidation of the JV for accounting depends on control as defined by IFRS 10 / ASC 810, and so affects reported leverage and covenants.
- **Contributions.** In-kind contributions (land, licenses, brands, technology) must be valued, and disagreements over valuation of non-cash contributions are a classic source of later conflict.

- **Funding obligations.** Who must provide additional capital if the JV needs it? Dilution formulae for the partner that does not contribute protect the others but can be used opportunistically to squeeze out a cash-constrained partner (a particular danger when a foreign partner faces capital controls or the local partner faces a banking crisis).
- **Dividend policy.** Local partners often prefer reinvestment (to grow the JV and their position) while the foreign partner needs cash flows to the parent. A minimum payout rule can reduce conflict.
- **Related-party transactions.** Fees, royalties and transfer prices between the JV and its partners are the principal channels through which value can be shifted from one partner to the other. They require clear arm's-length rules and approval by the unconflicted directors.
- **Deadlock resolution.** For 50/50 or veto-heavy structures, provisions for resolving deadlock (escalation, mediation, shotgun clauses or buy-sell) are essential.
- **Exit and transfer.** Rights of first refusal, tag-along and drag-along rights, lock-ups, and put and call options determine how each party can leave and at what price.

Local Partner Dynamics

The local partner is simultaneously the foreign investor's greatest asset and greatest risk. The following patterns recur:

1. **Asymmetric information.** The partner knows the local market, the regulators and the true state of the books; the investor does not. Due diligence on the partner, including its reputation, ownership, political exposure and other businesses, is as important as diligence on the target.
2. **Misaligned time horizons and objectives.** A family business may value control and succession over return on capital; a foreign strategic investor values consolidated market share and global brand protection.
3. **Hold-up and opportunism.** Once the foreign investor has committed capital and transferred know-how, its bargaining position weakens (the obsolescing bargain again, at the level of the partner). The partner may demand renegotiation, divert corporate opportunities to its other businesses, or compete using the transferred know-how.
4. **Tunneling.** Controlling shareholders may extract value through related-party transactions, asset sales at off-market prices, loans on favorable terms, or preferential dealing with affiliates. Tunneling is a central agency problem in markets with concentrated ownership and pyramid structures, and the minority protections in local law may be weak.
5. **Political connection as a double-edged asset.** A partner with strong political ties may speed licenses and protect the business from adverse treatment, but is exposed to changes of government and may expose the investor to anti-corruption liability.
6. **Evolution of the relationship.** Partnerships frequently begin with the foreign partner needing the local partner for access, and end with the local partner needing the foreign partner much less, as it acquires capabilities. Many JVs end in a buy-out by one partner. It is therefore wise to design the exit at the start.

Valuation Issues for JV Stakes

The value of a minority or JV stake is not a simple pro rata share of enterprise value. Adjustments include a *discount for lack of control* (if the stake carries no control over dividends, strategy or sale) and a *discount for lack of marketability* (restricted transferability), partially offset by the value of negotiated rights (vetoes, put options, tag-along). The magnitude of these adjustments depends on the governance terms: a stake with strong protective rights is worth more than one without.

Put and call options between partners are significant contingent claims. A put option written by the foreign partner allowing the local partner to sell its stake at a pre-agreed price is a contingent liability; if the strike is a floor price, the put is deep in the money precisely when the business has done badly, and the foreign partner will have to pay the most when it is least

able to. The value of such options can be estimated with the option methods of Volume 1, Chapter 18.

Example 18.5 – Fee leakage and the value of a put option in a JV. Part 1: Management fee. Solano holds 49 percent of a JV in Aurelia and the local partner holds 51 percent. The JV's pre-fee, pre-tax profit is $100 million, the local tax rate is 25 percent, and Solano proposes a management fee of $15 million per year payable to it, taxed in its hands at 21 percent (net of credits, simplified).

Without fee: JV net income = 100 × 0.75 = 75.0; Solano's share 0.49 × 75.0 = **36.75**; partner's share 0.51 × 75.0 = **38.25**.

With fee: JV pre-tax profit = 85; net income = 85 × 0.75 = 63.75; Solano's share of dividends = 0.49 × 63.75 = 31.24; Solano's fee after tax = 15 × 0.79 = 11.85; total to Solano = **43.09**. Partner's share = 0.51 × 63.75 = **32.51**.

Solano gains 43.09 − 36.75 = **$6.34 million**; the partner loses 38.25 − 32.51 = **$5.74 million**. The joint gain of $0.60 million equals the net tax effect: the fee saves 15 × 0.25 = 3.75 of local tax and costs 15 × 0.21 = 3.15 of home tax.

Interpretation: the fee mostly transfers value from the local partner to Solano; it is, in effect, the foreign partner collecting 100 percent of $15 million but bearing only 49 percent of its cost to the JV (the partner bears the other 51 percent). A local partner who understands this will insist on a cap, on arm's-length benchmarking of the fee, and on approval by a majority of the unconflicted directors. Fees which are really payment for services performed and costs incurred are legitimate; fees that exceed those values are a hidden dividend and a likely source of future conflict.

Part 2: Put option. The local partner (51 percent) holds a put option to sell its stake to Solano in three years at a floor price of $300 million. Business value at that time is 7.5 times JV EBITDA; the stake is worth 0.51 × 7.5 × EBITDA. Assume EBITDA scenarios of $50 million (probability 0.15), $65 million (0.25), $80 million (0.40) and $100 million (0.20).

Stake values: 0.51 × 7.5 × 50 = 191.25; × 65 = 248.63; × 80 = 306.00; × 100 = 382.50. Put payoffs max(300 − value, 0): **108.75**, **51.38**, 0, 0. Expected payoff = 0.15 × 108.75 + 0.25 × 51.375 = 16.31 + 12.84 = **$29.16 million**. PV at 10 percent over 3 years: 29.16 ÷ 1.331 = **$21.9 million**.

Interpretation: the put is a $21.9 million liability that does not appear in the purchase price but belongs in the valuation of Solano's position. Solano must also ask whether it will have the financial capacity, and the legal ability to move funds across the border, to honor the put in the scenario (low EBITDA, possibly a currency crisis) in which it is exercised. This *wrong-way* correlation between the exercise of the option and the investor's ability to pay is the main danger of floor-priced puts.

Table 18.2 – Entry Mode Comparison

Entry mode	Control	Capital at risk	Speed	Local knowledge	Key risks
Greenfield (wholly owned)	Full	High	Slow	Must build	Political risk, hold-up by government, slow ramp-up
Acquisition of local company	Full or majority	High	Fast	Inherited with the target	Integration, hidden liabilities, governance of target
Joint venture	Shared	Moderate	Medium	Partner provides	Partner conflict, tunneling, exit disputes
Minority stake	Limited	Moderate	Fast	Partner-led	Lack of control, valuation, exit
Licensing / franchising	None over operations	Low	Fast	Licensee	Leakage of IP, quality control, limited upside
Export / distributor	Limited	Low	Fast	Distributor	Dependence, thin margins, channel conflict

18.7 Capital Structure and Financing in Emerging Markets

Why Emerging-Market Capital Structure Differs

Volume 1, Chapter 15 developed the trade-off theory, pecking-order theory and agency perspectives on capital structure for developed markets. In emerging markets the same forces act, but the constraints and costs differ.

- **Shallow and short-term debt markets.** Corporate bond markets in local currency are small, with short maturities; banks dominate and prefer collateral and short tenors. Firms often face a trade-off between short-term local borrowing (refinancing risk) and long-term hard-currency borrowing (currency risk).
- **High local interest rates.** Local-currency borrowing costs include expected inflation and depreciation, as well as the sovereign risk premium; the local rate may look expensive relative to dollar debt.
- **Volatile cash flows and higher costs of distress.** Macroeconomic volatility raises the probability of distress at any leverage; weaker insolvency systems (slow, uncertain, favorable to insiders or to creditors, depending on the country) raise the cost of distress. The trade-off theory therefore points to *lower* target leverage, for a given tax shield.
- **Concentrated ownership and group structures.** Family and business-group ownership, pyramid structures and cross-holdings are common. Controlling shareholders may prefer debt to equity issuance to avoid dilution of control, and intra-group financing may substitute for weak external markets, with internal capital markets that can either support or exploit weaker members.
- **Information and governance problems.** Opaque accounting and weak creditor rights make lenders demand collateral and short maturity, and make equity investors demand a higher premium. Firms with strong governance (ADR listings, international auditors, foreign institutional investors) obtain better terms.
- **Tax shield variations.** Interest deductibility is often constrained by thin-capitalization rules, caps on deductible interest, withholding taxes on cross-border interest and inflation-related distortions.

- **Dollarization.** Many emerging-market firms borrow in dollars or euros because it is cheaper or available at longer tenors, creating the *currency mismatch* risk on balance sheets with local-currency revenues.

The Currency of Debt

The choice between local-currency and hard-currency debt is the crucial capital structure decision in emerging markets. Hard-currency debt looks cheaper because the nominal interest rate is lower, but the apparent saving is compensation for expected depreciation. Over any horizon, the borrower will pay less in hard-currency terms to the extent the local currency depreciates *less than* the interest differential implies, and more if it depreciates more. The break-even depreciation rate for a one-period loan is:

```
d* = (1 + k_local) ÷ (1 + k_hard) − 1
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If the actual depreciation exceeds d, *hard-currency debt turns out more expensive. Interest rate parity implies that a* hedged* hard-currency loan (using a forward or cross-currency swap) costs approximately the local rate, apart from market frictions. The unhedged choice is therefore a bet on the exchange rate, with downside concentrated in the states of the world in which the firm is already stressed: depreciation tends to come with recessions, falling local revenues and credit tightening (*wrong-way* risk). That asymmetry is why the currency-mismatch problem has been a recurring feature of emerging-market crises, and why a risk-neutral comparison of expected costs understates the true cost of unhedged foreign-currency debt.

Example 18.6 – Local versus hard-currency debt. Aurelia Refrescos can borrow the equivalent of $100 million in US dollars at 7.0 percent, or in local currency at 14.0 percent. Spot is LC 20 per dollar. Local inflation is 9 percent, US inflation 2.5 percent. Operating cash flow (EBIT) is LC 300 million per year, EBITDA is LC 400 million, and the debt is LC 2,000 million at spot.

Step 1 – Break-even depreciation: 1.14 ÷ 1.07 − 1 = **6.54 percent** per year. If the local currency depreciates by less than 6.54 percent a year, dollar debt costs less than local debt in local-currency terms.

Step 2 – Expected cost under PPP depreciation. The PPP-implied depreciation is 1.09 ÷ 1.025 − 1 = 6.34 percent. The expected local-currency cost of the dollar loan is 1.07 × 1.0634 − 1 = **13.79 percent**, close to the 14.0 percent of the local loan (the difference is the market's friction and credit spread). In expectation, the two loans cost almost the same: the dollar loan's low coupon is not a free lunch.

Step 3 – Stress test. With a dollar loan, interest in dollars is $7 million per year. Consider coverage and leverage in local currency terms after depreciation of the local currency against the dollar:

Local-currency depreciation	Spot (LC/$)	Interest (LC m)	EBIT cover	Debt (LC m)	Debt / EBITDA
0%	20	140	2.14x	2,000	5.0x
10%	22	154	1.95x	2,200	5.5x
20%	24	168	1.79x	2,400	6.0x
40%	28	196	1.53x	2,800	7.0x
60%	32	224	1.34x	3,200	8.0x

(Assuming revenues and EBITDA stay constant in local currency. If depreciation is accompanied by a recession, the cover is lower still, and if inputs are imported, the cost base rises with depreciation.)

Step 4 – Interpretation. A 60 percent depreciation, comparable to the largest observed in emerging-market crises, raises leverage from 5.0 to 8.0 times EBITDA and lowers interest cover from 2.14 to 1.34 times, even before any operating deterioration. With the local loan at 14 percent (interest LC 280 million, cover 300 ÷ 280 = 1.07x at the start) the firm is more strained at the outset, but its liabilities do not increase with depreciation. The choice is between a high fixed cost with no wrong-way exposure and a lower cost with convex downside.

Managerial implication: match the currency of debt to the currency of cash flows unless the firm earns hard-currency revenues (exporters), has natural offsets (dollar-priced inputs sold at dollar-linked prices), or has hedged. A beverage company selling in local currency should hold local-currency debt for the base of its capital structure, using hard-currency debt only up to the amount of its natural dollar exposure, or when hedged and the hedge cost is acceptable. Parent guarantees and cross-currency swaps can allow access to cheaper hard-currency funds while transferring the currency risk to the party better able to bear it.

Other Considerations

Parent funding can be equity (repatriated by dividends or capital reduction) or intercompany loans (deductible interest, principal returned without withholding, subject to thin-capitalization rules), a choice that must follow the repatriation analysis of Section 18.5. Local borrowing offers a political hedge, since a government is less likely to seize a major borrower from local banks; multilateral lenders provide long tenors and some protection. Because volatility, distress costs and refinancing difficulty are higher, target leverage should generally be lower than for a developed-market peer, and stress tests should combine currency, interest-rate and revenue shocks.

Case Study: Danube Grid Services in Karvania

Illustrative case; the company, country and figures are fictional.

Danube Grid Services (DGS) is a European provider of metering and grid-management services with a home-market WACC of 7 percent. It is bidding to acquire 70 percent of Karvania Metering, the leading supplier of smart meters to the national electricity distributors of Karvania, a middle-income country of 45 million people. The remaining 30 percent would stay with the Rostov family, the founders, who have excellent relationships with the energy ministry.

The numbers. Karvania Metering’s free cash flow is expected to be KC 900 million (Karvanian crowns) in year 1, growing at 12 percent a year for four years and then at 5 percent in nominal terms. Spot is KC 36 per euro. Expected inflation is 8 percent in Karvania and 2 percent in the euro area. Karvania’s euro-denominated sovereign bonds yield 4.5 percentage points over euro swaps; the volatility ratio of equities to bonds is 1.5. The euro risk-free rate is 3 percent, the mature-market equity risk premium 5.5 percent, the target’s beta 0.9, and target debt-to-capital 30 percent. The pre-tax cost of euro-denominated debt for the company would be the risk-free rate plus the sovereign spread plus 1.5 percent; tax rate 20 percent. DGS estimates that about 60 percent of the target’s profits depends on the domestic distribution companies, the remainder on exports of components.

Complications.

- The distribution companies are state-controlled, and payments have lagged by up to nine months during budget stress. A change of government in eighteen months could renegotiate meter tariffs. The finance ministry has twice imposed temporary limits on dividend transfers during currency stress, once for six months and once for fourteen months.

- The Rostov family has proposed a shareholders' agreement in which it holds a veto over budget, any new debt and the appointment of the CEO, and receives a put option at 6 times EBITDA exercisable from year 4. It has also asked that DGS provide a EUR 60 million parent guarantee for local bank loans in Karvanian crowns at 16 percent.
- A bank offers a EUR 80 million loan at 5.5 percent, which DGS finds attractive compared with a crown loan at 16 percent, and proposes that the target service it from crown revenues.
- DGS's chief executive has been told that one of the family's companies supplies components to the target at prices above those of independent suppliers.

Task. The DGS board has asked for a recommendation on whether to proceed, on what terms and at what price. The finance team has been asked to produce a valuation in euros, a risk assessment and a proposal on governance and financing.

Discussion Questions

1. Estimate DGS's cost of capital for the investment under the additive, beta-scaled and lambda approaches (using $\lambda = 0.8$ for the exporter share and domestic exposure as given), and recommend one. Explain which risks you would deal with through scenario analysis instead of the discount rate, and why.
2. Describe the method you would use to value the cash flows, in crowns or in euros. Compute the local-currency WACC for the case and the forecast exchange rate path implied by parity. What would happen to value if DGS discounted crown cash flows at the euro WACC?
3. Evaluate the proposed euro loan versus a crown loan, the parent guarantee, and the put option. What is the break-even depreciation for the loan, and how would you protect DGS against wrong-way risk from the put?
4. Design the governance and protective structure of the investment: reserved matters, related-party transaction rules (including the component supplier), dividend policy, exit and deadlock provisions, and measures to mitigate payment delay, dividend-transfer restrictions and tariff renegotiation. What conditions would make you walk away?

Key Takeaways

- Cross-border risk comprises country, currency, political, transfer, legal and liquidity risks; each should be placed once, in either the discount rate or the cash flows, and not both.
- Country risk premiums are usually built from sovereign spreads scaled by the equity-to-bond volatility ratio; the additive, beta-scaled and lambda approaches differ by about a percentage point, less than the difference from ignoring the premium.
- Lambda, the exposure of a firm's cash flows to the country, matters: exporters and multinationals with diversified revenue deserve a smaller premium than domestically dependent firms.
- Currency consistency is the central rule: local cash flows with local-currency discount rates, or hard-currency flows at forecast exchange rates with hard-currency rates; mixing the two can overstate value by 30 percent or more.
- Purchasing power parity and interest-rate parity give a neutral base-case forecast; real exchange-rate views should be explicit scenarios.
- Discrete political events such as expropriation are best modeled with probabilities, recovery rates and scenarios, and then mitigated through treaties, insurance, financing structure and relationships.
- Cash that reaches the parent is what has value; repatriation channels, withholding taxes and controls create leakage, and trapped funds should be discounted for delay and risk.

- Joint ventures need governance designed for the eventual conflict: reserved matters, related-party rules, funding obligations, deadlock mechanisms and exit options.
- Floor-priced put options and fee arrangements are contingent claims and value transfers that belong in valuation; wrong-way risk arises when the option is exercised just when the writer cannot pay.
- Match the currency of debt to the currency of cash flows; hard-currency borrowing looks cheaper only because the market already prices expected depreciation, and the risk concentrates in crises.

Review Questions

1. Why might a global-CAPM investor argue that a country risk premium is unnecessary, and why do practitioners nevertheless include one?
2. Explain how the country risk premium is derived from a sovereign spread, and why the spread is multiplied by an equity-to-bond volatility ratio.
3. Compare the additive, beta-scaled and lambda approaches. In what circumstances would you prefer each?
4. Explain why discounting local-currency cash flows at a hard-currency rate overstates value. What is the relationship between the local and hard-currency costs of capital?
5. What role does purchasing power parity play in valuation, and what are its limits?
6. How should expropriation risk be handled: in the discount rate, in the cash flows or through insurance? What are the advantages and weaknesses of each?
7. What is creeping expropriation and why is it harder to insure than outright seizure?
8. List the channels for repatriating cash from a subsidiary and explain how tax and legal features affect the choice.
9. Why is trapped cash worth less than its face value, and how would you value it?
10. Describe three ways a local partner can extract value from a joint venture, and how contracts can limit each.

Problems

1. **CRP.** The sovereign spread of a country is 4.2 percent and the equity-to-bond volatility ratio is 1.3. The risk-free rate is 3.5 percent, mature-market ERP 5.5 percent, and beta 0.9. Compute the CRP and the cost of equity under the additive, beta-scaled and lambda ($\lambda = 0.7$) approaches.
2. **Exchange-rate forecast.** Spot is LC 40 per dollar; inflation is 12 percent locally and 3 percent in the United States. Forecast the exchange rate for years 1 to 3 under relative PPP.
3. **Local WACC.** A project has a dollar WACC of 10 percent. With local inflation of 12 percent and US inflation of 3 percent, compute the local-currency WACC.
4. **Political risk.** A project yields \$50 million in years 1 and 2 and \$550 million in year 3 (including terminal value). The discount rate is 11 percent. There is a 5 percent annual probability of loss with no compensation. Compute the unadjusted PV, the survival-weighted PV, and the equivalent discount rate.
5. **Repatriation.** A subsidiary earns 200 of pre-tax profit; local tax is 30 percent, dividend withholding tax is 15 percent, and the home country exempts foreign dividends. How much reaches the parent and what is the total leakage rate?
6. **Blocked funds.** 150 of funds are released in three equal annual installments. The discount rate is 18 percent. Compute the PV and the haircut.

7. **Debt currency.** Dollar debt costs 6.5 percent and local debt 12.5 percent. Compute the break-even depreciation. If the currency depreciates 8 percent a year, compute the local-currency cost of the dollar loan.
8. **JV fee.** A JV with pre-tax profit before fee of 100 is owned 40 percent by the foreign partner and 60 percent by the local partner. Local tax is 30 percent. The foreign partner charges a fee of 10, taxed at 20 percent in its hands. Compute each partner's income with and without the fee.

Answers to Selected Problems

1. CRP = 4.2% × 1.3 = **5.46%**. Mature cost of equity = 3.5% + 0.9 × 5.5% = 8.45%. Additive: **13.91%**. Beta-scaled: 3.5% + 0.9 × (5.5% + 5.46%) = **13.36%**. Lambda: 8.45% + 0.7 × 5.46% = **12.27%**.
2. 40 × (1.12 ÷ 1.03)^t: year 1 **43.50**, year 2 **47.30**, year 3 **51.43** LC per dollar.
3. 1.10 × (1.12 ÷ 1.03) − 1 = **19.61%**.
4. Unadjusted PV = 50 ÷ 1.11 + 50 ÷ 1.2321 + 550 ÷ 1.3676 = 45.05 + 40.58 + 402.16 = **$487.78 million**. Survival-weighted PV = **$424.22 million** (13.0 percent lower). Equivalent discount rate = 1.11 ÷ 0.95 − 1 = **16.84%**.
5. After local tax: 140; after withholding: 119 = **119** reaches the parent; leakage = 81 ÷ 200 = **40.5%**.
6. PV = 50 × (0.8475 + 0.7182 + 0.6086) = **$108.71**; haircut = 1 − 108.71 ÷ 150 = **27.5%**.
7. Break-even = 1.125 ÷ 1.065 − 1 = **5.63%**. Cost at 8% depreciation: 1.065 × 1.08 − 1 = **15.02%**, higher than the 12.5% local loan.
8. Without fee: JV net income 70; foreign partner **28.0**, local partner **42.0**. With fee: JV net income = 90 × 0.7 = 63; foreign partner = 0.4 × 63 + 10 × 0.8 = 25.2 + 8.0 = **33.2**; local partner = 0.6 × 63 = **37.8**. The foreign partner gains 5.2; the local partner loses 4.2.

Chapter 19: Distress, Bankruptcy and Workouts

Learning Objectives

After studying this chapter you should be able to:

- Distinguish financial distress from economic distress, estimate direct and indirect costs of distress, and use scoring and structural models to assess default risk.
- Apply the absolute priority rule to build a recovery waterfall, and explain how secured claims, deficiency claims, and superpriority financing alter the order of payment.
- Value a firm in bankruptcy using going-concern and liquidation approaches, and explain why the reorganization value is the pivotal number in plan negotiation.
- Model equity as a call option on the firm's assets to explain why shareholders of distressed firms retain value, and why they behave as they do.
- Evaluate debtor-in-possession (DIP) financing, a negotiated plan of reorganization, and an out-of-court exchange offer from the perspectives of each stakeholder, including holdout problems.
- Analyze a distressed investment by identifying the fulcrum security, estimating scenario recoveries, and computing expected returns and break-even prices; and describe, cautiously, how insolvency regimes differ across jurisdictions.

Opening Vignette

(Illustrative scenario. Meridian Home Stores is a fictional company.)

Meridian Home Stores operates 140 mid-market home furnishing stores across the Midwest. Three years ago a sponsor-backed recapitalization left the company with a 400 million secured term loan, 300 million of senior unsecured notes, and 150 million of subordinated notes. Sales were 1.9 billion. EBITDA, which had been 190 million at the time of the deal, has slid to 80 million as e-commerce competitors take share and a shift in housing turnover cuts demand for big-ticket furniture.

In January the CFO, Dana Okafor, faces a calendar that is no longer under her control. A 60 million interest payment on the senior notes is due in six weeks. The revolver is fully drawn. Suppliers, who ship on 45-day terms, are quietly asking for cash in advance, which accelerates the liquidity drain. Same-store sales are falling at 8 percent. The term loan lenders have hired advisers; a group of noteholders holding about 40 percent of the senior notes has organized behind its own counsel.

Dana's board asks three questions. Is Meridian's problem a balance sheet problem, in which a viable business carries too much debt, or an operating problem, in which the business is worth more dead than alive? If it is a balance sheet problem, can the company restructure out of court, with the cooperation of hundreds of dispersed bondholders, or must it file for Chapter 11? And if it files, who will lend the cash to keep stores stocked and payroll paid while the case is pending?

This chapter provides the tools to answer these questions, returning to Meridian to build its waterfall, value the reorganized business, price its equity as an option, evaluate a DIP loan, negotiate a plan, test an exchange offer, and price the senior notes for a distressed investor.

19.1 The Anatomy of Financial Distress

19.1.1 Two kinds of trouble

Distress is not one phenomenon. **Economic distress** means the firm's assets are worth more in an alternative use than in their current use: the business model is broken, and value is destroyed every year it continues. **Financial distress** means the firm cannot meet its contractual obligations to creditors, even though the underlying business may be worth far more as a going concern than its debt. The policy and managerial responses differ sharply. An economically distressed firm should usually shrink or be liquidated, and the question for creditors is how to maximize recovery. A financially distressed but economically viable firm should be recapitalized, and the value at stake is the going-concern value that a chaotic default would destroy.

Most real cases mix the two. A useful diagnostic compares going-concern value (GCV), liquidation value (LV) and total claims:

```
Economic distress:      GCV < LV            (stop operating)
Financial distress:     LV < GCV < Claims   (restructure the balance sheet)
Solvent:                GCV > Claims
```

Meridian sits in the middle category: its going-concern value, estimated in Section 19.4 at about 620 million, exceeds a liquidation value of roughly 330 million but is far below its claims. The 290 million gap is the value at risk in a disorderly collapse, and the rationale for every institution in this chapter.

19.1.2 Why distress is costly

Volume 1, Chapter 15 introduced the trade-off theory: debt provides a tax shield but raises expected distress costs. Here we decompose those costs.

Direct costs are legal, advisory and court fees plus management time. Empirical work has generally found them modest relative to the value of large firms and proportionally heavier for small ones; they alone rarely justify avoiding leverage.

Indirect costs are larger and harder to measure. They include:

- *Lost sales:* customers doubt that a distressed firm will honor warranties or support products (furniture, airline tickets, software).
- *Tighter supplier terms:* trade credit is cut, converting a profitability problem into a liquidity problem.
- *Employee attrition:* the best people leave first.
- *Debt overhang* (Myers): management skips positive-NPV projects because benefits accrue to creditors.
- *Risk shifting* (Jensen and Meckling): equity holders, who hold a call option, may favor negative-NPV high-variance projects.
- *Strategic exploitation:* competitors price aggressively against a weak rival.

The central managerial point is that the expected cost of distress equals the probability of distress times the loss given distress. Both terms are driven by leverage, asset tangibility, the nature of the product, and the flexibility of the capital structure.

Example 19.1 – Expected distress costs and optimal leverage. A firm has an unlevered value of 1,000 million. The corporate tax rate is 25 percent, and debt is assumed permanent so that the present value of the tax shield is $0.25 \times D$. Distress costs, if distress occurs, equal 35 percent of firm value (1,000 million). The probability of distress rises with debt:

Debt D (million)	Probability of distress	PV tax shield = 0.25D	Expected distress cost = p × 0.35 × 1,000	Net benefit of debt
0	0%	0.0	0.0	0.0
200	2%	50.0	7.0	43.0
400	6%	100.0	21.0	79.0
600	15%	150.0	52.5	97.5
800	35%	200.0	122.5	77.5

Step by step for D = 600: shield = 0.25 × 600 = 150.0; expected cost = 0.15 × 0.35 × 1,000 = 52.5; net = 150.0 − 52.5 = 97.5. The net benefit peaks near D = 600 among these grid points and declines at 800 because the probability of distress is rising faster than the shield. The precise figure depends on unknowable assumptions; the lesson is that the marginal cost of distress rises convexly while the shield rises linearly. A prudent treasurer sits left of the peak because the benefit curve is flat near the optimum while distress costs are catastrophic and partly irreversible.

19.1.3 Warning signals and default prediction

Practitioners use two broad families of default predictors. **Accounting-based scores** map financial ratios into a single number. The best-known is Altman’s Z-score for public manufacturing firms:

```
Z = 1.2·X1 + 1.4·X2 + 3.3·X3 + 0.6·X4 + 1.0·X5
X1 = Working capital / Total assets
X2 = Retained earnings / Total assets
X3 = EBIT / Total assets
X4 = Market value of equity / Book value of total liabilities
X5 = Sales / Total assets
```

In Altman’s original calibration, scores below about 1.8 signaled high distress risk, scores above about 3.0 signaled safety, and the range between was a grey zone. Coefficients differ for private firms and non-manufacturers, and the model was estimated on a particular era; it is a screening tool, not a forecast.

Example 19.2 – Z-score for Meridian. From the accounts: X1 = 0.15, X2 = 0.08, X3 = 0.10, X4 = 0.60, X5 = 1.10.

```
Z = 1.2(0.15) + 1.4(0.08) + 3.3(0.10) + 0.6(0.60) + 1.0(1.10)
  = 0.180 + 0.112 + 0.330 + 0.360 + 1.100
  = 2.082
```

A score of 2.08 is in the grey zone, and X5 contributes 1.10 of it: a retailer’s high asset turnover flatters a business whose margins are collapsing, a standard criticism of the model.

Market-based measures use equity prices and volatility. The Merton model (Section 19.5) yields a *distance to default*, and credit spreads and CDS quotes embed the market’s view of default probability and recovery (Volume 1, Chapter 4). Warning signs include wide spreads, falling equity, covenant waivers, going-concern audit qualifications, restructuring advisers and deferred coupons. Three early decisions matter: diagnose economic versus financial distress, protect liquidity with a rigorous 13-week cash forecast, and act early, because the option set narrows as the runway shortens.

19.2 The Legal Framework: Priority, Process and Incentives

19.2.1 Why bankruptcy law exists

When a debtor cannot pay everyone, each creditor's rational response is to grab assets first. This common-pool problem dismembers a going business, exactly as a bank run destroys a bank. Bankruptcy law substitutes a collective procedure built on three devices: a **stay** on individual collection, a **priority scheme** that fixes who is paid first, and a **voting mechanism** that lets a majority bind dissenters to a restructuring. The description below follows the United States Bankruptcy Code, the best-known model; Section 19.10 addresses other jurisdictions cautiously. Nothing here is legal advice, and statutory detail and case law evolve.

19.2.2 Chapter 7 and Chapter 11

Chapter 7 is liquidation by a trustee. **Chapter 11** is reorganization: management typically remains in control as "debtor in possession," operations continue, and the firm proposes a plan that, if confirmed, binds all parties. An **automatic stay** halts collection and most litigation. The debtor has an initial *exclusivity period* in which only it may propose a plan; it is extendable by the court up to statutory ceilings, after which competing plans give creditors leverage.

Chapter 11 gives the debtor tools: it can **assume or reject contracts and leases** (rejection damages become unsecured claims, sometimes capped for real property leases); it can **sell assets free and clear** under Section 363, often to a *stalking horse* bidder whose offer sets a floor for an auction; it can borrow with priority (Section 19.6); and the estate can pursue **avoidance actions** against preferences (payments to creditors shortly before filing, 90 days for most) and fraudulent transfers (transfers for less than reasonably equivalent value while insolvent). Leveraged buyouts and special dividends that left a firm insolvent are frequent targets, a link back to Volume 1, Chapter 23.

19.2.3 Classes, voting and confirmation

A plan places claims in **classes** of substantially similar claims: administrative expenses and priority claims are paid in full unless holders agree otherwise; secured claims form one or more classes; unsecured claims, subordinated claims, and equity interests form their own classes. A class **accepts** the plan if holders of at least two-thirds in amount and more than one-half in number of the claims actually voting vote yes. For confirmation, the court examines several requirements, the most important of which for finance are:

1. **Best interests of creditors.** Each dissenting creditor must receive at least what it would receive in a Chapter 7 liquidation. This puts a floor under every class: the liquidation value.
2. **Feasibility.** The plan must not be likely to be followed by liquidation or further reorganization. The reorganized firm must be able to service its post-emergence capital structure, which is why exit leverage is carefully tested.
3. **At least one impaired accepting class,** excluding insiders, if any class is impaired.
4. **Cramdown** (Section 1129(b)): if some impaired class rejects, the court may still confirm the plan if it is *fair and equitable* and does *not discriminate unfairly* as to the rejecting class. For a dissenting class of unsecured creditors, "fair and equitable" means either that the class is paid in full or that no junior class (including equity) receives or retains any property on account of its junior interest. That is the **absolute priority rule** (APR).

The APR is thus a statutory requirement for cramdown, not an inviolable property right in every negotiated outcome. When every class accepts, the plan can deviate from strict priority because classes may waive their rights. The rule's practical effect is to set the *bargaining baseline* against which consensual deviations are negotiated: "the shadow of the waterfall."

19.2.4 The absolute priority waterfall

The standard ordering of payment is: (1) **secured claims**, to the value of their collateral (any shortfall is an unsecured *deficiency claim*); (2) **administrative expenses**, including professional fees, post-petition payables and DIP financing, which with superpriority can rank ahead of prepetition secured claims on the collateral it is granted; (3) **priority unsecured claims** (certain wages, benefits and taxes); (4) **general unsecured claims** (bonds, trade, lease rejection damages), pari passu; (5) **subordinated claims**; and (6) **preferred, then common equity**.

Courts may *equitably subordinate* claims for inequitable conduct and rarely *substantively consolidate* affiliates, but legal-entity structure is of first-order importance: a holding company bondholder without guarantees is *structurally subordinated* to creditors of operating subsidiaries even if its debt is called "senior."

Example 19.3 – Meridian's recovery waterfall. Claims at filing: DIP financing 100; secured term loan 400 (all-asset lien); general unsecured claims 400 (senior notes 300 plus trade and lease rejection claims 100, pari passu); subordinated notes 150. Equity is junior to everything. Assume the distributable value is the enterprise value (EV) in each scenario, ignoring professional fees for simplicity (they would reduce recoveries further).

Rule: pay each class in order until value runs out; recovery % = payment ÷ claim.

Distributable value	DIP	Secured TL	General unsecured	Subordinated	Equity
400	100 (100%)	300 (75.0%)	0 (0%)	0	0
500	100 (100%)	400 (100%)	0 (0%)	0	0
620	100 (100%)	400 (100%)	120 (30.0%)	0	0
800	100 (100%)	400 (100%)	300 (75.0%)	0	0
1,000	100 (100%)	400 (100%)	400 (100%)	100 (66.7%)	0

Step by step at 620: after the DIP, 620 − 100 = 520; after the term loan, 520 − 400 = 120; the 120 goes to general unsecured claims of 400, a recovery of 120 / 400 = 30.0 percent; nothing remains for subordinated holders or equity. At 800: 800 − 500 = 300 to the unsecured pool, 75 percent. At 1,000: 1,000 − 500 = 500; unsecured claims take their 400 in full, leaving 100 for the subordinated notes, a recovery of 100 / 150 = 66.7 percent.

Two lessons follow. First, recoveries are a piecewise-linear function of distributable value: each class is a "call spread" on the enterprise value, which is the intuition for the option pricing of Section 19.5. Second, the **fulcrum security** is the class in which value runs out. At 620 the fulcrum is the general unsecured class, and it is the class that will receive the new equity of the reorganized firm. The fulcrum holders are the real owners of the firm economically, even before the plan says so. The senior secured lender is "money good" at 620 and has little interest in the valuation fight unless value falls toward 500; subordinated holders and old equity are out of the money and have only nuisance leverage and the option value of delay.

19.2.5 Mistakes to avoid when building a waterfall

Include accrued interest, make-whole premia and lease rejection damages in claims; allocate collateral so that secured creditors are secured only to the value of their collateral (the remainder is a deficiency claim that competes with unsecured claims); build the waterfall entity by entity, with guarantees and intercompany claims; deduct professional fees and DIP interest before prepetition creditors; and value non-cash recoveries (new notes, illiquid equity) at market value rather than face.

19.3 Out-of-Court Workouts and the Collective Action Problem

19.3.1 The case for staying out of court

Chapter 11 is costly and public and can trigger customer flight. An **out-of-court restructuring** (workout) avoids many of those costs through maturity extensions, covenant amendments, debt-for-equity swaps, **exchange offers** (old bonds for new bonds with lower principal, lower coupons, longer maturity, higher seniority or equity kickers), asset sales and new equity. Workouts work best when creditors are few and sophisticated, the capital structure is simple, and the business is viable.

19.3.2 Why bond restructurings are hard: holdouts

A bank group can be gathered in a room; a bond issue held by hundreds of dispersed investors presents a free-rider problem. Each holder prefers that others concede while it holds out and collects full value if the firm is rescued, a prisoner's dilemma analyzed by Gertner and Scharfstein and by Bulow and Shoven. Design features that address it:

- **Collective action clauses** let a supermajority amend payment terms for all. In many US corporate indentures, changes to core payment terms need each affected holder's consent (a requirement associated with the Trust Indenture Act of 1939), whereas many other bond contracts and most modern sovereign bonds allow supermajority amendment. Check the actual indenture.
- **Exit consents.** Tendering holders vote to strip covenants or liens from the *old* bonds, leaving holdouts with a weakened instrument. Their permissible strength depends on contract and jurisdiction.
- **Seniority enhancement.** New securities are secured or senior, so holdouts become effectively subordinated. Such "uptier" or liability-management transactions, when offered to only a subset of creditors, have generated significant litigation over the rights of excluded holders.
- **Participation conditions and a prepackaged backstop.** A *prepackaged* bankruptcy solicits votes before filing and then obtains rapid confirmation if thresholds are met; its credible threat, which binds holdouts by class vote, is the stick behind many exchange offers conditioned on, say, 90 percent participation.

19.3.3 Evaluating an exchange offer

An exchange offer is a take-it-or-leave-it choice for each holder, who compares the value of the new package, V_new, with the value of holding out: V_old,succ if enough others tender, V_old,fail (the bankruptcy recovery) if the exchange collapses. A tenderer's outcome in the success state is V_new; a holdout's is V_old,succ. The offer must therefore be designed so that V_new materially exceeds V_old,succ, so that the package *pays* participants for the benefit they confer on the group.

Example 19.4 – Valuing the exchange for a bondholder. Meridian proposes this to its holders of 300 million of 5-year 8 percent senior unsecured notes (annual coupons, par 100). For each 100 face tendered, a holder receives 70 face of new 5-year 8 percent *secured* notes. The market yield for a secured note of this credit quality, after the restructuring, is 11 percent. If the exchange fails, the firm files for Chapter 11 and the general unsecured recovery is 30 cents on the dollar (Example 19.3). If the exchange succeeds and holders do not tender, the old notes would be subordinated in effect to the new secured notes; assume their price reflects a yield of 25 percent.

Value of 100 face of new secured notes per 100 tendered, with 70 face received:

```
Price per 100 face of new note = Σ 8/(1.11)^t (t=1..5) + 100/(1.11)^5
  = 8 × 3.6959 + 100 × 0.59345
  = 29.567 + 59.345 = 88.91
Value of package = 0.70 × 88.91 = 62.24
```

Value of holding out if the exchange succeeds: price at 25 percent yield:

```
Σ 8/(1.25)^t + 100/(1.25)^5 = 8 × 2.6893 + 100 × 0.32768 = 21.51 + 32.77 = 54.28
```

Value of holding out if the exchange fails: 30.0 (bankruptcy recovery).

Tendering gives 62.24 with certainty if the exchange goes through (and if it fails, the tendered holder's position reverts and receives 30, as in the failure branch). Holding out gives 54.28 on success and 30 on failure. Tendering therefore dominates in this example: 62.24 > 54.28 in the success state and equal in the failure state. The structure (a smaller face amount but with higher priority) has been designed so that the exchange converts a holdout's free ride into a penalty of 62.24 − 54.28 = 7.96 per 100 face. For the firm, debt on these notes falls to 0.70 × 300 = 210 million of face (a 90 million reduction), at the price of granting security that primes other unsecured creditors. Whether that is wise depends on what other creditors do, which is why liability management exchanges are among the most contested transactions in credit markets.

19.3.4 Tax wrinkles

Debt forgiveness generally creates cancellation-of-debt income for the issuer. Many tax systems give relief to insolvent borrowers, often by reducing tax attributes such as loss carryforwards, and ownership changes may limit use of those losses. Details differ by country and change over time, so tax advice should come early in plan design.

19.4 Valuing the Firm in Bankruptcy

Everything in a restructuring depends on one number: the value available for distribution. Valuation in bankruptcy is applied corporate finance with an adversarial twist: each class wants a valuation that puts value to it at the fulcrum or above, so senior creditors argue low and junior creditors and old equity argue high. A judge may have to choose.

19.4.1 Going-concern versus liquidation value

Liquidation value is what the assets would fetch if sold piecemeal, net of liquidation costs, in a forced or semi-forced sale. For asset-heavy firms (inventory, receivables, real estate, equipment) it is estimated with *recovery rates* by asset class (for example, a high fraction of face for receivables, a lower fraction for inventory, a lower still for specialized equipment). It provides the floor under the best-interests test.

Going-concern value is the value of the reorganized business. There are three standard methods:

1. **Discounted cash flow (DCF)** on the post-restructuring business plan, using a WACC appropriate to the *emerging* capital structure and risk (Volume 1, Chapters 11 and 21). It requires projecting free cash flow after the operational restructuring (closing stores, shedding unprofitable leases).
2. **Comparable companies**: apply multiples of EBITDA from healthy peers to normalized EBITDA, with a discount for distress or size.
3. **Precedent transactions**: multiples paid in comparable acquisitions (Volume 1, Chapter 22).

In a plan context, the figure is called the **plan value** or *reorganization value*. It includes debt and equity of the reorganized firm. Because the post-reorganization capital structure matters, circularity must be avoided: use an unlevered DCF or consistent peer multiples and then subtract the post-emergence net debt to get equity.

19.4.2 DCF for the emerging business

Example 19.5 – Meridian's reorganized enterprise value. After closing 35 underperforming stores and renegotiating leases, management projects unlevered free cash flow (after tax, capex and working capital) for five years of 40, 48, 56, 61 and 63 (million). Terminal growth is 2 percent. The WACC of the reorganized firm is 11 percent. Compute the enterprise value.

Discount each year's cash flow at 1.11^t:

Year	FCF	Discount factor	Present value
1	40	1/1.11 = 0.9009	36.04
2	48	0.8116	38.96
3	56	0.7312	40.95
4	61	0.6587	40.18
5	63	0.5935	37.39

Sum of PV of explicit cash flows = 36.04 + 38.96 + 40.95 + 40.18 + 37.39 = 193.51 (with exact discounting, 193.51).

Terminal value at the end of year 5 (Gordon growth):

```
TV = FCF5 × (1 + g) / (WACC − g) = 63 × 1.02 / (0.11 − 0.02) = 64.26 / 0.09 = 714.0
PV(TV) = 714.0 / 1.11^5 = 714.0 / 1.68506 = 423.7
EV = 193.5 + 423.7 = 617.2
```

The DCF gives about 617 million; we round to 620 million as the working plan value. The terminal value is 68.6 percent of the total, which is typical and is the weak point of the valuation. The sensitivity is large:

WACC	EV (g = 2%)
9%	801.1
10%	697.6
11%	617.2
12%	553.0
13%	500.5

A one-point change in WACC moves the EV by 50 to 100 million, or 8 to 15 percent. Now recall Example 19.3: at EV = 500 the term lender is just paid in full after the DIP, and the unsecured creditors recover zero; at 700 they recover about 50 percent (700 − 500 = 200, 200/400); at 800, 75 percent. The *recovery of the fulcrum class is therefore extremely sensitive to WACC*, which explains why valuation fights are the central contest in contested cases. Unsecured creditors argue for 10 percent or lower (they will say the post-restructuring firm is less risky because leases and debt were shed); the secured lender argues for 12 or 13 percent (execution risk, thin liquidity, a distressed retail sector).

Compare with liquidation value of 330: the best-interests test is satisfied for every class so long as the plan recovery exceeds what Chapter 7 would pay. In a Chapter 7 liquidation at 330, after the 100 DIP, 230 is available for the term loan, a recovery of 230 / 400 = 57.5 percent; unsecured creditors get nothing. Under the plan, the term loan receives 100 percent

and unsecured creditors receive 30 percent. Both beat liquidation.

19.4.3 Valuation traps in bankruptcy

- **Using the pre-distress WACC.** Use an unlevered cost from healthy comparables, relevered at the exit capital structure (Volume 1, Chapter 15).
- **Management's projections.** Management has an incentive to project a hockey stick to justify equity retention; creditors' advisers have the opposite. Stress-test against history and benchmark margins.
- **Terminal value dominance.** Cross-check the DCF with an exit multiple: 620 / 80 = 7.75 times normalized post-restructuring EBITDA of 80 million, a modest discount if healthy peers trade at 8 to 9 times.
- **Double-counting.** Excess cash and tax assets (NOLs) are valued separately if not in the cash flows.

19.5 Equity as an Option: Why Shareholders Fight

19.5.1 The Merton framework

Merton (1974) applied Black and Scholes to corporate liabilities. Suppose the firm has a single zero-coupon debt issue with face value D due at time T, and the firm's asset value V follows a lognormal diffusion with volatility σ. At maturity equity holders receive max(V − D, 0), a call option on the firm's assets with strike D; creditors receive min(V, D) = D − max(D − V, 0), which is risk-free debt less a put option on the assets. In Volume 1, Chapter 18's notation:

```
E = V·N(d1) − D·e^(−rT)·N(d2)
d1 = [ln(V/D) + (r + σ²/2)T] / (σ√T),   d2 = d1 − σ√T
Debt value = V − E
Risk-neutral probability of default = N(−d2) = 1 − N(d2)
Yield on debt:  y = (D/Debt value)^(1/T) − 1
```

The model is a stylized single-period description; it ignores coupons, multiple seniorities, strategic default, and the cost of the process, but it clarifies three facts about distress.

Example 19.6 – Meridian's equity as a call option. Treat all financial debt as a single zero-coupon claim of 1,000 million (face; simplified to include the term loan and unsecured and subordinated notes, each treated at face value plus accrued items for illustration) due in three years. The risk-free rate is 4 percent, asset volatility is 35 percent. Compute equity value and debt yield at V = 700, 1,000 and 1,400.

For V = 700, D = 1,000, r = 4%, σ = 35%, T = 3:

```
σ√T = 0.35 × 1.7321 = 0.6062
d1 = [ln(0.7) + (0.04 + 0.06125) × 3] / 0.6062
   = [−0.35667 + 0.30375] / 0.6062 = −0.0873
d2 = −0.0873 − 0.6062 = −0.6935
N(d1) = 0.4652,  N(d2) = 0.2440
PV of strike = 1,000 × e^(−0.12) = 886.92
E = 700 × 0.4652 − 886.92 × 0.2440 = 325.6 − 216.4 = 109.2
```

(Figures are rounded; the exact computation yields E = 109.25.) Debt value = 700 − 109.2 = 590.8; the debt's promised yield is (1,000 / 590.75)^(1/3) − 1 = 19.2 percent; the risk-neutral default probability is N(−d2) = 1 − 0.244 = 75.6 percent.

Results at the three asset values (computed with the formulas above):

V	d1	d2	N(d2)	Equity	Debt	Debt yield	P(default)
700	−0.087	−0.694	0.244	109.2	590.8	19.2%	75.6%
1,000	0.501	−0.105	0.458	285.5	714.5	11.9%	54.2%
1,400	1.056	0.450	0.674	598.9	801.1	7.7%	32.6%

The three lessons. First, equity is worth 109 million when the assets (700) are worth far less than the 1,000 of debt: the **option value of equity** is positive even deep out of the money, because volatility gives a chance of recovery. That is why equity trades above zero in distressed firms and why old shareholders demand a "tip" in the plan. Second, at V = 700 equity gains 285.5 − 109.2 = 176.3 when V rises by 300, a delta effect of roughly 0.59 per dollar over this range, while creditors capture only 123.7 of that increase: equity and debt share the upside according to their deltas. Third, equity's value *increases* with volatility (vega is positive), so shareholders prefer riskier strategies; creditors' value falls. In distress this incentive conflict is acute: management tied to equity will favor gambles ("resurrection" strategies), while creditors will insist on covenants and restrict operations.

19.5.2 Managerial and negotiation implications

Old equity resists liquidation because its option has value; deviations from absolute priority in negotiated cases can be read as creditors paying for equity's *nuisance value* and *delay option*, which grows with the case's duration because value becomes more volatile. Each class can be viewed as a *call spread* on enterprise value: the fulcrum class is long a call struck at the sum of senior claims and short a call struck at the sum of claims through its own class, so senior classes are short volatility and the fulcrum class is most exposed to valuation arguments.

19.6 Debtor-in-Possession Financing

19.6.1 Why a bankrupt firm needs new money

A firm in Chapter 11 faces an immediate squeeze: prepetition trade credit disappears and professional fees are paid currently. **DIP financing** is credit extended after filing. The Code offers inducements in ascending order of aggressiveness: **administrative expense priority**; **superpriority** over other administrative claims; **liens on unencumbered assets** and junior liens on encumbered ones; and **priming liens**, a first lien on assets already pledged to prepetition lenders, available only if the debtor cannot otherwise obtain credit and the prepetition lenders are adequately protected (for example by a value cushion or replacement liens).

DIP loans carry tight **milestones** (deadlines to file a plan, approve sale procedures or close a sale), strict budgets and reporting. Milestones give the lender effective control over the pace and sometimes the outcome of the case, and incumbent lenders can use the loan to steer toward "loan-to-own" outcomes, so courts scrutinize excessive control, releases and *roll-ups* that convert prepetition debt into post-petition debt.

19.6.2 Evaluating a DIP loan

From the debtor's perspective, DIP financing is an *option-preserving* investment: it buys time and keeps the going-concern value from collapsing. The question is whether the incremental value preserved exceeds the cost. From the prepetition lender's perspective, priming means being primed ahead by the new lender; the lender will accept if the DIP preserves enough value that its recovery rises, or else will offer to provide the DIP itself.

Example 19.7 – DIP economics for Meridian. Meridian needs 100 million of new money for the first nine months of the case: the 13-week cash forecast shows a cumulative funding gap of about 80 million with a 20 million cushion. A DIP lender offers a 100 million facility, fully drawn for simplicity, with a 3 percent upfront fee and an interest rate of 12 percent per year, with a nine-month expected case.

Cost of DIP:

```
Fee      = 3% × 100 = 3.0
Interest = 12% × 100 × (9/12) = 9.0
Total cost = 12.0 million
```

Benefit. Without DIP financing, Meridian must liquidate at about 330. The prepetition secured lenders' recovery would be 330 / 400 = 82.5 percent, with no DIP outstanding. (The 57.5 percent figure in Example 19.5 applies when the DIP is repaid first, as happens if the case is financed and then fails.) With DIP financing and a successful reorganization at 620, the term loan is paid in full (100 percent) after repaying the DIP's 100, and unsecured creditors recover 30 percent rather than zero.

Value comparison to the estate: going-concern value 620 versus liquidation 330 = 290 million of value preserved, against a financing cost of 12 million plus professional fees. Even if the probability that the case succeeds is only 60 percent and a failed case ends in liquidation at 330 (the DIP being repaid out of these totals), the expected value to the estate is:

```
E[value] = 0.60 × 620 + 0.40 × 330 = 372 + 132 = 504
Less DIP cost (certain): 12.0
Net = 492.0 versus liquidation immediately at 330
```

The DIP raises expected distributable value by 162 million (before professional fees). That is a large positive NPV even with a double-digit financing cost. Note that for the term lender the key issue is priming: with the DIP ahead of it, the term lender recovers 100 percent in the success state (620 − 100 = 520 ≥ 400) but only (330 − 100) / 400 = 57.5 percent in the failure state; its expected recovery is 0.6 × 100% + 0.4 × 57.5% = 83.0 percent, versus 82.5 percent if the firm liquidated at once without DIP. The DIP is marginally better for it (83.0 versus 82.5), which is why such lenders typically consent, and why, if the success probability were any lower, they would object. The break-even probability p* solves p × 400 + (1 − p) × 230 = 330, so p* = 100 / 170 = 58.8 percent; below this the secured lender would prefer immediate liquidation, and a priming fight would result.

A priming DIP is consensual when the probability-weighted improvement in the collateral's value exceeds the amount primed. Without sufficient cushion, priming is contested, and the debtor's best alternative may be a negotiated roll-up in which prepetition lenders provide the DIP themselves.

19.6.3 Related features

Critical vendor payments, administrative priority for goods delivered shortly before filing, and court-approved use of cash collateral are other places where the Code adjusts priority to protect going-concern value; in practice they are settled in negotiated first-day orders.

19.7 Negotiating and Confirming a Plan

19.7.1 The shape of a plan

A plan has four parts: the **classification** of claims; the **treatment** of each class (cash, new debt, new equity, warrants, litigation trust interests); the **new capital structure** and governance of the emerged firm; and the **releases and injunctions** that protect parties. The plan must be feasible: the emerged firm's leverage must be sustainable. Exit financing (term loan or notes issued at emergence) often repays the DIP in cash and provides distributions to secured claims.

19.7.2 Worked plan: who gets the equity?

Example 19.8 – Plan construction for Meridian. Plan value is 620 million. The exit financing is a new 250 million term loan (about 3.1 times normalized EBITDA of 80, a leverage level the firm can service with interest around 9 to 10 percent and an interest cover of roughly 3.3 times). Proceeds of the new loan repay the 100 million DIP in cash and provide 150 million cash to the existing secured lenders. The remaining equity value of the reorganized firm is:

```
Equity value = Plan value − Exit debt = 620 − 250 = 370
```

Claims by priority:

- DIP: 100 (cash from exit facility): 100%.
- Secured term loan 400: receives 150 cash plus new equity worth 250 to reach par. Equity share to term lenders = 250 / 370 = 67.6%.
- General unsecured (400): the residual equity, 370 − 250 = 120, which is 120 / 370 = 32.4% of the new equity. Recovery = 120 / 400 = 30.0%.
- Subordinated notes and old equity: nothing (value runs out at the unsecured class).

Check: total to prepetition creditors = DIP 100 + term lenders 400 + unsecured 120 = 620, equals plan value. The plan complies with the absolute priority rule: no junior class receives property while a senior dissenting class is unpaid; the unsecured class is the fulcrum.

Negotiation. The subordinated holders (150 face) threaten to contest valuation and delay; old equity calls for a higher WACC to be rejected. Suppose the term lenders, who bear the cost of delay, offer to "gift" 5 percent of the new equity to the subordinated class to secure acceptance. Cost of gift = 0.05 × 370 = 18.5 million. If the gift comes from the term lenders' share, their recovery falls from 100 percent to (400 − 18.5) / 400 = 95.4 percent. If it comes from the unsecured class, which is the fulcrum, its recovery falls from 30.0 to (120 − 18.5) / 400 = 25.4 percent. Which class bears the gift is therefore a central question: a gift by the *senior* class to a junior class past an intervening unsecured class would raise a **"gifting"** concern, since the unsecured class would see value flow around it; the gift from the unsecured class to the subordinated class is more standard, being permitted by the intervening class itself. Courts have scrutinized gifting in contested cases.

Every deviation from the waterfall is therefore a transfer measurable in dollars, to be compared with the expected cost of fighting: if litigation and delay would cost the estate 3 percent of value (18.6 million at 620), a gift of equal size is roughly break-even for the class that pays.

19.7.3 Dynamics of the negotiation

- **Valuation is the battleground.** As Example 19.5 showed, one point of WACC shifts value by 8 to 15 percent. Classes fully covered at any plausible value are indifferent; the fulcrum class gains from a higher valuation.
- **The waterfall is the bargaining shadow.** Senior classes accept less than par only for speed; junior classes accept little if the alternative is worse; old equity receives a tip when creditors prefer to avoid litigation.
- **Exclusivity and competing plans.** Debtors use exclusivity as a lever; the ability of creditors to propose plans or sponsor a sale reduces it.
- **Sales as an alternative.** A Section 363 sale, often to a stalking horse or a creditor group *credit-bidding* its secured debt, provides market testing of value.
- **New value plans.** Old equity may try to retain ownership by contributing capital; case law has required that such offers be exposed to competing bids.
- **Management incentives.** Retention and post-emergence equity grants are negotiated, with statutory limits on retention payments to insiders.

19.7.4 After emergence

Many emerged firms are overleveraged again or fail to fix operations, and return as a "Chapter 22." The CFO should align the exit capital structure with post-plan business risk: a liquidity cushion, covenants that normal volatility will not trip, and a credible operating plan.

19.8 Distressed Investing

19.8.1 Strategies

Distressed investors pursue several strategies: **passive trading** (buy where market-implied recovery is below one's own estimate); **fulcrum or "loan-to-own"** (buy the class that will receive the equity, often with a blocking position of more than one-third of a class's amount, and negotiate control); **capital structure arbitrage** (long one layer, short another when prices are inconsistent with the Merton relationships of Section 19.5); **rescue financing** (DIP or exit loans earning fees, spreads and warrants); and **claims trading**. Holders who receive non-public information through creditor committees may be restricted from trading, and purchased claims can be challenged if they carry avoidance liabilities, so the strategy requires specialized legal capability.

19.8.2 Valuing a distressed claim

The method is scenario analysis. For each outcome (consensual restructuring, prepack, contested case, sale, liquidation) estimate the plan value, run it through the waterfall, convert the recovery into cash and securities at expected value, assign a timing, and discount at a required return. A key discipline: the *expected* recovery depends on the plan value distribution, not a single point estimate, and since recoveries are piecewise-linear in value (Example 19.3), the expected recovery is convex in uncertainty only within a class's range.

Example 19.9 – Pricing Meridian's senior notes for a distressed investor. The senior notes (300 face) trade at 40 per 100 face. An investor builds three scenarios over an expected 18-month path to resolution (1.5 years). Recoveries, valued at the time of resolution including the market value of the new equity received:

Scenario	Probability	Recovery per 100 face
Consensual prepack, strong turnaround	25%	70
Base plan, as in Example 19.8 with a somewhat larger unsecured pool	45%	45
Contested case, deteriorating sales	30%	15

(The recoveries are analyst estimates per 100 face, reflecting judgment about the senior notes' treatment; they are not computed mechanically from the waterfall of Example 19.3, which treats all unsecured claims alike. A real analyst would document this choice.)

Expected recovery:

```
E[R] = 0.25 × 70 + 0.45 × 45 + 0.30 × 15 = 17.5 + 20.25 + 4.5 = 42.25
```

Annualized returns for each scenario from an entry price of 40 over 1.5 years: (R / 40)^(1/1.5) − 1.

```
Strong:   (70/40)^(0.6667) − 1 = 1.75^0.6667 − 1 = 45.2%
Base:     (45/40)^(0.6667) − 1 = 8.2%
Contested:(15/40)^(0.6667) − 1 = −48.0%
Expected: (42.25/40)^(0.6667) − 1 = 3.7%
```

The expected annualized return of 3.7 percent is far below a distressed investor's hurdle of, say, 18 percent. The *maximum price* for an 18 percent return is:

```
P* = 42.25 / 1.18^1.5 = 42.25 / 1.2818 = 32.96
```

So the investor should buy only at about 33 or below, or must have a reason to believe the recovery distribution is better than the market assumes (for example, a blocking position that influences the plan). The break-even recovery at a price of 40 is 40 × 1.2818 = 51.3, above every scenario but the strong one. The return distribution is skewed: the base case earns a modest 8 percent while the contested case loses almost half.

19.8.3 Common mistakes in distressed analysis

Ignoring a claim's rank within the legal-entity structure; valuing illiquid new equity at face; underestimating time (an extra year at a 15 percent hurdle costs about 13 percent of value); assuming the waterfall is mechanical when gifts, new-value plans and management incentive plans can redirect value; and failing to model fraudulent transfer, lien perfection and intercreditor litigation.

19.9 Strategy for the CFO: Choosing the Process

19.9.1 A decision framework

The choice among paths depends on business viability (going-concern versus liquidation value), the dispersion of the capital structure, liquidity runway, availability of DIP financing, contractual constraints such as cross-defaults and change-of-control clauses, and stakeholder positions. In broad terms: amend-and-extend suits mild stress with few lenders; an exchange offer suits a viable business with dispersed bonds; a prepackaged Chapter 11 suits agreed terms with holdout risk; a free-fall Chapter 11 suits complex disputes needing court tools such as lease rejection; a Section 363 sale suits time-sensitive assets with a strong buyer; and liquidation suits economic distress.

19.9.2 Governance and people

As insolvency nears, creditors' interests attract greater attention under the duties of directors in many jurisdictions, and directors may face liability for wrongful trading or preferences. Boards add independent restructuring expertise, document their process and obtain solvency opinions. Retention plans, clear communication to customers, and a reliable cash forecast are the CFO's levers; forecast accuracy, not optimism, earns flexibility.

19.10 Cross-Jurisdiction Differences (Stated Cautiously)

Insolvency laws vary and are frequently reformed. The contrasts below are general and must be checked against current law and local counsel.

- **United States.** Debtor-in-possession model, automatic stay, class voting, cramdown with the absolute priority rule, strong DIP and priming tools, Section 363 sales.
- **United Kingdom.** Administration (a collective process led by an insolvency practitioner, often used for business sales) and **schemes of arrangement**, which bind creditors on approval by a majority in number and 75 percent in value of each voting class. A **restructuring plan** procedure introduced in 2020 permits cross-class cramdown if dissenting classes are no worse off than in the relevant alternative and at least one in-the-money class approves, with court discretion over how benefits are shared; priority is thus applied by a different test from the US rule.
- **Continental Europe.** Regimes vary by country. Influenced by a European Union directive on preventive restructuring, many states now allow plans to bind dissenters through class voting and cross-class cramdown subject to a best-interests test and some priority rule (absolute in some states, relative in others).
- **Asia and elsewhere.** Systems range from court-supervised rehabilitation to liquidation-oriented regimes; the reliability of security enforcement and the speed of courts affect recoveries, and several jurisdictions have recently modernized.
- **Cross-border cases.** The UNCITRAL Model Law on Cross-Border Insolvency, adopted in many jurisdictions (in the US as Chapter 15), enables recognition of foreign proceedings, but group-wide solutions are treated differently and the debtor's center of main interests matters.

Research associated with La Porta, Lopez-de-Silanes, Shleifer and Vishny links legal origin to creditor protection and debt markets, though the evidence is nuanced and should not be applied mechanically. For a multinational CFO the practical point is to align governing law and the location of assets with the likely restructuring venue.

Case Study: Harlow Marine Logistics

(Illustrative scenario. Harlow Marine Logistics is a fictional company.)

Harlow Marine Logistics operates a fleet of 22 container feeder vessels and two port terminals serving regional trade routes. In the last four years the company borrowed heavily to expand: 520 million in a senior secured credit facility (secured by first mortgages on the vessels and a pledge of the terminal operating companies), 260 million of senior unsecured notes issued by the parent, and 90 million of convertible subordinated notes. Trade payables and accrued claims total about 70 million. A lease on one terminal expires in 18 months, and a customer concentration problem emerged when a major shipping line, accounting for 28 percent of revenue, announced it would move its feeder services in-house at the end of next year.

Charterhire rates, which had been strong, have fallen by about a third, and Harlow's EBITDA is now 95 million against 140 million two years ago. The company has 55 million in cash, a 70 million interest payment due in three months, and a covenant on leverage that will be breached at the next test date. The firm's CFO, Mateo Brandt, reports that independent appraisals value the vessels at 610 million on an orderly sale basis and 430 million in a forced sale; terminal assets are worth about 180 million as going-concern businesses but would fetch only 110 million in a break-up.

A bondholder committee holding 55 percent of the senior unsecured notes has proposed an exchange offer: holders would receive new notes with 65 percent of the face value, secured by a second lien on the terminals, plus 10 percent of the equity. The secured lenders object, arguing that the pledge of the terminal operating companies was given to them and that their lien ranks first. The convertible noteholders have threatened to sue, claiming that the exchange would amount to an improper transfer of value. Employees' representatives have asked for assurances on severance if vessels are sold.

Mateo estimates that a free-fall Chapter 11 would take about 14 months, with professional fees of roughly 45 million, and that a DIP of 80 million would be needed. Harlow's parent is a holding company incorporated in one country while the vessels are flagged elsewhere and the terminals operate in two further jurisdictions; the loan documents are governed by New York law, while the notes' indenture is governed by English law.

Mateo must advise the board whether to proceed with the exchange, prepare a prepackaged filing, or seek a sale of the terminals and a negotiated runoff of the fleet. He must also decide how to value the business: the senior lenders' advisers use 8 times 2026 forecast EBITDA with a distress discount, while the bondholders argue for a higher multiple based on the new cost structure and the fleet's appraisals.

Discussion Questions

1. Construct a preliminary waterfall for going-concern value of 700 million and for a forced-sale value of 540 million (430 vessels plus 110 terminals). Ignore the DIP and fees first, then include a DIP of 80 million ranking first and fees of 45 million. Identify the fulcrum security in each case, and explain why the bondholders' proposal to receive a second lien on terminal assets is contentious.
2. Evaluate the exchange offer from the perspective of (a) a senior unsecured bondholder, (b) a secured lender, and (c) a convertible noteholder. What holdout problems are likely, and what contractual provisions would you check (including governing law of the indenture)?
3. How would you test the claim that the business is worth more as a going concern than in liquidation, and which variables (charter rates, customer loss, terminal lease renewal) most affect the valuation? Construct two sensitivity cases for the WACC and the discount rate for the terminal value.
4. Discuss the cross-border dimension: how might the location of assets, the governing law, and the jurisdiction of incorporation affect the choice between an out-of-court exchange, a US Chapter 11 and a local procedure? What additional information would you want from counsel?

Key Takeaways

- Financial distress (a viable firm with the wrong balance sheet) and economic distress (a firm worth more dead than alive) call for different responses; comparing going-concern and liquidation values with total claims is the first diagnostic.
- Indirect costs of distress (lost customers, supplier tightening, underinvestment, risk shifting) typically exceed direct professional costs; expected distress cost equals probability times loss and rises convexly with leverage.

- The absolute priority rule sets the baseline order of payment (secured, administrative, unsecured, subordinated, equity) and is the statutory test for cramdown; actual negotiated outcomes often deviate, but always in its shadow.
- The fulcrum security is the class in which value runs out; it receives the equity of the reorganized firm and is the true economic owner during the case.
- Reorganization value is the pivotal number: because terminal value dominates, a one-point change in WACC can move enterprise value by 8 to 15 percent and shift the fulcrum class's recovery by tens of percentage points.
- Equity is a call option on the firm's assets (Merton); it retains value even when assets are below debt, which explains the nuisance value that old equity commands and its preference for risk.
- DIP financing is a high-priority, short-duration loan that buys going-concern value; priming is consensual when the probability-weighted improvement in collateral value exceeds the amount primed.
- Out-of-court exchanges suffer from holdout incentives; exit consents, seniority enhancement, participation conditions and a prepackaged-filing backstop restructure incentives so that tendering dominates.
- Distressed investing is scenario analysis: compute expected recovery, time to resolution, and the maximum price for the required return; most distressed trades are unattractive at observed prices unless the investor has an informational or control advantage.
- Insolvency regimes differ in process, priority tests, and creditor protection; a lender's expected recovery depends on the efficiency of the regime as well as on contractual rank, and cross-border groups must plan the venue early.

Review Questions

1. Distinguish economic distress from financial distress. How would a CFO test which one applies, and why does the distinction matter for the choice of process?
2. List the direct and indirect costs of financial distress. Which are most important for a consumer durables retailer and which for a software company with long-term customer contracts?
3. Explain debt overhang and risk shifting. How does each distort investment decisions of a distressed firm's management, and how do creditors respond?
4. State the absolute priority rule. In what sense is it a rule for cramdown rather than a requirement for every plan, and why does it matter even for consensual plans?
5. What is a fulcrum security? Using the waterfall in Example 19.3, identify the fulcrum at distributable values of 450, 620 and 900 million.
6. Why does a deficiency claim arise, and how does it change the recoveries of the unsecured class?
7. Why do bond exchange offers face holdout problems more than bank restructurings? Describe two contractual devices that mitigate them.
8. Explain, using the Merton model, why equity can have positive value when firm assets are less than the face value of debt, and why equity prefers higher volatility.
9. What protections does the Bankruptcy Code give a DIP lender? Under what conditions would prepetition secured lenders consent to being primed?
10. Describe three ways in which insolvency regimes differ across jurisdictions that matter to a lender's expected recovery, and explain why such statements should be treated cautiously.

Problems

1. **Distress trade-off.** A firm with unlevered value 800 million, tax rate 30 percent (permanent debt), and distress cost 30 percent of firm value has the following probabilities of distress: debt 100 → 1%; 200 → 3%; 300 → 8%; 400 → 18%. Compute the net benefit of debt at each level and identify the optimal level among these.
2. **Z-score.** A manufacturer has X1 = 0.10, X2 = 0.05, X3 = 0.06, X4 = 0.35, X5 = 0.90. Compute Altman's Z using the coefficients in Section 19.1.3 and classify the result using the 1.8 and 3.0 cut-offs.
3. **Waterfall.** A firm has a DIP of 60, a first-lien loan of 250, a second-lien loan of 100 (secured only on the same collateral, ranking behind the first lien), general unsecured claims of 300 and equity. Compute recoveries (amount and percent) at distributable values of 280, 400 and 520, and identify the fulcrum class in each case.
4. **Reorganization value.** Free cash flows are 20, 25, 30, 33 and 35 for years 1 to 5 with terminal growth 2.5 percent and WACC 12 percent. Compute enterprise value, the terminal value share, and the EV at WACC 10 percent.
5. **Merton equity.** Assets V = 500, debt face D = 600 due in 2 years, r = 3%, σ = 40%. Compute d1, d2, equity value (call) and debt value. (Use N(x) values from a standard normal table or software.)
6. **DIP economics.** A DIP of 50 carries a 2.5 percent fee and 11 percent interest for 12 months. Liquidation yields 180; going concern value 300 with probability 70 percent of success (else liquidation at 180). Prepetition secured debt is 220. Compute the DIP cost, the expected distributable value with and without the DIP (ignoring the DIP cost in the failure branch), and the prepetition lender's expected recovery in each case, assuming that in the failure branch the DIP of 50 is repaid first.
7. **Exchange offer.** Old notes: 5-year 6 percent annual coupon, par 100. Exchange: 60 face of new 5-year 6 percent secured notes per 100 old, new notes discounted at 9 percent. A holdout's value if the exchange succeeds: yield of 22 percent. Compute the value of tendering, the value of holding out, and the premium for tendering.
8. **Distressed claim.** A bond trades at 35. Scenarios over 2 years: 20% probability recovery 80; 50% probability recovery 40; 30% probability recovery 5. Compute expected recovery, expected annualized return from price 35, and the maximum price for a 20 percent annual return.

Answers to Selected Problems

1. Net benefit = 0.30·D − p × 0.30 × 800 (cost = 240p). D = 100: 30 − 2.4 = 27.6; D = 200: 60 − 7.2 = 52.8; D = 300: 90 − 19.2 = 70.8; D = 400: 120 − 43.2 = 76.8. The highest net benefit among these grid points is at D = 400.
2. Z = 1.2(0.10) + 1.4(0.05) + 3.3(0.06) + 0.6(0.35) + 1.0(0.90) = 0.120 + 0.070 + 0.198 + 0.210 + 0.900 = 1.498, below 1.8: high distress risk.
3. Value 280: DIP 60 (100%), first lien 220 of 250 (88.0%), nothing below; fulcrum is the first lien. Value 400: DIP 60, first lien 250 (100%), second lien 90 of 100 (90%), unsecured 0; fulcrum is the second lien. Value 520: 60 + 250 + 100 = 410, leaving 110 for unsecured of 300: 36.7%; fulcrum is general unsecured.
4. At 12%: PVs 17.857, 19.930, 21.353, 20.972, 19.860 sum to 99.97; TV = 35 × 1.025 / 0.095 = 377.63; PV(TV) = 214.28; EV = 314.2; terminal share 68.2%. At 10%: PVs sum to 105.65; TV = 35.875 / 0.075 = 478.33; PV = 297.0; EV = 402.7.
5. σ√T = 0.5657; d1 = [ln(500/600) + (0.03 + 0.08) × 2] / 0.5657 = (−0.18232 + 0.22) / 0.5657 = 0.0666; d2 = −0.4991. N(d1) ≈ 0.5266, N(d2) ≈ 0.3088. PV(D) = 600 × e^(−0.06) = 565.06. E = 500 × 0.5266 − 565.06 × 0.3088 = 263.3 − 174.5 = 88.8 (approximately). Debt = 500 − 88.8 = 411.2.

6. DIP cost = 50 × 0.025 + 50 × 0.11 × 1 = 1.25 + 5.50 = 6.75. Without DIP: liquidation 180. With DIP: 0.70 × 300 + 0.30 × 180 = 210 + 54 = 264 (before the DIP cost; net of 6.75 is 257.25). Prepetition lender with DIP: success: 300 − 50 = 250 ≥ 220, so 100%; failure: 180 − 50 = 130, recovery 130 / 220 = 59.1%. Expected = 0.7 × 100 + 0.3 × 59.1 = 87.7%. Without DIP: 180 / 220 = 81.8%.
7. Package: 0.60 × [6 × 3.8897 + 100 × 0.64993] = 0.60 × (23.34 + 64.99) = 0.60 × 88.33 = 53.0. Holdout if the exchange succeeds: 6 × 2.8636 + 100 × 0.37000 = 17.18 + 37.00 = 54.18. Holding out beats tendering by about 1.2 per 100, so this offer is too weak to dominate; the issuer would need a larger new face amount or an equity sweetener.
8. E[R] = 0.2 × 80 + 0.5 × 40 + 0.3 × 5 = 16 + 20 + 1.5 = 37.5. Annualized return = (37.5 / 35)^(1/2) − 1 = 1.0714^0.5 − 1 = 3.5%. Maximum price = 37.5 / 1.2² = 37.5 / 1.44 = 26.04.

Chapter 20: Divestitures, Spin-offs, SPACs and Dual-Class Structures

Learning Objectives

After studying this chapter you should be able to:

- Compare the main forms of corporate separation (asset sale, spin-off, split-off, equity carve-out, tracking stock) and choose among them using value, tax, control and financing criteria.
- Quantify the value created or destroyed by a separation using a sum-of-the-parts analysis net of dis-synergies and one-time costs, and allocate debt and tax basis in a spin-off.
- Explain, in general terms, the logic of tax-free spin-offs, the conditions typically attached to them, and the restrictions on subsequent transactions.
- Analyze SPAC economics: the sponsor promote, warrants, redemption rights, and dilution, and compute net cash and value per share under different redemption scenarios.
- Evaluate dual-class share structures, measuring the wedge between voting and cash-flow rights and summarizing the governance arguments for and against them, including sunset provisions.
- Compare IPOs, direct listings, SPAC mergers and sales as routes to public markets, including the full cost of each.

Opening Vignette

(Illustrative scenario. Calder Industries and its subsidiaries are fictional.)

Calder Industries is a diversified manufacturer with two divisions that share a corporate headquarters, a name and a balance sheet. Its Industrial division makes pumps and valves, a mature, cyclical business with EBITDA of 300 million. Its Consumer division makes premium kitchen appliances and has EBITDA of 150 million, growing faster, with higher margins and heavy marketing needs. Calder's consolidated enterprise value is 3.9 billion, about 8.7 times EBITDA, and net debt is 1.5 billion.

For three years the CEO, Priya Raman, has heard the same message from investors: the Consumer division deserves a multiple like that of branded-goods peers, and the Industrial division a multiple like that of industrial peers, yet the market blends them into a discount. An activist fund has bought 4 percent of the shares and published a letter proposing a separation. Raman's board has three alternatives: sell Consumer to a strategic buyer who has indicated an interest at a price that would trigger a large taxable gain; spin Consumer off to shareholders as a separate listed company; or sell a minority stake in Consumer in an IPO and keep control.

Meanwhile a founder-led company in a different industry, one Calder's venture arm has invested in, is deciding how to go public. Its founder wants to retain control and has been courted by a SPAC sponsor promising speed and certainty, while its bankers argue for a traditional IPO, and a third adviser suggests a direct listing. The founder wants a share structure in which her shares carry ten votes each.

Both situations raise the questions of this chapter. What does a separation do to value, and how much of the apparent gain is real? What are the tax logic and the economics of each route? What does a SPAC actually cost the company relative to an IPO? When does a dual-class structure create value and when does it entrench? We will use Calder throughout the first half

of the chapter and the founder's company in the second half.

20.1 The Map of Corporate Separations

20.1.1 Why firms separate

Volume 1, Chapter 22 treated acquisitions as a way to create value through synergy. Separations are the mirror image: they are justified when a business is worth more to someone else, or on its own, than inside the parent. Standard motivations are:

- **Focus.** Attention, capital allocation and incentive pay can be tailored to one business. Research by Lang and Stulz and by Berger and Ofek found that diversified firms tended on average to trade at a discount to the sum of their parts, though later work debates how much is measurement artifact or selection.
- **Investor clienteles and transparency.** Pure plays are easier to value, cover and compare.
- **Capital allocation.** Internal capital markets may cross-subsidize weak divisions (Scharfstein and Stein); separation makes each business compete for capital at its own cost.
- **Incentives and currency.** A listed business can grant equity pay tied to its own performance and use its shares for acquisitions.
- **Fit and regulation.** The business may drag on the parent's rating, face antitrust limits or need a different capital structure.
- **Monetization.** The parent may need cash to repay debt or fund a better opportunity.

Against these stand lost synergies, **dis-synergies** from duplicated functions, one-time separation costs (IT, legal entities, stranded costs), loss of cash flow diversification (which may raise each part's cost of debt) and, in a taxable transaction, tax.

20.1.2 The menu

Form	What happens	Cash to parent?	Control after	Typical tax outcome (general)
Asset or subsidiary sale (divestiture)	Sold to a strategic or financial buyer for cash or stock	Yes	Buyer	Taxable to the seller on gain over tax basis
Spin-off	Subsidiary shares distributed pro rata to parent shareholders	No (except via debt raised by subsidiary)	Separate public company, same shareholders	Can be tax-free if conditions are met
Split-off	Parent shareholders offered subsidiary shares in exchange for parent shares	No	Subset of holders	Can be tax-free if conditions are met
Equity carve-out	Minority stake in subsidiary sold to public investors in an IPO	Yes (subsidiary or parent receives proceeds)	Parent keeps control	Often taxable only to the extent of gain on the shares sold
Tracking stock	Parent issues a class of shares whose dividends track a division	No	Parent retains legal ownership	Typically no tax event on issuance
Sale via reverse merger (e.g., Reverse Morris Trust)	Spin-off followed by merger of the spun entity with a third party	Possibly (via debt)	Third party plus old holders	Can be tax-free if ownership tests are met

The tax column is a generalization. Tax-free treatment depends on detailed statutory conditions that vary by country and change over time; readers must consult tax counsel before relying on any of it.

20.1.3 Evidence in brief

Studies of spin-offs (Schipper and Smith, Cusatis, Miles and Woolridge, and others) and of asset sales generally report positive average announcement returns, and some find positive subsequent performance where separation increases focus. These are averages; individual transactions vary widely, and some of the post-spin gains are associated with takeovers of either company. A separation is not valuable merely because it is a separation, and the rest of this chapter shows how to test a specific one.

20.2 Divestitures: Selling a Business

20.2.1 The hold-sell decision

A divestiture is a capital budgeting decision in reverse. The parent should sell if the after-tax proceeds exceed the value of retaining the business, where value of retention is the present value of the after-tax cash flows the business will generate *to this owner*, including synergies. Since the buyer's value includes their own synergies, a sale happens when the buyer can generate more than the seller (Volume 1, Chapter 22). The parent's analysis should therefore compare:

```
Sell if:  Net after-tax proceeds  >  PV of retained cash flows (to this owner)
Net after-tax proceeds = Price − Tax on gain − Transaction costs − Stranded costs (PV)
Tax on gain = t × (Price − Tax basis)
```

Two points deserve emphasis. First, **tax basis** matters. A business bought long ago, or built organically, usually has a low tax basis, so a sale triggers tax on nearly the whole price. The same business separated in a qualifying tax-free spin-off, by contrast, may trigger no corporate-level tax. That is one reason spin-offs are popular. Second, **stranded costs**, the shared overhead formerly absorbed by the divested business, remain with the parent unless eliminated. Ignoring them overstates the benefit.

20.2.2 Consideration and structure

Cash gives certainty but crystallizes tax; stock can sometimes defer tax but exposes the seller to the buyer's performance. Common contract elements include earn-outs (Volume 1, Chapter 22), transition services agreements, indemnities and escrows, and non-compete clauses. In a sale to a private equity buyer (Volume 1, Chapter 23) financing is largely debt, so the price depends on credit conditions.

20.2.3 Worked comparison: sell or spin

Example 20.1 – Sale versus tax-free spin-off of Calder's Consumer division. A strategic buyer offers 1,800 million in cash for the Consumer division. Consumer's tax basis is 600 million, the corporate tax rate on the gain is 25 percent, and transaction costs are ignored. The alternative is a spin-off that would be tax-free to Calder and to its shareholders if conditions are met, in which case Consumer would be worth 1,704 million as a stand-alone company (derived in Example 20.2).

Sale:

```
Gain            = 1,800 − 600 = 1,200
Tax on gain     = 25% × 1,200 = 300
After-tax cash  = 1,800 − 300 = 1,500
```

Spin-off: shareholders receive Consumer's stand-alone value, 1,704, with no corporate tax and no current shareholder tax in the stylized case, but Calder receives no cash.

At first sight the spin-off (1,704) beats the sale (1,500) by 204. The two are comparable on an enterprise-value basis; in practice the parent would also have Consumer borrow and pay a dividend to Calder before the spin, so that Calder can reduce its own debt, which shifts value between the two equity stubs without changing the total.

The taxable-sale alternative would be preferable only if the buyer's price exceeds the spin value by at least the tax: to match 1,704 after tax, the sale price P must solve $P - 0.25(P - 600) = 1{,}704$, so $0.75P + 150 = 1{,}704$, $P = 2{,}072$. The buyer would have to pay about 2,072 million, 15 percent above its 1,800 offer, to leave Calder's shareholders as well off as the spin. Equivalently, the tax cost of 300 is a hurdle the sale price must clear. If the buyer's synergies are large enough to justify 2,072, the sale is better; if not, the spin is. Note that this ignores the possibility that a spin's value would be lower or higher than 1,704 in the market, which is uncertain, whereas a cash price is certain: certainty has value, as does a competitive auction.

20.3 Spin-offs

20.3.1 Mechanics

In a spin-off, the parent distributes shares of a subsidiary pro rata to its own shareholders. Each parent shareholder ends up with shares of both companies in proportion to the original holding, and no cash changes hands. The steps are: the parent establishes the subsidiary as a separate legal entity (the "separation"), contributes assets, negotiates a separation agreement (assets, employees, contracts, intellectual property, and the allocation of liabilities), allocates debt, arranges a tax ruling or opinion where appropriate, files a registration or information statement with the securities regulator, obtains stock exchange listing, declares a distribution with a record date, and settles the distribution. A "when-issued" market typically emerges before the distribution date, allowing price discovery.

Several decisions deserve care:

- **Capital structure.** Parents often have the spun entity borrow and pay a *cash dividend to the parent* just before the spin, reducing the parent's leverage while giving the new firm a stand-alone structure; each entity's leverage should reflect its own business risk (Volume 1, Chapter 15).
- **Governance and commercial arrangements.** Each firm needs its own board, management and incentive plans; transition services, supply and intellectual property agreements govern the relationship afterward, and a poorly designed agreement can destroy value through conflict.
- **Pensions.** Pension liabilities must be allocated with an actuarial view.
- **Flowback.** The new company may be too small, or in the wrong sector, for funds that hold the parent, and their selling often creates temporary price pressure, a source of the short-term underperformance that spin-off specialists exploit.

20.3.2 Tax-free logic, described generally

In many tax systems, a distribution of shares of a subsidiary can be tax-free to both the distributing corporation and its shareholders if it meets conditions designed to ensure that the transaction is a genuine reorganization and not a disguised cash distribution or a way to sell a business without tax. In the United States, the relevant provisions are in Section 355 of the Internal Revenue Code. Broadly, and without attempting to state every requirement, the typical conditions include:

- the parent controls the subsidiary and distributes all or substantially all of its stock;
- both businesses are engaged in an active trade or business that has been conducted for a minimum period;
- the transaction has a genuine corporate business purpose other than tax avoidance, and is not a “device” for distributing earnings;
- the historic shareholders maintain a continuing interest in both firms; and
- there are limits on acquisitions of either company, or of sizeable stakes, shortly after the distribution, since a spin followed by a taxable sale would otherwise let the parent sell the subsidiary without paying tax (the so-called Morris Trust issue, and the reason for the Reverse Morris Trust ownership thresholds).

Other countries have different rules, and some tax distributions of shares as dividends in kind unless relief applies. The managerial points are these. First, qualification is valuable (in Example 20.1, it avoids 300 million of tax), so the company will plan the transaction to satisfy the conditions and will often obtain an external tax opinion or a ruling. Second, qualification restricts what the parent and the spun-off company can do for some period afterward, notably a sale to a third party, and these restrictions are normally enforced through a tax matters agreement that allocates liability if a later action causes the spin to be taxable. Third, for the *shareholder* the typical consequence of a qualifying spin is deferral: no tax at distribution, with the original tax basis allocated between the two holdings, usually in proportion to their relative market values immediately after the spin, and the holding period carried over.

20.3.3 Valuing the separation: sum of the parts

The central analysis is a sum-of-the-parts (SOTP) valuation, in which each division is valued separately using appropriate multiples or DCF (Volume 1, Chapter 21), the values are added, and the total is compared with the current consolidated value. The *gross* gain is the difference. The *net* gain subtracts dis-synergies and one-time costs. The analysis is only as good as its inputs: the multiples applied to each division must be those of truly comparable stand-alone peers, and the dis-synergies must be realistic.

Example 20.2 – Calder’s sum of the parts. Calder’s consolidated enterprise value is 3,900 million, net debt 1,500 million and shares 100 million. Industrial EBITDA is 300 and Consumer EBITDA is 150, so the blended multiple is 3,900 / 450 = 8.67 times. Pure-play peers trade at 9 times EBITDA for industrial businesses and 12 times for branded consumer businesses. Separation would create stand-alone dis-synergies of 20 million of annual EBITDA (12 in Industrial, 8 in Consumer) and one-time costs of 90 million.

Step 1 – Gross SOTP:

```
Industrial: 300 × 9  = 2,700
Consumer:   150 × 12 = 1,800
SOTP                 = 4,500
Conglomerate discount = 4,500 − 3,900 = 600  (13.3% of 4,500)
```

Step 2 – Capitalize the dis-synergies at each division’s multiple:

```
Industrial: 12 × 9  = 108
Consumer:    8 × 12 =  96
Total               = 204
```

Step 3 – Net enterprise value after separation:

```
Post-separation EV = 4,500 − 204 = 4,296
Less one-time costs = 4,296 − 90 = 4,206
Net gain vs. current 3,900 = 306  (7.8% of 3,900)
```

Step 4 – Equity value per share. Net debt of 1,500 is allocated as 1,000 to Industrial and 500 to Consumer (3.5 times each division's post-dis-synergy EBITDA: 1,000 / 288 = 3.47 and 500 / 142 = 3.52). Equity values are:

```
Industrial equity: (2,700 − 108) − 1,000 = 1,592  → 15.92 per share
Consumer equity:   (1,800 −  96) −   500 = 1,204  → 12.04 per share
Combined                                  = 2,796 → 27.96 per share
Less one-time costs (90 / 100 shares)     = 0.90
Net value per original share              = 27.06
```

Current equity value is 3,900 − 1,500 = 2,400, or 24.00 per share. The spin therefore adds 3.06 per share, a 12.75 percent equity gain, from an enterprise value gain of 7.8 percent. This illustrates how leverage amplifies the equity effect: the same 306 million of enterprise value gain is spread over an equity base of 2,400.

Several cautions apply. The 13.3 percent conglomerate discount is the *gross* figure; 204 + 90 = 294 of it, or nearly half, is consumed by dis-synergies and costs, leaving a net gain of 306. A 1-turn lower multiple for Consumer (11 instead of 12) would reduce the Consumer value by 150, partly offset by 8 lower capitalized dis-synergies, cutting the net gain to about 164, so the conclusion is sensitive to the peer multiple. And, importantly, the *market* may already reflect the expected separation (activist pressure raises the share price), in which case the announcement return will be smaller. Finally, the figures above are values the market *could* assign; there is no guarantee. Separation changes the information environment and the investor base, but it does not by itself create cash flows.

20.3.4 Basis and per-share allocation

Example 20.3 – Allocating tax basis and checking leverage in the spin. Calder distributes one Consumer share for each Calder share held (a 1-for-1 distribution). Immediately after the spin the shares trade at 15.92 (Industrial, the continuing "RemainCo") and 12.04 (Consumer), per Example 20.2's values before the one-time cost, which we ignore for allocation. A shareholder who owned one Calder share with a tax basis of 18.00 allocates basis in proportion to market values:

```
Total post-spin value per original share = 15.92 + 12.04 = 27.96
Industrial share of value = 15.92 / 27.96 = 56.94%
Consumer share of value   = 12.04 / 27.96 = 43.06%
Basis allocated to Industrial = 18.00 × 0.5694 = 10.25
Basis allocated to Consumer   = 18.00 × 0.4306 =  7.75
```

If the shareholder later sells the Consumer share at 15.00, the gain is 15.00 − 7.75 = 7.25 (rather than the 15.00 − 18.00 = −3.00 loss that would result if she mistakenly assigned the whole basis to Consumer). The allocation matters for later tax. Leverage check: Industrial's net debt / post-dis-synergy EBITDA is 1,000 / 288 = 3.47 times and Consumer's 500 / 142 = 3.52 times, similar at the design stage, though Consumer's higher growth and need for marketing investment would argue for lower leverage, and Industrial's cyclicality argues for lower leverage too. A careful design would run each entity through a coverage stress test (Volume 1, Chapter 14) and might shift debt to the more stable cash generator. Credit ratings of both entities, and the cost of refinancing the parent's existing bonds (which may have change-of-control or asset-sale covenants), must be checked as well.

20.3.5 What goes wrong

Common failures: separating interdependent businesses so that dis-synergies exceed the multiple arbitrage; overloading the spun company with debt; weak governance at the spun company; stranded costs that leave RemainCo with overhead too large for its size; and misreading flowback selling as a verdict on the strategy. Creditors may also object if value is transferred away from a borrowing entity, which can raise fraudulent transfer concerns if the parent later becomes insolvent (Chapter 19).

20.4 Carve-Outs, Split-Offs, Tracking Stocks and Related Structures

20.4.1 Equity carve-outs

In an **equity carve-out**, the parent sells a minority stake (typically 10 to 30 percent) in a subsidiary to public investors, often through an IPO, and keeps control. The proceeds may go to the subsidiary (a primary offering, funding growth) or to the parent (a secondary offering, a monetization). Carve-outs create a market price for the subsidiary, which can reveal value and permit equity-linked compensation. They have two natural uses: as a *stage* before a full separation (the parent sells a stake now and distributes the rest later), and as a way to raise capital against a business the market values more highly than the parent. Empirical evidence, associated with Schipper and Smith and with Allen and McConnell, finds positive average announcement returns on average, particularly where proceeds are used to reduce debt.

The costs are the IPO costs discussed in Section 20.5, plus the *governance conflicts* of a controlled company: minority shareholders of the subsidiary depend on a parent that may favor its own interests in transfer pricing, shared services and capital allocation. The parent also gives up part of the cash flows of a business without receiving the full price of control.

Example 20.4 – Cost of carving out 20 percent of Consumer. Calder considers listing 20 percent of Consumer in an IPO instead of a spin. Consumer's equity is worth 1,204 million at fair value (Example 20.2). Investors require a 10 percent IPO discount to the fair price, and the gross spread is 5 percent of proceeds.

```
Fair value of 20% stake   = 0.20 × 1,204          = 240.8
Offer price discount 10%  → gross proceeds         = 216.7
Gross spread 5%           = 0.05 × 216.7           =  10.8
Net proceeds to Calder    = 216.7 − 10.8           = 205.9
Retained 80% (fair value) = 0.80 × 1,204           = 963.2
Value to Calder           = 205.9 + 963.2          = 1,169.1
Cost of the carve-out     = 1,204 − 1,169.1        =  34.9 (2.9% of Consumer equity)
```

The transaction costs 34.9 million, equal to 14.5 percent of the 240.8 million of fair value sold. The 24.1 million discount is a transfer to new investors that may or may not be recovered through re-rating of the retained stake, while the 10.8 million spread is permanent. Carve-outs are therefore most valuable when there is a specific reason to sell now: to fund a high-return investment, to retire costly debt, or to create the price discovery needed for a later spin.

20.4.2 Split-offs, tracking stocks and Reverse Morris Trusts

In a **split-off**, parent shareholders may *exchange* parent shares for subsidiary shares, typically at a premium. Because the parent's share count falls, it resembles a repurchase paid in subsidiary stock (Volume 1, Chapter 16), with similar tax logic to a spin-off, but each participant ends up with one company or the other.

A **tracking stock** is a class of parent equity whose dividends and price are meant to track a division. It gives a market price and tracking-linked compensation without legal separation, preserving synergies, but boards owe duties to all classes, resources can be shifted between divisions, and creditors have recourse to all of the parent's assets. Many have been

unwound, suggesting the structure is often a half-measure.

In a **Reverse Morris Trust**, a parent spins off a subsidiary that immediately merges with a third party, with the parent's old shareholders keeping a substantial minority of the combined business. If the ownership thresholds are met and the spin otherwise qualifies, the parent can effectively sell a business tax-free at the corporate level and extract cash through debt raised by the subsidiary. The ownership requirement limits how large the partner can be relative to the spun business.

20.5 Going Public: IPOs, Direct Listings and Their Costs

20.5.1 The traditional IPO

In a firm-commitment IPO, underwriters buy the shares, run a bookbuilding process to find the price, and resell to investors for a gross spread (often in the mid-single-digit percent range for mid-sized offerings). The *indirect* cost is **underpricing**: shares are on average priced below their first-day close. Explanations from Volume 1, Chapter 13 include Rock's winner's curse, Benveniste and Spindt's information production through bookbuilding, and rewards to investors who reveal information; Ritter's data show that average first-day returns have varied substantially across periods. For the issuer, money left on the table is a real cost equal to the first-day gain on the shares *sold*; retained shares benefit from the pop. The IPO also brings lock-ups (often 180 days), stabilization through a greenshoe over-allotment option (normally up to 15 percent of the base deal), analyst coverage and a defined price discovery event.

20.5.2 Direct listings

In a **direct listing**, shares are listed without a traditional underwritten offering. Existing holders may sell into the opening auction, the exchange publishes a reference price, and the opening price is set by the auction. No underwriting spread is paid (advisers charge fees), there is typically no mandatory lock-up, and there is no bookbuilding discount, although the opening price can differ from fundamental value in either direction. Rules on whether a direct listing may include a primary capital raise have developed over time and vary, so check current rules. The disadvantages are no committed capital or buyer base, potentially greater early volatility, and no stabilization. It suits well-known firms with no immediate capital need.

20.5.3 Cost comparison

Example 20.5 – IPO versus direct listing for the founder's company. The company wants to raise 300 million of primary capital. In an IPO it would sell 15 million shares at 20. Gross spread is 6 percent, other expenses are 6 million, and the first-day return is 15 percent. A direct listing costs 18 million in advisory, legal and listing fees and raises capital by selling shares in the opening auction (assume a primary-capable listing is available) at an opening price assumed equal to fair value, then adjusted in a sensitivity.

IPO route:

```
Gross proceeds        = 15M × 20            = 300.0
Gross spread          = 6% × 300            =  18.0
Other expenses                              =   6.0
Money left on table   = 15% × 300           =  45.0   (shares sold at 20, worth 23)
Total economic cost   = 18.0 + 6.0 + 45.0   =  69.0   (23.0% of the 300 raised)
```

Direct listing route:

```
Fees                                          =  18.0
Pricing error (if opening price is 10% below fair value): 0.10 × 300 = 30.0
Total, base (no pricing error)                =  18.0  (6.0% of 300)
Total, 10% below fair value                   =  48.0  (16.0% of 300)
Total, 10% above fair value                   = −12.0  (a gain)
```

In the base case the direct listing saves 51.0 million (69.0 − 18.0). The direct listing wins as long as the opening price is not more than about 17 percent below fair value: 18 + x × 300 = 69, x = 17.0 percent. The comparison depends on two things: whether an expected 15 percent first-day return is representative (the IPO's underpricing is a *systematic* expected cost), and whether the direct listing brings pricing *risk* both ways. The main benefit of the IPO is not captured in the numbers: certainty of capital raised and a book of long-term investors. A risk-averse founder who needs the money may rationally pay for it.

20.5.4 Dual-track and other alternatives

Many companies run a **dual-track** process, preparing for an IPO while soliciting private buyers; the credible IPO alternative disciplines bidders, and the sale alternative provides a floor. Other routes include a growth-equity or crossover round, a sale to a private equity sponsor (Volume 1, Chapter 23), listing on a smaller exchange or tier, reduced-disclosure regimes for small offerings (whose thresholds vary by country), and the SPAC merger, to which we now turn.

20.6 Special Purpose Acquisition Companies

20.6.1 How a SPAC works

A **special purpose acquisition company (SPAC)** is a shell company that raises cash in an IPO with the purpose of merging with a private operating company, which thereby becomes public. The standard structure, which dominated the market in 2020 and 2021 and has since evolved, has these features:

- **IPO of units.** Investors buy units, typically at 10.00, each consisting of one share and a fraction of a warrant to buy a share at a strike such as 11.50.
- **Trust account.** The proceeds are placed in a trust account invested in short-term government securities. They can be released only to fund the merger, to redeem public shareholders, or on liquidation.
- **Sponsor and founder shares.** The sponsor pays a nominal amount (such as 25,000) for *founder shares*, typically equal to 20 percent of the post-IPO shares (25 percent of the public shares). This is the **promote**. The sponsor often also buys *private placement warrants* to fund the IPO costs and working capital.
- **Deadline.** The SPAC usually has 18 to 24 months to complete a merger, after which it must liquidate and return the trust to public holders (the sponsor's founder shares and warrants then expire worthless).
- **Vote and redemption right.** Public shareholders can vote on the merger and, *whether they vote for or against it*, can redeem their shares for a pro rata share of the trust (typically about 10.00 plus interest). They keep their warrants.
- **PIPE and other financing.** To fill the gap left by redemptions and to demonstrate demand, SPACs often arrange a private investment in public equity (PIPE) at 10.00 per share.
- **Underwriting fees.** A part of the fee (often about 2 percent of the IPO) is paid at the IPO, and a deferred portion (often 3.5 percent) is paid at the merger.

In the merger the target's holders typically receive shares valued at 10.00, and the combined company lists in the SPAC's place. Seen from the target, the SPAC is a way to obtain a *negotiated price*, a faster timeline, and the ability to market a forward-looking story (historically, projections in merger materials faced a different liability regime than in an IPO prospectus, which regulators have since sought to narrow), and possibly more cash via the trust and the PIPE. From the SPAC investor's perspective, the structure offers a free put at the trust value plus a free option (the warrant) on the merged firm.

20.6.2 The dilution and the cost of capital

The headline price of 10.00 per share is misleading as a measure of value delivered, because several claims share in the combined company without contributing cash equal to their stake: the sponsor's founder shares, the warrants (public and private), the deferred underwriting fee, and transaction expenses. Moreover, redemptions reduce the trust, so the *cash delivered per share* falls while the *dilution is fixed*. The result is that the effective cost of the cash raised can be very high, especially in a high-redemption deal.

A convenient way to think about it is the **net cash per share**:

```
Net cash delivered = Trust remaining after redemptions + PIPE − deferred fee − transaction costs
Shares outstanding after merger = Public non-redeeming + PIPE shares + Founder shares + Target holders' shares
Net cash per share = Net cash delivered / Shares outstanding after merger
Value per share ≈ (Target standalone equity value + Net cash − Value of warrants) / Shares outstanding
```

The warrant value should come from an option model (Volume 1, Chapter 18), with the caveat that standard Black-Scholes ignores the redemption feature, early-exercise and redemption provisions of warrants, and the dilution effect of exercise.

Example 20.6 – SPAC dilution under different redemption levels. A SPAC raised 200 million with 20 million public shares at 10.00, founder shares of 5 million (20 percent of the post-IPO total of 25 million shares), 6.67 million public warrants (one-third of a warrant per unit) and 6 million private placement warrants (12.67 million warrants in all, strike 11.50, five-year life). The target's agreed pre-money equity value is 800 million (80 million shares at 10.00). A PIPE of 50 million (5 million shares at 10.00) is arranged. The deferred underwriting fee is 7 million and other transaction costs are 15 million (assume the initial IPO costs were paid from the private placement). The risk-free rate is 4 percent and volatility 40 percent for the warrant valuation.

Net cash and share count by redemption level:

Redemptions	Public shares left	Trust cash	Net cash (trust + 50 PIPE − 22)	Total shares after merger	Net cash per share
0%	20.0M	200.0	228.0	110.0M	2.07
50%	10.0M	100.0	128.0	100.0M	1.28
90%	2.0M	20.0	48.0	92.0M	0.52

Total shares = public left + 5 PIPE + 5 founder + 80 target. For 50 percent: 10 + 5 + 5 + 80 = 100 million. Net cash is 100 + 50 − 7 − 15 = 128.

The headline price is 10.00, but only 1.28 of cash per share arrives when half the trust is redeemed. Now estimate value per share. Let the post-merger share value be S, solved with the warrant value w(S) from a Black-Scholes call (K = 11.50, T = 5, r = 4%, σ = 40%):

```
S = (800 + Net cash − 12.67 × w(S)) / Shares
```

Iterating, with w(S) from the call formula (for example, w(10) = 3.65, w(9) = 3.0), the solution at 50% redemption is S ≈ 8.91, with w ≈ 2.92 and warrant dilution of 12.67 × 2.92 = 36.9 million. Results:

Redemptions	Share value S	Warrant value w	Value of warrants (12.67M)	Shortfall vs 10.00
0%	9.00	2.98	37.7	10.0%
50%	8.91	2.92	36.9	10.9%
90%	8.82	2.86	36.2	11.8%

The true value per share is about 11 percent below the 10.00 reference price because the target is valued at 800 (its actual worth, assumed) while shares are issued at 10.00 to the sponsor (5 million shares for nearly nothing), and to warrant holders. The *cost of capital* perspective is clearer still. With 50 percent redemptions, the target receives 128 million of net cash from 150 million of gross cash (100 trust plus 50 PIPE), and gives to the sponsor 5 million founder shares worth 5 × 8.91 = 44.6 million and to warrant holders 36.9 million: total non-cash and cash costs of 22 + 44.6 + 36.9 = 103.5 million, or 69 percent of the 150 million of gross cash raised and 81 percent of the 128 million of net cash received. (Part of the cost of the warrants is compensation for the redemption put that public holders were granted.) By comparison, the IPO in Example 20.5 cost 23 percent of proceeds. High redemptions make the SPAC an expensive source of cash, which is why targets increasingly demand minimum cash conditions, and why sponsors forfeit or restructure part of their promote in difficult deals.

The result is not an indictment of the SPAC route in all conditions: for a target that does not need the cash and primarily values a negotiated price, speed or the ability to market its projections, the *promote and warrants* are a price paid for a service. The point of the analysis is that the 10.00 headline and the 800 million valuation are not comparable to an IPO valuation until the dilution is stated.

20.6.3 Incentives: sponsor versus public shareholder

The founder shares are valuable even if the merged company performs poorly, because the sponsor paid almost nothing for them. This creates an incentive to complete *any* deal before the deadline, including deals that destroy value for public shareholders who do not redeem. The public shareholder can protect himself by redeeming, but then he forgoes the upside, so those who remain are exposed to the weakest incentives. This is a textbook agency problem, discussed in Volume 1, Chapter 24.

Example 20.7 – Sponsor versus public shareholder payoffs. Using the data of Example 20.6 (5 million founder shares, 6 million private warrants bought for 1.00 each, so the sponsor invests 6.025 million including the 25,000 for founder shares), compute payoffs at different post-merger share prices P. Warrant values w come from the Black-Scholes call (K = 11.50, T = 5, r = 4%, σ = 40%).

Post-merger price P	Warrant value w	Sponsor value 5P + 6w	Sponsor profit (value − 6.025)	Public non-redeemer profit per unit (P + w/3 − 10)	Public return
4	0.47	22.8	+16.8	−5.84	−58%
6	1.26	37.5	+31.5	−3.58	−36%
8	2.35	54.1	+48.0	−1.22	−12%
10	3.65	71.9	+65.9	+1.22	+12%
12	5.12	90.7	+84.7	+3.71	+37%
15	7.52	120.1	+114.1	+7.51	+75%

For example at P = 8: sponsor value = 5 × 8 + 6 × 2.35 = 40 + 14.1 = 54.1 million; profit = 54.1 − 6.025 = 48.0 million; the public holder who held a unit (one share plus one-third of a warrant) has 8 + 2.35/3 − 10 = −1.22.

The sponsor profits at every price shown, even at 4.00, a 60 percent decline from the reference price. The sponsor's founder shares alone break even at a price of 6.025 / 5 = 1.21. The public shareholder breaks even only at a price a little below 10.00 (about 9.0, where the warrant's contribution of w/3 makes up for the shortfall). The asymmetry shows why sponsors are accused of favoring completion over quality, and why changes to the structure (partial forfeiture of the promote, performance-based vesting of founder shares at higher prices, and tighter disclosure) have been proposed and adopted in various forms. It also explains why the public shareholder's redemption right, which converts him into a holder of a risk-free bond plus a free warrant, has been so heavily used.

20.6.4 The redemption option as a free put

Redemption gives each holder a put at trust value while keeping the warrants. A holder who buys a unit at 10.00 and redeems at about 10.00 plus interest earns the trust yield plus the retained warrant value, a nearly risk-free profit, and specialist investors used the strategy extensively. Sponsors responded with PIPE inducements, forward purchase agreements and extension fees, each adding dilution. An operating company evaluating a SPAC should therefore plan for high redemptions, with committed cash to cover its minimum requirement.

20.6.5 Regulation and the post-boom market

After a surge in 2020 and 2021 the SPAC market contracted sharply, with many mergers trading well below 10.00 and many SPACs liquidating without a deal. US regulators issued a 2021 accounting statement that led many SPACs to treat warrants as liabilities at fair value, and later adopted rules aligning disclosure and liability for SPAC mergers more closely with IPOs, including projections and sponsor conflicts. Details should be checked against current rules. The general lesson: a structure that hands a party a valuable option at negligible cost attracts investors who exercise it, and the cost falls on whoever supplies the cash. ## 20.7 Dual-Class Share Structures

20.7.1 The structure and the wedge

A **dual-class** (or multi-class) structure issues shares with unequal voting rights. The usual form gives founders or insiders a class with several votes per share (often ten) while public investors hold a class with one vote, or in some cases none. Cash-flow rights (dividends, liquidation) are typically equal per share. The consequence is a **wedge** between control and ownership: insiders can hold a majority of votes while owning a minority of the economic claims.

If a fraction f of the total shares carries v votes per share and is held by the insider, and the remainder carries one vote per share, the insider's voting share is:

```
Voting share = v·f / (v·f + (1 − f))
Economic share = f
Wedge = Voting share − Economic share
```

The economic ownership needed for 50 percent of votes is f = 1 / (v + 1): about 9.09 percent when v = 10.

Example 20.8 – Voting power and the private-benefits threshold. After going public, the founder's company has 100 million shares: 20 million Class B (10 votes each, all held by the founder) and 80 million Class A (1 vote each, public).

```
Total votes   = 20 × 10 + 80 × 1 = 280 million
Voting share  = 200 / 280        = 71.4%
Economic share = 20 / 100        = 20.0%
Wedge         = 71.4% − 20.0%    = 51.4 percentage points
```

If the founder sells Class B shares until she holds 10 million (converting the rest to Class A on transfer, a typical feature), then f = 0.10: votes = 10 × 10 + 90 = 190; voting share = 100 / 190 = 52.6 percent, still a majority from a 10 percent economic stake. She loses majority control when f < 9.09 percent: at f = 0.0909, voting share = 0.909 / (0.909 + 0.9091) = 50.0 percent.

Incentive effect. Suppose the founder can take an action that gives her private benefits of 1 million but costs the firm x million of value. She bears only her economic share f of the cost. She prefers the action if f × x < 1, that is, x < 1/f.

```
f = 20%:  x < 5.0 million   (she will destroy up to 5 of firm value to gain 1)
f = 10%:  x < 10.0 million
f = 100%: x < 1.0 million   (she internalizes everything)
```

At f = 15 percent (a 15 million Class B stake) her threshold is 1 / 0.15 = 6.67. The fall in her economic stake therefore *raises* the amount of value she is willing to destroy for a unit of private benefit, while her voting power (which makes her immune to a takeover or a shareholder vote) stays high. This is the agency-cost argument against large wedges; it implicitly assumes that private benefits are material and that she will take value-reducing actions, which is far from universal.

20.7.2 The debate

Arguments for. *Long-term vision:* founders insulated from quarterly pressure and activist campaigns can invest in projects with a long payoff and a story that the market does not yet appreciate (consistent with the theory of Stein on myopic investors and with the view that founders have private information about their vision). *Founder-specific human capital:* when a founder's skill is the firm's main asset, protecting control can protect value (a point associated with Aghion and Bolton's work on control rights). *Voluntary and priced:* investors know the structure at the IPO, so any entrenchment cost should be reflected in the IPO price, and many such firms have performed well. *Takeover bargaining:* a concentrated holder can bargain for a higher takeover price (but also can refuse a premium offer).

Arguments against. *Agency costs and entrenchment:* the wedge reduces accountability, as in Example 20.8. *Value effects:* research by Gompers, Ishii and Metrick on US dual-class firms found that firm value tended to fall as the wedge between votes and cash-flow rights widened, though they, like others, acknowledged issues of causality. *Governance fragility over time:* the founder who designed the structure may not be the one who runs the company decades later, and the structure may pass control to heirs or successors without the original justification (Bebchuk and Kastiel argued against *perpetual* dual-class structures on this ground). *Exclusion from certain indices or restrictions by some investors* have, at various times, made dual-class shares less attractive to some funds; index-provider policies on multi-class shares have changed over time.

What the evidence says. The evidence is mixed. Some studies find positive effects in the early years after the IPO, particularly for innovative firms, and negative effects as time passes and the founder's vision becomes less distinctive, a pattern that supports the design of *sunset provisions*. No honest summary can claim that dual-class structures are uniformly good or bad; the relevant questions are about *design*.

20.7.3 Design features that matter

- **Sunset clauses.** A *time-based* sunset converts high-vote shares after a set number of years; an *event-based* sunset converts them when the founder leaves, dies or falls below a minimum economic stake (say 10 percent); a *transfer-based* provision converts them on sale or gift.
- **Coattail provisions.** In a takeover all classes receive equal consideration.
- **Director and approval rights.** Class A holders elect a minimum number of directors, or related-party transactions need a majority-of-the-minority vote.
- **Regulatory conditions.** Several Asian exchanges began permitting weighted voting rights in the late 2010s subject to conditions on ratios, eligible holders and governance, and other markets' rules have evolved, so check locally. Non-voting classes raise the wedge further and are controversial among institutions.

20.7.4 Valuing the vote

If voting rights carried no value, both classes would trade at the same price. In practice the high-vote class often trades at a modest premium, which is a market-based estimate of the value of control. Care is needed: the premium also reflects differences in liquidity and conversion rights. Where the vote premium is large, it indicates that controlling shareholders can extract private benefits; where it is small, it suggests the market is not worried (Zingales; Nenova). For an acquirer, the structure means the deal can be approved only with the controller's consent, so the control holder's willingness to sell, and the price relative to public holders, are central to the negotiation.

20.8 Choosing the Route: A Managerial Framework

Table 20.1 summarizes how the alternatives compare on the dimensions most often decisive.

Route	Cash raised	Cost profile	Control retained	Key risk
Asset sale	Yes (after tax)	Tax on gain; transaction costs	None over the business	Tax leakage; buyer certainty
Spin-off	No (except via pre-spin debt)	One-time separation costs; dis-synergies	Shareholders own both	Flowback pressure; stranded costs
Equity carve-out	Yes (minority)	IPO discount and spread	Parent keeps control	Governance conflicts
IPO	Yes	Spread plus underpricing (about 20% or more of proceeds in Example 20.5)	Founder control with dual-class	Market window; price risk
Direct listing	Limited or none	Low fees; pricing risk	High	No committed capital
SPAC	Depends on redemptions	Promote and warrants (about 70% of gross cash in Example 20.6 at 50% redemptions)	Negotiated	High redemptions; sponsor incentives

A practical sequence for a CFO is: (1) establish the *objective* (raise cash, create a pure play, achieve control, gain speed); (2) estimate the *value* of each route net of tax and costs, as in Examples 20.1 to 20.7; (3) test *robustness* to market conditions, since many routes depend on windows of investor appetite; (4) design *governance* to deal with the conflicts each route creates (controlling shareholders, sponsors, spun-off companies' takeover defences); and (5) plan *communication and execution*, including investor days, transition services and separation of IT and finance functions. Boards should resist the temptation to select a route merely because it is fashionable in a given year.

Case Study: Halvorsen Mobility

(Illustrative scenario. Halvorsen Mobility and all people and figures are fictional.)

Halvorsen Mobility is a privately held developer of electric delivery vehicles and charging software, founded eight years ago by Ingrid Halvorsen, who owns 62 percent of the shares. The remaining shares are held by employees (10 percent) and three venture investors (28 percent), a total of 100 million shares before any offering. The company has revenue of 420 million, growing 40 percent per year, negative free cash flow of 90 million, and cash of 130 million, enough for about 18 months. It needs 350 million of new capital over the next two years to build a second plant and scale software.

Three routes are on the table. Bankers propose an IPO of 17.5 million shares at 20 (350 million primary), with a gross spread of 6 percent, other costs of 7 million and, based on comparable recent IPOs, an expected first-day return of 18 percent. A SPAC sponsor offers a merger at an agreed pre-money equity value of 1.6 billion (160 million shares at 10.00) in a SPAC with 25 million public shares and a 250 million trust, founder shares of 6.25 million, 8.33 million public warrants and 7 million private warrants (strike 11.50), a deferred underwriting fee of 8.75 million, and expected transaction costs of 18 million. The sponsor will arrange a PIPE of 100 million at 10.00 per share, but has warned that redemptions in recent comparable deals have exceeded 70 percent. A third adviser proposes a direct listing, which would raise no primary capital unless the exchange permits it, but would cost about 20 million in fees.

Ingrid also wants to control the company after listing. She proposes a dual-class structure: all her shares become Class B with ten votes each, public shares carrying one vote, with no sunset clause. The venture investors accept the structure only if a ten-year sunset and automatic conversion on transfer are included. Meanwhile an anchor institution that is willing to buy 100 million of shares in the IPO has told the bankers that it will not participate in a company with perpetual dual-class stock.

The board also notes a trade-off in the software business: management believes a subsidiary, Halvorsen Charge, whose revenues are largely recurring, could be valued by software investors at a much higher multiple than the vehicle business and could be spun or carved out after the listing. They are unsure whether the investors who buy the vehicle business would accept a later carve-out of the most valuable part.

Finally, the CFO has noticed that the SPAC sponsor would receive founder shares equal to about 20 percent of the post-IPO SPAC shares (6.25 million shares of 31.25 million) for a nominal price, and private warrants that cost 7 million. She wants to know what the true cost of capital of the SPAC route is.

Discussion Questions

1. Compute the total cost of the IPO (spread, expenses and money left on the table) as a percentage of the 350 million raised. Compare it with the direct listing at 20 million, and determine how far below fair value the opening price would have to be for the direct listing to cost as much as the IPO.

2. For the SPAC, compute the net cash delivered and the number of shares outstanding at 0 percent and 70 percent redemptions with the 100 million PIPE, and the net cash per share. Show how the sponsor's economics (6.25 million founder shares, 7 million of private warrants) change the cost of capital as redemptions rise. What changes to the structure would you negotiate?
3. Ingrid holds 62 percent of the company. If she converts all her shares to Class B and the post-IPO share count is the pre-IPO count plus the new IPO shares, compute her economic ownership and voting power after the IPO. What combination of sunset provisions, coattails and director rights would you propose to reconcile her wish for control with the anchor investor's objection?
4. Discuss the proposed later separation of Halvorsen Charge. Which route (spin, carve-out, sale) would you recommend, what information would you provide to IPO investors now, and what tax and governance issues does the dual-class structure create for a later carve-out?

Key Takeaways

- Separations create value only if the business is worth more to someone else, or on its own, than inside the parent after dis-synergies, stranded costs and one-time costs; a headline conglomerate discount is a gross, not a net, figure.
- In Calder's sum-of-the-parts, a 600 million gross discount shrinks to a 306 million net gain, and the equity gain (12.75 percent) is larger than the enterprise gain (7.8 percent) because of leverage.
- Tax basis drives the sale-versus-spin choice: a taxable sale must clear a price hurdle equal to the tax cost, whereas a qualifying spin can avoid corporate tax but restricts later transactions.
- The logic of tax-free spin-offs is that they are genuine reorganizations rather than disguised sales or dividends; the conditions (control, active business, business purpose, continuity, limits on subsequent acquisitions) differ by jurisdiction and must be verified with tax counsel.
- Spin-offs require deliberate allocation of debt, basis, pensions and contracts, and often suffer temporary flowback selling.
- Equity carve-outs raise cash and create a market price while retaining control, but impose IPO costs and a controlled-company governance conflict.
- IPOs impose spread plus underpricing, which together can cost a fifth or more of proceeds; direct listings reduce explicit costs but offer no committed capital and carry pricing risk.
- SPAC economics are driven by the promote, warrants and redemptions; at high redemptions the effective cost of cash can far exceed that of an IPO, and sponsors profit even when public holders lose.
- Dual-class structures create a wedge between votes and cash-flow rights; the evidence is mixed, and design features (sunsets, coattails, minimum stakes, independent directors) determine whether they protect vision or entrench control.
- The choice among routes should follow from the objective, a value comparison net of tax and costs, robustness to market conditions and a governance design that addresses the conflicts each route creates.

Review Questions

1. List four motivations for corporate separation and four costs that may offset them.
2. Why does tax basis matter so much in the sale-versus-spin decision? Derive the sale price that makes a taxable sale equivalent to a tax-free spin.

3. In general terms, why do many tax systems allow spin-offs to be tax-free, and what kinds of conditions do they impose?
4. Why is the “conglomerate discount” a gross rather than a net measure of value creation from separation? What adjustments are needed?
5. What is flowback, and how should management interpret weak early trading after a spin?
6. Compare a spin-off, an equity carve-out and a split-off in terms of cash raised, control and tax treatment.
7. Explain underpricing in IPOs and why a direct listing avoids part, but not all, of that cost.
8. Describe the roles of the founder shares, warrants, trust account and redemption right in a SPAC. Why do redemptions raise the effective cost of cash to the target?
9. Why does the sponsor profit at a post-merger price far below 10.00? What changes to the structure address this?
10. Define the wedge in a dual-class structure and summarize two arguments for and two against such structures. What are sunset provisions?

Problems

1. **Sale versus spin.** A subsidiary has a tax basis of 400 million. A buyer offers 1,500 million in cash; the tax rate on gains is 21 percent. A tax-free spin would deliver stand-alone value of 1,350 million. Compute the after-tax proceeds of the sale, which route is better, and the sale price that would equate the two routes.
2. **Sum of the parts.** A conglomerate has Division A with EBITDA 200 (peer multiple 8x) and Division B with EBITDA 120 (peer multiple 11x); its current enterprise value is 2,500, net debt 1,000 and 80 million shares. Separation causes dis-synergies of 10 EBITDA (capitalize at 9x) and one-time costs of 60. Compute the SOTP, the gross discount, net enterprise gain, and value per share before and after.
3. **Basis allocation.** A shareholder holds one share with a tax basis of 30. After a spin-off, the continuing company trades at 40 and the spun-off company at 20 per original share. Allocate basis and compute the gain if she sells the spun-off share at 25.
4. **SPAC net cash.** A SPAC has 10 million public shares at 10.00 (trust 100 million), founder shares of 2.5 million, no PIPE, a deferred fee of 3.5 million and other costs of 6 million. The target’s pre-money equity value is 400 million (40 million shares at 10.00). With 70 percent redemptions, compute net cash, total shares, and value per share ignoring warrants.
5. **Sponsor payoff.** The sponsor in Problem 4 owns 2.5 million founder shares (cost 25,000) and 3 million private warrants (cost 1.00 each). If the post-merger share price is 8.00 and each warrant is worth 1.00, compute the sponsor’s value and profit, and the share price at which the founder shares alone break even.
6. **Dual-class.** A company has 100 million shares, of which 15 million are Class B with 10 votes held by the founder, and 85 million are Class A with one vote. Compute the founder’s economic and voting shares, the wedge, and the value-destruction threshold $x < 1/f$ for a unit of private benefit.
7. **IPO versus direct listing.** An IPO sells 10 million shares at 25; the gross spread is 5.5 percent, other expenses 4 million, and the first-day return 18 percent. A direct listing costs 14 million. Compute the total IPO cost and the percentage of proceeds, and the saving from the direct listing at fair opening price.
8. **Warrant dilution.** Using Black-Scholes, value a warrant with $S = 10$, $K = 11.50$, $T = 5$ years, $r = 4$ percent, $\sigma = 35$ percent. Compute the value of 12 million warrants.

Answers to Selected Problems

1. Tax = 0.21 × (1,500 − 400) = 231; after-tax proceeds = 1,269. The spin (1,350) is better by 81. Equivalence price P solves P − 0.21(P − 400) = 1,350, so 0.79P + 84 = 1,350 and P = 1,602.5.

2. SOTP = 200 × 8 + 120 × 11 = 1,600 + 1,320 = 2,920; gross discount = 2,920 − 2,500 = 420. Net EV = 2,920 − 90 − 60 = 2,770; gain = 270 (10.8% of 2,500). Equity before = 1,500 → 18.75 per share; after = (2,770 − 1,000) / 80 = 22.13 per share.

3. Total value = 60; weights 2/3 and 1/3; basis 20.00 (continuing) and 10.00 (spun-off). Sale of the spun-off share at 25 gives gain 15.00.

4. Public shares remaining = 3.0 million; trust cash = 30.0; net cash = 30 − 3.5 − 6 = 20.5. Total shares = 3 + 2.5 + 40 = 45.5 million. Value per share = (400 + 20.5) / 45.5 = 9.24.

5. Sponsor value = 2.5 × 8 + 3 × 1.00 = 20 + 3 = 23.0 million; investment = 3.025 million; profit = 19.975 million. Founder shares alone break even at 3.025 / 2.5 = 1.21 per share.

6. Economic share = 15%; votes = 150 / (150 + 85) = 63.8%; wedge = 48.8 points; threshold 1 / 0.15 = 6.67 (she acts if x < 6.67 per unit of private benefit).

7. Gross proceeds 250; spread 13.75; other 4.0; money left on table 0.18 × 250 = 45.0; total 62.75 = 25.1% of proceeds. Direct listing at fair opening price costs 14.0; saving = 48.75.

8. d1 = [ln(10/11.5) + (0.04 + 0.06125) × 5] / (0.35 × 2.2361) = (−0.13976 + 0.50625) / 0.78262 = 0.4683; d2 = −0.3143; N(d1) = 0.6802, N(d2) = 0.3766; warrant = 10 × 0.6802 − 11.5 × e^(−0.2) × 0.3766 = 6.802 − 9.4155 × 0.3766 = 6.802 − 3.546 = 3.26. Value of 12 million warrants = about 39.1 million.

Part VI — Strategy, Tax, Climate and Quantitative Methods

Chapter 21: Corporate Strategy and Capital Allocation

Learning Objectives

After studying this chapter you should be able to:

- Explain how strategy, financial performance and enterprise value are linked through the value-driver formula, and use it to show why growth creates value only when returns exceed the cost of capital.
- Calculate Economic Value Added (EVA), invested capital, ROIC and the ROIC–WACC spread from financial statements, including common accounting adjustments such as the capitalization of R&D.
- Evaluate growth-versus-return trade-offs quantitatively, and decompose ROIC into margin and capital-turnover drivers to identify the most valuable operating levers.
- Apply portfolio-management tools: divisional hurdle rates, sum-of-the-parts valuation, the conglomerate discount, and the "parenting advantage" test for whether a business is better off inside the corporation.
- Analyze how internal capital markets work, including their typical biases, and allocate scarce capital across competing projects when capital is rationed.
- Design a capital allocation framework and a set of KPIs and incentives that are tied to value creation yet resistant to manipulation and short-termism.

Opening Vignette

Illustrative scenario. Halden Industries is a fictional company; all figures are invented.

Halden Industries is a diversified manufacturer with three divisions: Consumer Products, Industrial Components and a young Digital Services unit. Revenue has grown nine percent a year for six years and the board is proud of it. Earnings per share have grown too. Yet the share price has gone sideways for three years, and an activist investor has written a polite but pointed letter asking a single question: "Where does Halden earn its cost of capital, and where does it not?"

The CFO, Marguerite Okoye, realizes that nobody at the company can answer. Capital is allocated through an annual budget negotiation in which every division head asks for more than last year and receives roughly the same percentage increase as everyone else. All divisions are evaluated against a single corporate hurdle rate of nine percent. Bonuses are paid on divisional operating profit growth. Consumer Products, the most stable and least risky business, sits on cash and under-invests because its managers can hit their targets without trying. The Digital unit, the riskiest, is quietly approving projects that clear nine percent while actually earning less than its true cost of capital. Industrial Components has been acquiring smaller rivals at prices that make its reported profit grow but that dilute returns on the capital employed.

Okoye commissions a project with three aims: measure return on invested capital division by division against a risk-appropriate cost of capital; replace the budget negotiation with an explicit capital allocation framework; and redesign the bonus plan so that managers are paid for creating value rather than for growing profit. Within a quarter the analysis shows that one division is destroying value in every year of the data, that one is a modest creator, and that the best opportunity set lies in a division with the smallest budget. The remainder of this chapter develops the tools she used and the traps she had to avoid.

21.1 Strategy, Returns and Enterprise Value

21.1.1 What strategy means in financial terms

Strategy is a set of choices about where a company will compete and how it will win. Finance does not replace those choices, but it gives them a common language and a test. A strategy creates value if, and only if, it allows the firm to earn returns on the capital it deploys that exceed what investors could earn elsewhere on investments of comparable risk. All the sophisticated vocabulary of competitive advantage (scale, switching costs, network effects, brand, proprietary technology, regulation) is, in the end, an explanation of why a particular firm can earn a durable return above its cost of capital, and for how long.

This framing is useful because it disciplines two common failures. The first is the strategy that is attractive but uneconomic: a large and fast-growing market in which competition is so intense that nobody earns more than the cost of capital. Growth in such a market is a treadmill, not a source of value. The second is the strategy that is economically strong but starved of capital: a small business with excellent returns that is held back by an allocation process unable to recognize it. Capital allocation is the mechanism by which strategy meets resources, and it is where most corporate value is made or lost over a decade.

21.1.2 The value-driver formula

Volume 1, Chapter 21 valued an enterprise by discounting free cash flow. Here we exploit a particular algebraic form of that valuation, the *value-driver formula*, that exposes the economic logic. Free cash flow equals net operating profit after tax (NOPAT) less net investment:

```
FCF = NOPAT − Net Investment = NOPAT × (1 − g / ROIC)
```

The second equality holds in steady state because net investment equals the growth rate g times the invested capital base, and ROIC is NOPAT divided by invested capital; hence net investment = g × IC = g × NOPAT / ROIC. The reinvestment rate is therefore g / ROIC. If a firm grows at 4 percent and earns 16 percent on new capital, it must reinvest one quarter of NOPAT; if it earns only 8 percent on new capital, it must reinvest half.

Treat a constant-growth firm whose next-year NOPAT is NOPAT_1. Its enterprise value is

```
V₀ = NOPAT₁ × (1 − g / RONIC) / (WACC − g)
```

where RONIC is the return on *new* invested capital (the distinction between the return on existing capital and on incremental capital matters, and we return to it below). Three observations follow immediately.

First, value depends on three things only: the level of operating profit, the growth rate, and the return earned on new capital relative to the cost of capital. Everything else, from brand to supply chain, acts through these.

Second, if RONIC equals WACC the formula collapses to $V_0 = \text{NOPAT}_1 / \text{WACC}$: growth neither adds nor subtracts value. If RONIC exceeds WACC growth adds value; if RONIC is below WACC growth destroys it. This is the single most important idea in the chapter, and we will test it numerically in Section 21.3.

Third, the formula makes explicit why accounting growth is not value creation. Earnings per share can rise every year while value falls, as long as the firm is investing at returns below its cost of capital and financing it with retained earnings or cheap debt.

21.1.3 Value drivers and the strategic hierarchy

It is helpful to arrange the drivers in a hierarchy so that managers at different levels can see how their decisions connect to value.

Level	Driver	Typical decisions
Corporate	WACC, portfolio mix, capital structure	Which businesses to own; financing; payout
Business unit	Growth, ROIC, spread vs. unit WACC	Market entry; pricing; M&A; capacity
Operating	Margin, capital turnover, working-capital days	Cost programs; asset utilization; inventory
Front line	Yield, uptime, cycle time, customer retention	Day-to-day execution

The hierarchy encodes a principle of consistency: any measure used at a lower level should be traceable to value at the top, otherwise managers will optimize something that does not matter. We return to this in Section 21.7 on KPI design.

21.1.4 Competitive advantage and the fade of excess returns

A theoretical sticking point is that competition erodes excess returns. In a perfectly competitive market, ROIC converges to WACC and nobody creates value. The empirical regularity, which should be treated as a tendency rather than a law, is that high returns tend to fade toward the cost of capital over time, while very low returns tend to rise as weak competitors exit or restructure. This has a direct modeling consequence. A valuation that assumes today's spread persists forever with growth continuing indefinitely will systematically overvalue strong businesses. A more disciplined approach models an explicit forecast period in which the spread is sustained by identifiable advantages, followed by a terminal period in which RONIC fades toward WACC (in the limit, V = NOPAT / WACC at that point, ignoring growth).

> **Common mistake.** Assuming that a firm can earn 25 percent on new capital in perpetuity while growing faster than the economy. This is not a forecast; it is an assertion that the firm will eventually become larger than the economy it operates in. Always ask what supports the spread, and for how long.

21.2 Value-Based Management and Economic Value Added

21.2.1 The idea of value-based management

Value-based management (VBM) is a management system in which the objective of creating shareholder value, measured by the spread between returns and the cost of capital, drives strategy, planning, capital allocation, performance measurement and compensation. The word *system* is important. VBM is not a metric. Companies that adopt EVA or a similar measure merely as a reporting item, without changing the way capital is allocated and people are paid, typically find that nothing changes. VBM has four interlocking components:

1. **Strategy and objectives:** value creation is the stated goal, expressed as returns and growth by business.
2. **Processes:** planning, budgeting, capital approval and portfolio reviews are all conducted using value metrics and risk-adjusted hurdle rates.
3. **Measures:** a small set of consistent measures (ROIC, EVA, free cash flow) with clear definitions.
4. **Mindset and incentives:** managers understand the metrics and are rewarded on them.

21.2.2 Defining EVA

Economic Value Added is a residual-income measure. The trademark belongs to a consulting firm, and similar measures go by other names (economic profit, residual income, economic value added). The definition is simple:

```
EVA = NOPAT − WACC × Invested Capital
    = (ROIC − WACC) × Invested Capital
```

NOPAT is operating profit after taxes computed as if the firm had no debt: NOPAT = EBIT × (1 − tax rate), where the tax rate is the cash or operating tax rate on operating profit (see Chapter 22 for how to estimate it). Invested capital is the operating capital the firm has tied up: net working capital plus net fixed assets plus other operating assets, equivalently the sum of debt and equity (including debt-like items) less excess cash and non-operating assets. The capital charge, WACC × IC, is the dollar amount investors require in order to provide that capital.

EVA is the periodic counterpart of NPV. Indeed, a result that is both beautiful and practically important is that, with consistent definitions, the present value of all future EVA discounted at WACC plus current invested capital equals enterprise value:

```
Enterprise Value = Invested Capital₀ + Σ EVA_t / (1 + WACC)^t
```

The difference between market value and invested capital is called Market Value Added (MVA), and MVA equals the present value of expected future EVA. A firm's market value exceeds its invested capital only to the extent that investors expect it to earn positive EVA in the future. This reconciliation of a performance measure with a valuation technique is why EVA is attractive: it tells managers year by year whether they are creating the value implied by the stock price, while the DCF model delivers the same total in a single present value.

Example 21.1 – Computing EVA and the spread. Halden's Industrial Components division had the following figures last year (all in millions): revenue 2,400; EBIT 360; operating tax rate 25 percent; invested capital 2,000, financed with 1,200 of equity and 800 of debt at market values. The risk-free rate is 4.0 percent, the equity market risk premium 5.5 percent, the division's equity beta 1.10 and its pre-tax cost of debt 6.0 percent.

Step 1 – NOPAT. NOPAT = 360 × (1 − 0.25) = 270.

Step 2 – ROIC. ROIC = 270 / 2,000 = 13.5 percent.

Step 3 – cost of equity. Ke = 4.0% + 1.10 × 5.5% = 10.05%.

Step 4 – after-tax cost of debt. Kd(1 − t) = 6.0% × 0.75 = 4.50%.

Step 5 – WACC. Weights are 1,200/2,000 = 60% equity and 40% debt. WACC = 0.60 × 10.05% + 0.40 × 4.50% = 6.03% + 1.80% = 7.83%.

Step 6 – spread and EVA. Spread = 13.5% − 7.83% = 5.67 percentage points. Capital charge = 0.0783 × 2,000 = 156.6. EVA = 270 − 156.6 = 113.4. Cross-check: 0.0567 × 2,000 = 113.4. ✓.

Interpretation. The division generates 113.4 million of value over and above its capital charge. If this spread persisted and no growth were assumed, the MVA of the division would be 113.4 / 0.0783 ≈ 1,448 million, so enterprise value would be about 3,448 million against 2,000 of invested capital. The assumption of no growth with all NOPAT paid out is exactly what the perpetuity implies; the actual figure will differ once growth and fading spreads are modeled.

21.2.3 Why EVA is not the same as accounting profit

Net income deducts the cost of debt (interest) but not the cost of equity. A firm can report a handsome net income and still earn less than its total cost of capital. This is why companies with positive earnings can destroy value. EVA corrects this by charging for *all* capital. A second correction is the treatment of investment: expensed items that actually build long-lived assets (R&D, brand-building advertising, training) are treated as costs in accounting but can be treated as investments in a value-based view.

21.2.4 Adjustments to accounting numbers

Accounting rules are designed for creditors and general-purpose reporting, not for economic measurement. Practitioners of VBM make a limited number of adjustments, always weighing accuracy against complexity. Common candidates are:

- **Capitalizing R&D** and amortizing it over its useful life, so that the economic return on R&D programs is visible and so that managers are not rewarded for cutting R&D to lift current profit.
- **Operating leases:** treating leased assets as invested capital financed by lease liabilities. Under IFRS 16 this is built into the statements; under older US GAAP presentation analysts reconstructed it.
- **Goodwill:** including acquired goodwill in invested capital, at least for acquisition accountability, so that acquirers are evaluated on the return on what they *paid*, not just on the book value of acquired assets. Showing ROIC both with and without goodwill is a useful diagnostic.
- **Restructuring charges and one-offs:** treating recurring "non-recurring" items as operating costs.
- **Deferred taxes:** using cash operating taxes rather than the book provision, and treating the deferred tax liability as quasi-equity (see Chapter 22).
- **Excess cash and non-operating assets:** removing them from invested capital and their income from NOPAT.

The principle is to adjust only when the item is material and when the adjustment will change decisions. Fifteen adjustments will produce a measure that nobody understands and everyone distrusts.

Example 21.2 – Capitalizing R&D. A technology business reports EBIT of 300, a tax rate of 25 percent and invested capital of 1,500 (before any R&D adjustment). R&D spending over the past five years, oldest first, was 40, 45, 50, 55 and 60 (the 60 being the current year), all expensed. Management believes R&D has a five-year life and amortizes straight-line, with the first amortization charge in the year of spending. The WACC is 9 percent.

Step 1 – current-year amortization. Five vintages of R&D are each being amortized at one-fifth per year, so amortization = (40 + 45 + 50 + 55 + 60) / 5 = 250 / 5 = 50.

Step 2 – adjusted EBIT. Add back current R&D expense and deduct amortization: 300 + 60 − 50 = 310.

Step 3 – adjusted NOPAT. 310 × 0.75 = 232.5. (Unadjusted NOPAT is 300 × 0.75 = 225.)

Step 4 – unamortized R&D asset at year-end. The current vintage has four of five years remaining: 60 × 4/5 = 48. The prior year: 55 × 3/5 = 33. The next: 50 × 2/5 = 20. The next: 45 × 1/5 = 9. The oldest: 40 × 0 = 0. Total = 48 + 33 + 20 + 9 + 0 = 110.

Step 5 – adjusted invested capital. 1,500 + 110 = 1,610.

Step 6 – ROIC and EVA. Unadjusted ROIC = 225 / 1,500 = 15.0 percent; adjusted ROIC = 232.5 / 1,610 = 14.44 percent. Unadjusted EVA = 225 − 0.09 × 1,500 = 90.0; adjusted EVA = 232.5 − 0.09 × 1,610 = 232.5 − 144.9 = 87.6.

Interpretation. Because R&D is growing, capitalizing it raises profit and capital, but capital rises proportionally more, so ROIC falls slightly. The more important effect is behavioral: under the unadjusted measure a manager could raise this year's EVA by 60 × 0.75 = 45 simply by eliminating R&D, whereas under the adjusted measure the saving is 60 × 0.75 less the capital charge and amortization path, and the cut shows up as a smaller asset base only slowly. The adjustment reduces the reward for harvesting the future.

21.2.5 EVA versus alternative measures

Several measures compete with EVA, and each has strengths. Cash flow return on investment (CFROI) uses gross cash flows relative to gross investment, which reduces the distortions from depreciation policy; it is more complex to compute. Total shareholder return (TSR) is the ultimate outcome measure but is dominated by market sentiment over short horizons and is unsuitable below the corporate level. Return on equity is easily manipulated by leverage and buybacks. Free cash flow ignores investment quality: a firm can maximize this year's cash flow by cutting growth investment. EVA's strengths are that it charges for all capital, is conceptually tied to NPV, and can be computed for any unit with a defined capital base. Its weaknesses are historical-cost distortions (old assets, heavily depreciated, flatter ROIC and make new investment look unattractive) and a bias toward short-term harvesting if not paired with forward-looking measures.

> **Managerial implication.** The "old assets" problem is real. A division with a fully depreciated plant shows a very high ROIC, and its manager will resist replacing the plant because the new asset will cut ROIC even though the replacement has a positive NPV. Where this matters, evaluate investment proposals on NPV or IRR rather than on the first-year ROIC, and use measures such as EVA with a sinking-fund depreciation schedule that mimics economic depreciation.

21.3 The ROIC–WACC Spread: Growth versus Return

21.3.1 Why growth is not automatically good

Executives are surrounded by pressure to grow: analysts ask about the top line, compensation plans reward revenue and profit growth, and bigger firms enjoy prestige. The value-driver formula shows that growth is a *multiplier on the spread*. When the spread is positive, more growth means more value; when it is zero, growth is irrelevant; when it is negative, growth is destructive, and the faster the firm grows the faster it destroys value.

The intuition is that growth requires capital. A firm that grows NOPAT by one dollar must invest 1/RONIC dollars to do so. That dollar of investment is worth a dollar only if the firm earns exactly its cost of capital on it. Investing at a return above WACC makes the dollar worth more than a dollar; investing at a lower return makes it worth less. The value of a unit of growth is thus approximately proportional to (RONIC − WACC) / WACC per dollar of capital invested.

Example 21.3 – The value of growth at different returns. A business will earn $NOPAT_1$ of 100 next year. WACC is 9 percent. We compute enterprise value for four values of RONIC and three growth rates using $V_0 = NOPAT_1 \times (1 - g / RONIC) / (WACC - g)$.

Zero growth (g = 0): $V_0 = 100 / 0.09 = 1{,}111.1$ for every RONIC (no investment is required).

Growth of 3 percent: - RONIC = 5%: 100 × (1 − 0.03/0.05) / 0.06 = 100 × 0.40 / 0.06 = 666.7. - RONIC = 9%: 100 × (1 − 0.3333) / 0.06 = 100 × 0.6667 / 0.06 = 1,111.1. - RONIC = 15%: 100 × (1 − 0.20) / 0.06 = 1,333.3. - RONIC = 25%: 100 × (1 − 0.12) / 0.06 = 1,466.7.

Growth of 6 percent: - RONIC = 5%: 100 × (1 − 1.20) / 0.03 = −666.7 (the firm would need to reinvest more than 100 percent of NOPAT; free cash flow is negative forever and the formula says value is negative, a warning that the plan is infeasible or absurd). - RONIC = 9%: 100 × (1 − 0.6667) / 0.03 = 1,111.1. - RONIC = 15%: 100 × (1 − 0.40) / 0.03 = 2,000.0. - RONIC = 25%: 100 × (1 − 0.24) / 0.03 = 2,533.3.

RONIC	g = 0%	g = 3%	g = 6%
5%	1,111.1	666.7	−666.7
9%	1,111.1	1,111.1	1,111.1
15%	1,111.1	1,333.3	2,000.0
25%	1,111.1	1,466.7	2,533.3

Interpretation. Read down a column to see that higher returns always help; read across a row to see the role of growth. The 9 percent row is flat at 1,111.1: when RONIC equals WACC, growth is value-neutral. In the 5 percent row, growth reduces value from 1,111.1 to 666.7 at 3 percent growth. The value of growth at RONIC = 15 percent and g = 3 percent is 1,333.3 − 1,111.1 = 222.2, or 20 percent of the no-growth value. In the 25 percent row, doubling growth from 3 to 6 percent quadruples the premium over no-growth value, from 355.6 to 1,422.2, because the capital charge on each dollar of growth is small relative to its return.

> **Common mistake.** Treating the formula as a recipe: “grow at 6 percent at a 25 percent return.” Growth at high returns requires opportunities, and opportunities at high returns are finite. The table is a map of sensitivities, not a menu of options.

21.3.2 Returns on existing versus incremental capital

A firm’s average ROIC mixes returns on old capital (earned from past investments, possibly protected by a durable advantage) with returns on new capital. Valuation of growth turns on RONIC, not on average ROIC. A firm with a historic ROIC of 30 percent due to a legacy brand may have a RONIC of only 8 percent on its attempts to expand into adjacent categories. Treating 30 percent as the return on the next dollar will give an inflated valuation and justify misguided investment. Managers should therefore estimate RONIC separately by opportunity, by asking “What would a new entrant earn if it made the same investment?” The honest answer is often close to WACC.

Equally important, high average ROIC can be a sign of *under-investment* when opportunity exists. If a division earns 30 percent on its capital and has projects available at 25 percent, it should invest more. If the division is rewarded for ROIC alone, its manager will reject the 25 percent project because it lowers the average. This is an example of a point we revisit in the section on KPI design: a ratio measure encourages managers to stop investing when the marginal return falls to the average, not to the cost of capital; a dollar measure such as EVA encourages investment as long as the marginal return exceeds WACC.

21.3.3 Decomposing ROIC: margin and capital turnover

ROIC is the product of operating margin after tax and capital turnover:

```
ROIC = NOPAT / IC = (NOPAT / Revenue) × (Revenue / IC)
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The decomposition is the starting point for operational value creation. Capital turnover can be further broken down into working-capital days and fixed-asset intensity. Industries differ greatly in how they achieve a given ROIC: a grocer earns a modest margin and turns capital many times; a luxury goods firm earns a wide margin and turns capital slowly. It is the

combination that matters. Within an industry, differences in either factor identify competitive position.

Example 21.4 – ROIC tree and operational levers. A division has revenue of 1,000, EBIT margin of 12 percent, tax rate of 25 percent and invested capital of 666.7 (so that turnover is 1.5×). WACC is 9 percent.

Step 1 – base ROIC. NOPAT = 1,000 × 0.12 × 0.75 = 90. ROIC = 90 / 666.7 = 13.5 percent. Via the tree: 0.12 × 0.75 × 1.5 = 0.135 ✓. EVA = 90 − 0.09 × 666.7 = 90 − 60 = 30.

Step 2 – Lever A: lift EBIT margin by one percentage point to 13 percent. NOPAT = 1,000 × 0.13 × 0.75 = 97.5; ROIC = 97.5 / 666.7 = 14.625 percent; EVA = 97.5 − 60 = 37.5, an increase of 7.5.

Step 3 – Lever B: raise capital turnover from 1.5 to 1.6 (reduce invested capital to 625). NOPAT is unchanged at 90; ROIC = 90 / 625 = 14.4 percent; capital charge = 0.09 × 625 = 56.25; EVA = 90 − 56.25 = 33.75, an increase of 3.75.

Interpretation. A one-point margin improvement is worth twice as much EVA (7.5 versus 3.75) as a 0.1× turnover gain in this case. But the right question is feasibility and cost: trimming 41.7 of invested capital through tighter inventory management may be much easier than finding 1 percent of margin. The tree gives managers a common language; the dollar EVA effect tells them what each lever is worth so that they can compare it with its cost. It also warns that levers interact: pushing turnover by cutting capacity can later cost margin.

21.3.4 When the spread is negative: the discipline of not growing

The corollary of the spread logic is that businesses with RONIC below WACC should not grow and should consider returning capital. This is counterintuitive for managers rewarded on growth, and boards sometimes need to impose it.

Example 21.5 – Reinvesting at a sub-WACC return. A mature business earns $NOPAT_1$ of 100 and could either pay out all of it or reinvest part at a RONIC of 8 percent. WACC is 9 percent.

Pay out everything (reinvestment rate 0, g = 0): V = 100 / 0.09 = 1,111.1.

Reinvest 25 percent of NOPAT: g = 0.25 × 0.08 = 2.0 percent. V = 100 × (1 − 0.25) / (0.09 − 0.02) = 75 / 0.07 = 1,071.4.

Reinvest 50 percent of NOPAT: g = 0.50 × 0.08 = 4.0 percent. V = 100 × 0.50 / (0.09 − 0.04) = 50 / 0.05 = 1,000.0.

Interpretation. Reinvesting the first quarter of NOPAT at 8 percent costs shareholders 39.7 of value (1,111.1 − 1,071.4); reinvesting the second quarter costs a further 71.4, even though reported profit growth rises from zero to 2 percent to 4 percent. A board that applauds “growth” in this business is applauding value destruction. The remedy is to pay out the cash through dividends or buybacks (Volume 1, on payout policy), provided the firm has no better use, or to fix returns through restructuring. This is also why a high payout can be consistent with value maximization: for a firm without positive-spread opportunities, payout *is* the value-maximizing policy.

21.3.5 Strategic positions on the growth–return map

A simple two-by-two of growth against spread is a useful classification, but the quadrant labels need careful interpretation.

Quadrant	Characteristics	Value-maximizing stance
High spread, high growth opportunities	Strong franchise with room to expand	Invest aggressively; accept lower payout; protect advantage
High spread, low growth	Cash cow with a moat but a saturated market	Maximize cash generation; look for adjacent RONIC > WACC; pay out surplus
Low spread, high growth	Capital-hungry growth chasing share	Test whether scale will raise RONIC; otherwise slow growth or exit
Low or negative spread, low growth	Value trap	Fix returns (cost, pricing, capital cuts), sell, or liquidate

The most important decisions concern the transitions: whether a low-spread growth business can become a high-spread one as it scales (this requires an evidence-based theory, such as falling unit costs or network effects), and when a high-spread cash cow's excess cash should leave the firm rather than be redeployed in the low-spread quadrants. The second transition is the hardest, because the cash cow's managers have the power in the budget meeting.

21.4 Portfolio Management and the Role of the Corporate Parent

21.4.1 Why own a portfolio at all?

The theoretical case for a diversified corporation is surprisingly weak. Shareholders can diversify themselves at low cost, so diversification *per se* creates no value (this is a corollary of the Modigliani–Miller logic and the CAPM, Volume 1, Chapters 10 and 15). The corporation can add value only through advantages that individual shareholders cannot replicate:

- **Better governance and control** of the business than the market would exercise (monitoring, replacement of weak managers, active strategy).
- **Shared resources:** economies of scope in brand, distribution, technology or purchasing.
- **Internal capital markets:** cheaper and better-informed allocation of capital than external markets provide, particularly when external financing is costly or information is asymmetric.
- **Tax and financing advantages:** the debt capacity of a combined entity or use of losses in one unit against profits in another (Chapter 22).
- **Competencies in specific activities,** such as acquisition integration or operational improvement, that the parent can apply across units.

Against these benefits are well-known costs: corporate overhead, cross-subsidization of poor performers, slower decision-making, managerial empire-building, and the conglomerate discount that investors often apply to complex diversified firms. The empirical evidence on that discount is mixed and sensitive to methodology, so it would be wrong to treat it as a law, but its persistent appearance in practice is a prompt for managers to demonstrate benefits explicitly.

21.4.2 Divisional hurdle rates

Using one corporate cost of capital for all divisions is among the most expensive mistakes in corporate finance. Risk differs across businesses, and WACC should reflect the risk of the *asset* being financed, not of the company that owns it (Volume 1, Chapter 11). A single hurdle rate leads to systematic bias: low-risk divisions look less attractive than they truly are relative to their cost of capital, while high-risk divisions look attractive and receive too much capital.

Example 21.6 – Divisional WACC and EVA. Halden's divisions have the following data (millions). The risk-free rate is 4.0 percent, the market risk premium 5.5 percent, the tax rate 25 percent and the pre-tax cost of debt 6.0 percent for all divisions. Betas are asset (unlevered) betas of comparable pure-play firms; target debt-to-capital ratios reflect each division's debt capacity.

Division	NOPAT	Invested capital	Asset beta	Target D/V
Consumer Products	90	800	0.70	25%
Industrial Components	60	700	1.00	35%
Digital Services	40	300	1.50	15%

Step 1 – relever betas. Equity beta = βA × [1 + (1 − t) × D/E]. D/E = (D/V)/(1 − D/V). - Consumer: D/E = 0.25/0.75 = 0.3333; βE = 0.70 × (1 + 0.75 × 0.3333) = 0.70 × 1.25 = 0.875. - Industrial: D/E = 0.35/0.65 = 0.5385; βE = 1.00 × (1 + 0.75 × 0.5385) = 1.4038. - Digital: D/E = 0.15/0.85 = 0.1765; βE = 1.50 × (1 + 0.75 × 0.1765) = 1.6985.

Step 2 – cost of equity. Ke = 4.0% + βE × 5.5%: Consumer 8.81 percent; Industrial 11.72 percent; Digital 13.34 percent.

Step 3 – WACC. After-tax cost of debt 4.5 percent. Consumer: 0.75 × 8.81% + 0.25 × 4.5% = 6.609% + 1.125% = 7.73 percent. Industrial: 0.65 × 11.72% + 0.35 × 4.5% = 7.619% + 1.575% = 9.19 percent. Digital: 0.85 × 13.34% + 0.15 × 4.5% = 11.340% + 0.675% = 12.02 percent.

Step 4 – ROIC, spread and EVA.

Division	ROIC	WACC	Spread	EVA at own WACC	EVA at flat 9%
Consumer	11.25%	7.73%	+3.52 pts	+28.1	+18.0
Industrial	8.57%	9.19%	−0.62 pts	−4.4	−3.0
Digital	13.33%	12.02%	+1.32 pts	+4.0	+13.0

Interpretation. Under the flat 9 percent hurdle, Digital looks like the star (EVA of 13.0 on a capital base of only 300, against 18.0 for Consumer on 800), and Consumer is undervalued. With risk-appropriate rates Consumer's creation of value is 28.1, over seven times Digital's, and Digital's spread is only 1.3 points: a thin cushion for an unstable business. Industrial is a modest destroyer under both tests, but the flat rate understates the shortfall. A project-level illustration: a Digital project with an IRR of 11 percent is accepted under the flat 9 percent hurdle (a perpetual 100 investment returning 11 per year has an apparent NPV of +22.2 at 9 percent) but should be rejected at 12.02 percent (NPV ≈ −8.5). A Consumer project with an IRR of 8.5 percent is rejected at 9 percent (NPV ≈ −5.6) but is value-creating at 7.73 percent (NPV ≈ +10.0). The flat rate systematically moves capital from the safe to the risky.

> **Practical note.** Divisional betas should come from pure-play comparables, de-levered and re-levered at target structures. The alternative of using the company's own divisional historical betas is unreliable because there are no traded equity prices for divisions. And be explicit about what a hurdle rate is not: it is not a place to hide strategic preferences. If the firm wants to invest in a strategically important area at a return below the risk-adjusted hurdle, say so and treat the shortfall as an explicit subsidy.

21.4.3 Sum-of-the-parts valuation and the conglomerate discount

A sum-of-the-parts (SOTP) valuation values each division at its own cost of capital, with its own growth and returns, and sums the results. Comparing the SOTP with the market's valuation of the whole indicates whether the market is applying a discount (or premium) and whether breaking up the firm could release value. The discount may reflect real corporate costs

or poor allocation, or it may reflect nothing but the market's skepticism about opacity.

Example 21.7 – SOTP for Halden. Using the WACCs from Example 21.6 and the value-driver formula, assume the following next-year NOPAT, growth and RONIC (all illustrative).

Division	$NOPAT_1$	g	RONIC	WACC
Consumer	90	3.0%	11.5%	7.73%
Industrial	60	2.0%	8.57%	9.19%
Digital	40	6.0%	14.0%	12.02%

Consumer: V = 90 × (1 − 0.03/0.115) / (0.0773 − 0.03) = 90 × (1 − 0.2609) / 0.0473 = 90 × 0.7391 / 0.0473 ≈ 1,406.4.

Industrial: V = 60 × (1 − 0.02/0.0857) / (0.0919 − 0.02) = 60 × (1 − 0.2334) / 0.0719 = 60 × 0.7666 / 0.0719 ≈ 639.7. (Note that this is below the no-growth value 60/0.0919 = 652.9, because growth at RONIC < WACC destroys value.)

Digital: V = 40 × (1 − 0.06/0.14) / (0.1202 − 0.06) = 40 × (1 − 0.4286) / 0.0602 = 40 × 0.5714 / 0.0602 ≈ 379.7.

Sum of the parts: 1,406.4 + 639.7 + 379.7 = 2,425.8.

Suppose the market currently values Halden's operations (enterprise value) at 2,100. The implied conglomerate discount is (2,425.8 − 2,100) / 2,425.8 = 13.4 percent.

Interpretation. A 13 percent gap does not by itself justify a break-up. The questions are: how much of it reflects corporate overhead (which could be reduced), how much reflects cross-subsidization (which could be stopped), how much reflects investors' difficulty in valuing a complex firm, and how much would reappear as standalone costs, dis-synergies and tax leakage on separation? A fair test is to compare the discount with a realistic estimate of the value of separation after those costs. Also note the sensitivity: Industrial's value depends heavily on whether it can lift RONIC above 9.19 percent. If a restructuring raised its RONIC to 12 percent, with the same 2 percent growth, its value would rise to 60 × (1 − 0.02/0.12) / 0.0719 = 60 × 0.8333 / 0.0719 ≈ 695.4, an increase of about 56. The fix is worth more than a dispute over the discount.

21.4.4 The parenting-advantage test

A useful test for each business in the portfolio is the *parenting advantage* test: is this business worth more under our ownership than under the best alternative owner? The answer should come from specific, observable sources of benefit (shared procurement savings, distribution access, technology, management capability) net of costs the parent imposes (overhead, constraints, bureaucracy, inappropriate hurdle rates). Businesses that fail the test are candidates for sale, spin-off or carve-out. Businesses that pass but earn below WACC are candidates for restructuring. Businesses that pass and earn above WACC are the core.

A related point is that the best owner may pay more than the current owner's valuation. Divestiture proceeds are an under-appreciated source of capital, and a disciplined parent regularly sells businesses that are worth more to someone else, not only those that are failing. In terms of the portfolio matrix, the question that matters is not "is this a good business?" but "are we the best owners of it?"

21.5 Internal Capital Markets

21.5.1 How internal capital markets work

In a multi-business firm, the corporate center acts as a financial intermediary: it collects cash generated by some units and allocates it to others, through the budget, the capital-expenditure approval process and the setting of targets. The center can in principle do what the external market cannot. It sees detailed operating information, can monitor and intervene, and is not subject to the information asymmetries that make external equity expensive for a firm with private knowledge of its prospects. These advantages are most valuable when external capital is expensive or unavailable, for example in countries with underdeveloped markets or for unlisted subsidiaries with limited access to credit.

The theory of "winner picking" says that an efficient internal market takes capital away from low-return units and gives it to high-return units, more aggressively than an external market would. The firm's cash flows are thus fungible across divisions, and the center's discretion is a source of value.

21.5.2 Typical distortions

Observation of actual corporate behavior suggests that internal markets often fall short of this ideal. The documented patterns, which should be treated as tendencies, include:

- **Socialist or "spreading-thin" allocation:** capital is distributed relatively evenly across divisions regardless of opportunity, because of negotiation dynamics, fairness norms or the political power of division heads. High-opportunity units are under-funded; low-opportunity units are over-funded.
- **Cross-subsidization:** cash flow from strong units subsidizes weak ones, delaying restructuring or exit.
- **Influence activities:** managers spend time and distort information to lobby for capital, and the center may favor those with best access rather than best projects.
- **Optimistic bias and strategic misrepresentation:** division managers overstate benefits and understate risks of projects they want funded, because their reward depends on the size of their empire or because rejection is costly.
- **Anchoring on last year's budget:** incremental budgeting makes the allocation a function of history, not opportunity.
- **Hurdle-rate distortion:** the flat-rate problem of Section 21.4.2.

The corrective measures follow from the diagnosis: use risk-adjusted divisional hurdle rates; separate the *information-gathering* role from the *advocacy* role, for example through independent project review by a corporate planning group; require explicit strategic rationale and a post-audit; and tie compensation to value rather than to the size of the capital budget (Section 21.7).

21.5.3 Capital rationing

When capital is limited, whether because of a hard budget set by the board, a debt covenant or an internal limit, the NPV rule must be adapted. The correct objective is to maximize total NPV subject to the budget. Ranking by profitability index (PI = NPV / initial investment) is a good heuristic, but because projects are indivisible it does not always select the optimal set, and ranking by IRR is worse because IRR ignores scale.

Example 21.8 – Allocating a capital budget. Halden's capital committee has six candidate projects (millions). Present values are of future cash flows discounted at risk-appropriate rates, so NPV = PV − investment.

Project	Investment	PV of cash flows	NPV	PI
B	30	48	18	0.60
E	20	32	12	0.60
A	50	70	20	0.40
C	80	100	20	0.25
F	60	66	6	0.10
D	40	41	1	0.025

The projects are indivisible. The budget is 150.

Step 1 – PI ranking. Order: B, E, A, C, F, D (as in the table). Adding in order: B (30) + E (20) + A (50) = 100 invested, NPV = 18 + 12 + 20 = 50. Next is C, which needs 80 but only 50 remains; skip. F needs 60; skip. D needs 40 and fits: total investment 140, NPV = 50 + 1 = 51. Unused budget: 10.

Step 2 – exhaustive search. There are $2^6 = 64$ combinations; enumerating those that fit within 150 shows that the best is A + C + E, with investment 50 + 80 + 20 = 150 and NPV = 20 + 20 + 12 = 52.

Step 3 – compare. The PI-ranking heuristic delivers 51 versus the optimum of 52. The difference is small here, but the *composition* is quite different: the heuristic funds B and D and not C, while the optimum funds C and not B or D.

Interpretation. Two lessons. First, when projects are lumpy, the heuristic can leave capital idle and lose value; the loss can be larger with other data. Second, the shadow price of capital is important: if the committee could raise the budget by 20 to 170, what would be gained? With budget 170 the best set is A + B + C with NPV 58 (investment 160), an increase of 6 for 20 more capital: a marginal return on the extra capital of 30 percent in NPV terms. A capital budget that is too tight is itself a source of lost value, and a rationing constraint should be justified by something real (financing market access, execution capacity, or risk appetite), not by habit. If the constraint is "soft" (imposed by management for discipline) the right response may be to raise the budget and test whether the projects at the margin still create value.

21.6 Capital Allocation Frameworks

21.6.1 The order of priorities

A capital allocation framework is a ranked, explicit set of principles for using the firm's cash flow and borrowing capacity. A defensible hierarchy, adapted to the particular firm, runs as follows:

1. **Maintain the franchise:** fund the capital expenditure and working capital needed to sustain existing advantaged businesses. This spending is often non-discretionary and should be tested for whether it truly sustains returns.
2. **Invest in organic growth with RONIC > WACC:** projects in the existing portfolio where the spread is positive and defensible.
3. **Acquire selectively:** M&A where the return on the full price paid, including synergies that the buyer can actually capture, exceeds WACC.
4. **Maintain the target capital structure and financial flexibility:** keep a balance sheet consistent with the firm's debt capacity (Volume 1, Chapter 15).
5. **Return surplus cash to shareholders:** dividends for the stable component and repurchases for the variable component (Volume 1, on payout policy).

The point of the hierarchy is that payout is the residual after value-creating investment, rather than an entitlement or a target in itself, and that each use competes on the same measure, the spread over the risk-adjusted cost of capital. The framework should also state what the firm will *not* do: for example, no acquisitions outside defined core adjacencies, no growth projects below the divisional hurdle, no buybacks financed with debt that would take leverage beyond the target.

21.6.2 Buyback versus reinvestment

Share repurchases are a legitimate use of capital when the firm has no better alternative, but they are often used as a tool to boost EPS. The relevant comparison is the expected return on repurchased shares, which is the cost of equity if the shares are fairly priced, against the return from alternatives. A buyback of fairly priced shares has an NPV of zero for continuing shareholders; it creates value only if the shares are undervalued, if it sends a credible signal, or if it moves the firm toward a more efficient capital structure. Buying back overvalued shares transfers value to those who sell. Boards therefore should require an explicit view of intrinsic value before authorizing repurchases, and treat the decision as an investment with a stated expected return.

21.6.3 Stage-gating and real options

Many strategic investments are uncertain, and staging them can raise value by allowing the firm to learn and abandon. This is the real-options logic of Volume 1, Chapter 18, applied to capital allocation: commit a small amount to resolve the largest uncertainty, and release further capital against milestones.

Example 21.9 – The value of staging. A product requires a 20 million pilot now, after which in one year the firm will learn whether the product is viable. With probability 0.60 the pilot succeeds, and a full roll-out requires 100 million at that point and is then worth 160 million (present value at that date). With probability 0.40 it fails and the roll-out would be worthless. The discount rate is 10 percent.

Staged plan. Invest 20 now. In the success state, invest 100 and receive 160, a net gain of 60 at year 1; in the failure state, abandon. NPV = −20 + 0.60 × 60 / 1.10 = −20 + 32.73 = **+12.73**.

Unstaged plan. Commit 120 now (pilot and roll-out together). With probability 0.60 receive 160 at year 1; otherwise nothing. NPV = −120 + 0.60 × 160 / 1.10 = −120 + 87.27 = **−32.73**.

Difference. 12.73 − (−32.73) = 45.45. This equals the roll-out cost saved by not spending 100 in the failure state and by deferring it in the success state: 100 − 0.60 × 100 / 1.10 = 100 − 54.55 = 45.45 ✓.

Interpretation. The same project is a bad bet if committed all at once and a good one if staged. In practice, organizations often undermine the option by sinking costs early in a way that makes abandonment politically impossible; the framework should therefore define explicit kill criteria in advance and put someone other than the project sponsor in charge of the stage-gate decision.

21.6.4 Post-audits and learning

A capital allocation process that never looks back learns nothing. A post-audit compares realized outcomes with the investment case several years later, focusing less on blame than on identifying systematic biases: Are forecasts of volumes, prices and synergies consistently too high? Do large projects overrun? Are acquisition premiums justified by realized synergies? The results should feed back into the hurdle-rate and forecasting guidelines. Without this loop, optimism bias goes uncorrected, because project sponsors are rewarded at approval and the bill arrives in someone else’s tenure.

21.7 KPI Design and Incentives

21.7.1 What makes a good KPI

Performance measures shape behavior. The chosen KPIs are the firm's real strategy as experienced by employees, whatever the strategy deck says. A good value-oriented KPI set has the following properties:

- **Linked to value:** a causal path from the KPI to value creation (through growth, returns or risk).
- **Controllable:** the manager measured can influence it; otherwise the measure rewards luck.
- **Balanced between short and long term:** pair a lagging financial outcome (EVA, ROIC) with leading indicators (customer retention, backlog, product pipeline) that signal future value.
- **Hard to manipulate:** based on audited data where possible, with definitions that cannot be changed to hit a target.
- **Few in number:** five to seven at business-unit level; beyond this, nothing has priority.
- **Cascading:** each level translates top-level value drivers into measures meaningful at that level.

A well-known adage, often attributed to Charles Goodhart, states that when a measure becomes a target it ceases to be a good measure. The lesson is not to avoid targets but to expect gaming, design for it, and rotate or supplement measures when gaming appears.

21.7.2 Typical failure modes

KPI	Intended effect	Typical unintended behavior
Revenue growth	Market share gain	Unprofitable discounting; channel stuffing
EPS growth	Shareholder return	Buybacks at any price; under-investment; accounting choices
ROE	Efficient equity use	Excess leverage; rejection of positive-NPV projects
ROIC (ratio)	Capital efficiency	Under-investment when marginal return < average; harvesting assets
Operating margin	Cost discipline	Cuts in R&D and marketing; outsourcing to flatter margin
Free cash flow	Cash discipline	Stretching payables; deferring capex and maintenance
EVA (single-year)	Value creation	Short-termism if unadjusted; sandbagging targets

21.7.3 EVA-based incentives and the bonus bank

Because a single-year measure invites short-termism (a manager can increase this year's EVA by cutting investment that pays off in the future, and leave before the cost appears), well-designed EVA plans use a **bonus bank**. Bonus credits earned in a year, positive or negative, are added to a bank balance; only a fraction (commonly one-third) of the positive balance is paid out each year, and the remainder stays at risk against future negative EVA. The effect is to align the manager's horizon with the horizon of the decision: a value-destroying move that looks good in year one is penalized when its consequences arrive. Another important feature is that targets are set as *improvements* in EVA that are independent of the manager's own forecast, so that managers are rewarded for creating value, not for negotiating a low target.

Example 21.10 – A bonus bank in action. A division's EVA is +10, −6, +14 and +20 (millions) over four years. The plan pays out one-third of the positive bank balance each year; the balance (after payout) carries forward. (Treat the bonus pool as equal to EVA for simplicity; a real plan would use a percentage of EVA improvement.)

Year 1: Bank = 0 + 10 = 10. Payout = 10 / 3 = 3.33. Carry-forward = 10 − 3.33 = 6.67.

Year 2: Bank = 6.67 + (−6) = 0.67. Payout = 0.67 / 3 = 0.22. Carry-forward = 0.67 − 0.22 = 0.44.

Year 3: Bank = 0.44 + 14 = 14.44. Payout = 14.44 / 3 = 4.81. Carry-forward = 14.44 − 4.81 = 9.63.

Year 4: Bank = 9.63 + 20 = 29.63. Payout = 29.63 / 3 = 9.88. Carry-forward = 29.63 − 9.88 = 19.75.

Total paid = 3.33 + 0.22 + 4.81 + 9.88 = 18.24, against cumulative EVA of 38 and a remaining bank of 19.75.

Interpretation. The −6 in year 2 wipes out most of the year-1 credit, so the manager cannot "bank and run": the bad year is borne by earlier awards. A manager who is considering a risky investment that raises this year's EVA but will likely lose value later knows that the downstream loss will be charged against the bank. The cost is complexity, and the cost of retaining a large bank balance if the manager leaves (usually forfeited or paid with restrictions), which must be managed through the design of vesting rules. Notice that payout lags behind performance: only 18.24 of the 38 created has been paid. This is deliberate.

21.7.4 Linking KPIs to the strategic cascade

The cascade works as follows. At the corporate level: TSR relative to peers and absolute ROIC–WACC spread. At the business unit: EVA or EVA improvement, with growth and return targets agreed with the center. At the functional level: drivers such as inventory days, price realization or plant uptime, each converted into its EVA effect (as we did for margin and turnover in Example 21.4). Nonfinancial KPIs (safety, quality, customer metrics) belong in a scorecard when they are leading indicators of value or constraints on how value may be pursued. The scorecard should have a clear statement of weights, and thresholds below which the bonus is zero regardless of EVA (for example, safety or compliance gates).

21.7.5 Implementation: the change-management reality

Value-based management initiatives most commonly fail not because the arithmetic is wrong but because the organization does not change behavior. The usual causes are lack of senior sponsorship; metrics owned by finance rather than by line managers; training limited to finance staff; incentive plans not actually changed; and a drift back to old habits once the first difficult trade-off (such as selling a historical flagship business) arises. Successful programs start with a few businesses and a few decisions, demonstrate value quickly, and extend gradually. A simple test of whether VBM has taken hold: could a plant manager explain how reducing inventory by a given amount changes EVA, and would they be rewarded for doing it?

Case Study: Redrawing the Capital Map at Norvane Holdings

Illustrative case. Norvane Holdings is a fictional company; all figures are invented.

Norvane Holdings is a diversified group with three business units. **Norvane Foods** is a branded packaged-food business with stable cash flow and a loyal customer base. **Norvane Logistics** operates a regional freight network with heavy asset requirements. **Norvane Analytics** is a ten-year-old data and software unit that has grown 18 percent a year and is still reinvesting nearly all its cash.

The group's CFO, Tomas Reyes, has been in office for six months. His predecessor ran an annual process in which each unit submitted a capital request, the executive committee trimmed every request by a similar percentage and a single 8.5 percent hurdle rate was applied. Bonuses depended on operating profit growth and were paid in full when units hit 95

percent of plan. Reyes assembled the following picture for the most recent year (millions of currency units):

Unit	NOPAT	Invested capital	Capex request next year	Asset beta	Target D/V
Foods	150	1,100	80	0.60	30%
Logistics	70	1,000	220	0.90	40%
Analytics	55	350	130	1.40	10%

Using a risk-free rate of 4 percent, a market premium of 5.5 percent, a tax rate of 25 percent and a pre-tax cost of debt of 6 percent, Reyes computed unit WACCs of roughly 7.2 percent for Foods, 8.7 percent for Logistics and 11.6 percent for Analytics. Unit ROICs were 13.6 percent, 7.0 percent and 15.7 percent respectively.

The results shocked the executive committee. Logistics, the largest capital user and requesting the largest increase, earned less than its cost of capital and had done so for years; its managers argued that the network would eventually reach scale and the returns would rise, although the data showed no trend. Foods, with a spread of more than 6 points, had asked for little because its leaders believed the “pie was fixed”, and was sitting on 120 million in cash it could not deploy productively. Analytics had a modest positive spread of about 4 points, and its projects were uneven: some earned well above 20 percent, but a series of recent small acquisitions had been priced at premiums that, on a full-cost basis including goodwill, lowered the unit’s return on invested capital by two points.

Reyes proposed four changes. First, replace the single hurdle with unit-specific rates and require every project of more than 25 million to show NPV at the unit rate and a five-year post-audit plan. Second, impose a stage-gated process on Logistics’ network expansion, releasing capital only when measured route-level ROIC reached the unit WACC. Third, direct Foods to acquire or invest in adjacent categories only where RONIC exceeded 9 percent, and otherwise to raise its dividend and repurchase shares. Fourth, move the group to an EVA-based bonus bank with unit-level targets expressed as improvements in EVA, and a safety and compliance gate.

The Logistics head objected that EVA would “punish investment in the future.” The CEO asked Reyes how he would answer that, how he would deal with the legacy asset-base problem (Logistics’ oldest trucks were fully depreciated, flattering returns) and whether the board should consider selling the unit.

Discussion Questions

1. Using the data in the case, verify the unit WACCs and ROICs in the text (at least for Foods and Analytics), and compute each unit’s EVA. Which unit creates most value, and how does a flat 8.5 percent hurdle change the ranking?
2. Evaluate the argument that Logistics will “reach scale” and earn its cost of capital. What evidence would you require, and how would you structure a stage-gate for the network expansion?
3. How would you advise Foods on the use of its surplus cash? Frame your answer in terms of RONIC, buybacks versus dividends, and the risk of poorly priced acquisitions.
4. Design the EVA-based bonus plan for Analytics. How would you treat goodwill from past acquisitions, R&D and the unit’s growth, and how would you answer the objection that EVA discourages investment?

Key Takeaways

- Value comes from returns above the cost of capital on invested capital; growth multiplies the spread, so it creates value only when RONIC exceeds WACC and destroys value when it does not.

- The value-driver formula, $V = NOPAT \times (1 - g/RONIC) / (WACC - g)$, shows that profit growth by itself is neutral at best; reinvestment at sub-WACC returns lowers value while raising reported earnings.
- $EVA = NOPAT - WACC \times IC$ is the periodic counterpart of NPV; the present value of future EVA plus invested capital equals enterprise value. Its use as a management system requires consistent definitions, a few well-chosen adjustments (R&D, goodwill, leases) and attention to historical-cost distortions.
- ROIC decomposes into margin and capital turnover; translating each operational lever into dollars of EVA makes trade-offs transparent and comparable.
- A single corporate hurdle rate shifts capital from low-risk to high-risk businesses; use divisional rates built from pure-play betas and target capital structures.
- A diversified firm must justify its portfolio with specific parenting advantages; use sum-of-the-parts valuation and the best-owner test to decide what to keep, fix or sell.
- Internal capital markets are prone to spreading-thin allocation, cross-subsidization, lobbying and optimism bias; counter them with independent review, explicit rationing and post-audits.
- Under capital rationing, maximize total NPV subject to the budget; PI ranking is a heuristic that can fail with indivisible projects, and the shadow price of capital shows whether the constraint itself is destroying value.
- A capital allocation framework ranks uses of cash (maintain, invest organically, acquire selectively, preserve flexibility, return surplus) on the same value metric, and uses staging to preserve the option to abandon.
- KPIs shape behavior: pick few, controllable, value-linked measures; use dollar measures like EVA rather than ratios for investment decisions; use bonus banks and balanced scorecards to counter short-termism and gaming.

Review Questions

1. Explain why the value-driver formula implies that growth is value-neutral when RONIC equals WACC. What does this imply for an executive whose bonus depends on revenue growth?
2. Distinguish average ROIC from RONIC. Why is the second more relevant to the valuation of growth, and how could using the first lead to a poor investment?
3. Describe the relationship between EVA, MVA and enterprise value. Why does it matter that this relationship holds?
4. Why might a company capitalize R&D for internal performance measurement even if accounting standards require expensing? What are the drawbacks of such adjustments?
5. Why does a ratio-based measure such as ROIC discourage investment in some divisions where a dollar-based measure such as EVA would not?
6. What are the arguments for and against using a single corporate WACC for all divisions? Illustrate how it can misallocate capital.
7. What does the “parenting advantage” test require, and how does it differ from asking whether a business is attractive?
8. List three typical distortions in internal capital markets and propose one governance mechanism to counter each.
9. When capital is rationed, why can ranking by profitability index fail to give the optimal portfolio? Why is IRR ranking worse?
10. Explain how a bonus bank reduces short-termism, and describe two design choices (such as payout fraction and target setting) that determine its effectiveness.

Problems

Problem 21.1 – EVA. A business has NOPAT of 180 and invested capital of 1,500. Its WACC is 9.5 percent. Compute ROIC, the spread, the capital charge and EVA.

Problem 21.2 – Value of growth. $NOPAT_1$ is 80 and WACC is 10 percent. Compute enterprise value for g = 5 percent when RONIC is (a) 20 percent, (b) 10 percent and (c) 5 percent, and compare each with the no-growth value of 80 / 0.10.

Problem 21.3 – Capitalizing R&D. A firm has EBIT of 200, a tax rate of 30 percent, invested capital of 1,000 and a WACC of 10 percent. R&D expense for the last five years (oldest first) was 20, 24, 28, 32 and 36. R&D is amortized over three years on a straight-line basis, beginning in the year of spending (so only the last three vintages remain). Compute (a) current-year amortization, (b) the year-end unamortized R&D asset, (c) adjusted NOPAT and invested capital, (d) ROIC and EVA before and after adjustment.

Problem 21.4 – Divisional WACC. A division has an asset beta of 0.80 and a target D/V of 30 percent. The risk-free rate is 3.5 percent, the market risk premium 5.0 percent, the pre-tax cost of debt 5.5 percent and the tax rate 25 percent. Compute the equity beta, cost of equity and WACC.

Problem 21.5 – Capital rationing. Use the six projects of Example 21.8 with a budget of 170. Find the PI-ranking selection and the optimal selection by enumeration, and state each NPV.

Problem 21.6 – Bonus bank. A plan pays out one-half of the positive bank balance each year. A division's EVA over four years is +8, −12, +10 and +16. Compute the bank balance before payout, the payout and the carry-forward each year.

Problem 21.7 – ROIC tree. A division has an EBIT margin of 10 percent, a tax rate of 30 percent and capital turnover of 2.0×. (a) Compute ROIC. (b) Compute ROIC if the margin rises to 11 percent. (c) Compute ROIC if turnover rises to 2.2×. (d) Which lever raises ROIC more?

Problem 21.8 – Sum of the parts. Division X has $NOPAT_1$ = 60, g = 3 percent, RONIC = 10 percent and WACC = 8 percent. Division Y has $NOPAT_1$ = 30, g = 4 percent, RONIC = 15 percent and WACC = 11 percent. Compute each division's value and the sum, and the conglomerate discount if the market values the firm at 1,200.

Answers to Selected Problems

21.1. ROIC = 180 / 1,500 = 12.0 percent; spread = 12.0 − 9.5 = 2.5 points; capital charge = 0.095 × 1,500 = 142.5; EVA = 180 − 142.5 = 37.5.

21.2. (a) 80 × (1 − 0.05/0.20) / (0.10 − 0.05) = 80 × 0.75 / 0.05 = 1,200; (b) 80 × (1 − 0.5) / 0.05 = 800; (c) 80 × (1 − 1) / 0.05 = 0. The no-growth value is 800. Growth adds 400 at RONIC 20 percent, adds nothing at 10 percent and destroys 800 at 5 percent (all NOPAT is reinvested at a return below WACC, leaving no free cash flow).

21.3. (a) Amortization = (28 + 32 + 36) / 3 = 32. (b) Unamortized asset = 36 × 2/3 + 32 × 1/3 + 28 × 0 = 24 + 10.67 = 34.67. (c) Adjusted EBIT = 200 + 36 − 32 = 204; adjusted NOPAT = 204 × 0.70 = 142.8; adjusted invested capital = 1,034.67. (d) Before: ROIC = 140 / 1,000 = 14.0 percent, EVA = 140 − 100 = 40.0. After: ROIC = 142.8 / 1,034.67 = 13.80 percent, EVA = 142.8 − 103.47 = 39.33.

21.4. D/E = 0.30/0.70 = 0.4286; βE = 0.80 × (1 + 0.75 × 0.4286) = 0.80 × 1.3214 = 1.057; Ke = 3.5% + 1.057 × 5.0% = 8.79%; after-tax Kd = 4.125%; WACC = 0.70 × 8.79% + 0.30 × 4.125% = 6.15% + 1.24% = 7.39%.

21.5. PI ranking: B (30), E (20), A (50) = 100; C (80) does not fit; F (60) fits, total 160, NPV = 18 + 12 + 20 + 6 = 56. D (40) would exceed 170 (200), so no more. Optimal: A + B + C = 160, NPV = 20 + 18 + 20 = 58 (other combinations such as B + C + E = 130 with NPV 50, or A + B + E + F = 160 with NPV 56, are inferior). The heuristic loses 2.

21.6. Year 1: bank 8, payout 4, carry 4. Year 2: bank 4 − 12 = −8, payout 0, carry −8. Year 3: bank −8 + 10 = 2, payout 1, carry 1. Year 4: bank 1 + 16 = 17, payout 8.5, carry 8.5. Total paid 13.5 against cumulative EVA of 22. Note the negative balance carried forward from year 2 suppresses payouts in year 3.

21.7. (a) 0.10 × 0.70 × 2.0 = 14.0 percent. (b) 0.11 × 0.70 × 2.0 = 15.4 percent. (c) 0.10 × 0.70 × 2.2 = 15.4 percent. (d) Equal here: a 10 percent relative improvement in either factor raises ROIC by 10 percent (14.0 to 15.4); the lever with the lower cost should be chosen.

21.8. X = 60 × (1 − 0.03/0.10) / (0.08 − 0.03) = 60 × 0.70 / 0.05 = 840. Y = 30 × (1 − 0.04/0.15) / (0.11 − 0.04) = 30 × 0.7333 / 0.07 = 314.3. Sum = 1,154.3. Discount = (1,154.3 − 1,200) / 1,154.3 = −4.0 percent, that is, a 4.0 percent premium: the market values the firm above the sum of the parts.

Chapter 22: Corporate Tax Strategy and International Tax

Learning Objectives

After studying this chapter you should be able to:

- Distinguish statutory, marginal, average, effective and cash tax rates, reconcile an effective rate to the statutory rate, and choose the right rate for each financial decision.
- Explain the origin of deferred tax assets and liabilities, compute them from temporary differences, and decide how to treat them in valuation and performance measurement.
- Value interest tax shields and tax loss carryforwards, and explain how limits on interest deductibility, personal taxes and distress costs alter the textbook benefit of debt.
- Describe the main features of international tax systems, the arm's-length principle in transfer pricing, and quantify the effect of transfer prices and repatriation choices on after-tax cash flow.
- Explain, with appropriate caution, the logic of global minimum tax rules and illustrate a simplified top-up tax computation.
- Incorporate taxes correctly into valuation (APV, WACC, NOPAT) and assess the value, and the risks, of tax strategy.

Opening Vignette

Illustrative scenario. Brightwater Devices is a fictional company; all figures are invented.

Brightwater Devices designs and sells diagnostic equipment. It was founded in a high-tax country, which is still home to its headquarters, its engineering team and most of its intellectual property. Over fifteen years it has built a manufacturing plant in a lower-tax country and sales subsidiaries in a dozen markets. Its reported effective tax rate has drifted down from 27 percent to 17 percent, and the board has been told that this is a point of pride.

Then three things happen in one quarter. The tax authority of one sales country opens an audit, challenging the price at which the manufacturing subsidiary sells devices to the local distributor and arguing that too little profit is left in the country. Brightwater's home government announces a limit on interest deductions at a fixed share of earnings, which will bite on the acquisition debt the company took on two years ago. And the CFO's office learns that several of the countries in which it operates are adopting a minimum tax on large multinational groups, so that the low rate in the manufacturing country may no longer yield the full benefit expected.

The CFO, Priya Venkataraman, discovers that the group's tax planning has been run as a cost-minimization exercise within the tax department, with little connection to valuation, financing or operations. The effective rate is a target in the bonus plan. The deferred tax balances, which are large, have never been examined for what they say about cash. Cash is trapped in subsidiaries because management is uncertain about the tax cost of bringing it home. Nobody can say how much value a one-point change in the effective rate is worth, or how much risk is hiding behind the favourable rate.

Venkataraman decides to rebuild the framework: measure taxes correctly for decisions, understand which parts of the rate are durable and which are fragile, value the tax shields and the risks, and align tax with the operating and financing strategy rather than allowing it to drive them. This chapter walks through that framework.

22.1 Tax Rates and Their Proper Use

22.1.1 A taxonomy of rates

"The tax rate" is not one number. Using the wrong one is among the most frequent technical errors in corporate finance.

- **Statutory rate:** the headline rate set by law for a jurisdiction (often combined with sub-national taxes to give a combined statutory rate).
- **Marginal rate:** the tax rate on the next unit of taxable income, including all layers of tax (national, regional, any surtaxes) and taking into account whether the firm actually pays tax at the margin (a firm with large losses has a marginal rate near zero today).
- **Average (book) effective rate:** total income tax expense (current plus deferred) divided by pre-tax accounting income.
- **Cash tax rate:** income taxes actually paid in the period divided by pre-tax income (or operating income). It differs from the effective rate by the change in deferred taxes and timing of payments.
- **Operating tax rate:** the rate applied to operating profit when computing NOPAT, which strips out the effect of interest and non-operating items.

The right rate depends on the question. For evaluating an incremental project in a jurisdiction where the firm is profitable, use the *marginal* rate in that jurisdiction, because the project's taxes will be the extra tax paid. For valuing the firm's operations, use the expected *cash operating* tax rate on operating profit over time (which may rise towards the marginal rate as tax credits expire or loss carryforwards are used). For valuing the interest tax shield, use the marginal rate at which interest is deducted, subject to limitations on deductibility. For comparing performance across years, the effective rate is informative but it is the noisiest of all.

> **Common mistake.** Applying a low effective rate (driven by one-time credits, valuation allowance releases or a favorable mix) to the terminal value of a business. The effective rate in the terminal year should converge to the marginal rate unless there is a structural, durable reason for it to remain low.

22.1.2 Reconciling statutory and effective rates

Financial statements include a rate reconciliation showing how the statutory rate is transformed into the effective rate. It is one of the best sources of insight into the quality of the rate. Items fall into groups:

- **Permanent differences** that never reverse: non-deductible expenses (certain fines, entertainment, a portion of compensation), tax-exempt income, and the effect of different rates abroad.
- **Credits and incentives:** R&D credits, investment incentives, tax holidays.
- **Changes in uncertain tax positions and valuation allowances,** which are accounting judgments about the likelihood of obtaining or preserving a benefit.
- **One-time items:** effects of tax law changes, settlements of audits.

An analyst or CFO should ask of each item: is it recurring or not, is it tied to real operations or just structure, and is it at risk of change in law or interpretation?

Example 22.1 – Reconciling the rate and finding the right operating tax rate. Brightwater reports EBIT of 550 and interest expense of 50, giving pre-tax income of 500 (millions). The combined statutory rate is 25 percent. Items affecting tax expense: the lower rate on foreign income reduces tax by 20; R&D credits reduce it by 10; non-deductible expenses increase it by 8; an increase in the valuation allowance on deferred tax assets increases it by 2. Current tax expense is 95 and deferred tax expense is 10.

Step 1 – statutory tax. 500 × 25% = 125.

Step 2 – reconciliation. 125 − 20 − 10 + 8 + 2 = 105. Check: current + deferred = 95 + 10 = 105 ✓.

Step 3 – rates. Effective rate = 105 / 500 = 21.0 percent. Cash tax rate = 95 / 500 = 19.0 percent (taxes actually due for the year; this ignores timing of payment).

Step 4 – operating tax rate for NOPAT. The firm's tax expense reflects the shield on interest: 50 × 25% = 12.5. Taxes on operating profit as if unlevered = 105 + 12.5 = 117.5, a rate of 117.5 / 550 = 21.4 percent. To convert to a cash basis, subtract the increase in the deferred tax liability (10), since that portion of the expense was not paid: cash operating taxes = 117.5 − 10 = 107.5, a rate of 107.5 / 550 = 19.5 percent. NOPAT on a cash-tax basis = 550 − 107.5 = 442.5.

Step 5 – what is durable? Of the 20-point difference in tax (125 − 105), the foreign rate differential (−20) is structural but exposed to minimum-tax rules and audit risk; the R&D credit (−10) is tied to operations but depends on legislation; the non-deductible items (+8) recur; and the valuation allowance (+2) is an accounting judgment. A sustainable operating rate for forecasting beyond the near term might therefore be between 21 percent and 25 percent depending on assumptions about the first two items, not the 19.5 percent shown on a cash basis today.

Interpretation. The distinction between 19.5 percent and 25 percent is worth 5.5 percentage points on a taxable base of 550: about 30 per year. Capitalized at a WACC less growth of 6 percent, that is about 500 of enterprise value. The tax rate used in forecasting is therefore a major valuation assumption and should be defended like any other.

22.2 Deferred Taxes

22.2.1 Temporary differences

Deferred taxes arise because the timing of recognizing revenues and expenses differs between the tax return and the financial statements. Under the balance-sheet liability method (IAS 12 under IFRS and ASC 740 under US GAAP follow the same basic approach with differences in detail), a *temporary difference* is the difference between the carrying amount of an asset or liability in the financial statements and its tax base. A taxable temporary difference gives rise to a **deferred tax liability (DTL)**: the firm will pay more tax in the future when the difference reverses. A deductible temporary difference gives rise to a **deferred tax asset (DTA)**: the firm will pay less in the future, provided it expects to have taxable profit to use it. Tax loss carryforwards and unused credits also generate DTAs.

Typical sources of DTLs: accelerated tax depreciation relative to book depreciation; capitalized development costs deducted for tax when paid; unrealized gains recognized in accounts but taxed on realization; fair-value step-ups in acquisitions that are not recognized for tax. Typical sources of DTAs: warranty and pension provisions deductible only when paid; impairments not yet deductible; deferred revenue taxed when received; tax losses.

DTAs are recognized only to the extent that realization is probable (IFRS) or more likely than not (US GAAP); where this is not the case, a **valuation allowance** (or non-recognition) reduces the asset. Releasing such an allowance when profitability improves produces a one-time reduction in tax expense that is not a part of recurring earnings power.

22.2.2 The cash effect of deferral

A DTL is not an ordinary debt. It is a liability only if the timing difference will actually reverse, and for a growing firm that keeps investing it may never reverse in aggregate: each year's new accelerated deductions offset the reversal of old ones. Yet the *time value* of the deferral is real: paying tax later rather than sooner is an interest-free loan from the government. Valuation should capture this by using cash taxes in the forecast, not book taxes.

Example 22.2 – Accelerated depreciation and the deferred tax liability. Brightwater buys equipment for 1,000 (millions). Book depreciation is straight-line over five years: 200 a year. For tax, depreciation is accelerated: 400, 300, 200, 100 and 0 in years 1 to 5. The tax rate is 25 percent and the discount rate is 9 percent.

Step 1 – cumulative temporary difference (tax depreciation minus book depreciation, cumulative). Year 1: 400 − 200 = 200. Year 2: 200 + (300 − 200) = 300. Year 3: 300 + (200 − 200) = 300. Year 4: 300 + (100 − 200) = 200. Year 5: 200 + (0 − 200) = 0.

Step 2 – DTL at 25 percent. Year 1: 50. Year 2: 75. Year 3: 75. Year 4: 50. Year 5: 0.

Step 3 – annual deferred tax expense (change in DTL). Year 1: +50; Year 2: +25; Year 3: 0; Year 4: −25; Year 5: −50. The total over five years is zero: the deferral is purely a matter of timing.

Step 4 – value of the deferral. The present value of the tax saved on tax depreciation is Σ 25% × tax dep_t / 1.09^t = 0.25 × (400/1.09 + 300/1.09² + 200/1.09³ + 100/1.09⁴) = 0.25 × (366.97 + 252.50 + 154.44 + 70.84) = 0.25 × 844.75 = 211.19. The present value on the straight-line schedule is 0.25 × 200 × (1/1.09 + 1/1.09² + 1/1.09³ + 1/1.09⁴ + 1/1.09⁵) = 50 × 3.8897 = 194.48. The value of acceleration is 211.19 − 194.48 = 16.71, 1.7 percent of the asset cost.

Interpretation. The DTL peaks at 75 and then reverses, and the benefit of deferral is 16.7 in present value. If the firm keeps buying equipment every year, the aggregate DTL on the balance sheet stays positive and growing, which is why analysts debate whether to treat it as debt, as equity or to ignore it. A practical convention in valuation: include deferred taxes in cash taxes in the free cash flow forecast, so that the benefit is counted once; do not additionally treat the balance-sheet DTL as equity or deduct it as debt, unless the firm is expected to stop investing.

22.2.3 Treatment in valuation and measures

Three consistent treatments exist. (1) *Cash-tax FCF approach:* forecast cash operating taxes (including the effect of deferral) and exclude the DTL from the debt-like items in the bridge from enterprise value to equity value. (2) *Book-tax approach:* forecast taxes at the book effective rate and treat the DTL as a liability to be settled at its present value, which requires an assumed reversal pattern. (3) *Equity-equivalent approach in EVA:* add the DTL to invested capital as quasi-equity and use cash taxes in NOPAT (as in Chapter 21). The error to avoid is mixing: using cash taxes in NOPAT *and* deducting the DTL as debt double-counts the deferral. DTAs deserve skepticism: a DTA from a tax loss is only as valuable as the probability and timing of future taxable profits (see Section 22.2.4).

22.2.4 Tax loss carryforwards

Net operating losses (NOLs) can usually be carried forward to offset future taxable profits, subject to rules that may limit the annual amount used, the number of years, or the use after an ownership change. Their value is the present value of tax savings, which depends on when profits arise.

Example 22.3 – Valuing an NOL. A company has a tax loss carryforward of 300. Expected taxable income before NOL use over the next four years is 50, 80, 120 and 150. The tax rate is 25 percent and the discount rate is 9 percent. (Assume no annual limitation.)

Step 1 – usage. Year 1: use 50 (remaining 250). Year 2: use 80 (remaining 170). Year 3: use 120 (remaining 50). Year 4: use 50 (remaining 0), leaving 100 of income taxable.

Step 2 – annual tax saving. 50 × 25% = 12.5; 80 × 25% = 20.0; 120 × 25% = 30.0; 50 × 25% = 12.5.

Step 3 – present value. $12.5/1.09 + 20/1.09^2 + 30/1.09^3 + 12.5/1.09^4 = 11.47 + 16.83 + 23.17 + 8.86 = 60.32$.

Step 4 – compare with face. The undiscounted value is 300 × 25% = 75. The NOL is worth 60.32, or about 80 percent of face, because of timing. If profits were expected to arrive more slowly or the discount rate were higher, the percentage would fall; if an ownership-change rule capped annual use, the value would fall further and could approach zero if profits never come.

Interpretation. NOLs should be valued by scenario, and typically discounted at a rate that reflects the risk of the underlying taxable income, which is higher than the cost of debt. Acquirers who pay full face value for NOLs usually overpay, because the NOL is subject to restrictions after a change of control and to the target's own profitability.

22.3 Tax Shields and Capital Structure

22.3.1 The interest tax shield

Interest is generally tax deductible while dividends are not, so debt financing reduces the total tax paid by the firm and its investors' claims together. Volume 1, Chapter 15 introduced the Modigliani–Miller proposition with corporate taxes: for permanent debt D at marginal tax rate t, the value of the levered firm is

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V_L = V_U + t × D
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where t × D is the present value of the tax shield when the shield is discounted at the cost of debt (appropriate when the debt amount is fixed and the shield is as certain as the debt's interest). If the firm instead maintains a constant leverage ratio, the shield's riskiness is more like that of the firm's assets, and a higher discount rate (such as the unlevered cost of equity) is appropriate, which lowers the value of the shield. The choice of assumption matters and should be explicit.

Example 22.4 – Value of the shield under different assumptions. Brightwater has 800 of debt at a 6 percent interest rate, and a marginal tax rate of 25 percent.

Step 1 – annual shield. Interest = 800 × 6% = 48; shield = 48 × 25% = 12.

Step 2 – permanent debt. PV = 12 / 0.06 = 200, equal to t × D = 0.25 × 800 = 200 ✓.

Step 3 – five-year bullet loan, repaid at the end of year 5. The shield is 12 for five years: $PV = 12 \times [1 - 1.06^{-5}] / 0.06 = 12 \times 4.2124 = 50.55$.

Step 4 – debt that grows at 2 percent a year, with shields discounted at an unlevered cost of 9 percent: PV = 12 / (0.09 − 0.02) = 171.4 for the first-year shield of 12. If the shield is still discounted at 6 percent, 12 / (0.06 − 0.02) = 300; the difference between 171.4 and 300 shows how sensitive the result is to the discount rate.

Interpretation. The same debt gives a shield worth anywhere from about 50 (temporary) to 300 (growing and certain). The permanent-debt textbook figure of 200 is an intermediate case. The shield has value only as long as the firm has taxable profit to deduct against and the debt is actually maintained.

22.3.2 What limits the benefit of debt

Several forces lower the net tax advantage of debt.

Personal taxes. Miller's (1977) analysis recognized that investors pay tax on interest income and on equity income at possibly different rates. With a corporate rate t_c, a personal tax rate on equity income t_e and a personal tax rate on interest t_d, the gain from leverage per unit of debt is

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Gain = [1 − (1 − t_c)(1 − t_e) / (1 − t_d)] × D
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With t_c = 25 percent, t_e = 15 percent and t_d = 30 percent, the term (0.75 × 0.85) / 0.70 = 0.9107, and the gain is 1 − 0.9107 = 8.9 percent of D, much lower than the 25 percent corporate shield alone. Where equity income is lightly taxed relative to interest, the net advantage of debt shrinks, and in extreme cases disappears. The empirical size of the personal-tax offset is debated and varies by country and investor type, but the direction is clear.

Limits on deductibility. Many countries restrict the deduction of net interest to a fixed share of earnings (a rule often of the order of 30 percent of a defined earnings measure such as EBITDA or EBIT, though the exact percentage, base and exceptions vary by jurisdiction and change over time; check current local rules). Disallowed interest may be carried forward but may not be deductible at the margin. Transfer-pricing and "thin capitalization" rules also limit intra-group interest. Highly leveraged buyouts and infrastructure-style structures are most affected.

Non-debt tax shields. Depreciation, credits and losses reduce taxable income and thereby reduce the value of interest deductions at the margin: a firm with large non-debt shields may have a low or zero marginal rate for debt, as DeAngelo and Masulis pointed out.

Financial distress and agency costs. Debt raises the probability of financial distress, with direct and indirect costs, and the trade-off theory sets the marginal tax benefit against the marginal expected distress cost (Volume 1, Chapter 15).

Example 22.5 – An interest limitation rule. Brightwater's EBITDA is 1,000 and its interest expense is 400. A rule allows net interest deduction up to 30 percent of EBITDA. The tax rate is 25 percent.

Step 1 – deductible cap. 0.30 × 1,000 = 300.

Step 2 – disallowed interest. 400 − 300 = 100.

Step 3 – shield realized. Allowed: 300 × 25% = 75. Unconstrained: 400 × 25% = 100. Lost current shield: 25.

Step 4 – marginal view. On the margin, an extra 1 of interest saves no tax this year. The *effective marginal shield rate* on additional debt is zero until EBITDA rises or the rules change, apart from the carried-forward disallowed amount (valued at its probability and timing of use). The cost of debt for the next financing decision is therefore the pre-tax rate, not the after-tax rate: 6 percent, not 4.5 percent.

Interpretation. The after-tax cost of debt in the WACC formula, Kd(1 − t), assumes the full marginal shield is available. For a constrained company it overstates the benefit of debt, understates WACC, and encourages too much leverage. In practice, an analyst should test whether the cap binds, in each forecast year, and adjust either the shield or the tax rate on interest.

22.3.3 Managerial implications for financing

Debt policy should be informed by tax but not driven by it. Check marginal rates at the borrower: debt should be placed in the entity that has taxable income against which to deduct it and in the jurisdiction with the highest effective marginal rate, subject to interest-limitation and thin-capitalization rules. Consider the stability of the shield: governments change rules, and a tax-driven structure that exceeds the firm's operating debt capacity often fails in the first downturn. And avoid the "free lunch" illusion: tax savings on debt are real but modest relative to the swings in value from operating decisions.

22.4 International Tax Structure and Transfer Pricing

22.4.1 How countries tax cross-border income

A multinational group is a collection of legal entities in different jurisdictions, each subject to local tax law. Several concepts organize the landscape.

Residence and source. A country may tax the worldwide income of its residents (residence-based taxation) and the income arising within its borders, whoever earns it (source-based taxation). Overlap creates the risk of double taxation, which countries mitigate through unilateral relief and bilateral tax treaties.

Worldwide versus territorial systems. Under a *worldwide* system the home country taxes foreign profits of its resident companies, usually allowing a credit for foreign taxes paid, so that the total tax rate on foreign income is at least the home rate when the profit is brought home. Under a *territorial* (participation exemption) system the home country exempts most foreign-source dividends or profits, so that the foreign rate largely determines the final burden. Real systems are hybrids: many "territorial" systems include anti-abuse rules and tax on passive or low-taxed income earned by controlled foreign subsidiaries, and many "worldwide" systems allow deferral until repatriation. Features and rates change frequently, so for decisions one must consult current law in each jurisdiction.

Withholding taxes. Source countries often withhold tax on dividends, interest and royalties paid to non-residents, at rates reduced by treaty. Withholding taxes are a real cost of moving profit through the group and a driver of holding-company structures.

Foreign tax credits. A credit reduces home tax by foreign tax paid, usually limited to the home tax attributable to the foreign income. Excess credits (where foreign tax exceeds the limit) cannot ordinarily be refunded and may be carried over or lost.

Permanent establishment and nexus. A foreign business is generally taxed on income attributable to a fixed place of business or similar presence. Digital and remote business models have put pressure on this concept, and some countries have introduced additional taxes on digital activities.

22.4.2 Transfer pricing: the arm's-length principle

Intercompany transactions (sale of goods, services, loans, licences of IP) must be priced for tax. Because the group can shift profit by changing prices, tax authorities require that prices be consistent with what independent parties would have agreed under comparable circumstances: the **arm's-length principle**, set out in the OECD Transfer Pricing Guidelines and incorporated into most countries' laws. Taxpayers are expected to document the analysis. Common methods include:

Method	Idea	Typical use
Comparable uncontrolled price (CUP)	Compare price with that charged between independent parties for the same product	Commodities; licences with close comparables
Resale price	Start from the resale price to independent customers and deduct an arm's-length gross margin	Distribution entities
Cost plus	Add an arm's-length markup to costs	Contract manufacturing; services
Transactional net margin method (TNMM)	Compare the net profit margin of the tested party with that of comparable independent companies	The most widely used for routine entities
Profit split	Divide combined profit according to contributions	Highly integrated operations with unique contributions on both sides

Economic substance is increasingly central: under current guidelines the allocation of returns from intangibles is tied to who performs and controls the functions of developing, enhancing, maintaining, protecting and exploiting them (and bears the associated risk and funding), not merely to legal ownership. A paper owner with no staff and no ability to bear risk tends to earn only a routine return.

22.4.3 Quantifying transfer pricing

Example 22.6 – Transfer pricing and total tax. Brightwater's home company (country A, tax rate 30 percent) manufactures a device at a cost of 60 per unit and sells it to its foreign distribution subsidiary (country B, rate 12 percent), which incurs selling costs of 15 per unit and sells to external customers at 100. Volume is 1 million units a year.

Step 1 – effect of the intercompany price. Let P be the transfer price. Profit in A = (P − 60) × 1m; profit in B = (100 − P − 15) × 1m. - P = 70: A = 10m, B = 15m. Tax = 0.30 × 10 + 0.12 × 15 = 3.0 + 1.8 = 4.8m. - P = 80: A = 20m, B = 5m. Tax = 6.0 + 0.6 = 6.6m. - P = 90: A = 30m, B = −5m. Tax = 9.0 − 0.6 = 8.4m (assuming that B's loss yields a tax benefit, which may be doubtful).

Step 2 – the arm's-length test using TNMM. Suppose comparable independent distributors earn an operating margin of 4 to 8 percent of sales, with a median of 6 percent. B's arm's-length profit is therefore 4 to 8 per unit (6 at the median). The corresponding transfer price is P = 100 − 15 − margin: 81 (at 4 percent), 79 (at 6 percent) or 77 (at 8 percent). At the median, A earns (79 − 60) × 1m = 19m and B earns 6m. Tax = 0.30 × 19 + 0.12 × 6 = 5.70 + 0.72 = 6.42m.

Step 3 – the cost of aggressive pricing. At P = 70, B earns 15m (15 percent of sales), far above the comparables' range, and A earns less than the arm's-length level. The nominal tax saving relative to the median arm's-length price is 6.42 − 4.80 = 1.62m a year. Suppose country A's authority adjusts the price to 79: A's profit becomes 19m and its tax 5.70m. If B's authority does not grant a corresponding adjustment, B's profit of 15m remains taxed at 12 percent: 1.80m. Total tax = 5.70 + 1.80 = 7.50m, which exceeds the arm's-length figure of 6.42m by 1.08m a year, before interest and penalties.

Interpretation. Pricing outside the arm's-length range yields a small expected saving in the best case and creates a double-taxation exposure when authorities disagree. The expected value of an aggressive position must be calculated net of the probability of challenge, double taxation, interest and penalties, and reputation costs. Many groups use advance pricing agreements (APAs) with the relevant authorities to obtain certainty, and mutual agreement procedures under tax treaties to resolve double taxation when it arises.

22.4.4 Operating strategy and tax

A durable approach is to locate real activities where they make economic sense, with tax as one of the inputs: access to talent, proximity to customers, supply chain costs, political risk. Tax-driven structures that lack economic substance are fragile: they are the ones that anti-avoidance rules, the OECD's base erosion and profit shifting (BEPS) work, and minimum tax rules are designed to counter. A good test is whether the structure would be adopted if the tax benefit were zero. If not, assume the benefit is temporary and price in a risk of loss.

22.5 Repatriation and Financing Foreign Operations

22.5.1 The repatriation decision

Cash earned by foreign subsidiaries can be returned to the parent through dividends, interest on intercompany loans, royalties and service fees, or capital reductions. The tax cost depends on the home system, source-country withholding taxes and any residual home-country tax. Where there is a significant cost to bringing cash home and no immediate need for it, groups have accumulated cash abroad. This "trapped cash" matters for valuation (the cash is worth less than its book amount if the repatriation tax is material) and for capital allocation (cash that cannot be deployed at home may fund foreign acquisitions that would not otherwise meet the hurdle rate: an example of tax-induced distortion).

Under IFRS and US GAAP, a deferred tax liability is generally recognized on unremitted earnings of subsidiaries unless the parent controls the timing of the reversal and it is probable that the difference will not reverse in the foreseeable future (US GAAP uses the phrase "indefinitely reinvested" for the equivalent exception). The disclosure of unrecognized amounts is useful for analysis.

Example 22.7 – Repatriation under different systems. A foreign subsidiary earns 100 of pre-tax profit. The local tax rate is 12 percent. Dividends are subject to 5 percent withholding. The home rate is 25 percent, and the home country grants a credit for foreign tax paid, including the withholding tax and the underlying local tax, limited to the home tax on the foreign income.

Step 1 – local tax and distributable profit. Local tax = 12; profit after local tax = 88.

Step 2 – withholding. Dividend of 88 incurs 5% × 88 = 4.4 of withholding. Cash received by the parent before home tax = 83.6.

Step 3 – home tax with credit (worldwide system). Home tax on grossed-up income of 100 = 25. Credit = 12 (underlying) + 4.4 (withholding) = 16.4. Residual home tax = 25 − 16.4 = 8.6. Cash retained = 83.6 − 8.6 = 75.0. Total tax = 12 + 4.4 + 8.6 = 25.0 (25 percent, the home rate).

Step 4 – territorial system with full exemption of foreign dividends. Residual home tax = 0. Cash retained = 83.6. Total tax = 12 + 4.4 = 16.4 percent of the original pre-tax profit.

Step 5 – high-tax subsidiary under the worldwide system. If the local rate were 35 percent, profit after local tax would be 65, withholding 3.25, and the foreign tax of 38.25 exceeds the home tax of 25. The credit is limited to 25, so there is no residual home tax; the excess of 13.25 is a credit carried over, if permitted, or lost. Cash retained = 65 − 3.25 = 61.75; total tax = 38.25 percent.

Step 6 – the value of deferral (worldwide system, low-tax subsidiary). If the dividend is deferred for five years, the residual home tax of 8.6 is also deferred. At a 9 percent discount rate its present value is $8.6 / 1.09^5 = 8.6 / 1.5386 = 5.59$, so deferral reduces the effective residual tax by 8.6 − 5.59 = 3.01 per 100 of pre-tax profit, plus whatever the subsidiary earns on the retained cash.

Interpretation. Under a worldwide system with credits, the final tax rate on repatriated profit equals the higher of the home and foreign rates, so the low foreign rate is a benefit only for as long as profits remain abroad, and its value is therefore the value of deferral. Under a territorial system the foreign rate governs and there is no tax cost of repatriation beyond withholding. For valuation, the right approach is to apply the expected *total* tax on the profit in each jurisdiction (including the expected residual cost of repatriation weighted by when and whether it will occur) and to value trapped cash at its expected after-tax amount.

22.5.2 Intra-group financing

Groups fund subsidiaries through equity and intercompany loans. Interest payments reduce the borrower's taxable profit and are taxable to the lender, subject to withholding, so loans from low-tax to high-tax entities shift profit. The tools used by tax authorities include the arm's-length test on the interest rate and amount of debt, thin-capitalization rules, interest limitation rules (Section 22.3.2) and anti-hybrid rules that deny deductions for payments that do not produce a corresponding income inclusion. Treasury should design cash pooling and internal funding for liquidity first and tax second, since policies that are effective on paper but fragile under audit create more risk than value.

22.5.3 Currency, hedging and tax

Foreign-exchange gains and losses, and hedging results, may be taxed in a different period or at a different rate than the underlying exposures. A hedge of a net investment in a foreign subsidiary can produce gains in the parent's taxable income while the offsetting translation loss lies in equity and is not currently deductible. Analysis of hedge programs should therefore be on an after-tax basis (see the discussion of derivatives in Volume 1 and the risk-management chapters of this volume).

22.6 Global Minimum Tax: Concepts

The following describes a rapidly evolving area. The rules and their local implementation are complex and change often. This section explains the logic and gives an illustrative calculation; it does not substitute for current law or professional advice, and the details (rates, carve-outs, transitional relief, status in each country) should be verified for any specific decision.

22.6.1 Motivation and architecture

For decades, countries competed on corporate tax rates and incentives, and multinational groups arranged profits to arise in low-tax jurisdictions. In response, a large group of countries agreed, under the OECD/G20 Inclusive Framework, on a two-pillar approach. Pillar One concerns the allocation of a part of the profits of the very largest and most profitable groups to market jurisdictions; its implementation has been uncertain. Pillar Two, commonly called the **global minimum tax** or GloBE rules (Global Anti-Base Erosion), is designed to ensure that large multinational groups, above a revenue threshold of the order of 750 million euros in consolidated revenue, pay a minimum effective tax rate of 15 percent in each jurisdiction in which they operate.

The key mechanism is the computation of a jurisdictional effective tax rate (ETR): covered taxes paid or accrued by the group's entities in a jurisdiction divided by the GloBE income there (a measure derived from financial accounting income with specified adjustments). If the ETR is below 15 percent, a **top-up tax** is due, equal to the shortfall in percentage points times the excess profit, where excess profit is the GloBE income less a **substance-based income exclusion** (a formulaic return on payroll and tangible assets, intended to protect genuine activity). The top-up can be collected by the jurisdiction

itself through a *qualified domestic minimum top-up tax* (which allows the local government to capture the revenue rather than leave it to another country) or by other jurisdictions through the *income inclusion rule* (applied by the parent's country) and an *undertaxed profits rule* as a backstop. Transitional safe harbors and simplifications exist for the early years.

22.6.2 An illustrative calculation

Example 22.8 – A simplified top-up tax. A group's manufacturing subsidiaries in one jurisdiction have GloBE income of 200 and covered taxes of 14. The minimum rate is 15 percent. The group's payroll and tangible assets in the jurisdiction give a substance-based income exclusion of 30 (for illustration: 5 percent of a carve-out base of 600; actual percentages and transitional rules differ).

Step 1 – jurisdictional ETR. 14 / 200 = 7.0 percent.

Step 2 – top-up percentage. 15% − 7.0% = 8.0 percentage points.

Step 3 – excess profit. 200 − 30 = 170.

Step 4 – top-up tax. 170 × 8.0% = 13.6. (Without the exclusion it would be 200 × 8.0% = 16.0.)

Step 5 – total tax. 14 + 13.6 = 27.6, or 13.8 percent of GloBE income of 200. The effective rate is still below 15 percent because of the exclusion: the carve-out effectively exempts 30 of income from top-up.

Interpretation. The jurisdiction's rate advantage over 15 percent is lost; in this example the group pays 13.6 in additional tax to some government. If the local jurisdiction had introduced a qualified domestic top-up tax, the 13.6 would be paid locally and the other countries would not claim it. The group's incentive therefore changes from "where is the lowest rate?" to "where do we have real activity that earns the substance-based exclusion, and who collects the top-up?"

22.6.3 Strategic implications

Three broad implications follow, stated cautiously. First, planning that relies on a low rate with little substance loses value for groups within the scope of the rules; the remaining benefit of low-tax locations depends on real activity (payroll and assets), on credits whose treatment within the rules is favorable (some refundable credits are treated more favorably than others), and on the allocation of top-up revenue. Second, a corporate tax rate below 15 percent is no longer a reliable tool for a government to attract investment from in-scope groups, which will likely shift competition towards other instruments, such as investment-related incentives. Third, compliance burdens increase significantly: groups need data processes to compute jurisdictional ETRs. A prudent valuation approach is to model scenarios for a group's tax rate, with and without the minimum tax, and to treat any residual rate advantage below 15 percent as at risk.

22.7 Tax and Valuation

22.7.1 Where taxes enter the valuation

Taxes appear in a valuation in four places: in the cash taxes deducted to compute free cash flow (the operating tax rate and deferral pattern), in the cost of capital (through the after-tax cost of debt), as the explicit value of financing-related shields (in APV), and in specific tax assets such as NOLs. The principal risk is double counting. If WACC is computed with an after-tax cost of debt, the interest tax shield is already in the discount rate, so free cash flow must be computed from taxes on operating profit *without* the interest shield (NOPAT-based FCF). If instead the model discounts at the unlevered cost of capital (APV), the shield must be added separately.

22.7.2 Adjusted present value with taxes

The APV approach separates operating value from financing side effects, making each tax effect visible and allowing different assumptions for each (Volume 1, Chapter 21 on enterprise valuation). It is particularly useful when the capital structure changes over time, as in leveraged transactions, or when interest limitations make the shield uncertain.

Example 22.9 – APV with an interest tax shield. Brightwater's unlevered free cash flows are 100, 110, 121 and 133.1 in years 1 to 4. After year 4 they grow at 3 percent a year. The unlevered cost of capital is 9 percent. The firm will maintain permanent debt of 600 at 6 percent. The marginal tax rate is 25 percent.

Step 1 – terminal value at year 4. TV = 133.1 × 1.03 / (0.09 − 0.03) = 137.093 / 0.06 = 2,284.88.

Step 2 – PV of explicit flows. $100/1.09 + 110/1.09^2 + 121/1.09^3 + 133.1/1.09^4 = 91.74 + 92.59 + 93.44 + 94.29 = 372.06$.

Step 3 – PV of terminal value. $2{,}284.88 / 1.09^4 = 2{,}284.88 / 1.41158 = 1{,}618.66$.

Step 4 – unlevered value. V_U = 372.06 + 1,618.66 = 1,990.72.

Step 5 – tax shield. Annual interest = 600 × 6% = 36; shield = 36 × 25% = 9. PV of a permanent shield discounted at the cost of debt = 9 / 0.06 = 150 (= t × D = 0.25 × 600).

Step 6 – levered value. V_L = 1,990.72 + 150 = 2,140.72. Equity value = 2,140.72 − 600 = 1,540.72.

Step 7 – sensitivity: binding interest limitation. Suppose only 60 percent of the interest is deductible because of a limitation rule. Shield = 0.60 × 150 = 90, so V_L = 1,990.72 + 90 = 2,080.72 and equity = 1,480.72. The loss relative to the base is 60, or 3.9 percent of equity value.

Interpretation. The shield is 7.0 percent of the levered value in the base case; its size is no trivial matter but it is dwarfed by the operating value. This is why taxes should be modeled carefully, but not made the centerpiece of strategy. Note too that the shield must be discounted at a rate consistent with its risk: using 6 percent assumes a fixed, permanent amount of debt; for debt that tracks firm value, a rate closer to the unlevered cost of capital is appropriate (Example 22.4).

22.7.3 The sensitivity of value to the tax rate

Because tax is a cost on a perpetual stream, value is highly sensitive to the long-run rate. A change in the statutory rate affects value through cash flow directly and, for levered firms, through the shield, which partially offsets.

Example 22.10 – A change in the tax rate. An unlevered business has EBIT of 133.33 (constant growth of 3 percent in all years), a discount rate of 9 percent and a 25 percent tax rate, so that next year's NOPAT is 100. Assume no net investment cost beyond what is embedded in free cash flow equal to NOPAT (that is, growth is costless, a simplification to isolate the tax effect).

Base: V = 100 / (0.09 − 0.03) = 1,666.7.

Rate rises to 28 percent: NOPAT = 133.33 × 0.72 = 96.0; V = 96.0 / 0.06 = 1,600.0. Change: −66.7, or −4.0 percent.

Rate falls to 21 percent: NOPAT = 133.33 × 0.79 = 105.33; V = 105.33 / 0.06 = 1,755.6. Change: +88.9, or +5.3 percent.

Interpretation. A three-point rise costs 4.0 percent of value and a four-point cut adds 5.3 percent; the asymmetry in percentages comes from the base. For a firm with a large share of value in the terminal period, assumptions about long-run tax rates deserve scrutiny similar to assumptions about the terminal growth rate. For a levered firm the net effect is smaller because a higher rate raises the value of the interest shield: the combined effect depends on the leverage and the deductibility of interest.

22.7.4 Tax in M&A

Taxes affect the structure and price of acquisitions. In an asset purchase (or a share purchase with a tax election to treat it as an asset purchase), the buyer obtains a stepped-up tax basis in the assets and can amortize goodwill and other assets for tax, which has value; the seller may pay more tax. In a share purchase without a step-up, the buyer inherits the old tax basis and any deferred taxes and exposure; the sale may be taxed at lower rates for the seller. Acquired tax attributes (NOLs, credits) face restrictions. The difference in structure shifts value between the parties and should be modeled explicitly as part of the deal valuation (Volume 1, Chapter 22). Buyers should also diligence uncertain tax positions, transfer-pricing exposures and withholding obligations that transfer with the company, and use indemnities or price adjustments.

22.7.5 Uncertain tax positions and tax risk

Many corporate tax positions are interpretations of ambiguous law. Under both IFRS and US GAAP, firms must assess whether an uncertain tax treatment is likely to be accepted and recognize a provision or reduce the benefit when it is not (IFRIC 23 under IFRS; ASC 740 uncertain tax position guidance in the US). These provisions are estimates and the disclosures (reserves, the range of reasonably possible changes) are valuable to analysts. As a valuation matter, a low effective tax rate backed by a thick layer of uncertain positions has a different quality from one backed by statutory credits.

22.8 Governance of Tax Strategy

22.8.1 Tax as one stakeholder claim among several

Tax is a real cost, and reducing it within the law is a legitimate objective. But it is not costless to pursue: aggressive positions produce audits, penalties, reputational cost, and sometimes political backlash. Increasingly, investors, customers and regulators view tax conduct as part of corporate responsibility, and some jurisdictions require public country-by-country reporting. A sensible governance position is to make the board accountable for a tax policy that states the group's risk appetite, its principles on structures without substance, its approach to transparency, and the relationship with tax authorities (cooperative compliance programs, advance rulings).

22.8.2 Aligning incentives

Setting an effective tax rate target in executive compensation encourages the pursuit of rate over value: a lower rate achieved through a structure that costs more in operations, or through positions that later reverse, is a net loss. Better measures are after-tax cash flow, the *sustainable* rate, the volatility and quality of the rate, the amount of uncertain positions, and the number of disputes. Place the tax function in the valuation conversation: it should be able to say how much value is at stake in any proposed restructuring, in present-value terms and net of risk, as well as in the next year's effective rate.

22.8.3 A practical checklist

When evaluating a tax-driven proposal, it helps to ask: Is there economic substance, and would we do this without the tax benefit? What is the present value of the benefit, net of costs of implementation, ongoing compliance and the probability-weighted cost of challenge? Is the benefit durable in light of minimum tax rules, interest limitation rules and anti-avoidance rules? Does it constrain operating flexibility, cash access or financing? What does it do to the effective rate *quality* and to transparency? And finally: how would we explain it, in public, to a skeptical reader?

Case Study: Tax Strategy at Coralis Instruments

Illustrative case. Coralis Instruments is a fictional company; all figures are invented.

Coralis Instruments makes precision measurement devices. The company is headquartered in Country H (combined statutory rate 28 percent), with a manufacturing and IP-holding subsidiary in Country L (statutory rate 9 percent) and sales subsidiaries in five other countries. Over the past decade the group's effective tax rate fell from 27 percent to 14 percent. Group revenue is 4.2 billion, comfortably above the threshold for global minimum tax rules, and Country L has announced that it will introduce a domestic minimum top-up tax at 15 percent from next year.

The manufacturing subsidiary in Country L earns profit before tax of 600 million, taxed at an effective 9 percent. Its payroll and tangible assets are modest: the company transferred ownership of the key patents to Country L ten years ago, but most R&D staff remain in Country H and the senior engineers who decide the research agenda are there. Local employees number about 300. An illustrative substance-based exclusion for the subsidiary is 45 million. The tax authority in Country H has opened an inquiry into the royalty-free licensing of the patents to the Country L entity and the price at which Country L sells to the distribution subsidiaries.

Meanwhile Coralis holds 1.8 billion in cash abroad. The group's treasurer says that bringing it home would cost 8 percent in residual tax and withholding. The CFO, Lena Marchetti, wants the cash for a share repurchase, and the CEO prefers an acquisition in Country L's industry cluster, which would provide additional local assets and payroll but is priced richly.

Coralis also has acquisition debt of 2.5 billion at 6.5 percent, giving interest expense of 162.5 million against EBITDA of 900 million in Country H. A limit that allows net interest deductions up to 30 percent of EBITDA applies in Country H starting next year.

The finance team is asked by the board to assess the situation and to recommend a strategy. Its first-pass estimates are as follows. If the transfer-pricing audit in Country H results in a reallocation of 200 million of annual profit from L to H, the group would pay roughly 28 percent less 9 percent, or 19 points, on the shifted profit, plus interest and penalties, with a risk that Country L would not grant a corresponding adjustment. Under minimum tax rules, the top-up on Country L's profit, using the illustrative exclusion above and an ETR of 9 percent, would equal (600 − 45) × (15% − 9%) = 33.3 million a year, if collected by Country L's domestic tax, roughly. The interest limitation does not bind today: the cap is 30 percent of 900, or 270 million, so interest of 162.5 million is fully deductible. But the CFO fears that EBITDA could fall in a downturn, and that further debt would then be constrained.

The board asks whether the 14 percent effective rate should be treated as the base for the five-year plan, whether to pursue the share repurchase or the acquisition, and how the group's tax organization should be governed.

Discussion Questions

1. Compute the expected top-up tax on Country L's profit in the case and the group's resulting effective rate on that profit. What does the result imply for the long-term sustainable effective tax rate for the group and for the valuation of the business?
2. Estimate the annual tax cost of a successful transfer-pricing challenge by Country H for 200 million of profit, and the additional cost if Country L does not give relief. How would you set a provision or risk-adjust the value of the existing structure?
3. Evaluate the choices between using offshore cash for a share repurchase, the acquisition, or leaving it abroad. What is the after-tax cost of repatriation on 1.8 billion at 8 percent, and how should the tax cost enter the comparison with the acquisition?

4. Does the interest-limitation rule bind today? Under what EBITDA level would it bind with 162.5 million of interest, and how would that affect the group's debt capacity and its after-tax cost of debt?

Key Takeaways

- Different decisions need different tax rates: marginal for incremental projects and interest shields, cash operating rates for NOPAT and valuation, effective rates for performance analysis with attention to quality.
- The reconciliation of the statutory and effective rates reveals which items are durable and which are temporary; the terminal-year rate should converge to the marginal rate unless a durable reason exists.
- Deferred taxes reflect timing differences; value comes from the time value of deferral and is captured by using cash taxes in free cash flow. Avoid double counting by not also treating the DTL as debt.
- NOLs and other tax attributes are valued as present values of expected tax savings, discounted at a rate reflecting the risk of the taxable income, and discounted for restrictions after ownership changes.
- The interest tax shield is worth $t \times D$ only under strong assumptions; personal taxes, interest-limitation rules, non-debt shields and distress costs reduce it, and the shield's discount rate depends on the debt policy.
- Transfer prices must follow the arm's-length principle; pricing outside the range yields small expected savings and exposure to double taxation, penalties and reputational damage.
- Under a worldwide system with credits, the tax on repatriated profit tends to the higher of the home and foreign rates, and the value of a low foreign rate lies largely in deferral; territorial systems shift the burden to the foreign rate and withholding.
- Global minimum tax rules, stated cautiously, aim to ensure a 15 percent effective rate per jurisdiction for large groups, with a substance-based exclusion; planning that depends on low rates without substance is at risk.
- APV makes tax effects explicit and is well suited for changing leverage and uncertain shields; with WACC, make sure the shield is counted once.
- Tax strategy should follow business substance, be evaluated net of risk and costs, and be governed by the board; effective-rate targets in incentive plans can produce value-destroying behavior.

Review Questions

1. Distinguish the statutory, marginal, effective and cash tax rates. For each of the following decisions, state which rate you would use and why: a new plant, the terminal value of a business, a debt financing, a performance comparison.
2. Why might a firm's effective tax rate be sustainably lower than its statutory rate, and why might it not be?
3. Explain how a deferred tax liability arises from accelerated depreciation, and why a growing firm's aggregate DTL may never reverse.
4. Describe two consistent treatments of deferred taxes in valuation and explain how mixing them leads to double counting.
5. Why are NOLs worth less than their face value multiplied by the tax rate? Name three factors that affect their value.
6. State the MM proposition with corporate taxes. How do personal taxes, interest limitation rules and non-debt tax shields alter the conclusion?
7. What is the arm's-length principle? Describe two transfer pricing methods and the type of transaction each suits.

8. Compare the tax consequences of repatriating foreign profit under a worldwide system with credits and a territorial system. Why does the “trapped cash” phenomenon arise?
9. Explain the logic of a global minimum tax, the role of a substance-based income exclusion, and the reason for a qualified domestic top-up tax.
10. Why can setting an effective-tax-rate target in executive incentives be counterproductive?

Problems

Problem 22.1 – Rate reconciliation. A firm has pre-tax income of 800 and a statutory rate of 25 percent. Lower rates on foreign income reduce tax by 30, credits reduce it by 12 and non-deductible items increase it by 10. Deferred tax expense is 20. Compute total tax expense, the effective rate, current tax and the cash tax rate (current tax divided by pre-tax income).

Problem 22.2 – Deferred tax. An asset costs 600. Book depreciation is 200 a year for three years. Tax depreciation is 300, 180 and 120. The tax rate is 30 percent and the discount rate is 8 percent. Compute the cumulative temporary difference and DTL at each year-end, and the present value gain from accelerated tax depreciation.

Problem 22.3 – Value of shield. A firm has debt of 500 at 7 percent, with a marginal tax rate of 21 percent. Compute the annual shield, its present value if the debt is permanent (discounted at 7 percent), and its present value if the debt is a ten-year bullet loan.

Problem 22.4 – Interest limitation. A firm has EBITDA of 600, interest expense of 250 and a tax rate of 25 percent. Interest deductions are limited to 30 percent of EBITDA. Compute the deductible interest, the disallowed amount and the lost tax shield for the year. What EBITDA is needed for the entire interest to be deductible?

Problem 22.5 – Transfer pricing. A parent in a country with a 28 percent rate makes a product at a cost of 40 and sells it to a subsidiary in a country with a 10 percent rate. The subsidiary has selling costs of 10 per unit and resells at 80 to third parties. Volume is 2 million units. Arm’s-length profit for the subsidiary under TNMM is 5 percent of sales. (a) Compute the arm’s-length transfer price, the profit of each entity and total tax. (b) Compute total tax if the transfer price is 55. (c) What is the apparent tax saving?

Problem 22.6 – Repatriation. A subsidiary earns 200 of pre-tax profit, is taxed locally at 15 percent, and pays a dividend subject to 10 percent withholding. The home rate is 25 percent with a credit for foreign tax including the withholding tax. Compute cash retained and total tax under (a) the worldwide system with credit and (b) a territorial exemption.

Problem 22.7 – Top-up tax. A jurisdiction has GloBE income of 500 and covered taxes of 40. The minimum rate is 15 percent, and a substance-based income exclusion is 80. Compute the ETR, the top-up percentage, the excess profit, the top-up tax and the total tax rate on income after top-up.

Problem 22.8 – NOL value. A firm has an NOL of 400. Taxable income before NOL over the next three years is 100, 150 and 200. The tax rate is 30 percent and the discount rate is 10 percent. Compute the NOL used in each year, the tax savings and the present value of the NOL compared with its undiscounted tax value.

Answers to Selected Problems

22.1. Statutory tax = 800 × 25% = 200. Total tax = 200 − 30 − 12 + 10 = 168; effective rate = 168 / 800 = 21.0 percent. Current tax = 168 − 20 = 148; cash tax rate = 148 / 800 = 18.5 percent.

22.2. Cumulative difference: year 1: 300 − 200 = 100; year 2: 100 + (180 − 200) = 80; year 3: 80 + (120 − 200) = 0. DTL at 30 percent: 30, 24, 0. PV of tax on tax-depreciation shield = 0.30 × (300/1.08 + 180/1.08^2 + 120/1.08^3) = 0.30 × 527.36 = 158.21; on book schedule = 0.30 × 200 × 2.5771 = 154.63; gain = 3.58.

22.3. Annual shield = 500 × 7% × 21% = 7.35. Permanent: 7.35 / 0.07 = 105.0 (= 0.21 × 500). Ten-year bullet: 7.35 × [1 − 1.07^{-10}] / 0.07 = 7.35 × 7.0236 = 51.62.

22.4. Cap = 0.30 × 600 = 180. Disallowed = 250 − 180 = 70. Lost shield = 70 × 25% = 17.5. Full deductibility requires EBITDA ≥ 250 / 0.30 = 833.3.

22.5. (a) Subsidiary arm's-length profit = 5% × 80 = 4 per unit; transfer price = 80 − 10 − 4 = 66. Parent profit = (66 − 40) × 2m = 52m; subsidiary profit = 4 × 2m = 8m. Tax = 0.28 × 52 + 0.10 × 8 = 14.56 + 0.80 = 15.36m. (b) At 55: parent profit = 15 × 2m = 30m; subsidiary profit = (80 − 55 − 10) × 2m = 30m; tax = 8.4 + 3.0 = 11.4m. (c) Apparent saving = 15.36 − 11.4 = 3.96m a year, at risk of adjustment, penalties and double taxation.

22.6. Local tax 30; profit after tax 170; withholding 17; cash before home tax 153. (a) Home tax on 200 = 50; credit = 30 + 17 = 47; residual = 3. Cash retained = 153 − 3 = 150; total tax = 50 (25 percent). (b) Territorial: cash retained = 153; total tax = 47 (23.5 percent).

22.7. ETR = 40 / 500 = 8.0 percent; top-up percentage = 15% − 8% = 7.0 points; excess profit = 500 − 80 = 420; top-up = 420 × 7.0% = 29.4; total tax = 40 + 29.4 = 69.4, or 13.9 percent of 500. (Without the exclusion the top-up would be 35.)

22.8. Usage: year 1: 100; year 2: 150; year 3: 150 (remaining NOL 400 − 250 = 150). Tax savings: 30, 45, 45. PV = 30/1.10 + 45/1.21 + 45/1.331 = 27.27 + 37.19 + 33.81 = 98.27, compared with an undiscounted value of 400 × 30% = 120 (82 percent).

Chapter 23: Climate Finance, the Energy Transition and ESG Investing

Learning Objectives

- Explain how physical and transition climate risks reach a firm's cash flows, discount rate and balance sheet, and why they belong inside the capital budgeting process rather than in a separate sustainability report.
- Build and defend a carbon price assumption (explicit, shadow or internal) and run an NPV analysis under multiple carbon-price scenarios, including pass-through, break-even prices and the value of the option to retire an asset early.
- Evaluate stranded-asset exposure using scenario-weighted valuation, accelerated depreciation and impairment logic, and distinguish a stranded asset from a merely less profitable one.
- Rank decarbonization projects with a marginal abatement cost curve and decide how much abatement to buy internally at a given carbon price.
- Compare green, sustainability-linked, transition and social financing instruments, quantify the "greenium" and step-up economics for an issuer, and recognize the credibility risks (greenwashing) that attach to labelled finance.
- Discuss the theory and the mixed empirical evidence on ESG investing carefully, separating the question "does ESG raise returns?" from "does ESG change the cost of capital?", and describe in general terms the main climate disclosure frameworks (TCFD, ISSB, GHG Protocol scopes and regional regimes).

Opening Vignette

Illustrative scenario. Meridian Cement & Aggregates is a fictional mid-sized building-materials group with eleven plants in three countries, EBITDA of 410 million, and net debt of 1.1 billion.

In March, the board's investment committee receives two proposals. The first is a 380 million replacement of an aging clinker kiln with a modern, more efficient one: same process, same fuel, about 12 percent less carbon per tonne of product. Its internal rate of return, calculated the way the company has always calculated it, is 13.5 percent against a hurdle rate of 9 percent. The second is a 260 million package to switch part of the fuel mix to biomass and waste-derived fuel, install waste-heat recovery, and prepare a site for future carbon capture. Its conventional IRR is 7.4 percent. On the old rules, the first proposal wins and the second is declined.

The newly hired head of strategy asks three awkward questions. What carbon price does the first project's 13.5 percent assume? (Answer: today's free-allowance regime, effectively near zero for the next decade.) What happens to that project if free allocations are phased out, as the local regulator has signalled, and the price of a tonne reaches 80? (The IRR falls to roughly 8 percent, below the hurdle.) And what would a lender or investor in 2035 say about a kiln that is expected to operate until 2060?

The CFO realizes the committee has been valuing cash flows from a world that will not exist. The company also has a sustainability-linked loan that steps up its margin if emissions intensity targets are missed, a green bond programme under discussion, and investors who ask whether the share price embeds "transition risk". The finance team must now translate climate science, policy uncertainty and investor sentiment into discount rates, cash flows, scenarios and financing choices.

This chapter provides the toolkit. It treats climate as a corporate finance problem: a set of uncertain, correlated, policy-driven cash-flow shocks, a set of new financing instruments with explicit economics, and an investor landscape in which the evidence is more nuanced than either advocates or critics suggest.

23.1 Climate as a Corporate Finance Problem

23.1.1 Physical risk and transition risk

Climate-related financial risk is conventionally divided into two families.

Physical risk is damage from climate hazards. *Acute* physical risks are discrete events such as floods, storms, wildfires and heat waves that damage assets and interrupt operations. *Chronic* physical risks are gradual shifts such as sea-level rise, changing rainfall, persistent heat stress and water scarcity that change the productivity or insurability of locations. For a corporate, the channels are direct (repair costs, lost production days), indirect (supply-chain disruption, labour productivity), and financial (insurance premiums and deductibles, collateral values, covenant breaches).

Transition risk arises from the process of moving to a lower-carbon economy: carbon pricing and regulation, technology shifts that make incumbent assets uncompetitive, changes in customer preference, litigation and reputational pressure, and changes in the availability and cost of capital. Transition risk is the main subject of this chapter because it is the form most directly manipulable through corporate capital allocation, and because it is intrinsically a *policy-and-technology* uncertainty rather than a physical-science one.

The two risks interact. A fast, orderly transition raises transition risk and lowers physical risk in the long run; a slow transition does the reverse; and a late, disorderly transition can raise both. This is why scenario analysis, not a single forecast, is the dominant method.

23.1.2 How climate reaches the financial statements

It helps to trace each risk through the standard valuation identity from Volume 1, Chapter 6 and Chapter 21:

```
Value = Σ  FCFF_t / (1 + WACC)^t  +  Terminal value / (1 + WACC)^T
FCFF  = EBIT × (1 − tax) + Depreciation − Capex − ΔNWC
```

Valuation driver	Transition channel	Physical channel
Revenue	Demand shifts, product bans, green premium for low-carbon products	Plant outages, crop and demand shocks
Operating costs	Carbon prices, energy cost changes, compliance cost	Cooling/water cost, repair cost
Capex	Abatement investment, retrofit, electrification	Adaptation, hardening, relocation
Asset lives / depreciation	Earlier retirement, impairment	Shortened useful life of exposed sites
Working capital	Inventory and supply-chain re-sourcing	Safety stocks against disruption
WACC	Changes in investor demand, spreads, beta	Insurance availability, credit spreads
Terminal value	Does the business exist, and at what margin, after 2050?	Is the location viable?

The table makes a central point: *climate risk is not a separate asset class that sits beside the discounted cash-flow model; it is a set of perturbations to the inputs of that model*. The discipline of climate finance is therefore mostly the discipline of making the perturbations explicit, consistent across projects, and linked to a coherent scenario.

23.1.3 Do not put climate risk only into the discount rate

A common shortcut is to "add 100 basis points to the WACC for carbon-intensive projects". This is attractive because it is easy, but it is a poor tool for three reasons.

1. **Timing.** A discount-rate add-on penalizes distant cash flows exponentially and near-term cash flows hardly at all. Carbon costs, however, are usually *back-loaded*: they are small now and large later as prices rise and free allowances disappear. A constant add-on therefore penalizes the wrong years and can be too harsh on a project that earns back its capital before the policy bites, and too lenient on one that depends on decades of high-carbon output.
2. **Asymmetry.** Carbon-pricing risk is a cash-flow risk with a specific, modelable shape (a price path times an emissions quantity). Burying it in a rate hides the thing management can actually act on: emissions per unit, pass-through, abatement options and closure triggers.
3. **Diversifiability confusion.** The CAPM logic in Volume 1, Chapter 10 prices *systematic* risk. A policy shock may be systematic for the carbon-intensive sector without being systematic for the market, and may be positively or negatively correlated with market returns depending on the shock. Mechanically adding spread can double-count (if cash flows already contain expected carbon costs) or under-count (if cash flows contain none).

The principle we adopt is: put *expected* climate cash-flow effects in the cash flows, scenario by scenario; reserve discount-rate adjustments for genuine changes in systematic risk or in the investor base (for example, a firm that is excluded from many portfolios may face a higher cost of capital, as discussed in Section 23.6).

23.2 Carbon Pricing in Capital Budgeting

23.2.1 Explicit, shadow and internal carbon prices

Three distinct concepts are often conflated.

- **An explicit carbon price** is a cost the firm actually pays: a carbon tax, the price of emission allowances under a cap-and-trade (emissions trading) system, or a fuel levy. It is a cash cost, deductible in most tax systems, and it enters operating costs directly.
- **A shadow carbon price** is an analytical value used in project evaluation to reflect a cost the firm *expects or fears* but does not yet pay, such as future regulation. It is applied in the NPV spreadsheet but never invoiced.
- **An internal carbon price (ICP)** is a price a company sets for internal transfers or charges to business units, sometimes actually collected into a fund (an "internal carbon fee") and sometimes used only for decision support. It can serve as a hurdle-rate-like device: every project's emissions are multiplied by the ICP and the result is charged against the project.

A well-designed system uses the explicit price where the firm is already covered by a pricing scheme, the shadow price for the portion of the horizon beyond what is currently legislated, and an ICP to convey a consistent signal across divisions. The choice of the price path is the single most important modelling decision, and it should be *scenario-based*. Many

companies disclose price paths that rise over time; a path that increases with inflation plus a real growth rate (for example 4 to 6 percent real) is common as a stylized assumption, though real-world outcomes depend on policy design and are highly uncertain.

23.2.2 The mechanics of a carbon-adjusted NPV

For a project with annual emissions `E` (tonnes of CO2-equivalent), a carbon price path `P_t` (per tonne) and a pass-through fraction `φ` (the share of the carbon cost recovered from customers through higher prices), the after-tax carbon cash flow in year t is:

```
Carbon cost_t (after tax) = E × P_t × (1 − φ) × (1 − tax rate)
```

The carbon-adjusted free cash flow is the conventional free cash flow minus this term. Three practical issues arise.

Pass-through is endogenous. If competitors face the same carbon price, the price of the product rises and part of the cost is passed on; if the firm competes with imports from jurisdictions without carbon pricing, pass-through may be near zero or even negative (lost volume). Border carbon adjustment mechanisms attempt to level this, but design details differ and should be modelled as scenarios, not assumed.

Free allocation phases out. Under many emissions trading systems, industries judged exposed to international competition received free allowances, which reduced the effective carbon cost to near zero. When free allocation is reduced, the effective price paid per tonne rises even if the market price stays constant. Treat the *effective* price as the product of the market price and the share of emissions that must be covered by purchased allowances.

Tax deductibility and timing. In most jurisdictions allowance purchases are deductible, so the after-tax cost is `(1 − τ)` times the pre-tax cost. Allowances may be purchased in advance, producing a working-capital effect that can be modelled as a financing cost.

23.2.3 Worked examples

Example 23.1 – Carbon-adjusted NPV of an industrial plant. A firm considers a 500 million plant with a 20-year life. Before carbon, it generates EBITDA of 95 million a year. It emits 0.6 million tonnes of CO2e a year. Depreciation is straight-line over 20 years, the tax rate is 25 percent and WACC is 8 percent. Carbon cost is not passed through (φ = 0).

Step 1: Conventional free cash flow. Depreciation = 500 / 20 = 25. EBIT = 95 − 25 = 70. NOPAT = 70 × (1 − 0.25) = 52.5. FCF = 52.5 + 25 = 77.5.

Step 2: Annuity factor. The 20-year annuity factor at 8 percent is `[1 − 1.08^−20] / 0.08 = 9.8181`.

Step 3: NPV without carbon. PV of cash flows = 77.5 × 9.8181 = 760.9. NPV = 760.9 − 500 = **+260.9 million**.

Step 4: Flat carbon price of 50. Annual pre-tax carbon cost = 0.6 × 50 = 30. After-tax = 30 × 0.75 = 22.5. Adjusted FCF = 77.5 − 22.5 = 55.0. PV = 55.0 × 9.8181 = 540.0. NPV = **+40.0 million**.

Step 5: Break-even flat price. The NPV is zero when the PV of after-tax carbon cost equals 260.9: `0.6 × P × 0.75 × 9.8181 = 260.9`, so `P = 260.9 / (0.45 × 9.8181) = 59.05`. The project survives a flat price up to about **59 per tonne**.

Step 6: Rising price. Suppose the price starts at 30 and grows 6 percent a year (escalating from year 1). Discounting each year’s after-tax cost 0.45 × 30 × 1.06^(t−1) at 8 percent gives a total PV of 210.5, and NPV = **+50.4 million**. A starting price of 40.4 growing at 5 percent a year is the break-even for that pattern.

Interpretation. The same plant looks like a 260.9 million winner, a 40 million marginal project, or a loser depending on the carbon assumption. The break-even price (59) is the useful managerial number: it converts an unanswerable forecasting question ("what will carbon cost?") into a comparison ("is 59 plausible within our project life?").

Example 23.2 – Pass-through sensitivity. Using the plant in Example 23.1 with a carbon price starting at the stated level and growing 5 percent a year, compute NPV for 0 percent and 50 percent pass-through.

Method. Pass-through scales the carbon cost by $(1 - \varphi)$. NPV is therefore linear in the starting price for any given growth rate: $NPV = 260.9 - k \times P0 \times (1 - \varphi)$, where k is the PV of the after-tax escalating emission cost per unit of starting price. From the computed results (price starting at 25, 50, 75, 100, and zero pass-through) NPVs are 99.4 (at 25), −62.1 (at 50), −223.7 (at 75) and −385.2 (at 100); so k = (260.9 − 99.4)/25 = 6.46 per unit of starting price.

Starting price (5% growth)	NPV, φ = 0%	NPV, φ = 50%
0	260.9	260.9
25	99.4	180.1
50	−62.1	99.4
75	−223.7	18.6
100	−385.2	−62.1

Interpretation. Halving the effective cost by 50 percent pass-through doubles the carbon price the project can tolerate (the NPV at 50 with φ = 50 percent equals the NPV at 25 with φ = 0). A manager evaluating competitive position should therefore treat the pass-through coefficient with the same attention as the price path: it is a strategic variable, shaped by market structure, regulation and the competitors' cost position.

23.2.4 Scenario weighting and what *not* to average

Scenario analysis in climate typically uses three archetypes, in the spirit of the scenario sets published by central-bank networks and energy agencies, though the exact names and numbers vary by publisher:

- **Delayed or limited action:** current policies continue, low carbon prices, high physical risk later.
- **Orderly transition:** gradual, credible tightening of policy; moderate-to-high prices.
- **Disorderly transition:** late and abrupt tightening, with prices rising sharply and technology and demand shifting quickly.

Example 23.3 – Probability-weighted NPV and the option to close early. For the plant in Example 23.1, assign three scenarios with illustrative probabilities and price paths.

Scenario	Probability	Starting price	Annual growth	NPV (million)
Delayed action	40%	15	3%	+178.2
Orderly transition	40%	50	5%	−62.1
Disorderly transition	20%	90	7%	−426.6

Step 1: Expected NPV if the plant must run for 20 years. $0.4 \times 178.2 + 0.4 \times (-62.1) + 0.2 \times (-426.6) = 71.28 - 24.84 - 85.32 = -38.9$ million.

Step 2: The naive shortcut. A manager who averages the *starting prices* ($0.4 \times 15 + 0.4 \times 50 + 0.2 \times 90 = 44$) and applies a flat path obtains NPV = **+66.5 million**, a positive number. The shortcut is wrong by 105 million because it removes the escalation embedded in the high-price scenarios, where the losses accumulate. The lesson: average *outcomes* (NPVs, or cash-flow vectors), not scenario *inputs*, whenever the model is nonlinear in the path.

Step 3: Value of flexibility. In the disorderly scenario the after-tax annual cash flow is `77.5 − 0.75 × 0.6 × P_t = 77.5 − 0.45 × P_t` (the depreciation tax shield is retained while the firm is operating and has taxable profit elsewhere to absorb losses). It turns negative once `P_t > 172.2`. With a start of 90 growing 7 percent, the price is `90 × 1.07^9 = 165.5` in year 10 (cash flow +3.0) and `90 × 1.07^10 = 177.0` in year 11 (cash flow −2.1). A rational operator therefore closes after year 10. Computing NPV with operations stopping after year 10 gives −339.7 instead of −426.6 for running to year 20, an improvement of 86.9 million (we ignore closure costs and the tax write-off of remaining book value). Expected NPV with this abandonment option: `0.4 × 178.2 + 0.4 × (−62.1) + 0.2 × (−339.7) = 71.28 − 24.84 − 67.94 = −21.5 million`.

Interpretation. Both numbers are negative: on these assumptions the plant is a poor investment as specified, despite a 260.9 million "no-carbon" NPV. The abandonment option reduces but does not eliminate the loss. Two managerial responses follow: reduce capex or emissions intensity (making the project survive more scenarios), or stage the investment so the high-carbon commitment is smaller. This is the real-options logic of Volume 1, Chapter 18 applied to climate.

23.3 Transition Risk, Stranded Assets and Valuation

23.3.1 What is a stranded asset?

An asset is **stranded** when, because of changes associated with the transition, it suffers unanticipated or premature write-downs, devaluation or conversion to a liability. The word *unanticipated* matters: an asset that was always expected to run for twelve years and does so is not stranded. Examples include fossil-fuel reserves that cannot economically be extracted under a given policy path, thermal power plants whose utilization falls below break-even as renewable capacity expands, internal-combustion vehicle platforms and tooling, and buildings that fail evolving efficiency standards.

Three features distinguish stranding from ordinary cyclical weakness:

1. **Structural, not cyclical.** The shock is policy- or technology-driven, so it will not reverse when the cycle turns.
2. **Asset-specific and long-lived.** Heavy industry, power generation and infrastructure have lives of 30 to 60 years, much longer than typical planning horizons, so a decision made today is exposed to many policy cycles.
3. **Skewed.** The loss distribution is typically negatively skewed: the upside from a slow transition is modest (the asset earns what it was expected to earn), while the downside from a rapid one is large.

Stranding is also an *interaction* of asset and balance sheet. A stranded plant financed 70 percent with long-dated project debt can wipe out equity while leaving the lenders short; a plant financed with short-dated equity-like capital can be run off. The capital-structure analysis of Volume 1, Chapter 15 therefore matters: leverage converts asset stranding into equity destruction and sometimes distress.

23.3.2 Accounting and valuation hooks

Two accounting mechanisms translate expected stranding into reported numbers, although neither anticipates all of it.

- **Impairment testing.** Under IFRS (IAS 36) and, with different mechanics, under US GAAP, an asset (or cash-generating unit) is written down when its carrying amount exceeds its recoverable amount. Under IFRS the

recoverable amount is the higher of fair value less costs of disposal and *value in use*, the present value of expected future cash flows. If expected cash flows incorporate a rising carbon cost and a shorter life, value in use falls and impairment can be triggered. A climate-related change in policy or market expectations is an impairment indicator in its own right.

- **Useful lives and decommissioning.** Revising a useful life shortens depreciation and raises annual depreciation expense. Decommissioning or restoration obligations are recognized as provisions (IAS 37 under IFRS) at the present value of expected outflows, and an earlier closure date raises the present value of that provision by shortening the discount period.

Neither mechanism produces a market-value measure; both rely on management's own forecasts. This is why analysts and stress-testers prefer to build scenario-weighted market valuations independently.

Example 23.4 – Scenario-weighted value in use and impairment. A power producer carries a generating asset at 800 million. After-tax cash flow is expected to be 100 million a year while it runs. The discount rate for value in use is 9 percent. Decommissioning cost is 60 million (paid at closure). Management considers three closure scenarios with illustrative probabilities.

Scenario	Closure after	Probability	PV of 100/yr	PV of 60 decommissioning	Value in use
Continues (delayed policy)	25 years	25%	982.3	7.0	975.3
Orderly phase-out	12 years	45%	716.1	21.3	694.7
Early forced closure	8 years	30%	553.5	30.1	523.4

Step 1: PV of 100 a year. For 25 years at 9 percent the annuity factor is 9.8226, so PV = 982.3. For 12 years it is 7.1607, so PV = 716.1. For 8 years it is 5.5348, so PV = 553.5.

Step 2: PV of decommissioning. `60 / 1.09^25 = 6.96`; `60 / 1.09^12 = 21.33`; `60 / 1.09^8 = 30.11`.

Step 3: Probability-weighted value. `0.25 × 975.3 + 0.45 × 694.7 + 0.30 × 523.4 = 243.8 + 312.6 + 157.0 = 713.5`.

Step 4: Impairment. Carrying amount 800 − 713.5 = **86.5 million** shortfall. The asset would be written down by roughly 11 percent, and, if the *continues* scenario alone had been used, there would have been no impairment at all.

Step 5: Depreciation consequence. Had the carrying amount been depreciated straight-line over 25 years, the charge is 32.0 per year. If the board adopts a 12-year expected life, the annual charge on the same carrying amount would be 800 / 12 = 66.7, an increase of 34.7 million a year in reported expense *without any change in cash flow*.

Interpretation. The cash flows did not change; the beliefs did. The write-down is the accounting recognition of a shift in the probability distribution. Managers who anticipate this can stage disclosure, adjust leverage and plan replacement capital before the accounting event forces the issue, and treasury should check whether covenants are based on book equity, EBITDA or both.

23.3.3 Transition risk beyond carbon prices

Carbon prices are the most modelable transition driver but not the only one. A complete assessment should consider:

- **Technology risk.** Falling costs of renewables and storage, or electrification of heat and transport, can reduce demand for incumbent products faster than policy alone would.

- **Demand and customer risk.** Corporate buyers increasingly require low-carbon inputs (for example, low-carbon steel or cement for certain public and private projects) and may impose supplier requirements regardless of regulation.
- **Legal and litigation risk.** Claims related to misleading climate statements, failure to disclose, or damages attributed to emitters create contingent liabilities.
- **Market and financing risk.** Banks, insurers and asset owners may set sector policies that limit new financing of certain assets; this raises spreads or removes refinancing options. This channel links directly to Section 23.6.
- **Reputation and talent.** Harder to quantify but often reflected in customer retention, recruitment costs and license-to-operate.

Opportunities mirror these risks: low-carbon products with price premia, new markets for efficiency, grid services, and green subsidies and tax credits. A balanced analysis values both, because a decarbonization project that merely avoids a loss can still have a negative standalone NPV and yet be value-creating at the firm level through the avoided downside.

23.3.4 Climate scenario analysis as a corporate process

A defensible process has five steps: (1) *Choose scenarios* with a coherent set of policy, technology and macro assumptions, drawing on published reference scenarios rather than internal guesses; (2) *Map each scenario to drivers*: carbon price, energy prices, demand, cost of capital, physical hazards; (3) *Translate into the financial model*: cash flows, balance sheet, covenants, liquidity; (4) *Assess resilience*: where does the strategy break, and what are the triggers; and (5) *Decide actions* with trigger-based capital plans and report the results in a decision-useful format. Scenario analysis is not forecasting; its output is a map of vulnerabilities and options, not a number to optimize.

A useful discipline is to compute, for each major asset, the **break-even carbon price** (the price at which its NPV is zero, as in Example 23.1) and compare it to the price paths in each scenario. Assets whose break-even is below the lowest-scenario price in the investment horizon are candidates for retirement or conversion; assets whose break-even exceeds the highest-scenario price are robust. The assets in between are where management attention, hedging and phased capital belong.

23.4 Abatement Economics and Capital Allocation

23.4.1 The marginal abatement cost curve

A **marginal abatement cost (MAC) curve** ranks abatement options by cost per tonne of CO2e avoided, from the cheapest (often negative, meaning that the option saves money) to the most expensive, with the width of each bar showing the tonnes it abates. The cost per tonne is computed on an annualized, usually after-tax basis:

```
Abatement cost per tonne = (Annualized capex + ΔOpex − Energy savings) / Annual tonnes abated
Annualized capex         = Capex × r / [1 − (1 + r)^−n]      (the capital recovery factor)
```

The decision rule is straightforward: *carry out every option whose cost per tonne is below the relevant carbon price*, where the relevant price may be the explicit price, the shadow price, or the internal carbon price. At the margin, the firm abates until the cost of the next tonne equals the price. This is the same logic as the textbook condition for efficient pollution control: equalize marginal abatement cost across sources.

MAC curves have well-known limitations. They are static, so they ignore interaction effects (insulating a building reduces the savings from a more efficient boiler), learning curves (costs fall as deployment grows), option value (waiting may reveal better technology) and non-financial barriers (capital rationing, split incentives, operational disruption). Negative-

cost options that are not undertaken ("the efficiency gap") often reflect hidden costs, risk, or management attention, not irrationality. Finally, the choice of discount rate for the capital recovery factor shifts the ordering: capital-intensive options (CCS, hydrogen) look much worse at 10 percent than at 5 percent.

Example 23.5 – Building and using a MAC curve. A manufacturer lists seven options (the figures are illustrative):

Rank	Option	Abatement (Mt/yr)	Cost (per tonne)	Cumulative abatement (Mt)	Annual net cost (million)
1	LED retrofit	0.20	−40	0.20	−8.0
2	Heat recovery	0.35	−15	0.55	−5.2
3	Process electrification	0.50	25	1.05	12.5
4	Renewable PPA	0.80	35	1.85	28.0
5	Fuel switch	0.60	60	2.45	36.0
6	CCS retrofit	0.90	110	3.35	99.0
7	Green hydrogen feed	0.40	180	3.75	72.0

Step 1: Annual net cost = abatement (Mt) × cost per tonne, since Mt × currency/tonne = currency millions. Option 1: 0.20 × (−40) = −8.0; Option 2: 0.35 × (−15) = −5.25 (shown as −5.2); Option 3: 0.50 × 25 = 12.5; Option 4: 0.80 × 35 = 28.0; Option 5: 0.60 × 60 = 36.0; Option 6: 0.90 × 110 = 99.0; Option 7: 0.40 × 180 = 72.0.

Step 2: Apply an internal carbon price of 70. Accept options 1 to 5 (cost below 70). Total abatement = 0.20 + 0.35 + 0.50 + 0.80 + 0.60 = **2.45 Mt a year**. Total net annual cost = −8.0 − 5.25 + 12.5 + 28.0 + 36.0 = **63.25 million**.

Step 3: Avoided carbon charge. 2.45 Mt × 70 = 171.5 million. Net benefit = 171.5 − 63.25 = **108.25 million a year**.

Step 4: Sensitivity to price. At a carbon price of 120 the CCS retrofit also becomes attractive: abatement rises to 3.35 Mt and total net cost to 162.25 million; avoided charge is 3.35 × 120 = 402.0 million, giving a net benefit of 239.75 million. At a price of 50 only options 1 to 4 pass, abating 1.85 Mt at a net cost of 27.25 million (benefit 92.5 − 27.25 = 65.25 million).

Interpretation. The MAC curve turns a price assumption into a capital plan: the *amount* of abatement the firm should buy rises as the price rises. But the plan should be robust: investing now in the cheaper options is a no-regret move; the expensive options (CCS, hydrogen) are better handled as options whose exercise depends on a trigger price and on technology learning. Resist the temptation to implement all options simply to reach a headline target; the incremental cost of the last tonne can exceed any plausible carbon price.

23.5 Green and Transition Finance Instruments

23.5.1 A taxonomy of labelled debt

"Labelled" instruments are debt (or loans) whose terms or use of proceeds refer to sustainability. The main families are as follows; the market conventions are voluntary guidelines published by industry bodies, and regional regulators have in some cases layered official standards on top.

Instrument	What is promised	Pricing mechanism	Main credibility risk
Green bond / loan	Proceeds allocated to eligible environmental projects	Same issuer credit; possible small “greenium”	Eligibility of projects; allocation and impact reporting
Social bond	Proceeds to social projects (housing, health, education)	Same as above	Defining and measuring social outcomes
Sustainability-linked bond (SLB) / loan (SLL)	General-purpose proceeds; coupon or margin depends on achieving sustainability performance targets	Step-up (or step-down) in coupon if targets missed (met)	Ambition of targets; materiality; size of financial penalty
Transition bond / loan	Proceeds or targets tied to decarbonizing a hard-to-abate business	Varies; often target-based	Credibility of the transition plan; risk of lock-in
Green securitization, green covered bond	Asset pool or covered assets have environmental characteristics	Structure plus label	Eligibility tests of underlying assets
Blue, catastrophe and resilience bonds	Marine, climate-adaptation or catastrophe-risk transfer	Instrument-specific	Basis risk; parametric triggers

Use-of-proceeds (UoP) instruments such as green bonds follow, in the widely used market guidelines, four core components: (1) a defined use of proceeds, (2) a process for project evaluation and selection, (3) management of proceeds (tracking, often via a sub-account), and (4) reporting, typically annual allocation and, where possible, impact reporting. An external review (a “second-party opinion”) is common but optional. Legally, a UoP bond is usually a full-recourse obligation of the issuer: if the green project fails, the bondholders do not bear that loss. This is why the credit spread tends to follow the issuer’s credit, with at most a small premium or discount attributable to the label.

Sustainability-linked instruments are different in design: the proceeds are unrestricted, and the financial terms depend on the issuer reaching pre-specified **Sustainability Performance Targets (SPTs)** measured by **Key Performance Indicators (KPIs)**. The widely used principles for such instruments emphasize five components: KPI selection (material, measurable, benchmarkable), calibration of SPTs (ambitious relative to a baseline and to peers), bond or loan characteristics (the financial variation), reporting, and verification. Transition instruments are less standardized; their appeal is that they finance firms that cannot yet be called “green” but are credibly moving, but the risk is that the label is attached to a plan that is not credible.

Transition finance has an additional conceptual challenge. A taxonomy that classifies activities as “green” or “not green” has no category for the cement kiln that cuts emissions 40 percent but remains emissions-intensive. Official taxonomies differ in how they handle this; some regimes allow “transitional” activities that meet declining thresholds, while others focus on technologies consistent with long-run neutrality. From a corporate-finance standpoint the priority is to link financing terms to *verifiable pathway milestones* rather than to the label itself.

23.5.2 The economics of the “greenium”

The **greenium** is the yield difference between a green bond and a comparable conventional bond from the same issuer (same maturity, seniority and currency). Empirical estimates in the literature are generally small (a few basis points in many studies), vary by market segment, region and period, are sensitive to the matching method, and in some samples are statistically indistinguishable from zero. A reasonable theoretical basis exists: if some investors have a mandate or

preference for green assets, they will bid up the price, and the issuer enjoys a lower yield. But the benefit is limited by arbitrage between green and conventional bonds of the same issuer, and the issuer also bears costs: verification, reporting systems, and the risk of reputational damage if the use of proceeds is challenged.

Evidence on *other* benefits is also mixed but intriguing: some studies have found a positive stock-price reaction to green bond announcements and improved environmental performance after issuance for certain issuers, consistent with a *signalling* role (issuing a green bond is a costly, credible commitment). Because these are observational studies, causality is debated.

Example 23.6 – Valuing a greenium net of costs. An issuer sells a 10-year, 500 million green bond with annual coupons. A comparable conventional bond would price at a 5.08 percent yield; the green bond prices at 5.00 percent, a greenium of 8 basis points. The issuer incurs annual verification and reporting costs of 0.15 million.

Step 1: Annual interest saving. 500 million × 0.0008 = **0.40 million a year**.

Step 2: Present value. The 10-year annuity factor at 5 percent is `[1 − 1.05^−10] / 0.05 = 7.7217`. PV of savings = 0.40 × 7.7217 = **3.089 million** (0.62 percent of face value).

Step 3: PV of annual costs. 0.15 × 7.7217 = **1.158 million**.

Step 4: Net benefit. 3.089 − 1.158 = **1.930 million**, or about 0.39 percent of face value.

Step 5: Check by pricing. A 10-year bond with a 5.00 percent coupon priced at a 5.08 percent yield would sell at 99.38 per 100 face; at a yield of 5.00 percent it sells at par. The 0.62 points difference is the market's valuation of the greenium; on 500 million it is 3.09 million, matching Step 2.

Interpretation. The net benefit is small relative to the 500 million issued. A treasurer should not issue a green bond solely for pricing; the case rests on investor diversification (a broader, longer-term investor base), alignment of financing with strategy, signalling, and the option value of being able to return to the market in a stress period. Note also that the same 1.158 million of annual costs would wipe out the benefit if the greenium were only 3 basis points: `500 × 0.0003 = 0.15 million`, exactly the cost. The break-even greenium is therefore 3 basis points.

23.5.3 Sustainability-linked structures and the step-up

An SLB typically adds a coupon **step-up** (for example, +25 basis points) if the issuer fails to meet an SPT on an observation date. The step-up is a *contingent cost* to the issuer and a *contingent benefit* to investors; it is economically a bet on the KPI.

Two design questions determine whether the instrument is more than a label. First, is the **target ambitious**: does it go beyond business-as-usual and would it require capital expenditure the firm would not otherwise make? Second, is the **financial penalty large enough** to influence behaviour? A 25 basis point step-up on a 500 million bond for five years costs the issuer 1.25 million a year, small compared with the 100 million capex needed to cut emissions meaningfully. If the penalty is tiny, rational management will accept it; if it is huge, the issuer will not issue.

Example 23.7 – Does the step-up change behaviour? An issuer sells a 10-year, 500 million SLB at 5 percent with a 25 basis point step-up for years 6 to 10 if an emissions-intensity SPT is missed at the end of year 5. Discount rate 5 percent.

Step 1: Annual cost of a miss. 500 million × 0.0025 = **1.25 million a year** for years 6 to 10.

Step 2: PV at issuance. The PV factor for an annuity of years 6 to 10 at 5 percent is `[1 − 1.05^−5] / 0.05 × 1.05^−5 = 4.3295 × 0.7835 = 3.3923`. PV of the step-up = 1.25 × 3.3923 = **4.240 million**.

Step 3: Expected cost for different miss probabilities. At 20 percent: 0.848 million; at 50 percent: 2.120 million; at 80 percent: 3.392 million.

Step 4: Would extra effort be rational? Suppose an additional abatement programme costing 2.0 million in present value would raise the probability of hitting the target from 40 percent to 85 percent (so the probability of a miss falls from 60 percent to 15 percent). Expected saving = (0.60 − 0.15) × 4.240 = **1.908 million**. This is slightly *less* than the 2.0 million cost, so purely on the penalty a rational manager would not spend the money.

Interpretation. The step-up is too small to be the driving force; it works only if the firm independently values the target (because of carbon cost, customers or strategy). Investors and regulators should therefore look at target ambition and the firm's overall transition plan, not the existence of a step-up. For the issuer, the SLB is best seen as a commitment device and a signal; if the firm has no intention of hitting the target, the issuance is a reputational liability. Note also the asymmetric information: management knows the probability better than the lender, so a low-ambition target with a small step-up is cheap for the issuer and misleading for the buyer.

23.5.4 Greenwashing risk, legal exposure and governance

"Greenwashing" is the practice of making unsupported or misleading sustainability claims. For issuers of labelled debt the exposures are: *regulatory and legal* (misleading statements to investors can attract enforcement or litigation in many jurisdictions, and consumer or advertising regulators have also acted on environmental claims), *market* (investors may exclude an issuer after a controversy, and secondary-market spreads can widen), and *covenant or documentation* (some instruments allow investors to claim remedies if reporting commitments are not met, though most do not treat a missed label as a default). The corporate remedy is to apply the same controls to sustainability claims as to financial statements: defined methodology, internal audit, independent assurance of key figures, and board-level review. Consistency between what the company *funds* and what it *says* is the key test: a firm that issues a green bond while lobbying against the policy that makes its projects viable, or investing most capex in the opposite direction, invites the charge.

23.6 ESG Investing: Theory and Evidence

23.6.1 What "ESG investing" means

"ESG investing" covers several distinct practices that deserve separate analysis.

- **Exclusionary (negative) screening:** do not hold certain sectors or companies.
- **Best-in-class or ESG-tilted:** overweight firms with high ESG ratings, underweight low-rated firms.
- **Thematic:** invest in themes such as clean energy.
- **Impact investing:** aim for measurable environmental or social outcomes alongside financial return, often accepting lower return.
- **Engagement and stewardship:** hold the shares and use voting and dialogue to change corporate behaviour.
- **ESG integration:** incorporate ESG factors as *financial risk information* into otherwise conventional valuation.

The first five are partly *preference-driven* (the investor wants something besides money), whereas integration is *information-driven* (ESG data predict cash flows). Confusing the two is the root of much confusion in the debate: the empirical question of whether ESG data predict returns is separate from the normative question of whether investors may want to sacrifice return for values.

23.6.2 Theory: how ESG preferences affect prices and the cost of capital

The theoretical benchmarks come from asset pricing.

Segmentation and exclusion. In the model by Merton (1987) of investors with incomplete information, stocks held by fewer investors have a higher required return because investors who hold them bear risk that cannot be spread across a wider holder base. Heinkel, Kraus and Zechner (2001) applied the same logic to ethical exclusion: if a fraction of investors refuses to hold "brown" stocks, those stocks are held by a smaller group, so the price is lower and the expected return higher; "green" stocks have a lower expected return. The cost-of-capital effect depends on the *size* of the excluding group and on the ability of remaining investors to absorb the shares.

Preferences in equilibrium. Pástor, Stambaugh and Taylor (2021) model investors who gain utility from holding green assets. In equilibrium, green assets have *negative alphas* (lower expected return) because investors accept lower returns for the non-pecuniary benefit; brown assets have positive alphas. But there is a twist: if concerns about climate strengthen unexpectedly, green assets can *outperform* during the period of rising concern, even though their *expected* return is lower. Therefore, a period of strong green-stock performance is not evidence of a permanent return advantage; it may reflect a one-time repricing as preferences shift. This explains why historical returns to ESG strategies in the recent past can overstate the future.

Information. If ESG factors proxy for material, previously unpriced risks (for example, litigation or regulation), then incorporating them should earn a return only until the market prices the information, after which expected returns converge. In an efficient market, there is no free lunch from public ESG ratings.

Corporate-level view. Economists contrast the view that firms should maximize shareholder *value* with the view that they should reflect shareholder *welfare*, which may include social preferences (a distinction developed by Hart and Zingales). Under the first view, ESG expenditures should be justified by their effect on cash flows or risk; under the second, firms might invest in social goods if shareholders would otherwise spend their own wealth on those goods less efficiently. For capital budgeting, the practical implication is simple: every ESG investment should state *which* of these arguments it relies on, because the two imply different hurdle rates.

23.6.3 Evidence on returns and cost of capital: what we can and cannot say

It is easy to find studies supporting either side. A careful reader should apply the following filters.

1. **Large reviews tend to find a non-negative association** between ESG measures and financial performance. For example, a well-known meta-analysis of many studies (by Friede, Busch and Bassen) reported that the majority of studies found a non-negative relationship. But such reviews aggregate heterogeneous measures and study quality, and association does not demonstrate causation.
2. **Good firms may simply be profitable firms.** Companies with high cash flows can afford ESG programmes, so ESG can be a *consequence* of performance. Controls for size, profitability and sector reduce but do not eliminate this bias.
3. **Ratings disagree.** Research on ESG ratings by Berg, Koelbel and Rigobon documents that the correlations across major rating providers are far lower than correlations across credit rating agencies (broadly around 0.5 for ESG versus very high for credit). Differences arise from scope (what attributes are included), measurement (different indicators for the same attribute) and weights. Any study using one provider's rating, therefore, measures that provider's view. Rating disagreement is itself related to higher return volatility in some studies, suggesting that it is a form of risk.

4. **Exclusion premia.** Evidence on "sin" stocks (alcohol, tobacco, gambling) by Hong and Kacperczyk (2009) found that they earned higher returns than comparable stocks over the sample, which they interpreted as the price of social norms restricting investor demand. Work on carbon emissions (for example, Bolton and Kacperczyk) found that stocks of higher-emitting firms earned higher returns in some samples, consistent with a carbon risk premium; but other studies find the sign depends on period and measure, in line with the Pástor-Stambaugh-Taylor point about unexpected shifts in preferences.
5. **Time variation.** The performance of ESG-tilted portfolios has varied over time, in part because of exposures to sectors (technology tilt, underweight energy) and interest-rate sensitivity (growth-style tilt). A fund that appears to have positive ESG alpha may be earning a sector or style premium; factor regressions (market, size, value, profitability, investment, momentum, from Volume 1, Chapter 10 and its extensions) are necessary to test this.
6. **Cost of capital.** Studies of credit spreads and cost of equity find that firms with strong environmental profiles tend to face somewhat lower financing costs in some markets and periods, and that firms facing high exclusion or large controversies may face higher costs. The magnitude is uncertain and depends on the sample; and the direction is consistent with the theory above (exclusion raises the cost of capital for the excluded firm).

The fair conclusion is: **ESG, as a broad label, has not been shown to produce reliable risk-adjusted outperformance; specific, material factors may inform risk and cash-flow forecasts; and the cost of capital effect operates through investor demand and may be small for most firms but meaningful for firms at the extremes.** A manager should not promise investors that sustainability will raise the share price, nor claim that it must lower it.

Example 23.8 – What a small cost-of-capital change does to value. Take a firm whose dividends or free cash flow to equity are expected to grow at 3 percent forever. Next year's cash flow is 1.00 per share. Initially the cost of equity is 8 percent.

Step 1: Initial value. `V0 = 1.00 / (0.08 − 0.03) = 20.00`.

Step 2: Exclusion raises the cost of equity by 50 basis points. New value `V1 = 1.00 / (0.085 − 0.03) = 18.18`. Change = (18.18 − 20.00) / 20.00 = **−9.1 percent**.

Step 3: Other sizes. For +25 basis points: `1.00 / 0.0525 = 19.05`, a **−4.8 percent** change. For +100 basis points: `1.00 / 0.06 = 16.67`, a **−16.7 percent** change.

Interpretation. Long-duration cash flows make value highly sensitive to small changes in the discount rate. This is why even a modest exclusion premium can matter for a long-lived, capital-intensive firm. It also shows the *inverse* effect: if investors' preferences caused green firms' costs of capital to fall by 25 to 50 basis points, their share prices would rise 5 to 10 percent in a one-time repricing, an effect that shows up as outperformance in the sample but is *not* repeated, and implies lower subsequent expected returns.

Example 23.9 – How long to detect ESG alpha? An ESG-tilted portfolio earns an average excess return versus its benchmark of 0.8 percent a year, with a tracking error (volatility of active return) of 3.0 percent a year. How strong is the evidence?

Step 1: t-statistic for n years. The standard error of the mean active return is `TE / √n`; the t-statistic is `α / (TE / √n)`. After 5 years: `0.8 / (3.0 / 2.236) = 0.60`. After 10 years: `0.8 / (3.0 / 3.162) = 0.84`. After 20 years: `0.8 / (3.0 / 4.472) = 1.19`.

Step 2: Years needed for t = 2. `n = (2 × TE / α)^2 = (2 × 3.0 / 0.8)^2 = 7.5^2 = 56.25 years`.

Interpretation. An annual alpha of 0.8 percent against 3 percent tracking error cannot be statistically distinguished from luck for roughly 56 years. Typical fund track records, 5 to 10 years, reveal almost nothing. The result generalizes: the claim "this ESG strategy outperforms" is very hard to establish from realized returns, and just as hard to refute. Rely on economic reasoning about mechanisms (are the cash-flow effects material and mispriced?), not on short track records.

23.6.4 Implications for corporate managers

Four practical points follow. Do not manage to ratings, which are inconsistent and change methodology; manage and disclose the underlying risk. Know your investor base, since concentrated holdings by excluding investors affect your marginal cost of equity in stress. Evaluate ESG capex like any capex, stating any non-financial rationale explicitly. And avoid overpromising on technologies that are not yet commercial, which creates later credibility and liability problems.

23.7 Disclosure Frameworks (Described in General Terms)

Climate disclosure has converged on a family of frameworks. The details of regimes change and differ between jurisdictions, so treat the following as an orientation and check current rules before preparing a report.

The TCFD architecture. The Task Force on Climate-related Financial Disclosures, created by the Financial Stability Board, recommended disclosures organized around four pillars that remain the backbone of most subsequent regimes: **Governance** (board oversight and management role), **Strategy** (risks and opportunities over short, medium and long term, and the resilience of strategy under different scenarios, including a 2°C-or-lower scenario), **Risk management** (processes to identify, assess and manage climate risks and their integration into enterprise risk management) and **Metrics and targets** (including Scope 1, 2 and, where appropriate, Scope 3 emissions, and targets used to manage performance).

ISSB standards. The International Sustainability Standards Board issued IFRS S1 (general requirements for disclosure of sustainability-related financial information) and IFRS S2 (climate-related disclosures) which incorporate the TCFD pillars and add industry-based metrics. The orientation is **financial materiality**: information that could reasonably be expected to influence investors' decisions. Adoption is a matter for each jurisdiction, and several have moved to adopt or align with the standards, with variations.

Other regimes. The European Union's Corporate Sustainability Reporting Directive and its reporting standards apply the concept of **double materiality**, that is, both the financial impact of sustainability matters on the firm and the firm's impact on people and environment. In the United States, climate-disclosure rules have been the subject of regulatory and legal contention and of state-level initiatives; a manager of a multinational should map the requirements of each jurisdiction in which the company is listed or operates. Many national regimes also include requirements for banks and insurers.

Greenhouse gas accounting. The GHG Protocol is the most widely used standard for measuring emissions. It defines **Scope 1** (direct emissions from owned or controlled sources), **Scope 2** (indirect emissions from purchased electricity, steam, heat and cooling, reportable on a *location-based* or *market-based* method) and **Scope 3** (all other indirect emissions in the value chain, upstream and downstream, divided into fifteen categories such as purchased goods, business travel, use of sold products). Scope 3 is often the largest portion of a company's footprint and the hardest to measure, because it depends on suppliers' and customers' data and on estimation (spend-based, average-data or supplier-specific methods).

Quality of numbers. Because emissions data are constructed rather than directly observed, the controls that apply to financial figures should apply here: documented boundaries (operational control versus equity share), emission factors with sources and dates, base-year recalculation policy for acquisitions and divestitures, and assurance (limited, then reasonable). For finance teams, three connections to financial reporting are important: *consistency* (the assumptions in climate scenarios

should be consistent with the assumptions in impairment tests and going-concern assessments), *connectivity* (disclosed targets and capex plans should reconcile to the financial plan) and *forward-looking statements* (provide risk language and keep a record of assumptions).

For financial institutions, **financed emissions** (the emissions attributed to loans and investments) are measured by allocating the borrower's emissions in proportion to the lender's share of the borrower's financing: `Financed emissions = Σ (Outstanding amount_i / EVIC_i) × Emissions_i`, where EVIC is enterprise value including cash. This metric is useful but sensitive to market valuations: a rising share price lowers the attribution.

23.8 Embedding Climate in the Corporate Finance Function

A practical programme has five elements: a governed carbon price and scenario set approved annually by the finance committee; appraisal templates that require emissions, effective carbon price, pass-through and break-even price for every sizeable project; scenario-weighted valuation of major assets with trigger-based plans (for example, convert or close a unit if the carbon price exceeds a stated level for two consecutive years); a financing strategy that matches instrument to purpose and stress-tests covenants, ratings and liquidity against transition scenarios; and incentives and disclosure that link pay to a few verifiable metrics that cannot be met by moving emissions outside the reporting boundary. Hedging tools (forward allowance purchases, power purchase agreements, insurance) can reduce price and physical exposures, but watch the basis risk between hedge and exposure. Residual emissions that cannot be abated are sometimes offset with carbon credits; treat credits as a last resort whose quality (additionality, permanence, leakage, verification) varies widely, and disclose gross emissions separately.

Common mistakes include: using a single carbon price for all years and assets; applying a discount-rate add-on instead of modelling cash flows; averaging scenario inputs instead of outcomes (Example 23.3); treating a decarbonization project as value-destroying because standalone NPV is negative while ignoring the avoided-loss value to the rest of the firm; counting offsets as abatement; and presenting targets without a funded capital plan.

Case Study: Nordvik Steel and the Transition Plan

Illustrative case; Nordvik Steel is a fictional company.

Nordvik Steel operates two blast-furnace sites that supply automotive and construction customers. Revenue is 3.2 billion, EBITDA 480 million, net debt 1.3 billion (2.7 times EBITDA), and the shares trade at 6.0 times EBITDA, a discount to peers. Site A, built 35 years ago, has a furnace that needs a 700 million relining within four years to keep running for another 20. Site B was refurbished recently. Together the sites emit 6.0 million tonnes of CO2e a year; in the company's jurisdiction free allowances currently cover about 85 percent of those emissions, and the regulator has announced a phase-out of free allocation over the next ten years.

The strategy team has proposed replacing Site A's blast furnace with a direct-reduced-iron plant fed initially by natural gas and later by hydrogen, plus an electric arc furnace. Capex is 1.9 billion over five years; emissions would fall by roughly 60 percent initially and more later, if low-carbon hydrogen becomes available at an acceptable price. The relining alternative costs 700 million but locks in the current technology until the 2050s. The CFO has assembled the following scenario assumptions for the evaluation:

Scenario	Probability	Carbon price in 10 years	Free allocation after 10 years	Low-carbon steel premium
Delayed action	30%	35	50%	negligible
Orderly	45%	90	0%	moderate
Disorderly	25%	160	0%	large and rising

Under these inputs the relining option has a higher NPV than the new plant in the delayed scenario, a lower NPV in the orderly scenario, and a sharply lower NPV in the disorderly scenario, where relined Site A would face carbon costs exceeding its EBITDA contribution by year 12. The expected-NPV comparison favours the new plant narrowly, but the new plant requires external capital of about 1.2 billion. The banks have indicated they will lend, but on a sustainability-linked basis with step-ups tied to emissions intensity, and the board is split. One director argues that investors will not reward a decarbonization programme that depresses near-term free cash flow; another notes that large automotive customers have begun to ask for low-carbon steel and favour suppliers with credible plans. A third points out that the hydrogen supply is uncertain and the plant could become stranded if hydrogen costs do not fall, since the gas-fired mode is only partially decarbonized.

The treasurer favours a staged plan: build the DRI plant “hydrogen-ready” for natural gas first, hedge part of the carbon exposure with forward allowance purchases, issue a green bond for the eligible portion of the capex with a second-party opinion, and negotiate the sustainability-linked loan with an ambitious but achievable emissions-intensity target. She also proposes disclosure aligned to the ISSB climate standard, including Scope 3 for purchased iron ore and coking coal, and a break-even carbon price analysis for each site.

The board must decide on the capital plan, the financing mix, and what to say to shareholders. The CFO wants to present to the board a recommendation with explicit triggers rather than a single yes-or-no decision.

Discussion Questions

1. Using the scenario table, outline how you would compute the NPV of each option, including the effect of free-allocation phase-out on the effective carbon price. Which inputs would you average, and which would you keep as scenario-specific?
2. Evaluate the argument that the relining option is cheaper and therefore preferable. How does the option to retire the furnace early, and the leverage of 2.7 times EBITDA, change the assessment?
3. Design a staged investment plan with trigger conditions for the hydrogen conversion. What information would you want before committing the second stage, and how does this resemble a real option?
4. Compare financing the new plant with a green bond, a sustainability-linked loan and plain debt. What are the likely pricing and credibility trade-offs, and how would you avoid greenwashing allegations?

Key Takeaways

- Climate risk reaches the firm through cash flows, asset lives, financing costs and terminal value; the right place for most of it is in the explicit cash-flow forecast, scenario by scenario, not in an ad hoc discount-rate add-on.
- Distinguish explicit, shadow and internal carbon prices; the effective carbon cost depends on free allocation, tax deductibility and pass-through, not only the headline price.
- Break-even carbon price is the most useful single metric: it turns an unforecastable price into a comparison with scenario paths.

- Average scenario outcomes (NPVs or cash-flow vectors), not scenario inputs; averaging inputs can hide the loss concentration in the tail (Example 23.3).
- Stranded assets arise from unanticipated, structural, policy or technology-driven loss of value; leverage and long-dated debt convert asset stranding into equity losses and covenant stress.
- Marginal abatement cost curves translate a carbon price into an abatement plan: undertake all options costing less than the price, treat expensive options as triggered options, and remember the curve's static nature.
- Green bonds are full-recourse use-of-proceeds instruments with small and uncertain pricing benefits; sustainability-linked instruments depend on target ambition and a meaningful penalty. Credibility, not the label, creates value.
- ESG is a bundle of practices; the evidence on returns is mixed, ratings disagree, realized outperformance can reflect one-time preference shifts, and statistical detection of small alphas takes decades.
- The cost-of-capital effect of exclusion or investor preference can be modest but is magnified by long cash-flow duration; know your investor base.
- Disclosure frameworks (TCFD pillars, ISSB standards, GHG Protocol scopes, regional regimes) are converging, but climate numbers need the same governance and assurance as financial numbers.

Review Questions

1. Why is adding a fixed spread to WACC an inadequate way to account for carbon pricing risk? Give at least two reasons.
2. Distinguish a shadow carbon price from an internal carbon price. In what circumstances would a firm use both?
3. Explain how free allocation of emission allowances affects the effective carbon cost, and why its phase-out matters to long-lived assets.
4. Why does averaging scenario inputs (such as carbon prices) differ from averaging scenario outputs (such as NPVs)? Illustrate with the logic of Example 23.3.
5. What is a stranded asset, and why does the way it is financed matter?
6. Describe how a decline in expected useful life and a rise in decommissioning costs each affect impairment and depreciation.
7. What does a MAC curve show, and what are three reasons why it can mislead?
8. Compare the use-of-proceeds structure of a green bond with the performance-linked structure of a sustainability-linked bond. Which one creates stronger incentives for the issuer, and why might neither be sufficient?
9. Summarize the theoretical argument that ESG preferences lower expected returns on green assets, and explain why green stocks may nevertheless have outperformed in a given sample.
10. What are Scope 1, 2 and 3 emissions, and why is Scope 3 hard to measure? What does "double materiality" add to financial materiality?

Problems

1. **Carbon-adjusted NPV.** A 300 million plant has a 15-year life, straight-line depreciation, EBITDA of 60 million a year, emissions of 0.3 million tonnes a year, tax rate 30 percent, WACC 9 percent. (a) Compute NPV without carbon. (b) Compute NPV with a flat carbon price of 40 and no pass-through. (c) Find the break-even flat carbon price.

2. **Pass-through.** For the plant in Problem 1, compute NPV at a flat price of 60 if 40 percent of the carbon cost is passed through, and find the break-even price with 40 percent pass-through.
3. **Green bond economics.** A firm issues a 7-year, 300 million green bond with a 6 basis point greenium against a conventional bond; the yield is 4 percent. Annual reporting and verification cost is 0.1 million. Compute the PV of the interest saving, the PV of the costs and the net benefit.
4. **Step-up.** A 400 million, 8-year SLB carries a 30 basis point step-up in years 4 to 8 if the SPT is missed. Discount rate 4.5 percent. Compute the annual cost of a miss and its present value at issuance.
5. **Impairment.** An asset is carried at 500 million. It generates 70 million a year after tax. The discount rate is 10 percent. There is a 40 percent chance it runs 20 years and a 60 percent chance it closes after 10 years. Ignore decommissioning. Compute the expected value in use and any impairment.
6. **MAC.** Options (abatement in Mt, cost per tonne): A (0.3, −30), B (0.5, 20), C (0.7, 45), D (0.4, 70), E (0.6, 95). At a carbon price of 55, which options are accepted? Compute total abatement, net cost and net benefit. Repeat at a price of 100.
7. **Cost of equity.** A firm's cash flow next year is 1.00 per share, growing 2.5 percent a year forever; cost of equity is 9 percent. Exclusion by some investors raises the cost of equity by 30 basis points. Compute the change in value.
8. **Detecting alpha.** A strategy has annual active return 1.0 percent and tracking error 4.0 percent. Compute the t-statistic after 10 years and the number of years needed for t = 2.

Answers to Selected Problems

1. a. Depreciation 20; FCF = (60 − 20) × 0.70 + 20 = 48.0; annuity factor (15 years, 9%) = 8.0607; NPV = 48.0 × 8.0607 − 300 = **+86.9 million**. (b) After-tax carbon cost = 0.3 × 40 × 0.70 = 8.4 a year; PV = 8.4 × 8.0607 = 67.7; NPV = **+19.2 million**. (c) Break-even = 86.9 / (0.3 × 0.70 × 8.0607) = **51.3 per tonne**.
2. At 60 with 40 percent pass-through, the net cost = 0.3 × 60 × 0.6 × 0.7 = 7.56 a year, PV 60.9; NPV = **+26.0 million**. Break-even = 51.3 / 0.6 = **85.6 per tonne**.
3. Annual saving = 300 × 0.0006 = 0.18 million; annuity factor (7 years, 4%) = 6.0021; PV = **1.080 million**. PV of costs = 0.1 × 6.0021 = **0.600 million**. Net = **+0.480 million**.
4. Annual cost = 400 × 0.0030 = **1.2 million**. PV factor for years 4 to 8 at 4.5 percent = 3.8469; PV = **4.616 million**.
5. PV(20 years) = 70 × 8.5136 = 596.0; PV(10 years) = 70 × 6.1446 = 430.1; expected = 0.4 × 596.0 + 0.6 × 430.1 = **496.5**; impairment = 500 − 496.5 = **3.5 million**.
6. At 55: accept A, B, C (costs below 55): abatement 1.5 Mt; net cost = −9 + 10 + 31.5 = 32.5 million; avoided charge = 82.5; net benefit = **50.0 million**. At 100: accept A to E: abatement 2.5 Mt; net cost = −9 + 10 + 31.5 + 28 + 57 = 117.5; avoided charge = 250; net benefit = **132.5 million**.
7. V0 = 1 / 0.065 = 15.38; V1 = 1 / 0.068 = 14.71; change = **−4.4 percent**.
8. t after 10 years = 1.0 / (4.0 / √10) = **0.79**. Years for t = 2: (2 × 4.0 / 1.0)^2 = **64 years**.

Chapter 24: Quantitative Methods: Simulation, Optimization and Machine Learning for Corporate Finance

Learning Objectives

- Design and interpret a Monte Carlo simulation of project value or cash flow, including the choice of stochastic drivers, the number of trials, the standard error of the estimate and simple variance-reduction techniques.
- Model correlated risks with Cholesky factorization, explain why correlation changes both the spread and the mean of outcomes, and recognize the limits of linear correlation in the tails.
- Build a stochastic forecast of prices and cash flows (random-walk and mean-reverting processes) and translate it into cash-flow-at-risk, liquidity buffers and covenant-breach probabilities.
- Formulate capital allocation as a linear, integer or convex optimization problem, interpret shadow prices, and explain why greedy rules such as “rank by profitability index” can fail.
- Apply regression and classification (OLS, ridge, logistic regression, boosted trees) to forecasting and credit decisions, evaluate them out of sample, and convert predicted probabilities into economically optimal decisions.
- Describe model risk, validation, monitoring and governance practices that make quantitative tools safe for decision-making in a finance function.

Opening Vignette

Illustrative scenario. Halcyon Industrial Gases is a fictional company with 14 air-separation plants, 2.1 billion in revenue, and a three-person “analytics” team inside the treasury.

The Halcyon capital committee meets every autumn to allocate roughly 400 million across 31 proposed projects. For years the process was a spreadsheet: each project had a single base-case NPV, a profitability index, and a traffic-light sensitivity table prepared by the sponsor. The portfolio was assembled by sorting on the index until the money ran out. Three things went wrong in the same year. A cluster of projects that all depended on the same industrial customers’ volumes underperformed together, so a downturn that each sensitivity table had treated as survivable became a firm-wide cash squeeze. The treasury’s liquidity plan, built on a single energy-price assumption, missed the cash trough by a wide margin. And the credit control team’s scorecard, which had been quietly re-fitted by a vendor, started approving customer limits that produced a spike in bad debts.

The new CFO’s diagnosis is blunt: “We have been using point estimates for decisions whose value comes from uncertainty and dependence.” She asks the analytics team for three things within a quarter. First, a simulation of the major projects that captures how energy prices, volumes and costs move together. Second, an optimization of the project portfolio that respects two years of budget constraints and tells the committee what an extra unit of capital is worth. Third, a disciplined review of the customer credit model with an independent validation. Her instructions end with a warning: “Anything that produces a number in a model is only as good as the assumptions behind it, so I want to see where each one can fail.”

This chapter builds that toolkit in the order the team used it: Monte Carlo simulation, correlation, stochastic forecasting, optimization, machine learning, and finally the governance that keeps all of it honest. Every code listing was run, and the printed output is the real output of that code (random seeds are fixed for reproducibility). The numbers are illustrative: they illustrate method, not any real company or market.

24.1 Why Quantitative Methods, and What They Cannot Do

24.1.1 From point estimates to distributions

Volume 1 taught valuation as a deterministic exercise: forecast cash flows, discount at a risk-adjusted rate, compare to cost (Chapter 6), then adjust for risk by sensitivity and scenario analysis. These tools answer "what if this input is different?" They do not answer "what is the probability that value is destroyed?", "how do risks interact?" or "what is the best way to spend limited capital across many uncertain options?" Quantitative methods address three kinds of questions:

1. **Propagation of uncertainty (simulation).** Given distributions for inputs and a model that maps inputs to outputs, what is the distribution of outputs? This is the domain of Monte Carlo simulation.
2. **Decision under constraints (optimization).** Given objectives and constraints, which action is best, and how sensitive is the answer to the constraints? This is the domain of linear, integer and convex programming.
3. **Prediction from data (statistical and machine learning).** Given historical data, how can we forecast a quantity or classify a case (for example, will this borrower default?) accurately out of sample?

The three are complements. A simulation without an optimizer tells you the risk of a given plan but not the best plan; an optimizer fed with point estimates ignores risk; a predictive model supplies the input distributions that both need.

24.1.2 The discipline of modelling

A model is a deliberate simplification. Useful models are *wrong in known and bounded ways*. Three principles guide this chapter.

- **Start from the decision.** The model exists to support a specific choice (invest or not, how much liquidity to hold, whether to approve credit). If you cannot name the decision, you are not ready to build the model.
- **Prefer transparent models until complexity earns its place.** A regression with five interpretable variables that you can explain to a board is usually better than a black box that improves accuracy by one percent. We will test that claim on data in Section 24.6.
- **Distrust precision.** A simulation reporting "probability of loss 46.7 percent" does not know the probability to three digits; the inputs are not that well known. Report ranges and show which assumptions drive them.

Garbage in, garbage out is a cliché but also a quantitative fact: the output uncertainty from *model* error (wrong distribution, wrong correlation, omitted risk factors) often exceeds the *sampling* uncertainty that simulation statistics report. We return to this in Section 24.7.

24.2 Monte Carlo Simulation

24.2.1 The method in five steps

Monte Carlo simulation replaces an analytical calculation with an experiment. The procedure is:

1. **Define the output** (for example, project NPV, year-3 EBITDA, minimum cash balance).
2. **Specify the model** that links inputs to the output, usually the same cash-flow model you would use deterministically.
3. **Choose input distributions** (and their dependence), calibrated from history, market data or expert judgement.
4. **Draw N random scenarios**, compute the output in each, and store the results.
5. **Summarize** the distribution: mean, standard deviation, percentiles, probability of falling below a threshold, expected shortfall, and the standard error of those estimates.

The central statistical fact is the **standard error of the Monte Carlo mean**:

```
SE = σ / √N
```

where σ is the standard deviation of the output across trials. To halve the error you need four times as many trials; to gain one more decimal digit of accuracy, one hundred times as many. Estimates of tail probabilities converge more slowly than means, because few trials land in the tail: the standard error of an estimated probability p is `√(p(1 − p) / N)`.

24.2.2 Choosing and calibrating distributions

- **Normal** distributions suit quantities that can be positive or negative and are roughly symmetric (annual changes in margin). They are unsuitable for prices or volumes that cannot go below zero.
- **Lognormal** distributions suit prices, sales and asset values: the logarithm is normal, so the variable is positive and right-skewed. In a geometric Brownian motion model, `P_t = P_0 × exp[(μ − σ²/2) t + σ √t × Z]`, with Z standard normal. The `−σ²/2` term (the Itô correction) keeps the *expected* price consistent with the stated drift.
- **Triangular or PERT** distributions are practical when only a minimum, most likely and maximum are available from a manager's judgement.
- **Mean-reverting** processes suit commodity prices, margins and interest rates, which are pulled toward a long-run level (Section 24.4).
- **Empirical (bootstrap)** distributions resample historical outcomes. They avoid parametric assumptions but inherit the history's blind spots.

Calibration is the weak point. A common error is to use the volatility of a short, calm sample for a long-horizon project. Another is to use *independent* distributions where drivers co-move (Section 24.3). Always compare the simulated distribution of an easily observed quantity (say, annual revenue growth) to history as a sanity check.

24.2.3 A worked simulation

Example 24.1 – Simulating a project NPV. A project costs 100 and runs for 5 years. In year 1 the expected volume is 1.0 million units, expected price 100, expected variable cost 60 per unit, fixed costs 8 per year, straight-line depreciation (20 a year), a 25 percent tax rate and a WACC of 9 percent. Price and unit cost follow annual geometric random walks (volatilities 15 percent and 8 percent), and volume is subject to a persistent level shock (volatility 12 percent). We consider three correlation structures for the shocks. The code below runs 100,000 trials per case.

Step 1: Deterministic benchmark. EBITDA = 1.0 × (100 − 60) − 8 = 32. EBIT = 32 − 20 = 12. NOPAT = 12 × 0.75 = 9. FCF = 9 + 20 = 29. The 5-year annuity factor at 9 percent is 3.8897, so PV = 29 × 3.8897 = 112.8 and NPV = 112.8 − 100 = **+12.8**.

Step 2: The simulation.

```
import numpy as np
rng = np.random.default_rng(42)
N, T = 100_000, 5
capex, wacc, tax = 100.0, 0.09, 0.25
vol0, price0, cost0 = 1.0, 100.0, 60.0      # million units, per unit
sig_p, sig_c, sig_v = 0.15, 0.08, 0.12      # annual vol of price, cost; volume level sd

def simulate(rho_pc, rho_pv, n=N, seed=42):
    rng = np.random.default_rng(seed)
    corr = np.array([[1, rho_pc, rho_pv],
                     [rho_pc, 1, 0.0],
                     [rho_pv, 0.0, 1]])
    L = np.linalg.cholesky(corr)
    z = rng.standard_normal((n, T, 3)) @ L.T      # correlated shocks, each year
    price = price0 * np.exp(np.cumsum(sig_p * z[:, :, 0] - 0.5 * sig_p**2, axis=1))
    cost  = cost0  * np.exp(np.cumsum(sig_c * z[:, :, 1] - 0.5 * sig_c**2, axis=1))
    vol_level = vol0 * np.exp(sig_v * z[:, 0, 2] - 0.5 * sig_v**2)   # persistent level
    volume = vol_level[:, None] * np.ones(T)
    ebitda = volume * (price - cost) - 8.0          # 8 fixed cost
    dep = capex / T
    fcf = (ebitda - dep) * (1 - tax) + dep
    disc = (1 + wacc) ** -np.arange(1, T + 1)
    return fcf @ disc - capex

det = sum(((1.0 * (100 - 60) - 8 - 20) * 0.75 + 20) / 1.09**t for t in range(1, 6)) - capex
print(f"Deterministic NPV: {det:.2f}")
for label, rpc, rpv in [("independent", 0.0, 0.0), ("natural hedge (rho_pc=+0.6, rho_pv=-0.4)", 0.6,
-0.4), ("adverse (rho_pc=-0.3, rho_pv=+0.5)", -0.3, 0.5)]:
    npv = simulate(rpc, rpv)
    q5, q95 = np.percentile(npv, [5, 95])
    print(f"{label:42s} mean={npv.mean():7.2f} sd={npv.std():6.2f} P(NPV<0)={np.mean(npv<0):.3f} 5%=
{q5:7.2f} 95%={q95:7.2f} SE={npv.std()/np.sqrt(len(npv)):.3f}")
# convergence
npv = simulate(0.6, -0.4)
for n in (100, 1000, 10_000, 100_000):
    print(n, round(npv[:n].mean(), 2), round(npv[:n].std()/np.sqrt(n), 2))
```

Output.

```
Deterministic NPV: 12.80
independent                                mean=  12.81 sd= 68.45 P(NPV<0)=0.467 5%= -84.39 95%= 136.40
SE=0.216
natural hedge (rho_pc=+0.6, rho_pv=-0.4)   mean=  10.74 sd= 50.70 P(NPV<0)=0.457 5%= -61.18 95%= 102.27
SE=0.160
adverse (rho_pc=-0.3, rho_pv=+0.5)         mean=  15.46 sd= 79.20 P(NPV<0)=0.472 5%= -93.91 95%= 160.48
SE=0.250
100 11.94 5.6
1000 9.95 1.55
10000 10.06 0.51
100000 10.74 0.16
```

Step 3: Reading the results. The deterministic NPV (12.80) matches the independent-case mean (12.81), a useful check that the model is coded correctly. The distribution, however, is wide: the standard deviation is 68.5, the 5th percentile is −84.4, and the project loses money in 46.7 percent of trials, even though the point estimate is positive. The “probability of a

negative NPV" is a more honest summary of the opportunity than the 12.8 mean. Because the mean is only 0.19 standard deviations above zero, the NPV is close to a coin flip with a large payoff range.

Step 4: Standard error and convergence. With σ = 68.45 and N = 100,000, the standard error of the mean is 68.45 / √100,000 = 0.216, matching the printed 0.216. The convergence table shows the running estimate settling as N grows (with the natural-hedge correlation structure): about 11.9 ± 5.6 at 100 trials, 9.95 ± 1.55 at 1,000, 10.06 ± 0.51 at 10,000 and 10.74 ± 0.16 at 100,000. A reader who ran only 100 trials would not even know the sign with confidence.

Step 5: Managerial interpretation. The decision question is not "is the NPV positive?" but "is the upside worth the downside and can we survive the downside?" The 5th percentile of −84 is nearly the entire initial investment. If the firm cannot absorb that loss, or if the downside is correlated with other exposures in the portfolio, the project should be restructured, staged or hedged even though the expected NPV is positive.

24.2.4 Variance reduction

Because accuracy improves only with √N, practitioners use techniques that reduce σ for a given N. The most accessible is **antithetic variates**: for each random draw Z also use −Z, so that the paired outcomes are negatively correlated and their average has lower variance. Other methods include **control variates** (adjust the estimate using a similar quantity with a known expectation), **stratified** or **Latin hypercube sampling** (force draws to cover the input space evenly), and **quasi-random (low-discrepancy) sequences**. As an illustration, the following code values a call-like payoff, an expansion option with strike 110 on an asset worth 100 with 30 percent volatility over two years, and compares plain and antithetic estimates against the closed-form Black-Scholes value (Volume 1, Chapter 18). Each configuration is estimated 200 times with 10,000 draws each, and the spread across runs is reported.

```
import numpy as np
# Estimating E[max(S_T - K,0)] (a call-like payoff, e.g. an expansion option) with plain vs antithetic MC
from math import log, sqrt, exp, erf
S0, K, r, sig, T = 100.0, 110.0, 0.04, 0.30, 2.0
Phi = lambda x: 0.5 * (1 + erf(x / sqrt(2)))
d1 = (log(S0 / K) + (r + 0.5 * sig**2) * T) / (sig * sqrt(T)); d2 = d1 - sig * sqrt(T)
bs = S0 * Phi(d1) - K * exp(-r * T) * Phi(d2)
print("Black-Scholes value: %.4f" % bs)
def est(n, anti, seed):
    rng = np.random.default_rng(seed)
    z = rng.standard_normal(n // 2 if anti else n)
    if anti:
        z = np.concatenate([z, -z])
    ST = S0 * np.exp((r - 0.5 * sig**2) * T + sig * sqrt(T) * z)
    return exp(-r * T) * np.maximum(ST - K, 0).mean()
for anti in (False, True):
    e = np.array([est(10_000, anti, s) for s in range(200)])
    print("antithetic=%s: mean of 200 estimates %.4f, std error across runs %.4f" % (anti, e.mean(),
e.std()))
```

```
Black-Scholes value: 16.1686
antithetic=False: mean of 200 estimates 16.1408, std error across runs 0.3405
antithetic=True: mean of 200 estimates 16.1450, std error across runs 0.2602
```

Both methods are unbiased (their averages are close to the exact 16.17), but the antithetic version has a standard error about 24 percent lower (0.260 versus 0.341), equivalent to a 70 percent increase in the number of trials at no extra cost. The gain depends on the payoff: monotone payoffs benefit most, and payoffs that are symmetric in Z benefit least.

24.2.5 Common pitfalls

Treating the output as truth (show sensitivity to the *distributional assumptions*, not just the seed); too few trials for tail statistics (a 1 percent tail probability with 10 percent relative error needs about (0.99 / 0.01) / 0.01 = 9,900 trials); ignoring management response such as closing a plant when margins turn negative (compare Chapter 23, Example 23.3); mixing real-world cash-flow simulation with risk-free discounting, or risk-neutral simulation with risk-adjusted discounting, which double-counts or omits risk; and sloppy random-number practice (unfixed seeds, reused seeds for independent processes, incorrectly built correlation).

24.3 Correlated Risks

24.3.1 Why correlation matters more than the average

Risk factors in corporations rarely move independently: input costs and product prices share an energy driver; volumes across divisions depend on the same macroeconomy; credit losses cluster in downturns. Dependence affects the *distribution* of outcomes in two ways.

First, it changes *dispersion*. For two variables X and Y with standard deviations σ_X, σ_Y and correlation ρ, the variance of a margin `M = X − Y` is:

```
Var(X − Y) = σ_X² + σ_Y² − 2 ρ σ_X σ_Y
```

Positive correlation between a price and a cost *reduces* margin variance (a natural hedge); negative correlation increases it. For a sum, the sign of the correlation term flips: `Var(X + Y) = σ_X² + σ_Y² + 2ρσ_Xσ_Y`, so positively correlated divisional profits make the sum riskier, which is why diversification across negatively or weakly correlated units reduces risk.

Second, when the model is nonlinear, dependence also changes the *mean*: `E[XY] = E[X]E[Y] + Cov(X, Y)`. In Example 24.1, the simulated mean NPV differed across correlation assumptions (12.8, 10.7 and 15.5) because revenue is the product of price and volume, and the covariance enters the expectation.

24.3.2 Generating correlated shocks: Cholesky factorization

To simulate correlated normal shocks from independent standard normals, factor the correlation matrix R as `R = L Lᵀ`, where L is lower-triangular (the **Cholesky factor**). Then if Z is a vector of independent N(0,1) variables, `X = L Z` has correlation matrix R. For two variables:

```
L = [ 1          0          ]        X₁ = Z₁
    [ ρ    √(1 − ρ²)        ]        X₂ = ρ Z₁ + √(1 − ρ²) Z₂
```

Check: `Var(X₂) = ρ² + (1 − ρ²) = 1`, and `Cov(X₁, X₂) = ρ`. In code, `np.linalg.cholesky(corr)` produces L, and `shocks @ L.T` applies it to a matrix of independent draws (as in the listing in Example 24.1). The correlation matrix must be positive semi-definite; estimated matrices from short or incomplete data may not be, and need repair (for instance, by shrinkage toward a structured target or by adjusting negative eigenvalues).

Example 24.2 – How correlation changes the risk of a margin. In Example 24.1, the first-year unit margin is `M = P − C`, with price volatility 15 (15 percent of 100) and cost volatility 4.8 (8 percent of 60). Compute the standard deviation of unit margin for three correlations between price and cost shocks and relate it to the simulated NPV spreads.

Step 1: Independent (ρ = 0). Var = 15² + 4.8² − 0 = 225 + 23.04 = 248.04; σ = **15.75**.

Step 2: Cost follows price (ρ = +0.6). Var = 248.04 − 2 × 0.6 × 15 × 4.8 = 248.04 − 86.40 = 161.64; σ = **12.71**, a reduction of 19 percent.

Step 3: Price falls while cost does not follow (ρ = −0.3). Var = 248.04 + 2 × 0.3 × 15 × 4.8 = 248.04 + 43.20 = 291.24; σ = **17.07**, an increase of 8 percent.

Step 4: Link to the simulation. In Example 24.1 the NPV standard deviations are 68.5 (independent), 50.7 (price-cost +0.6 and price-volume −0.4) and 79.2 (price-cost −0.3, price-volume +0.5). The simulation also incorporates the volume correlation and compounding across years, so the ratios are larger than the single-year margin ratios, but the ordering is identical: the cost-follows-price and offsetting-volume structure cut the NPV standard deviation by 26 percent, whereas "bad" dependence (price and volume falling together, cost not following) raises it by 16 percent. The 5th-percentile NPV moves accordingly: −61.2, −84.4 and −93.9.

Interpretation. A manager can sometimes change the correlation structure instead of just measuring it: indexing input-cost contracts to output price (increasing ρ_pc), choosing customers whose volumes are counter-cyclical, or using hedges whose payoffs offset the exposures. The simulation reveals which dependency is worth paying to alter.

24.3.3 Limits of linear correlation

Pearson correlation measures *linear* dependence in the bulk of the distribution. Two problems matter for corporate risk.

- **Tail dependence.** Variables with modest correlation in normal times can crash together in a crisis. A multivariate normal distribution has no tail dependence: extreme joint moves are assumed to be exceedingly rare. Distributions with heavier joint tails (for example, the multivariate Student-t, or a t-copula) allow extremes to cluster. The 2007-2009 financial crisis popularized the lesson that using a Gaussian copula for credit portfolios understated joint default risk; the point of principle is that the *copula* (dependence structure) is a separate modelling choice from the *marginal* distributions.
- **Unstable correlations.** Estimated correlations change with the regime and typically rise in stress. Practical safeguards: stress the correlation matrix (add 0.2 to all off-diagonals, capped), use rank correlations (Spearman, Kendall) that are robust to outliers, and run a "everything goes wrong together" scenario in addition to the statistical model.

A practical compromise for corporate finance is a **factor model**: let each risk driver depend on a small number of common factors (macro activity, energy prices, credit spreads) plus idiosyncratic noise. This yields a structured, positive-definite correlation matrix with an economic interpretation, and supports scenario design ("what if the macro factor falls two standard deviations?").

24.4 Stochastic Forecasting

24.4.1 Three families of processes

A forecast is a distribution over future paths, not a line. Which stochastic process to use depends on the economics of the variable.

Geometric Brownian motion (random walk in logs). Appropriate for asset values, revenues of growth businesses and, with caution, equity prices. Shocks are permanent, and uncertainty grows with the square root of time: the standard deviation of log value at horizon t is $\sigma\sqrt{t}$.

Mean reversion (Ornstein-Uhlenbeck in logs). Appropriate for commodity prices, margins, utilization and interest rates, where economic forces (entry, substitution, policy) push the variable back toward a long-run level. In discrete time with quarterly steps:

```
x_{t+1} = x_t + κ (m − x_t) + σ ε_{t+1},   x = ln(price),  m = ln(long-run price)
E[x_t] − m = (x_0 − m) (1 − κ)^t
```

The parameter κ is the speed of reversion; the half-life is `ln(0.5) / ln(1 − κ)` periods. For κ = 0.15 per quarter, the half-life is 4.3 quarters. Unlike a random walk, the uncertainty does not grow without bound: the long-run standard deviation of x is `σ / √(1 − (1 − κ)²)`.

Regime-switching and jump models. Appropriate when the world has distinct states (expansion and recession, regulated and deregulated) or when rare large events are central (a regulatory ban, a supply disruption). A two-state Markov chain with different drifts and volatilities per state, plus a jump process with intensity and size distribution, are the standard building blocks.

Forecasters must also separate **parameter uncertainty** (we do not know κ or σ exactly) from **process uncertainty** (the shocks themselves). A careful analysis samples parameters from their estimated distribution and then simulates paths for each parameter set; otherwise the resulting fan chart is too narrow.

24.4.2 From paths to risk measures

Once paths are simulated, corporate risk measures follow directly.

- **Cash-flow-at-risk (CFaR).** The shortfall of cash flow (or EBITDA) over a horizon relative to its expected value at a given confidence level: `CFaR_95 = E[CF] − 5th percentile of CF`. It is the corporate analogue of value-at-risk but is defined on flows, which is how a non-financial firm experiences risk.
- **Probability of breaching a covenant or liquidity floor** at any time in the horizon, computed from the *minimum* over the path rather than the end value (a path-dependent measure).
- **Liquidity buffer.** The amount of undrawn facilities and cash needed so that the probability of running out is below a tolerance, for example 1 percent over the next 12 quarters.
- **Expected shortfall.** The average outcome in the worst tail, which captures the severity of tail outcomes that a percentile ignores.

These measures link naturally to hedging decisions (Volume 1, Chapter 19) and to capital structure (Volume 1, Chapter 15): the higher the operating leverage, the greater the probability of covenant stress at a given debt level.

Example 24.3 – Liquidity and cash-flow-at-risk for a commodity-exposed firm. A producer sells 4.0 million units a quarter at a market price that follows a mean-reverting process in logs: current price 80, long-run price 70, κ = 0.15 per quarter, quarterly volatility σ = 6 percent. Variable cost is 40 per unit, fixed cost is 100 per quarter, capex 6 and interest 9 per quarter, opening cash 40. The firm wants a minimum cash floor of 20 and a minimum rolling four-quarter interest cover (EBITDA/interest) of 1.5x. Simulate 12 quarters with 50,000 paths.

```
import numpy as np
rng = np.random.default_rng(7)
N, Q = 50_000, 12                 # 12 quarters
# log-price Ornstein-Uhlenbeck: x_{t+1} = x_t + k*(m - x_t) + s*eps
p0, pbar = 80.0, 70.0
k, s = 0.15, 0.06                 # quarterly reversion speed and vol
m, x = np.log(pbar), np.full(N, np.log(p0))
price = np.empty((N, Q))
for q in range(Q):
    x = x + k * (m - x) + s * rng.standard_normal(N)
    price[:, q] = np.exp(x)
units, varcost, fixed = 4.0, 40.0, 100.0          # million units per quarter; per unit; fixed per
quarter
ebitda = units * (price - varcost) - fixed
capex, interest, tax_rate = 6.0, 9.0, 0.0       # simplified: pre-tax, quarterly
fcf = ebitda - capex - interest
cash = 40.0 + np.cumsum(fcf, axis=1)
min_cash = cash.min(axis=1)
print("Expected price path (q1,q4,q8,q12):", np.round(price.mean(axis=0)[[0,3,7,11]], 1))
print("Std of price (q1,q4,q8,q12):       ", np.round(price.std(axis=0)[[0,3,7,11]], 1))
print("Mean 12q cumulative FCF:", round(fcf.sum(axis=1).mean(), 1))
print("Cumulative FCF 5th/50th/95th pct:", np.round(np.percentile(fcf.sum(axis=1), [5, 50, 95]), 1))
print("P(cash < 0 at any quarter):", round(np.mean(min_cash < 0), 4))
print("P(cash < 20 at any quarter):", round(np.mean(min_cash < 20), 4))
print("Liquidity buffer needed for 95% / 99% confidence (min cash shortfall vs 20 floor):")
need = np.maximum(20 - min_cash, 0)
print(round(np.percentile(need, 95), 1), round(np.percentile(need, 99), 1))
# Cash-flow-at-risk: shortfall of 12q cumulative EBITDA versus its mean
cum = ebitda.sum(axis=1)
print("Mean cumulative EBITDA:", round(cum.mean(), 1), " 5th pct:", round(np.percentile(cum, 5), 1),
      " CFaR(95%):", round(cum.mean() - np.percentile(cum, 5), 1))
# Covenant: rolling 4-quarter EBITDA / interest at least 1.5x in every quarter from q4
ltm = np.stack([ebitda[:, i-3:i+1].sum(axis=1) for i in range(3, Q)], axis=1)
cover = ltm / (4 * interest)
print("P(LTM interest cover < 1.5x in any quarter):", round(np.mean((cover < 1.5).any(axis=1)), 4))
```

```
Expected price path (q1,q4,q8,q12): [78.5 75.4 73.  71.8]
Std of price (q1,q4,q8,q12):        [4.7 7.4 8.  8.1]
Mean 12q cumulative FCF: 267.1
Cumulative FCF 5th/50th/95th pct: [-142.3  256.9  710. ]
P(cash < 0 at any quarter): 0.131
P(cash < 20 at any quarter): 0.1563
Liquidity buffer needed for 95% / 99% confidence (min cash shortfall vs 20 floor):
127.6 271.4
Mean cumulative EBITDA: 447.1  5th pct: 37.7  CFaR(95%): 409.4
P(LTM interest cover < 1.5x in any quarter): 0.4397
```

Step 1: Check the price process against theory. The expected log-price after 4 quarters is `ln 70 + (ln 80 − ln 70) × 0.85⁴ = 4.2485 + 0.1335 × 0.5220 = 4.3182`, which corresponds to roughly 75.0 in log terms; the simulated mean price at quarter 4 is 75.4, a bit higher because the mean of a lognormal exceeds the exponential of the mean log (Jensen's inequality; the gap is about ½ × variance × price ≈ 0.4). By quarter 12, expected price is 71.8, approaching the long-run level 70, and the standard deviation (8.1) is flattening, as mean reversion implies.

Step 2: Operating leverage. At the long-run price EBITDA is 4.0 × (70 − 40) − 100 = **20 per quarter**, while the margin per unit is 30 against price volatility of about 7 to 8, so a one-standard-deviation price move (8) shifts EBITDA by 4.0 × 8 = 32, more than the whole expected EBITDA. This extreme operating leverage is why the outputs below are so dispersed.

Step 3: Risk measures. The 12-quarter cumulative EBITDA has a mean of 447.1 and a 5th percentile of 37.7, so **CFaR(95%) = 447.1 − 37.7 = 409.4**. The probability that cash goes negative at some point is 13.1 percent and the probability that it falls below the 20 floor is 15.6 percent. To limit the floor-breach probability to 5 percent the firm needs an additional liquidity buffer of about **127.6**, and for a 1 percent tolerance about **271.4**. The probability of breaching the interest-cover covenant at least once is 44.0 percent.

Step 4: Interpretation. The point forecast says "EBITDA 447 over three years, comfortably above debt service." The distribution says the firm has a one-in-seven chance of running out of cash and a four-in-ten chance of covenant stress. Management's choices are now concrete: arrange a committed facility of roughly 130 to 270, hedge part of the volume (reducing σ), renegotiate the covenant, or lower fixed costs. A hedge that fixes the price on half the volume cuts the exposure by half and converts a large share of tail outcomes into manageable ones; the simulation can be rerun with the hedge to price its benefit against its cost.

Common mistakes in stochastic forecasting. Using a random walk for a mean-reverting variable (overstating long-run uncertainty), or mean reversion for a variable with a structural break (understating it); calibrating volatility to a short calm window; ignoring parameter uncertainty; simulating variables independently when they share drivers; and omitting management response such as cutting capex, drawing facilities or suspending dividends, which makes the model a poor representation of actual decision-making.

24.5 Optimization for Capital Allocation

24.5.1 The structure of an optimization problem

Every optimization problem has three parts: **decision variables** (which projects, how much of each), an **objective** (maximize NPV or risk-adjusted value) and **constraints** (budgets over several years, mutual exclusivity, minimum scale, strategic requirements). Problem types matter because they determine how reliably the solution can be found:

Type	Decision variables	Typical corporate use	Tractability
Linear program (LP)	Continuous	Divisible investments, product mix, funding mix	Very high; optimum guaranteed; shadow prices available
Integer / mixed-integer program (MIP)	Whole-project yes/no	Capital rationing with indivisible projects	Good for dozens to thousands of projects; can be hard in general
Convex program (e.g., quadratic)	Continuous	Mean-variance allocation across businesses, least-squares fitting, hedge ratios	High; local optimum is global
Non-convex / stochastic	Mixed	Staging with real options, sequential decisions	Needs heuristics, dynamic programming or simulation-optimization

24.5.2 Capital rationing as a linear and integer program

When capital is limited, the NPV rule of Volume 1, Chapter 6 (accept all positive-NPV projects) must be replaced by the choice that maximizes total NPV subject to budgets. With binary variables `x_i ∈ {0, 1}` for each of n projects:

```
maximize   Σ_i NPV_i · x_i
subject to Σ_i Outlay_{i,t} · x_i ≤ Budget_t       for each year t
           x_i ∈ {0, 1}
```

The **LP relaxation** replaces `x_i ∈ {0,1}` by `0 ≤ x_i ≤ 1` (projects are divisible). It gives an *upper bound* on the achievable NPV and a **shadow price** (dual value) for each budget constraint, the increase in total NPV from one additional unit of budget. The shadow price is the economically correct *cost of capital for internal rationing*: a project should be accepted only if its NPV per unit of scarce budget exceeds the shadow price. When the budget is not binding the shadow price is zero, and when it is binding it equals the profitability index of the marginal project.

The popular heuristic “rank by profitability index (NPV per unit of outlay) and fund until the money runs out” is optimal only when there is a single constraint and projects are divisible. With several periods, indivisible projects, or mutually exclusive alternatives, it can leave capital unused or pick a poor combination.

Example 24.4 – Capital rationing with two budgets. Eight projects have the NPVs and outlays shown below (all in millions). The year-0 budget is 220 and the year-1 budget is 90.

Project	NPV	Year-0 outlay	Year-1 outlay	PI (NPV / yr-0 outlay)
A	14	40	10	0.350
B	22	65	20	0.338
C	9	25	0	0.360
D	31	90	30	0.344
E	18	55	15	0.327
F	12	30	25	0.400
G	27	80	40	0.338
H	7	20	5	0.350

```
import numpy as np, itertools
from scipy.optimize import linprog, minimize
# --- Capital rationing: 8 projects, two budget years
names = list("ABCDEFGH")
npv   = np.array([14, 22, 9, 31, 18, 12, 27, 7.0])
cap0  = np.array([40, 65, 25, 90, 55, 30, 80, 20.0])    # year-0 outlay
cap1  = np.array([10, 20,  0, 30, 15, 25, 40,  5.0])    # year-1 outlay
B0, B1 = 220.0, 90.0
print("Profitability index (NPV / yr-0 outlay):", {n: round(float(v), 3) for n, v in zip(names, npv /
cap0)})
# LP relaxation (projects divisible): maximize NPV
res = linprog(-npv, A_ub=[cap0, cap1], b_ub=[B0, B1], bounds=[(0, 1)] * 8, method="highs")
print("LP relaxation  NPV = %.2f" % -res.fun, " x =", np.round(res.x, 3))
print("Shadow prices (NPV per extra unit of budget): yr0 = %.4f, yr1 = %.4f" % tuple(-
res.ineqlin.marginals))
# Integer solution by enumeration (2^8 = 256 combinations)
best = (0, None)
for bits in itertools.product([0, 1], repeat=8):
    x = np.array(bits)
    if x @ cap0 <= B0 and x @ cap1 <= B1 and x @ npv > best[0]:
        best = (x @ npv, bits)
sel = [n for n, b in zip(names, best[1]) if b]
print("Integer optimum NPV = %.1f, projects %s, yr0 spend = %.0f, yr1 spend = %.0f" % (best[0], sel,
np.array(best[1]) @ cap0, np.array(best[1]) @ cap1))
# Greedy by PI for comparison
order = np.argsort(-(npv / cap0)); x = np.zeros(8); u0 = u1 = 0
for i in order:
    if u0 + cap0[i] <= B0 and u1 + cap1[i] <= B1: x[i] = 1; u0 += cap0[i]; u1 += cap1[i]
print("Greedy by PI: projects %s, NPV = %.1f" % ([n for n, b in zip(names, x) if b], x @ npv))
# Budget sensitivity (LP bound)
for b in (200, 220, 240):
    r = linprog(-npv, A_ub=[cap0, cap1], b_ub=[b, B1], bounds=[(0, 1)] * 8, method="highs")
    print("yr0 budget", b, "LP NPV %.2f" % -r.fun)
```

```
Profitability index (NPV / yr-0 outlay): {'A': 0.35, 'B': 0.338, 'C': 0.36, 'D': 0.344, 'E': 0.327, 'F':
0.4, 'G': 0.338, 'H': 0.35}
LP relaxation  NPV = 78.08  x = [1.    0.231 1.    1.    0.    1.    0.    1.   ]
Shadow prices (NPV per extra unit of budget): yr0 = 0.3385, yr1 = 0.0000
Integer optimum NPV = 77.0, projects ['C', 'D', 'E', 'F', 'H'], yr0 spend = 220, yr1 spend = 75
Greedy by PI: projects ['A', 'C', 'D', 'F', 'H'], NPV = 73.0
yr0 budget 200 LP NPV 71.28
yr0 budget 220 LP NPV 78.08
yr0 budget 240 LP NPV 84.85
```

Step 1: LP relaxation. The relaxed problem funds A, C, D, F and H fully and 23.1 percent of B, for a total NPV of **78.08**. The year-0 budget binds (40 + 25 + 90 + 30 + 20 + 0.231 × 65 = 220) while the year-1 budget does not (10 + 0 + 30 + 25 + 5 + 0.231 × 20 = 74.6 < 90). Hence the year-1 shadow price is zero and the year-0 shadow price is 0.3385, exactly the PI of the marginal project B (22 / 65).

Step 2: Meaning of the shadow price. Each extra unit of year-0 capital is worth 0.3385 of NPV (at the margin), so a 20 increase in the budget raises the bound from 71.28 (budget 200) to 78.08 (220) to 84.85 (240), steps of about 6.8, equal to 20 × 0.3385 as the output confirms. A manager can ask: “Is it worth raising 20 more of capital at a cost below 0.34 of NPV per unit?” If raising capital costs less than that in lost value, the budget is too tight.

Step 3: Integer optimum. Projects cannot be split, so enumerating the $2^8 = 256$ combinations finds the best feasible portfolio: **C, D, E, F, H**, with NPV **77.0**, year-0 spend 220 and year-1 spend 75. The LP bound is 78.08, so the integrality gap is only 1.08, or 1.4 percent.

Step 4: The greedy rule. Ranking by PI picks F (0.400), C (0.360), A and H (0.350), then D (0.344), giving A, C, D, F, H with NPV **73.0**: it spends only 205 of the 220 available (and cannot afford B or E in the remaining 15) and gives up 4.0 of NPV relative to the integer optimum, because it did not consider the combination E plus the rest. The optimal portfolio excludes A, a project with a higher PI than E, because E fits the budget better when combined with the others.

Interpretation. The optimizer gives the committee three things the spreadsheet did not: a provably best portfolio (77.0), a price of capital (0.34 per unit of year-0 budget), and a quantification of what the rule of thumb costs (4.0, about 5 percent of portfolio NPV). In practice, add constraints for mutually exclusive projects, minimum viable scale, strategic must-do investments and diversification limits, and run the optimization under simulated NPV distributions to pick a portfolio that is robust, not just optimal on point estimates.

Caveats. Optimizers exploit errors: if one project's NPV is overstated by an optimistic sponsor, the optimizer will pick it ("optimizer's curse"), so apply haircuts or shrinkage to sponsor forecasts and require independent review. Optimal portfolios can be unstable: small changes in inputs may change the chosen projects, so examine a range of near-optimal solutions. And NPVs derived under different risk assumptions are not comparable; make sure each project's NPV is computed at a discount rate consistent with its risk (Volume 1, Chapter 11).

24.5.3 Convex optimization: allocating across risky businesses

When returns are uncertain and correlated, the question is how much to allocate to each business unit, geography or technology. The mean-variance framework of Markowitz (Volume 1, Chapter 10) gives a convex quadratic problem:

```
minimize   ½ λ wᵀ Σ w − μᵀ w
subject to Σ w_i = 1,   0 ≤ w_i ≤ w_max
```

where μ is the vector of expected returns, Σ the covariance matrix ($\Sigma_{ij} = \rho_{ij}\,\sigma_i\,\sigma_j$), λ the risk-aversion parameter and w_{max} a concentration limit. Because the objective is convex and the constraints are linear, any local optimum found by a solver is the global optimum.

For two assets the minimum-variance weight has a closed form:

$$w_1^* = \frac{\sigma_2^2 - \rho\,\sigma_1\,\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\,\rho\,\sigma_1\,\sigma_2}$$

Example 24.5 – Allocation across four business units. A conglomerate has four units with expected returns 8, 11, 14 and 9 percent, volatilities 10, 16, 25 and 12 percent and the correlation matrix shown in the code (unit pairs 1-2: 0.3, 1-3: 0.1, 1-4: 0.4, 2-3: 0.5, 2-4: 0.2, 3-4: 0.1). Risk aversion $\lambda = 3$; no unit may exceed 50 percent of capital. The code below is the second half of the listing in Example 24.4 (it reuses the imports at the top of that listing).

```
# --- Convex: minimum-variance allocation of a fixed capital pot across 4 business units
mu = np.array([0.08, 0.11, 0.14, 0.09])
sd = np.array([0.10, 0.16, 0.25, 0.12])
C = np.array([[1, .3, .1, .4], [.3, 1, .5, .2], [.1, .5, 1, .1], [.4, .2, .1, 1]])
S = np.outer(sd, sd) * C
lam = 3.0   # risk aversion
obj = lambda w: 0.5 * lam * w @ S @ w - mu @ w
cons = [{"type": "eq", "fun": lambda w: w.sum() - 1}]
r = minimize(obj, np.full(4, .25), bounds=[(0, 0.5)] * 4, constraints=cons, method="SLSQP")
w = r.x
print("Mean-variance weights (max 50% per unit):", np.round(w, 3))
print("Expected return %.4f, risk %.4f" % (mu @ w, np.sqrt(w @ S @ w)))
r2 = minimize(lambda w: w @ S @ w, np.full(4, .25), bounds=[(0, 0.5)] * 4, constraints=cons,
method="SLSQP")
print("Min-variance weights:", np.round(r2.x, 3), "return %.4f risk %.4f" % (mu @ r2.x, np.sqrt(r2.x @ S @
r2.x)))
print("Equal weights: return %.4f risk %.4f" % (mu @ np.full(4, .25), np.sqrt(np.full(4, .25) @ S @
np.full(4, .25))))
```

```
Mean-variance weights (max 50% per unit): [0.059 0.263 0.285 0.393]
Expected return 0.1089, risk 0.1185
Min-variance weights: [0.5   0.099 0.055 0.346] return 0.0897 risk 0.0868
Equal weights: return 0.1050 risk 0.1091
```

Step 1: Maximum-utility portfolio. The solver allocates 5.9 percent, 26.3 percent, 28.5 percent and 39.3 percent to units 1 to 4, for an expected return of 10.89 percent and risk of 11.85 percent. The highest-return unit (unit 3, 14 percent, but 25 percent volatility) gets only 28.5 percent: the model trades return against risk and correlation. Unit 4 gets the largest weight, 39.3 percent, because it combines moderate return, low volatility and low correlation with units 2 and 3.

Step 2: Compare with alternatives. The minimum-variance portfolio holds 50 percent in unit 1 (the cap binds), returning 8.97 percent at 8.68 percent risk. The equal-weight portfolio returns 10.50 percent at 10.91 percent risk. The optimal portfolio has 0.39 more return than equal-weight for 0.94 more risk. A return-per-risk comparison (return divided by risk) is 0.92 for the optimum, 0.96 for equal weights and 1.03 for minimum variance; note that these ratios are not the objective, since the investor's risk aversion $\lambda = 3$ determines the preferred trade-off.

Step 3: Practical warnings. Mean-variance solutions are extremely sensitive to the inputs, especially expected returns. Rerunning the solver with unit 1's expected return raised by just one percentage point (8 to 9 percent) moves its weight from 5.9 percent to 33.4 percent and cuts unit 4's weight by 18 points. Remedies are shrinkage of expected returns toward a common value, concentration caps (as here), resampling, or specifying views in terms of scenarios (Black-Litterman style). Internal "business units" are also not freely tradable assets: capital cannot be shifted overnight and the diversification benefit assumes correlations are stable. Use the output as a guide to direction (reduce reliance on the highest-risk unit), not an exact prescription.

24.6 Regression and Classification for Credit and Forecasting

24.6.1 Prediction is not explanation

Supervised learning fits a function from inputs (features) to a target using historical examples. For a finance function two tasks dominate: **regression** (predict a continuous quantity such as next-quarter revenue growth, a recovery rate or a collection delay) and **classification** (predict a category or probability, such as default versus no default). The goal is

generalization, accuracy on data the model has not seen, rather than a good fit to the past.

The central tension is the **bias-variance trade-off**. A model that is too simple (high bias) misses real structure. A model that is too flexible (high variance) fits noise: with many candidate features and limited history it will find patterns that disappear out of sample. Corporate data sets are typically small (60 quarters of history, a few hundred defaults), so variance is the dominant danger. Remedies are regularization (ridge or lasso penalties), feature selection driven by economic reasoning, simpler model families and, above all, honest out-of-sample evaluation.

24.6.2 Honest evaluation: train, test and walk-forward

- **Hold-out test set.** Split the data randomly for cross-sectional problems (credit scoring across borrowers), keeping the test data untouched until the end.
- **Walk-forward (rolling or expanding window) validation.** For time series, train on data up to time t, forecast t+1, move forward. Random splits leak future information into training and overstate accuracy.
- **Cross-validation** for tuning hyperparameters (the ridge penalty, tree depth), always nested inside the training portion.
- **Leakage.** Any feature that would not be known at decision time (a variable updated after default, a restated financial figure) contaminates the model. Leakage is the most common cause of a model that looks superb in development and fails in production.

24.6.3 Regression for forecasting: ordinary least squares and ridge

OLS chooses coefficients β to minimize the sum of squared residuals. With p features and n observations the variance of out-of-sample prediction error rises roughly in proportion to p/n; with p close to n it explodes. **Ridge regression** minimizes $\Sigma\ \text{residual}^2 + \alpha\ \Sigma\ \beta_j^2$, shrinking all coefficients toward zero and trading a little bias for a large reduction in variance. **Lasso** uses $\alpha\ \Sigma\ |\beta_j|$, which sets some coefficients exactly to zero (feature selection). The penalty α is chosen by cross-validation.

Example 24.6 – Overfitting in a revenue-growth forecast. A firm tries to forecast quarterly revenue growth. Suppose that growth truly depends on two drivers (a macro indicator and an order-book indicator) and has noise, while the analyst has also collected 18 other indicators that are pure noise but look plausible. There are 120 quarters of data. We compare four forecasters using walk-forward validation: train on the first 60 quarters (then expand by one quarter each step), forecast the next quarter, and score the 60 forecasts.

```
import numpy as np
from sklearn.linear_model import LinearRegression, RidgeCV
from sklearn.metrics import mean_squared_error
rng = np.random.default_rng(11)
# Quarterly revenue growth forecasting: 2 true drivers + 18 noise indicators, 60 quarters of history
T, k_noise = 120, 18
macro = rng.normal(0, 1, T)
orders = rng.normal(0, 1, T)
noise = rng.normal(0, 1, (T, k_noise))
growth = 0.02 + 0.015 * macro + 0.020 * orders + rng.normal(0, 0.02, T)
X = np.column_stack([macro, orders, noise])
# walk-forward: train on first 60 quarters, forecast the next 60 one-step-ahead using expanding window
def rmse(a, b): return float(np.sqrt(mean_squared_error(a, b)))
preds = {"naive (train mean)": [], "OLS 2 drivers": [], "OLS all 20": [], "Ridge all 20": []}
actual = []
for t in range(60, T):
    Xtr, ytr = X[:t], growth[:t]
    preds["naive (train mean)"].append(ytr.mean())
    preds["OLS 2 drivers"].append(LinearRegression().fit(Xtr[:, :2], ytr).predict(X[t:t+1, :2])[0])
    preds["OLS all 20"].append(LinearRegression().fit(Xtr, ytr).predict(X[t:t+1])[0])
    preds["Ridge all 20"].append(RidgeCV(alphas=np.logspace(-2, 3, 30)).fit(Xtr, ytr).predict(X[t:t+1])
[0])
    actual.append(growth[t])
for name, p in preds.items():
    print("%-20s out-of-sample RMSE = %.4f" % (name, rmse(actual, p)))
ols_all = LinearRegression().fit(X[:60], growth[:60])
print("In-sample R^2, OLS all 20 (first 60q): %.3f ; OLS 2 drivers: %.3f" % (ols_all.score(X[:60],
growth[:60]), LinearRegression().fit(X[:60, :2], growth[:60]).score(X[:60, :2], growth[:60])))
```

```
naive (train mean)   out-of-sample RMSE = 0.0345
OLS 2 drivers        out-of-sample RMSE = 0.0204
OLS all 20           out-of-sample RMSE = 0.0223
Ridge all 20         out-of-sample RMSE = 0.0218
In-sample R^2, OLS all 20 (first 60q): 0.686 ; OLS 2 drivers: 0.597
```

Step 1: In-sample versus out-of-sample. In the first 60 quarters, OLS with all 20 features fits better ($R^2 = 0.686$) than OLS with the two true drivers (0.597). A naive reading would call the 20-feature model superior.

Step 2: Out-of-sample results. Out of sample the ranking reverses. The two-driver model has the lowest RMSE (0.0204). The all-feature OLS is 9.3 percent worse (0.0223) because it has fitted noise. Ridge with cross-validated penalty recovers part of the loss (0.0218) but cannot match the model that used only the right variables. The naive forecast (training mean) has RMSE 0.0345; so the predictors are genuinely useful, and the best model cuts forecast error by 41 percent.

Step 3: Lessons. (a) In-sample R^2 rewards complexity; (b) regularization helps but is not a substitute for economic judgement about which variables belong; (c) always compare with a naive benchmark, because a complicated model that does not beat “last year’s growth” or “the historical mean” adds cost without value.

24.6.4 Classification for credit: logistic regression and tree ensembles

For binary outcomes the standard model is **logistic regression**:

```
P(default) = 1 / (1 + exp(−z)),    z = β₀ + β₁ x₁ + … + β_k x_k
```

Coefficients are estimated by maximum likelihood, and each β_j is the change in the log-odds of default per unit of x_j. Logistic regression is transparent, produces calibrated probabilities if the model is well-specified, and is widely accepted by regulators and auditors. **Tree ensembles** (random forests, gradient-boosted trees) capture nonlinearities and interactions automatically but are harder to explain and more prone to overfit on small samples.

Evaluate classifiers on two distinct dimensions.

- **Discrimination:** does the model rank risky cases above safe ones? The common measure is the area under the ROC curve (**AUC**), the probability that a randomly chosen defaulter receives a higher score than a randomly chosen non-defaulter (0.5 = random, 1.0 = perfect). A related measure is the Gini coefficient, `2 × AUC − 1`, and the KS statistic.
- **Calibration:** do predicted probabilities match observed frequencies? A model can rank perfectly yet systematically over- or under-state PD, which would distort pricing and provisions. Measure with a calibration table (predicted versus observed by decile) and the **Brier score**, the mean squared difference between predicted probability and the 0/1 outcome.

Neither measure tells you whether the model is *economically* useful. That requires a decision rule tied to costs and benefits.

The economic decision rule. Suppose a loan of exposure EAD earns a margin m if repaid and loses `LGD × EAD` if the borrower defaults. Expected profit is `(1 − PD) m − PD × LGD × EAD`, which is positive when

```
PD < m / (m + LGD × EAD)
```

This **break-even PD** is the optimal approval cut-off for a calibrated model, and it makes the point that a threshold should come from the economics of the decision and not from the statistics (an arbitrary 0.5 probability threshold is meaningless for rare events).

Example 24.7 – Building and using a credit model. A lender has 20,000 commercial borrowers with five features: net debt/EBITDA, interest cover, current ratio, return on assets and log of assets. We simulate data in which the true default process includes an interaction (high leverage together with low cover raises risk beyond the linear effects), so a linear model is slightly misspecified. We fit logistic regression and a gradient-boosted tree model on 70 percent of the data and evaluate on the remaining 30 percent (6,000 borrowers). Exposure is 100, margin 4 and LGD 60 percent.

```
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import GradientBoostingClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import roc_auc_score, brier_score_loss, confusion_matrix
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline
rng = np.random.default_rng(2024)
n = 20_000
lev   = np.clip(rng.normal(3.0, 1.5, n), 0, None)          # net debt / EBITDA
cover = np.clip(rng.lognormal(1.6, 0.6, n), 0.2, None)     # EBITDA / interest
curr  = np.clip(rng.normal(1.4, 0.4, n), 0.3, None)        # current ratio
roa   = rng.normal(0.06, 0.05, n)
logsz = rng.normal(6.0, 1.2, n)
noise = rng.normal(0, 1, n)
# true default log-odds: includes an interaction (high leverage AND low cover) that a linear model misses
z = (-3.9 + 0.55 * lev - 0.45 * np.log(cover) - 0.9 * (curr - 1.4) - 9 * roa - 0.15 * (logsz - 6)
     + 1.2 * (lev > 4) * (cover < 3) + 0.3 * noise)
p = 1 / (1 + np.exp(-z))
y = rng.binomial(1, p)
X = np.column_stack([lev, cover, curr, roa, logsz])
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.3, random_state=1, stratify=y)
print("Default rate: %.3f  (train %d, test %d)" % (y.mean(), len(ytr), len(yte)))
logit = make_pipeline(StandardScaler(), LogisticRegression(C=1.0, max_iter=1000)).fit(Xtr, ytr)
gbm = GradientBoostingClassifier(n_estimators=150, max_depth=3, learning_rate=0.05, subsample=0.8,
random_state=0).fit(Xtr, ytr)
for name, m in [("Logistic", logit), ("GradBoost", gbm)]:
    ph = m.predict_proba(Xte)[:, 1]
    print("%-9s AUC=%.3f  Brier=%.4f  mean pred PD=%.3f" % (name, roc_auc_score(yte, ph),
brier_score_loss(yte, ph), ph.mean()))
print("Logistic coefficients (standardized):", dict(zip(["lev", "cover", "curr", "roa", "logsz"],
[round(float(c), 2) for c in logit[-1].coef_[0]])))
# decile calibration for logistic
ph = logit.predict_proba(Xte)[:, 1]
dec = np.digitize(ph, np.quantile(ph, np.linspace(0.1, 0.9, 9)))
print("Decile  mean PD  observed")
for d in (0, 4, 9):
    print("  %2d     %.3f    %.3f" % (d + 1, ph[dec == d].mean(), yte[dec == d].mean()))
# decision threshold: lend or decline. Loss given default 60% of 100 exposure; margin earned if no default
4
LGD, EAD, margin = 0.6, 100.0, 4.0
def exp_profit(ph, thr):
    lend = ph < thr
    return np.sum(np.where(yte[lend] == 1, -LGD * EAD, margin))
for thr in (0.02, 0.05, 0.065, 0.10, 0.20, 1.01):
    lend = ph < thr
    print("thr %.3f  approve %.1f%%  realized defaults in approved %.2f%%  profit %.0f" % (thr, 100 *
lend.mean(), 100 * yte[lend].mean(), exp_profit(ph, thr)))
print("Break-even PD = margin/(margin+LGD*EAD) = %.4f" % (margin / (margin + LGD * EAD)))
```

```
Default rate: 0.057  (train 14000, test 6000)
Logistic  AUC=0.794  Brier=0.0482  mean pred PD=0.059
GradBoost AUC=0.803  Brier=0.0472  mean pred PD=0.058
Logistic coefficients (standardized): {'lev': 0.87, 'cover': -0.51, 'curr': -0.42, 'roa': -0.36, 'logsz':
-0.23}
Decile  mean PD  observed
   1     0.005    0.005
   5     0.031    0.020
  10     0.227    0.247
thr 0.020  approve 31.4%  realized defaults in approved 1.22%  profit 6060
thr 0.050  approve 62.2%  realized defaults in approved 1.96%  profit 10264
thr 0.065  approve 70.3%  realized defaults in approved 2.23%  profit 10852
thr 0.100  approve 83.0%  realized defaults in approved 3.11%  profit 10012
thr 0.200  approve 95.0%  realized defaults in approved 4.37%  profit 6876
thr 1.010  approve 100.0%  realized defaults in approved 5.73%  profit 1984
Break-even PD = margin/(margin+LGD*EAD) = 0.0625
```

Step 1: Base rate and discrimination. The default rate is 5.7 percent. Logistic regression achieves AUC 0.794; boosted trees 0.803. The difference, of order 0.01, is of the same size as the sampling noise of an AUC computed on roughly 340 test defaults, so it should not be over-interpreted. Brier scores (0.0482 and 0.0472) are similarly close. The boosted model captures the interaction but pays for it in complexity; the improvement is real but small.

Step 2: Interpreting the logistic model. Standardized coefficients show leverage as the dominant driver (+0.87 per standard deviation, in log-odds), followed by interest cover (−0.51), current ratio (−0.42), ROA (−0.36) and size (−0.23). Signs match economic intuition. A reviewer can discuss each with the credit officers, a major governance advantage over a black box.

Step 3: Calibration. The mean predicted PD (5.9 percent) matches the observed rate (5.7 percent). In the riskiest decile the model predicts 22.7 percent versus an observed 24.7 percent. The middle decile shows a modest over-prediction (3.1 percent versus 2.0 percent). A calibration table makes visible a defect that AUC hides.

Step 4: The break-even cut-off. `PD* = 4 / (4 + 0.6 × 100) = 4/64 = 6.25%`. Check the profit for different cut-offs (profit measured in the same currency units as exposure, over the 6,000 test loans): approving everyone earns **1,984**; approving only borrowers with predicted PD below 2 percent earns 6,060 on 31.4 percent of the book; below 5 percent earns 10,264 (62.2 percent approved); below 6.5 percent earns **10,852** (70.3 percent approved, realized default rate in the approved group 2.2 percent); below 10 percent earns 10,012; and below 20 percent earns 6,876. Profit peaks almost exactly at the theoretical break-even of 6.25 percent, as the theory predicts.

Step 5: Interpretation. The model converts a book with a 5.7 percent default rate that barely earns anything (1,984) into a book earning more than five times as much by refusing the riskiest 30 percent. The marginal value of the more complex tree model has to be weighed against its explanation and governance cost; with AUC gains of about 0.01, a bank might reasonably keep the logistic model as the primary scorecard and use the tree model as a challenger. In practice, borrowers rejected by the model never reveal whether they would have defaulted (selection bias or “reject inference”), and the realized default rates will drift as the economy changes, so ongoing monitoring (Section 24.7) is not optional.

24.6.5 Beyond credit

The same toolkit applies to receivables collection, churn, anomaly detection in payments and journal entries, and nowcasting sales. Three cautions apply. Prediction does not identify causation, so counterfactual questions (“what if we change the price?”) need experiments or causal methods. Finance data are non-stationary and actors adapt, so a model

trained in a benign credit cycle may fail in a recession. And small samples (ten years of annual defaults for 200 firms) cannot support complex models; prefer simple models, shrinkage and expert priors.

24.7 Model Risk and Governance

24.7.1 What is model risk?

Model risk is the potential for adverse consequences from decisions based on incorrect or misused model outputs. It has two sources: *fundamental errors* (a wrong specification, wrong data, a coding bug, a misestimated parameter) and *misuse* (applying a model outside its intended domain, ignoring its limits, or treating its output as certainty). The 2011 supervisory guidance on model risk management issued in the United States (the Federal Reserve's SR 11-7, adopted also by the banking regulator OCC) has become a reference for good practice well beyond banks. Its core ideas are accessible and widely applicable:

- **Effective challenge:** critical analysis by objective, informed parties who can identify limitations and force changes.
- **Sound development, implementation and use:** documented purpose, design, data, assumptions, testing and limitations.
- **Independent validation:** review by people who did not build the model, covering conceptual soundness, ongoing monitoring and outcomes analysis (back-testing).
- **Governance:** a model inventory, assigned ownership, risk tiering, policies, and board and senior management oversight.

24.7.2 A practical governance framework for a finance team

Stage	Key controls	Questions to ask
Scoping	Written decision purpose; materiality tier	What decision will this support? What is the loss if it is wrong?
Data	Source lineage, quality checks, treatment of missing values and outliers, leakage review	Would this feature be known at decision time? Is the sample representative of future conditions?
Development	Benchmark against simple model; out-of-sample testing; sensitivity and stress tests	Does it beat the naive forecast? What breaks it?
Validation	Independent replication; challenger models; review of assumptions and code	Can someone else reproduce the numbers? Are results stable to reasonable changes?
Implementation	Version control; access controls; change log; reconciliation of production to development	Is the deployed model the validated model?
Monitoring	Drift metrics, back-testing, performance triggers, periodic revalidation	Is the population changing? Are predicted and realized outcomes aligned?
Use and override	Documented limitations; human override with logged reasons; escalation	Who can override, and how are overrides tracked and reviewed?

Spreadsheet models deserve special attention because they are the most common "models" in corporate finance and are often poorly controlled. Errors in spreadsheets are frequent; controls include separating inputs, calculations and outputs, using named ranges and consistent formulas, built-in checks (balance sheet balances, cash reconciles), peer review and version control.

24.7.3 Monitoring for drift

Models degrade when the population or relationships change. Two forms are important: **data (covariate) drift**, when the distribution of inputs changes, and **concept drift**, when the relationship between inputs and outcomes changes. A standard monitoring statistic for data drift is the **Population Stability Index (PSI)**, which compares the share of observations in each bin (typically deciles of the development sample) between development (a_i) and current (b_i) populations:

```
PSI = Σ_i (b_i − a_i) × ln(b_i / a_i)
```

A common rule of thumb treats PSI below 0.10 as stable, 0.10 to 0.25 as a moderate shift needing review, and above 0.25 as a significant shift. These thresholds are conventions, not laws of nature.

Example 24.8 – Detecting portfolio drift with PSI. A credit model was developed when borrower leverage averaged 3.0 (standard deviation 1.5). The current portfolio averages 3.6 (standard deviation 1.8). Compute the PSI on decile bins of the development data.

```
import numpy as np
rng = np.random.default_rng(5)
dev = rng.normal(3.0, 1.5, 20_000)              # leverage at model development
cur = rng.normal(3.6, 1.8, 5_000)               # leverage in current portfolio (drifted)
edges = np.quantile(dev, np.linspace(0, 1, 11)); edges[0], edges[-1] = -np.inf, np.inf
a = np.histogram(dev, edges)[0] / len(dev)
b = np.histogram(cur, edges)[0] / len(cur)
psi = np.sum((b - a) * np.log(b / a))
print("Bin shares (dev):", np.round(a, 3))
print("Bin shares (cur):", np.round(b, 3))
print("PSI = %.3f" % psi)
same = rng.normal(3.0, 1.5, 5_000)
c = np.histogram(same, edges)[0] / len(same)
print("PSI (no drift, fresh sample) = %.3f" % np.sum((c - a) * np.log(c / a)))
```

```
Bin shares (dev): [0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1]
Bin shares (cur): [0.074 0.079 0.069 0.074 0.081 0.081 0.099 0.106 0.125 0.212]
PSI = 0.130
PSI (no drift, fresh sample) = 0.002
```

Step 1: Bin shares. By construction each development decile holds 10 percent. In the current portfolio the highest-leverage bin holds 21.2 percent, more than double its development share, while the lowest bins have shrunk to 7 to 8 percent.

Step 2: PSI. Summing `(b − a) ln(b / a)` across the ten bins gives **0.130**. For comparison, a fresh sample drawn from the *development* distribution gives 0.002, which shows that sampling noise alone is negligible here.

Step 3: Interpretation. A PSI of 0.13 falls in the “review” zone. It does not prove the model is wrong, but it says the book now has twice as many borrowers in the riskiest leverage bucket as the model saw during development, exactly where its calibration is hardest. The proper response is to run the calibration table on recent outcomes, check whether the highest-risk bins are over- or under-predicting, and document the review. Drift monitoring is an early-warning sign, not a verdict.

24.7.4 Explainability, fairness and organizational risk

Where a model affects individuals, many jurisdictions require reasons for adverse decisions, restrict protected characteristics and test for disparate impact; proxy variables (such as geography) can produce discriminatory outcomes even when inputs look neutral. For corporate lending the legal constraints are lighter, but the need to explain a decision to a

client, auditor or regulator remains, so prefer models whose drivers can be named and treat post-hoc explanation tools as supplements. Quantitative methods also fail through organizational causes: automation bias (trusting a number because a computer produced it), gaming of forecasts and scorecards, model monoculture when everyone uses the same vendor model, and key-person risk. Counter these with challenger models from a different family, reporting of model limits alongside results, and clear accountability: a model has an owner, a validator and a user, and the user remains responsible for the decision.

A practical checklist for any quantitative analysis presented to a committee:

1. State the decision and the metric, not just the output.
2. Show the benchmark (naive forecast, deterministic NPV, current rule).
3. Show uncertainty: intervals, scenarios and the assumptions driving them.
4. Show out-of-sample or back-tested evidence, not just in-sample fit.
5. Name the three assumptions that would most change the answer, and test them.
6. State the model's domain of validity and what would trigger a review.

Case Study: Brightfield Packaging's Capital Committee

Illustrative case; Brightfield Packaging is a fictional company.

Brightfield Packaging is a mid-sized manufacturer of corrugated and flexible packaging with 1.4 billion in revenue, 11 plants and 190 million of EBITDA. Resin and paper-pulp costs are volatile, and about 40 percent of revenue is under customer contracts with only partial cost pass-through. The capital committee has 150 million of capex to allocate this year (and 60 million next year) across 12 proposals: plant automation, new converting lines, a recycling facility, a digital-printing line and several maintenance replacements.

Historically the committee ranked proposals by profitability index. This year the CFO, Adaeze Okonkwo, has asked the finance analytics team to supply simulation and optimization support. The team built a Monte Carlo model for each project with three stochastic drivers (pulp price, resin price, and regional volume), calibrated on ten years of data. They fitted a factor model so that all projects share a macro volume factor and an energy-cost factor. For the largest project, a new 60 million converting line, the simulation gives an expected NPV of 11 million and a standard deviation of 34 million, with a 38 percent probability of a negative NPV; with independent drivers the standard deviation falls to 22 million. The team admits that the correlation between pulp and resin prices is estimated from only 40 quarterly observations and was much higher in the last two years than in the full sample.

The optimization, run on expected NPVs and two budget years, selected seven projects with total NPV of 63 million; the profitability-index ranking would have delivered 57 million. The year-1 budget has a shadow price of zero but the year-0 budget has a shadow price of 0.31. Two sponsors have objected: one says her project was excluded only because of a "black-box" model; another says the optimizer favours projects with the most optimistic forecasts, which the sponsors' own incentives encourage.

Separately, the credit team presented a machine-learning model of customer default that raised AUC from 0.76 (the existing logistic scorecard) to 0.81. The model was built on five years of data, of which two include the 2020s inflation spike. A vendor supplied the model and an independent validator has only been able to replicate the headline AUC, not the feature engineering. The head of risk is reluctant to approve it for production and wants the old scorecard retained as a challenger; the head of sales wants the new model because it would approve 8 percent more volume at the same predicted loss.

The CFO now needs to decide how to use the quantitative work in the committee, how to adjust for its weaknesses, and how to govern the credit model.

Discussion Questions

1. For the 60 million converting line, the standard deviation of NPV is 34 million with the estimated correlations and 22 million with independence. Explain the likely source of the difference, what you would need to check about the 40-observation correlation estimate, and how you would present the project's risk to the committee.
2. The optimizer's solution beats the profitability-index ranking by 6 million NPV. Explain why a PI ranking can fail with two budgets and indivisible projects, interpret the 0.31 shadow price, and propose how to respond to the sponsor's concern about optimistic forecasts (the "optimizer's curse").
3. Evaluate whether the credit model with AUC 0.81 should be approved. Which additional evidence (calibration, out-of-time test, replication of feature engineering, stability, explainability, economic cut-off analysis) would you require, and would you accept the sales head's argument about 8 percent more volume?
4. Design a governance framework (roles, validation, monitoring triggers, overrides) for the finance analytics models at Brightfield, including a metric such as PSI and an escalation path if drift exceeds a threshold.

Key Takeaways

- Point estimates hide the quantities that matter for decisions under uncertainty: the probability of loss, the interaction of risks and the value of flexibility. Simulation, optimization and prediction address these in complementary ways.
- Monte Carlo error falls with √N; always report the standard error, and check the model first by reproducing the deterministic result.
- Dependence changes both the spread and the mean of outcomes: positive price-cost correlation reduces margin risk, while shocks that hit price and volume together increase it. Cholesky factorization generates correlated normal shocks, but linear correlation understates tail dependence.
- Choose stochastic processes by economics: random walks for permanent shocks, mean reversion for commodities and margins, regimes and jumps for structural shifts, and report parameter uncertainty.
- Corporate risk measures such as cash-flow-at-risk, minimum-liquidity probability and covenant-breach probability are path-dependent and are best computed from simulated paths.
- Capital rationing is an optimization problem. The LP relaxation provides an upper bound and shadow prices; integer solutions can beat profitability-index ranking; shadow prices measure the internal cost of capital.
- Mean-variance allocation is convex and tractable but highly sensitive to expected-return inputs; use caps, shrinkage and scenario views.
- Evaluate predictive models out of sample with walk-forward validation; in-sample fit rewards complexity, and a model must beat a naive benchmark to justify itself.
- Judge a credit model on discrimination (AUC), calibration and economics: the approval cut-off should come from the break-even PD, `m / (m + LGD × EAD)`, not a default threshold.
- Model risk is managed by effective challenge, independent validation, monitoring for drift (for example PSI), clear ownership, and human accountability for the decision.

Review Questions

1. Why is the probability of a negative NPV often more informative than the mean NPV of a simulation? Under what circumstances might the two lead to different decisions?
2. How does the standard error of a Monte Carlo estimate depend on the number of trials, and why are tail probabilities harder to estimate than means?
3. Explain how antithetic variates reduce variance and when they help least.
4. Write the variance of the difference of two variables in terms of their correlation. How would you change a firm's contracts to reduce margin risk, in terms of this formula?
5. Describe how Cholesky factorization produces correlated random shocks, and why an estimated correlation matrix might fail the factorization.
6. Why is a Gaussian (multivariate normal) dependence assumption dangerous for tail risk, and what practical remedies exist?
7. Contrast random-walk and mean-reverting price models. Which would you use for a commodity input, and how does the choice change the 10-year cash-flow-at-risk?
8. What is a shadow price in a capital rationing problem, and how should a manager use it?
9. Why can ranking projects by profitability index be suboptimal, and what does the integer program add?
10. Distinguish discrimination from calibration in a credit model. Why does a lender need both, and how should the approval cut-off be set?

Problems

1. **Monte Carlo accuracy.** A simulation of project NPV has a standard deviation of 50 across trials. (a) How many trials give a standard error of the mean of 0.25? (b) How many are needed to halve that standard error? (c) Compute the number of trials needed so that the estimated probability of loss, believed to be about 5 percent, has a relative standard error of 10 percent.
2. **Correlation and margin risk.** A product's price has a standard deviation of 12 and its unit cost 5. Compute the standard deviation of unit margin (price minus cost) if the correlation is (a) +0.5 and (b) −0.2.
3. **Cholesky.** Two shocks should have a correlation of −0.4. (a) Write the 2 × 2 Cholesky factor. (b) If independent draws are $Z_1 = 1.0$ and $Z_2 = -0.5$, compute the correlated shocks X_1 and X_2.
4. **Mean reversion.** Log-price follows `x_{t+1} = x_t + 0.2 (m − x_t) + noise` with long-run price 60. Current price is 90. Compute the expected log-price (and the price implied by exponentiating it) after 4 quarters and after 8 quarters, ignoring the Jensen correction.
5. **Capital rationing.** Four projects (NPV, outlay in millions): A (12, 50), B (20, 80), C (9, 30), D (15, 60). Budget 140. (a) Find the best whole-project portfolio. (b) Find the LP relaxation value and the shadow price of the budget.
6. **Minimum variance.** Two business units have volatilities of 10 percent and 20 percent and correlation 0.3. Compute the minimum-variance weight of the first unit and the resulting portfolio volatility.
7. **Credit cut-off.** A loan has margin 3, LGD 50 percent, exposure 100. (a) Find the break-even PD. (b) Compute the expected profit per loan for PD of 4 percent and 8 percent.
8. **PSI.** A model's development sample has four equal bins (25 percent each). The current portfolio shares are 15, 20, 30 and 35 percent. Compute the PSI and classify it by the rule of thumb in Section 24.7.3.

Answers to Selected Problems

1. a. $N = (\sigma/SE)^2 = (50 / 0.25)^2 =$ **40,000**. (b) Halving SE requires four times as many trials: **160,000**. (c) Relative $SE = \sqrt{((1 - p) / (pN))} = 0.10$, so $N = (1 - p) / (p \times 0.01) = 0.95 / (0.05 \times 0.01) =$ **1,900** trials.

2. $Var = 12^2 + 5^2 - 2\rho(12)(5)$. (a) $\rho = +0.5$: $144 + 25 - 60 = 109$, $\sigma =$ **10.44**. (b) $\rho = -0.2$: $144 + 25 + 24 = 193$, $\sigma =$ **13.89**.

3. a. $L = [[1, 0], [-0.4, \sqrt{(1 - 0.16)}]] = [[1, 0], [-0.4, 0.9165]]$. (b) $X_1 = 1.0$; $X_2 = -0.4 \times 1.0 + 0.9165 \times (-0.5) =$ **−0.858**.

4. $\ln 90 = 4.4998$, $\ln 60 = 4.0943$, gap 0.4055. After 4 quarters: $0.4055 \times 0.8^4 = 0.1661$, so $x = 4.2604$, price = **70.8**. After 8 quarters: $0.4055 \times 0.8^8 = 0.0680$, $x = 4.1623$, price = **64.2**.

5. a. Feasible portfolios within 140: A+C+D = 140 (NPV 36); B+D = 140 (35); A+B = 130 (32); best is **A, C, D with NPV 36**. (b) PIs: C 0.30, B 0.25, D 0.25, A 0.24. LP: fund C (30), B (80), and half of D (30 of 60): NPV = 9 + 20 + 7.5 = **36.5**; the shadow price is the marginal PI, **0.25** per unit of budget.

6. $w_1 = (0.20^2 - 0.3 \times 0.10 \times 0.20) / (0.10^2 + 0.20^2 - 2 \times 0.3 \times 0.10 \times 0.20) = (0.04 - 0.006) / (0.01 + 0.04 - 0.012) = 0.034 / 0.038 =$ **0.8947**. Variance $= 0.8947^2 \times 0.01 + 0.1053^2 \times 0.04 + 2 \times 0.8947 \times 0.1053 \times 0.3 \times 0.02 = 0.008006 + 0.000443 + 0.001130 = 0.009579$; volatility = **9.79 percent**.

7. a. $PD^* = 3 / (3 + 50) =$ **5.66 percent**. (b) At PD 4 percent: $0.96 \times 3 - 0.04 \times 50 = 2.88 - 2.00 =$ **+0.88**. At PD 8 percent: $0.92 \times 3 - 0.08 \times 50 = 2.76 - 4.00 =$ **−1.24**.

8. $PSI = (0.15 - 0.25) \ln(0.15/0.25) + (0.20 - 0.25) \ln(0.20/0.25) + (0.30 - 0.25) \ln(0.30/0.25) + (0.35 - 0.25) \ln(0.35/0.25) = 0.05108 + 0.01116 + 0.00912 + 0.03365 =$ **0.105**, in the 0.10 to 0.25 “review” range.